IB CHEMISTRY HL

Past-Paper Review Package

Questions - only the topics of Lessons 1.1 to 9.4

Every question below comes from an IB Chemistry HL past paper (May 2010 - May 2025: Papers 1, 2 and 3; 2025: Papers 1A, 1B and 2) and was kept only if it tests the content of these lessons.
Each item shows its source (session, time zone, paper, question). Answers are in the companion markscheme under the same labels, e.g. 9.4-M3 (multiple choice) or 9.4-S2 (structured).
Dashed rules mark places where out-of-scope parts of a structured question were removed.
A copy of the IB Chemistry data booklet is needed for many questions.

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LESSON 1.1

Inorganic Nomenclature

11 multiple-choice questions · 5 structured questions

Part A - Multiple choice

1.1-M1 May 2025 TZ3 · Paper 1A · Q6

6. What is the formula of the compound formed by aluminium ions and sulfate(VI) ions?

A. $AlSO_3$

B. $Al_2(SO_3)_3$

C. $AlSO_4$

D. $Al_2(SO_4)_3$

1.1-M2 May 2022 TZ1 · Paper 1 · Q9

9. A compound consists of the ions Ca^{2+} and PO_4^{3-}. What are the name and formula of the compound?

	Name	Formula
A.	calcium phosphorus oxide	$CaPO_4$
B.	calcium phosphorus oxide	$Ca_3(PO_4)_2$
C.	calcium phosphate	$CaPO_4$
D.	calcium phosphate	$Ca_3(PO_4)_2$

1.1-M3 May 2019 TZ1 · Paper 1 · Q10

10. What is the IUPAC name of $NiCO_3$?

A. nickel(II) carbonate

B. nickel carbonate

C. nickel(I) carbonate

D. nitrogen(I) carbonate

1.1-M4 May 2018 TZ2 · Paper 1 · Q9

9. What is the formula of magnesium nitride?

A. MgN

B. Mg_2N_3

C. Mg_3N

D. Mg_3N_2

1.1-M5 May 2016 · Paper 1 · Q30

30. Applying IUPAC rules, what is the name of MnO_2?

A. Magnesium(II) oxide

B. Manganese(II) oxide

C. Magnesium(IV) oxide

D. Manganese(IV) oxide

1.1-M6 May 2015 TZ2 · Paper 1 · Q9

9. The formula of gallium phosphate is $GaPO_4$. What is the correct formula of gallium sulfate?

A. $GaSO_4$

B. GaS

C. $Ga_2(SO_4)_3$

D. Ga_2S_3

1.1-M7 Nov 2013 · Paper 1 · Q9

9. What is the formula of calcium nitride?

A. Ca_3N_2

B. Ca_2N_3

C. $Ca(NO_2)_2$

D. $Ca(NO_3)_2$

1.1-M8 Nov 2013 · Paper 1 · Q31

31. What is the name of MnO_2?

A. Manganese(II) oxide

B. Magnesium(II) oxide

C. Manganese(IV) oxide

D. Magnesium(IV) oxide

1.1-M9 May 2013 TZ1 · Paper 1 · Q9

9. What are the correct formulas of the following ions?

	Nitrate	Sulfate	Phosphate	Hydrogencarbonate
A.	NO_3^-	SO_4^{2-}	PO_4^{3-}	HCO_3^-
B.	NO_3^-	SO_4^{2-}	PO_3^{3-}	HCO_3^{2-}
C.	NO_2^-	SO_4^-	PO_4^{3-}	HCO_3^-
D.	NO_2^-	SO_3^{2-}	PO_3^{3-}	HCO_3^{2-}

1.1-M10 Nov 2012 · Paper 1 · Q30

30. What is the correct systematic name of MnO_2?

A. Manganese(II) oxide

B. Manganese(IV) oxide

C. Magnesium(II) oxide

D. Magnesium(IV) oxide

1.1-M11 Nov 2011 · Paper 1 · Q13

13. What are the correct formulas of the following ions?

	Nitrate	Phosphate	Carbonate	Ammonium
A.	NO_3^-	PO_4^{3-}	CO_3^-	NH_3^+
B.	NO_3^{2-}	PO_3^{2-}	CO_3^{2-}	NH_3^+
C.	NO_3^-	PO_4^{3-}	CO_3^{2-}	NH_4^+
D.	NO_3^{2-}	PO_3^{2-}	CO_3^{2-}	NH_4^+

Part B - Structured questions

1.1-S1 May 2019 TZ1 · Paper 2 · Q4 parts (b)(v)

4. This question is about the decomposition of hydrogen peroxide.

(b) Hydrogen peroxide decomposes to water and oxygen when a catalyst such as potassium iodide, KI, is added.

$$2H_2O_2(aq) \xrightarrow{KI(aq)} O_2(g) + 2H_2O(l)$$

(v) MnO_2 is another possible catalyst for the reaction. State the IUPAC name for MnO_2. [1]

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. . . . . .

1.1-S2 May 2015 TZ1 · Paper 3 · Q17 parts (b)(i)

17. The oxygen levels in water can change for a number of reasons.

(b) The use of phosphate fertilizers can also produce changes in the oxygen concentrations in a river.

(i) Phosphate ions can be removed from a solution by adding calcium ions. State the ionic equation for the reaction of calcium ions with phosphate ions. [1]

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1.1-S3 May 2014 TZ2 · Paper 3 · Q19 parts (a)

19. One method of removing heavy metal ions from a solution is by precipitation.

(a) State an ionic equation, including state symbols, for the reaction taking place when an aqueous solution containing chloride ions is added to an aqueous solution containing lead(II) ions. *[2]*

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1.1-S4 May 2013 TZ1 · Paper 3 · QD1 parts (b)(i), (b)(iii) · also 6.1

D1. Depressants can have different effects depending on their doses.

(b) A breathalyser containing crystals of potassium dichromate(VI) can be used by the police to detect whether a driver has consumed alcohol.

(i) State the chemical formula for potassium dichromate(VI). *[1]*

. . . . . .

(iii) State the oxidation number of chromium in the product. *[1]*

. . . . . .

1.1-S5 May 2010 TZ1 · Paper 3 · QE3 parts (c)

(c) Phosphate ions can be removed by adding calcium or aluminium ions. Give the ionic equation for the reaction of aluminium ions with phosphate ions. *[1]*

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LESSON 1.2

Stoichiometry

77 multiple-choice questions · 55 structured questions

Part A - Multiple choice

1.2-M1 May 2025 TZ2 · Paper 1A · Q23

23. What are the limiting reactant and theoretical yield of hydrogen when 0.20 mol of magnesium reacts with 0.20 mol of hydrochloric acid?

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

	Limiting reactant	Theoretical yield of H_2 / mol
A.	Mg	0.10
B.	Mg	0.20
C.	HCl	0.05
D.	HCl	0.10

1.2-M2 May 2025 TZ3 · Paper 1A · Q1

1. How many oxygen atoms are present in 0.0200 mol $CuSO_4 \cdot 5H_2O$?

A. 1.20×10^{22}

B. 1.08×10^{22}

C. 6.02×10^{23}

D. 1.08×10^{23}

1.2-M3 May 2025 TZ3 · Paper 1A · Q30

30. How many moles of oxygen are needed for the complete combustion of 11.5 g of $C_3H_8(g)$?

M_r propane $= 44.1\,g\,mol^{-1}$

A. 10.4

B. 5.20

C. 2.60

D. 1.30

1.2-M4 Nov 2024 · Paper 1 · Q1

1. How many moles of phosphate ions, PO_4^{3-}, are there in 103.39 g of $Ca_3(PO_4)_2$?
$M_r = 310.18$

A. 0.11

B. 0.33

C. 0.67

D. 2.00

1.2-M5 Nov 2024 · Paper 1 · Q2

2. What is the sum of the coefficients for the balanced equation of the combustion of iron(II) sulphide using the smallest whole numbers?

$$_FeS(s) + _O_2(g) \rightarrow _Fe_2O_3(s) + _SO_2(g)$$

A. 6

B. 7

C. 14

D. 17

1.2-M6 May 2024 TZ1 · Paper 1 · Q38

38. A 50.0 cm^3 sample of sodium hydroxide solution, NaOH(aq), weighed 53.894 g at 25 °C. What is the density of the solution in g cm^{-3}?

A. 1.08

B. 1.078

C. 1.0779

D. 1.07788

1.2-M7 May 2024 TZ2 · Paper 1 · Q3

3. 6.00 mol of copper, Cu, are mixed with 12.00 mol of dilute nitric acid, $HNO_3(aq)$. The equation for the reaction that occurs is shown below.

$$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$$

What is the amount, in mol, of nitrogen(II) oxide, NO, produced assuming that the reaction goes to completion?

A. 3.00

B. 4.00

C. 8.00

D. 18.00

1.2-M8 Nov 2023 TZ1 · Paper 1 · Q1

1. Ammonia reacts with oxygen to produce nitrogen (II) oxide and water.

$$___\ NH_3(g) + ___\ O_2(g) \rightarrow ___\ NO(g) + ___\ H_2O(l)$$

What is the NH_3:O_2 ratio in the balanced equation?

A. 2:5

B. 4:5

C. 1:1

D. 2:1

1.2-M9 Nov 2023 TZ1 · Paper 1 · Q2

2. Metal M reacts with 16.0 g of sulfur to produce 26.0 g of the compound MS_2. What is the relative atomic mass of M?

A. 5

B. 10

C. 20

D. 40

1.2-M10 Nov 2023 TZ1 · Paper 1 · Q3

3. 64 g of methane and 96 g of oxygen are reacted according to the equation.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

What would be found in the reaction vessel at completion of the reaction?

A. $CO_2(g)$ and $H_2O(l)$ only

B. $O_2(g)$, $CO_2(g)$ and $H_2O(l)$ only

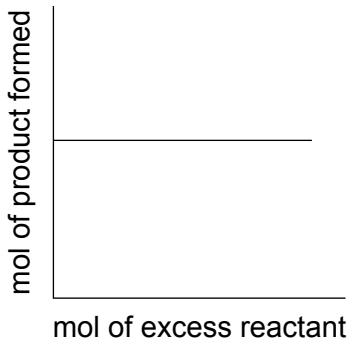
C. $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ only

D. $CH_4(g)$, $O_2(g)$, $CO_2(g)$ and $H_2O(l)$

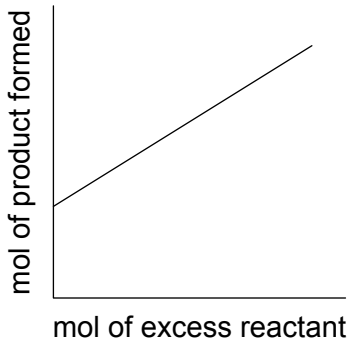
1.2-M11 Nov 2023 TZ1 · Paper 1 · Q38

38. Which graph shows the relationship between quantity of product formed and quantity of excess reactant after the limiting reactant is consumed?

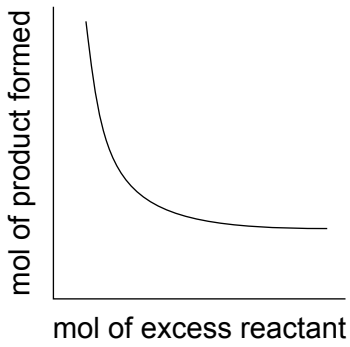
A.



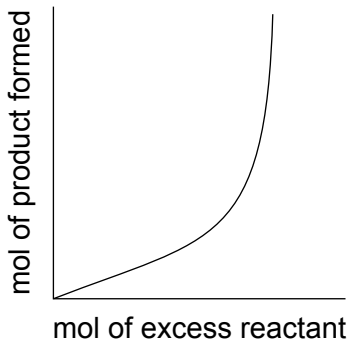
B.



C.



D.



1.2-M12 May 2023 TZ2 · Paper 1 · Q1

1. What is the mass of one molecule of C_{60}?

$$N_A = 6.0 \times 10^{23}$$

A. $1.0 \times 10^{-22}\,g$

B. $2.0 \times 10^{-23}\,g$

C. $8.3 \times 10^{-24}\,g$

D. $1.2 \times 10^{-21}\,g$

1.2-M13 Nov 2022 · Paper 1 · Q1

1. How many oxygen atoms are present in $0.0500\,mol\ Ba(OH)_2 \cdot 8H_2O$?

$$N_A = 6.02 \times 10^{23}$$

A. 3.01×10^{23}

B. 6.02×10^{23}

C. 3.01×10^{24}

D. 6.02×10^{24}

1.2-M14 Nov 2022 · Paper 1 · Q3

3. How many moles of carbon dioxide are produced by the complete combustion of $7.0\,g$ of ethene, $C_2H_4(g)$?

$$M_r = 28$$

A. 0.25

B. 0.5

C. 0.75

D. 1.0

1.2-M15 May 2022 TZ1 · Paper 1 · Q1

1. 2.67 g of lead(II) carbonate is decomposed by heating until constant mass.

$$PbCO_3(s) \rightarrow PbO(s) + CO_2(g)$$

What is the final mass of solid?

A. $0.44\,g$

B. $2.23\,g$

C. $2.67\,g$

D. $3.11\,g$

1.2-M16 May 2022 TZ1 · Paper 1 · Q2

2. 0.02 mol of zinc is added to 10.0 cm^3 of 1.0 mol dm^{-3} hydrochloric acid.

$$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

How many moles of hydrogen are produced?

A. 0.005

B. 0.01

C. 0.02

D. 0.04

1.2-M17 May 2022 TZ2 · Paper 1 · Q1

1. What is the concentration of chloride ions, in mol dm^{-3}, in a solution formed by mixing 200 cm^3 of 1 mol dm^{-3} HCl with 200 cm^3 of 5 mol dm^{-3} NaCl?

A. 1

B. 2

C. 3

D. 6

1.2-M18 Nov 2021 · Paper 1 · Q1

1. How much ethanol contains 1.20×10^{24} atoms of carbon?

Avogadro's constant, L or N_A: 6.02×10^{23} mol^{-1}

A. 0.333 mol

B. 0.500 mol

C. 1.00 mol

D. 2.00 mol

1.2-M19 Nov 2021 · Paper 1 · Q2

2. 3.00 mol of C_3H_8 is mixed with 20.00 mol of O_2. Which quantity is present at the end of the reaction

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

A. 1.00 mol of C_3H_8

B. 5.00 mol of O_2

C. 12.00 mol of CO_2

D. 16.00 mol of H_2O

1.2-M20 May 2021 TZ1 · Paper 1 · Q1

1. Which contains the most atoms of oxygen?

A. 64 g of O_2

B. 1.2×10^{24} molecules of O_2

C. 64 g of $C_3H_5O_3$

D. 1.2×10^{24} molecules of $C_3H_5O_3$

1.2-M21 May 2021 TZ2 · Paper 1 · Q1

1. 0.20 mol of magnesium is mixed with 0.10 mol of hydrochloric acid.

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Which is correct?

	Limiting reagent	Maximum yield of H_2 / mol
A.	HCl	0.10
B.	Mg	0.20
C.	HCl	0.05
D.	Mg	0.10

1.2-M22 May 2021 TZ2 · Paper 1 · Q2

2. Which amount, in mol, of sodium chloride is needed to make 250 cm^3 of 0.10 mol dm^{-3} solution?

A. 4.0×10^{-4}

B. 0.025

C. 0.40

D. 25

1.2-M23 May 2021 TZ2 · Paper 1 · Q3

3. What is the sum of the coefficients when the equation is balanced with whole numbers?

$$__Sn(OH)_4(aq) + __NaOH(aq) \rightarrow __Na_2SnO_3(aq) + __H_2O(l)$$

A. 4

B. 5

C. 6

D. 7

1.2-M24 Nov 2020 · Paper 1 · Q1

1. What is the number of carbon atoms in 12 g of ethanoic acid CH_3COOH, $M_r = 60$?

A. 0.20

B. 2.0

C. 1.2×10^{23}

D. 2.4×10^{23}

1.2-M25 Nov 2019 · Paper 1 · Q1

1. 0.10 mol of hydrochloric acid is mixed with 0.10 mol of calcium carbonate.

$$2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Which is correct?

	Limiting reagent	Maximum yield of CO_2 / mol
A.	$HCl(aq)$	0.10
B.	$CaCO_3(s)$	0.05
C.	$HCl(aq)$	0.05
D.	$CaCO_3(s)$	0.10

1.2-M26 Nov 2019 · Paper 1 · Q2

2. What is the sum of the coefficients when the equation is balanced with whole numbers?

$$__MnO_2(s) + __HCl(aq) \rightarrow __MnCl_2(aq) + __H_2O(l) + __Cl_2(g)$$

A. 6

B. 7

C. 8

D. 9

1.2-M27 Nov 2019 · Paper 1 · Q3

3. Which contains the greatest number of moles of oxygen atoms?

A. 0.05 mol $Mg(NO_3)_2$

B. 0.05 mol $C_6H_4(NO_2)_2$

C. 0.1 mol H_2O

D. 0.1 mol NO_2

1.2-M28 May 2019 TZ2 · Paper 1 · Q1

1. How many moles of magnesium hydroxide are produced with 0.50 mol of ammonia?

$$Mg_3N_2(s) + 6H_2O(l) \rightarrow 3Mg(OH)_2(aq) + 2NH_3(aq)$$

A. 0.25

B. 0.33

C. 0.75

D. 1.5

1.2-M29 May 2019 TZ2 · Paper 1 · Q38

38. The following data were recorded for determining the density of three samples of silicon, Si.

Mass / g ±0.01 g	**Volume / cm^3 ±0.1 cm^3**
5.61	2.8
4.32	1.7
6.37	2.8

Which average density value, in $g\,cm^{-3}$, has been calculated to the correct number of significant figures?

A. 2

B. 2.3

C. 2.27

D. 2.273

1.2-M30 Nov 2018 · Paper 1 · Q1

1. How many moles of FeS_2 are required to produce 32 g of SO_2? (A_r: S = 32, O = 16)

$$4FeS_2(s) + 11O_2(g) \rightarrow 2Fe_2O_3(s) + 8SO_2(g)$$

A. 0.25

B. 0.50

C. 1.0

D. 2.0

1.2-M31 Nov 2018 · Paper 1 · Q3

3. An antacid tablet containing 0.50 g of $NaHCO_3$ ($M_r = 84$) is dissolved in water to give a volume of 250 cm^3. What is the concentration, in mol dm^{-3}, of HCO_3^- in this solution?

A. $\frac{0.250 \times 84}{0.50}$

B. $\frac{0.50}{84 \times 0.250}$

C. $\frac{250 \times 84}{0.50}$

D. $\frac{0.50}{84 \times 250}$

1.2-M32 Nov 2018 · Paper 1 · Q18

18. Samples of sodium carbonate powder were reacted with separate samples of excess hydrochloric acid.

$$Na_2CO_3(s) + 2HCl(aq) \rightarrow CO_2(g) + 2NaCl(aq) + H_2O(l)$$

Reaction I: 1.0 g $Na_2CO_3(s)$ added to 0.50 mol dm^{-3} $HCl(aq)$
Reaction II: 1.0 g $Na_2CO_3(s)$ added to 2.0 mol dm^{-3} $HCl(aq)$

What is the same for reactions I and II?

A. Initial rate of reaction

B. Total mass of CO_2 produced

C. Total reaction time

D. Average rate of production of CO_2

1.2-M33 May 2018 TZ1 · Paper 1 · Q3

3. What is the percentage yield when 7 g of ethene produces 6 g of ethanol?

M_r (ethene) = 28 and M_r (ethanol) = 46

$$C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(g)$$

A. $\frac{6 \times 7 \times 100}{28 \times 46}$

B. $\frac{6 \times 46 \times 100}{7 \times 28}$

C. $\frac{6 \times 28}{7 \times 46 \times 100}$

D. $\frac{6 \times 28 \times 100}{7 \times 46}$

1.2-M34 May 2018 TZ2 · Paper 1 · Q1

1. What is the number of atoms of oxygen in 2.0 mol of hydrated sodium carbonate, $Na_2CO_3 \cdot 10H_2O$? Avogadro's constant, L or N_A : $6.02 \times 10^{23}\,mol^{-1}$

A. 6

B. 26

C. 3.6×10^{24}

D. 1.6×10^{25}

1.2-M35 May 2018 TZ2 · Paper 1 · Q2

2. What is the volume, in cm^3, of the final solution if 100 cm^3 of a solution containing 1.42 g of sodium sulfate, Na_2SO_4, is diluted to the concentration of 0.020 mol dm^{-3}?
$M_r\,(Na_2SO_4) = 142$

A. 50

B. 400

C. 500

D. 600

1.2-M36 May 2018 TZ2 · Paper 1 · Q3

3. What is the percentage yield when 2.0 g of ethene, C_2H_4, is formed from 5.0 g of ethanol, C_2H_5OH?
M_r (ethene) = 28; M_r (ethanol) = 46

A. $\frac{2.0}{28} \times \frac{5.0}{46} \times 100$

B. $\frac{\frac{2.0}{28}}{\frac{5.0}{46}} \times 100$

C. $\frac{28}{2.0} \times \frac{5.0}{46} \times 100$

D. $\frac{\frac{28}{2.0}}{\frac{5.0}{46}} \times 100$

1.2-M37 Nov 2017 · Paper 1 · Q1

1. How many atoms of nitrogen are there in 0.50 mol of $(NH_4)_2CO_3$?

A. 1

B. 2

C. 3.01×10^{23}

D. 6.02×10^{23}

1.2-M38 May 2017 TZ1 · Paper 1 · Q39

39. What is the density, in $g\,cm^{-3}$, of a 34.79 g sample with a volume of 12.5 cm^3?

A. 0.359

B. 0.36

C. 2.783

D. 2.78

1.2-M39 Nov 2016 · Paper 1 · Q1

1. Which volume, in cm^3, of 0.20 mol dm^{-3} NaOH (aq) is needed to neutralize 0.050 mol of H_2S (g)?

$$H_2S\,(g) + 2NaOH\,(aq) \rightarrow Na_2S\,(aq) + 2H_2O\,(l)$$

A. 0.25

B. 0.50

C. 250

D. 500

1.2-M40 Nov 2016 · Paper 1 · Q3

3. 5.0 mol of Fe_2O_3 (s) and 6.0 mol of CO (g) react according to the equation below. What is the limiting reactant and how many moles of the excess reactant remain unreacted?

$$Fe_2O_3\,(s) + 3CO\,(g) \rightarrow 2Fe\,(s) + 3CO_2\,(g)$$

	Limiting reactant	Moles of excess reactant remaining
A.	CO	2.0
B.	CO	3.0
C.	Fe_2O_3	1.0
D.	Fe_2O_3	2.0

1.2-M41 Nov 2015 · Paper 1 · Q3

3. What is the percentage yield when 1.1 g of ethanal, CH_3CHO, is obtained from 4.6 g of ethanol, CH_3CH_2OH? $M_r(CH_3CH_2OH) = 46$; $M_r(CH_3CHO) = 44$

$$CH_3CH_2OH(l) + [O] \rightarrow CH_3CHO(l) + H_2O(l)$$

A. $\frac{1.1 \times 46 \times 100}{44 \times 4.6}$

B. $\frac{1.1 \times 100}{4.6}$

C. $\frac{4.6 \times 44 \times 100}{4.6 \times 1.1}$

D. $\frac{1.1 \times 46}{44 \times 4.6}$

1.2-M42 May 2015 TZ1 · Paper 1 · Q1

1. 4.0 g of solid sodium hydroxide is added to 0.10 dm^3 of 1.0 mol dm^{-3} aqueous sulfuric acid.

$$2NaOH(s) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Which statement is correct?

A. Neither reactant is in excess.

B. 0.10 mol Na_2SO_4 is formed.

C. Excess H_2SO_4 remains in solution.

D. Excess NaOH remains in solution.

1.2-M43 May 2015 TZ1 · Paper 1 · Q3

3. Which solution contains the biggest amount, in mol, of chloride ions?

A. 20 cm^3 of 0.50 mol dm^{-3} NH_4Cl

B. 60 cm^3 of 0.20 mol dm^{-3} $MgCl_2$

C. 70 cm^3 of 0.30 mol dm^{-3} NaCl

D. 100 cm^3 of 0.30 mol dm^{-3} $ClCH_2COOH$

1.2-M44 Nov 2014 · Paper 1 · Q1

1. 0.040 mol of $(NH_4)_2Ni(SO_4)_2 \cdot 6H_2O$ is dissolved in water to give 200 cm^3 of aqueous solution. What is the concentration, in mol dm^{-3}, of ammonium ions?

A. 0.00040

B. 0.0080

C. 0.20

D. 0.40

1.2-M45 May 2014 TZ1 · Paper 1 · Q3

3. 100.0 cm^3 of a 0.50 mol dm^{-3} solution of $BaCl_2$ is added to 50.0 cm^3 of a 0.10 mol dm^{-3} solution of Na_2SO_4. A precipitate of $BaSO_4$ is formed according to the equation below.

$$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$$

What is the amount, in mol, of $BaSO_4$ produced?

A. 0.0050

B. 0.010

C. 0.050

D. 0.10

1.2-M46 May 2014 TZ2 · Paper 1 · Q1

1. What is the mass, in g, of one mole of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, given the following relative atomic mass values?

Element	Cu	S	H	O
Relative atomic mass	64	32	1	16

A. 160

B. 178

C. 186

D. 250

1.2-M47 May 2014 TZ2 · Paper 1 · Q2

2. An excess of calcium carbonate is added to a solution containing $0.10\,\text{mol}$ of HCl(aq). What mass of calcium carbonate reacts, and what mass of carbon dioxide is formed?

$$\text{Mass of one mole of } \text{CaCO}_3 = 100\,\text{g}$$

$$\text{Mass of one mole of } \text{CO}_2 = 44\,\text{g}$$

$$\text{CaCO}_3\text{(s)} + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{H}_2\text{O(l)} + \text{CO}_2\text{(g)}$$

	$\text{CaCO}_3\text{(s)}$ / g	$\text{CO}_2\text{(g)}$ / g
A.	10	4.4
B.	10	2.2
C.	5.0	2.2
D.	5.0	4.4

1.2-M48 Nov 2013 · Paper 1 · Q1

1. What is the total number of oxygen atoms in $0.200\,\text{mol}$ of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$?

A. 1.20

B. 6.00

C. 1.20×10^{23}

D. 7.22×10^{23}

1.2-M49 Nov 2013 · Paper 1 · Q2

2. What are the coefficients of $\text{H}_2\text{SO}_4\text{(aq)}$ and $\text{H}_3\text{PO}_4\text{(aq)}$ when the following equation is balanced using the smallest possible whole numbers?

$$___\,\text{Ca}_3(\text{PO}_4)_2\text{(s)} + ___\,\text{H}_2\text{SO}_4\text{(aq)} \rightarrow ___\,\text{CaSO}_4\text{(s)} + ___\,\text{H}_3\text{PO}_4\text{(aq)}$$

	Coefficient of $\text{H}_2\text{SO}_4\text{(aq)}$	Coefficient of $\text{H}_3\text{PO}_4\text{(aq)}$
A.	1	2
B.	2	3
C.	3	1
D.	3	2

1.2-M50 Nov 2013 · Paper 1 · Q3

3. 7.102 g of Na_2SO_4 ($M = 142.04\,g\,mol^{-1}$) is dissolved in water to prepare $0.5000\,dm^3$ of solution. What is the concentration of Na_2SO_4 in $mol\,dm^{-3}$?

A. 2.500×10^{-2}

B. 1.000×10^{-1}

C. 1.000×10

D. 1.000×10^{2}

1.2-M51 Nov 2013 · Paper 1 · Q40

40. A student measured the mass and volume of a piece of silver and recorded the following values.

Mass of empty weighing bottle	1.0800 g
Mass of weighing bottle with piece of silver	11.5700 g
Volume of silver	$1.00\,cm^3$

Which value, in $g\,cm^{-3}$, for the density of silver should the student report in her laboratory notebook?

A. 10.49

B. 10.4900

C. 10.5

D. 10.500

1.2-M52 May 2013 TZ1 · Paper 1 · Q1

1. What is the whole number ratio of the coefficients of ammonia to oxygen when the following equation is balanced correctly?

$$___\,NH_3(g) + ___\,O_2(g) \rightarrow ___\,NO(g) + ___\,H_2O(l)$$

A. 1 : 2

B. 2 : 1

C. 4 : 5

D. 5 : 4

1.2-M53 May 2013 TZ2 · Paper 1 · Q1

1. How many atoms are present in 0.10 mol of $PtCl_2(NH_3)_2$?

A. 6.0×10^{22}

B. 3.0×10^{23}

C. 6.6×10^{23}

D. 6.6×10^{24}

1.2-M54 May 2013 TZ2 · Paper 1 · Q2

2. What mass of carbon dioxide, $CO_2(g)$, in g, is produced when 5.0 g of calcium carbonate, $CaCO_3(s)$, reacts completely with hydrochloric acid, $HCl(aq)$?

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

A. 0.050

B. 2.2

C. 4.4

D. 5.0

1.2-M55 Nov 2012 · Paper 1 · Q1

1. What is the number of ions in 0.20 mol of $(NH_4)_3PO_4$?

A. 8.0×10^{-1}

B. 1.2×10^{23}

C. 4.8×10^{23}

D. 2.4×10^{24}

1.2-M56 Nov 2012 · Paper 1 · Q2

2. The equation for the reduction of iron(III) oxide is:

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$$

What mass of carbon dioxide, in g, is produced by the complete reduction of 80 g of iron(III) oxide?

A. 44

B. 66

C. 88

D. 132

1.2-M57 May 2012 TZ1 · Paper 1 · Q1

1. How many atoms of hydrogen are in 0.500 mol of CH_3OH molecules?

A. 1.20×10^{23}

B. 3.01×10^{23}

C. 6.02×10^{23}

D. 1.20×10^{24}

1.2-M58 May 2012 TZ1 · Paper 1 · Q2

2. Calcium carbonate reacts with hydrochloric acid according to the following equation.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

What is the theoretical yield, in mol, of calcium chloride if 0.10 mol $CaCO_3$ is added to 100 cm^3 of 1.0 mol dm^{-3} HCl?

A. 0.050

B. 0.10

C. 0.20

D. 0.50

1.2-M59 May 2012 TZ2 · Paper 1 · Q1

1. What is the total number of atoms in 0.100 mol of $[Pt(NH_3)_2Cl_2]$?

A. 11

B. 6.02×10^{22}

C. 3.01×10^{23}

D. 6.62×10^{23}

1.2-M60 May 2012 TZ2 · Paper 1 · Q2

2. Nitroglycerine, $C_3H_5N_3O_9$, can be used in the manufacture of explosives. What is the coefficient of $C_3H_5N_3O_9(l)$ when the equation for its decomposition reaction is balanced using the lowest whole numbers?

$$___\, C_3H_5N_3O_9(l) \rightarrow ___\, CO_2(g) + ___\, H_2O(l) + ___\, N_2(g) + ___\, O_2(g)$$

A. 2

B. 4

C. 20

D. 33

1.2-M61 Nov 2011 · Paper 1 · Q1

1. How many oxygen **atoms** are in 0.100 mol of $CuSO_4 \cdot 5H_2O$?

A. 5.42×10^{22}

B. 6.02×10^{22}

C. 2.41×10^{23}

D. 5.42×10^{23}

1.2-M62 Nov 2011 · Paper 1 · Q2

2. What is the sum of the coefficients when the following equation is balanced using whole numbers?

$$___\, Fe_2O_3(s) + ___\, CO(g) \rightarrow ___\, Fe(s) + ___\, CO_2(g)$$

A. 5

B. 6

C. 8

D. 9

1.2-M63 Nov 2011 · Paper 1 · Q5

5. What is the amount, in moles, of sulfate ions in 100 cm^3 of 0.020 $mol\,dm^{-3}$ $FeSO_4(aq)$?

A. 2.0×10^{-3}

B. 2.0×10^{-2}

C. 2.0×10^{-1}

D. 2.0

1.2-M64 Nov 2011 · Paper 1 · Q40

40. A student heated a solid in a crucible. The student measured the mass of the solid and crucible before and after heating and recorded the results.

$$\text{Mass of crucible and solid before heating} = 101.692\ \text{g}$$

$$\text{Mass of crucible and solid after heating} = 89.312\ \text{g}$$

What value should the student record for the mass lost in grams?

A. 12.4

B. 12.38

C. 12.380

D. 12.3800

1.2-M65 May 2011 TZ1 · Paper 1 · Q1

1. 1.7 g of $NaNO_3$ ($M_r = 85$) is dissolved in water to prepare 0.20 dm^3 of solution. What is the concentration of the resulting solution in mol dm^{-3}?

A. 0.01

B. 0.1

C. 0.2

D. 1.0

1.2-M66 May 2011 TZ1 · Paper 1 · Q2

2. What mass, in g, of hydrogen is formed when 3 mol of aluminium react with excess hydrochloric acid according to the following equation?

$$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$$

A. 3.0

B. 4.5

C. 6.0

D. 9.0

1.2-M67 May 2011 TZ1 · Paper 1 · Q4

4. What is the sum of all coefficients when the following equation is balanced using the smallest possible whole numbers?

$$__ C_2H_2 + __O_2 \rightarrow __CO_2 + __H_2O$$

A. 5

B. 7

C. 11

D. 13

1.2-M68 May 2011 TZ2 · Paper 1 · Q1

1. What is the total number of hydrogen atoms in 1.0 mol of benzamide, $C_6H_5CONH_2$?

A. 7

B. 6.0×10^{23}

C. 3.0×10^{24}

D. 4.2×10^{24}

1.2-M69 May 2011 TZ2 · Paper 1 · Q2

2. Chloroethene, C_2H_3Cl, reacts with oxygen according to the equation below.

$$2C_2H_3Cl(g) + 5O_2(g) \rightarrow 4CO_2(g) + 2H_2O(g) + 2HCl(g)$$

What is the amount, in mol, of H_2O produced when 10.0 mol of C_2H_3Cl and 10.0 mol of O_2 are mixed together, and the above reaction goes to completion?

A. 4.00

B. 8.00

C. 10.0

D. 20.0

1.2-M70 May 2011 TZ2 · Paper 1 · Q3

3. What is the concentration of NaCl, in mol dm^{-3}, when 10.0 cm^3 of 0.200 mol dm^{-3} NaCl solution is added to 30.0 cm^3 of 0.600 mol dm^{-3} NaCl solution?

A. 0.450

B. 0.300

C. 0.500

D. 0.800

1.2-M71 May 2011 TZ2 · Paper 1 · Q40

40. A piece of metallic aluminium with a mass of 10.044 g was found to have a volume of 3.70 cm^3. A student carried out the following calculation to determine the density.

$$\text{Density (g cm}^{-3}) = \frac{10.044}{3.70}$$

What is the best value the student could report for the density of aluminium?

A. 2.715 g cm^{-3}

B. 2.7 g cm^{-3}

C. 2.71 g cm^{-3}

D. 2.7146 g cm^{-3}

1.2-M72 Nov 2010 · Paper 1 · Q2

2. 300 cm^3 of water is added to a solution of 200 cm^3 of 0.5 mol dm^{-3} sodium chloride. What is the concentration of sodium chloride in the new solution?

A. 0.05 mol dm^{-3}

B. 0.1 mol dm^{-3}

C. 0.2 mol dm^{-3}

D. 0.3 mol dm^{-3}

1.2-M73 May 2010 TZ1 · Paper 1 · Q1

1. What is the approximate molar mass, in g mol^{-1}, of $MgSO_4 \cdot 7H_2O$?

A. 120

B. 130

C. 138

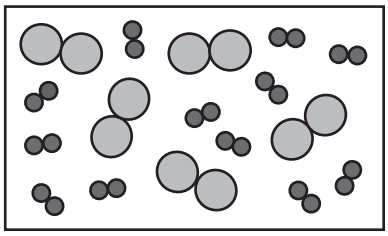
D. 246

1.2-M74 May 2010 TZ1 · Paper 1 · Q3

3. 12 molecules of hydrogen gas, $H_2(g)$, and 5 molecules of oxygen gas, $O_2(g)$, were mixed together under conditions which allowed the reaction to go to completion, according to the following equation.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

The following diagram represents the mixture of reactants.



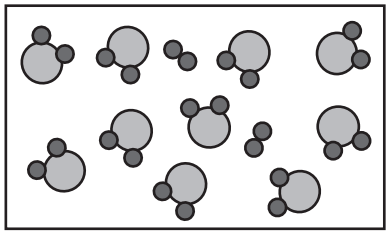
Key:

Oxygen atom

Hydrogen atom

Which diagram represents the reaction mixture when the reaction was complete?

A.



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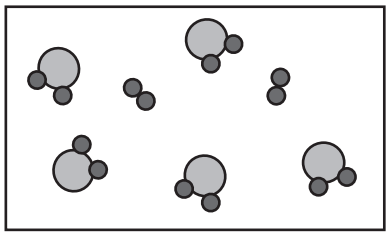

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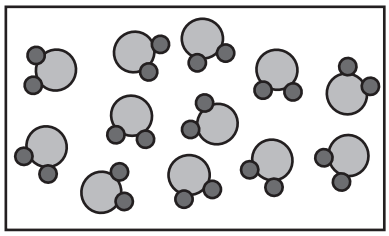

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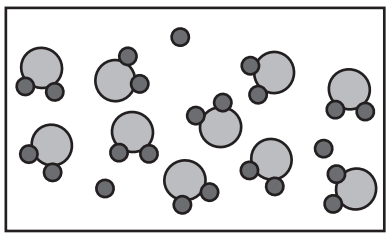


1.2-M75 May 2010 TZ2 · Paper 1 · Q1

1. What is the mass, in g, of one molecule of ethane, C_2H_6?

A. 3.0×10^{-23}

B. 5.0×10^{-23}

C. 30

D. 1.8×10^{25}

1.2-M76 May 2010 TZ2 · Paper 1 · Q2

2. 6.0 mol of aluminium reacts with oxygen to form aluminium oxide. What is the amount of oxygen, in mol, needed for complete reaction?

$$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$$

A. 1.5

B. 3.0

C. 4.5

D. 6.0

1.2-M77 May 2010 TZ2 · Paper 1 · Q39

39. How many significant figures are there in 0.00370?

A. 2

B. 3

C. 5

D. 6

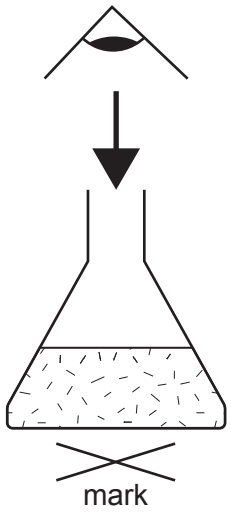
Part B - Structured questions

1.2-S1 May 2025 TZ1 · Paper 1B · Q1 parts (a)

1. A student investigated the effect of concentration on the rate of reaction between sodium thiosulfate, $Na_2S_2O_3$, and hydrochloric acid, HCl.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow S(s) + 2NaCl(aq) + SO_2(g) + H_2O(l)$$

Since the solid sulfur product is insoluble, the rate can be determined by measuring the time it takes for the clear solution to turn off-white or pale yellow until the X mark on a white tile below the flask can no longer be seen.



(a) Determine the mass of sodium thiosulfate needed to make 500.0 cm^3 of a 0.1500 mol dm^{-3} solution. [2]

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1.2-S2 May 2025 TZ2 · Paper 1B · Q1 parts (d)(iv), (d)(v)

1. The concentration of calcium ions, Ca^{2+}, in aqueous solution can be determined by titration with ethylenediaminetetraacetic acid (EDTA), or by atomic absorption spectrophotometry.

(d) A standard stock solution of calcium chloride was diluted to produce several solutions with accurately known concentrations. The absorbance of these solutions was measured in a spectrophotometer that was calibrated with distilled water.

(iv) The 4.00 ppm solution was made by diluting 2.00 cm^3 of the stock solution to 25.00 cm^3. Calculate the concentration, in ppm, of the stock solution. [1]

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(v) Calculate the concentration, in mol dm^{-3}, of Ca^{2+} in the 4.00 ppm solution. 1 ppm = 1 mg dm^{-3}. [1]

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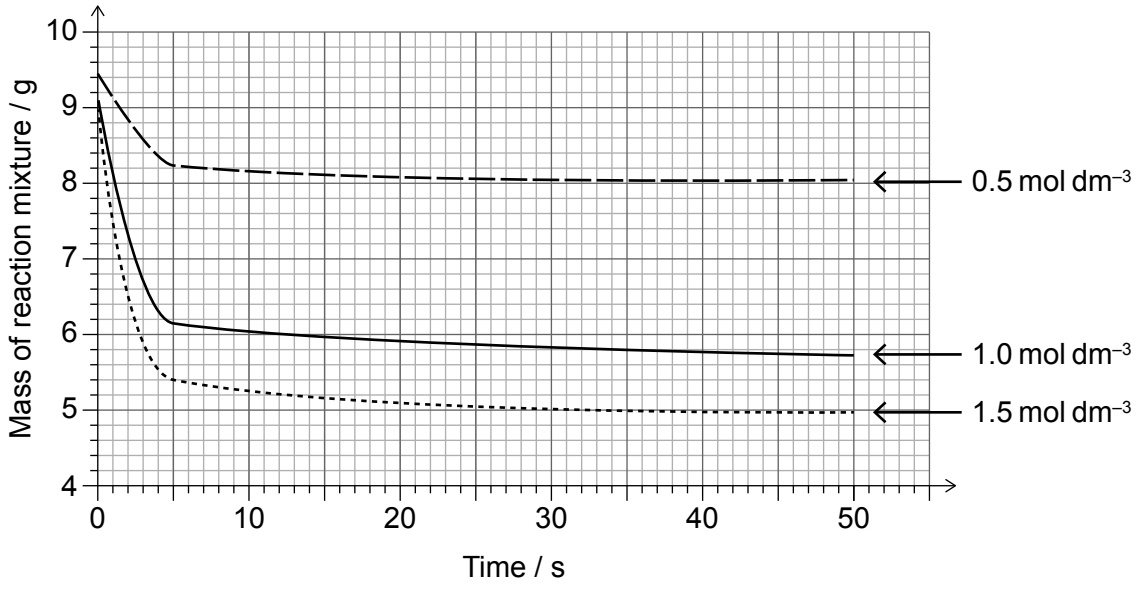
1.2-S3 May 2025 TZ2 · Paper 1B · Q3 parts (b)

3. The reaction of solid sodium hydrogencarbonate, $NaHCO_3(s)$, with hydrochloric acid, $HCl(aq)$, was investigated using the same mass of solid and a constant volume of different concentrations of acid. A data logger was used to measure the mass of the reaction mixture every five seconds.

The results were graphed.



(b) Suggest, with a reason, whether $NaHCO_3(s)$ **or** $HCl(aq)$ was the limiting reagent in the reactions using 0.5 and $1.0\,mol\,dm^{-3}$ $HCl(aq)$. [1]

Limiting reagent: . . . . . .

Reason: . . . . . .

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1.2-S4 May 2025 TZ2 · Paper 2 · Q4 parts (b)

4. Strychnine, $C_{21}H_{22}N_2O_2$ ($M_r = 334.4$), is a white crystalline solid obtained from plants.

(b) 48.73 g of strychnine was converted into its sulfate by the reaction:

$$2C_{21}H_{22}N_2O_2(aq) + H_2SO_4(aq) \rightarrow (C_{21}H_{23}N_2O_2)_2SO_4(aq)$$

Determine the percentage yield if 51.41 g of product was obtained. Use section 7 of the data booklet. [2]

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1.2-S5 May 2025 TZ3 · Paper 1B · Q1 parts (a)(ii), (a)(iii)

1. Nitrogen dioxide, NO_2, is a brown, toxic and corrosive gas. It can be made in a school laboratory by heating a group II metal nitrate or by the reaction of copper, Cu, with concentrated nitric acid, HNO_3.

(ii) Deduce the coefficients in the equation for the reaction of Cu with concentrated HNO_3. [1]

$$___Cu(s) + ___HNO_3(aq) \rightarrow ___Cu(NO_3)_2(aq) + ___NO_2(g) + ___H_2O(l)$$

(iii) Calculate the mass, in g, of Cu required to make 0.0100 moles of NO_2. Use section 7 of the data booklet. [1]

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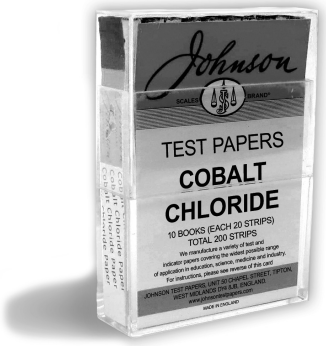
1.2-S6 May 2025 TZ3 · Paper 1B · Q3 parts (b)(i), (b)(v)

3. Cobalt(II) chloride has been used for many years as a qualitative test for water, H_2O.



Hydrated cobalt(II) chloride has the formula $CoCl_2 \cdot \mathbf{X}H_2O$ where **X** can be 1, 2, 6 or 9.

A teacher set a task to determine the value of **X**. Hydrated cobalt(II) chloride is pink in colour and anhydrous cobalt(II) chloride is blue.

A group of students weighed 10.000 g of hydrated cobalt(II) chloride and left it overnight in an oven at 30 °C. They observed a slight colour change from the original pink as a result.

Each student heated the hydrated cobalt(II) chloride until no further observable change was seen, allowed it to cool and recorded the mass.

The following equation illustrates the process occurring:

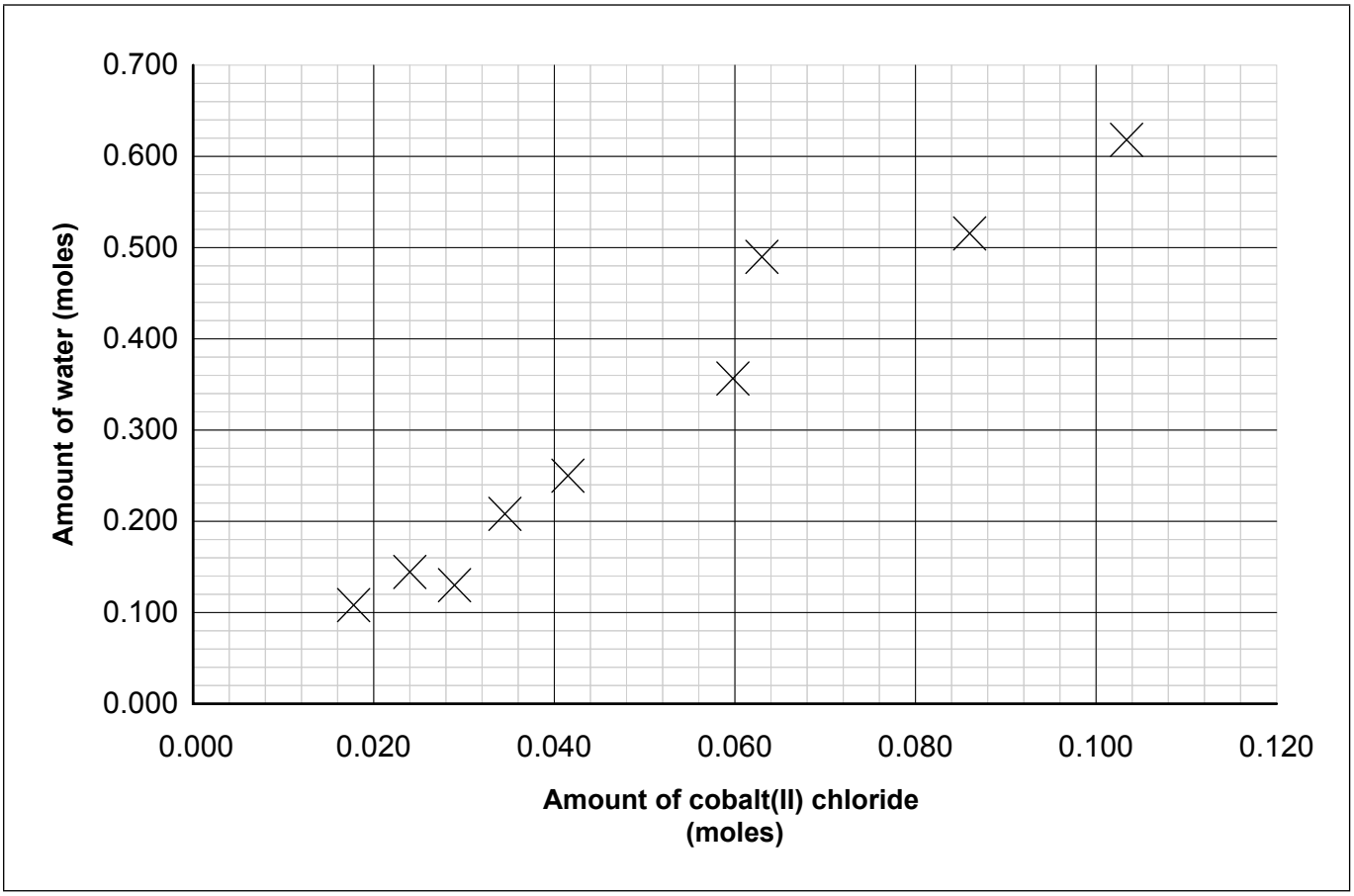
$$CoCl_2 \cdot \mathbf{X}H_2O(s) \rightarrow CoCl_2(s) + \mathbf{X}H_2O(g)$$

(b) Each of the students carried out the same experiment using different masses of hydrated cobalt(II) chloride. The shared results are shown below.

Experiment number	Mass of $CoCl_2 \cdot XH_2O$	Mass of $CoCl_2$	Loss in mass (g)	Amount of $CoCl_2$ / mol	Amount of H_2O / mol
1	4.256	2.299	1.957	0.018	0.109
2	5.711	3.117	2.594	0.024	0.144
3	6.853	4.501	2.352	0.029	0.131
4	8.224	4.470	3.754	0.035	0.208
5	9.868	5.372	4.496	0.041	0.250
6	11.842	6.481		0.050	
7	14.210	7.761	6.449	0.060	0.358
8	17.052	8.222	8.830	0.063	0.490
9	20.463	11.172	9.291	0.086	0.516
10	24.555	13.396	11.159	0.103	0.619

(i) Determine the missing values in the table above (M_r $H_2O = 18.02\,g\,mol^{-1}$). [1]

A graph of the results was plotted.



(v) Determine the value of **X**. [1]

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1.2-S7 Nov 2024 · Paper 2 · Q7 parts (a), (b), (c) · also 6.1

7. The Winkler Method is used to find the concentration of oxygen in water.

200 cm^3 of water was taken from a river and analysed using this method. The reactions taking place are summarized.

Step 1: $2Mn^{2+}(aq) + 4OH^-(aq) + O_2(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$

Step 2: $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$

Step 3: $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

(a) Identify, with a reason, the species which is reduced in step 1. [1]

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(b) 4.8×10^{-3} moles of I^- were formed in step 3. Determine the number of moles of O_2 in the water sample. [1]

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(c) Calculate the concentration of oxygen, in g dm^{-3}, in the water sample. If you did not obtain an answer in (b), use 2.0×10^{-3} mol although this is not the correct answer. [1]

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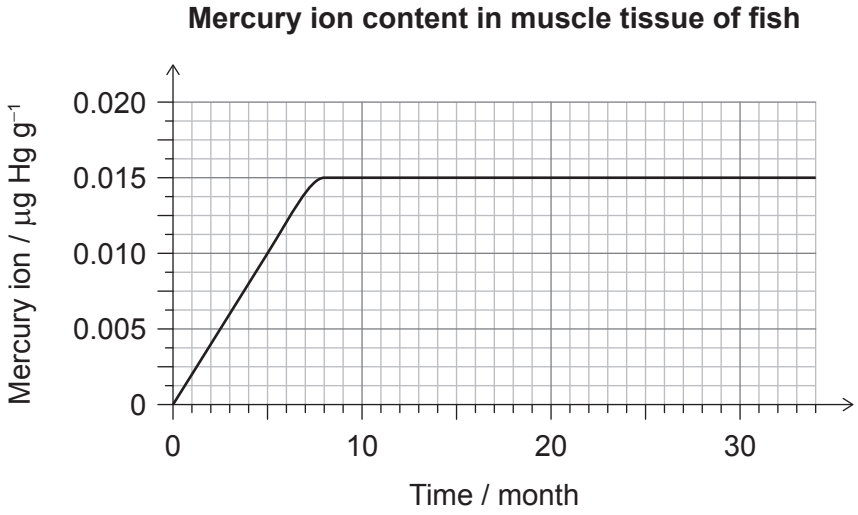
1.2-S8 Nov 2024 · Paper 3 · Q1 parts (b)(i)

1. Water contaminated with mercury contains methylmercury ions, CH_3Hg^+. These ions are absorbed by living organisms, then slowly metabolized and excreted.

Young fish were taken at regular intervals from a mercury contaminated lake and tested to determine mercury ion content.



(b) The concentration of mercury ion in a sample of the fish is $0.0052 \pm 0.0001\,\mu g\,Hg\,g^{-1}$.

(i) Calculate the mass of Hg, in μg, in 3.723 g of the sample. [1]

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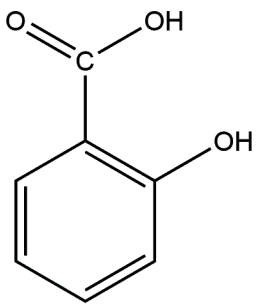
1.2-S9 Nov 2024 · Paper 3 · Q16 parts (a)(i), (a)(ii), (a)(iii) · also 5.3

16. Aspirin and morphine are analgesics.

(a) Aspirin was synthesized by mixing 0.897 g of salicylic acid with excess ethanoic anhydride.

M_r (salicylic acid) = 138.13. M_r (aspirin) = 180.17.

Salicylic acid



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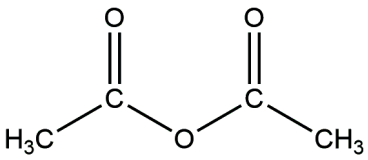
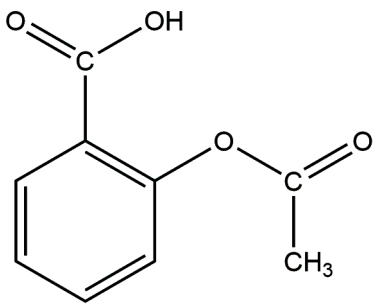


Ethanoic anhydride

H^+

Aspirin



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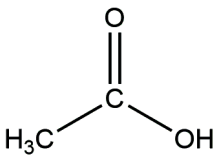

Ethanoic acid

(i) Calculate the theoretical yield of aspirin, in g. [1]

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(ii) Water was added and 1.31 g of solid product was isolated. Suggest, giving a reason, the identity of **one** possible impurity. [1]

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(iii) Outline how the aspirin sample could be purified by recrystallization. [3]

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1.2-S10 May 2024 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(iii)

1. A powder has the following percentage composition by mass:

30.0 % sucrose, $C_{12}H_{22}O_{11}$

45.0 % citric acid, $C_6H_8O_7$

25.0 % sodium hydrogencarbonate, $NaHCO_3$

In the presence of water, the powder effervesces as the citric acid reacts with the sodium hydrogencarbonate:

$$3NaHCO_3(s) + C_6H_8O_7(aq) \rightarrow Na_3(C_6H_5O_7)(aq) + 3CO_2(g) + 3H_2O(l)$$

(a) (i) Determine the limiting reactant when 1.00 g of this powder reacts. [3]

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(iii) Calculate the percentage yield obtained by a student who collected 0.043 dm^3 of carbon dioxide from 1.00 g of the powder.

If you did not obtain an answer to (a)(ii), use 0.068 dm^3, but this is not the correct value. [1]

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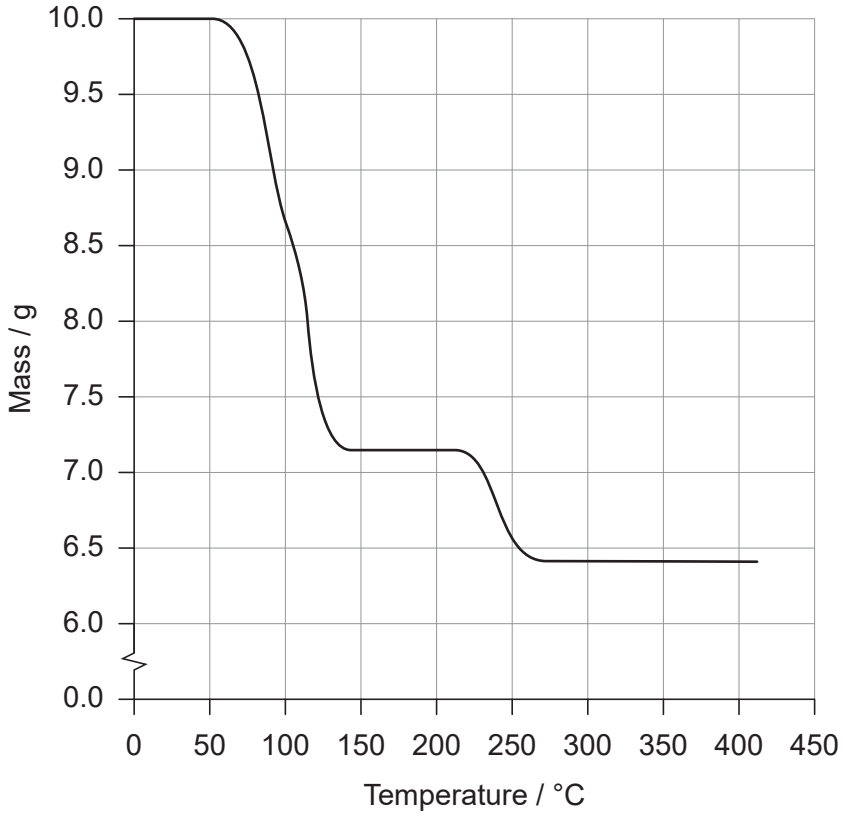
1.2-S11 May 2024 TZ1 · Paper 3 · Q2 parts (a)

2. The **Figure 3** shows the change in mass when 10.0 g of copper (II) sulfate crystals are heated from 20 °C to 400 °C. Only water vapour is lost during this process and no further water vapour is lost on heating to higher temperatures.

Figure 3



(a) Demonstrate that this data is consistent with the formula $CuSO_4 \cdot 5H_2O$. [2]

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1.2-S12 Nov 2023 TZ1 · Paper 2 · Q3 parts (b)

3. Methanoic acid can be converted into methyl methanoate, $HCOOCH_3$.

(b) 1.72 g of methyl methanoate is produced from 2.83 g of methanoic acid and excess of the other reagent. Determine the percentage yield. [2]

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1.2-S13 May 2023 TZ1 · Paper 3 · Q14 parts (c)

14. Energy production often has associated costs. Global warming potential (GWP) is a measure of heat absorbed by a greenhouse gas relative to the heat absorbed by the same mass of carbon dioxide.

(c) The GWP of methane is approximately 21. Estimate the heat absorbed by one mole of methane relative to a mole of carbon dioxide. [2]

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1.2-S14 May 2023 TZ2 · Paper 2 · Q5 parts (a)(iii)

5. Methanoic acid is a monoprotic weak acid.

(a) The concentration of methanoic acid was found by titration with a 0.200 mol dm^{-3} standard solution of sodium hydroxide, NaOH (aq), using an indicator to determine the end point.

(iii) 22.5 cm^3 of NaOH (aq) neutralized 25.0 cm^3 of methanoic acid. Determine the concentration of the methanoic acid. [1]

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1.2-S15 May 2023 TZ2 · Paper 3 · Q2 parts (a)(ii), (c)(iii)

2. To investigate how much kale would supply the daily recommended intake of iron a student:

1. weighed 79.6 g of kale leaves and blended with 500 cm^3 of water
2. boiled, filtered and cooled
3. pipetted 10.0 cm^3 of the filtrate into 20.0 cm^3 of 2.00 mol dm^{-3} sulfuric acid in a flask
4. titrated with 0.00100 mol dm^{-3} potassium manganate(VII).

The reaction taking place is:

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

(a) All species are almost colourless except for MnO_4^-, which has an intense purple colour, though the kale extract is coloured by the chlorophyll present.

(ii) Outline how the addition of distilled water to the 10.0 cm^3 aliquot before titration will affect the titrant volume at the end point. [1]

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(c) The end point occurred when 3.1 ± 0.1 cm^3 of the titrant had been added.

(iii) The solution in the titration flask contained 8.66×10^{-4} g of iron. Determine, to three significant figures, the percentage of iron, by mass, in the kale leaves. [2]

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1.2-S16 Nov 2022 · Paper 2 · Q5 parts (a), (c), (f) · also 3.1, 5.1

5. Lignite, a type of coal, contains about 0.40 % sulfur by mass.

(a) Calculate the amount, in mol, of sulfur dioxide produced when 500.0 g of lignite undergoes combustion. [2]

$$S(s) + O_2(g) \rightarrow SO_2(g)$$

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(c) Deduce the Lewis (electron dot) structure for sulfur dioxide. [1]

(f) SF_4Cl_2 can form two isomers, one which is polar and another non-polar. Deduce the 3-dimensional representations of both isomers of SF_4Cl_2. [2]

Non-polar isomer:

Polar isomer:

1.2-S17 May 2022 TZ1 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(iii), (c)(i), (c)(ii), (d)(i), (d)(iii), (e)(iv) · also 2.1, 6.1, 7.3

1. When heated in air, magnesium ribbon reacts with oxygen to form magnesium oxide.

(a) (i) Write a balanced equation for the reaction that occurs. [1]

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(b) The reaction in (a)(i) was carried out in a crucible with a lid and the following data was recorded:

Mass of crucible and lid = 47.372 ±0.001 g

Mass of crucible, lid and magnesium ribbon before heating = 53.726 ±0.001 g

Mass of crucible, lid and product after heating = 56.941 ±0.001 g

(i) Calculate the amount of magnesium, in mol, that was used. [1]

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(iii) Assume the reaction in (a)(i) is the only one occurring and it goes to completion, but some product has been lost from the crucible. Deduce the percentage yield of magnesium oxide in the crucible. [2]

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(c) When magnesium is burnt in air, some of it reacts with nitrogen to form magnesium nitride according to the equation:

$$3\,Mg\,(s) + N_2\,(g) \rightarrow Mg_3N_2\,(s)$$

(i) Evaluate whether this, rather than the loss of product, could explain the yield found in (b)(iii). [1]

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(ii) Suggest an explanation, other than product being lost from the crucible or reacting with nitrogen, that could explain the yield found in (b)(iii). [1]

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(d) The presence of magnesium nitride can be demonstrated by adding water to the product. It is hydrolysed to form magnesium hydroxide and ammonia.

(i) Calculate coefficients that balance the equation for the following reaction. [1]

$$__\ Mg_3N_2\,(s) + __\ H_2O\,(l) \rightarrow __\ Mg(OH)_2\,(s) + __\ NH_3\,(aq)$$

(iii) Determine the oxidation state of nitrogen in Mg_3N_2 and in NH_3. [1]

Mg_3N_2: . . . . . .

NH_3: . . . . . .

(e) Most nitride ions are $^{14}N^{3-}$.

(iv) Suggest, giving a reason, whether magnesium or nitrogen would have the greater sixth ionization energy. [1]

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1.2-S18 May 2022 TZ2 · Paper 2 · Q1 parts (a), (b)(i), (c) · also 6.1

1. Lithium reacts with water to form an alkaline solution.

(a) Determine the coefficients that balance the equation for the reaction of lithium with water. [1]

$$\ldots \mathrm{Li(s)} + \ldots \mathrm{H_2O(l)} \rightarrow \ldots \mathrm{LiOH(aq)} + \ldots \mathrm{H_2(g)}$$

(b) A 0.200 g piece of lithium was placed in 500.0 cm^3 of water.

(i) Calculate the molar concentration of the resulting solution of lithium hydroxide. [2]

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(c) The reaction of lithium with water is a redox reaction. Identify the oxidizing agent in the reaction giving a reason. [1]

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1.2-S19 Nov 2021 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 6.2

6. Biochemical oxygen demand (BOD) can be determined by the Winkler Method.

(c) A 25.00 cm^3 sample of water was treated according to the Winkler Method.

Step I: $2Mn^{2+}(aq) + O_2(g) + 4OH^-(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$

Step II: $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$

Step III: $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

The iodine produced was titrated with 37.50 cm^3 of 5.000×10^{-4} mol dm^{-3} $Na_2S_2O_3$.

(i) Calculate the amount, in moles of $Na_2S_2O_3$ used in the titration. [1]

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(ii) Deduce the mole ratio of O_2 consumed in step I to $S_2O_3^{2-}$ used in step III. [1]

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(iii) Calculate the concentration of dissolved oxygen, in mol dm^{-3}, in the sample. [2]

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(iv) The three steps of the Winkler Method are redox reactions.

Deduce the reduction half-equation for step II. [1]

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1.2-S20 Nov 2020 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(ii), (b)(iii), (b)(v), (b)(vi) · also 2.1, 6.1

1. Chlorine undergoes many reactions.

(a) (i) State the full electron configuration of the chlorine atom. [1]

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(b) 2.67 g of manganese(IV) oxide was added to 200.0 cm^3 of 2.00 mol dm^{-3} HCl.

$$MnO_2(s) + 4HCl(aq) \rightarrow Cl_2(g) + 2H_2O(l) + MnCl_2(aq)$$

(i) Calculate the amount, in mol, of manganese(IV) oxide added. [1]

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(ii) Determine the limiting reactant, showing your calculations. [2]

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(iii) Determine the excess amount, in mol, of the other reactant. [1]

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(v) State the oxidation state of manganese in MnO_2 and $MnCl_2$. [2]

MnO_2:

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$MnCl_2$:

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(vi) Deduce, referring to oxidation states, whether MnO_2 is an oxidizing or reducing agent. [1]

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1.2-S21 Nov 2019 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii)

2. The biochemical oxygen demand of a water sample can be determined by the following series of reactions. The final step is titration of the sample with sodium thiosulfate solution, $Na_2S_2O_3(aq)$.

$$2Mn^{2+}(aq) + O_2(aq) + 4OH^-(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$$

$$MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$$

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

A student analysed two 300.0 cm^3 samples of water taken from the school pond: one immediately (day 0), and the other after leaving it sealed in a dark cupboard for five days (day 5). The following results were obtained for the titration of the samples with 0.0100 $mol\,dm^{-3}$ $Na_2S_2O_3(aq)$.

Sample	**Titre / cm^3 $\pm$0.1 cm^3**
Day 0	25.8
Day 5	20.1

(a) (i) Determine the mole ratio of $S_2O_3^{2-}$ to O_2, using the balanced equations. [1]

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(ii) Calculate the number of moles of oxygen in the day 0 sample. [2]

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(iii) The day 5 sample contained 5.03×10^{-5} moles of oxygen.

Determine the 5-day biochemical oxygen demand of the pond, in mg dm^{-3} ("parts per million", ppm). [2]

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1.2-S22 Nov 2019 · Paper 2 · Q7 parts (a)(i), (a)(ii)

7. Automobile air bags inflate by a rapid decomposition reaction. One typical compound used is guanidinium nitrate, $C(NH_2)_3NO_3$, which decomposes very rapidly to form nitrogen, water vapour and carbon.

(a) (i) Deduce the equation for the decomposition of guanidinium nitrate. [1]

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(ii) Calculate the total number of moles of gas produced from the decomposition of 10.0 g of guanidinium nitrate. [1]

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1.2-S23 May 2019 TZ2 · Paper 2 · Q5 parts (d)(iii)

5. Carbonated water is produced when carbon dioxide is dissolved in water under pressure. The following equilibria are established.

Equilibrium (1) $CO_2(g) \overset{H_2O(l)}{\rightleftharpoons} CO_2(aq)$

Equilibrium (2) $CO_2(aq) + H_2O(l) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$

(d) Soda water has sodium hydrogencarbonate, $NaHCO_3$, dissolved in the carbonated water.

(iii) $100.0\,cm^3$ of soda water contains 3.0×10^{-2} g $NaHCO_3$.

Calculate the concentration of $NaHCO_3$ in $mol\,dm^{-3}$. [2]

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1.2-S24 May 2019 TZ2 · Paper 3 · Q1 parts (c)(ii)

1. This question is about a mug made of a lead alloy.

The rate of lead dissolving in common beverages with various pH values was analysed.

Lead dissolving in beverages at various times and temperatures

Experiment	Beverage	pH	Time / min	Temp. / °C	Lead concentration / mg dm^{-3}
1	Cola	2.5	5	16	6
2	Cola	2.5	30	16	14
3	Cola	2.5	60	16	23
4	Cola	2.5	5	18	11
5	Lemonade	2.9	5	18	14
6	Orange juice	3.7	5	18	18
7	Beer	4.2	5	18	2.3
8	Tap water	5.9	5	18	15

[Source: first published in *Chemistry in Australia*, chemaust.raci.org.au]

(ii) A mean daily lead intake of greater than 5.0×10^{-6} g per kg of body weight results in increased lead levels in the body.

Calculate the volume, in dm^3, of tap water from experiment 8 which would exceed this daily lead intake for an 80.0 kg man. [2]

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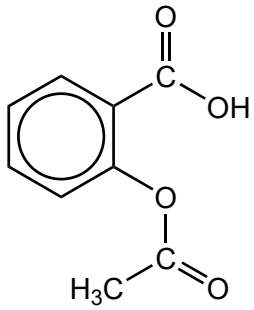
1.2-S25 May 2019 TZ2 · Paper 3 · Q22 parts (b)(i), (b)(ii)

22. A student synthesized aspirin, acetylsalicylic acid, in a school laboratory.



Aspirin
$M_r = 180.17$

(b) 0.300 g of crude aspirin was dissolved in ethanol and titrated with sodium hydroxide solution, NaOH (aq).

$$NaOH(aq) + C_9H_8O_4(\text{in ethanol}) \rightarrow NaC_9H_7O_4(aq) + H_2O(l)$$

(i) Determine the mass of aspirin which reacted with 16.25 cm^3 of 0.100 $mol\,dm^{-3}$ NaOH solution. [2]

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(ii) Determine the percentage purity of the synthesized aspirin. [1]

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1.2-S26 Nov 2018 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (d) · also 7.2, 8.2

1. 3.26 g of iron powder are added to 80.0 cm^3 of 0.200 mol dm^{-3} copper(II) sulfate solution. The following reaction occurs:

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$

(a) (i) Determine the limiting reactant showing your working. [2]

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(ii) The mass of copper obtained experimentally was 0.872 g. Calculate the percentage yield of copper. [2]

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(b) (i) The reaction was carried out in a calorimeter. The maximum temperature rise of the solution was 7.5 °C.

Calculate the enthalpy change, ΔH, of the reaction, in kJ, assuming that all the heat released was absorbed by the solution. Use sections 1 and 2 of the data booklet. [2]

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(ii) State another assumption you made in (b)(i). [1]

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(d) A student electrolyzed aqueous iron(II) sulfate, $FeSO_4(aq)$, using platinum electrodes. State half-equations for the reactions at the electrodes, using section 24 of the data booklet. [2]

Anode (positive electrode):

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Cathode (negative electrode):

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1.2-S27 Nov 2018 · Paper 3 · Q3 parts (b)(i)

3. The presence of very small amounts of lead in calcium-based antacids can be determined using inductively coupled plasma-mass spectroscopy (ICP-MS).

(b) An unknown antacid sample has a lead ion concentration of $0.50\,\mu g\,dm^{-3}$.

(i) Calculate the concentration of lead ions in the sample in $mol\,dm^{-3}$. [2]

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1.2-S28 Nov 2018 · Paper 3 · Q18 parts (c)

18. Buffer systems control pH in the body.

(c) Suggest a technique for measuring the percentage mass of calcium carbonate in this type of antacid tablet. [1]

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1.2-S29 May 2018 TZ1 · Paper 3 · Q2 parts (a)

2. Students were asked to investigate how a change in concentration of hydrochloric acid, HCl, affects the initial rate of its reaction with marble chips, $CaCO_3$.

They decided to measure how long the reaction took to complete when similar chips were added to 50.0 cm^3 of 1.00 mol dm^{-3} acid and 50.0 cm^3 of 2.00 mol dm^{-3} acid.

Two methods were proposed:
(1) using small chips, keeping the acid in excess, and recording the time taken for the solid to disappear
(2) using large chips, keeping the marble in excess, and recording the time taken for bubbles to stop forming.

(a) Annotate the balanced equation below with state symbols. [1]

$$CaCO_3(__) + 2HCl(__) \rightarrow CaCl_2(__) + CO_2(__) + H_2O(__)$$

1.2-S30 May 2018 TZ2 · Paper 2 · Q1 parts (a)

1. A student determined the percentage of the active ingredient magnesium hydroxide, $Mg(OH)_2$, in a 1.24 g antacid tablet.

The antacid tablet was added to 50.00 cm^3 of 0.100 mol dm^{-3} sulfuric acid, which was in excess.

(a) Calculate the amount, in mol, of H_2SO_4. [1]

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1.2-S31 May 2018 TZ2 · Paper 3 · Q4 parts (a)(iii)

4. Vanadium forms a body centred cubic (BCC) crystal structure with an edge length of 303 pm, (303×10^{-12} m).

(iii) Calculate the average mass, in g, of a vanadium atom by using sections 2 and 6 of the data booklet. [1]

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1.2-S32 May 2017 TZ2 · Paper 2 · Q8 parts (b)(i), (b)(ii)

8. Soluble acids and bases ionize in water.

(b) A solution containing 0.510 g of an unknown monoprotic acid, HA, was titrated with $0.100\,mol\,dm^{-3}$ NaOH (aq). $25.0\,cm^3$ was required to reach the equivalence point.

(i) Calculate the amount, in mol, of NaOH (aq) used. [1]

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(ii) Calculate the molar mass of the acid. [1]

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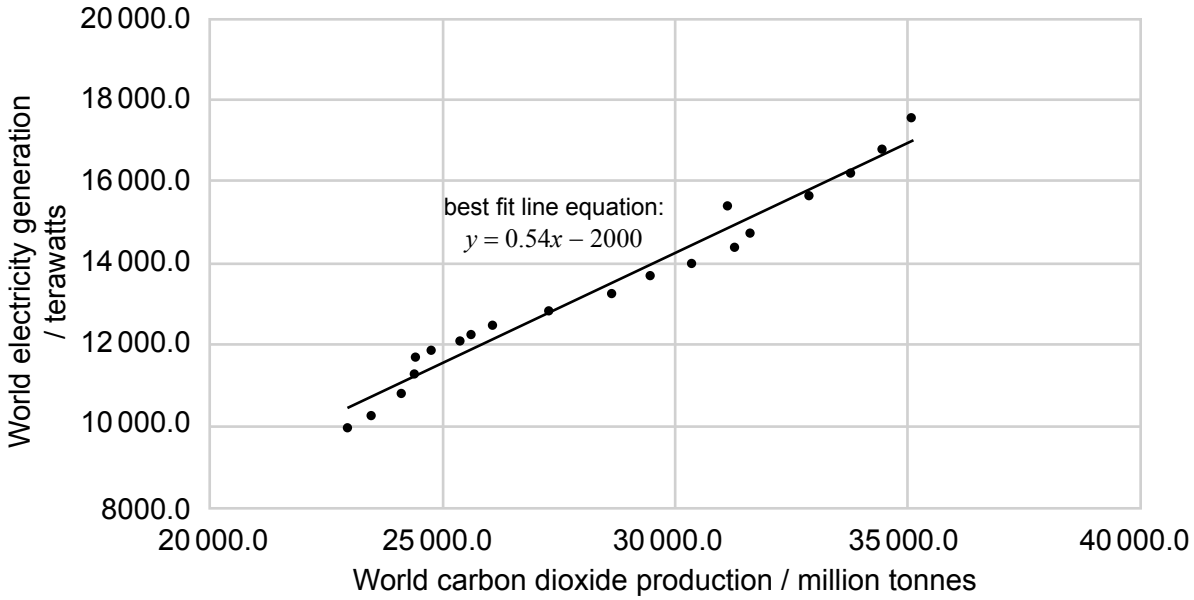
1.2-S33 May 2017 TZ2 · Paper 3 · Q1 parts (b)

1. There is a link between world energy consumption and carbon dioxide production.

(b) O_2 is consumed in producing CO_2 for electricity generation. The graph shows the relationship between the world's electricity generation and CO_2 production between 1994 and 2013.



[Source: BP statistical review of world energy, www.bp.com]

Calculate the mass, in million tonnes, of oxygen gas ultimately found in CO_2 which is consumed in generating 18 000 terawatts of electricity using the equation given for the best fit line. Give your answer to 2 significant figures.
Assume coal is the only energy source. [2]

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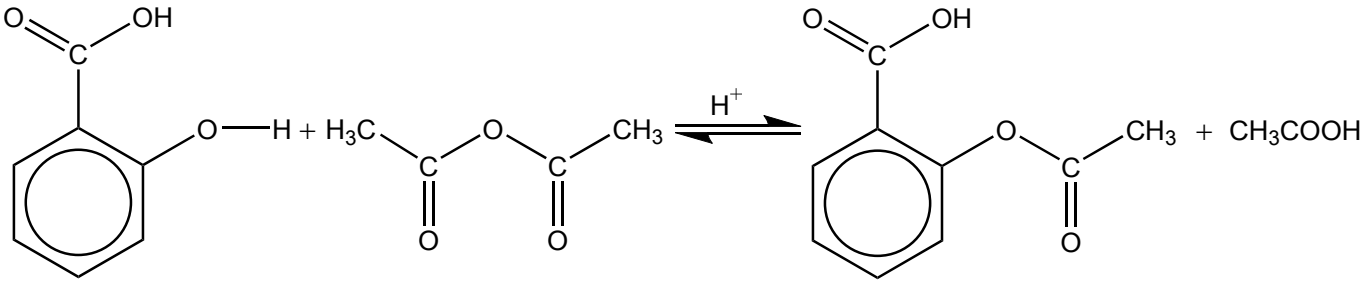
1.2-S34 May 2017 TZ2 · Paper 3 · Q20 parts (a)(i), (a)(ii)

20. Aspirin is one of the most widely used drugs in the world.

(a) Aspirin was synthesized from 2.65 g of salicylic acid (2-hydroxybenzoic acid) ($M_r = 138.13$) and 2.51 g of ethanoic anhydride ($M_r = 102.10$).



Salicylic acid Ethanoic anhydride Aspirin Ethanoic acid

(i) Calculate the amounts, in mol, of each reactant. [1]

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(ii) Calculate, in g, the theoretical yield of aspirin. [1]

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1.2-S35 Nov 2016 · Paper 2 · Q2 parts (c)(ii), (c)(iii), (c)(iv), (d), (e) · also 3.1, 4.1

2. The concentration of a solution of a weak acid, such as ethanedioic acid, can be determined by titration with a standard solution of sodium hydroxide, NaOH (aq).

(c) 5.00 g of an impure sample of hydrated ethanedioic acid, $(COOH)_2 \cdot 2H_2O$, was dissolved in water to make 1.00 dm^3 of solution. 25.0 cm^3 samples of this solution were titrated against a 0.100 mol dm^{-3} solution of sodium hydroxide using a suitable indicator.

$$(COOH)_2(aq) + 2NaOH(aq) \rightarrow (COONa)_2(aq) + 2H_2O(l)$$

The mean value of the titre was 14.0 cm^3.

(ii) Calculate the amount, in mol, of NaOH in 14.0 cm^3 of 0.100 mol dm^{-3} solution. [1]

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(iii) Calculate the amount, in mol, of ethanedioic acid in each 25.0 cm^3 sample. [1]

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(iv) Determine the percentage purity of the hydrated ethanedioic acid sample. [3]

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(d) Draw the Lewis (electron dot) structure of the ethanedioate ion, $^{-}OOCCOO^{-}$. [1]

(e) Outline why all the C–O bond lengths in the ethanedioate ion are the same length and suggest a value for them. Use section 10 of the data booklet. [2]

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1.2-S36 Nov 2016 · Paper 2 · Q3 parts (a), (c)

3. Sodium thiosulfate solution reacts with dilute hydrochloric acid to form a precipitate of sulfur at room temperature.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow S(s) + SO_2(g) + 2NaCl(aq) + X$$

(a) Identify the formula and state symbol of X. [1]

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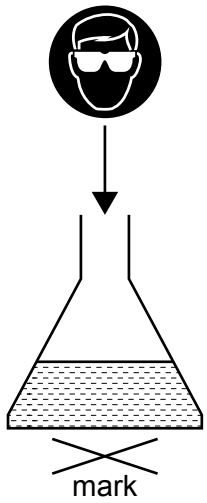
(c) The precipitate of sulfur makes the mixture cloudy, so a mark underneath the reaction mixture becomes invisible with time.



$10.0\,cm^3$ of $2.00\,mol\,dm^{-3}$ hydrochloric acid was added to a $50.0\,cm^3$ solution of sodium thiosulfate at temperature, T_1. Students measured the time taken for the mark to be no longer visible to the naked eye. The experiment was repeated at different concentrations of sodium thiosulfate.

Experiment	**$[Na_2S_2O_3(aq)]$ / mol dm^{-3}**	**Time, t, for mark to disappear / s ± 1 s**	**$\frac{1}{t}$* / 10^{-3} s^{-1}**
1	0.150	23	43.5
2	0.120	27	37.0
3	0.090	36	27.8
4	0.060	60	16.7
5	0.030	111	9.0

* The reciprocal of the time in seconds can be used as a measure of the rate of reaction.

[Source: Adapted from http://www.flinnsci.com/]

Show that the hydrochloric acid added to the flask in experiment 1 is in excess. [2]

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1.2-S37 Nov 2016 · Paper 3 · Q25 parts (a), (b) · also 5.3

25. The mild analgesic aspirin can be prepared in the laboratory from salicylic acid.

$$(CH_3CO)_2O + HOC_6H_4COOH \rightarrow CH_3CO_2C_6H_4COOH + CH_3COOH$$

Salicylic acid Aspirin

After the reaction is complete, the product is isolated, recrystallized, tested for purity and the experimental yield is measured. A student's results in a single trial are as follows.

	Mass / g ±0.001	**Melting point / °C ±1**
Initial salicylic acid	1.552	
Crude product	1.398	106–114
Product after recrystallization	1.124	122–125

Literature melting point data: aspirin = 138–140 °C

(a) Determine the percentage experimental yield of the product after recrystallization. The molar masses are as follows: M(salicylic acid) = 138.13 g mol^{-1}, M(aspirin) = 180.17 g mol^{-1}. (You do not need to process the uncertainties in the calculation.) [2]

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(b) Suggest why isolation of the crude product involved the addition of ice-cold water. [1]

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1.2-S38 May 2016 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iv), (a)(v), (b)(i), (b)(iii), (b)(iv), (c)(i), (c)(ii), (c)(iii), (d)(i) · also 3.1, 4.2, 5.1, 5.2, 6.1, 8.2

1. Phosphine (IUPAC name phosphane) is a hydride of phosphorus, with the formula PH_3.

(a) (i) Draw a Lewis (electron dot) structure of phosphine. [1]

(ii) State the hybridization of the phosphorus atom in phosphine. [1]

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(iv) Outline whether you expect the bonds in phosphine to be polar or non-polar, giving a brief reason. [1]

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(v) Phosphine has a much greater molar mass than ammonia. Explain why phosphine has a significantly lower boiling point than ammonia. [2]

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(b) Phosphine is usually prepared by heating white phosphorus, one of the allotropes of phosphorus, with concentrated aqueous sodium hydroxide. The equation for the reaction is:

$$P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

(i) The first reagent is written as P_4, not 4P. Describe the difference between P_4 and 4P. [1]

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(iii) State the oxidation state of phosphorus in P_4 and $H_2PO_2^-$. [2]

P_4:

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$H_2PO_2^-$:

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(iv) Oxidation is now defined in terms of change of oxidation number. Explore how earlier definitions of oxidation and reduction may have led to conflicting answers for the conversion of P_4 to $H_2PO_2^-$ and the way in which the use of oxidation numbers has resolved this. [3]

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(c) 2.478 g of white phosphorus was used to make phosphine according to the equation:

$$P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

(i) Calculate the amount, in mol, of white phosphorus used. [1]

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(ii) This phosphorus was reacted with 100.0 cm^3 of 5.00 mol dm^{-3} aqueous sodium hydroxide. Deduce, showing your working, which was the limiting reagent. [1]

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(iii) Determine the excess amount, in mol, of the other reagent. [1]

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(d) Impurities cause phosphine to ignite spontaneously in air to form an oxide of phosphorus and water.

(i) 200.0 g of air was heated by the energy from the complete combustion of 1.00 mol phosphine. Calculate the temperature rise using section 1 of the data booklet and the data below. [1]

Standard enthalpy of combustion of phosphine, $\Delta H_c^\ominus = -750\,kJ\,mol^{-1}$

Specific heat capacity of air $= 1.00\,J\,g^{-1}K^{-1} = 1.00\,kJ\,kg^{-1}K^{-1}$

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1.2-S39 Nov 2015 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b), (c)(iii), (d), (e) · also 6.1

A student used the technique of titration to determine the concentration of ascorbic acid ($C_6H_8O_6$) in a sample of orange juice. Excess potassium iodide, $KI(aq)$, was added to acidified orange juice. The resulting solution was titrated with potassium iodate, $KIO_3(aq)$, in the presence of starch as an indicator. The end-point of the titration was shown by a blue-black colour.

Step 1 $IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \rightarrow 3I_2(aq) + 3H_2O(l)$

Iodine is only slightly soluble in water; but in the presence of excess iodide ions, $I^-(aq)$, it forms the soluble tri-iodide ion, $I_3^-(aq)$.

Step 2 $I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$

Ascorbic acid reacts with tri-iodide ions as follows.

Step 3 $C_6H_8O_6(aq) + I_3^-(aq) \rightarrow C_6H_6O_6(aq) + 2H^+(aq) + 3I^-(aq)$

(a) (i) Deduce the changes in oxidation number of iodine in step 1. [2]

IO_3^- to I_2:

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I^- to I_2:

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(ii) Identify the oxidizing and reducing agents in step 1. [1]

Oxidizing agent:

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Reducing agent:

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(b) Calculate the mass, in g, of potassium iodate, $KIO_3(s)$, which was required to prepare $0.250\,dm^3$ of a $2.00 \times 10^{-3}\,mol\,dm^{-3}$ solution. [2]

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(c) The concentration of KIO_3 used in the titration was $2.00 \times 10^{-3}\,mol\,dm^{-3}$.
The titration produced the following results.

	Titration 1	**Titration 2**	**Titration 3**
Final volume of KIO_3 ($\pm\,0.05\,cm^3$)	7.10	14.40	21.60
Initial volume of KIO_3 ($\pm\,0.05\,cm^3$)	0.00	7.10	14.40
Volume added of KIO_3 ($\pm\,0.10\,cm^3$)	7.10	7.30	7.20
Mean volume added of KIO_3 ($\pm\,0.10\,cm^3$)	7.20		

(iii) Determine the amount, in mol, of $KIO_3(aq)$, in the mean volume. [1]

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(d) Determine the amount, in mol, of ascorbic acid, $C_6H_8O_6(aq)$, in the sample of acidified orange juice. [2]

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(e) Calculate the mass, in g, of ascorbic acid, $C_6H_8O_6(aq)$, present in the sample of acidified orange juice. [1]

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1.2-S40 Nov 2015 · Paper 2 · Q9 parts (a)(ii)

(a) A 0.842 g sample of a liquid halogenoalkane, RBr(l), was heated under reflux with 1.35×10^{-2} mol of aqueous sodium hydroxide, NaOH(aq). After cooling the mixture, the excess NaOH was titrated with hydrochloric acid, HCl(aq), and required 7.36×10^{-3} mol of the acid.

(ii) Calculate the amount, in mol, of sodium hydroxide that reacted with the halogenoalkane. [1]

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1.2-S41 May 2015 TZ1 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d), (e) · also 3.1, 4.1, 5.2

1. Ethanedioic acid is a diprotic acid. A student determined the value of x in the formula of hydrated ethanedioic acid, HOOC–COOH•xH_2O, by titrating a known mass of the acid with a 0.100 mol dm^{-3} solution of NaOH(aq).

0.795 g of ethanedioic acid was dissolved in distilled water and made up to a total volume of 250 cm^3 in a volumetric flask.

25 cm^3 of this ethanedioic acid solution was pipetted into a flask and titrated against aqueous sodium hydroxide using phenolphthalein as an indicator.

The titration was then repeated twice to obtain the results below.

Volume of 0.100 mol dm^{-3} NaOH / cm^3	**Titration 1**	**Titration 2**	**Titration 3**
Final burette reading (± 0.05)	13.00	25.70	38.20
Initial burette reading (± 0.05)	0.00	13.00	25.70
Volume added			

(a) Calculate the average volume of NaOH added, in cm^3, in titrations 2 and 3, and then calculate the amount, in mol, of NaOH added. [2]

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(b) (i) The equation for the reaction taking place in the titration is:

$$HOOC{-}COOH\,(aq) + 2NaOH\,(aq) \rightarrow NaOOC{-}COONa\,(aq) + 2H_2O\,(l)$$

Determine the amount, in mol, of ethanedioic acid that reacts with the average volume of NaOH (aq). [1]

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(ii) Determine the amount, in mol, of ethanedioic acid present in 250 cm^3 of the original solution. [1]

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(iii) Determine the molar mass of hydrated ethanedioic acid. [1]

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(iv) Determine the value of x in the formula $HOOC{-}COOH{\bullet}xH_2O$. [2]

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(c) Identify the strongest intermolecular force in solid ethanedioic acid. [1]

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(d) Deduce the Lewis (electron dot) structure of ethanedioic acid, HOOC–COOH. [1]

(e) Predict and explain the difference in carbon-oxygen bond lengths in ethanedioic acid and its conjugate base, $^{-}OOC–COO^{-}$. [3]

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1.2-S42 Nov 2014 · Paper 2 · Q4 parts (a), (b)

Phosphorus(V) oxide, P_4O_{10} ($M_r = 283.88$), reacts vigorously with water ($M_r = 18.02$), according to the equation below.

$$P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$$

(a) A student added 5.00 g of P_4O_{10} to 1.50 g of water. Determine the limiting reactant, showing your working. *[2]*

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(b) Calculate the mass of phosphoric(V) acid, H_3PO_4, formed in the reaction. *[2]*

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1.2-S43 May 2014 TZ2 · Paper 2 · Q1 parts (a), (c)(ii), (c)(iii), (g) · also 5.2

1. A class studied the equilibrium established when ethanoic acid and ethanol react together in the presence of a strong acid, using propanone as an inert solvent. The equation is given below.

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

One group made the following **initial mixture**:

Liquid	Volume / cm^3
Ethanoic acid	5.00 ± 0.05
Ethanol	5.00 ± 0.05
$6.00\,mol\,dm^{-3}$ aqueous hydrochloric acid	1.00 ± 0.02
Propanone	39.0 ± 0.5

(a) The density of ethanoic acid is $1.05\,g\,cm^{-3}$. Determine the amount, in mol, of ethanoic acid present in the initial mixture. *[3]*

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(c) After one week, a $5.00 \pm 0.05\,cm^3$ sample of the final equilibrium mixture was pipetted out and titrated with $0.200\,mol\,dm^{-3}$ aqueous sodium hydroxide to determine the amount of ethanoic acid remaining. The following titration results were obtained:

Titration number	1	2	3
Initial reading / cm^3 ± 0.05	1.20	0.60	14.60
Final reading / cm^3 ± 0.05	28.80	26.50	40.70
Titre / cm^3	27.60	25.90	26.10

(ii) Suggest the average volume of alkali, required to neutralize the $5.00\,cm^3$ sample, that the student should use. *[1]*

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(iii) $3.00\,cm^3$ of the $0.200\,mol\,dm^{-3}$ aqueous sodium hydroxide reacted with the hydrochloric acid present in the $5.00\,cm^3$ sample. Determine the concentration of ethanoic acid in the final equilibrium mixture. *[2]*

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(g) Propanone is used as the solvent because one compound involved in the equilibrium is insoluble in water. Identify this compound and explain why it is insoluble in water. *[2]*

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1.2-S44 Nov 2013 · Paper 2 · Q5 parts (a), (c), (g), (h), (i) · also 3.1, 4.1, 5.1, 9.3

5. In December 2010, researchers in Sweden announced the synthesis of N,N–dinitronitramide, $N(NO_2)_3$. They speculated that this compound, more commonly called trinitramide, may have significant potential as an environmentally friendly rocket fuel oxidant.

(a) Methanol reacts with trinitramide to form nitrogen, carbon dioxide and water. Deduce the coefficients required to balance the equation for this reaction. *[1]*

$$___\ N(NO_2)_3(g) + ___\ CH_3OH(l) \rightarrow ___\ N_2(g) + ___\ CO_2(g) + ___\ H_2O(l)$$

(c) Calculate the enthalpy change, in $kJ\,mol^{-1}$, when one mole of trinitramide decomposes to its elements, using bond enthalpy data from Table 10 of the Data Booklet. Assume that all the N–O bonds in this molecule have a bond enthalpy of $305\,kJ\,mol^{-1}$. *[3]*

(g) Outline how the length of the N–N bond in trinitramide compares with the N–N bond in nitrogen gas, N_2. *[2]*

(h) Deduce the N–N–N bond angle in trinitramide and explain your reasoning. *[3]*

(i) Predict, with an explanation, the polarity of the trinitramide molecule. *[2]*

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1.2-S45 May 2013 TZ1 · Paper 2 · Q1 parts (b)

1. A student decided to determine the molecular mass of a solid monoprotic acid, HA, by titrating a solution of a known mass of the acid.

The following recordings were made.

Mass of bottle / g ± 0.001 g	1.737
Mass of bottle + acid HA / g ± 0.001 g	2.412

(b) This known mass of acid, HA, was then dissolved in distilled water to form a 100.0 cm^3 solution in a volumetric flask. A 25.0 cm^3 sample of this solution reacted with 12.1 cm^3 of a 0.100 mol dm^{-3} NaOH solution. Calculate the molar mass of the acid. *[3]*

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1.2-S46 May 2013 TZ2 · Paper 2 · Q1 parts (e)(i), (e)(ii), (f)(i), (f)(ii), (f)(iii), (f)(iv) · also 6.1

1. Iron tablets are often prescribed to patients. The iron in the tablets is commonly present as iron(II) sulfate, $FeSO_4$.

Two students carried out an experiment to determine the percentage by mass of iron in a brand of tablets marketed in Cyprus.

Experimental Procedure:

- The students took five iron tablets and found that the **total mass** was 1.65 g.
- The five tablets were ground and dissolved in 100 cm^3 dilute sulfuric acid, $H_2SO_4(aq)$. The solution and washings were transferred to a 250 cm^3 volumetric flask and made up to the mark with deionized (distilled) water.
- 25.0 cm^3 of this $Fe^{2+}(aq)$ solution was transferred using a pipette into a conical flask. Some dilute sulfuric acid was added.
- A titration was then carried out using a 5.00×10^{-3} mol dm^{-3} standard solution of potassium permanganate, $KMnO_4(aq)$. The end-point of the titration was indicated by a slight pink colour.

The following results were recorded.

	Rough titre	**First accurate titre**	**Second accurate titre**
Initial burette reading / $cm^3 \pm 0.05$	1.05	1.20	0.00
Final burette reading / $cm^3 \pm 0.05$	20.05	18.00	16.80

(e) This experiment involves the following redox reaction.

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

(i) Define the term *reduction* in terms of electrons. *[1]*

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(ii) Deduce the oxidation number of manganese in the $MnO_4^-(aq)$ ion. *[1]*

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(f) (i) Determine the amount, in mol, of MnO_4^-(aq), used in each accurate titre. *[2]*

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(ii) Calculate the amount, in mol, of Fe^{2+}(aq) ions in 250 cm^3 of the solution. *[1]*

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(iii) Determine the total mass of iron, in g, in the 250 cm^3 solution. *[1]*

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(iv) Determine the percentage by mass of iron in the tablets. *[1]*

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1.2-S47 Nov 2012 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b), (c)(ii), (c)(iii), (c)(iv), (c)(v), (c)(vi), (d), (e) · also 3.1, 5.2, 8.1

1. Two groups of students (Group A and Group B) carried out a project* on the chemistry of some group 7 elements (the halogens) and their compounds.

(a) In the first part of the project, the two groups had a sample of iodine monochloride (a corrosive brown liquid) prepared for them by their teacher using the following reaction.

$$I_2(s) + Cl_2(g) \rightarrow 2ICl(l)$$

The following data were recorded.

Mass of $I_2(s)$	10.00 g
Mass of $Cl_2(g)$	2.24 g
Mass of $ICl(l)$ obtained	8.60 g

(i) State the number of significant figures for the masses of $I_2(s)$ and $ICl(l)$. *[1]*

$I_2(s)$: . . . . . .

$ICl(l)$: . . . . . .

(ii) The iodine used in the reaction was in excess. Determine the theoretical yield, in g, of $ICl(l)$. *[3]*

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(iii) Calculate the percentage yield of $ICl(l)$. *[1]*

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(iv) Using a digital thermometer, the students discovered that the reaction was exothermic. State the sign of the enthalpy change of the reaction, ΔH. *[1]*

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(b) Although the molar masses of ICl and Br_2 are very similar, the boiling point of ICl is 97.4 °C and that of Br_2 is 58.8 °C. Explain the difference in these boiling points in terms of the intermolecular forces present in each liquid. *[2]*

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(c) The students reacted ICl(l) with CsBr(s) to form a yellow solid, $CsICl_2(s)$, as one of the products. $CsICl_2(s)$ has been found to produce very pure CsCl(s) which is used in cancer treatment.

To confirm the composition of the yellow solid, Group A determined the amount of iodine in 0.2015 g of $CsICl_2(s)$ by titrating it with 0.0500 mol dm^{-3} $Na_2S_2O_3(aq)$. The following data were recorded for the titration.

Mass of $CsICl_2(s)$ taken (in g ± 0.0001)	0.2015
Initial burette reading of 0.0500 mol dm^{-3} $Na_2S_2O_3(aq)$ (in cm^3 ± 0.05)	1.05
Final burette reading of 0.0500 mol dm^{-3} $Na_2S_2O_3(aq)$ (in cm^3 ± 0.05)	25.25

(ii) State the volume, in cm^3, of 0.0500 mol dm^{-3} $Na_2S_2O_3(aq)$ used in the titration. *[1]*

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(iii) Determine the amount, in mol, of 0.0500 mol dm^{-3} $Na_2S_2O_3(aq)$ added in the titration. *[1]*

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(iv) The overall reaction taking place during the titration is:

$$CsICl_2(s) + 2Na_2S_2O_3(aq) \rightarrow NaCl(aq) + Na_2S_4O_6(aq) + CsCl(aq) + NaI(aq)$$

Calculate the amount, in mol, of iodine atoms, I, present in the sample of $CsICl_2(s)$. *[1]*

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(v) Calculate the mass of iodine, in g, present in the sample of $CsICl_2(s)$. *[1]*

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(vi) Determine the percentage by mass of iodine in the sample of $CsICl_2(s)$, correct to **three** significant figures, using your answer from (v). *[1]*

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(d) Group B heated the yellow solid, $CsICl_2(s)$, which turned white and released a brown gas which condensed into a brown liquid.

Group B identified the white solid as $CsCl(s)$. Suggest the identity of the brown liquid. *[1]*

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(e) When iodine reacts with excess chlorine, ICl_3 can form. Deduce the Lewis (electron dot) structure of ICl_3 and ICl_2^- and state the name of the shape of each species. *[4]*

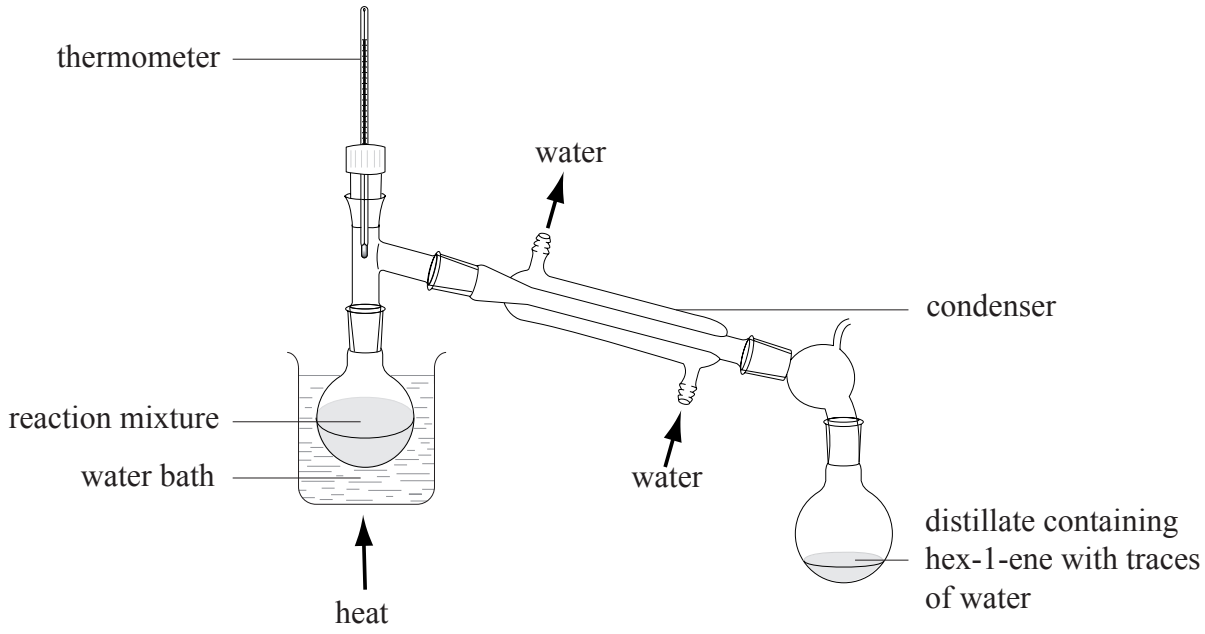
	ICl_3	**ICl_2^-**
Lewis structure		
Name of shape		

1.2-S48 May 2012 TZ1 · Paper 2 · Q7 parts (a)(i), (a)(ii), (a)(iii)

7. A student prepared hex-1-ene, C_6H_{12}, from hexan-1-ol, $C_6H_{13}OH$, by a dehydration reaction.

$$C_6H_{13}OH\,(l) \rightarrow C_6H_{12}\,(l) + H_2O\,(l)$$

The apparatus for this preparation is shown below. The reaction mixture contains 5.00 g of hexan-1-ol and an excess of concentrated sulfuric acid, which removes the water from the organic compound.



The distillate was dried to obtain 2.62 g of hex-1-ene.

(a) (i) Determine the amount, in mol, of hexan-1-ol present in the reaction mixture. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Calculate the percentage yield of hex-1-ene produced. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(iii) Another student repeated the experiment and reported a yield of 5.24 g of organic product. Comment on this result. *[2]*

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1.2-S49 May 2012 TZ2 · Paper 2 · Q2 parts (a), (b), (c)(i) · also 6.3

2. A student added 7.40×10^{-2} g of magnesium ribbon to 15.0 cm^3 of 2.00 $mol\,dm^{-3}$ hydrochloric acid. The hydrogen gas produced was collected using a gas syringe at 20.0 °C and 1.01 10^5 Pa.

(a) State the equation for the reaction between magnesium and hydrochloric acid. *[1]*

. . . . . .

(b) Determine the limiting reactant. *[3]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .
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. . . . . .

(c) Calculate the theoretical yield of hydrogen gas:

(i) in mol. *[1]*

. . . . . .
. . . . . .

1.2-S50 Nov 2011 · Paper 2 · Q1 parts (b)(i), (b)(ii), (c)(ii) · also 4.3

1. Airbags are an important safety feature in vehicles. Sodium azide, potassium nitrate and silicon dioxide have been used in one design of airbag.

[Source: www.hilalairbag.net]

Sodium azide, a toxic compound, undergoes the following decomposition reaction under certain conditions.

$$2NaN_3(s) \rightarrow 2Na(s) + 3N_2(g)$$

Two students looked at data in a simulated computer-based experiment to determine the volume of nitrogen generated in an airbag.

(b) Using the simulation programme, the students entered the following data into the computer.

Temperature (T) / °C	**Mass of NaN_3(s) (m) / kg**	**Pressure (p) / atm**
25.00	0.0650	1.08

(i) State the number of significant figures for the temperature, mass and pressure data. *[1]*

T: . . . . . .

m: . . . . . .

p: . . . . . .

(ii) Calculate the amount, in mol, of sodium azide present. *[1]*

. . . . . .

. . . . . .

(c) The chemistry of the airbag was found to involve three reactions. The first reaction involves the decomposition of sodium azide to form sodium and nitrogen. In the second reaction, potassium nitrate reacts with sodium.

$$2KNO_3(s) + 10Na(s) \rightarrow K_2O(s) + 5Na_2O(s) + N_2(g)$$

(ii) The metal oxides from the second reaction then react with silicon dioxide to form a silicate in the third reaction.

$$K_2O(s) + Na_2O(s) + SiO_2(s) \rightarrow Na_2K_2SiO_4(s)$$

Draw the structure of silicon dioxide and state the type of bonding present. *[2]*

Structure:

Bonding: . . . . . .

1.2-S51 May 2011 TZ1 · Paper 3 · QE2 parts (b)

E2. Organic matter present in water can be decomposed under aerobic and anaerobic conditions by bacteria.

(b) In a certain lake, anaerobic conditions exist due to the presence of 10 ppm by mass (0.010 g dm^{-3}) of organic matter. Determine the mass of oxygen required to oxidize the organic matter in 1 dm^3 of water. Assume that the aerobic decomposition process can be represented by the following equation. *[2]*

$$C_6H_{10}O_5 + 6O_2 \rightarrow 6CO_2 + 5H_2O$$

. . . . . .

. . . . . .

. . . . . .

. . . . . .

1.2-S52 Nov 2010 · Paper 2 · Q6 parts (d)(i)

6. The Haber process enables the large-scale production of ammonia needed to make fertilizers.

(d) A student decided to investigate the reactions of the two acids with separate samples of 0.20 mol dm^{-3} sodium hydroxide solution.

(i) Calculate the volume of the sodium hydroxide solution required to react exactly with a 15.0 cm^3 solution of 0.10 mol dm^{-3} nitric acid. *[1]*

1.2-S53 May 2010 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i) · also 2.1, 6.1

1. Brass is a copper containing alloy with many uses. An analysis is carried out to determine the percentage of copper present in three identical samples of brass. The reactions involved in this analysis are shown below.

Step 1: $Cu(s) + 2HNO_3(aq) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$

Step 2: $4I^-(aq) + 2Cu^{2+}(aq) \rightarrow 2CuI(s) + I_2(aq)$

Step 3: $I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

(a) (i) Deduce the change in the oxidation numbers of copper and nitrogen in step 1. *[2]*

Copper:

. . . . . .

Nitrogen:

. . . . . .

(ii) Identify the oxidizing agent in step 1. *[1]*

. . . . . .

(b) A student carried out this experiment three times, with three identical small brass nails, and obtained the following results.

Mass of brass = 0.456 g ± 0.001 g

Titre	**1**	**2**	**3**
Initial volume of 0.100 mol dm^{-3} $S_2O_3^{2-}$ (±0.05 cm^3)	0.00	0.00	0.00
Final volume of 0.100 mol dm^{-3} $S_2O_3^{2-}$ (±0.05 cm^3)	28.50	28.60	28.40
Volume added of 0.100 mol dm^{-3} $S_2O_3^{2-}$ (±0.10 cm^3)	28.50	28.60	28.40
Average volume added of 0.100 mol dm^{-3} $S_2O_3^{2-}$ (±0.10 cm^3)	28.50		

(i) Calculate the average amount, in mol, of $S_2O_3^{2-}$ added in step 3. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(ii) Calculate the amount, in mol, of copper present in the brass. *[1]*

. . . . . .

. . . . . .

(iii) Calculate the mass of copper in the brass. *[1]*

. . . . . .

. . . . . .

(iv) Calculate the percentage by mass of copper in the brass. *[1]*

. . . . . .

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(c) In step 1 the copper reacts to form a blue solution.

(i) State the full electronic configuration of Cu^{2+}. *[1]*

. . . . . .

1.2-S54 May 2010 TZ2 · Paper 2 · Q1 parts (a), (b), (c), (g)

1. The percentage by mass of calcium carbonate in eggshell was determined by adding excess hydrochloric acid to ensure that all the calcium carbonate had reacted. The excess acid left was then titrated with aqueous sodium hydroxide.

(a) A student added 27.20 cm^3 of 0.200 $mol\,dm^{-3}$ HCl to 0.188 g of eggshell. Calculate the amount, in mol, of HCl added. *[1]*

. . . . . .

. . . . . .

(b) The excess acid requires 23.80 cm^3 of 0.100 $mol\,dm^{-3}$ NaOH for neutralization. Calculate the amount, in mol, of acid that is in excess. *[1]*

. . . . . .

. . . . . .

(c) Determine the amount, in mol, of HCl that reacted with the calcium carbonate in the eggshell. *[1]*

. . . . . .

. . . . . .

(g) Deduce **one** assumption made in arriving at the percentage of calcium carbonate in the eggshell sample. *[1]*

. . . . . .

. . . . . .

1.2-S55 May 2010 TZ2 · Paper 2 · Q4 parts (a), (b), (c) · also 3.1, 9.3

4. One important property of a rocket fuel mixture is the large volume of gaseous products formed which provide thrust. Hydrazine, N_2H_4, is often used as a rocket fuel. The combustion of hydrazine is represented by the equation below.

$$N_2H_4(g) + O_2(g) \rightarrow N_2(g) + 2H_2O(g) \qquad \Delta H_c^{\ominus} = -585\,kJ\,mol^{-1}$$

(a) Hydrazine reacts with fluorine to produce nitrogen and hydrogen fluoride, all in the gaseous state. State an equation for the reaction. *[2]*

. . . . . .

. . . . . .

(b) Draw the Lewis structures for hydrazine and nitrogen. *[2]*

(c) Use the average bond enthalpies given in Table 10 of the Data Booklet to determine the enthalpy change for the reaction in part (a) above. *[3]*

. . . . . .

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. . . . . .

LESSON 2.1

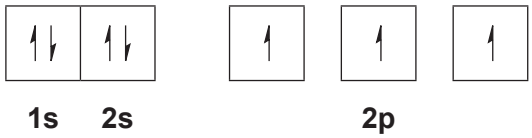
Ionization Energy

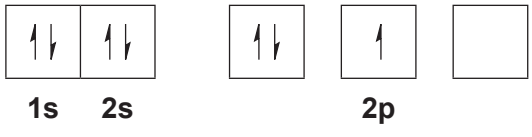
64 multiple-choice questions · 38 structured questions

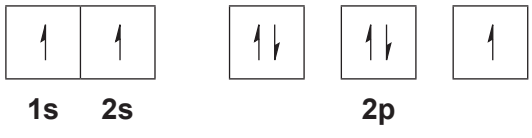
Part A - Multiple choice

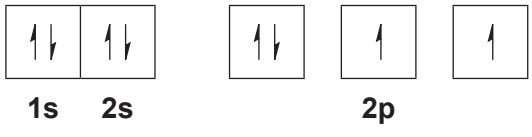
2.1-M1 May 2025 TZ1 · Paper 1A · Q4

4. Which diagram shows the correct electron configuration of a nitrogen atom in the ground state?

A. 

B. 

C. 

D. 

2.1-M2 May 2025 TZ1 · Paper 1A · Q7

7. Which row shows the correct formula and electron configuration of the species in calcium nitride?

	Formula	Electron configuration of the Ca species	Electron configuration of the N species
A.	Ca_2N_3	[Ar]	[Ne]
B.	Ca_2N_3	[Ar] $4s^2$	[He] $2s^2$ $2p^3$
C.	Ca_3N_2	[Ar] $4s^2$	[He] $2s^2$ $2p^3$
D.	Ca_3N_2	[Ar]	[Ne]

2.1-M3 May 2025 TZ1 · Paper 1A · Q15

15. Which statement explains the trend in first ionisation energy from sodium, Na, to chlorine,Cl?

A. Nuclear charge increases.

B. Electronegativity decreases.

C. Atomic radius increases.

D. Shielding decreases.

2.1-M4 May 2025 TZ2 · Paper 1A · Q3

3. What is the maximum number of electrons that can occupy the n = 3 main energy level?

A. 3

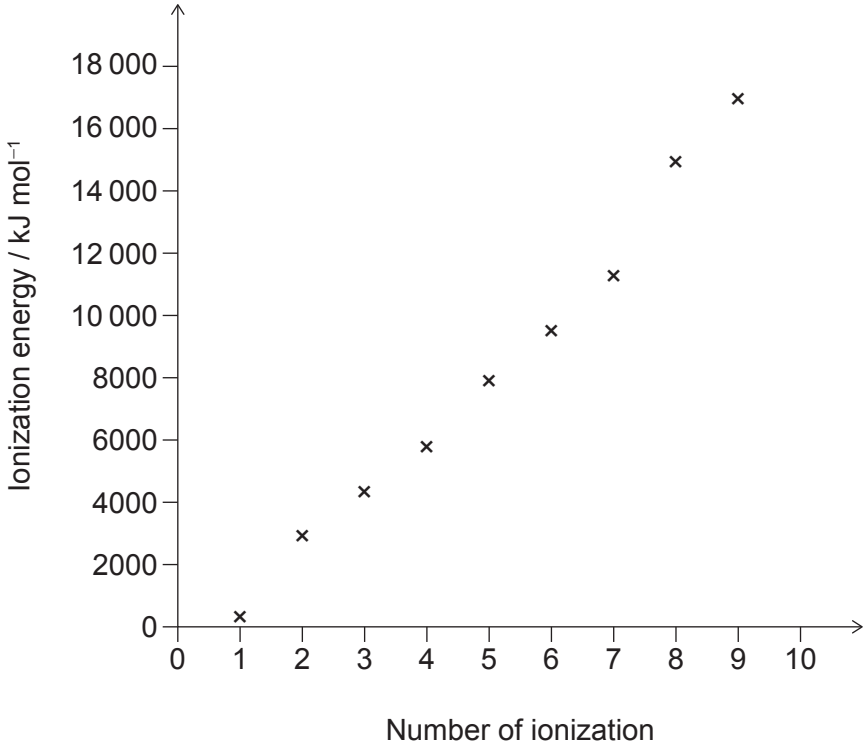
B. 8

C. 18

D. 28

2.1-M5 May 2025 TZ2 · Paper 1A · Q4

4. What is the group number of the element whose successive ionization energies are shown?



A. 1

B. 2

C. 13

D. 14

2.1-M6 May 2025 TZ2 · Paper 1A · Q6

6. What is the charge of the monatomic ion usually formed by the element with the following electron configuration?

$$[He]\,2s^2 2p^3$$

A. $3-$

B. $2+$

C. $3+$

D. $5+$

2.1-M7 May 2025 TZ2 · Paper 1A · Q12

12. Why do group 17 elements have higher first ionization energies than group 1 elements in the same period?

A. Group 17 elements have more electrons in the valence shell.

B. Group 17 elements have more filled energy levels.

C. Group 17 elements have more paired electrons.

D. Group 17 elements have more protons in the nucleus.

2.1-M8 May 2025 TZ2 · Paper 1A · Q16

16. What is the electron configuration of the Cr^{2+} ion?

A. $[Ar]\,4s^2 3d^2$

B. $[Ar]\,4s^1 3d^5$

C. $[Ar]\,3d^4$

D. $[Ar]\,3d^5$

2.1-M9 May 2025 TZ3 · Paper 1A · Q13

13. What is the electron configuration of an atom of Cu?

A. $[Ar]3d^9$

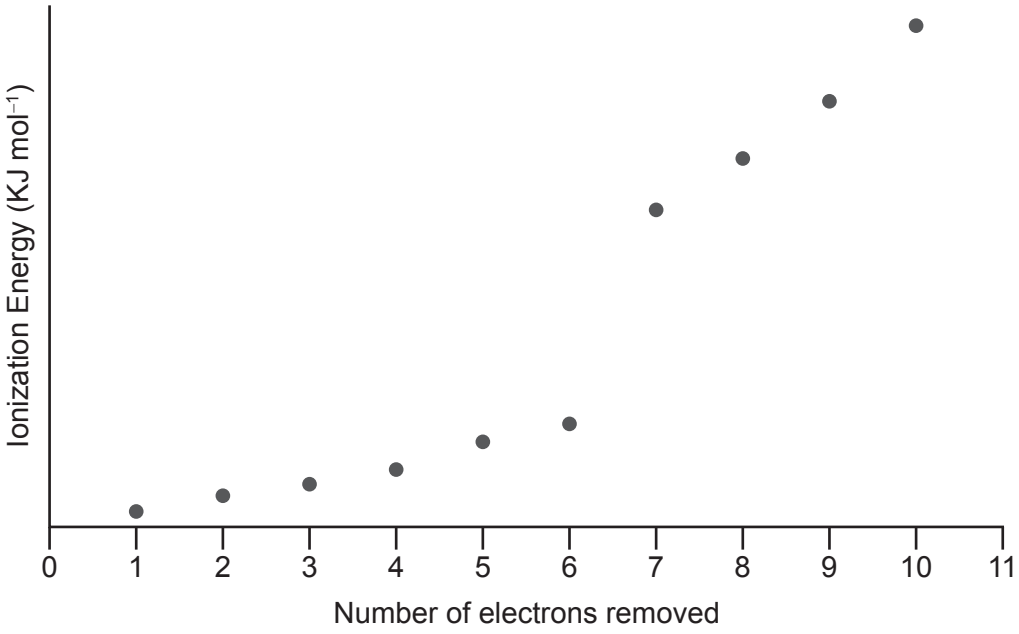
B. $[Ar]4s^2 3d^9$

C. $[Ar]4s^2 4d^9$

D. $[Ar]4s^1 3d^{10}$

2.1-M10 Nov 2024 · Paper 1 · Q7

7. The graph represents the first ten ionization energies (IE) of an element.



What is the element?

A. Cl

B. Ne

C. O

D. S

2.1-M11 Nov 2024 · Paper 1 · Q10

10. Which configuration is that of a transition metal atom in its ground state?

A. $[Ne]3s^2 3p^6$

B. $[Ar]3d^9$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

D. $[Ar]4s^1 3d^5$

2.1-M12 May 2024 TZ1 · Paper 1 · Q5

5. Which of the following can be used to determine the group of an element in the periodic table?

I. successive ionization energies
II. electron configuration
III. number of occupied energy levels

A. I and II only

B. I and III only

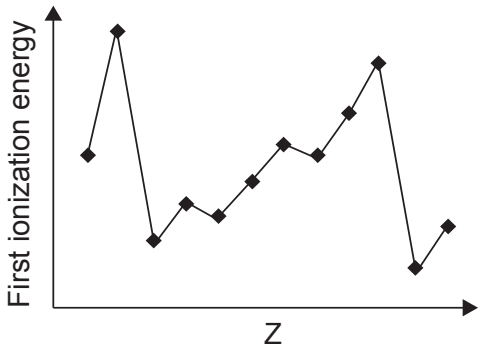
C. II and III only

D. I, II and III

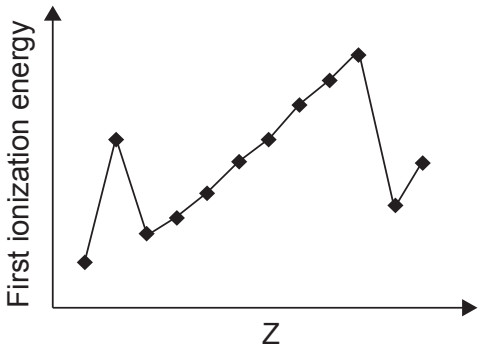
2.1-M13 May 2024 TZ2 · Paper 1 · Q6

6. Which graph shows the correct trend in the first ionization energies of the first twelve elements in the periodic table?

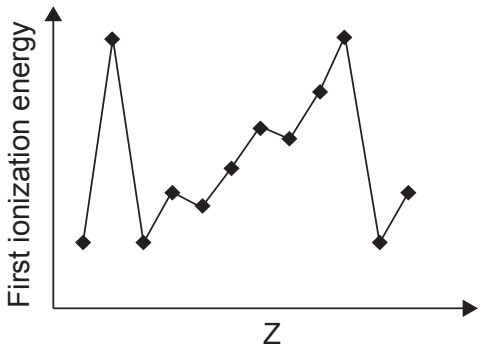
A.



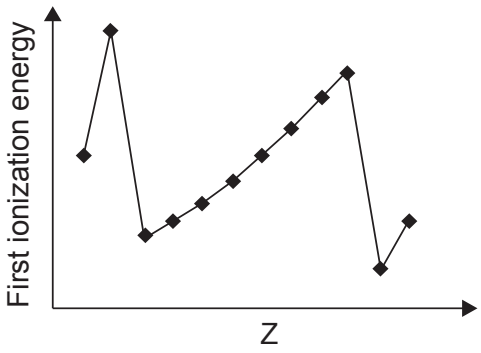
B.



C.

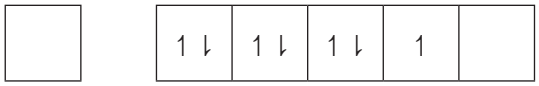


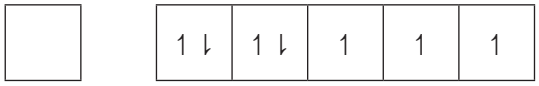
D.



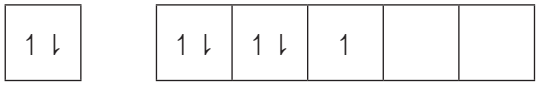
2.1-M14 May 2024 TZ2 · Paper 1 · Q9

9. In which diagram are the 4s and 3d orbitals of the Co^{2+} ion correctly filled with electrons?

A. 

B. 

C.

D. 

2.1-M15 Nov 2023 TZ1 · Paper 1 · Q5

5. The first three ionization energies for two elements, X and Y, are:

	Ionization energies ($kJ\,mol^{-1}$)		
	First	**Second**	**Third**
X	900	1757	14 849
Y	1086	2350	4620

Which pair of elements represent X and Y, respectively?

A. Lithium and beryllium

B. Lithium and carbon

C. Beryllium and carbon

D. Helium and beryllium

2.1-M16 May 2023 TZ1 · Paper 1 · Q5

5. Which statement best explains the first ionization energy of sulfur being lower than that of phosphorus?

A. Sulfur has more protons than phosphorus.

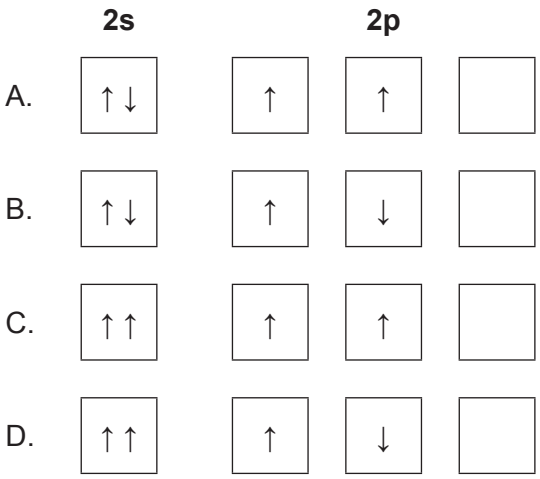
B. Phosphorus does not have paired electrons in the outer p sub-level.

C. Sulfur has an unpaired electron in the outer p sub-level.

D. Phosphorus is more reactive than sulfur.

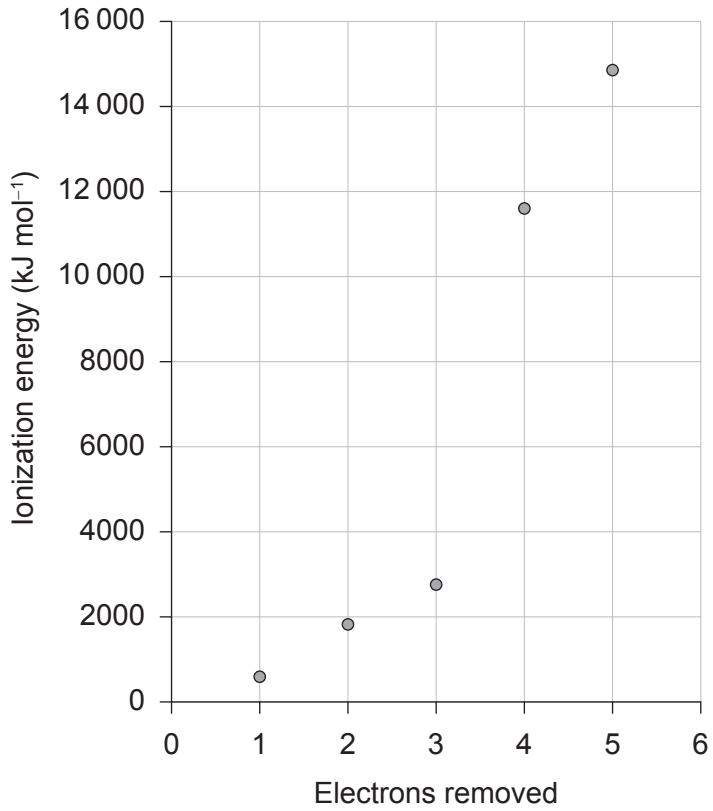
2.1-M17 May 2023 TZ2 · Paper 1 · Q4

4. What is the correct ground state electron orbital configuration for $2s^2 2p^2$?



2.1-M18 May 2023 TZ2 · Paper 1 · Q5

5. The successive ionization energies for an element in period three are shown.



Which element in period 3 has these successive ionization energies?

A. Na

B. Mg

C. Al

D. Si

2.1-M19 Nov 2022 · Paper 1 · Q4

4. Successive ionization energies of an element, **X**, are shown.

	1st	**2nd**	**3rd**	**4th**
Ionization energy (kJ mol^{-1})	740	1450	7730	10 540

What energy, in kJ mol^{-1}, is required for element **X** to reach its most stable oxidation state in ionic compounds?

A. 740

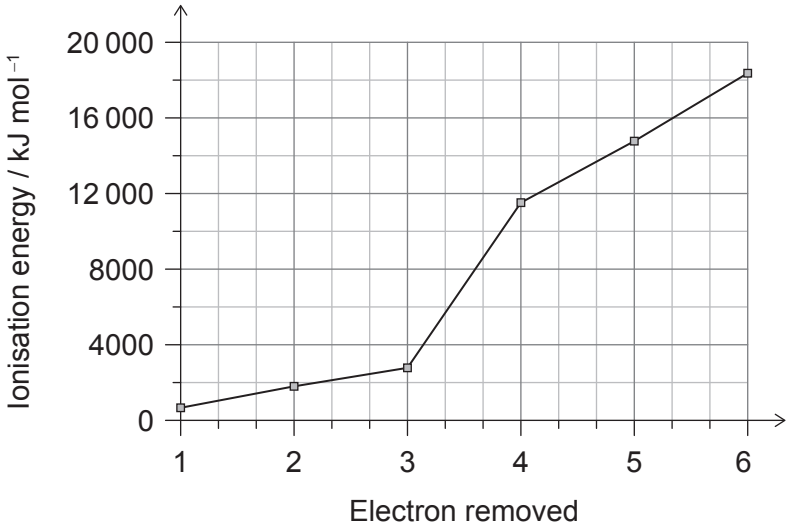
B. 1450

C. 2190

D. 7730

2.1-M20 May 2022 TZ1 · Paper 1 · Q5

5. The graph shows the first six ionization energies of an element.



In which group is the element?

A. 13

B. 14

C. 15

D. 16

2.1-M21 May 2022 TZ2 · Paper 1 · Q5

5. What is the correct order for **increasing** first ionization energy?

A. Na < Mg < Al

B. Na < Al < Mg

C. Al < Mg < Na

D. Al < Na < Mg

2.1-M22 Nov 2021 · Paper 1 · Q5

5. Which statement explains why the **second** ionization energy of aluminium is higher than the **first** ionization energy of magnesium?

A. Ionization energy increases along period 3.

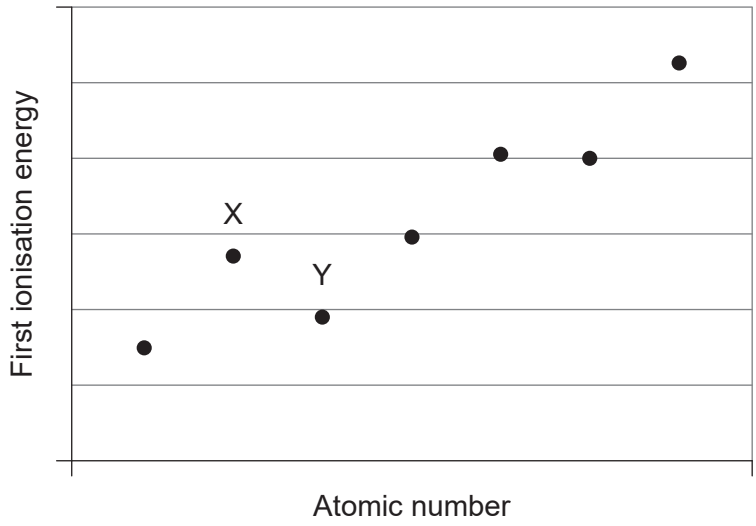
B. 3p electrons are at a higher energy level than 3s electrons.

C. 3p electrons are further away from the nucleus than 2p electrons.

D. Both have the same number of electrons and aluminium has one more proton.

2.1-M23 May 2021 TZ1 · Paper 1 · Q6

6. The diagram shows the first ionisation energies of consecutive elements in the same period of the periodic table.



Which factor explains why element X has a higher first ionisation energy than element Y?

A. Element Y loses an electron from a different sub-level.

B. Element Y has a smaller atomic radius.

C. Element X has a full octet.

D. Element Y has a greater nuclear charge.

2.1-M24 May 2021 TZ1 · Paper 1 · Q7

7. Which species has the same electron configuration as argon?

A. Br^-

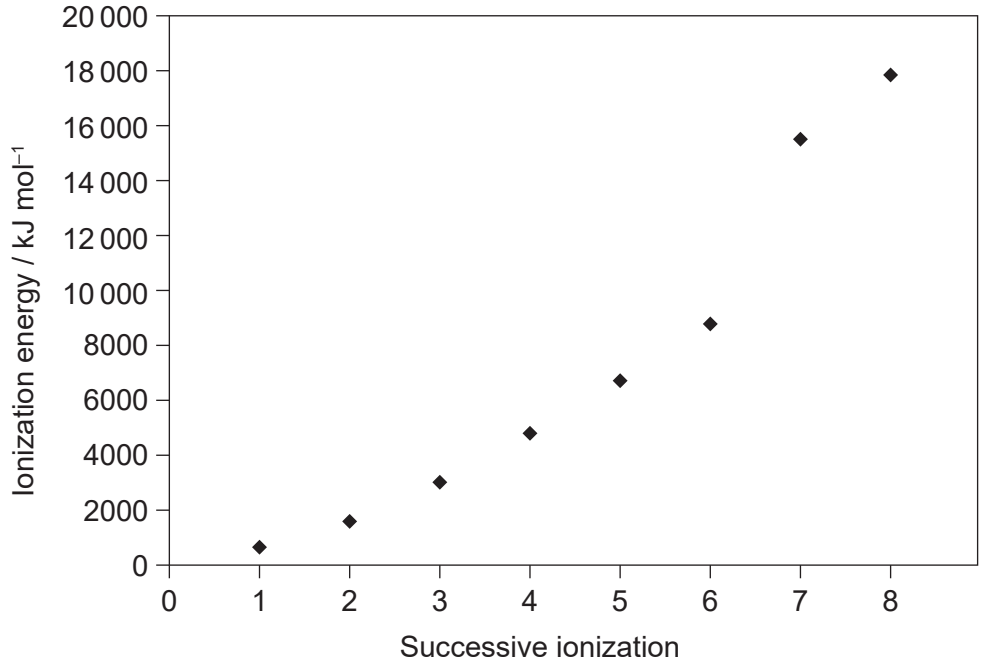
B. Ca^{2+}

C. Al^{3+}

D. Si^{4+}

2.1-M25 May 2021 TZ2 · Paper 1 · Q5

5. The first eight successive ionization energies for an element are shown. In which group is the element?



A. 6

B. 7

C. 8

D. 17

2.1-M26 Nov 2020 · Paper 1 · Q5

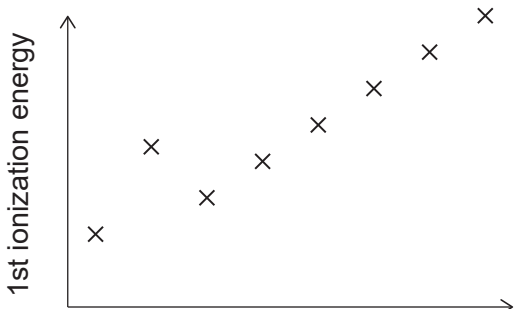
5. Which element is in group 13?

	Ionization energy / kJ mol^{-1}			
	1st	**2nd**	**3rd**	**4th**
A.	789	1580	3230	4360
B.	578	1820	2750	11 600
C.	738	1450	7730	10 500
D.	496	4560	6910	9540

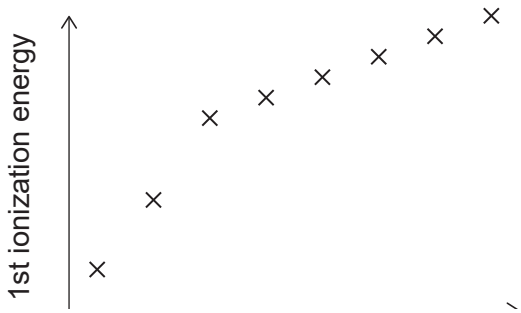
2.1-M27 Nov 2019 · Paper 1 · Q5

5. Which shows the first ionization energies of successive elements across period 2, from left to right?

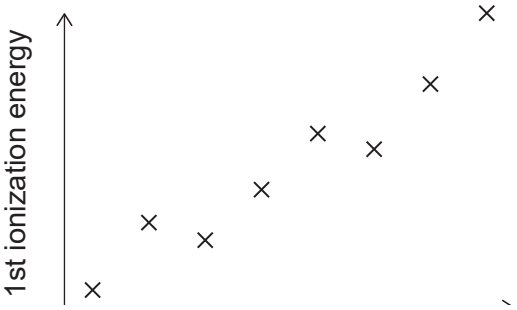
A.



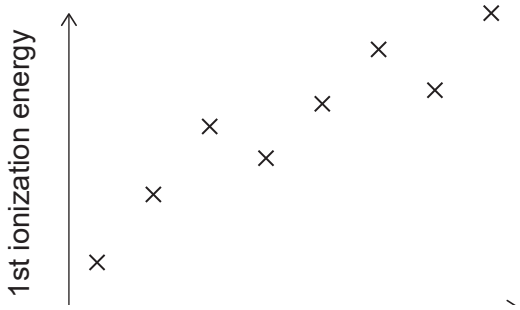
B.



C.



D.



2.1-M28 May 2019 TZ1 · Paper 1 · Q4

4. What is the ground state electron configuration of an atom of chromium, Cr (Z = 24)?

A. [Ar]3d^6

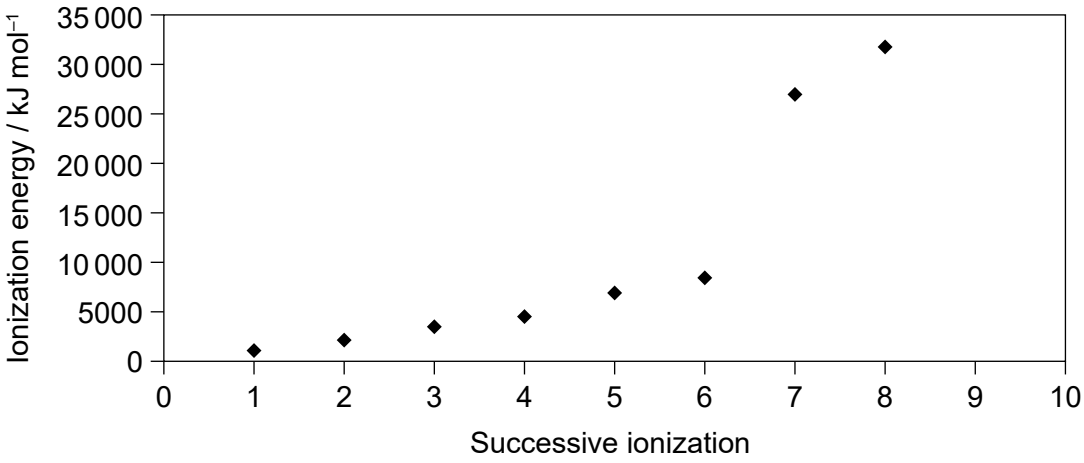
B. [Ar]4s^{2}3d^4

C. [Ar]4s^{1}3d^5

D. [Ar]4s^{2}4p^4

2.1-M29 May 2019 TZ1 · Paper 1 · Q5

5. Which element is represented by the first eight successive ionization energies on the graph?



A. Mg

B. S

C. Cl

D. Ar

2.1-M30 May 2019 TZ1 · Paper 1 · Q8

8. Which electrons are removed from iron (Z = 26) to form iron(II)?

A. two 3d electrons

B. two 4s electrons

C. one 4s electron and one 3d electron

D. two 4p electrons

2.1-M31 Nov 2018 · Paper 1 · Q5

5. The values for the first three successive ionization energies for two elements **X** and **Z** are given.

Element	First ionization energy / kJ mol^{-1}	Second ionization energy / kJ mol^{-1}	Third ionization energy / kJ mol^{-1}
X	520	7300	11 800
Z	1090	2350	4610

Which pair of elements represents **X** and **Z**?

	X	Z
A.	Li	Be
B.	Li	C
C.	Be	Li
D.	Be	C

2.1-M32 Nov 2018 · Paper 1 · Q7

7. Which species will require the least energy for the removal of one electron?

A. Na^+

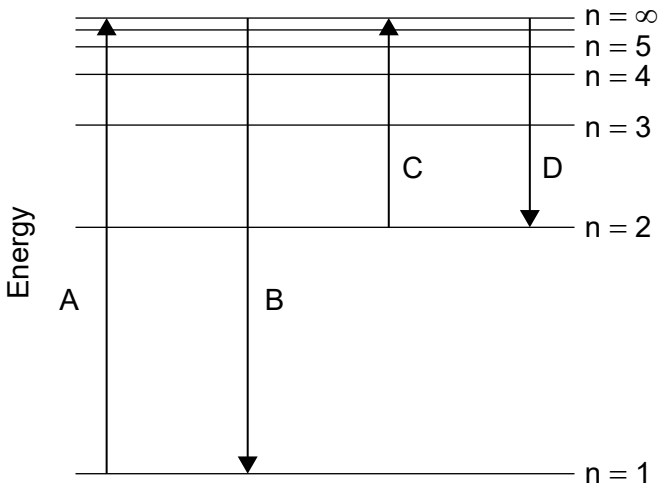
B. Mg^+

C. Al^{2+}

D. C^{3+}

2.1-M33 May 2018 TZ1 · Paper 1 · Q5

5. Which transition on the diagram corresponds to the ionization of hydrogen in the ground state?



2.1-M34 May 2018 TZ1 · Paper 1 · Q7

7. Which statement is correct?

A. Atomic radius decreases down group 17.

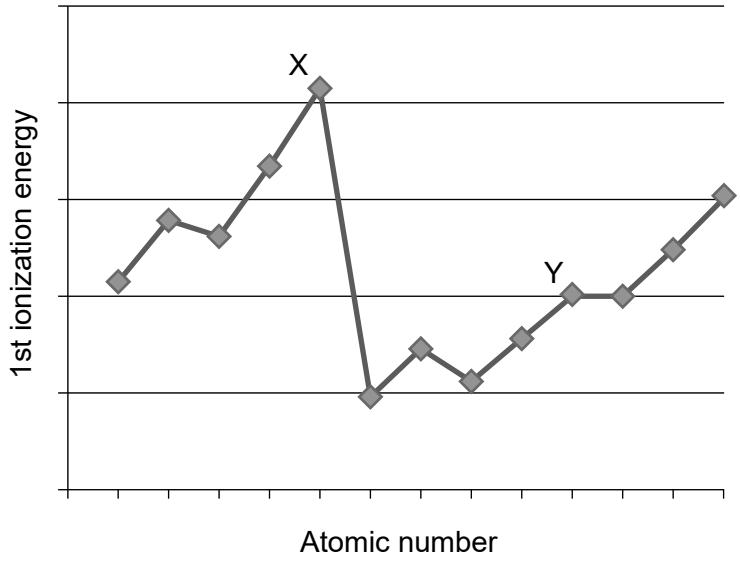
B. First ionization energy decreases down group 1.

C. Atomic radius increases across period 3 from Na to Cl.

D. First ionization energy decreases across period 3 from Na to Cl.

2.1-M35 May 2018 TZ2 · Paper 1 · Q5

5. The graph shows the first ionization energies of some consecutive elements.



Which statement is correct?

A. Y is in group 3

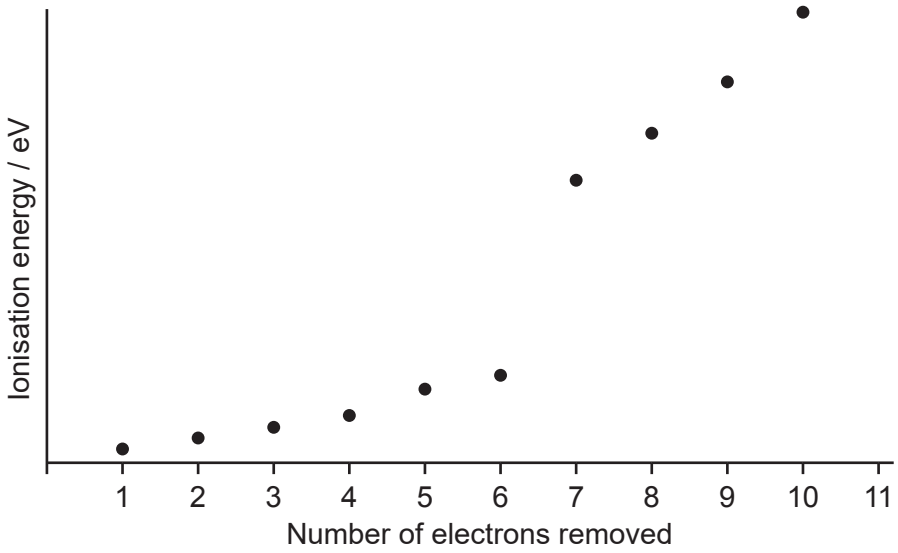
B. Y is in group 10

C. X is in group 5

D. X is in group 18

2.1-M36 Nov 2017 · Paper 1 · Q6

6. The graph represents the first ten ionisation energies (IE) of an element.



What is the element?

A. O

B. S

C. Ne

D. Cl

2.1-M37 Nov 2017 · Paper 1 · Q7

7. Which electron configuration is that of a transition metal atom in the ground state?

A. $[Ne]3s^2 3p^6 4s^1$

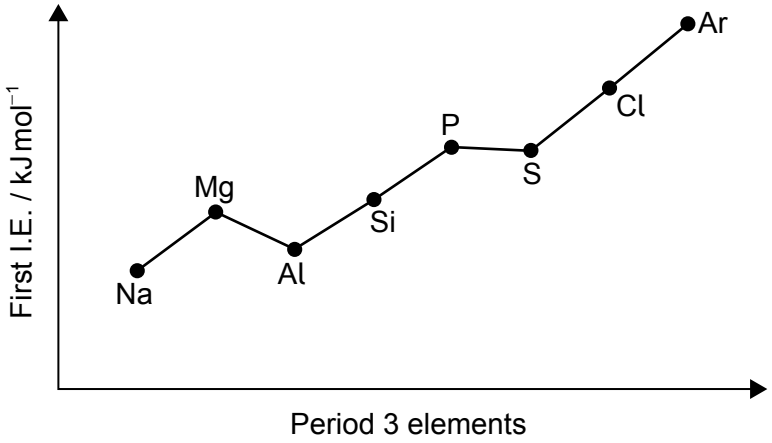
B. $[Ar]3d^9$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

D. $[Ar]4s^1 3d^5$

2.1-M38 May 2017 TZ1 · Paper 1 · Q5

5. Which statement explains one of the decreases in first ionization energy (I.E.) across period 3?



A. The nuclear charge of element Al is greater than element Mg.

B. The electron-electron repulsion is greater, for the electron with the opposite spin, in element S than in element P.

C. A new sub-level is being filled at element S.

D. The p orbital being filled in element Al is at a lower energy than the s orbital in element Mg.

2.1-M39 May 2017 TZ2 · Paper 1 · Q5

5. X, Y and Z represent the successive elements, Ne, Na and Mg, but not necessarily in that order.

	First ionization energy / kJ mol^{-1}
X	2081
Y	496
Z	738

What is the order of increasing atomic number?

A. X < Y < Z

B. X < Z < Y

C. Y < Z < X

D. Y < X < Z

2.1-M40 Nov 2016 · Paper 1 · Q6

6. A period 3 element, **M**, forms an oxide of the type $\mathbf{M}_2O$. Which represents the first four successive ionization energies of **M**?

	Ionization energy / kJ mol^{-1}			
	First	Second	Third	Fourth
A.	496	4563	6913	9544
B.	738	1451	7733	10541
C.	578	1817	2745	11578
D.	787	1577	3232	4356

2.1-M41 May 2016 · Paper 1 · Q5

5. Which electron configuration is correct for the selenide ion, Se^{2-}?

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 4d^{10}\ 4p^4$

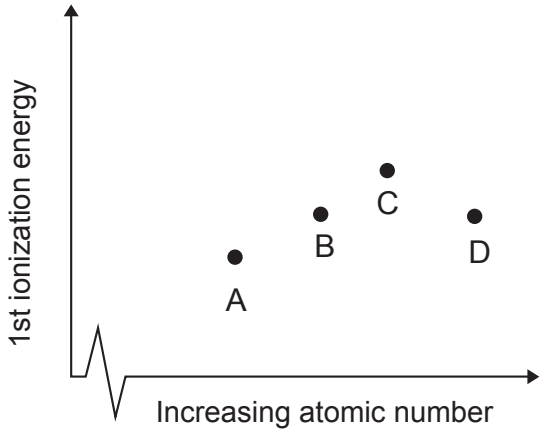
B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 4d^{10}\ 4p^6$

C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$

D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$

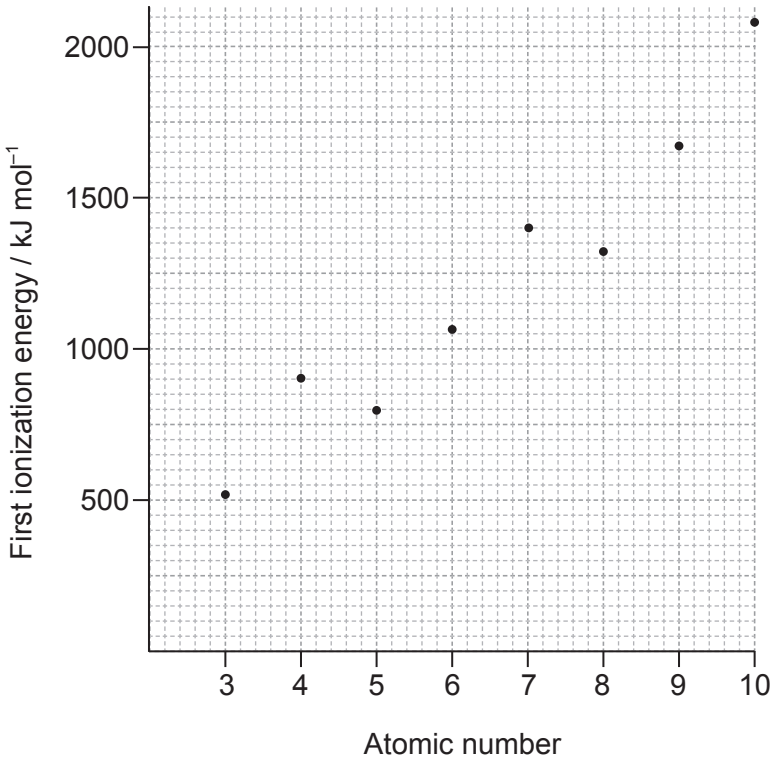
2.1-M42 May 2016 · Paper 1 · Q6

6. The diagram shows the first ionization energies of four consecutive elements in the periodic table. Which element is in Group 14?



2.1-M43 Nov 2015 · Paper 1 · Q5

5. Which statement is correct about the first ionization energies of consecutive elements shown in the graph?



[Source: Values from Nuffied Advance Science - Book of Data, Revised Edition (1984)]

A. The graph falls between Be and B because there is an electron in the third energy level.

B. The graph increases from B to N because the atomic radius is increasing.

C. The graph increases from Li to Ne because the number of electrons is increasing.

D. The graph falls between Be and B because the outer electron in B is in a p sub-level.

2.1-M44 Nov 2015 · Paper 1 · Q6

6. Which element has the greatest first ionization energy?

A. Al

B. Ar

C. Cl

D. Cs

2.1-M45 May 2015 TZ1 · Paper 1 · Q4

4. Ultraviolet radiation has a shorter wavelength than infrared radiation. How does the frequency and energy of ultraviolet radiation compare with infrared radiation?

	Frequency	Energy
A.	higher	higher
B.	higher	lower
C.	lower	higher
D.	lower	lower

2.1-M46 May 2015 TZ1 · Paper 1 · Q5

5. The first ionization energies (in $kJ\,mol^{-1}$) of five **successive** elements in the periodic table are:

1314, 1681, 2081, 496 and 738

What could these elements be?

A. d-block elements

B. The last two elements of one period and the first three elements of the next period

C. The last three elements of one period and the first two elements of the next period

D. The last five elements of a period

2.1-M47 May 2015 TZ1 · Paper 1 · Q7

7. What is the definition of the term *first ionization energy*?

A. The energy released when one mole of electrons is removed from one mole of gaseous atoms.

B. The energy required to remove one mole of electrons from one mole of gaseous atoms.

C. The energy released when one mole of gaseous atoms gains one mole of electrons.

D. The energy required to add one mole of electrons to one mole of gaseous atoms.

2.1-M48 May 2015 TZ2 · Paper 1 · Q5

5. What is the electron configuration of the copper(I) ion, Cu^+?

A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^9$

B. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

2.1-M49 Nov 2014 · Paper 1 · Q5

5. Successive ionization energies for an element, **Z**, are shown in the table below.

Electrons removed	1st	2nd	3rd	4th	5th
Ionization energy / kJ mol^{-1}	736	1450	7740	10 500	13 600

What is the most likely formula for the ion of **Z**?

A. Z^{+}

B. Z^{2+}

C. Z^{3+}

D. Z^{4+}

2.1-M50 Nov 2014 · Paper 1 · Q7

7. The elements argon, potassium, and calcium are consecutive in the periodic table. Which gives the correct order of **increasing** first ionization energies?

A. Ar < Ca < K

B. K < Ar < Ca

C. Ca < K < Ar

D. K < Ca < Ar

2.1-M51 May 2014 TZ1 · Paper 1 · Q5

5. Which electron configurations do **not** follow the Hund's rule?

	1s	**2s**	**2p**		
I.	↑ ↓	↑ ↓	↑	↑	↑
II.	↑ ↓	↑ ↓	↑ ↓	↑	
III.	↑ ↓	↑ ↓	↑	↓	↑

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

2.1-M52 May 2014 TZ2 · Paper 1 · Q4

4. What is the abbreviated electron configuration of the cobalt(II) ion, Co^{2+}?

A. [Ar] $3d^7$

B. [Ar] $4s^2\ 3d^5$

C. [Ar] $4s^2\ 3d^7$

D. [Ar] $4s^1\ 3d^6$

2.1-M53 May 2014 TZ2 · Paper 1 · Q6

6. Which equation represents the second ionization energy of potassium?

A. $K(g) \rightarrow K^{2+}(g) + 2e^-$

B. $K^+(g) \rightarrow K^{2+}(g) + e^-$

C. $K(s) \rightarrow K^{2+}(g) + 2e^-$

D. $K^+(s) \rightarrow K^{2+}(g) + e^-$

2.1-M54 May 2013 TZ1 · Paper 1 · Q5

5. What is the correct electron configuration of the Cu^+ ion?

A. [Ar] $3d^9\ 4s^1$

B. [Ar] $3d^7\ 4s^2$

C. [Ar] $3d^{10}$

D. [Ar] $3d^8\ 4s^1$

2.1-M55 May 2013 TZ2 · Paper 1 · Q5

5. Which species has the electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8$?

A. Ni

B. Ni^{2+}

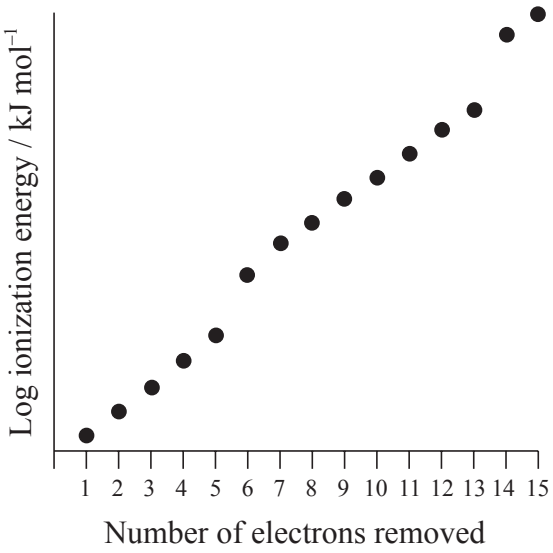
C. Fe

D. Cu^{2+}

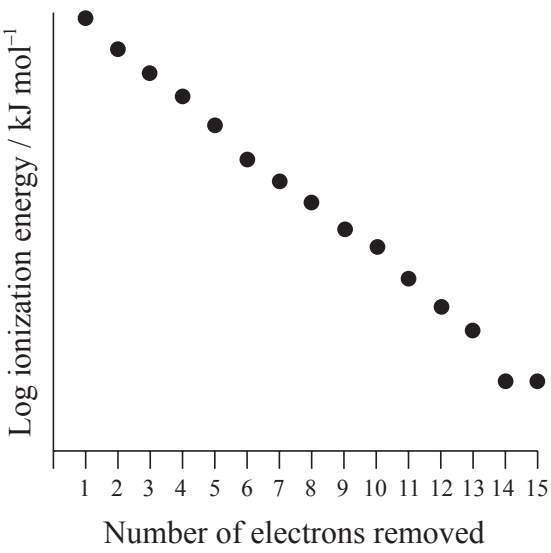
2.1-M56 Nov 2012 · Paper 1 · Q5

5. Which of the graphs below shows the successive logarithmic ionization energies of phosphorus?

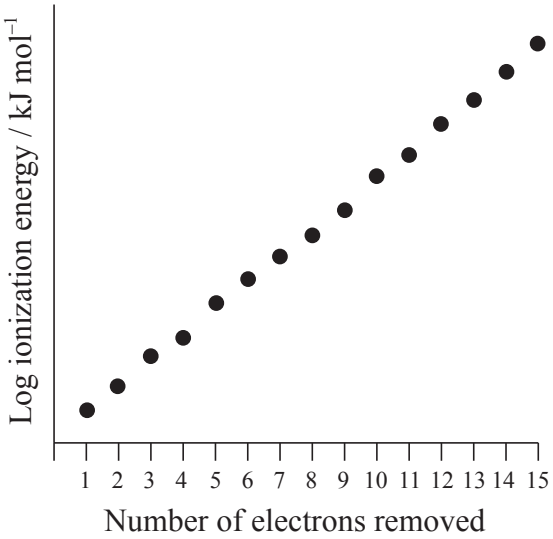
A.



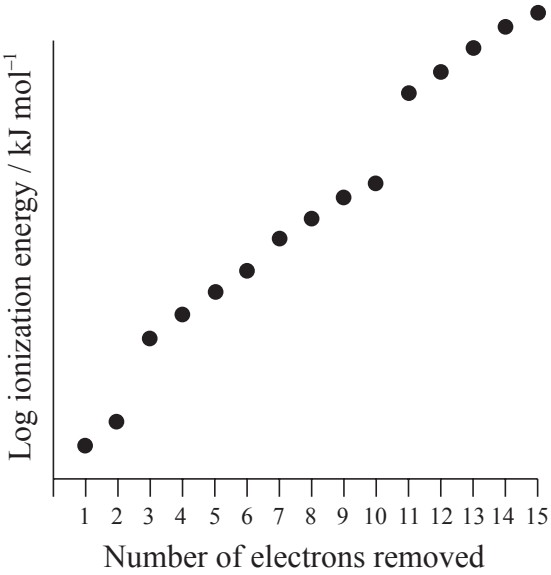
B.



C.



D.



2.1-M57 May 2012 TZ1 · Paper 1 · Q8

8. What are the electron configurations of Cu, Cu^+ and Cu^{2+}?

	Cu	**Cu^+**	**Cu^{2+}**
A.	[Ar] $4s^2$ $3d^9$	[Ar] $4s^2$ $3d^8$	[Ar] $4s^2$ $3d^7$
B.	[Ar] $4s^2$ $3d^9$	[Ar] $4s^1$ $3d^9$	[Ar] $3d^9$
C.	[Ar] $4s^2$ $3d^9$	[Ar] $3d^{10}$	[Ar] $3d^9$
D.	[Ar] $4s^1$ $3d^{10}$	[Ar] $3d^{10}$	[Ar] $3d^9$

2.1-M58 Nov 2011 · Paper 1 · Q6

6. Which shows the sub-levels in order of **increasing** energy in the fourth energy level of an atom?

A. f < d < p < s

B. p < d < f < s

C. d < f < p < s

D. s < p < d < f

2.1-M59 May 2011 TZ1 · Paper 1 · Q5

5. What is the electron configuration of vanadium?

A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^3$

B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$

C. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^1$

D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

2.1-M60 May 2011 TZ2 · Paper 1 · Q6

6. Values for the successive ionization energies for an unknown element are given in the table below.

First ionization energy / kJ mol^{-1}	Second ionization energy / kJ mol^{-1}	Third ionization energy / kJ mol^{-1}	Fourth ionization energy / kJ mol^{-1}
420	3600	4400	5900

In which group of the periodic table would the unknown element be found?

A. 1

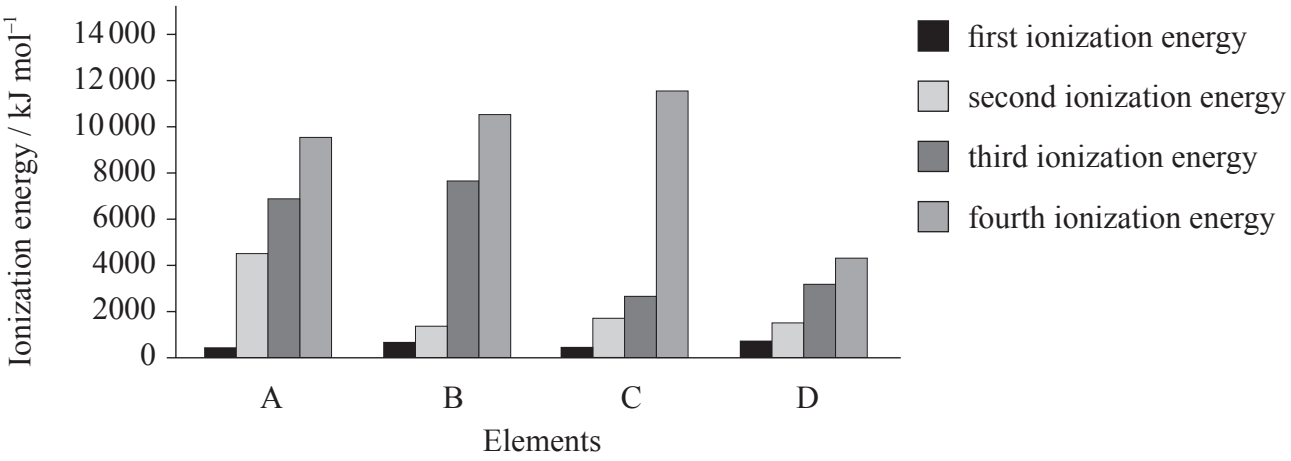
B. 2

C. 3

D. 4

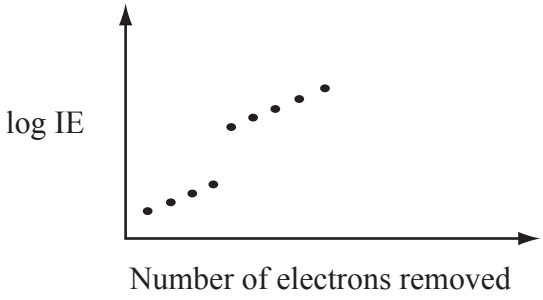
2.1-M61 Nov 2010 · Paper 1 · Q5

5. The graph below shows the first four ionization energies of four elements A, B, C and D (the letters are not their chemical symbols). Which element is magnesium?



2.1-M62 May 2010 TZ1 · Paper 1 · Q5

5. The graph represents the energy needed to remove nine electrons, one at a time, from an atom of an element. Not all of the electrons have been removed.



Which element could this be?

A. C

B. Si

C. P

D. S

2.1-M63 May 2010 TZ1 · Paper 1 · Q6

6. An ion has the electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$. Which ion could it be?

A. Ni^{2+}

B. Cu^{+}

C. Cu^{2+}

D. Co^{3+}

2.1-M64 May 2010 TZ2 · Paper 1 · Q5

5. What is the electron configuration of Fe^{3+}?

A. $[Ar]4s^2 3d^6$

B. $[Ar]3d^5$

C. $[Ar]4s^2 3d^5$

D. $[Ar]4s^2 3d^3$

Part B - Structured questions

2.1-S1 May 2025 TZ1 · Paper 2 · Q1 parts (d)

1. Iron(II) sulfide can be produced by heating powdered iron and sulfur together.

(d) Calculate the wavelength, in m, for the limit of convergence observed in the line spectrum of iron. Use sections 1, 2 and 9 of the data booklet. [3]

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2.1-S2 May 2025 TZ1 · Paper 2 · Q2 parts (a)

2. Cobalt ions form coloured compounds.

(a) Deduce the electron configuration of the Co^{2+} ion. [1]

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2.1-S3 May 2025 TZ2 · Paper 2 · Q2 parts (b)(iii), (b)(vii), (b)(viii), (c)(i), (c)(ii), (c)(iii) · also 6.3, 7.2, 9.4

2. Thallium is a heavy metal in group 13 of the periodic table.

(b) Thallium (I) sulfate has the formula Tl_2SO_4.

(iii) State the enthalpy term that characterises the strength of the bonding between the ions in an ionic solid. [1]

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(vii) Discuss how the relative reactivity of copper and thallium could be established using the metals and aqueous solutions of their sulfates. [2]

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(viii) Discuss the products at the electrodes when aqueous thallium (I) sulfate is electrolyzed.

Use standard reduction potential of $Tl^{+}(aq) + e^{-} \rightleftharpoons Tl(s) = -0.34\,V$ and section 19 of the data booklet. [2]

Anode (positive electrode): . . . . . .

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Cathode (negative electrode): . . . . . .

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(c) Thallium has an atomic emission spectrum with a strong green line.

(i) State the feature of the atomic emission spectrum that may be used to find the ionization energy of the element. [1]

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(ii) Calculate the wavelength that corresponds to the ionization energy of thallium. Use sections 1, 2 and 9 of the data booklet. [3]

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(iii) Explain why, in terms of nuclear charge and the shielding of the valence electrons, the first ionization energy of thallium is lower than that of lead. [2]

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2.1-S4 May 2024 TZ1 · Paper 2 · Q5 parts (a), (b)(i) · also 3.1

5. Successive ionization energies of an element (E) are shown. E is not the real symbol of the element.

Number of ionization energy	**first**	**second**	**third**	**fourth**	**fifth**	**sixth**	**seventh**
IE / kJ mol^{-1}	1000	2295	3375	4565	6950	8490	27 107

(a) Identify the group of the periodic table in which element E is located, giving a reason. [1]

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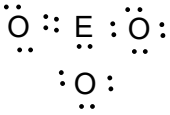
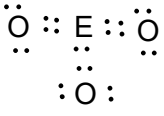
(b) Element E forms an oxide EO_3. Two possible Lewis structures are shown.

(i) Show which structure is most likely, using the concept of formal charge. [3]

Structure 1



Structure 2



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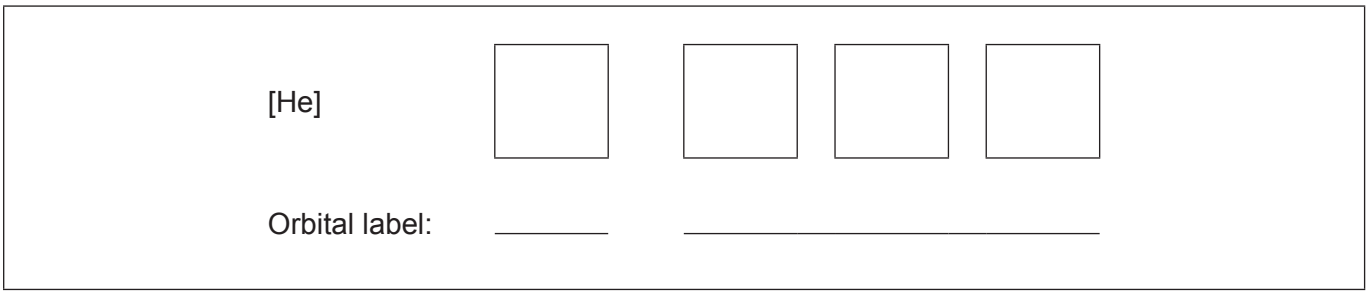
2.1-S5 May 2023 TZ1 · Paper 2 · Q2 parts (a)(i), (b), (c)(i), (c)(ii), (c)(iii) · also 3.1, 5.2

2. The periodic table is a useful tool in explaining trends of chemical behaviour.

(a) (i) Annotate and label the ground state orbital diagram of boron, using arrows to represent electrons. [1]



(b) Explain the decrease in first ionization energy from Li to Cs, group 1. [2]

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(c) (i) State the electron domain geometry of the ammonia molecule. [1]

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(ii) Deduce the Lewis (electron dot) structure of ammonia and sketch its 3D molecular shape. [2]

Lewis structure:

3D molecular shape:

(iii) Explain, with reference to the forces between molecules, why ammonia has a higher boiling point than phosphine (PH_3). [3]

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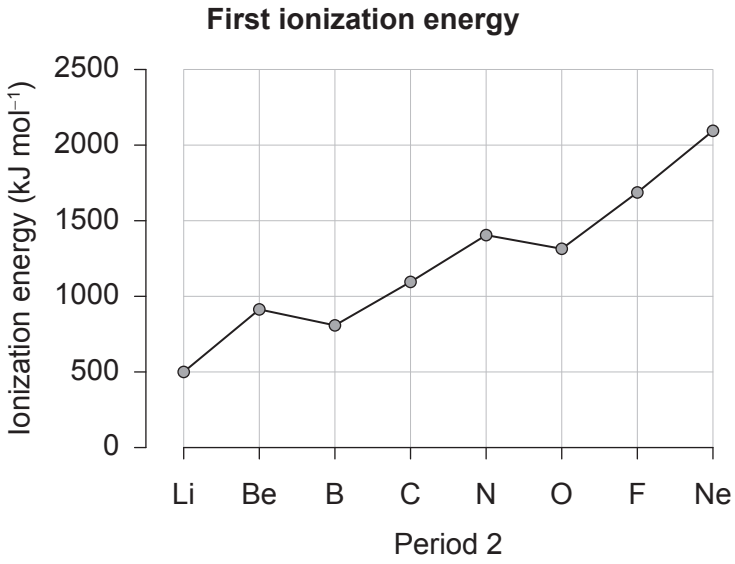
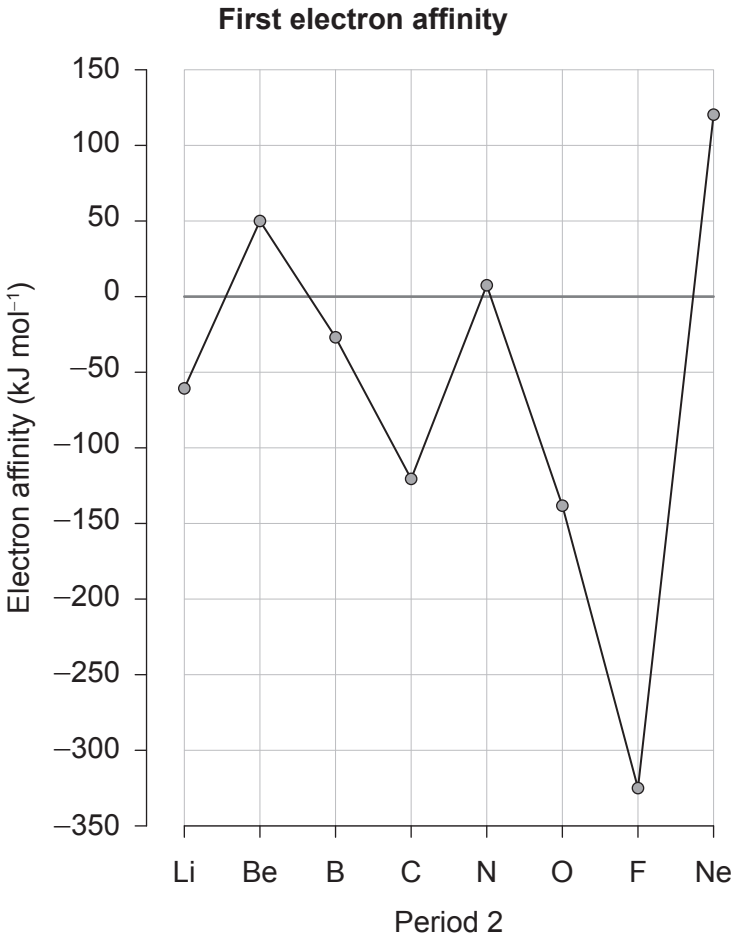
2.1-S6 May 2023 TZ1 · Paper 3 · Q1 parts (a), (b), (c) · also 9.4

1. Graphs showing the first ionization energy and first electron affinity of the elements in period 2 of the periodic table are shown.





(a) Outline why ionization energies have positive values but most electron affinities have negative values. [1]

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(b) First ionization energy tends to increase across the period. Explain the decrease in first ionization energy from beryllium to boron. [2]

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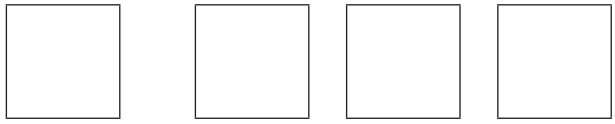
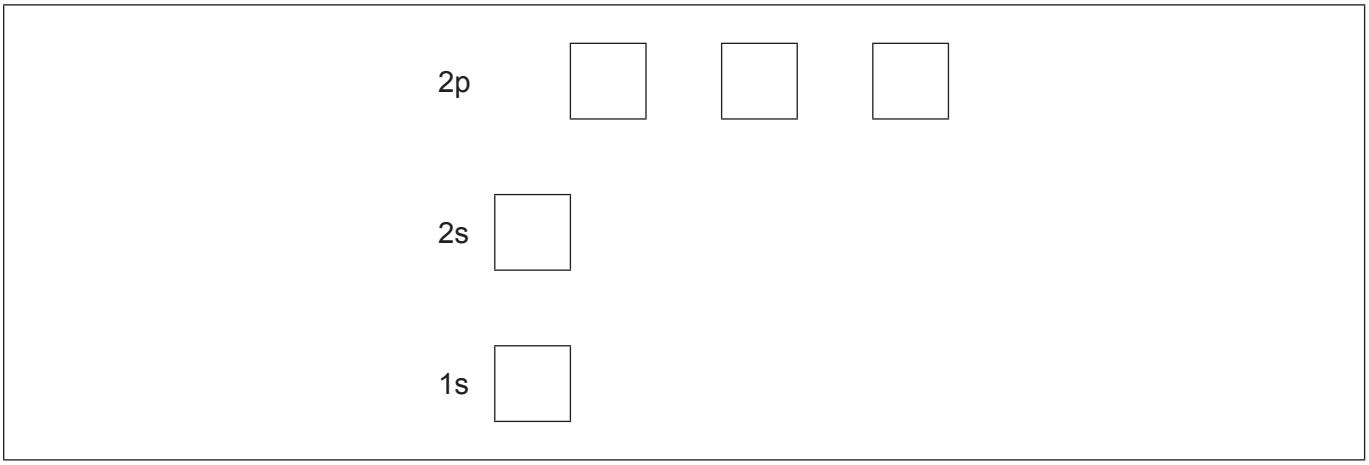
(c) The electron affinity of nitrogen is 6.8 kJ mol^{-1}. Sketch the 2s and 2p orbital filling diagram that represents the electron arrangement of the species produced. [1]



2.1-S7 May 2022 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii) · also 3.1, 4.1

6. Nitric acid is usually produced by the oxidation of ammonia.

(a) (i) Draw arrows in the boxes to represent the electron configuration of a nitrogen atom. [1]



(ii) Deduce a Lewis (electron dot) structure of the nitric acid molecule, HNO_3, that obeys the octet rule, showing any non-zero formal charges on the atoms. [2]

(iii) Explain the relative lengths of the three bonds between N and O in nitric acid. [3]

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2.1-S8 May 2022 TZ2 · Paper 2 · Q2 parts (a), (b)(ii)

2. Electrons are arranged in energy levels around the nucleus of an atom.

(a) Explain why the first ionization energy of calcium is greater than that of potassium. [2]

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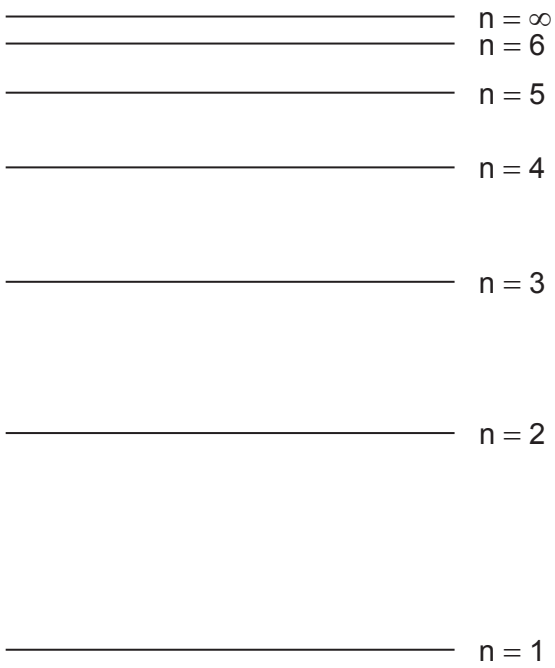
(b) The diagram represents possible electron energy levels in a hydrogen atom.



(ii) Draw an arrow, labelled **X**, to represent the electron transition for the ionization of a hydrogen atom in the ground state. [1]

2.1-S9 May 2022 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii), (b) · also 6.1

5. Iron(II) disulfide, FeS_2, has been mistaken for gold.

(a) (i) State the full electronic configuration of Fe^{2+}. [1]

. . . . . .

. . . . . .

(ii) Explain why there is a large increase from the 8th to the 9th ionization energy of iron. [2]

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(b) Calculate the oxidation state of sulfur in iron(II) disulfide, FeS_2. [1]

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2.1-S10 Nov 2021 · Paper 2 · Q2 parts (a)

2. Electron transitions are related to trends in the periodic table.

(a) Explain the general increase in trend in the first ionization energies of the period 3 elements, Na to Ar. [2]

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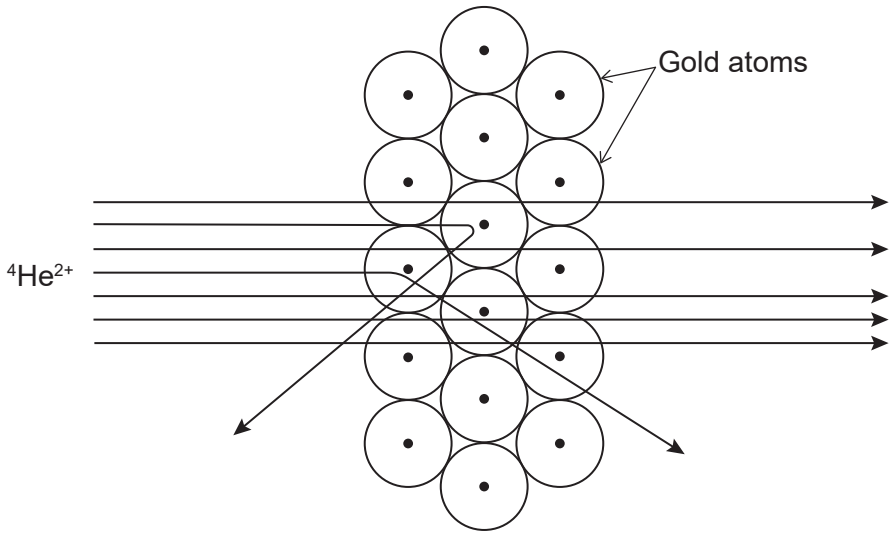
2.1-S11 Nov 2021 · Paper 2 · Q9 parts (b)(ii)

9. Fast moving helium nuclei ($^4He^{2+}$) were fired at a thin piece of gold foil with most passing undeflected but a few deviating largely from their path. The diagram illustrates this historic experiment.



(ii) State the electron configuration of copper. [1]

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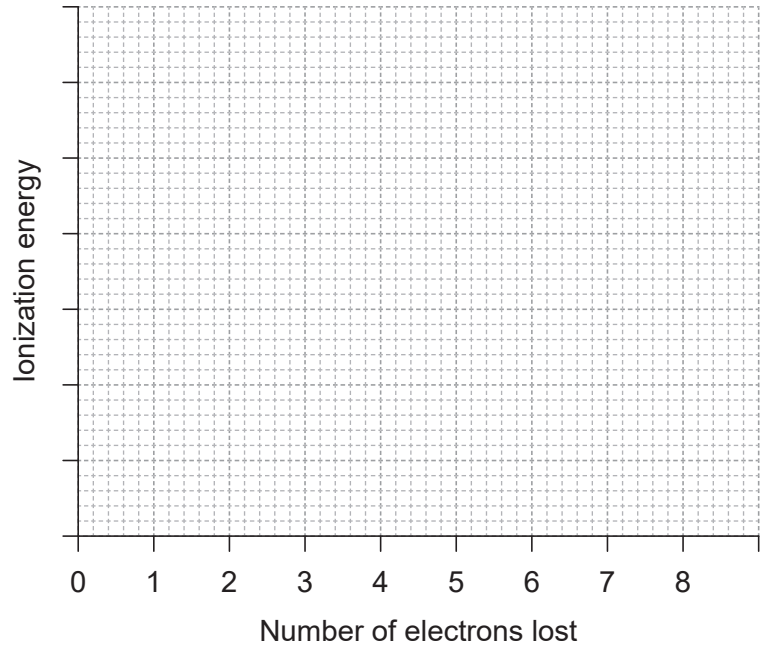
2.1-S12 May 2021 TZ1 · Paper 2 · Q1 parts (c), (d)(iii), (e)(i), (e)(ii) · also 1.2, 6.1

1. Iron may be extracted from iron (II) sulfide, FeS.

(c) Sketch the first eight successive ionisation energies of sulfur. [2]



(d) Iron (II) sulfide, FeS, is ionically bonded.

(iii) State the full electron configuration of the sulfide ion. [1]

. . . . . .

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(e) The first step in the extraction of iron from iron (II) sulfide is to roast it in air to form iron (III) oxide and sulfur dioxide.

(i) Write the equation for this reaction. [1]

. . . . . .

. . . . . .

(ii) Deduce the change in the oxidation state of sulfur. [1]

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2.1-S13 May 2021 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i), (c), (d) · also 3.1, 5.2

2. The properties of elements can be predicted from their position in the periodic table.

(ii) Explain why the first ionization energy of sulfur is lower than that of phosphorus. [2]

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(b) (i) State the condensed electron configurations for Cr and Cr^{3+}. [2]

Cr:

. . . . . .

Cr^{3+}:

. . . . . .

(c) Deduce the Lewis (electron dot) structure and molecular geometry of sulfur tetrafluoride, SF_4, and sulfur dichloride,SCl_2. [4]

Species	**SF_4**	**SCl_2**
Lewis structure		
Molecular geometry	. . . . .	. . . . .

(d) Suggest, giving reasons, the relative volatilities of SCl_2 and H_2O. [3]

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2.1-S14 Nov 2020 · Paper 2 · Q6 parts (a), (c) · also 7.2

6. The electron configuration of copper makes it a useful metal.

(a) Determine the frequency of a photon that will cause the first ionization of copper. Use sections 1, 2 and 8 of the data booklet. [2]

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(c) Copper plating can be used to improve the conductivity of an object.

State, giving your reason, at which electrode the object being electroplated should be placed. [1]

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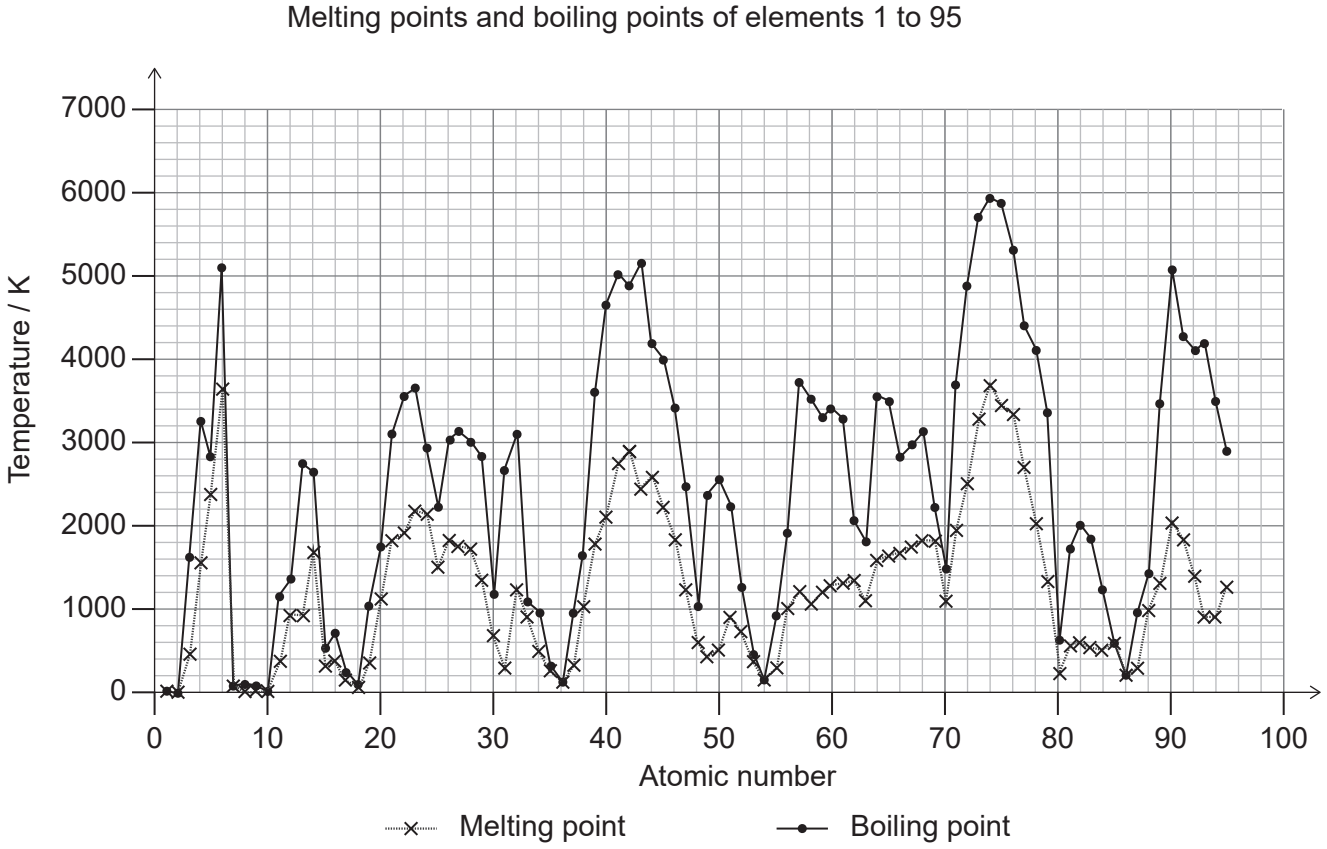
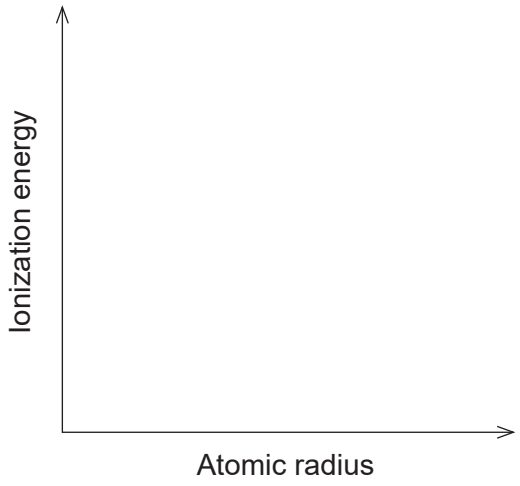
2.1-S15 May 2019 TZ1 · Paper 3 · Q1 parts (b)(iv)

1. Physical properties of elements vary according to atomic number. Sections 6 to 9 of the data booklet list some of these properties.



[Source: www.mrbigler.com/documents/Periodic-Table.xls, used with the kind permission of Jeff Bigler]

(iv) Sketch how the first ionization energies of elements vary with their atomic radius. [1]



2.1-S16 May 2019 TZ2 · Paper 2 · Q3 parts (c), (d)(i), (d)(ii), (d)(iii) · also 3.1, 4.1, 4.2

3. Dinitrogen monoxide, N_2O, causes depletion of ozone in the stratosphere.

(c) Explain why the first ionization energy of nitrogen is greater than both carbon and oxygen. [2]

Nitrogen and carbon:

. . . . . .

. . . . . .

. . . . . .

Nitrogen and oxygen:

. . . . . .

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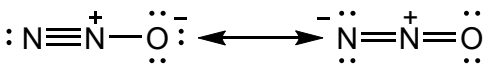
(d) The Lewis (electron dot) structure of the dinitrogen monoxide molecule can be represented as:



(i) State what the presence of alternative Lewis structures shows about the nature of the bonding in the molecule. [1]

. . . . . .

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(ii) State, giving a reason, the shape of the dinitrogen monoxide molecule. [1]

. . . . . .

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(iii) Deduce the hybridization of the central nitrogen atom in the molecule. [1]

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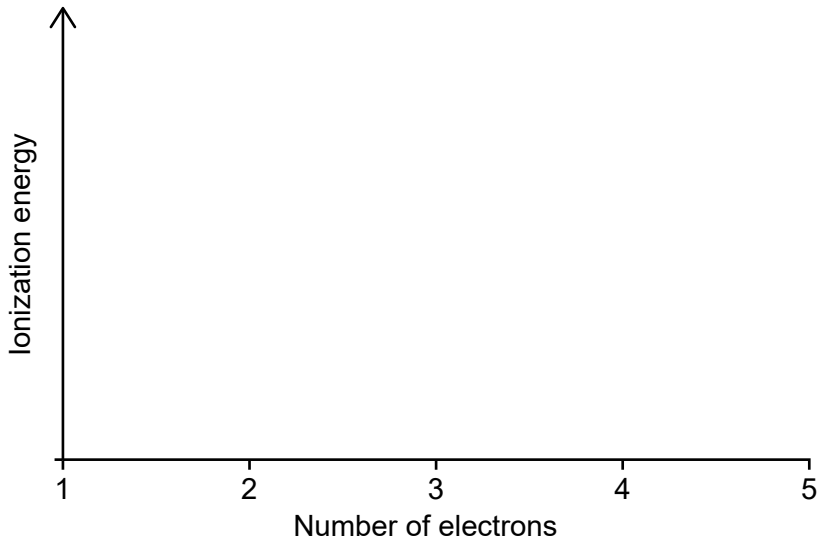
2.1-S17 Nov 2018 · Paper 2 · Q4 parts (c)

4. Properties of elements and their compounds can be related to the position of the elements in the periodic table.

(c) Sketch a graph to show the relative values of the successive ionization energies of boron. [2]



2.1-S18 Nov 2017 · Paper 2 · Q3 parts (a), (d)(ii) · also 6.1

3. Trends in physical and chemical properties are useful to chemists.

(a) Explain the general increasing trend in the first ionization energies of the period 3 elements, Na to Ar. [2]

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(d) Cobalt forms the transition metal complex $[Co(NH_3)_4(H_2O)Cl]Br$.

(ii) Deduce the charge on the complex ion and the oxidation state of cobalt. [2]

Charge on complex ion:

. . . . . .

Oxidation state of cobalt:

. . . . . .

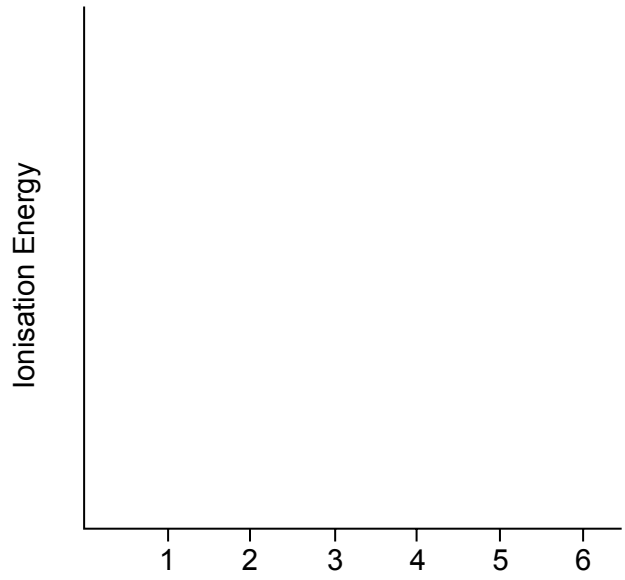
2.1-S19 May 2017 TZ1 · Paper 2 · Q2 parts (d)(i), (d)(iii)

2. Titanium and vanadium are consecutive elements in the first transition metal series.

(d) (i) State the full electron configuration of the $^{48}_{22}Ti^{2+}$ ion. [1]

. . . . . .

(iii) Sketch a graph of the first six successive ionization energies of vanadium on the axes provided. [1]



2.1-S20 May 2016 · Paper 2 · Q4 parts (a)(i), (b)(ii), (e)(i), (e)(ii), (e)(iii) · also 7.1, 9.4

4. Tin(II) chloride is a white solid that is commonly used as a reducing agent.

(a) (i) State why you would expect tin(II) chloride to have a similar lattice enthalpy to strontium chloride, using section 9 of the data booklet. [1]

. . . . . .

. . . . . .

(b) Tin can also exist in the +4 oxidation state.

$$Sn^{4+}(aq) + 2e^- \rightleftharpoons Sn^{2+}(aq) \qquad E^{\ominus} = +0.15\,V$$

Vanadium can be reduced from an oxidation state of +4 to +3 according to the equation:

$$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l) \qquad E^{\ominus} = +0.34\,V$$

(ii) Deduce, giving your reason, whether a reaction between $Sn^{2+}(aq)$ and $VO^{2+}(aq)$ would be spontaneous. [1]

. . . . . .

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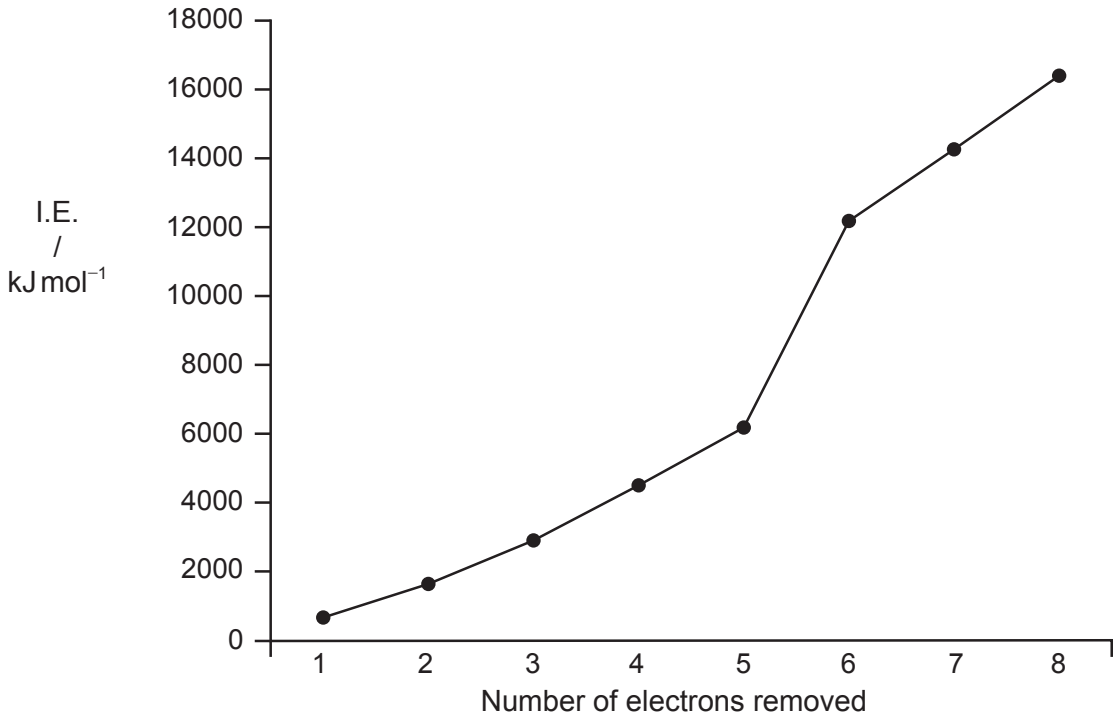
(e) Eight successive ionisation energies of vanadium are shown in the graph below:



(i) State the sub-levels from which each of the first four electrons are lost. [1]

First: . . . . Second: . . . . Third: . . . . Fourth: . . . .

(ii) Outline why there is an increase in ionization energy from electron 3 to electron 5. [1]

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(iii) Explain why there is a large increase in the ionization energy between electrons 5 and 6. [3]

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2.1-S21 Nov 2015 · Paper 2 · Q2 parts (a)

(a) State the **full** electron configurations of copper, Cu, and copper (II) ion, Cu^{2+}. [2]

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. . . . . .

2.1-S22 May 2015 TZ1 · Paper 2 · Q4 parts (a)

4. Copper is a metal that has been used by humans for thousands of years.

(a) State the full electron configuration of ^{65}Cu. [1]

. . . . . .

2.1-S23 May 2015 TZ1 · Paper 2 · Q8 parts (e), (f) · also 3.1

8. Acids can be described as strong or weak.

(e) Describe how chlorine's position in the periodic table is related to its electron arrangement. [2]

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. . . . . .

(f) SCl_2 and $SClF_5$ are two sulfur chloride type compounds with sulfur having different oxidation states. Predict the name of the shape, the bond angle and polarity of these molecules. [3]

Molecule	Shape	Bond angle	Polarity
SCl_2	. . . . .	. . . . .	. . . . .
$SClF_5$	. . . . .	. . . . .	. . . . .

2.1-S24 May 2015 TZ1 · Paper 3 · Q1 parts (a)

1. Electromagnetic waves can transfer energy and carry information.

(a) State the relationship between the energy of a wave and its wavelength. [1]

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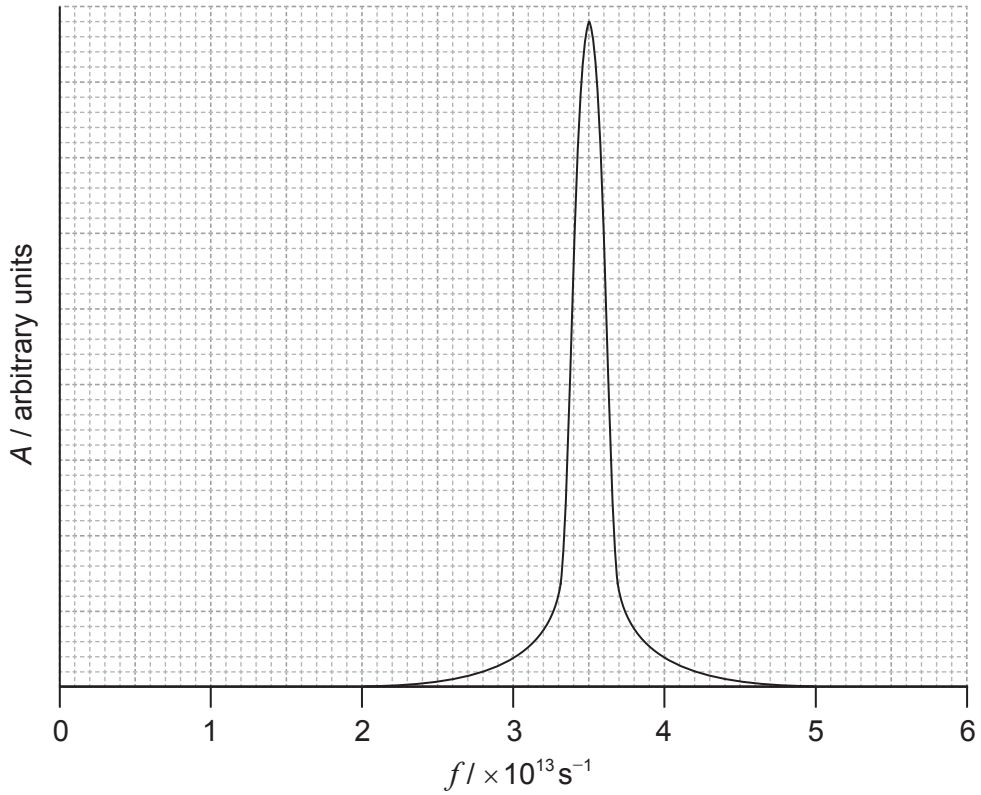
2.1-S25 May 2015 TZ1 · Paper 3 · Q18 parts (a)(i)

18. The atmospheric greenhouse effect is caused by several gases.

(a) The graph shows part of the absorption spectrum of an oxide of nitrogen in which the intensity of absorbed radiation *A* is plotted against frequency f.



(i) Determine the wavelength of the radiation absorbed, in m, using table 1 of the data booklet. (The speed of light, *c*, is 3.0×10^8.) [1]

. . . . . .

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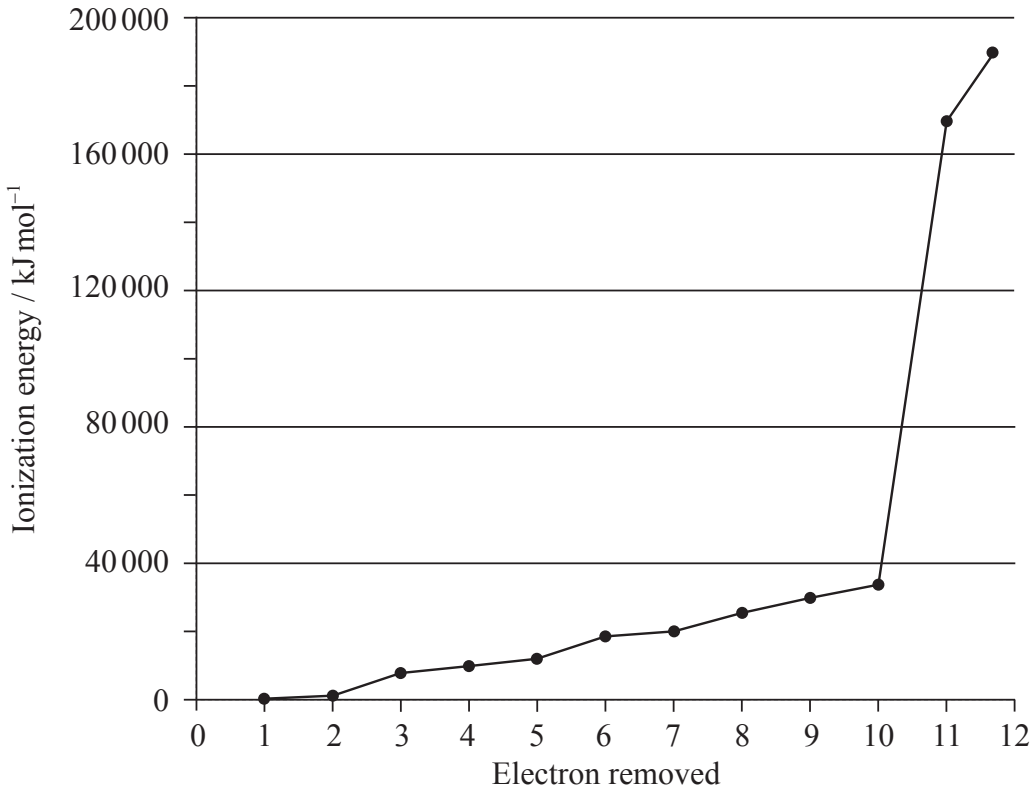
2.1-S26 Nov 2014 · Paper 2 · Q8 parts (b)(i), (b)(ii), (c)(ii), (d)(i), (d)(ii), (d)(iii) · also 3.1, 4.2, 5.1

A sample of magnesium contains three isotopes: magnesium-24, magnesium-25 and magnesium-26, with abundances of 77.44 %, 10.00 % and 12.56 % respectively.

(b) A graph of the successive ionization energies of magnesium is shown below.



(i) Explain the increase in ionization energy values from the 3rd to the 8th electrons. *[1]*

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. . . . . .

(ii) Explain the sharp increase in ionization energy values between the 10th and 11th electrons. *[2]*

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(ii) Carbon reacts with oxygen to form a covalent compound, carbon dioxide. Describe what is meant by a covalent bond. *[1]*

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. . . . . .

(d) (i) Predict the type of hybridization of the carbon and oxygen atoms in CO_2. *[2]*

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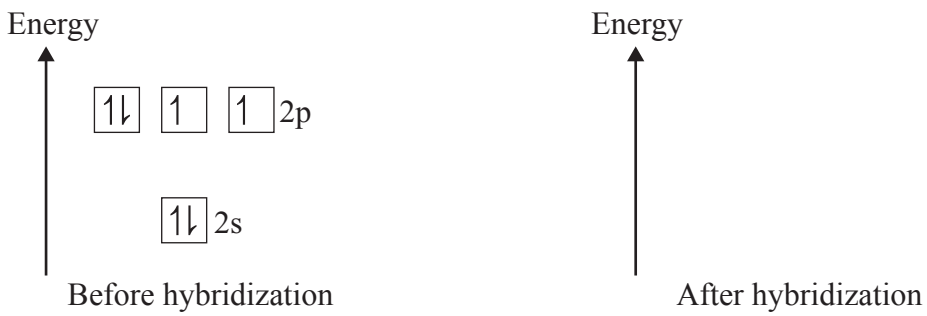
(ii) Sketch the orbitals of an oxygen atom in CO_2 on the energy level diagram provided, including the electrons that occupy each orbital. *[2]*



(iii) Define the term *electronegativity*. *[1]*

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2.1-S27 May 2014 TZ1 · Paper 2 · Q5 parts (b)(ii), (c)(i), (c)(ii), (d)(i), (e) · also 5.2, 6.3

(b) Chlorine gas, $Cl_2(g)$, is bubbled through separate solutions of aqueous bromine, $Br_2(aq)$, and potassium bromide, KBr(aq).

(ii) State the half-equations for the reactions that occur. *[2]*

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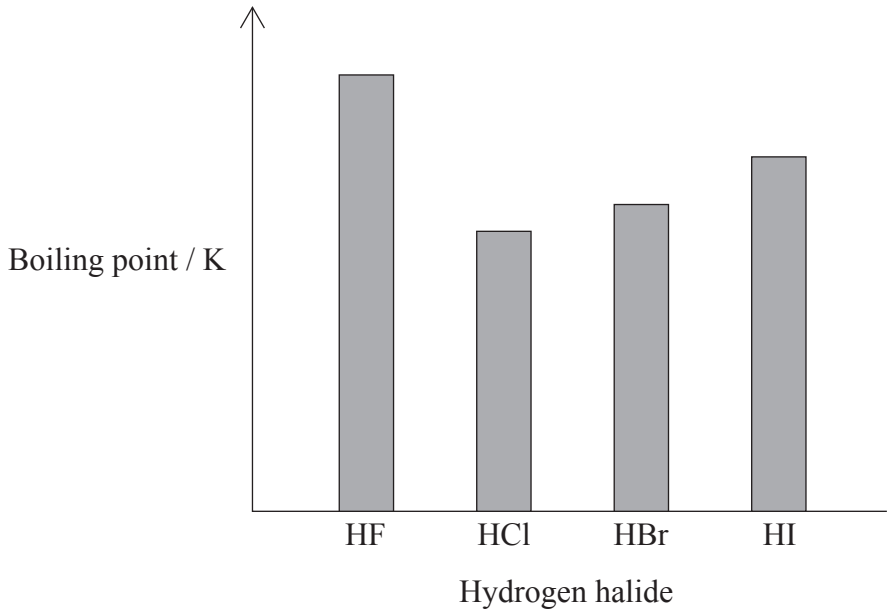
(c) The hydrogen halides do not show perfect periodicity. A bar chart of boiling points shows that the boiling point of hydrogen fluoride, HF, is much higher than periodic trends would indicate.



(i) Explain why the boiling point of HF is much higher than the boiling points of the other hydrogen halides. *[1]*

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. . . . . .

(ii) Explain the trend in the boiling points of HCl, HBr and HI. *[2]*

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. . . . . .

(d) Transition metals form complex ions which are usually coloured.

(i) State the full electron configurations of Cr and Cr^{3+}. *[2]*

Cr:

. . . . . .

Cr^{3+}:

. . . . . .

(e) Explain how the number of electrons in the outer main energy level of phosphorus, P, can be determined using the data of successive ionization energies. *[2]*

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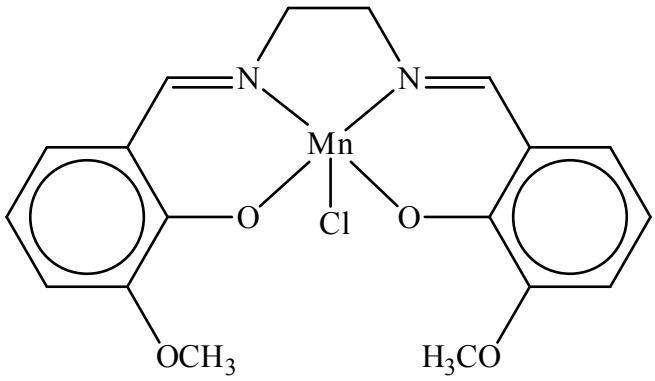
2.1-S28 Nov 2013 · Paper 2 · Q4 parts (a)

4. EUK-134, the structure of which is shown below, is a complex ion of manganese(III) that is used in expensive sun-protection products because of its powerful antioxidant properties.

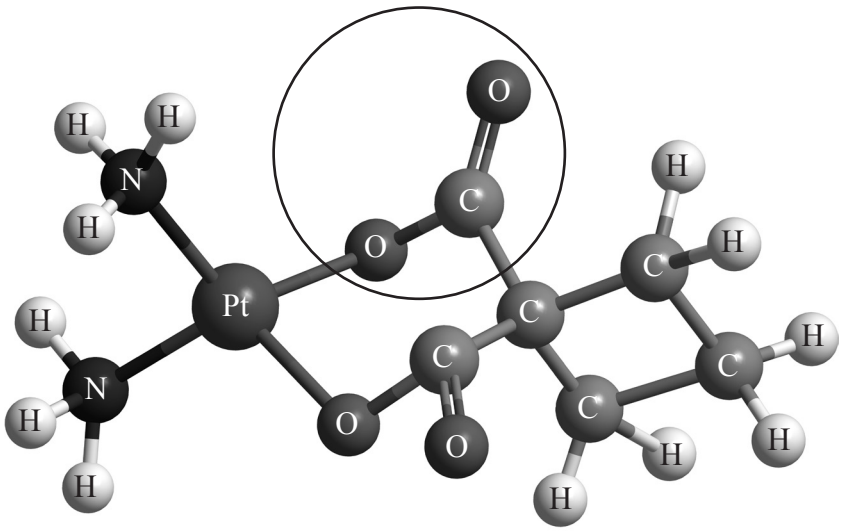


(a) State the electron configuration of the manganese ion in EUK-134. *[1]*

. . . . . .

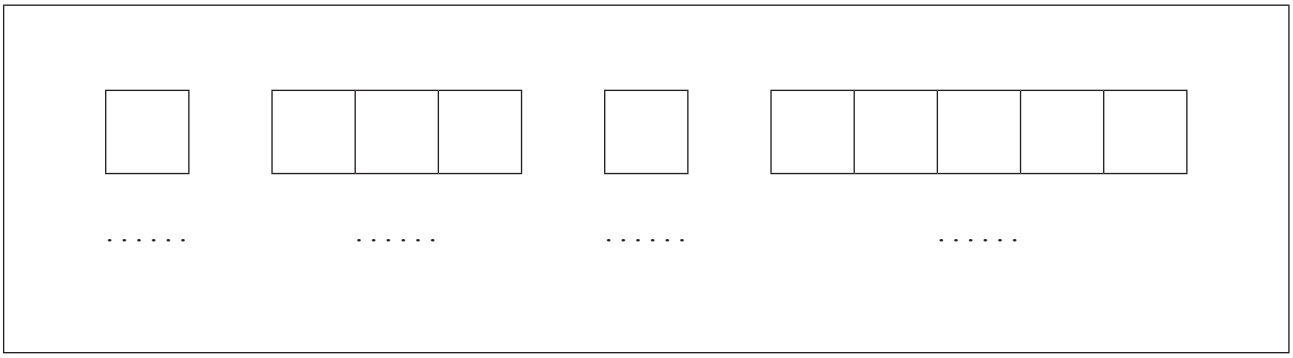
2.1-S29 May 2013 TZ2 · Paper 2 · Q3 parts (e)

3. Carboplatin used in the treatment of lung cancer has the following three-dimensional structure.



(e) Iron is in the same group in the periodic table as ruthenium.

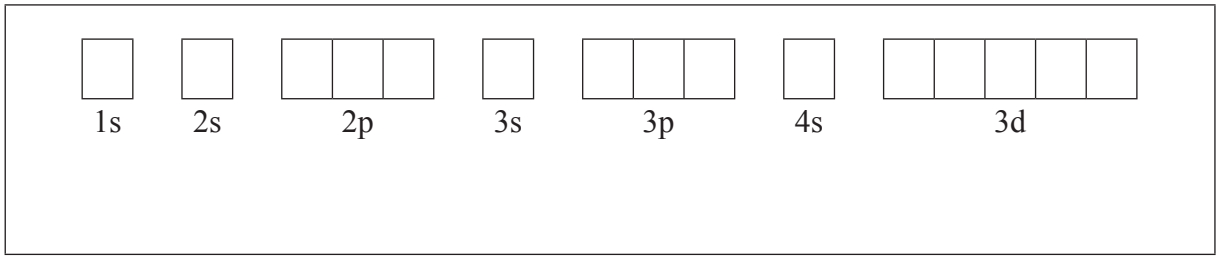
Construct the orbital diagram (using the arrow-in-box notation) for iron, showing the electrons in the $n = 3$ and $n = 4$ energy levels only **and** label each sub-level on the diagram. *[1]*



2.1-S30 Nov 2012 · Paper 2 · Q2 parts (b)(ii)

2. Lithium and boron are elements in period 2 of the periodic table. Lithium occurs in group 1 (the alkali metals) and boron occurs in group 3. Isotopes exist for both elements.

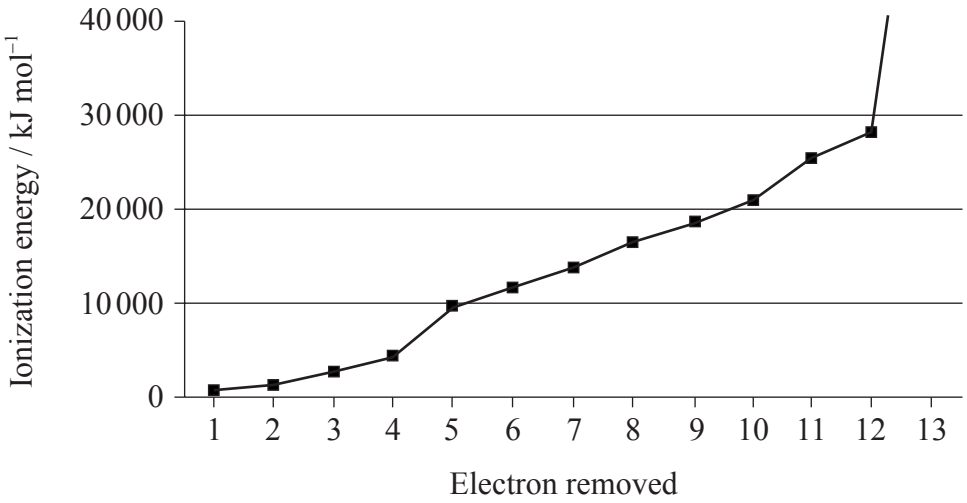
(ii) Cobalt is a transition metal. One common ion of cobalt is Co^{3+}. Draw the orbital diagram (using the arrow-in-box notation) for the Co^{3+} ion. *[1]*



2.1-S31 May 2012 TZ1 · Paper 2 · Q8 parts (b)(i), (b)(ii)

8. The element titanium is present in meteorites.

(b) The successive ionization energies of titanium are shown below.



(i) State the **full** electron configuration of an atom of titanium and identify the sub-level from which the electron is removed when the 1st ionization energy is measured. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Explain why there are relatively large differences between the 4th and 5th, **and** between the 10th and 11th ionization energies. *[3]*

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. . . . . .
. . . . . .

2.1-S32 May 2012 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii)

9. (a) Transition metals exhibit variable oxidation states in their complexes.

(i) State the **full** electron configuration of Fe. *[1]*

. . . . . .
. . . . . .

(ii) State the **abbreviated** electron configuration of Fe^{3+} ions. *[1]*

. . . . . .
. . . . . .

2.1-S33 Nov 2011 · Paper 2 · Q2 parts (c)

2. Isotopes are atoms of the same element with different mass numbers. Two isotopes of cobalt are Co-59 and Co-60.

(c) State the **full** electron configuration of $^{59}Co^{3+}$. *[1]*

. . . . . .

. . . . . .

2.1-S34 May 2011 TZ1 · Paper 2 · Q3 parts (c)

(c) Deduce the electron configuration for the ion Co^{2+}. *[1]*

. . . . . .

2.1-S35 May 2011 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (a)(iii)

3. (a) The electron configuration of chromium can be expressed as $[Ar]4s^x3d^y$.

(i) Explain what the square brackets around argon, [Ar], represent. *[1]*

. . . . . .

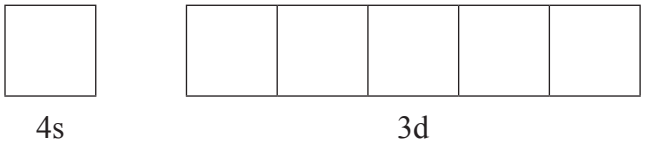
. . . . . .

(ii) State the values of *x* and *y*. *[1]*

x: . . . . . .

y: . . . . . .

(iii) Annotate the diagram below showing the 4s and 3d orbitals for a chromium atom using an arrow, ↿ and ⇂, to represent a spinning electron. *[1]*



2.1-S36 Nov 2010 · Paper 2 · Q3 parts (d)

3. Iron has three main naturally occurring isotopes which can be investigated using a mass spectrometer.

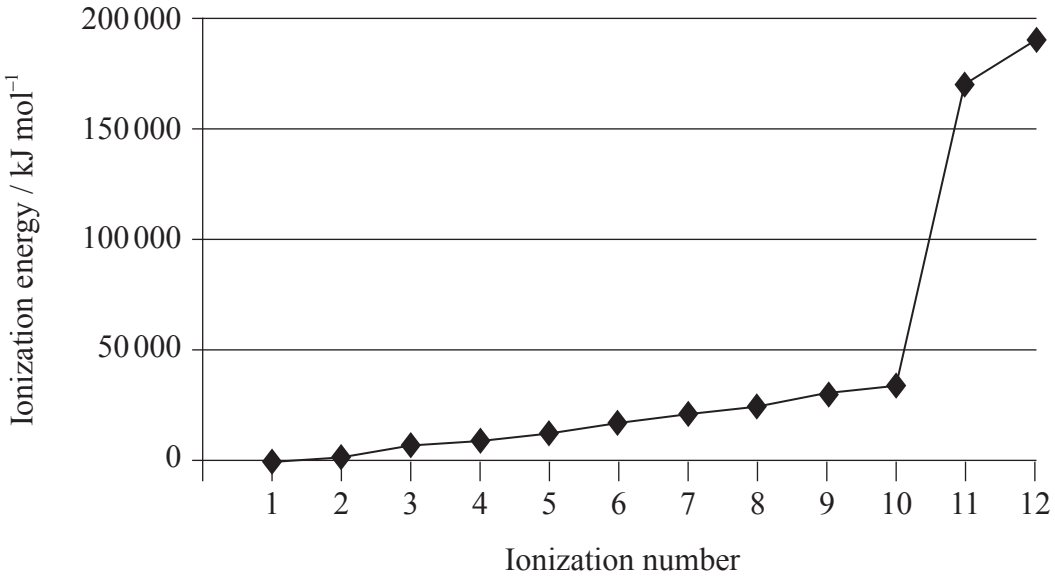
(d) State the full electronic configurations of a Cu atom and a Cu^+ ion. *[2]*

Cu: . . . . . .

Cu^+: . . . . . .

2.1-S37 Nov 2010 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (c)(iv) · also 7.2, 9.4

4. Magnesium is the eighth most abundant element in the earth's crust. The successive ionization energies of the element are shown below.



(a) (i) Define the term *first ionization energy* and state the equation for the first ionization of magnesium. *[3]*

(ii) Explain the general increase in successive ionization energies of the element. *[2]*

(iii) Explain the large increase between the tenth and eleventh ionization energies. *[3]*

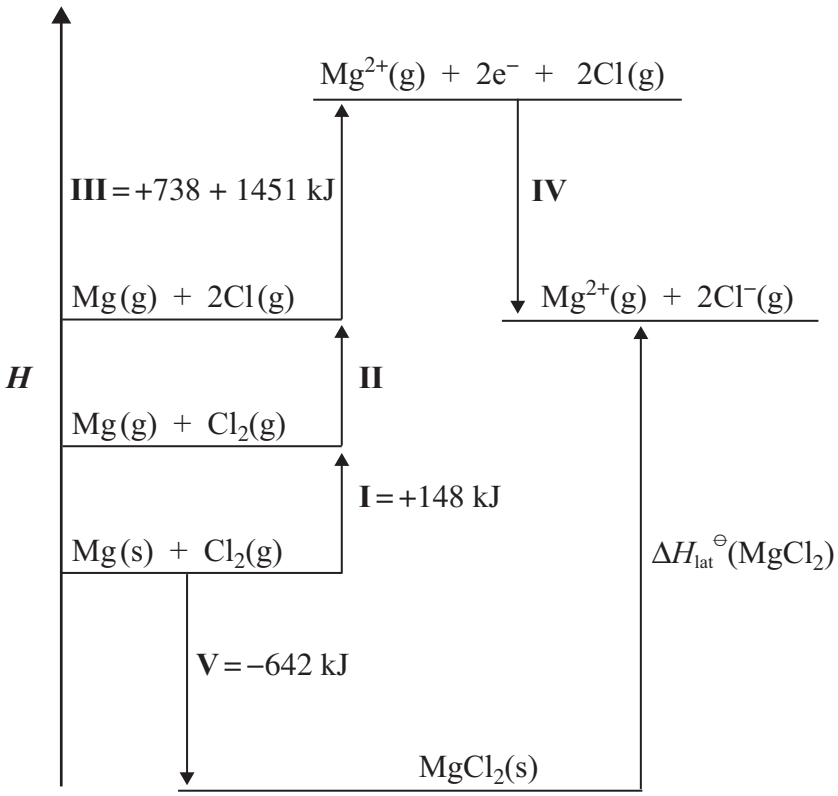
(b) Magnesium can be produced from the electrolysis of molten magnesium chloride, $MgCl_2$.

(i) Explain how molten magnesium chloride conducts an electric current. *[2]*

(ii) Identify the electrode where oxidation occurs during electrolysis of molten magnesium chloride and state an equation for the half-reaction. *[2]*

(iii) Explain why magnesium is not formed during the electrolysis of aqueous magnesium chloride solution. *[1]*

(c) The lattice enthalpy of magnesium chloride can be calculated from the Born-Haber cycle shown below.



(i) Identify the enthalpy changes labelled by **I** and **V** in the cycle. *[2]*

(ii) Use the ionization energies given in the cycle above and further data from the Data Booklet to calculate a value for the lattice enthalpy of magnesium chloride. *[4]*

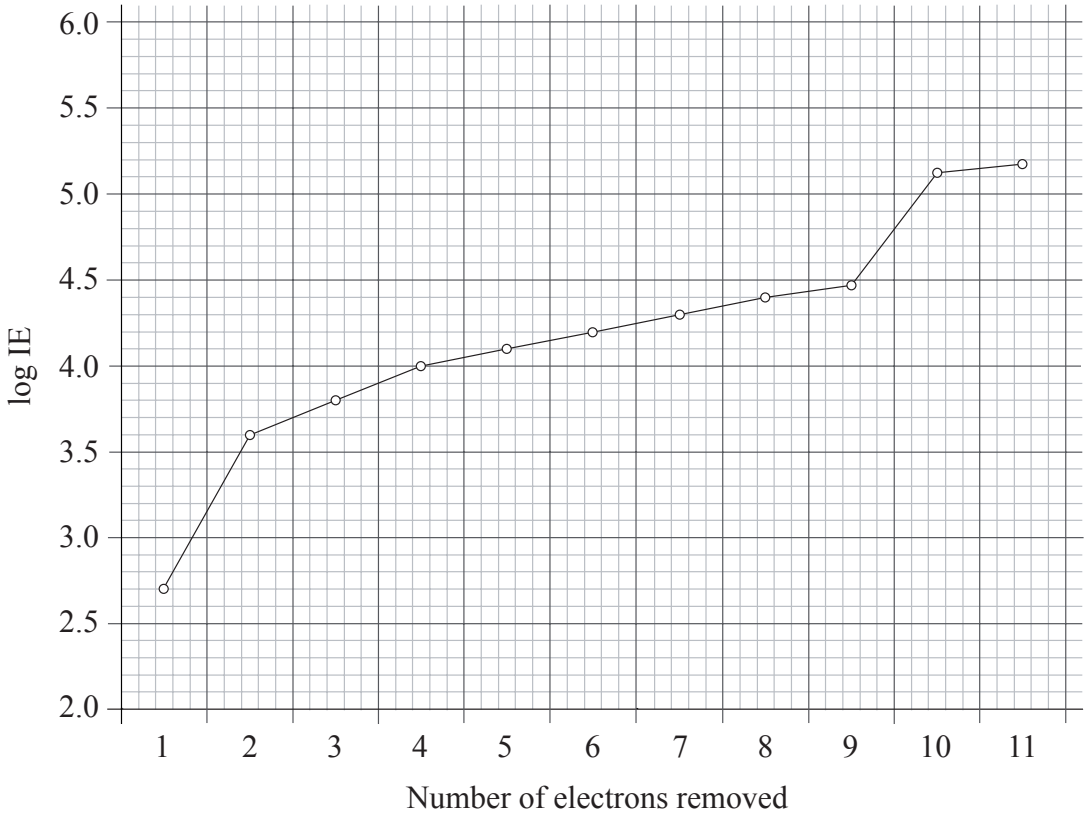
(iv) The experimental lattice enthalpy of magnesium oxide is given in Table 13 of the Data Booklet. Explain why magnesium oxide has a higher lattice enthalpy than magnesium chloride. *[2]*

2.1-S38 May 2010 TZ2 · Paper 2 · Q6 parts (b)(i), (d), (e)(i), (e)(ii) · also 5.1

6. The periodic table shows the relationship between electron configuration and the properties of elements and is a valuable tool for making predictions in chemistry.

(b) (i) Define the term *electronegativity*. *[2]*

(d) The graph below represents the successive ionization energies of sodium. The vertical axis plots log (ionization energy) instead of ionization energy to allow the data to be represented without using an unreasonably long vertical axis.



State the full electron configuration of sodium and explain how the successive ionization energy data for sodium are related to its electron configuration. *[4]*

(e) (i) Explain why the first ionization energy of aluminium is **lower** than the first ionization energy of magnesium. *[2]*

(ii) Explain why the first ionization energy of sulfur is **lower** than the first ionization energy of phosphorus. *[2]*

LESSON 3.1

VSEPR Theory

49 multiple-choice questions · 29 structured questions

Part A - Multiple choice

3.1-M1 May 2025 TZ1 · Paper 1A · Q9

9. In which of the following molecules do all atoms obey the octet rule?

I. NF_3

II. BF_3

III. PCl_3

A. I and II only

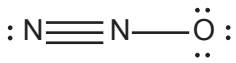
B. I and III only

C. II and III only

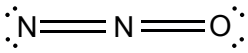
D. I, II and III

3.1-M2 May 2025 TZ1 · Paper 1A · Q10

10. Nitrogen(I) oxide, N_2O, can be represented by two Lewis formulas.

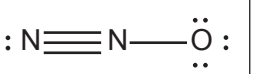
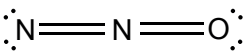


Structure X



Structure Y

Which row correctly identifies the formal charges and the more stable structure?

	Structure X 			Structure Y 			More stable
A.	0	+1	−1	−1	+1	0	X
B.	−1	+1	0	+1	0	−1	X
C.	0	+1	−1	−1	0	+1	Y
D.	−1	0	+1	0	+1	−1	Y

3.1-M3 May 2025 TZ1 · Paper 1A · Q11

11. Which row puts these molecules in order of decreasing bond angle?

A. NH_3, H_2O, C_2H_4, CH_4

B. H_2O, C_2H_4, CH_4, NH_3

C. C_2H_4, CH_4, NH_3, H_2O

D. CH_4, NH_3, H_2O, C_2H_4

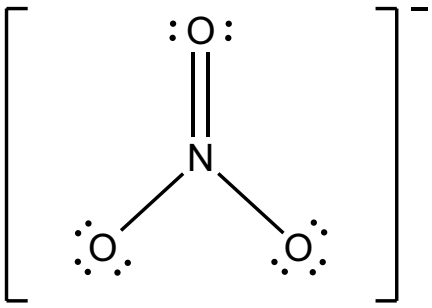
3.1-M4 May 2025 TZ2 · Paper 1A · Q9

9. What are the electron domain and molecular geometries of $XeCl_4$?

	Electron domain geometry	Molecular geometry
A.	octahedral	tetrahedral
B.	octahedral	square planar
C.	tetrahedral	tetrahedral
D.	tetrahedral	square planar

3.1-M5 May 2025 TZ3 · Paper 1A · Q9

9. What is the formal charge of the nitrogen atom in NO_3^-?



A. −3

B. −1

C. +1

D. +3

3.1-M6 May 2024 TZ1 · Paper 1 · Q11

11. Which combination best describes NO_2^-?

	Number of electron domains around N	Molecular geometry	Bond angle
A.	2	linear	180°
B.	3	bent	105°
C.	3	bent	117°
D.	4	trigonal planar	120°

3.1-M7 May 2024 TZ1 · Paper 1 · Q12

12. What is the molecular geometry of BrF_5?

A. trigonal bipyramidal

B. octahedral

C. square-based pyramidal

D. T-shaped

3.1-M8 May 2024 TZ2 · Paper 1 · Q12

12. What is the correct number of bonding pairs of electrons in ethanedioic acid, $(COOH)_2$?

A. 7

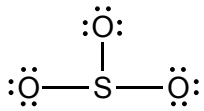
B. 8

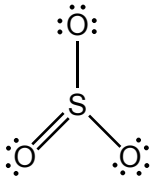
C. 9

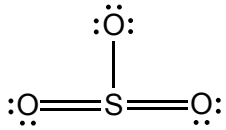
D. 18

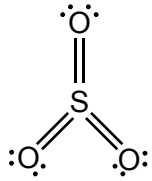
3.1-M9 Nov 2023 TZ1 · Paper 1 · Q12

12. Which is the correct structure of SO_3, based on the lowest formal charge?

A. 

B. 

C. 

D. 

3.1-M10 May 2023 TZ1 · Paper 1 · Q11

11. In which molecule does the central atom have an incomplete octet of electrons?

A. H_2Se

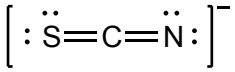
B. PH_3

C. OF_2

D. BF_3

3.1-M11 May 2023 TZ1 · Paper 1 · Q12

12. What are the formal charges on the atoms in this molecular ion?



	S	C	N
A.	−1	0	0
B.	0	0	−1
C.	−1	+1	0
D.	0	+1	−1

3.1-M12 May 2023 TZ2 · Paper 1 · Q9

9. Which compound has an element with an incomplete octet of electrons?

A. BF_3

B. CF_4

C. OF_2

D. ClF_3

3.1-M13 May 2023 TZ2 · Paper 1 · Q11

11. What is the correct comparison of H—N—H bond angles in NH_2^-, NH_3, and NH_4^+?

A. $NH_2^- < NH_3 < NH_4^+$

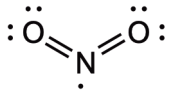
B. $NH_4^+ < NH_3 < NH_2^-$

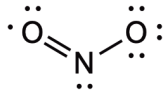
C. $NH_3 < NH_2^- < NH_4^+$

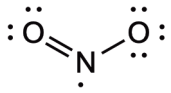
D. $NH_3 < NH_4^+ < NH_2^-$

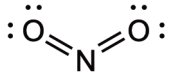
3.1-M14 May 2023 TZ2 · Paper 1 · Q12

12. Which is the valid and preferred Lewis structure of the NO_2• radical, based on formal charge?

A. 

B. 

C. 

D. 

3.1-M15 Nov 2022 · Paper 1 · Q10

10. Which elements are capable of forming expanded octets?

I. Nitrogen
II. Phosphorus
III. Arsenic

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

3.1-M16 Nov 2022 · Paper 1 · Q11

11. Which molecule has a tetrahedral molecular geometry?

A. HNO_3

B. SF_4

C. XeF_4

D. XeO_4

3.1-M17 May 2022 TZ1 · Paper 1 · Q13

13. What are the electron domain and molecular geometries of the XeF_4 molecule?

	Electron domain geometry	Molecular geometry
A.	tetrahedral	planar
B.	tetrahedral	tetrahedral
C.	octahedral	planar
D.	octahedral	tetrahedral

3.1-M18 May 2022 TZ2 · Paper 1 · Q11

11. What is the formal charge of the oxygen atom in H_3O^+?

A. −2

B. −1

C. 0

D. +1

3.1-M19 May 2022 TZ2 · Paper 1 · Q12

12. What is the molecular geometry of SF_4?

A. Tetrahedral

B. Trigonal bipyramidal

C. See-saw

D. Square planar

3.1-M20 May 2021 TZ2 · Paper 1 · Q12

12. Which atom has an expanded octet?

A. C in CO_2

B. S in SCl_4

C. O in H_2O_2

D. P in PCl_3

3.1-M21 Nov 2020 · Paper 1 · Q11

11. Which combination correctly describes the geometry of BrF_4^-?

	Electron domain geometry around Br	Molecular geometry around Br
A.	Octahedral	Tetrahedral
B.	Tetrahedral	Square planar
C.	Octahedral	Square planar
D.	Tetrahedral	Tetrahedral

3.1-M22 Nov 2019 · Paper 1 · Q13

13. Which atom does **not** obey the octet rule?

A. C in CO_2

B. F in BF_3

C. O in H_2O

D. S in SF_6

3.1-M23 May 2019 TZ2 · Paper 1 · Q11

11. Which species has a square planar molecular geometry?

A. SF_4

B. XeF_4

C. CF_4

D. PF_4^+

3.1-M24 Nov 2018 · Paper 1 · Q9

9. Which species has the same molecular geometry as SO_3^{2-}?

A. BF_3

B. SO_3

C. PF_3

D. CO_3^{2-}

3.1-M25 Nov 2018 · Paper 1 · Q10

10. How many lone pairs and bonding pairs of electrons surround the central chlorine atom in ClF_2^+?

	Lone pairs	Bonding pairs
A.	0	2
B.	0	4
C.	2	4
D.	2	2

3.1-M26 May 2018 TZ1 · Paper 1 · Q10

10. What is the molecular geometry and bond angle in the molecular ion NO_3^-?

	Molecular geometry	Bond angle
A.	tetrahedral	109.5°
B.	trigonal planar	120°
C.	trigonal pyramidal	107°
D.	trigonal planar	109.5°

3.1-M27 May 2018 TZ2 · Paper 1 · Q11

11. What are the predicted electron domain geometries around the carbon and both nitrogen atoms in urea, $(NH_2)_2CO$, applying VSEPR theory?

	Carbon atom	Nitrogen atoms
A.	trigonal planar	trigonal pyramidal
B.	trigonal planar	tetrahedral
C.	tetrahedral	tetrahedral
D.	trigonal pyramidal	trigonal planar

3.1-M28 May 2018 TZ2 · Paper 1 · Q12

12. Which molecule has an expanded octet?

A. CO

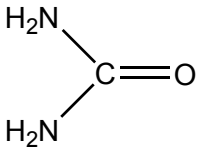
B. CO_2

C. SF_2

D. SF_4

3.1-M29 May 2017 TZ2 · Paper 1 · Q9

9. How many bonding electrons are there in the urea molecule?



A. 8

B. 16

C. 20

D. 24

3.1-M30 Nov 2016 · Paper 1 · Q11

11. How many electrons form the carbon–oxygen bond in methanal, HCHO?

A. 2

B. 4

C. 8

D. 12

3.1-M31 Nov 2016 · Paper 1 · Q14

14. Which species has bond angles of 90°?

A. $AlCl_4^-$

B. ICl_4^-

C. NH_4^+

D. $SiCl_4$

3.1-M32 May 2016 · Paper 1 · Q11

11. Which species breaks the octet rule?

A. PCl_3

B. BF_4^-

C. SCl_4

D. NH_4^+

3.1-M33 May 2015 TZ1 · Paper 1 · Q6

6. What is the total number of valence electrons in CH_3COO^-?

A. 16

B. 22

C. 23

D. 24

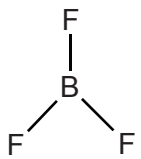
3.1-M34 May 2015 TZ1 · Paper 1 · Q12

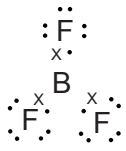
12. What is correct for PCl_5?

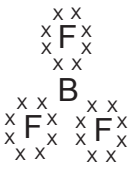
	Shape	Bond angle(s)
A.	Octahedral	90° and 180°
B.	Trigonal pyramidal	107°
C.	Square pyramidal	90° and 180°
D.	Trigonal bipyramidal	90°, 120° and 180°

3.1-M35 May 2015 TZ2 · Paper 1 · Q10

10. Which diagrams can be used to represent the Lewis (electron dot) structure of boron trifluoride?

I. 

II. 

III. 

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

3.1-M36 May 2015 TZ2 · Paper 1 · Q12

12. Which combination of shape and bond angle is correct for a molecule of xenon tetrafluoride, XeF_4?

	Shape	Bond angle
A.	square pyramid	90°
B.	square planar	90°
C.	tetrahedral	109.5°
D.	octahedral	90°

3.1-M37 Nov 2014 · Paper 1 · Q9

9. Which species contains a dative covalent (coordinate) bond?

A. HCN

B. C_2H_2

C. CO_2

D. CO

3.1-M38 May 2014 TZ2 · Paper 1 · Q12

12. Which molecule is trigonal bipyramidal in shape?

A. PCl_3

B. $SiCl_4$

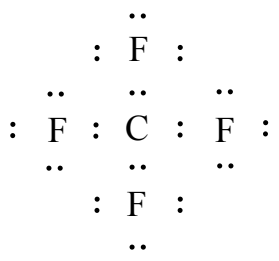
C. PCl_5

D. SF_6

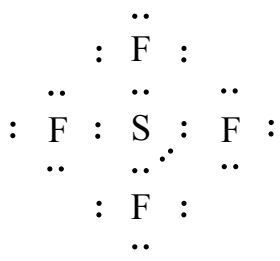
3.1-M39 May 2013 TZ1 · Paper 1 · Q12

12. Which species has a square planar shape?

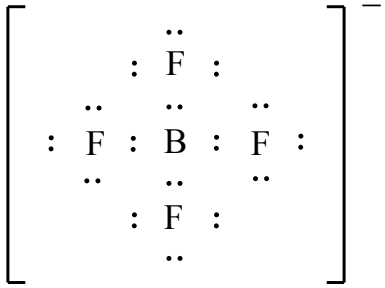
A.



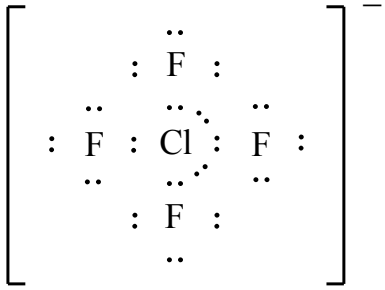
B.



C.

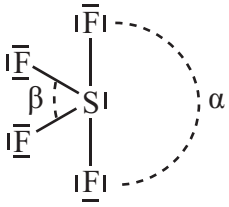


D.



3.1-M40 Nov 2012 · Paper 1 · Q12

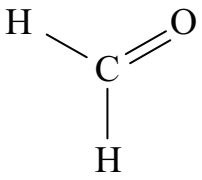
12. In the molecule SF_4, which are the correct bond angles?



	α / °	β / °
A.	180	120
B.	187	103
C.	187	120
D.	180	90

3.1-M41 May 2012 TZ1 · Paper 1 · Q9

9. What is the correct number of centres of negative charge for carbon and the shape of H_2CO?



	Centres of negative charge on C-atom	Shape
A.	3	trigonal pyramidal
B.	3	trigonal planar
C.	4	trigonal pyramidal
D.	4	trigonal planar

3.1-M42 May 2012 TZ2 · Paper 1 · Q9

9. Which species contain dative covalent bonds?

I. CO

II. NH_3

III. H_3O^+

A. I and II only

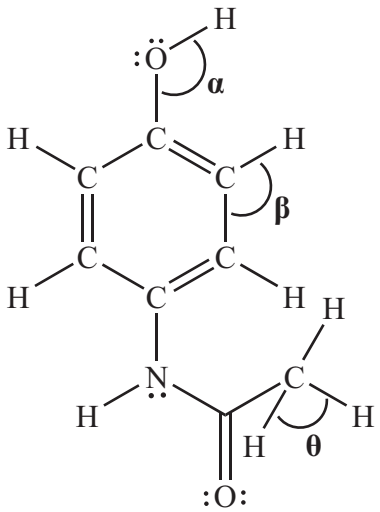
B. I and III only

C. II and III only

D. I, II and III

3.1-M43 May 2012 TZ2 · Paper 1 · Q11

11. The Lewis (electron dot) structure of paracetamol (acetaminophen) is:

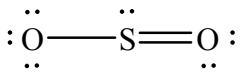


What are the approximate values of the bond angles?

	α	β	θ
A.	104.5°	120°	109.5°
B.	109.5°	109.5°	109.5°
C.	120°	120°	90°
D.	104.5°	120°	90°

3.1-M44 May 2011 TZ1 · Paper 1 · Q14

14. The Lewis structure of SO_2 is given below.



What is the shape of the SO_2 molecule?

A. Bent (V-shaped)

B. Linear

C. T-shaped

D. Triangular planar

3.1-M45 May 2011 TZ2 · Paper 1 · Q11

11. Which molecule has a non-bonding (lone) pair of electrons around the central atom?

A. BF_3

B. SO_2

C. PCl_5

D. SiF_4

3.1-M46 Nov 2010 · Paper 1 · Q10

10. Which species contain a dative covalent bond?

I. HCHO

II. CO

III. H_3O^+

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

3.1-M47 Nov 2010 · Paper 1 · Q12

12. Which molecule has an octahedral shape?

A. SF_6

B. PCl_5

C. XeF_4

D. BF_3

3.1-M48 May 2010 TZ2 · Paper 1 · Q9

9. Which species have a dative covalent bond?

I. CO

II. NH_3

III. H_3O^+

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

3.1-M49 May 2010 TZ2 · Paper 1 · Q12

12. The Lewis structure of XeF_2 contains two bonding pairs of electrons and three non-bonding pairs of electrons (lone pairs) around the central xenon atom. What is the shape of XeF_2 molecule?

A. Bent

B. Trigonal bipyramidal

C. Square planar

D. Linear

Part B - Structured questions

3.1-S1 May 2025 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(iv), (b)(ii), (b)(iii), (b)(iv) · also 4.1, 6.1, 6.2

6. The rate of reaction between solutions of iron(III) nitrate and sodium thiosulfate can be measured using the time it takes for the colour to change.

(a) The first step in the reaction produces an unstable dark violet complex.

$$[Fe(H_2O)_6]^{3+}(aq) + 2S_2O_3^{2-}(aq) \rightarrow [Fe(S_2O_3)_2(H_2O)_2]^{-}(aq) + 4H_2O(l)$$

(i) Deduce the oxidation state of S in $[Fe(S_2O_3)_2(H_2O)_2]^{-}$. [1]

. . . . . .

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(iv) The reaction continues until the violet colour disappears. The thiosulfate ion, $S_2O_3^{2-}$, is oxidized to SO_2, and Fe^{3+} is reduced to Fe^{2+}.
Deduce the oxidation half-equation, and the overall redox equation for this second step of the reaction. [2]

Oxidation half-equation: . . . . . .

. . . . . .

Overall redox equation: . . . . . .

. . . . . .

(b) Iron(III) nitrate is a compound that involves both ionic and covalent bonding.

(ii) Deduce a Lewis formula of the nitrate ion. [1]

(iii) State the molecular geometry of the nitrate ion. [1]

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(iv) Predict, with a reason, the bond lengths of the nitrate ion. Use section 11 of the data booklet. [2]

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3.1-S2 May 2025 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (a)(v), (b)(vi), (c)(ii), (c)(v) · also 4.2, 5.1, 5.2, 6.1, 7.1, 8.2

1. Hydrogen cyanide, HCN, is a very toxic compound.

(a) Pure HCN is a volatile liquid, boiling at 26 °C.

(i) Draw the Lewis formula of the HCN molecule. [1]

(ii) Deduce the hybridization of the carbon atom and the number of sigma and pi bonds that it forms. [2]

Hybridization: . . . . . .

Sigma bonds: . . . . . .

Pi bonds: . . . . . .

(iii) State and explain the molecular geometry of HCN without referring to hybridization. [2]

Molecular geometry: . . . . . .

Explanation: . . . . . .

. . . . . .

. . . . . .

(iv) HCN is a polar molecule. Deduce which atom carries a partial positive charge and which carries a partial negative charge. [1]

Partial positive charge: . . . . . .

Partial negative charge: . . . . . .

(v) Explain why nitrogen gas, N_2, has a much lower boiling point than HCN. [2]

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(b) HCN acts as a weak acid in aqueous solution.

(vi) Calculate the temperature rise of the solution when 30.0 cm^3 of 0.200 $mol\,dm^{-3}$ KOH is mixed with 20.0 cm^3 of 0.100 $mol\,dm^{-3}$ HCN. Use enthalpy of neutralization of HCN = $-13.3\,kJ\,mol^{-1}$ and sections 1 and 2 of the data booklet. [2]

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(c) The cyanide ion, CN^-, can form complex ions, such as $[Fe(CN)_6]^{4-}$.

(ii) Deduce the oxidation state of iron in the complex ion. [1]

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(v) Outline, giving a reason, whether $[Fe(CN)_6]^{3-}$ is a stronger or weaker oxidizing agent than $Fe^{3+}(aq)$.

Use $E^{\ominus}$ $[Fe(CN)_6]^{3-}(aq) + e^- \rightleftharpoons [Fe(CN)_6]^{4-}(aq) = +0.37\,V$ and section 19 of the data booklet. [2]

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3.1-S3 May 2025 TZ3 · Paper 2 · Q10 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b) · also 1.2, 2.1, 5.1

10. Sulfur dioxide, SO_2, is involved in the formation of acid rain.

(a) (i) State the electron configuration for S. [1]

. . . . . .

. . . . . .

(ii) Draw the Lewis formula of SO_2. [1]

(iii) Deduce the electron domain geometry and molecular geometry of SO_2. [2]

Electron domain geometry: . . . . . .

. . . . . .

Molecular domain geometry: . . . . . .

. . . . . .

(iv) Explain the polarity of the SO_2 molecule. [2]

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(b) Sulfur dioxide reacts with rain water and oxygen to form sulfuric acid, H_2SO_4.

State an equation for this reaction including the state symbols. [2]

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3.1-S4 Nov 2024 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (d)(i), (d)(iv) · also 4.1, 6.1, 8.1

4. Sulfur trioxide is an important compound in industry.

(a) Sulfur trioxide has more than one possible Lewis (electron dot) structure.

(i) Sketch **two** Lewis (electron dot) structures for SO_3, one of which obeys the octet rule and one of which does not. [2]

Obeys octet rule:

Does not obey octet rule:

(ii) State how chemists decide which Lewis (electron dot) structure is more stable. [1]

. . . . . .

(iii) Predict the length of each S to O bond in pm. Use section 10 of the data booklet. [1]

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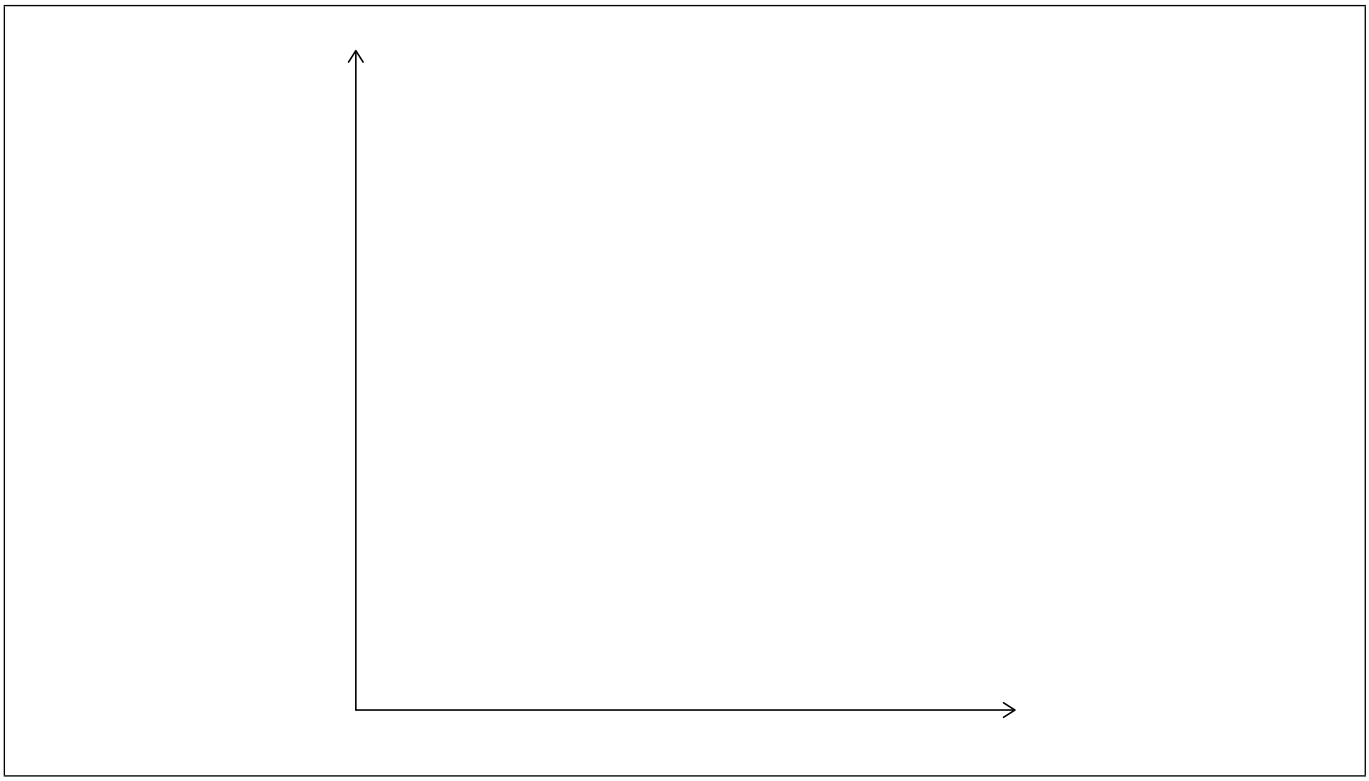
(d) $SO_3(g)$ is made using the contact process.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H^{\ominus} < 0$$

(i) Sketch a potential energy profile for this reaction on the axes provided. Label E_a and include labels on the axes. [3]



(iv) During the reaction, V_2O_5 changes to V_2O_4. Identify the oxidation states of vanadium in each compound. [1]

V_2O_5: . . . . . .

V_2O_4: . . . . . .

3.1-S5 May 2024 TZ2 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d) · also 2.1, 4.1, 4.2

2. In the stratosphere, ozone is decomposed by ultraviolet radiation.

$$O_3(g) \rightarrow O_2(g) + O(g)$$

(a) State the full electron configuration of an oxygen atom and the number of unpaired electrons in that atom. [2]

Electron configuration: . . . . . .

Unpaired electrons: . . . . . .

(b) (i) Draw a Lewis (electron dot) structure for the ozone molecule. [1]

(ii) Deduce the formal charge on each of the three oxygen atoms by adding them to your Lewis (electron dot) structure in (b)(i). [2]

(iii) Predict the shape and bond angle of the ozone molecule. [2]

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(iv) State the hybridization state of the central oxygen atom in ozone. [1]

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(c) Suggest a value, in pm, for the bond lengths in the ozone molecule and explain your answer. Use section 10 of the data booklet. [2]

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(d) Explain the dependence of the dissociation of diatomic oxygen, O_2, and ozone, O_3, on the wavelength of UV radiation. [2]

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3.1-S6 May 2023 TZ1 · Paper 2 · Q5 parts (a)(i), (b)(i) · also 1.2

5. Double salts are substances with two cations and one anion. A hydrated sulfate containing two cations has this percentage composition.

Element	Percentage (%)
Nitrogen (N)	7.09
Hydrogen (H)	5.11
Sulfur (S)	16.22
Cobalt (Co)	14.91
Oxygen (O)	—

(a) (i) Draw **one** Lewis (electron dot) structure of the sulfate ion. [1]

(b) 1.20 g of the double salt was dissolved in water and an excess of aqueous barium chloride was added, precipitating all the sulfate ions as barium sulfate.

(i) Formulate an ionic equation, including state symbols, for the reaction of barium ions with sulfate ions. [1]

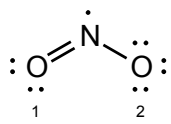
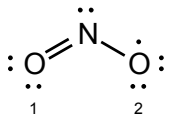
. . . . . .

3.1-S7 May 2023 TZ1 · Paper 2 · Q9 parts (b)(iii), (b)(iv)

(b) Nitrogen (II) oxide radicals (NO•) catalyse the decomposition of ozone (O_3).

Chlorine also forms free radicals; the bond enthalpy for Cl_2 is 4.02×10^{-19} J.

(iii) Calculate the formal charge on each atom in the **two** Lewis structures of the NO_2•(g) radical. [1]

	Structure A 	**Structure B** 
Oxygen 1	. . . . .	. . . . .
Nitrogen	. . . . .	. . . . .
Oxygen 2	. . . . .	. . . . .

(iv) Lewis structure A is more stable. Suggest, giving **one** reason, whether the formal charge model supports this. [1]

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3.1-S8 May 2023 TZ2 · Paper 2 · Q2 parts (a)(iv), (b), (c)(i), (c)(ii) · also 4.1, 9.2

2. Nitrogen (IV) oxide, NO_2, is a brown gas found in photochemical smog and has a pollutant causing acid deposition.

(a) Nitrogen (IV) oxide exists in equilibrium with dinitrogen tetroxide, $N_2O_4(g)$, which is colourless.

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

(iv) Calculate the standard enthalpy change, in kJ mol^{-1}, for the reaction: [1]

$$N_2O_4(g) \rightarrow 2NO_2(g)$$

	$\Delta H^{\ominus}_f$ (kJ mol^{-1})
NO_2	33.18
N_2O_4	9.16

. . . . . .

. . . . . .

. . . . . .

(b) Deduce the Lewis structure of N_2O_4. [1]

(c) The NO bond lengths in N_2O_4 are all 1.19×10^{-10} m.

(i) Suggest what the bond lengths indicate about the structure of N_2O_4. [1]

. . . . . .

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(ii) Predict the ONN bond angle in N_2O_4. [1]

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3.1-S9 May 2022 TZ2 · Paper 2 · Q6 parts (c)(i), (c)(ii)

6. Sulfur trioxide is produced from sulfur dioxide.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H = -196\,kJ\,mol^{-1}$$

(c) (i) Draw the Lewis structure of SO_3. [1]

(ii) Explain the electron domain geometry of SO_3. [2]

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3.1-S10 Nov 2021 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c)(i) · also 1.2, 5.1, 9.2

3. White phosphorus is an allotrope of phosphorus and exists as P_4.

(a) (i) Sketch the Lewis (electron dot) structure of the P_4 molecule, containing only single bonds. [1]

(ii) Write an equation for the reaction of white phosphorus (P_4) with chlorine gas to form phosphorus trichloride (PCl_3). [1]

. . . . . .

. . . . . .

(b) (i) Deduce the electron domain and molecular geometry using VSEPR theory, and estimate the Cl–P–Cl bond angle in PCl_3. [3]

Electron domain geometry:

. . . . . .

Molecular geometry:

. . . . . .

Bond angle:

. . . . . .

(ii) Outline the reason why PCl_5 is a non-polar molecule, while PCl_4F is polar. [3]

PCl_5 is non-polar:

. . . . . .

. . . . . .

. . . . . .

PCl_4F is polar:

. . . . . .

. . . . . .

. . . . . .

(c) An equilibrium exists between PCl_3 and PCl_5.

$$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$$

(i) Calculate the standard enthalpy change ($\Delta H^{\ominus}$) for the forward reaction in $kJ\,mol^{-1}$.

$$\Delta H^{\ominus}_f PCl_3(g) = -306.4\,kJ\,mol^{-1}$$

$$\Delta H^{\ominus}_f PCl_5(g) = -398.9\,kJ\,mol^{-1}$$ [1]

. . . . . .

. . . . . .

3.1-S11 May 2021 TZ1 · Paper 2 · Q2 parts (a)(i), (a)(ii)

2. Iron (II) sulfide reacts with hydrochloric acid to form hydrogen sulfide, H_2S.

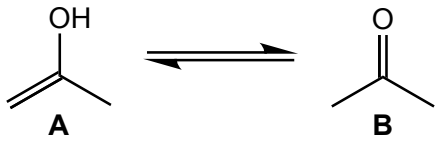
(a) (i) Draw the Lewis (electron dot) structure of hydrogen sulfide. [1]

(ii) Predict the shape of the hydrogen sulfide molecule. [1]

. . . . . .

3.1-S12 Nov 2020 · Paper 2 · Q2 parts (a), (b), (c) · also 4.1, 4.2

2. Compound **A** is in equilibrium with compound **B**.



(a) Predict the electron domain and molecular geometries around the **oxygen** atom of molecule **A** using VSEPR. [2]

Electron domain geometry:

. . . . . .

Molecular geometry:

. . . . . .

(b) State the type of hybridization shown by the central carbon atom in molecule **B**. [1]

. . . . . .

(c) State the number of sigma (σ) and pi (π) bonds around the central carbon atom in molecule **B**. [1]

σ-bonds:

. . . . . .

π-bonds:

. . . . . .

3.1-S13 May 2019 TZ1 · Paper 2 · Q5 parts (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 4.2

5. Both vinegar (a dilute aqueous solution of ethanoic acid) and bleach are used as cleaning agents.

(c) Bleach reacts with ammonia, also used as a cleaning agent, to produce the poisonous compound chloramine, NH_2Cl.

(i) Draw a Lewis (electron dot) structure of chloramine. [1]

(ii) State the hybridization of the nitrogen atom in chloramine. [1]

. . . . . .

(iii) Deduce the molecular geometry of chloramine and estimate its H–N–H bond angle. [2]

Molecular geometry:

. . . . . .

H–N–H bond angle:

. . . . . .

(iv) State the type of bond formed when chloramine is protonated. [1]

. . . . . .

. . . . . .

3.1-S14 Nov 2018 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 2.1, 6.2

3. Bromine can form the bromate(V) ion, BrO_3^-.

(a) (i) State the electron configuration of a bromine atom. [1]

. . . . . .

(ii) Sketch the orbital diagram of the **valence shell** of a bromine atom (ground state) on the energy axis provided. Use boxes to represent orbitals and arrows to represent electrons. [1]

(b) (i) Draw two Lewis (electron dot) structures for BrO_3^-. [2]

Structure I – follows octet rule:

Structure II – does not follow octet rule:

(ii) Determine the preferred Lewis structure based on the formal charge on the bromine atom, giving your reasons. [2]

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(c) Predict, using the VSEPR theory, the geometry of the BrO_3^- ion and the O–Br–O bond angles. [3]

Geometry:

. . . . . .

Reason:

. . . . . .

. . . . . .

O–Br–O angle:

. . . . . .

(d) (i) Bromate(V) ions act as oxidizing agents in acidic conditions to form bromide ions.

Deduce the half-equation for this reduction reaction. [2]

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(ii) Bromate(V) ions oxidize iron(II) ions, Fe^{2+}, to iron(III) ions, Fe^{3+}.

Deduce the equation for this redox reaction. [1]

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3.1-S15 May 2018 TZ2 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (d), (e) · also 4.1, 4.3, 5.1, 5.2

7. Some physical properties of molecular substances result from the different types of forces between their molecules.

(a) (i) Explain why the hydrides of group 16 elements (H_2O, H_2S, H_2Se and H_2Te) are polar molecules. [2]

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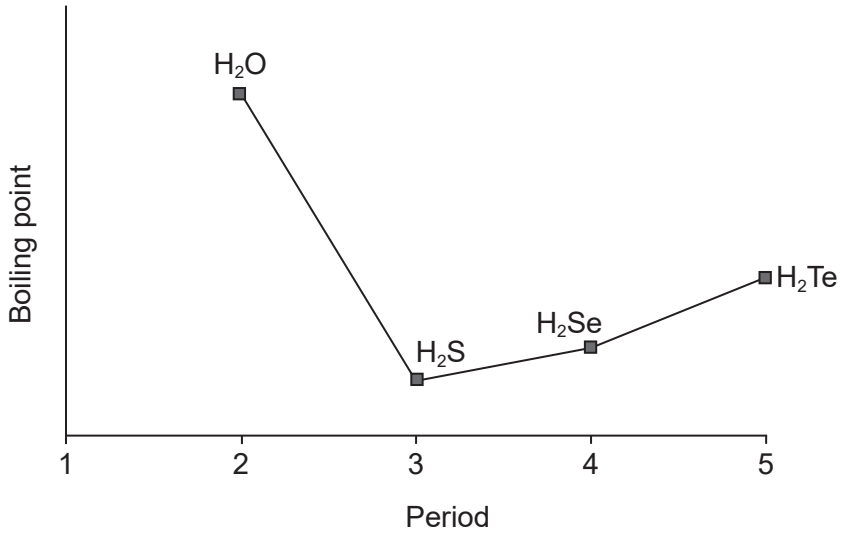
(ii) The graph shows the boiling points of the hydrides of group 16 elements.



Explain the increase in the boiling point from H_2S to H_2Te. [2]

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(b) Lewis structures show electron domains and are used to predict molecular geometry.

Deduce the electron domain geometry and the molecular geometry for the NH_2^- ion. [2]

Electron domain geometry:

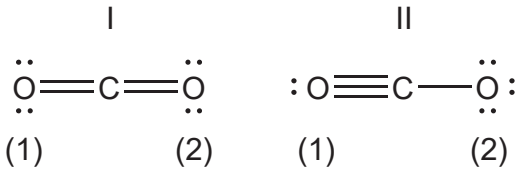
. . . . . .

Molecular geometry:

. . . . . .

(c) Resonance structures exist when a molecule can be represented by more than one Lewis structure.

(i) Carbon dioxide can be represented by at least two resonance structures, I and II.



Calculate the formal charge on each oxygen atom in the two structures. [2]

Structure	**I**	**II**
O atom labelled (1)	. . . . .	. . . . .
O atom labelled (2)	. . . . .	. . . . .

(ii) Deduce, giving a reason, the more likely structure. [1]

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(d) Absorption of UV light in the ozone layer causes the dissociation of oxygen and ozone.

Identify, in terms of bonding, the molecule that requires a longer wavelength to dissociate. [2]

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(e) Carbon and silicon are elements in group 14.

Explain why CO_2 is a gas but SiO_2 is a solid at room temperature. [2]

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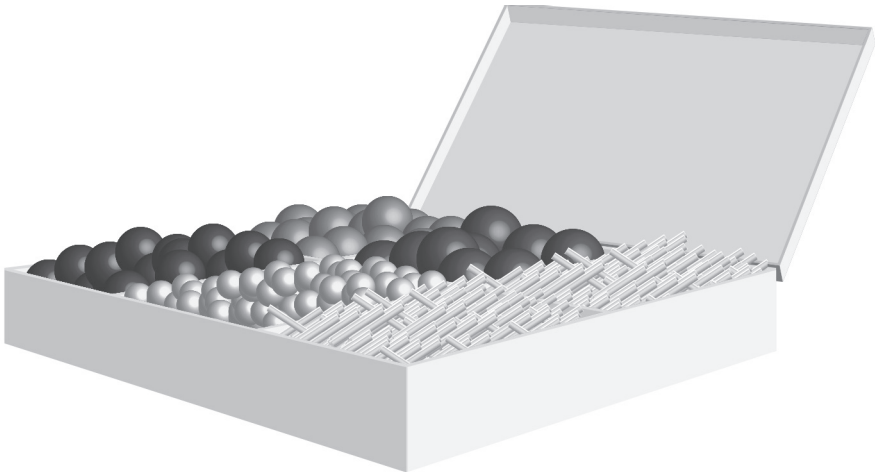
3.1-S16 May 2018 TZ2 · Paper 3 · Q2 parts (a)

2. Organic molecules can be visualized using three-dimensional models built from kits such as that pictured below.



[Source: © International Baccalaureate Organization 2018]

(a) Describe **two** differences, other than the number of atoms, between the models of ethane and ethene constructed from the kit shown. [2]

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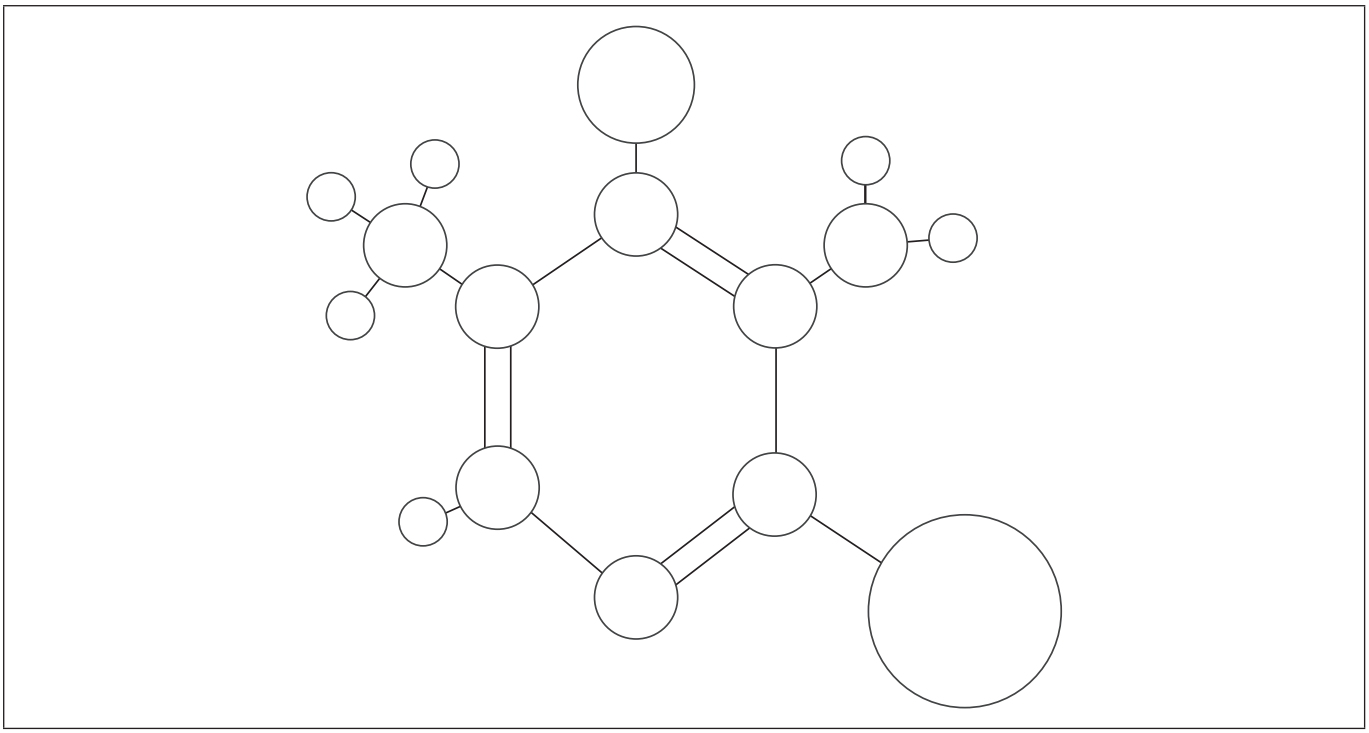

[Source: © International Baccalaureate Organization 2018]

3.1-S17 Nov 2017 · Paper 2 · Q4 parts (a), (b), (c) · also 4.2, 5.1

4. Lewis (electron dot) structures are useful models.

(a) Draw the Lewis (electron dot) structures of PF_3 and PF_5 and use the VSEPR theory to deduce the molecular geometry of each species including bond angles. [6]

	PF_3	PF_5
Lewis (electron dot) structure		
Molecular geometry	. . . . .	. . . . .
Bond angles	. . . . .	. . . . .

(b) Predict whether the molecules PF_3 and PF_5 are polar or non-polar. [1]

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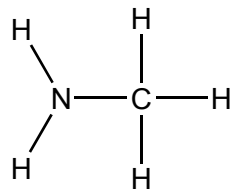
. . . . . .

(c) State the type of hybridization shown by the phosphorus atom in PF_3. [1]

. . . . . .

3.1-S18 May 2017 TZ1 · Paper 2 · Q5 parts (a), (b), (f), (g), (h)(i) · also 1.2, 4.2, 9.1, 9.3

5. Two hydrides of nitrogen are ammonia and hydrazine, N_2H_4. One derivative of ammonia is methanamine whose molecular structure is shown below.



(a) Estimate the H–N–H bond angle in methanamine using VSEPR theory. [1]

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(b) State the electron domain geometry around the nitrogen atom and its hybridization in methanamine. [2]

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(f) Determine the enthalpy change of reaction, ΔH, in kJ, when 1.00 mol of gaseous hydrazine decomposes to its elements. Use bond enthalpy values in section 11 of the data booklet. [3]

$$N_2H_4(g) \rightarrow N_2(g) + 2H_2(g)$$

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(g) The standard enthalpy of formation of $N_2H_4(l)$ is $+50.6\,kJ\,mol^{-1}$. Calculate the enthalpy of vaporization, ΔH_{vap}, of hydrazine in $kJ\,mol^{-1}$.

$$N_2H_4(l) \rightarrow N_2H_4(g)$$

(If you did not get an answer to (f), use $-85\,kJ$ but this is not the correct answer.) [2]

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(h) Hydrazine is used to remove oxygen from water used to generate steam or hot water.

$$N_2H_4(aq) + O_2(aq) \rightarrow N_2(g) + 2H_2O(l)$$

The concentration of dissolved oxygen in a sample of water is $8.0 \times 10^{-3}\,g\,dm^{-3}$.

(i) Calculate, showing your working, the mass of hydrazine needed to remove all the dissolved oxygen from $1000\,dm^3$ of the sample. [3]

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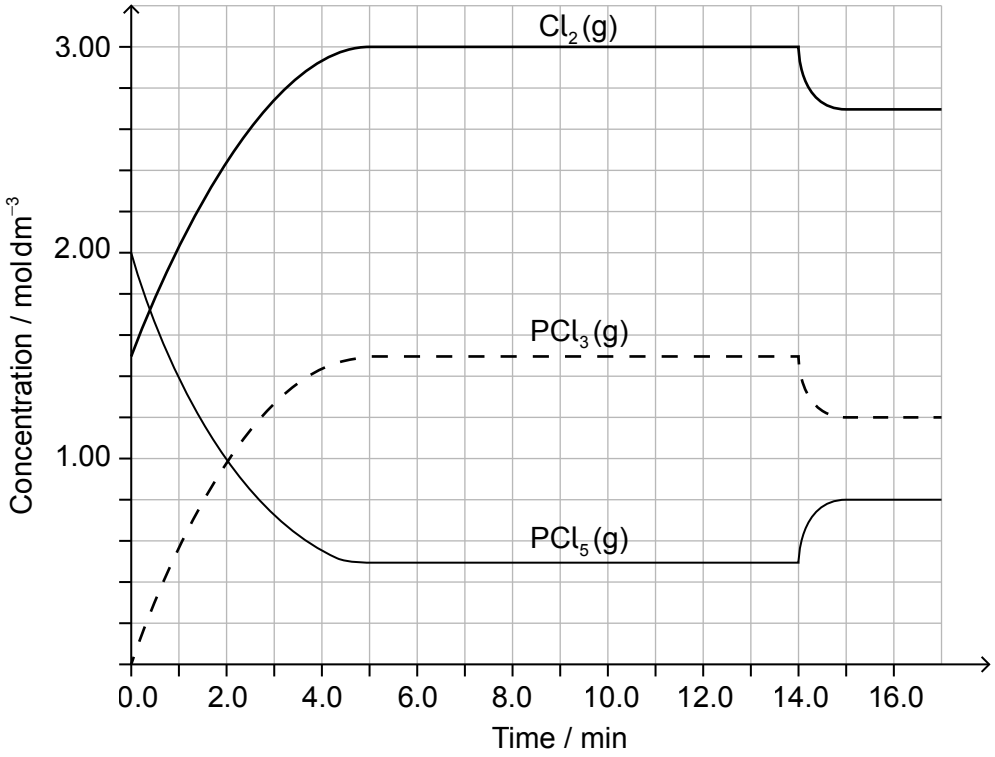
3.1-S19 May 2017 TZ2 · Paper 2 · Q3 parts (b)

3. $PCl_5(g)$ and $Cl_2(g)$ were placed in a sealed flask and allowed to reach equilibrium at 200°C. The enthalpy change, ΔH, for the decomposition of $PCl_5(g)$ is positive.



[Source: Source: http://education.alberta.ca/media]

(b) Deduce the Lewis (electron dot) structure, molecular geometry and the bond angles of PCl_3. [3]

Lewis structure:

Molecular geometry:

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Bond angles:

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3.1-S20 Nov 2016 · Paper 3 · Q22 parts (c)(i)

22. Nuclear reactions transform one nuclide into another. Fission, splitting a large nucleus into two smaller nuclei, releases vast amounts of energy.

(c) (i) Uranium hexafluoride, UF_6, is used in the uranium enrichment process that produces fuel for nuclear reactors.
State the molecular shape of uranium hexafluoride. [1]

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3.1-S21 May 2014 TZ2 · Paper 2 · Q4 parts (a), (b), (c), (d), (e) · also 4.1, 4.2

4. Ozone, O_3, in the upper atmosphere prevents harmful UV radiation reaching the surface of the Earth.

(a) Draw the Lewis structure for ozone. *[1]*

(b) State the shape of the ozone molecule and estimate the bond angle. *[2]*

Shape:

. . . . . .

Bond angle:

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(c) State the hybridization of the central oxygen atom. *[1]*

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(d) In terms of σ and π bonds, describe the two oxygen-oxygen bonds in the Lewis structure. *[1]*

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(e) The two oxygen-oxygen bonds in ozone are in fact of equal length. Deduce why this is the case and how the length of these would compare to oxygen-oxygen bond lengths in hydrogen peroxide, H_2O_2, and in the oxygen molecule, O_2. *[2]*

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3.1-S22 Nov 2013 · Paper 2 · Q3 parts (a)(ii), (b) · also 4.1

3. Calcium nitrate contains both covalent and ionic bonds.

(ii) State which atoms are covalently bonded. *[1]*

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(b) Bonding in the nitrate ion involves electron delocalization. Explain the meaning of electron delocalization and how it affects the ion. *[2]*

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3.1-S23 May 2013 TZ1 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v), (e) · also 2.1, 4.1, 4.2, 5.2

6. The element boron has two naturally occurring isotopes, ^{10}B and ^{11}B.

(c) Phosphorus forms two chlorides, PCl_3 and PCl_5.

(i) Apply the Aufbau principle to state the **full** electron configuration for an atom of phosphorus. *[1]*

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(ii) Deduce the Lewis structures for PCl_3 and PCl_5. *[2]*

PCl_3 PCl_5

(iii) Predict the shapes and the bond angles in the two molecules. *[4]*

	PCl_3	**PCl_5**
Shape	. . . . .	. . . . .
Bond angles	. . . . .	. . . . .

(iv) Identify the type of hybridization present in PCl_3. *[1]*

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(v) Compare the melting points of PCl_3 and PCl_5 and explain the difference. *[3]*

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(e) Explain the delocalization of π electrons using the O_3 molecule as an example, including **two** facts that support the delocalization. *[4]*

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3.1-S24 May 2013 TZ2 · Paper 2 · Q5 parts (a)(iii), (a)(iv), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 5.1, 9.2

5. Phosphoryl chloride, $POCl_3$, is a dehydrating agent.

(a) $POCl_3(g)$ decomposes according to the following equation.

$$2POCl_3(g) \rightarrow 2PCl_3(g) + O_2(g)$$

(iii) Define the term *standard enthalpy change of formation*, $\Delta H_f^{\ominus}$. *[1]*

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(iv) Calculate the standard enthalpy change for the reaction, $\Delta H^{\ominus}$, in kJ mol^{-1}, using the data below.

Substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$POCl_3(g)$	–542.2
$PCl_3(g)$	–288.1

[1]

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(b) (i) Deduce the Lewis (electron dot) structure of $POCl_3$ (with P as the central element) and PCl_3 and predict the shape of each molecule, using the valence shell electron pair repulsion theory (VSEPR). *[4]*

	$POCl_3$	PCl_3
Lewis (electron dot) structure		
Shape	. . . . .	. . . . .

(ii) State and explain the Cl–P–Cl bond angle in PCl_3. *[3]*

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(c) $POCl_3$ can be prepared by the reaction of phosphorus pentachloride, PCl_5, with tetraphosphorus decaoxide, P_4O_{10}.

(i) Deduce the Lewis (electron dot) structure of PCl_5. *[1]*

(ii) Predict the shape of this molecule, using the valence shell electron pair repulsion theory (VSEPR). *[1]*

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(iii) Identify all the different bond angles in PCl_5. *[1]*

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(iv) PCl_3Br_2 has the same molecular shape as PCl_5. Draw the three isomers of PCl_3Br_2 and deduce whether each isomer is polar or non-polar. *[3]*

	Isomer 1	Isomer 2	Isomer 3
Structure			
Molecular polarity	. . . . .	. . . . .	. . . . .

3.1-S25 May 2012 TZ1 · Paper 2 · Q5 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (a)(v), (a)(vi), (b)(i), (b)(ii), (b)(iii), (c) · also 4.2, 4.3, 5.1, 5.2, 9.2, 9.3

5. Ethane, C_2H_6, and disilane, Si_2H_6, are both hydrides of group 4 elements with similar structures but with different chemical properties.

(a) (i) Deduce the Lewis (electron dot) structure for Si_2H_6 showing all valence electrons. *[1]*

(ii) State and explain the H–Si–H bond angle in Si_2H_6. *[2]*

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(iii) Identify the type of hybridization shown by the silicon atoms in Si_2H_6. *[1]*

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(iv) State which of the bonds, Si–H or C–H, is more polar. Explain your choice. *[2]*

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(v) Predict, with an explanation, the polarity of the two molecules. *[2]*

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(vi) Explain why disilane has a higher boiling point than ethane. *[2]*

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(b) Disilane undergoes complete oxidation to form silicon dioxide and water.

$$2Si_2H_6(g)+7O_2(g) \rightarrow 4SiO_2(s)+6H_2O(l)$$

(i) The standard enthalpy of formation of the silicon compounds is given below.

	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$Si_2H_6(g)$	+80
$SiO_2(s)$	–911

Calculate the standard enthalpy change, in kJ, for this reaction using these data together with Table 12 of the Data Booklet. *[3]*

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(ii) Calculate the standard enthalpy change, in kJ, for the corresponding combustion reaction of 2 moles of ethane, using Table 12 of the Data Booklet. *[1]*

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(iii) Compare the structure and bonding in carbon dioxide and silicon dioxide. *[3]*

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(c) Disilane reacts with hydrogen to produce silane, SiH_4.

$$Si_2H_6(g) + H_2(g) \rightarrow 2SiH_4(g)$$

Use values from Table 10 of the Data Booklet to calculate the enthalpy change, $\Delta H^{\ominus}$, for this reaction. *[3]*

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3.1-S26 May 2012 TZ2 · Paper 2 · Q4

4. Draw the Lewis structures, predict the shape and deduce the bond angles for xenon tetrafluoride and the nitrate ion. *[6]*

Species	Lewis structure	Shape	Bond angle
XeF_4			
NO_3^-			

3.1-S27 Nov 2011 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (d)(i) · also 4.1, 5.1

(c) For each of the species PBr_3 and SF_6:

(i) deduce the Lewis structure.

(ii) predict the shape and bond angle.

(iii) predict and explain the molecular polarity. *[8]*

PBr_3	**SF_6**
(i) Lewis structure: (ii) Shape: . . . . . . Bond angle: . . . . . . (iii) Polarity: . . . . . . Explanation:	(i) Lewis structure: (ii) Shape: . . . . . . Bond angle: . . . . . . (iii) Polarity: . . . . . . Explanation: . . . .

(d) (i) Compare the formation of sigma (σ) and pi (π) bonds between the carbon atoms in a molecule of ethyne. *[2]*

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3.1-S28 May 2011 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii)

5. (a) Ammonia, NH_3, is a weak base. It has a pK_b value of 4.75.

(i) Draw the Lewis structure of ammonia and state the shape of the molecule and its bond angles. *[3]*

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(ii) The conjugate acid of ammonia is the ammonium ion, NH_4^+. Draw the Lewis structure of the ammonium ion and deduce its shape and bond angles. *[3]*

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3.1-S29 Nov 2010 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (d)(ii), (d)(iii) · also 4.2, 5.2, 9.2, 9.3

7. Hydrazine, N_2H_4, is a valuable rocket fuel.

(a) (i) Draw the Lewis (electron dot) structure for N_2H_4 showing all valence electrons. *[1]*

(ii) State and explain the H–N–H bond angle in hydrazine. *[3]*

(b) Hydrazine and ethene, C_2H_4, are hydrides of adjacent elements in the periodic table. The boiling point of hydrazine is much higher than that of ethene. Explain this difference in terms of the intermolecular forces in each compound. *[2]*

(c) The equation for the reaction between hydrazine and oxygen is given below.

$$N_2H_4(l) + O_2(g) \rightarrow N_2(g) + 2H_2O(l)$$

(i) The enthalpy change of formation, $\Delta H_f^{\ominus}$, of liquid hydrazine is $50.6\,kJ\,mol^{-1}$. Use this value, together with data from Table 12 of the Data Booklet, to calculate the enthalpy change for this reaction. *[3]*

(ii) Use the bond enthalpy values from Table 10 of the Data Booklet to determine the enthalpy change for this reaction. *[3]*

(iii) Identify the calculation that produces the most accurate value for the enthalpy change for the reaction given and explain your choice. *[3]*

(d) The reaction between $N_2H_4(aq)$ and $HCl(aq)$ can be represented by the following equation.

$$N_2H_4(aq) + 2HCl(aq) \rightarrow N_2H_6^{2+}(aq) + 2Cl^-(aq)$$

(ii) Predict the value of the H–N–H bond angle in $N_2H_6^{2+}$. *[1]*

(iii) Suggest the type of hybridization shown by the nitrogen atoms in $N_2H_6^{2+}$. *[1]*

LESSON 4.1

Sigma & Pi Bonds and Delocalization

54 multiple-choice questions · 30 structured questions

Part A - Multiple choice

4.1-M1 May 2025 TZ1 · Paper 1A · Q13

13. Kekulé suggested a structure of benzene with alternating single and double bonds. Which piece of evidence disproves this structure?

A. Benzene has the formula C_6H_6.

B. Benzene is an unsaturated molecule.

C. All C–C bonds have the same length.

D. Benzene can be hydrogenated.

4.1-M2 Nov 2024 · Paper 1 · Q13

13. How many sigma (σ) and pi (π) bonds are present in the molecule below?

$(NC)_2CC(CN)_2$

	σ	π
A.	7	5
B.	9	5
C.	9	9
D.	13	5

4.1-M3 May 2024 TZ1 · Paper 1 · Q10

10. Which species have resonance structures?

I. CH_2CHCH_3

II. O_3

III. NO_3^-

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M4 May 2024 TZ2 · Paper 1 · Q14

14. What are the numbers of sigma and pi bonds in propanenitrile, CH_3CH_2CN?

	Sigma	Pi
A.	6	2
B.	7	3
C.	8	2
D.	9	1

4.1-M5 Nov 2023 TZ1 · Paper 1 · Q9

9. For which molecule can resonance structures be used to describe the bonding?

A. HCN

B. H_2CO_3

C. PCl_3

D. SO_2

4.1-M6 May 2023 TZ1 · Paper 1 · Q15

15. Which allotrope, oxygen or ozone, has the stronger bond between O atoms, and which absorbs higher frequency UV radiation in the atmosphere?

	Stronger bond between O atoms	Absorbs higher frequency UV
A.	ozone	ozone
B.	ozone	oxygen
C.	oxygen	oxygen
D.	oxygen	ozone

4.1-M7 May 2023 TZ1 · Paper 1 · Q33

33. Why does benzene undergo substitution more readily than addition?

A. Benzene is unsaturated.

B. Addition could produce an alkane.

C. Resonance makes carbon–carbon bonds too strong to break.

D. A benzene molecule is planar.

4.1-M8 May 2023 TZ2 · Paper 1 · Q13

13. Which wavelength and energy of light will break bonds in ozone rather than oxygen molecules?

A. Shorter wavelength and lower energy

B. Shorter wavelength and higher energy

C. Longer wavelength and lower energy

D. Longer wavelength and higher energy

4.1-M9 Nov 2022 · Paper 1 · Q9

9. Which molecule can be represented by resonance structures?

A. H_2S

B. HNO_3

C. H_2O_2

D. HClO

4.1-M10 Nov 2022 · Paper 1 · Q13

13. Chlorofluorocarbons (CFCs) contain bonds of the following lengths:

$C{-}C = 1.54 \times 10^{-10}\,m$

$C{-}F = 1.38 \times 10^{-10}\,m$

$C{-}Cl = 1.77 \times 10^{-10}\,m$

What is the order of **increasing** bond strength in the CFC molecule?

A. $C{-}C < C{-}F < C{-}Cl$

B. $C{-}C < C{-}Cl < C{-}F$

C. $C{-}Cl < C{-}C < C{-}F$

D. $C{-}F < C{-}C < C{-}Cl$

4.1-M11 May 2022 TZ1 · Paper 1 · Q11

11. For which species can resonance structures be drawn?

A. HCOOH

B. $HCOO^-$

C. CH_3OH

D. H_2CO_3

4.1-M12 Nov 2021 · Paper 1 · Q12

12. Which molecules contain two pi (π) bonds?

I. HCN
II. H_2CO_3
III. $H_2C_2O_4$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

4.1-M13 May 2021 TZ1 · Paper 1 · Q12

12. Which contain delocalised electrons?

I. C_6H_5OH
II. CH_3COO^-
III. CO_3^{2-}

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

4.1-M14 May 2021 TZ2 · Paper 1 · Q10

10. Which compound has the shortest C to N bond?

A. HCN
B. $CH_3CH_2NH_2$
C. CH_3CHNH
D. $(CH_3)_2NH$

4.1-M15 Nov 2020 · Paper 1 · Q9

9. Which of these species contains the shortest carbon to oxygen bond length?

A. $CH_3CH_2O^-$
B. CH_3CH_2OH
C. CH_3COO^-
D. CH_3COOH

4.1-M16 Nov 2020 · Paper 1 · Q13

13. Which statement is correct?

A. O_3 bond dissociation occurs at a longer wavelength of light than O_2 bond dissociation.

B. O_3 bond dissociation occurs at a higher energy than O_2 bond dissociation.

C. O_3 bond lengths are shorter than O_2 bond lengths.

D. O_3 bond dissociation occurs at a higher frequency of light than O_2 bond dissociation.

4.1-M17 Nov 2019 · Paper 1 · Q9

9. Which compound has the shortest C to O bond?

A. CH_3CHO

B. CO

C. CO_2

D. $C_2H_5OC_2H_5$

4.1-M18 Nov 2019 · Paper 1 · Q10

10. Which describes a resonance structure?

A. Double bond can be drawn in alternative positions.

B. Bonds vibrate by absorbing IR radiation.

C. A double and a single bond in the molecule

D. A Lewis structure

4.1-M19 May 2019 TZ1 · Paper 1 · Q12

12. Which species has delocalized electrons?

A. OH^-

B. H_2CO

C. CO_2

D. CO_3^{2-}

4.1-M20 May 2019 TZ1 · Paper 1 · Q36

36. Which statement is **not** correct regarding benzene?

A. It is planar.

B. The ring contains delocalized electrons.

C. It always reacts in the same way as alkenes.

D. The carbon–carbon bond has a bond order of 1.5.

4.1-M21 May 2019 TZ2 · Paper 1 · Q12

12. How many sigma (σ) and pi (π) bonds are present in hydrogen cyanide, HCN?

	Sigma (σ)	Pi (π)
A.	1	1
B.	2	2
C.	2	1
D.	1	3

4.1-M22 Nov 2018 · Paper 1 · Q12

12. What is the number of sigma (**σ**) and pi (**π**) bonds in the molecule $(NC)_2C{=}C(CN)_2$?

	σ	**π**
A.	9	9
B.	5	9
C.	13	5
D.	9	5

4.1-M23 May 2018 TZ1 · Paper 1 · Q13

13. Which can be represented with only one Lewis structure?

A. CH_2O

B. C_6H_6

C. O_3

D. NO_3^-

4.1-M24 May 2018 TZ2 · Paper 1 · Q10

10. Which species has the longest carbon to oxygen bond length?

A. CO

B. CH_3OH

C. $CH_3CO_2^-$

D. H_2CO

4.1-M25 May 2018 TZ2 · Paper 1 · Q13

13. Which overlap of atomic orbitals leads to the formation of only a sigma (σ) bond?

I. s – p
II. p – p
III. s – s

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

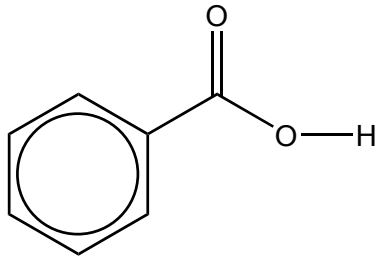
4.1-M26 Nov 2017 · Paper 1 · Q14

14. How many sigma (σ) and pi (π) bonds are present in this molecule?

	σ	π
A.	12	6
B.	14	5
C.	16	6
D.	17	5

4.1-M27 May 2017 TZ1 · Paper 1 · Q12

12. Which combination describes the bonding and structure in benzoic acid, C_6H_5COOH?



	Number of electron domains per carbon atom	Number of π-electrons	Number of σ-bonds
A.	3	6	6
B.	3	8	15
C.	4	6	6
D.	4	8	10

4.1-M28 May 2017 TZ1 · Paper 1 · Q13

13. Which species have resonance structures?

I. Ozone, O_3
II. Carbon dioxide, CO_2
III. Benzene, C_6H_6

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M29 May 2017 TZ2 · Paper 1 · Q10

10. Which does **not** show resonance?

A. PO_4^{3-}

B. C_6H_6

C. C_6H_{12}

D. O_3

4.1-M30 May 2017 TZ2 · Paper 1 · Q13

13. Which statement is correct?

A. Sigma bonds are formed only by the combination of s atomic orbitals.

B. Pi bonds can be formed in the absence of sigma bonds.

C. Pi bonds are formed parallel to the axis between atoms.

D. Pi bonds are formed only by the combination of hybrid orbitals.

4.1-M31 May 2017 TZ2 · Paper 1 · Q15

15. What can be deduced from the facts that ozone absorbs UV radiation in the region of 340 nm and molecular oxygen in the region of 242 nm?

A. The bond between atoms in molecular oxygen is a double bond.

B. The bonds in ozone are delocalized.

C. The bonds between atoms in ozone are stronger than those in molecular oxygen.

D. The bonds between atoms in molecular oxygen need more energy to break.

4.1-M32 May 2016 · Paper 1 · Q14

14. In which group do both compounds contain delocalized electrons?

A. C_6H_{10}, C_5H_{10}

B. Na_2CO_3, NaOH

C. $NaHCO_3$, C_6H_6

D. $NaHCO_3$, C_6H_{12}

4.1-M33 Nov 2015 · Paper 1 · Q13

13. Which set contains two or more species with delocalized π electrons?

A. CH_3CH_3, $H_2C{=}CH_2$, $H_2C{=}O$

B. NaCl, C_6H_6, $H_2C{=}O$

C. CO_3^{2-}, C_6H_6, C_6H_{12}

D. O_2, CH_3COCH_3, CH_3COOCH_3

4.1-M34 Nov 2014 · Paper 1 · Q10

10. Which sequence has the molecules in order of increasing nitrogen-nitrogen bond length?

A. $N_2 < N_2H_4 < N_2H_2$

B. $N_2 < N_2H_2 < N_2H_4$

C. $N_2H_4 < N_2H_2 < N_2$

D. $N_2H_2 < N_2H_4 < N_2$

4.1-M35 Nov 2014 · Paper 1 · Q12

12. What is the correct number of sigma (σ) and pi (π) bonds in prop-2-enenitrile, CH_2CHCN?

	σ bonds	**π bonds**
A.	7	2
B.	4	5
C.	6	3
D.	3	3

4.1-M36 Nov 2014 · Paper 1 · Q13

13. Which group of ions and molecules has delocalized electrons in **all** the species?

A. CH_3COCH_3, $C_2H_5COO^-$ and O_3

B. NO_3^-, NO_2^- and CO_2

C. C_6H_6, CO_3^{2-} and graphite

D. C_6H_6, CO_3^{2-} and C_2H_2

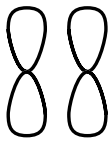
4.1-M37 May 2014 TZ1 · Paper 1 · Q10

10. Which combination of length and strength of the carbon–to–carbon bonds in C_2H_2 and C_2H_4 is correct?

	Bond length	**Bond strength**
A.	$C_2H_2 > C_2H_4$	$C_2H_2 < C_2H_4$
B.	$C_2H_2 > C_2H_4$	$C_2H_2 > C_2H_4$
C.	$C_2H_2 < C_2H_4$	$C_2H_2 < C_2H_4$
D.	$C_2H_2 < C_2H_4$	$C_2H_2 > C_2H_4$

4.1-M38 May 2014 TZ1 · Paper 1 · Q12

12. The diagrams below show *s* and *p* orbitals in different positions. Which combinations can form a σ-bond?

I. 

II.

III.

A. I and II only

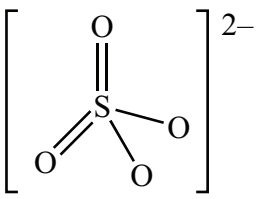
B. I and III only

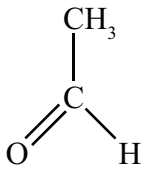
C. II and III only

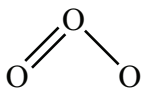
D. I, II and III

4.1-M39 May 2014 TZ1 · Paper 1 · Q13

13. Which species contain delocalized electrons?

I. 

II. 

III. 

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M40 Nov 2013 · Paper 1 · Q13

13. How many sigma (σ) and pi (π) bonds are there in $CH_3CH_2CCCH_2COOH$?

A. 13σ and 5π

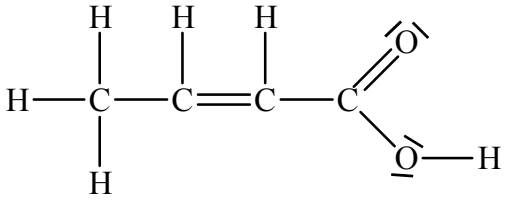
B. 15σ and 2π

C. 15σ and 3π

D. 15σ only

4.1-M41 May 2013 TZ2 · Paper 1 · Q12

12. How many sigma (σ) and pi (π) bonds are there in the following molecule?



	σ bonds	π bonds
A.	9	2
B.	9	4
C.	11	2
D.	11	4

4.1-M42 May 2013 TZ2 · Paper 1 · Q13

13. Which species have delocalized π electrons?

I. CH_3COCH_3

II. NO_2^-

III. CO_3^{2-}

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M43 Nov 2012 · Paper 1 · Q13

13. Which substances have delocalized electrons in their structure?

I. Ethanal
II. Ozone
III. Benzene

A. I and II only

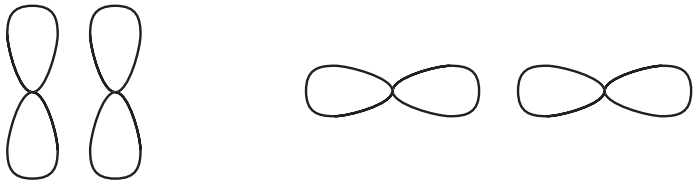
B. I and III only

C. II and III only

D. I, II and III

4.1-M44 May 2012 TZ1 · Paper 1 · Q12

12. Diagrams **I** and **II** show two p orbitals on adjacent atoms in different relative orientations.



I **II**

Which types of bonds are formed when the orbitals overlap?

	Orientation I	Orientation II
A.	σ	σ
B.	π	π
C.	π	σ
D.	σ	π

4.1-M45 May 2012 TZ1 · Paper 1 · Q13

13. Which molecules have delocalized π electrons?

I. C_6H_6

II. CH_3COOH

III. O_3

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M46 May 2011 TZ1 · Paper 1 · Q10

10. How many σ and π bonds are present in a molecule of propyne, CH_3CCH?

	σ	π
A.	5	3
B.	6	2
C.	7	1
D.	8	0

4.1-M47 May 2011 TZ1 · Paper 1 · Q11

11. Which species does **not** contain delocalized electrons?

A. $CH_3CH_2O^-$

B. $CH_3CO_2^-$

C. O_3

D. NO_3^-

4.1-M48 May 2011 TZ2 · Paper 1 · Q10

10. When C_2H_2, C_2H_4 and C_2H_6 are arranged in order of **increasing** carbon-carbon bond strength (weakest bond first), what is the correct order?

A. C_2H_2, C_2H_4, C_2H_6

B. C_2H_2, C_2H_6, C_2H_4

C. C_2H_6, C_2H_4, C_2H_2

D. C_2H_6, C_2H_2, C_2H_4

4.1-M49 May 2011 TZ2 · Paper 1 · Q13

13. How many sigma and pi bonds are there in propyne, CH_3CCH?

A. 2 sigma and 2 pi

B. 7 sigma and 1 pi

C. 6 sigma and 2 pi

D. 5 sigma and 3 pi

4.1-M50 May 2011 TZ2 · Paper 1 · Q14

14. Which species does **not** have delocalized electrons?

A. NO_3^-

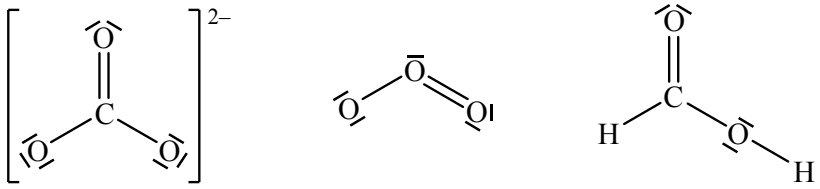
B. NO_2^-

C. O_3

D. C_3H_6

4.1-M51 Nov 2010 · Paper 1 · Q13

13. Which species have delocalized electrons?



I II III

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M52 May 2010 TZ1 · Paper 1 · Q10

10. Which molecule has the shortest carbon-oxygen bond length?

A. CH_3COOH

B. CH_3CH_2OH

C. CO_2

D. CO

4.1-M53 May 2010 TZ1 · Paper 1 · Q13

13. Which statements about σ and π bonds are correct?

I. σ bonds result from the axial overlap of orbitals.

II. σ bonds only form from s orbitals.

III. π bonds result from the sideways overlap of parallel p orbitals.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M54 May 2010 TZ2 · Paper 1 · Q13

13. How many sigma (σ) bonds are present in $(CH_3)_2C{=}CClCH_2CH_3$?

A. 1

B. 4

C. 16

D. 17

Part B - Structured questions

4.1-S1 May 2022 TZ2 · Paper 2 · Q7 parts (b)(i), (b)(ii), (b)(iii), (c) · also 4.2, 5.2

7. The overall equation for the production of hydrogen cyanide, HCN, is shown below.

$$CH_4(g) + NH_3(g) + \frac{3}{2}O_2(g) \rightarrow HCN(g) + 3H_2O(g)$$

(b) (i) Sketch the shape of one sigma (σ) and one pi (π) bond. [2]

Sigma (σ):

Pi (π):

(ii) Identify the number of sigma and pi bonds in HCN. [1]

Sigma (σ):

. . . . . .

Pi (π):

. . . . . .

(iii) State the hybridization of the carbon atom in HCN. [1]

. . . . . .

(c) Suggest why hydrogen chloride, HCl, has a lower boiling point than hydrogen cyanide, HCN. [1]

	M_r	Boiling point
HCN	27.03	26.00 °C
HCl	36.51	–85.05 °C

. . . . . .

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4.1-S2 Nov 2021 · Paper 2 · Q10 parts (a)(i), (a)(ii) · also 4.2

10. Hybridization of hydrocarbons affects their reactivity.

(a) (i) Distinguish between a sigma and pi bond. [2]

Sigma (σ) bond:

. . . . . .

. . . . . .

Pi (π) bond:

. . . . . .

. . . . . .

(ii) Identify the hybridization of carbon in ethane, ethene and ethyne. [1]

	Ethane	Ethene	Ethyne
Hybridization of carbon	. . . . .	. . . . .	. . . . .

4.1-S3 May 2021 TZ1 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b) · also 3.1

7. Oxygen exists as two allotropes, diatomic oxygen, O_2, and ozone, O_3.

(a) (i) Draw a Lewis (electron dot) structure for ozone. [1]

(ii) Discuss the relative length of the two O–O bonds in ozone. [2]

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(b) Explain why there are frequencies of UV light that will dissociate O_3 but not O_2. [2]

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4.1-S4 Nov 2019 · Paper 2 · Q1 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 3.1

1. The equations show steps in the formation and decomposition of ozone in the stratosphere, some of which absorb ultraviolet light.

Step 1 $O_2 \rightarrow 2O\bullet$

Step 2 $O\bullet + O_2 \rightarrow O_3$

Step 3 $O_3 \rightarrow O\bullet + O_2$

Step 4 $O\bullet + O_3 \rightarrow 2O_2$

(a) Draw the Lewis structures of oxygen, O_2, and ozone, O_3. [2]

(b) Outline why both bonds in the ozone molecule are the same length and predict the bond length in the ozone molecule. Refer to section 10 of the data booklet. [2]

Reason:

. . . . . .

. . . . . .

. . . . . .

Length:

. . . . . .

(c) Predict the bond angle in the ozone molecule. [1]

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(d) Discuss how the different bond strengths between the oxygen atoms in O_2 and O_3 in the ozone layer affect radiation reaching the Earth's surface. [2]

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(e) (i) Identify the steps which absorb ultraviolet light. [1]

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(ii) Determine, showing your working, the wavelength, in m, of ultraviolet light absorbed by a single molecule in one of these steps. Use sections 1, 2 and 11 of the data booklet. [2]

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4.1-S5 May 2019 TZ1 · Paper 2 · Q2 parts (c), (f)(ii), (g), (h) · also 5.2, 6.1, 7.3

2. Benzoic acid, C_6H_5COOH, is another derivative of benzene.

(c) Outline **one** piece of physical evidence for the structure of the benzene ring. [1]

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(ii) Formulate the equation for the complete combustion of benzoic acid in oxygen using only integer coefficients. [2]

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(g) The combustion reaction in (f)(ii) can also be classed as redox. Identify the atom that is oxidized and the atom that is reduced. [1]

Oxidized:

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Reduced:

. . . . . .

(h) Suggest how benzoic acid, $M_r = 122.13$, forms an apparent dimer, $M_r = 244.26$, when dissolved in a non-polar solvent such as hexane. [1]

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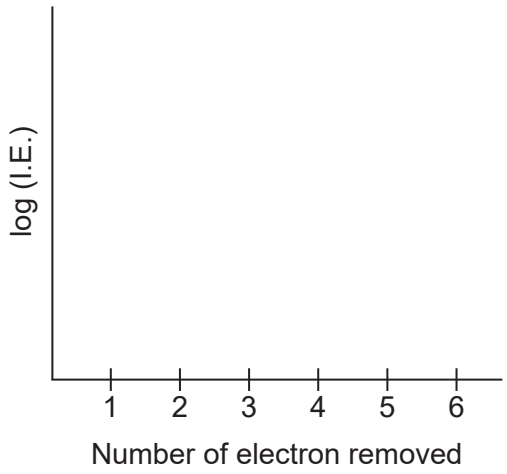
4.1-S6 May 2018 TZ1 · Paper 2 · Q2 parts (e), (g)(i), (g)(ii) · also 2.1

2. Calcium carbide, CaC_2, is an ionic solid.

(e) Sketch a graph of the first six ionization energies of calcium. [2]



(g) (i) Describe how sigma (σ) and pi (π) bonds are formed. [2]

sigma (σ):

. . . . . .

. . . . . .

pi (π):

. . . . . .

. . . . . .

(ii) Deduce the number of σ and π bonds in a molecule of ethyne. [1]

sigma (σ):

. . . . . .

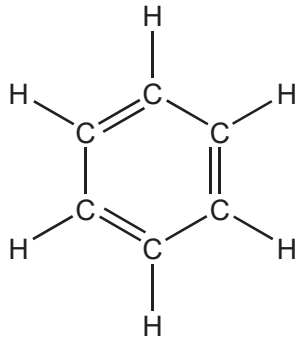
pi (π):

. . . . . .

4.1-S7 May 2018 TZ2 · Paper 2 · Q8 parts (a)

8. The structure of an organic molecule can help predict the type of reaction it can undergo.

(a) The Kekulé structure of benzene suggests it should readily undergo addition reactions.



Discuss two pieces of evidence, **one** physical and **one** chemical, which suggest this is not the structure of benzene. [2]

Physical evidence:

. . . . . .

. . . . . .

Chemical evidence:

. . . . . .

. . . . . .

4.1-S8 May 2017 TZ1 · Paper 2 · Q7 parts (a), (b)

7. Benzene is an aromatic hydrocarbon.

(a) Discuss the physical evidence for the structure of benzene. [2]

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(b) State the typical reactions that benzene and cyclohexene undergo with bromine. [1]

Benzene:

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Cyclohexene:

. . . . . .

4.1-S9 May 2017 TZ2 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (c) · also 2.1, 3.1, 5.2, 6.1

4. Bonds can be formed in many ways.

(a) The landing module for the Apollo mission used rocket fuel made from a mixture of hydrazine, N_2H_4, and dinitrogen tetraoxide, N_2O_4.

$$2N_2H_4(l) + N_2O_4(l) \rightarrow 3N_2(g) + 4H_2O(g)$$

(i) State and explain the difference in bond strength between the nitrogen atoms in a hydrazine and nitrogen molecule. [2]

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(ii) State why hydrazine has a higher boiling point than dinitrogen tetraoxide. [1]

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(iii) Determine the oxidation state of nitrogen in the two reactants. [1]

N_2H_4:

. . . . . .

N_2O_4:

. . . . . .

(iv) Deduce, giving a reason, which species is the reducing agent. [1]

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(b) (i) Discuss the bonding in the resonance structures of ozone. [3]

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. . . . . .
. . . . . .
. . . . . .

(ii) Deduce one resonance structure of ozone and the corresponding formal charges on each oxygen atom. [2]

(c) The first six ionization energies, in $kJ\,mol^{-1}$, of an element are given below.

IE_1	IE_2	IE_3	IE_4	IE_5	IE_6
578	1816	2744	11576	14829	18375

Explain the large increase in ionization energy from IE_3 to IE_4. [2]

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. . . . . .
. . . . . .
. . . . . .

4.1-S10 Nov 2016 · Paper 2 · Q5 parts (b)(i)

5. Propane and propene are members of different homologous series.

(b) (i) Draw diagrams to show how sigma (σ) and pi (π) bonds are formed between atoms. [2]

Sigma (σ):

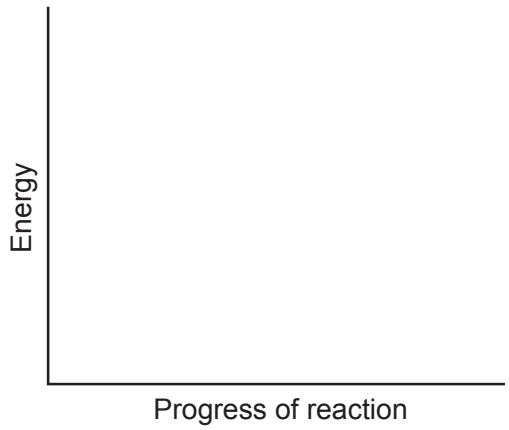
Pi (π):

4.1-S11 May 2016 · Paper 2 · Q3 parts (b)(i), (c)(i), (c)(ii) · also 8.1

3. The reaction between hydrogen and nitrogen monoxide is thought to proceed by the mechanism shown below.

$2NO(g) \rightleftharpoons N_2O_2(g)$	fast equilibrium
$N_2O_2(g) + H_2(g) \rightarrow N_2O(g) + H_2O(g)$	slow reaction
$N_2O(g) + H_2(g) \rightarrow N_2(g) + H_2O(g)$	fast reaction

(b) The enthalpy change for the reaction between nitrogen monoxide and hydrogen is −664 kJ and its activation energy is 63 kJ.



(i) Sketch the potential energy profile for the overall reaction, using the axes given, indicating both the enthalpy of reaction and activation energy. [2]

(c) One of the intermediates in the reaction between nitrogen monoxide and hydrogen is dinitrogen monoxide, N_2O. This can be represented by the resonance structures below:

(i) Analyse the bonding in dinitrogen monoxide in terms of σ-bonds and π-bonds. [3]

(ii) State what is meant by resonance. [1]

4.1-S12 May 2015 TZ1 · Paper 3 · Q20 parts (a)

20. Ozone and oxygen both dissociate in the ozone layer when they absorb ultraviolet light of different wavelengths.

$$O_3(g) \rightarrow O_2(g) + O\bullet(g) \qquad \lambda = 330\,\text{nm}$$

$$O_2(g) \rightarrow O\bullet(g) + O\bullet(g) \qquad \lambda = 242\,\text{nm}$$

(a) Explain, on a molecular level, why ozone dissociates with radiation of a longer wavelength than oxygen. [2]

4.1-S13 May 2015 TZ2 · Paper 2 · Q9 parts (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii), (d)(iii), (d)(iv) · also 3.1, 4.2, 5.1

(c) Consider the molecules PBr_3 and SF_4.

(i) Deduce the Lewis (electron dot) structure of both molecules. [2]

(ii) Predict the shapes of the two molecules, giving the Br–P–Br bond angle in PBr_3 and the F–S–F bond angles in SF_4. [4]

PBr_3	**SF_4**
Shape: . . . . Bond angle:	Shape: . . . . Bond angles: . . . .

(iii) Explain why both PBr_3 and SF_4 are polar. [2]

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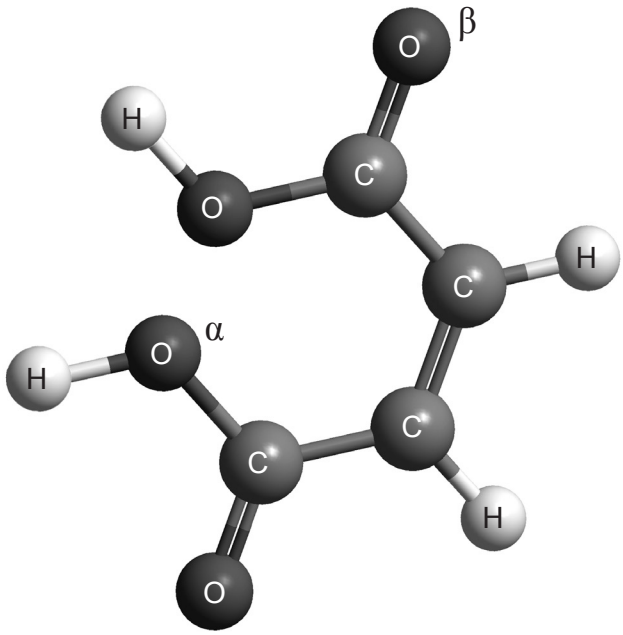
(d) The structure of *cis*-but-2-ene-1,4-dioic acid is shown below.



(i) Describe the covalent bond between carbon and hydrogen in the molecule above and how it is formed. [2]

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(ii) Deduce the hybridization of the oxygen atoms labelled α and β. [1]

α:

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β:

. . . . . .

(iii) Describe sigma (σ) and pi (π) bonds between atoms. [2]

σ bond:

. . . . . .
. . . . . .

π bond:

. . . . . .
. . . . . .

(iv) Identify the number of sigma (σ) and pi (π) bonds present in a molecule of *cis*-but-2-ene-1,4-dioic acid. [1]

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. . . . . .

4.1-S14 May 2015 TZ2 · Paper 3 · Q23 parts (a)

23. Another major source of concern is the depletion of ozone in the stratosphere as a result of human activity.

(a) Explain how the frequency of UV radiation absorbed by oxygen and ozone depends on the bonding in these molecules. [2]

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4.1-S15 May 2015 TZ2 · Paper 3 · Q30 parts (a), (b)

30. Benzene, C_6H_6, was once thought to contain alternate single and double bonds between the carbon atoms.

(a) Describe the currently accepted structure and bonding of the benzene molecule. [3]

Structure:

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Bonding:

. . . . . .

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(b) Outline **one** piece of thermochemical evidence that provides support for the bonds in the benzene ring **not** being alternately single and double. [1]

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4.1-S16 Nov 2014 · Paper 3 · Q22 parts (a)

22. The ozone layer protects us by absorbing ultraviolet (UV) radiation from the Sun during its natural formation and depletion.

(a) Explain how the bonding in O_2 and O_3 affects the wavelengths of UV light they absorb. *[3]*

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4.1-S17 May 2014 TZ1 · Paper 3 · Q16 parts (d)

16. Ozone prevents UV radiation emitted from the Sun reaching the surface of the Earth.

(d) Explain, with the aid of Lewis (electron-dot) structures, the difference between oxygen and ozone in terms of the energy required to dissociate both molecules. *[2]*

Oxygen: Ozone:

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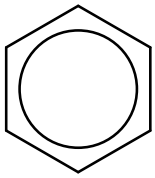
4.1-S18 Nov 2013 · Paper 3 · Q27 parts (a)(i), (a)(ii)

27. The bonding in benzene is one of the most challenging problems for chemists interested in explaining the bonding and structure of covalent molecules.



(a) (i) Outline what the circle represents in the diagram above. *[1]*

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(ii) Describe how the structure accounts for the C–C bond length in Table 9 of the Data Booklet. *[1]*

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. . . . . .

4.1-S19 May 2013 TZ1 · Paper 3 · QE1 parts (c)

E1. Increasing concentrations of greenhouse gases are considered to cause global warming. Ozone depletion is another environmental concern.

(c) Ozone and oxygen are in equilibrium in the stratosphere. Both gases absorb ultraviolet radiation and dissociate producing oxygen atoms.

Describe the dependence of ozone and oxygen dissociation on the wavelength of radiation absorbed, and explain how this is related to the bonding in each molecule. *[3]*

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4.1-S20 May 2013 TZ1 · Paper 3 · QG1 parts (a)

G1. Benzene, C_6H_6, is found naturally in crude oil.

(a) Describe the chemical structure of benzene. *[3]*

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4.1-S21 May 2013 TZ2 · Paper 3 · QG2 parts (a)

G2. (a) Discuss **three** separate pieces of evidence (physical or chemical) to show that the bonding between the carbon atoms in benzene is not simply alternate double and single carbon–carbon covalent bonds. *[3]*

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4.1-S22 May 2012 TZ1 · Paper 3 · QE1 parts (a)

E1. Ozone is a gas present in both the stratosphere and the troposphere.

(a) Oxygen absorbs much of the ultraviolet (UV) radiation from the sun, but ozone is important because it absorbs UV radiation not absorbed by oxygen. Explain, referring to the bonding in the two molecules, why this is the case. *[3]*

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4.1-S23 May 2012 TZ2 · Paper 2 · Q7 parts (a)(vi), (a)(vii), (b)(i), (b)(ii), (b)(iii) · also 4.2, 4.3

7. (a) An organic compound, **X**, with a molar mass of approximately 88 g mol^{-1} contains 54.5 % carbon, 36.3 % oxygen and 9.2 % hydrogen by mass.

(vi) The carboxylic acid contains two different carbon-oxygen bonds. Identify which bond is stronger and which bond is longer. *[2]*

Stronger bond:

. . . . . .

Longer bond:

. . . . . .

(vii) Predict and explain the bond lengths and bond strengths of the carbon-oxygen bonds in $CH_3CH_2COO^-$. *[3]*

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(b) (i) State the meaning of the term *hybridization*. *[1]*

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(ii) Describe the hybridization of the carbon atom in methane and explain how the concept of hybridization can be used to explain the shape of the methane molecule. *[3]*

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. . . . . .

(iii) Identify the hybridization of the carbon atoms in diamond and graphite and explain why graphite is an electrical conductor. *[3]*

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4.1-S24 May 2012 TZ2 · Paper 3 · QE3 parts (a)

E3. (a) Explain, in terms of their bonding, how the presence of oxygen and ozone in the ozone layer helps to prevent both higher and lower energy UV light from reaching the surface of the Earth. *[3]*

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4.1-S25 May 2011 TZ1 · Paper 3 · QE4 parts (a)

E4. (a) The following reactions take place in the ozone layer by the absorption of ultraviolet light.

I $O_3 \rightarrow O_2 + O\bullet$

II $O_2 \rightarrow O\bullet + O\bullet$

State and explain, by reference to the bonding, which of the reactions, **I** or **II**, requires a shorter wavelength. *[2]*

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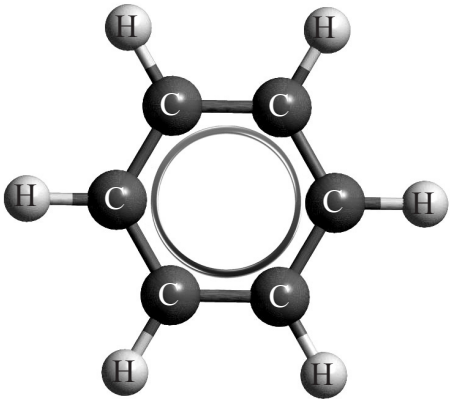
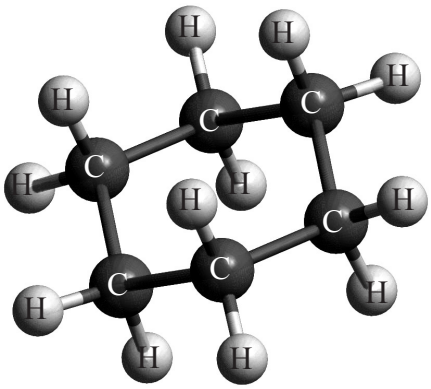
4.1-S26 May 2011 TZ2 · Paper 3 · QG1 parts (b)

G1. Benzene, C_6H_6, is a planar compound which differs from the non-planar structure of cyclohexane, C_6H_{12}. The structures of benzene and the most stable form of cyclohexane are represented below.



Benzene



Cyclohexane

(b) Explain why it is more common for benzene to undergo substitution reactions than addition reactions. *[1]*

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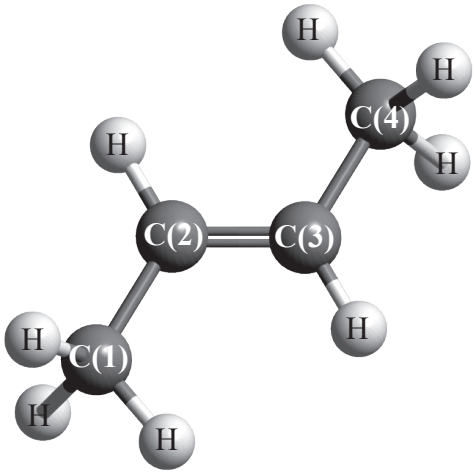
4.1-S27 Nov 2010 · Paper 2 · Q2 parts (a)(i), (a)(ii), (d)(iv) · also 4.2, 5.2

2. (a) But-2-ene is a straight-chain alkene with formula C_4H_8. The molecule contains both σ and π bonds.



(i) Explain the formation of the π bond. *[2]*

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(ii) For each of the carbon atoms, C(1) and C(2), identify the type of hybridization shown. *[1]*

C(1): . . . . . .

C(2): . . . . . .

(d) The polymerization of the alkenes is one of the most significant reactions of the twentieth century.

(iv) Explain why monomers are often gases or volatile liquids, but polymers are solids. *[2]*

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4.1-S28 Nov 2010 · Paper 3 · QE4 parts (a)(i)

E4. (a) (i) Explain the dependence of the dissociation of diatomic oxygen, O_2, and ozone, O_3, on the wavelength of light. *[2]*

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4.1-S29 Nov 2010 · Paper 3 · QG1 parts (a), (b)

G1. (a) Describe the structure of benzene, C_6H_6. *[3]*

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(b) State **two** pieces of evidence that support this description. *[2]*

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4.1-S30 May 2010 TZ2 · Paper 3 · QE3 parts (d)

E3. Chlorofluorocarbons (CFCs) and other ozone-depleting agents reduce the concentration of ozone in the Earth's stratosphere and increase the exposure of humans and other living organisms to dangerous UV radiation.

(d) Explain why UV radiation with $\lambda = 300 - 330\,\text{nm}$ contributes to photochemical decomposition of ozone but not to the formation of ozone from oxygen. *[3]*

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LESSON 4.2

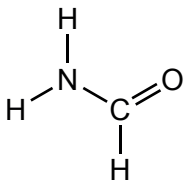
Hybridization

25 multiple-choice questions · 3 structured questions

Part A - Multiple choice

4.2-M1 May 2025 TZ2 · Paper 1A · Q10

10. What are the hybridizations of the N, C and O atoms in this molecule?



	N	C	O
A.	sp^3	sp^3	sp^2
B.	sp^3	sp^2	sp^2
C.	sp^2	sp^2	sp
D.	sp^2	sp^3	sp

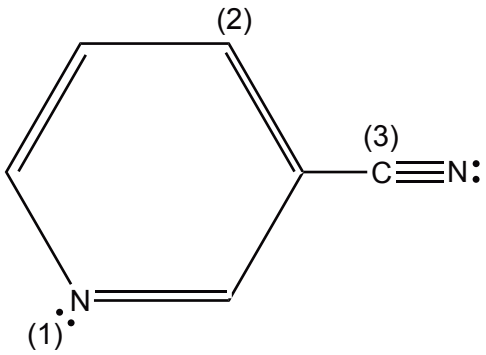
4.2-M2 May 2025 TZ3 · Paper 1A · Q10

10. What is the hybridization of carbon and oxygen in CO_3^{2-}?

	C	O
A.	sp^2	sp^3
B.	sp^2	sp^2
C.	sp^3	sp^2
D.	sp^3	sp

4.2-M3 Nov 2024 · Paper 1 · Q12

12. What is the molecular geometry and hybridization of the numbered atoms in the molecule shown below?



	N(1)	C(2)	C(3)
A.	sp^2 / bent	sp^2 / trigonal planar	sp / linear
B.	sp^2 / trigonal planar	sp^2 / bent	sp^2 / bent
C.	sp^3 / tetrahedral	sp^2 / trigonal planar	sp / bent
D.	sp / bent	sp^3 / tetrahedral	sp^3 / linear

4.2-M4 Nov 2023 TZ1 · Paper 1 · Q13

13. What bond angle is most likely found with an sp^2 hybridized carbon as the central atom?

A. 90°

B. 109.5°

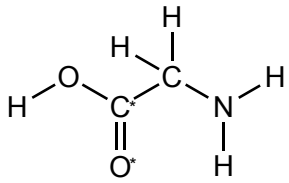
C. 120°

D. 180°

4.2-M5 May 2023 TZ1 · Paper 1 · Q13

13. Which types of hybridization are present in glycine?

Glycine



	C*	N	O*
A.	sp^2	sp^3	sp^2
B.	sp^2	sp^3	sp
C.	sp	sp^2	sp^2
D.	sp	sp^2	sp

4.2-M6 May 2022 TZ1 · Paper 1 · Q12

12. In which compound are all carbon atoms sp^3 hybridized?

A. C_2H_2

B. $C_2H_2Cl_2$

C. C_2Cl_4

D. C_2Cl_6

4.2-M7 Nov 2021 · Paper 1 · Q13

13. What is the hybridization of nitrogen and chlorine in NCl_3?

	N	Cl
A.	sp^2	sp^2
B.	sp^2	sp^3
C.	sp^3	sp^2
D.	sp^3	sp^3

4.2-M8 May 2021 TZ1 · Paper 1 · Q13

13. In which series are all carbon atoms sp^2 hybridized?

A. C_2H_2 H_2CO HCOOH

B. C_2H_4 H_2CO HCOOH

C. C_2H_2 CO HCN

D. C_2H_6 CH_3OH CH_3OCH_3

4.2-M9 Nov 2019 · Paper 1 · Q12

12. Which atom is sp^2 hybridized?

A. C in H_2CO

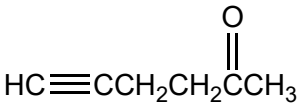
B. C in CO_2

C. N in CH_3NH_2

D. O in H_2O

4.2-M10 May 2019 TZ1 · Paper 1 · Q13

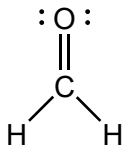
13. How many carbon atoms are sp^3, sp^2 and sp hybridized in the molecule?



	sp^3	sp^2	sp
A.	3	1	2
B.	2	1	3
C.	3	2	1
D.	3	2	2

4.2-M11 May 2019 TZ2 · Paper 1 · Q13

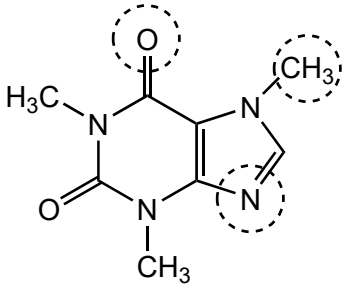
13. What is the hybridization of carbon and oxygen in methanal?



	Hybridization of C	Hybridization of O
A.	sp^2	sp^2
B.	sp^2	sp
C.	sp	sp^2
D.	sp^3	sp^3

4.2-M12 Nov 2018 · Paper 1 · Q13

13. What is the hybridization of the circled carbon, oxygen and nitrogen atoms?



	Carbon	Oxygen	Nitrogen
A.	sp^3	sp	sp
B.	sp^2	sp^2	sp
C.	sp^2	sp^3	sp^2
D.	sp^3	sp^2	sp^2

4.2-M13 May 2018 TZ1 · Paper 1 · Q12

12. Which molecules have at least one sp^2 hybridized atom?

I. CH_3COOH

II. CH_3COCH_3

III. CH_2CHCH_2OH

A. I and II only

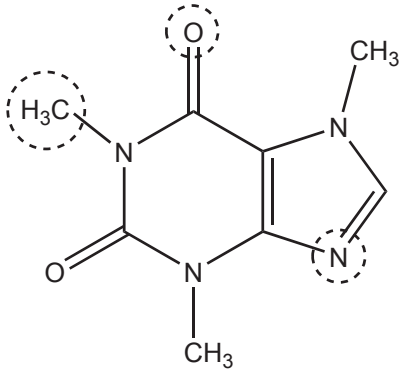
B. I and III only

C. II and III only

D. I, II and III

4.2-M14 Nov 2017 · Paper 1 · Q13

13. What is the hybridization state and electron domain geometry around the circled C, N and O atoms?



	C	O	N
A.	sp^3 and tetrahedral	sp^2 and trigonal planar	sp^2 and trigonal planar
B.	sp^2 and trigonal planar	sp and linear	sp^3 and tetrahedral
C.	sp^3 and tetrahedral	sp and linear	sp^2 and trigonal planar
D.	sp^3 and trigonal pyramidal	sp^2 and trigonal planar	sp^3 and trigonal pyramidal

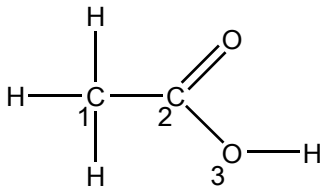
4.2-M15 May 2017 TZ1 · Paper 1 · Q11

11. Which combination describes the PH_4^+ ion?

	Molecular geometry	Central atom hybridization
A.	Tetrahedral	sp^3
B.	Square planar	sp^3
C.	Tetrahedral	sp^2
D.	Square planar	sp^2

4.2-M16 Nov 2016 · Paper 1 · Q15

15. What is the hybridization of the numbered atoms in ethanoic acid?



	Atom 1	Atom 2	Atom 3
A.	sp^3	sp	sp^2
B.	sp^3	sp^2	sp
C.	sp^2	sp^3	sp^2
D.	sp^3	sp^2	sp^3

4.2-M17 May 2016 · Paper 1 · Q15

15. Which of the following is correct?

	Atom	Number of electron domains	Molecular geometry	Hybridization
A.	C in C_2H_2	2	linear	sp
B.	C in C_2H_6	4	square planar	sp^3
C.	N in NH_3	3	trigonal pyramidal	sp^3
D.	O in H_2O	4	bent	sp^2

4.2-M18 May 2015 TZ1 · Paper 1 · Q13

13. Which molecules have sp^2 hybridization?

I. C_2H_4

II. C_4H_{10}

III. C_6H_6

A. I and II only

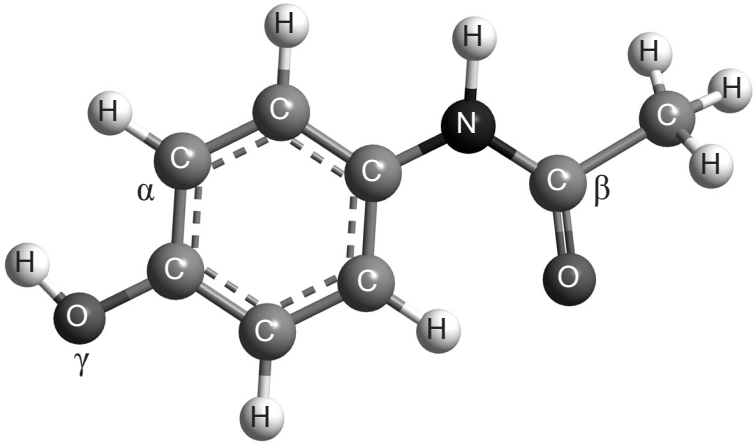
B. I and III only

C. II and III only

D. I, II and III

4.2-M19 May 2015 TZ2 · Paper 1 · Q13

13. Which combination correctly describes the types of hybridization shown by the two carbon atoms labelled α and β and the oxygen atom labelled γ in the molecule of paracetamol shown below?



Paracetamol

	α	β	γ
A.	sp^2	sp^2	sp^3
B.	sp^3	sp^2	sp^2
C.	sp^2	sp^2	sp^2
D.	sp^2	sp^3	sp^3

4.2-M20 May 2014 TZ2 · Paper 1 · Q11

11. Which allotropes of carbon show sp^2 hybridization?

I. Diamond

II. Graphite

III. C_{60} fullerene

A. I and II only

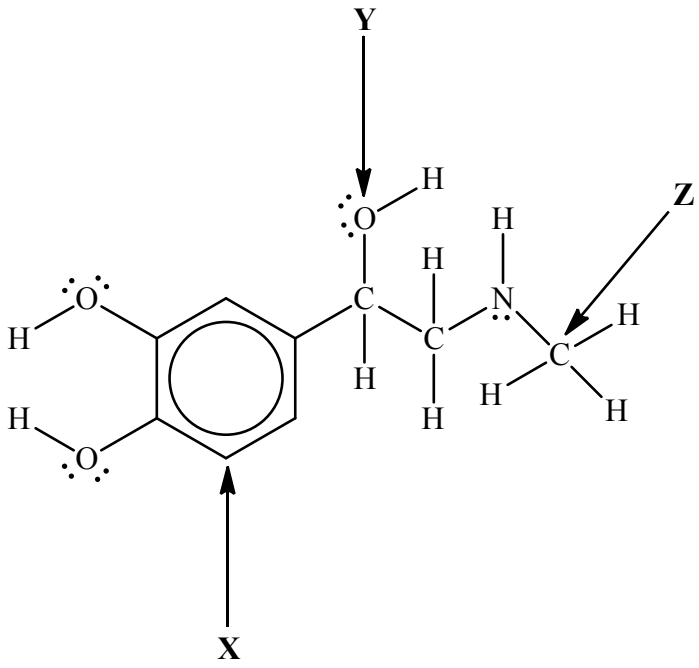
B. I and III only

C. II and III only

D. I, II and III

4.2-M21 Nov 2013 · Paper 1 · Q14

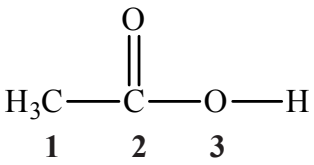
14. What is the hybridization of atoms **X**, **Y** and **Z** in epinephrine?



	X	Y	Z
A.	sp^2	sp^3	sp^3
B.	sp^2	sp	sp^3
C.	sp^3	sp^2	sp^2
D.	sp^3	sp^3	sp^3

4.2-M22 May 2013 TZ1 · Paper 1 · Q13

13. What are the hybridizations of the atoms labelled **1**, **2** and **3** in the molecule below?



	1	2	3
A.	sp^2	sp^2	sp
B.	sp^3	sp^2	sp^3
C.	sp^2	sp	sp^3
D.	sp^3	sp^2	sp

4.2-M23 Nov 2011 · Paper 1 · Q14

14. Which statements about hybridization are correct?

I. The hybridization of carbon in diamond is sp^3.

II. The hybridization of carbon in graphite is sp^2.

III. The hybridization of carbon in C_{60} fullerene is sp^3.

A. I and II only

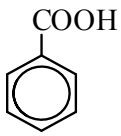
B. I and III only

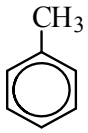
C. II and III only

D. I, II and III

4.2-M24 May 2011 TZ2 · Paper 1 · Q15

15. In which compound are all the carbon atoms sp^2 hybridized?

A. 

B. 

C. CH_2CHCH_3

D. $CH_3CH_2CHCHCH_2CH_3$

4.2-M25 May 2010 TZ1 · Paper 1 · Q14

14. In which substance does a carbon atom have sp^2 hybridization?

A. 2-methylbutan-1-ol

B. Propyne, CH_3CCH

C. C_{60} fullerene

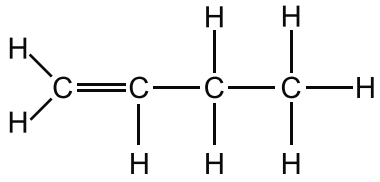
D. Diamond

Part B - Structured questions

4.2-S1 Nov 2024 · Paper 2 · Q2 parts (a)(i), (a)(ii) · also 4.1

2. Alkenes are useful in industry.

(a) But-1-ene contains sigma (σ) and pi (π) bonds.



A **B**

(i) Identify the hybridization of the carbon atoms labelled **A** and **B**. [1]

A: . . . . . .

B: . . . . . .

(ii) Describe how sigma (σ) and pi (π) bonds are formed. [2]

σ bonds: . . . . . .

. . . . . .

. . . . . .

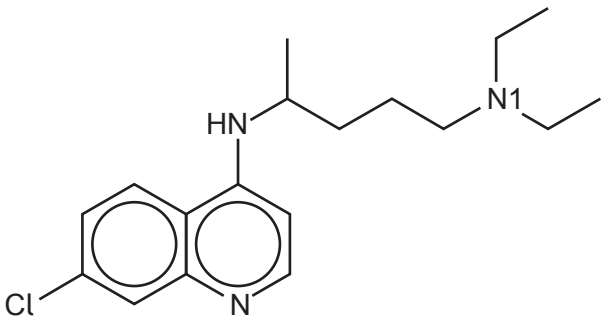
π bonds: . . . . . .

. . . . . .

. . . . . .

4.2-S2 Nov 2022 · Paper 2 · Q2 parts (b), (d), (e)(ii) · also 2.1, 4.1

2. Chloroquine is a medication used to prevent and treat malaria.



(b) State the number of sp^2 hybridized carbons in chloroquine. [1]

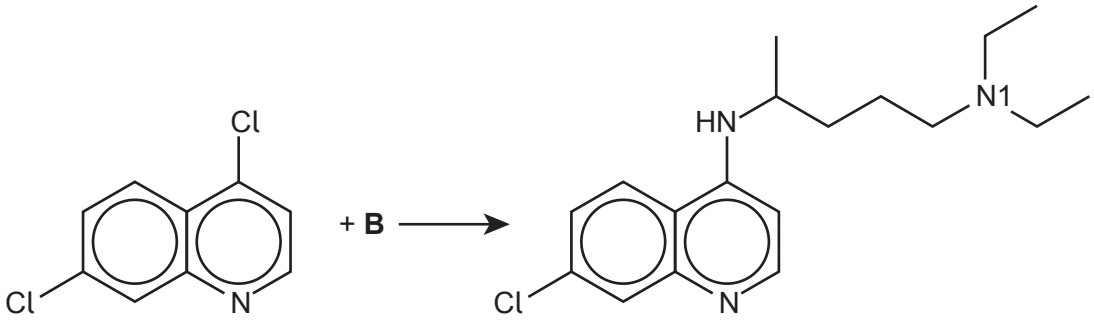
. . . . . .

(d) Compare, giving a reason, the length of the carbon-nitrogen bond in the ring to the length of the carbon-N1 bond. [1]

. . . . . .

. . . . . .

(e) Chloroquine can be synthesized by reacting 4,7-dichloroquinoline with another reactant, **B**.



4,7-dichloroquinoline **chloroquine**

(ii) This reaction can be done with a copper catalyst. State the ground-state electron configuration for copper. [1]

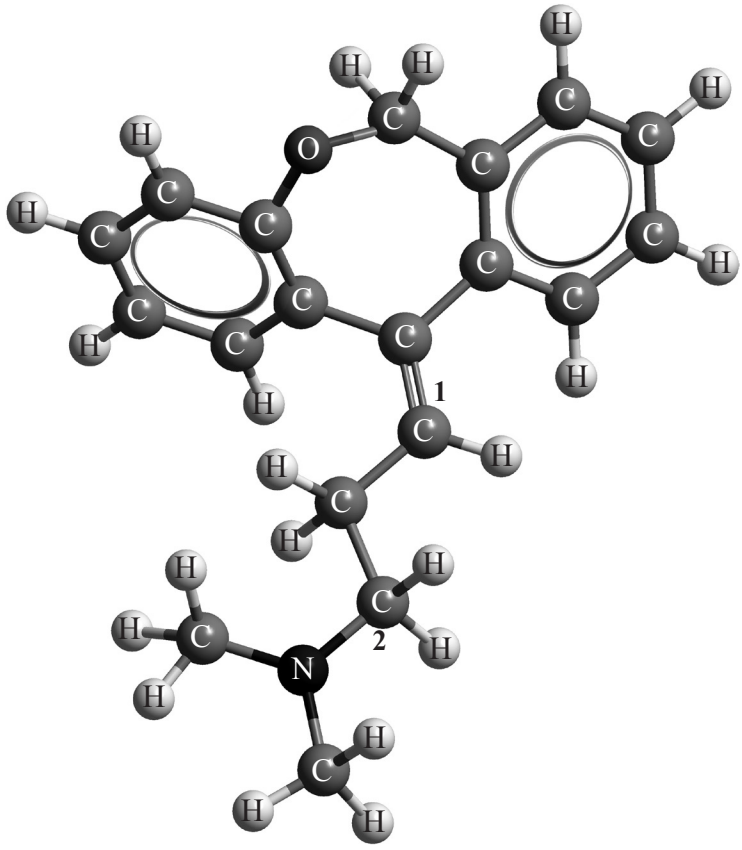
. . . . . .

. . . . . .

4.2-S3 May 2013 TZ2 · Paper 2 · Q8 parts (b)

8. Geometrical isomerism and optical isomerism are two sub-groups of stereoisomerism in organic chemistry.

(b) Geometrical isomers have different physical properties and many drugs, such as doxepin (which has antidepressant properties), have geometrical isomers.



Example of a geometrical isomer of doxepin

For each of the carbon atoms labelled **1** and **2** in doxepin, deduce the type of hybridization involved (sp, sp^2 or sp^3). *[1]*

1: . . . . . .

2: . . . . . .

LESSON 4.3

Covalent Network Structures

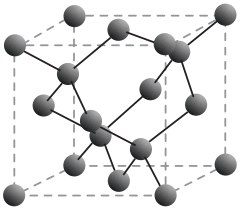
13 multiple-choice questions · 10 structured questions

Part A - Multiple choice

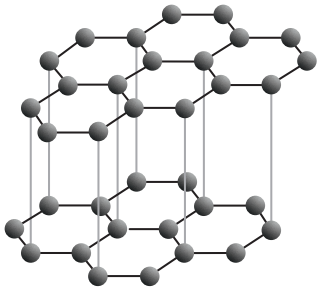
4.3-M1 May 2025 TZ1 · Paper 1A · Q8

8. Diagrams of four stable allotropes of carbon are shown.
Which row correctly identifies each structure?

W



X



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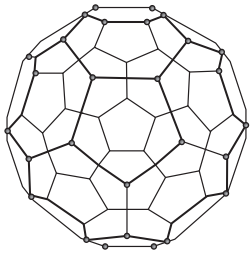
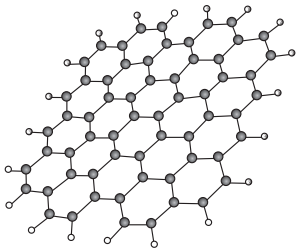


Z



	Graphene	Fullerene	Graphite	Diamond
A.	Z	Y	X	W
B.	W	Z	Y	X
C.	X	W	Z	Y
D.	Y	X	W	Z

4.3-M2 May 2024 TZ1 · Paper 1 · Q9

9. Which type of substance has the following properties?

Substance properties
high melting point
non-conductor of electricity as a liquid
insoluble in polar solvents

A. ionic

B. metallic

C. simple molecular

D. giant molecular

4.3-M3 May 2024 TZ2 · Paper 1 · Q11

11. What is the geometry around a carbon atom in graphene?

A. Hexagonal

B. Pyramidal

C. Tetrahedral

D. Trigonal planar

4.3-M4 May 2021 TZ2 · Paper 1 · Q13

13. What is the electron domain geometry of Si in SiO_2?

A. bent

B. linear

C. square planar

D. tetrahedral

4.3-M5 Nov 2019 · Paper 1 · Q11

11. What is the structure and bonding in $SiO_2(s)$?

	Structure	Bonding
A.	giant	covalent
B.	giant	ionic
C.	bent molecule	covalent
D.	linear molecule	covalent

4.3-M6 May 2018 TZ1 · Paper 1 · Q9

9. Which form of carbon is the poorest electrical conductor?

A. Graphite

B. Graphene

C. Diamond

D. Carbon nanotube

4.3-M7 May 2017 TZ1 · Paper 1 · Q9

9. A substance has the following properties:

Melting point / °C	Electrical conductivity	
	Molten	Solid
1414	poor	poor

What is the most probable structure of this substance?

A. Network covalent

B. Polar covalent molecule

C. Ionic lattice

D. Metallic lattice

4.3-M8 Nov 2016 · Paper 1 · Q13

13. Which substance has a giant covalent structure?

	Melting point / °C	Solubility in water	Electrical conductivity in the molten state
A.	186	high	none
B.	801	high	good
C.	1083	low	good
D.	1710	low	none

4.3-M9 May 2014 TZ1 · Paper 1 · Q11

11. A solid has a melting point of 1582 °C and does not dissolve in water. It does not conduct electricity in the molten state. What type of structure does the solid have?

A. Ionic

B. Metallic

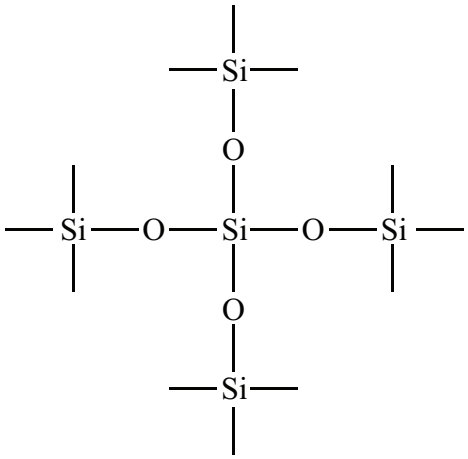
C. Giant molecular

D. Simple molecular

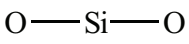
4.3-M10 May 2014 TZ2 · Paper 1 · Q13

13. Which diagram represents the bonding in SiO_2?

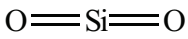
A.



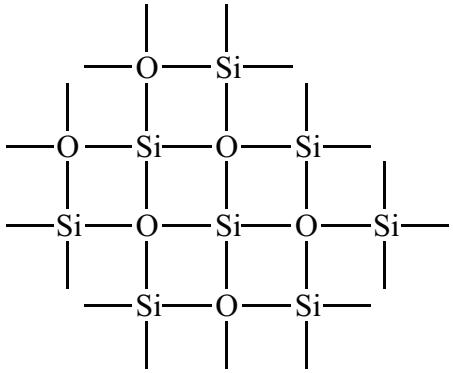
B.



C.



D.



4.3-M11 May 2013 TZ1 · Paper 1 · Q11

11. Which combination best describes the type of bonding present and the melting point of silicon and silicon dioxide?

	Silicon		Silicon dioxide	
A.	covalent bonding	high melting point	covalent bonding	high melting point
B.	metallic bonding	high melting point	covalent bonding	low melting point
C.	ionic bonding	high melting point	ionic bonding	low melting point
D.	covalent bonding	low melting point	ionic bonding	high melting point

4.3-M12 May 2013 TZ2 · Paper 1 · Q10

10. Which statements about the structure and bonding of silicon dioxide are correct?

	Structure	Bonding
A.	Silicon dioxide forms a giant covalent network.	Each oxygen atom is covalently bonded to two silicon atoms.
B.	Silicon dioxide molecules are V-shaped or bent.	Each silicon atom is covalently bonded to two oxygen atoms.
C.	Silicon dioxide molecules are linear.	A double covalent bond exists between silicon and oxygen atoms.
D.	Silicon dioxide forms a giant covalent network.	Each oxygen atom is covalently bonded to four silicon atoms.

4.3-M13 Nov 2011 · Paper 1 · Q11

11. Which row correctly describes the bonding type and melting point of carbon and carbon dioxide?

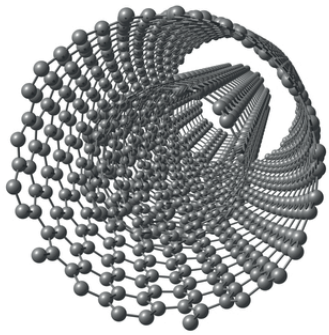
	Carbon		Carbon dioxide	
A.	covalent bonding	high melting point	covalent bonding	low melting point
B.	ionic bonding	low melting point	ionic bonding	high melting point
C.	ionic bonding	high melting point	ionic bonding	low melting point
D.	covalent bonding	low melting point	covalent bonding	high melting point

Part B - Structured questions

4.3-S1 Nov 2023 TZ1 · Paper 3 · Q3 parts (b)

3. Properties of materials are dependent upon their chemical structure.

(b) Suggest why double walled carbon nanotubes are excellent conductors of heat along the tube but poor conductors across the width of the tube. [2]



[Source: iStock.com/ollaweila.]

Good conductors along the length of the tube:

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Poor conductors across the width of the tube:

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4.3-S2 May 2023 TZ2 · Paper 3 · Q1 parts (c)

1. One definition of atomic volume is given by the formula:

$$\text{Atomic volume} = \frac{\text{atomic mass (g mol}^{-1})}{\text{density (g cm}^{-3})}$$

The table gives the atomic volumes of the first nineteen elements, in the form in which they occur at STP.

Key:

1 11 240							2 22 400
3 13.00	4 4.870	5 4.620	6 5.459 (3.419)	7 11 200	8 11 200 (7460)	9 11 200	10 22 420
11 23.70	12 13.97	13 9.993	14 12.06	15 16.99 (13.24)	16 15.49 (16.36)	17 11 080	18 22 390
19 43.93	20 ?						

(c) Suggest why some elements, such as carbon and oxygen, have more than one value for their atomic volume. [1]

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4.3-S3 May 2022 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (f)(i), (f)(ii) · also 8.1, 9.3

8. Carbon forms many compounds.

(a) C_{60} and diamond are allotropes of carbon.

(i) Outline **two** differences between the bonding of carbon atoms in C_{60} and diamond. [2]

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(ii) Explain why C_{60} and diamond sublime at different temperatures and pressures. [2]

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(f) Chlorine reacts with methane.

$$CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$$

(i) Calculate the enthalpy change of the reaction, ΔH, using section 11 of the data booklet. [3]

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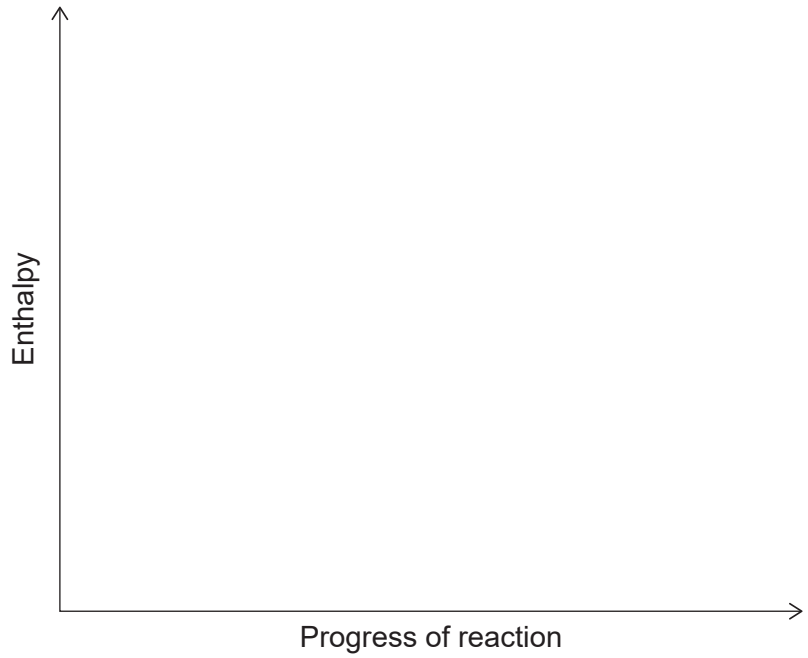
(ii) Draw and label an enthalpy level diagram for this reaction. [2]



4.3-S4 Nov 2020 · Paper 3 · Q4 parts (a)

4. There has been significant growth in the use of carbon nanotubes, CNT.

(a) Explain these properties of carbon nanotubes. [2]

Excellent strength:

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Excellent conductivity:

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4.3-S5 May 2019 TZ1 · Paper 3 · Q6 parts (a), (b)

6. Nanotechnology has allowed the manipulation of materials on the atomic level.

(a) Describe the structure and bonding of a carbon nanotube. [2]

Structure:

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Bonding:

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(b) Suggest **one** application for carbon nanotubes. [1]

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4.3-S6 May 2018 TZ2 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b), (c)

1. The table summarizes some properties of graphite and graphene.

Property	Graphite	Graphene
Delocalization (Hybridization)	Yes (sp^2)	Yes (sp^2)
Electron mobility / $cm^2\ V^{-1}\ s^{-1}$	1800	15 000–200 000
Average bond length / nm	0.142	0.142
Distance between layers / nm	0.335	Not applicable (N/A)
Tensile strength / Pascals	$4.8–76 \times 10^6$	1.3×10^{11}
Density / g cm^{-3}	1.80–2.23	(N/A)
Melting point at 1×10^6 kPa / K	4300	4510
Specific surface area / $m^2\ g^{-1}$	90	2630

[Source: © Graphenea. Used with permission]

(a) (i) Graphene is two-dimensional, rather than three-dimensional, material.

Justify this by using the structure of graphene and information from the table. [2]

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(ii) Show that graphene is over 1600 times stronger than graphite. [1]

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(b) Diamond, graphene, and graphite are all network solids.

Suggest, giving a reason, the electron mobility of diamond compared to graphene. [2]

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(c) The melting point of diamond at 1×10^6 kPa is 4200 K (in the absence of oxygen).

Suggest, based on molecular structure, why graphene has a higher melting point under these conditions. [2]

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4.3-S7 Nov 2015 · Paper 3 · Q13 parts (a)

13. Nanotechnology manipulates the properties of substances by positioning individual atoms or molecules in specific ways. Carbon nanotubes are a product of nanotechnology.

(a) Outline how the bonding in carbon nanotubes causes them to be much stronger than graphite. [2]

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4.3-S8 May 2015 TZ2 · Paper 3 · Q16 parts (c)

16. Nanotechnology has expanded in the past 30 years.

(c) Outline why bundles of carbon nanotubes have high tensile strength. [1]

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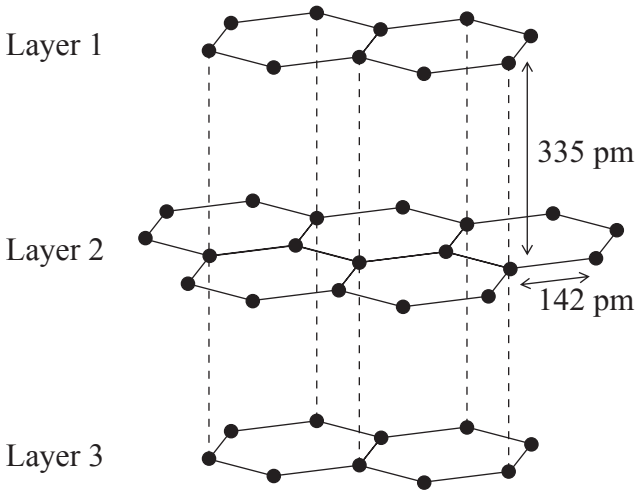
4.3-S9 Nov 2014 · Paper 2 · Q5 parts (a), (b)

Graphite has a layered structure of carbon atoms. A section of the structure is shown below.



(a) Identify the type of attraction represented by the dotted lines shown between the layers. *[1]*

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(b) Graphite is used as a lubricant. Discuss **two** other uses of graphite with reference to its layered structure. *[4]*

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4.3-S10 May 2011 TZ1 · Paper 2 · Q6 parts (a), (b)(i), (b)(ii), (c), (d), (e) · also 4.1, 4.2

6. Carbon and silicon belong to the same group of the periodic table.

(a) Describe and compare **three** features of the structure and bonding in the three allotropes of carbon: diamond, graphite and C_{60} fullerene. *[6]*

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(b) Both silicon and carbon form oxides.

(i) Describe the structure and bonding in SiO_2. *[2]*

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(ii) Explain why silicon dioxide is a solid and carbon dioxide is a gas at room temperature. *[2]*

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(c) Describe the bonding within the carbon monoxide molecule. *[2]*

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(d) Describe the delocalization of pi (π) electrons and explain how this can account for the structure and stability of the carbonate ion, CO_3^{2-}. *[3]*

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(e) Explain the meaning of the term *hybridization*. State the type of hybridization shown by the carbon atoms in carbon dioxide, diamond, graphite and the carbonate ion. *[5]*

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LESSON 5.1

Molecular Polarity

9 multiple-choice questions · 2 structured questions

Part A - Multiple choice

5.1-M1 May 2025 TZ2 · Paper 1A · Q7

7. Which molecule is polar?

A. CH_4

B. C_6H_6

C. SO_2

D. CO_2

5.1-M2 May 2024 TZ2 · Paper 1 · Q15

15. Which molecule is non-polar?

A. XeF_2

B. IF_5

C. SF_2

D. PF_3

5.1-M3 Nov 2020 · Paper 1 · Q10

10. Which molecule is most polar?

A. CHF_3

B. CF_4

C. $CClF_3$

D. CCl_4

5.1-M4 Nov 2017 · Paper 1 · Q12

12. The electronegativity values of four elements are given.

C	N	O	F
2.6	3.0	3.4	4.0

What is the order of **increasing** polarity of the **bonds** in the following compounds?

A. $CO < OF_2 < NO < CF_4$

B. $CF_4 < CO < OF_2 < NO$

C. $NO < OF_2 < CO < CF_4$

D. $CF_4 < NO < OF_2 < CO$

5.1-M5 Nov 2015 · Paper 1 · Q11

11. Which molecule is polar?

A. C_2H_6

B. CH_2Cl_2

C. CO_2

D. CCl_4

5.1-M6 May 2015 TZ2 · Paper 1 · Q7

7. What is the definition of electronegativity?

A. The relative measure of the tendency of an atom when bonded in a molecule to attract a shared pair of electrons towards itself.

B. The minimum energy required to remove a mole of electrons from a mole of gaseous atoms.

C. The enthalpy change occurring in $kJ\,mol^{-1}$ when a gaseous atom gains one electron to form a negative ion.

D. The strength of an atom measured in $kJ\,mol^{-1}$ to attract an electron to itself when bonded in a molecule.

5.1-M7 Nov 2012 · Paper 1 · Q10

10. Which molecule is polar?

	Molecule	Shape
A.	CO_2	linear
B.	SO_3	trigonal planar
C.	CCl_4	tetrahedral
D.	SO_2	bent (V-shaped)

5.1-M8 May 2012 TZ2 · Paper 1 · Q10

10. Which single covalent bond is the most polar, given the following electronegativity values?

Element	H	C	S	O
Electronegativity	2.2	2.6	2.6	3.4

A. C–O

B. S–H

C. C–H

D. O–H

5.1-M9 May 2010 TZ2 · Paper 1 · Q11

11. Which molecule is polar?

A. CH_2Cl_2

B. BCl_3

C. Cl_2

D. CCl_4

Part B - Structured questions

5.1-S1 Nov 2015 · Paper 2 · Q5 parts (a)(i)

(a) (i) Define the term electronegativity. [1]

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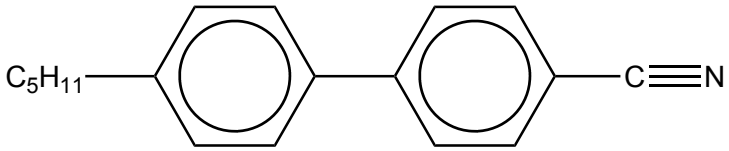
5.1-S2 May 2015 TZ1 · Paper 3 · Q10 parts (b)(i)

10. Thermotropic liquid crystals are widely used in display devices and sensors.

(b) The structure of a material used in electrical display devices is shown below.



(i) Suggest, with reference to the structure, why the molecule is able to change orientation in an electric field. [1]

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LESSON 5.2

Intermolecular Forces

39 multiple-choice questions · 26 structured questions

Part A - Multiple choice

5.2-M1 May 2025 TZ3 · Paper 1A · Q8

8. Which forces of attraction cause the boiling point of ammonia, NH_3, to be significantly higher than that of phosphine, PH_3?

A. Covalent bonds

B. Dipole–dipole forces

C. London (dispersion) forces

D. Hydrogen bonds

5.2-M2 Nov 2024 · Paper 1 · Q11

11. Which types of intermolecular force exist between CH_4, CH_3OH and CH_3Cl molecules?

	CH_4	CH_3OH	CH_3Cl
A.	London dispersion only	London dispersion, H-bonding, dipole-dipole	London dispersion, dipole-dipole
B.	London dispersion, H-bonding	London dispersion, H-bonding, dipole-dipole	London dispersion, H-bonding, dipole-dipole
C.	London dispersion only	London dispersion, dipole-dipole	London dispersion only
D.	London dispersion, H-bonding	London dispersion only	London dispersion, dipole-dipole

5.2-M3 Nov 2023 TZ1 · Paper 1 · Q11

11. Which substance exhibits only London (dispersion) forces between molecules?

A. PF_5

B. SF_4

C. SO_2

D. XeO_2

5.2-M4 May 2023 TZ2 · Paper 1 · Q10

10. What is the correct sequence if the compounds are arranged in order of increasing boiling point?

A. $CH_3OCH_3 < CH_3CH_2OH < CH_3CHO$

B. $CH_3OCH_3 < CH_3CHO < CH_3CH_2OH$

C. $CH_3CHO < CH_3CH_2OH < CH_3OCH_3$

D. $CH_3CHO < CH_3OCH_3 < CH_3CH_2OH$

5.2-M5 May 2022 TZ2 · Paper 1 · Q10

10. What is the main interaction between liquid CH_4 molecules?

A. London (dispersion) forces

B. Dipole–dipole forces

C. Hydrogen bonding

D. Covalent bonding

5.2-M6 Nov 2021 · Paper 1 · Q11

11. The following compounds have similar relative molecular masses. What is the order of increasing boiling point?

A. $CH_3CH_2CH_2OH < CH_3CH_2CHO < CH_3COOH$

B. $CH_3CH_2CHO < CH_3CH_2CH_2OH < CH_3COOH$

C. $CH_3CH_2CHO < CH_3COOH < CH_3CH_2CH_2OH$

D. $CH_3COOH < CH_3CH_2CHO < CH_3CH_2CH_2OH$

5.2-M7 May 2021 TZ1 · Paper 1 · Q32

32. Which series is in order of increasing boiling point?

A. $CH_2CH_2CH_3OH$ CH_3COCH_3 $CH_3CH_2CH_3$

B. $CH_3CH_2CH_3$ CH_3COCH_3 $CH_2CH_2CH_3OH$

C. CH_3COCH_3 $CH_2CH_2CH_3OH$ $CH_3CH_2CH_3$

D. $CH_3CH_2CH_3$ $CH_2CH_2CH_3OH$ CH_3COCH_3

5.2-M8 May 2021 TZ2 · Paper 1 · Q9

9. Which compound has the greatest volatility under the same conditions?

A. SO_2

B. SiO_2

C. SnO_2

D. SrO

5.2-M9 May 2021 TZ2 · Paper 1 · Q11

11. Which is the correct order based on **increasing** strength?

A. covalent bonds < hydrogen bonds < dipole–dipole forces < dispersion forces

B. dipole–dipole forces < dispersion forces < hydrogen bonds < covalent bonds

C. dispersion forces < dipole–dipole forces < hydrogen bonds < covalent bonds

D. dispersion forces < dipole–dipole forces < covalent bonds < hydrogen bonds

5.2-M10 May 2019 TZ1 · Paper 1 · Q9

9. What is the order of increasing boiling point?

A. $CH_3CH_2CH_2CH_3 < CH_3CH(OH)CH_3 < CH_3COCH_3 < CH_3CO_2H$

B. $CH_3CH_2CH_2CH_3 < CH_3COCH_3 < CH_3CH(OH)CH_3 < CH_3CO_2H$

C. $CH_3CO_2H < CH_3COCH_3 < CH_3CH(OH)CH_3 < CH_3CH_2CH_2CH_3$

D. $CH_3CH_2CH_2CH_3 < CH_3COCH_3 < CH_3CO_2H < CH_3CH(OH)CH_3$

5.2-M11 May 2019 TZ2 · Paper 1 · Q32

32. Which compound has the lowest boiling point?

A. $CH_3CH_2CH_2CH_2CH_2CH_3$

B. $CH_3CH_2CH_2CH_2CH_3$

C. $CH_3CH(CH_3)CH_2CH_3$

D. $CH_3C(CH_3)_2CH_3$

5.2-M12 Nov 2018 · Paper 1 · Q11

11. Which compound has the highest boiling point?

A. CH_3CHO

B. CH_3CH_2F

C. CH_3OCH_3

D. $CH_3CH_2NH_2$

5.2-M13 Nov 2017 · Paper 1 · Q11

11. Which of the following series shows increasing hydrogen bonding with water?

A. Propane < propanal < propanol < propanoic acid

B. Propane < propanol < propanal < propanoic acid

C. Propanal < propane < propanoic acid < propanol

D. Propanoic acid < propanol < propanal < propane

5.2-M14 Nov 2016 · Paper 1 · Q7

7. Which property increases down group 17, the halogens?

A. Electron affinity

B. Boiling point

C. First ionization energy

D. Reactivity

5.2-M15 Nov 2016 · Paper 1 · Q12

12. Between which pair of molecules can hydrogen bonding occur?

A. CH_4 and H_2O

B. CH_3OCH_3 and CF_4

C. CH_4 and HF

D. CH_3OH and H_2O

5.2-M16 May 2016 · Paper 1 · Q13

13. Which of the following are van der Waals' forces?

I. Dipole-dipole forces
II. Hydrogen bonds
III. London (dispersion) forces

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

5.2-M17 Nov 2015 · Paper 1 · Q35

35. Which compound is most soluble in water?

A. CH_3CH_2CHO

B. $CH_3CH_2CH_2CHO$

C. $CH_3CH_2CO_2H$

D. $CH_3CH_2CH_2CO_2H$

5.2-M18 May 2015 TZ1 · Paper 1 · Q10

10. The following compounds have similar molar masses:

$$CH_3CH_2COOH,\ CH_3CH_2CH_2CH_2OH \text{ and } CH_3CH_2CH_2CH_2CH_3$$

What is the order of **increasing** boiling points?

A. $CH_3CH_2CH_2CH_2OH < CH_3CH_2COOH < CH_3CH_2CH_2CH_2CH_3$

B. $CH_3CH_2COOH < CH_3CH_2CH_2CH_2CH_3 < CH_3CH_2CH_2CH_2OH$

C. $CH_3CH_2COOH < CH_3CH_2CH_2CH_2OH < CH_3CH_2CH_2CH_2CH_3$

D. $CH_3CH_2CH_2CH_2CH_3 < CH_3CH_2CH_2CH_2OH < CH_3CH_2COOH$

5.2-M19 May 2015 TZ1 · Paper 1 · Q11

11. Which substance has the following properties?

- Low melting point
- Very soluble in water
- Does not conduct electricity when molten

A. Glucose, $C_6H_{12}O_6$

B. Silicon dioxide, SiO_2

C. Sodium chloride, NaCl

D. Tetrachloromethane, CCl_4

5.2-M20 May 2015 TZ2 · Paper 1 · Q11

11. Which correctly lists butane ($M_r = 58$), propanone ($M_r = 58$), propan-1-ol ($M_r = 60$) and propan-2-ol ($M_r = 60$) in order of **increasing** boiling point?

A. $C_4H_{10} < CH_3COCH_3 < CH_3CH(OH)CH_3 < CH_3CH_2CH_2OH$

B. $CH_3CH_2CH_2OH < CH_3CH(OH)CH_3 < CH_3COCH_3 < C_4H_{10}$

C. $C_4H_{10} < CH_3CH(OH)CH_3 < CH_3CH_2CH_2OH < CH_3COCH_3$

D. $C_4H_{10} < CH_3COCH_3 < CH_3CH_2CH_2OH < CH_3CH(OH)CH_3$

5.2-M21 Nov 2014 · Paper 1 · Q11

11. Which process involves the breaking of hydrogen bonds?

A. $2HI(g) \rightarrow H_2(g) + I_2(g)$

B. $CH_4(g) \rightarrow C(g) + 4H(g)$

C. $H_2(l) \rightarrow H_2(g)$

D. $NH_3(l) \rightarrow NH_3(g)$

5.2-M22 May 2014 TZ2 · Paper 1 · Q24

24. Which combination of properties is correct?

	Enthalpy of vaporization	Boiling point	Intermolecular forces	Volatility
A.	large	high	strong	low
B.	large	low	weak	high
C.	small	low	weak	low
D.	small	high	weak	low

5.2-M23 Nov 2013 · Paper 1 · Q11

11. Which intermolecular forces exist between the following molecules?

	H_2Se	CO	H_2
A.	van der Waals' and dipole-dipole	van der Waals' and dipole-dipole	van der Waals' only
B.	van der Waals', dipole-dipole and hydrogen bonding	van der Waals' only	van der Waals' and hydrogen bonding
C.	van der Waals', dipole-dipole and hydrogen bonding	van der Waals' and dipole-dipole	van der Waals' only
D.	van der Waals' and dipole-dipole	van der Waals' and dipole-dipole	van der Waals' and hydrogen bonding

5.2-M24 Nov 2013 · Paper 1 · Q24

24. Which compound has the **highest** molar enthalpy of vaporization?

A. Ethane

B. Ethanoic acid

C. Propane

D. Propanoic acid

5.2-M25 May 2013 TZ2 · Paper 1 · Q11

11. Which series shows **increasing** boiling points?

A. $CH_3CH_2CH_3 < CH_3CH_2OH < CH_3CHO$

B. $CH_3CHO < CH_3CH_2CH_3 < CH_3CH_2OH$

C. $CH_3CH_2OH < CH_3CHO < CH_3CH_2CH_3$

D. $CH_3CH_2CH_3 < CH_3CHO < CH_3CH_2OH$

5.2-M26 May 2013 TZ2 · Paper 1 · Q25

25. Which compound has the highest enthalpy of vaporization?

A. CO_2

B. NH_3

C. H_2S

D. H_2O

5.2-M27 Nov 2012 · Paper 1 · Q11

11. Which intermolecular forces are present in HI (l)?

I. Hydrogen bonding

II. Dipole-dipole forces

III. Van der Waals' (London dispersion) forces

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

5.2-M28 Nov 2012 · Paper 1 · Q34

34. Which compound has the lowest boiling point?

A. $(CH_3)_3COH$

B. $CH_3(CH_2)_3OH$

C. $(CH_3)_4C$

D. $CH_3(CH_2)_3CH_3$

5.2-M29 May 2012 TZ1 · Paper 1 · Q10

10. Which statement about intermolecular forces is correct?

A. The intermolecular force between H_2 molecules is hydrogen bonding, because H_2 has temporary dipoles.

B. The intermolecular forces between PH_3 molecules are greater than the intermolecular forces between NH_3 molecules, because they have a greater mass.

C. The intermolecular force between H_2 molecules is hydrogen bonding, because H_2 has permanent dipoles.

D. The intermolecular forces between Br_2 molecules are van der Waals', because Br_2 has temporary dipoles.

5.2-M30 May 2012 TZ1 · Paper 1 · Q11

11. Which substances are soluble in hexane, C_6H_{14}?

I. C_8H_{18}

II. CH_4

III. H_2O

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

5.2-M31 May 2012 TZ2 · Paper 1 · Q12

12. Which types of intermolecular forces exist in HBr, Cl_2 and CH_3F?

	HBr	**Cl_2**	**CH_3F**
A.	van der Waals' and dipole-dipole	van der Waals' only	van der Waals' and dipole-dipole
B.	van der Waals' and dipole-dipole	van der Waals' only	van der Waals', dipole-dipole and hydrogen bonding
C.	van der Waals' only	van der Waals' only	van der Waals', dipole-dipole and hydrogen bonding
D.	van der Waals' and dipole-dipole	van der Waals' and dipole-dipole	van der Waals', dipole-dipole and hydrogen bonding

5.2-M32 May 2012 TZ2 · Paper 1 · Q34

34. Which compound has the **lowest** boiling point?

A. $CH_3CH_2CH_2OH$

B. $CH_3CH_2CH_2Br$

C. CH_3CH_2COOH

D. $CH_3CH_2CH_2CH_3$

5.2-M33 Nov 2011 · Paper 1 · Q12

12. What is the correct order of **increasing** boiling points?

A. $CH_3CH_3 < CH_3CH_2Cl < CH_3CH_2Br < CH_3CH_2I$

B. $CH_3CH_2Cl < CH_3CH_2Br < CH_3CH_3 < CH_3CH_2I$

C. $CH_3CH_2I < CH_3CH_2Br < CH_3CH_2Cl < CH_3CH_3$

D. $CH_3CH_2Br < CH_3CH_2Cl < CH_3CH_2I < CH_3CH_3$

5.2-M34 May 2011 TZ1 · Paper 1 · Q12

12. Which compound forms hydrogen bonds in the liquid state?

A. C_2H_5OH

B. $CHCl_3$

C. CH_3CHO

D. $(CH_3CH_2)_3N$

5.2-M35 May 2011 TZ1 · Paper 1 · Q25

25. Which is the correct relationship between enthalpy of vaporization, intermolecular forces and boiling point?

	Enthalpy of vaporization	Intermolecular forces	Boiling point
A.	small	weak	high
B.	small	strong	low
C.	large	weak	high
D.	large	strong	high

5.2-M36 May 2011 TZ2 · Paper 1 · Q24

24. Which combination of intermolecular forces, boiling point and enthalpy of vaporization is correct?

	Intermolecular forces	Boiling point	Enthalpy of vaporization
A.	strong	low	low
B.	strong	high	low
C.	weak	low	high
D.	weak	low	low

5.2-M37 May 2010 TZ1 · Paper 1 · Q11

11. Which pair of compounds is arranged in correct order of relative boiling points?

	Lower Boiling Point	Higher Boiling Point
A.	CH_3OCH_3	CH_3CH_2OH
B.	CH_3CHO	$CH_3CH_2CH_3$
C.	CH_3CH_2OH	CH_3CHO
D.	CH_3COOH	CH_3CH_2OH

5.2-M38 May 2010 TZ1 · Paper 1 · Q12

12. Which intermolecular forces exist between molecules of carbon monoxide, CO?

A. Hydrogen bonds and van der Waals' forces

B. Dipole-dipole attractions and van der Waals' forces

C. Van der Waals' forces only

D. Dipole-dipole attractions only

5.2-M39 May 2010 TZ2 · Paper 1 · Q10

10. Which substance can form intermolecular hydrogen bonds in the liquid state?

A. CH_3OCH_3

B. CH_3CH_2OH

C. CH_3CHO

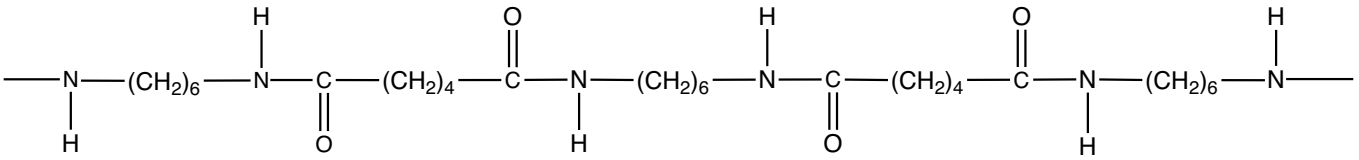
D. $CH_3CH_2CH_3$

Part B - Structured questions

5.2-S1 May 2025 TZ3 · Paper 2 · Q7 parts (b)(i), (b)(ii), (c) · also 4.3

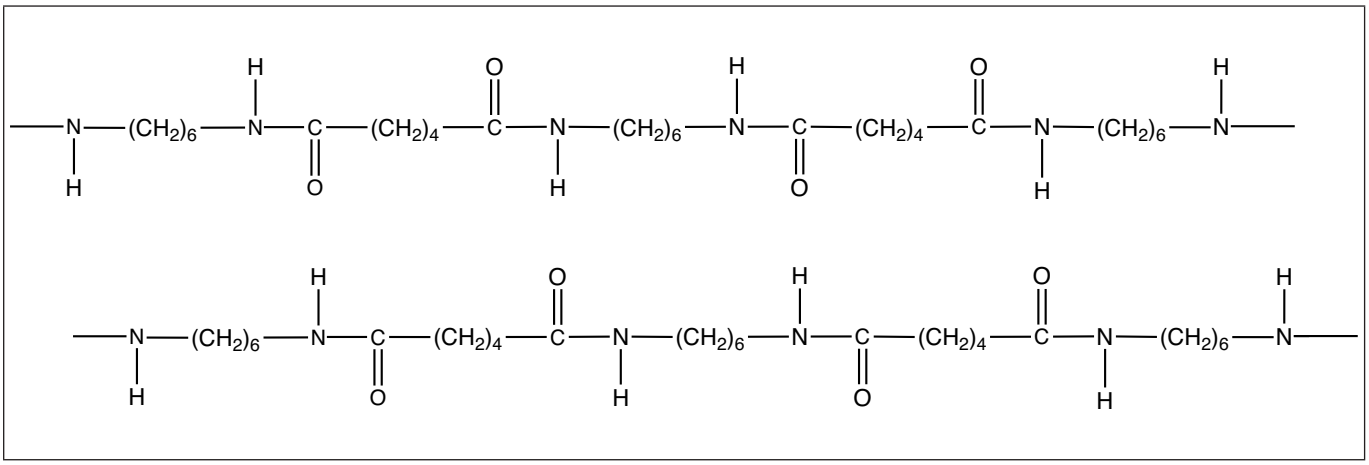
7. Nylon 6,6 is a polyamide used in many things, such as fishing lines.

A section of the polymer chain of nylon 6,6



(b) (i) The strength of nylon 6,6 is due to the hydrogen bonding between the polymer chains.

Annotate the diagram to show a hydrogen bond between the sections of the polymer chains. Use a dotted line to represent the hydrogen bond. [1]



(ii) Outline how a hydrogen bond is formed. [2]

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(c) Discuss the following data in terms of the bonding **and/or** intermolecular forces in each substance. [3]

Substance	**Melting point / K**
Silicon	1683
Poly(ethene)	393

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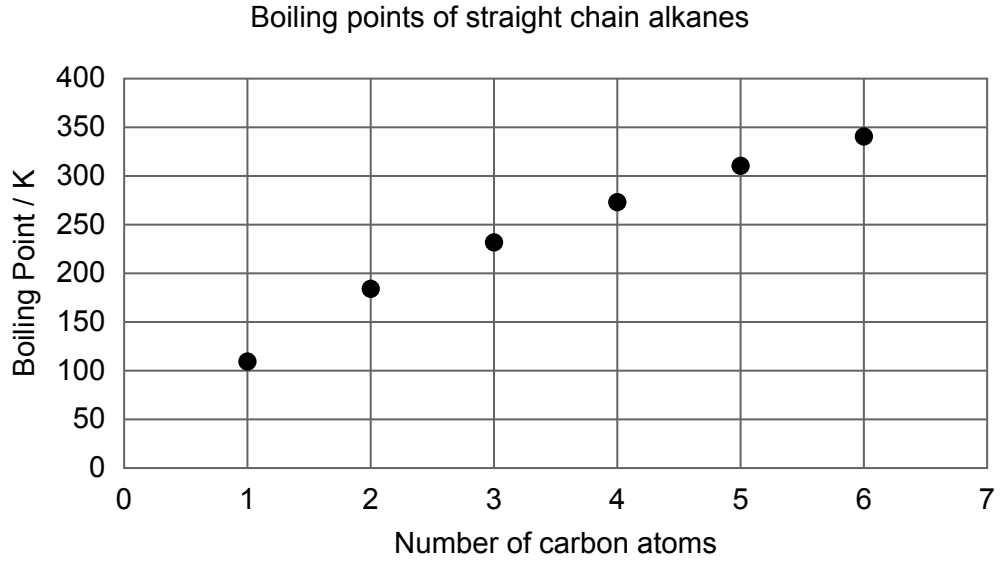
5.2-S2 Nov 2024 · Paper 2 · Q1 parts (a)(i), (a)(ii) · also 8.1

1. Alkanes are commonly occurring organic compounds.

(a) The first four straight chain alkanes are gases at room temperature.



(i) Explain why the boiling point increases from methane to propane. [2]

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(ii) Explain why the volume occupied by a sample of propane increases sharply when the sample is heated up from 200 to 250 K at constant pressure. [2]

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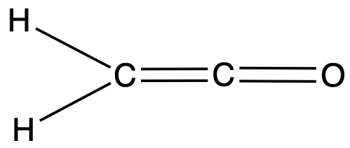
5.2-S3 May 2024 TZ2 · Paper 2 · Q5 parts (b), (c)(i), (c)(ii) · also 8.1, 9.2

5. Ethenone, CH_2CO, is used in the synthesis of pharmaceutical compounds.



(b) Compare and contrast the intermolecular forces that result in ethenone being less volatile than carbon dioxide. [2]

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(c) Ethenone can be made by the thermal decomposition of propanone.

$$CH_3COCH_3\ (l) \rightarrow CH_2CO\ (g) + CH_4\ (g)$$

(i) Calculate the standard enthalpy change for this reaction.
Use $\Delta H^{\ominus}_f$ ethenone $= -87.2\,kJ\ mol^{-1}$ and section 12 of the data booklet. [2]

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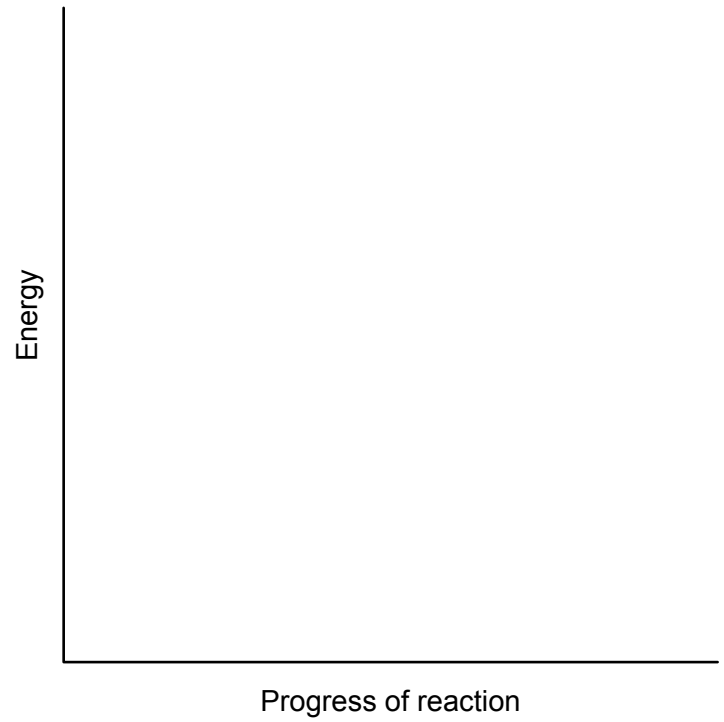
(ii) Sketch the potential energy diagram for the thermal decomposition of propanone from (c)(i). Use the axis given and indicate both the enthalpy of reaction and the activation energy. [2]



5.2-S4 Nov 2023 TZ1 · Paper 2 · Q1 parts (c)(i), (c)(ii), (d)(i), (d)(iv) · also 1.2, 4.1

1. Methanoic acid (HCOOH) is the first member of the homologous series of carboxylic acids.

(c) Methanoic acid and ethanal (CH_3CHO) both contain a carbonyl group and have similar molar masses.

(i) Explain why, in terms of the strongest intermolecular forces between the molecules, ethanal has a much lower boiling point than methanoic acid. [2]

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(ii) Outline why ethanal and methanoic acid are both fully miscible with water. [1]

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(d) Methanoic acid acts as a weak monobasic acid in aqueous solution.

(i) $2.00\,dm^3$ of a solution of methanoic acid was prepared, and $25.0\,cm^3$ of this was found to require exactly $20.7\,cm^3$ of $0.100\,mol\,dm^{-3}$ aqueous sodium hydroxide to completely convert it to sodium methanoate, HCOONa. Calculate the mass of methanoic acid used to make the solution. [2]

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(iv) Explain why the two carbon–oxygen bonds in the methanoate ion are of equal length, and compare their length to the carbon–oxygen bonds in methanoic acid. [2]

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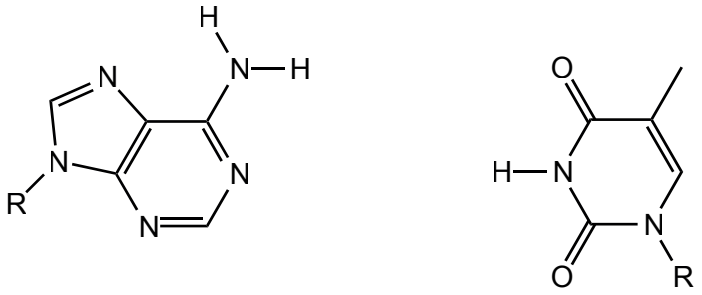
5.2-S5 May 2023 TZ1 · Paper 3 · Q9 parts (b)

9. DNA is an essential biological molecule.

(b) Draw the hydrogen bonds between adenine and thymine. [1]

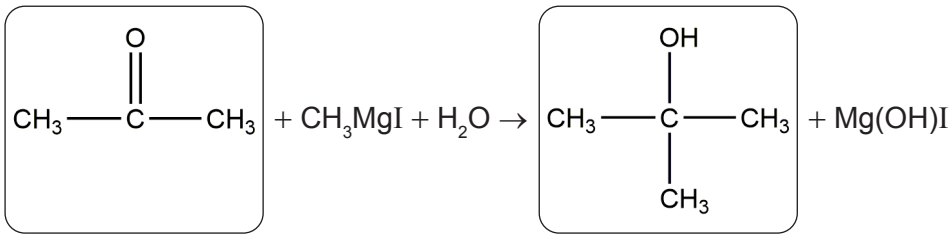
Adenine **Thymine**



5.2-S6 May 2022 TZ1 · Paper 2 · Q5 parts (a)(ii), (a)(iii), (a)(iv) · also 4.1, 4.2

5. Organomagnesium compounds can react with carbonyl compounds. One overall equation is:

Compound A **Compound B**



(ii) Compound A and Compound B are both liquids at room temperature and pressure. Identify the strongest intermolecular force between molecules of Compound A. [1]

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(iii) State the number of σ (sigma) and π (pi) bonds in Compound A. [1]

σ: . . . . . π: . . . . . .

(iv) Deduce the hybridization of the central carbon atom in Compound A. [1]

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5.2-S7 Nov 2019 · Paper 3 · Q15 parts (c)

(c) Explain why maltose, $C_{12}H_{22}O_{11}$, is soluble in water. [2]

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5.2-S8 May 2019 TZ1 · Paper 3 · Q10 parts (a)

10. The main fatty acid composition of cocoa butter and coconut oil is detailed below.

Source	**Saturated fatty acids / %**						**Mono-unsaturated / %**	**Poly-unsaturated / %**
	Octanoic ($C_8H_{16}O_2$)	Lauric ($C_{12}H_{24}O_2$)	Myristic ($C_{14}H_{28}O_2$)	Palmitic ($C_{16}H_{32}O_2$)	Stearic ($C_{18}H_{36}O_2$)	Others	Oleic ($C_{18}H_{34}O_2$)	Total
Cocoa butter	0.0	0.0	0.1	26.9	35.2	0.0	34.6	3.3
Coconut oil	7.5	46.2	18.4	9.4	2.8	6.8	7.0	1.9

[Source: U.S. Department of Agriculture, Agricultural Research Service. FoodData Central, 2019. fdc.nal.usda.gov.]

(a) The melting points of cocoa butter and coconut oil are 34 °C and 25 °C respectively.

Explain this in terms of their saturated fatty acid composition. [3]

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5.2-S9 May 2019 TZ2 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (c)(ii), (c)(iii), (d)(ii), (d)(iii) · also 3.1, 4.1, 6.1, 7.3, 8.1, 9.3

1. Ethyne, C_2H_2, reacts with oxygen in welding torches.

(a) Write an equation for the complete combustion of ethyne. [1]

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(b) (i) Deduce the Lewis (electron dot) structure of ethyne. [1]

(ii) Compare, giving a reason, the length of the bond between the carbon atoms in ethyne with that in ethane, C_2H_6. [1]

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(iii) Identify the type of interaction that must be overcome when liquid ethyne vaporizes. [1]

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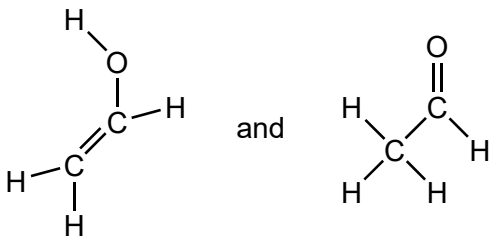
(c) Ethyne reacts with steam.

$$C_2H_2(g) + H_2O(g) \rightarrow C_2H_4O(g)$$

Two possible products are:



A **B**

(ii) Determine the enthalpy change for the reaction, in kJ, to produce **A** using section 11 of the data booklet. [3]

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(iii) The enthalpy change for the reaction to produce **B** is −213 kJ.

Predict, giving a reason, which product is the most stable. [1]

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(d) Product **B**, CH_3CHO, can also be synthesized from ethanol.

(ii) Deduce the average oxidation state of carbon in product **B**. [1]

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(iii) Explain why product **B** is water soluble. [3]

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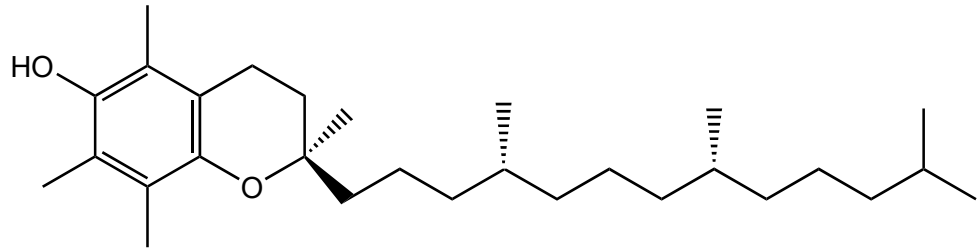
5.2-S10 May 2019 TZ2 · Paper 3 · Q11 parts (c)

11. Phosphatidylcholine is an example of a phospholipid found in lecithin.

(c) Lecithin aids the body's absorption of vitamin E.



The α-tocopherol form of vitamin E.

Suggest why vitamin E is fat-soluble. [1]

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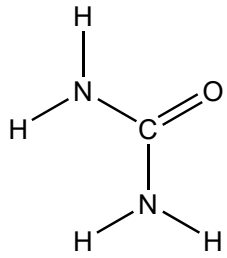
5.2-S11 May 2018 TZ1 · Paper 2 · Q1 parts (b), (c), (e)(i), (e)(ii), (f), (i) · also 1.2, 3.1, 4.1, 7.3

1. Urea, $(H_2N)_2CO$, is excreted by mammals and can be used as a fertilizer.

(b) The structural formula of urea is shown.



Predict the electron domain and molecular geometries at the nitrogen and carbon atoms, applying the VSEPR theory.

	Electron domain geometry	Molecular geometry
Nitrogen	. . . . .	. . . . .
Carbon	. . . . .	trigonal planar

[3]

(c) Urea can be made by reacting potassium cyanate, KNCO, with ammonium chloride, NH_4Cl.

$$KNCO\,(aq) + NH_4Cl\,(aq) \rightarrow (H_2N)_2CO\,(aq) + KCl\,(aq)$$

Determine the maximum mass of urea that could be formed from $50.0\,cm^3$ of $0.100\,mol\,dm^{-3}$ potassium cyanate solution. [2]

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(e) (i) Suggest one reason why urea is a solid and ammonia a gas at room temperature. [1]

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(ii) Sketch two different hydrogen bonding interactions between ammonia and water. [2]

(f) The combustion of urea produces water, carbon dioxide and nitrogen.

Formulate a balanced equation for the reaction. [2]

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(i) The C–N bonds in urea are shorter than might be expected for a single C–N bond. Suggest, in terms of electrons, how this could occur. [1]

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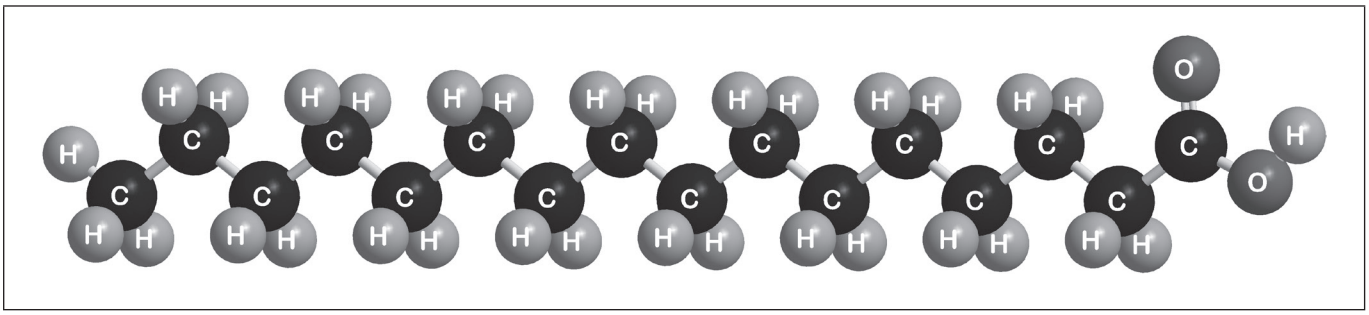
5.2-S12 May 2018 TZ1 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b)(ii) · also 1.2

1. Palmitic acid has a molar mass of 256.5 g mol^{-1}.



(a) (i) Part of this molecule is hydrophilic (bonds readily to water) and part hydrophobic (does not bond readily to water). Draw a circle around all of the hydrophilic part of the molecule. [1]

(ii) When a small amount of palmitic acid is placed in water it disperses to form a layer on the surface that is only one molecule thick. Explain, in terms of intermolecular forces, why this occurs. [2]

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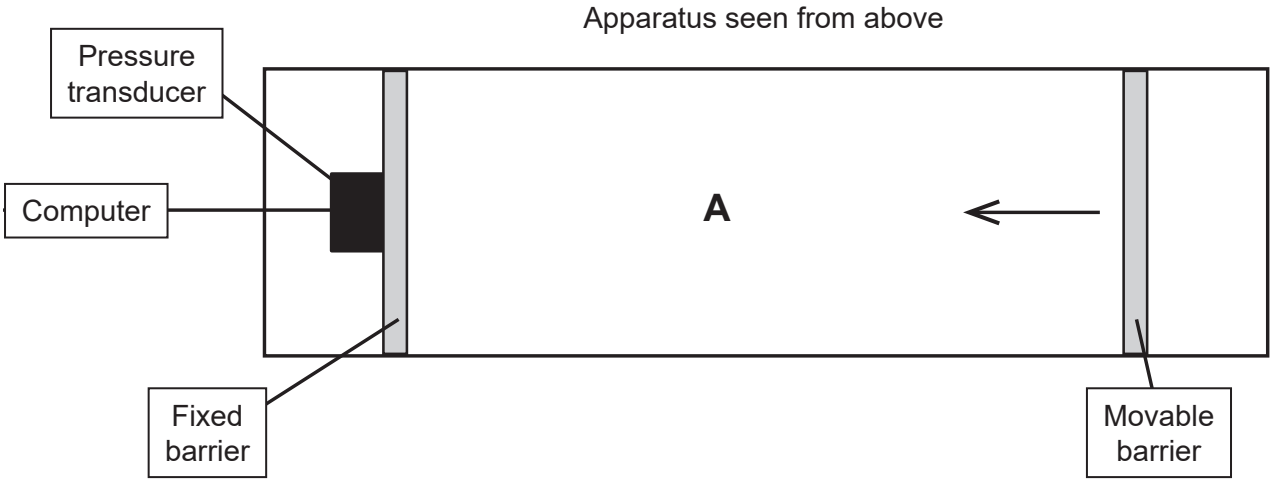
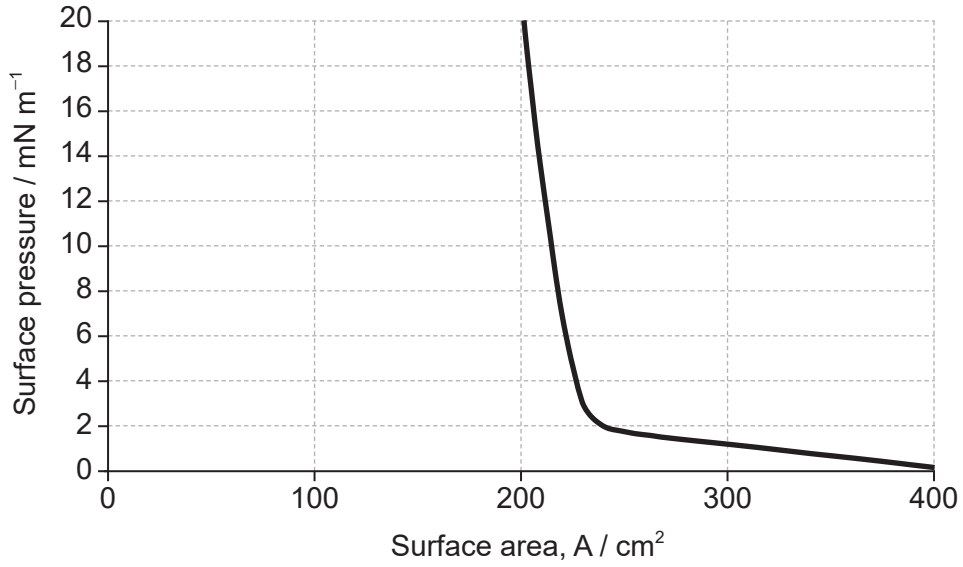
(b) The apparatus in the diagram measures the surface pressure created by palmitic acid molecules on the surface of water. This pressure is caused by palmitic acid molecules colliding with the fixed barrier. The pressure increases as the area, **A**, available to the palmitic acid is reduced by the movable barrier.



[Source: Physical Chemistry Chemical Physics, 2001, 3, 4774-4783 - Reproduced by permission of The Royal Society of Chemistry]

When a drop of a solution of palmitic acid in a volatile solvent is placed between the barriers, the solvent evaporates leaving a surface layer. The graph of pressure against area was obtained as the area **A** was reduced.



[Source: Influence of Lecithin on Structure and Stability of Parenteral Fat Emulsions, Christoph Wabel, 1998, Figure 34. Used with permission.]

(ii) The solution of palmitic acid had a concentration of $0.0034\,\text{mol}\,\text{dm}^{-3}$. Calculate the number of molecules of palmitic acid present in the $0.050\,\text{cm}^3$ drop, using section 2 of the data booklet. [2]

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5.2-S13 May 2017 TZ1 · Paper 3 · Q12 parts (a)(i)

12. Lipids and carbohydrates contain the same elements but have different properties.

The drain pipe of a kitchen sink can become clogged by fatty acids, such as linoleic acid, $C_{18}H_{32}O_2$, but not by the trisaccharide, raffinose, $C_{18}H_{32}O_{16}$, containing the same number of carbon atoms.

(a) (i) Explain why raffinose is far more water soluble than linoleic acid. [2]

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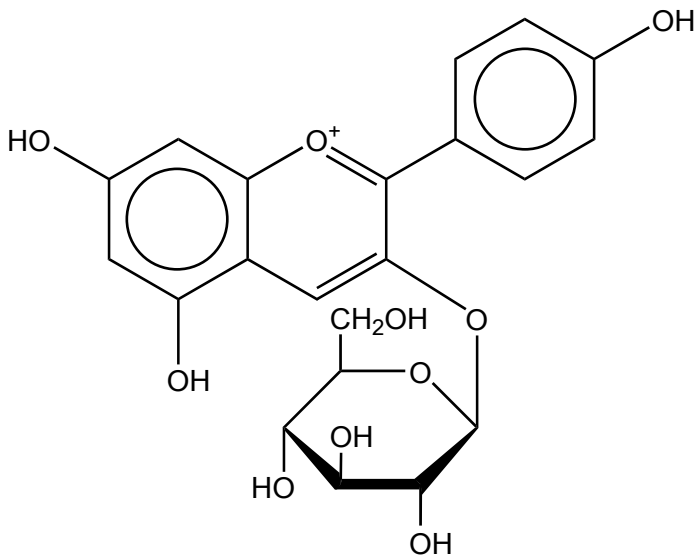
5.2-S14 Nov 2016 · Paper 3 · Q16 parts (b)

16. Anthocyanins are pigments that give colour to many flowers and fruits. The red colour of ripe strawberries is mainly due to the anthocyanin pigment whose structure is shown below.



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(b) With reference to its chemical structure, outline whether this pigment is found in aqueous solution in the cells or in the lipid-based membranes. [1]

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5.2-S15 Nov 2015 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii) · also 3.1, 4.1, 4.2

Iron (III) oxide is the main source of iron but the decomposition of $Fe_2O_3(s)$ into its elements is extremely difficult due to a large positive value of $\Delta G^{\ominus}$.

(c) (i) Deduce the type of hybridization shown by the nitrogen atoms in NF_4^+, N_2H_2 and N_2H_4. [3]

	NF_4^+	N_2H_2	N_2H_4
Hybridization	. . . . .	. . . . .	. . . . .

(ii) Describe how sigma (σ) and pi (π) bonds form. [2]

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(iii) Draw the Lewis (electron dot) structures of SF_4 and SF_6. Use the valence shell electron pair repulsion (VSEPR) theory to predict the name of the shape of each molecule. [4]

(d) (i) List the following compounds in order of **increasing** boiling point:
CH_3CHO, $CH_3CH_2CH_3$, CH_3COOH, CH_3CH_2OH. [2]

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(ii) Explain the order of boiling points in the compounds listed in part (d) (i), in terms of intermolecular forces. [4]

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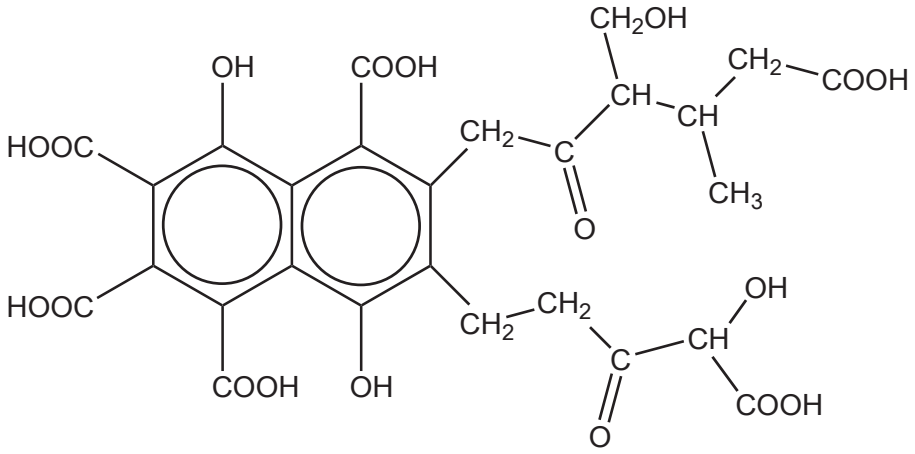
5.2-S16 May 2015 TZ1 · Paper 3 · Q19 parts (b)(i)

19. Intensive farming changes the composition of soils and may lead to soil degradation.

(b) Fulvic acid is an important constituent of soil organic matter (SOM). A representation of its structure is shown below.



(i) Outline how this structure of fulvic acid is able to retain water in soil. [2]

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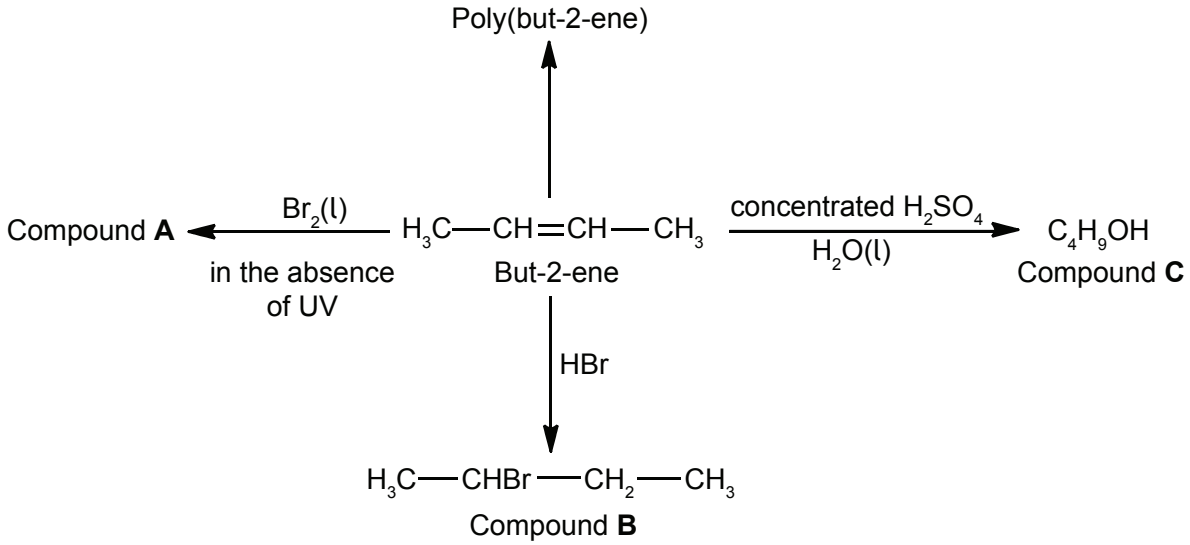
5.2-S17 May 2015 TZ2 · Paper 2 · Q10 parts (f), (g) · also 7.3

10. Some reactions of but-2-ene are given below.



(f) Explain why but-2-ene is more volatile than compound **C**. [2]

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(g) Deduce the equation for the complete combustion of compound **C**. [1]

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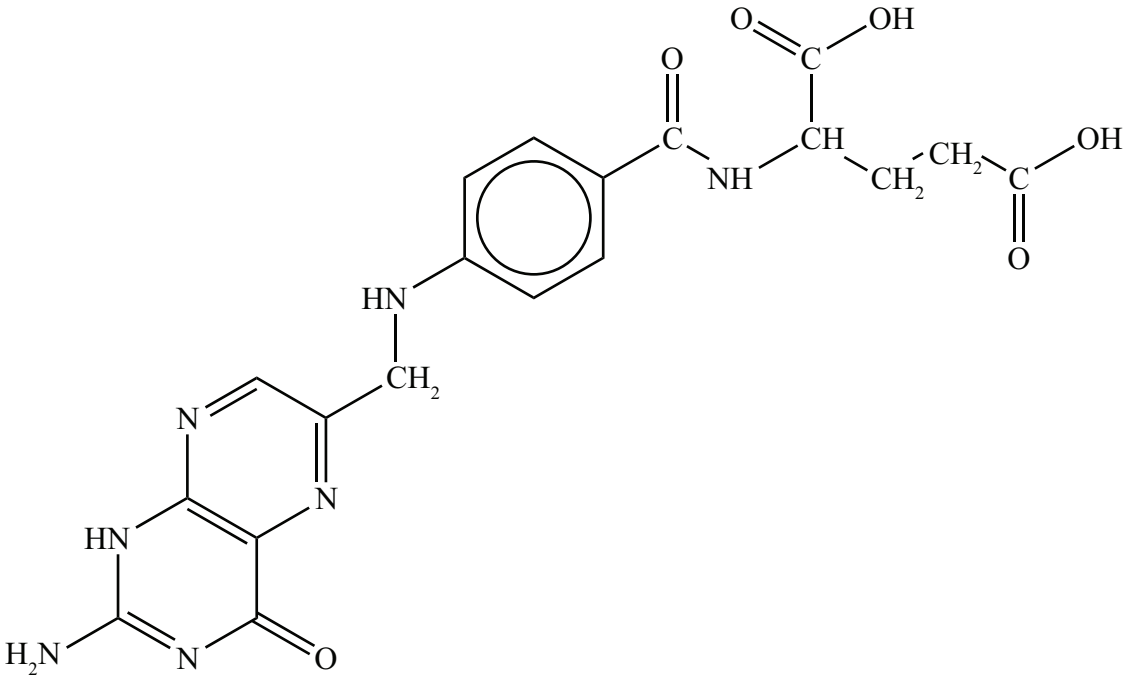
5.2-S18 May 2014 TZ1 · Paper 3 · Q6 parts (a)

6. Vitamins are organic compounds needed in small amounts for normal metabolism in the body. Vitamins can be classified as water-soluble or fat-soluble.

(a) Vitamin B_9 is water-soluble and is important in the repair of DNA. The structure of vitamin B_9 is given below.



Suggest why this vitamin is water-soluble. *[1]*

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5.2-S19 Nov 2013 · Paper 2 · Q8 parts (d)

8. 2-methylbutan-2-ol, $(CH_3)_2C(OH)CH_2CH_3$, is a liquid with a smell of camphor that was formerly used as a sedative. One way of producing it starts with 2-methylbut-2-ene.

(d) Explain why 2-methylbut-2-ene is less soluble in water than 2-methylbutan-2-ol. *[2]*

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5.2-S20 May 2013 TZ2 · Paper 2 · Q4 parts (b), (c)(i), (c)(ii)

(b) Identify the intermolecular forces present in hydrogen iodide in the liquid state, HI (l). *[1]*

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(c) Consider the compounds $(CH_3)_2NH$ and CH_4.

(i) State and explain which compound can form hydrogen bonds **with water**. *[2]*

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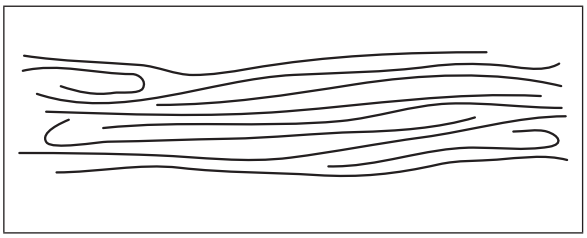
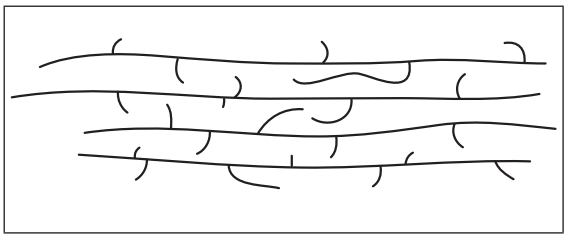
(ii) Draw a diagram showing the resulting hydrogen bonds between water and the compound chosen in (i). *[1]*

5.2-S21 Nov 2012 · Paper 3 · QC3 parts (a)(i)

C3. The two diagrams below show the arrangement of molecules in two different types of polyethene, labelled **A** and **B**.



A



B

(a) Compare the strength of the intermolecular forces, density and flexibility of polymers **A** and **B**.

(i) Intermolecular forces: *[1]*

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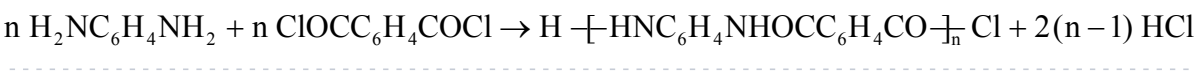
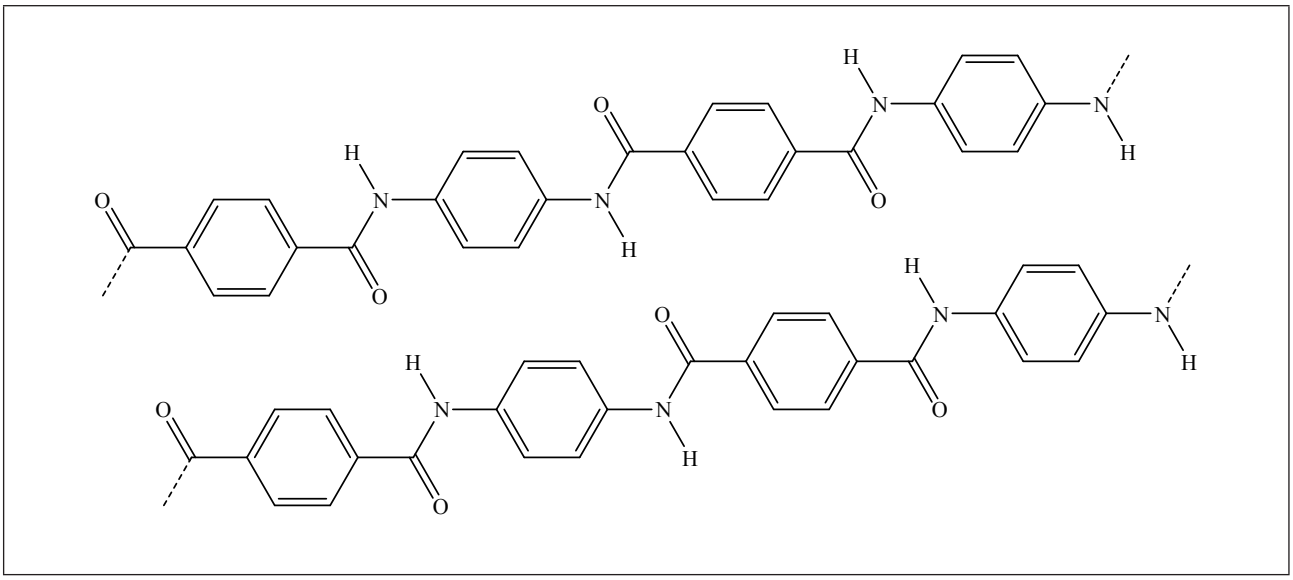
5.2-S22 Nov 2012 · Paper 3 · QC4 parts (b)(i), (b)(ii)

C4. Kevlar is a polyamide polymer which is made in a very similar way to nylon, by reacting two monomer species together.



(b) The following diagram shows two adjacent molecules in a sample of solid Kevlar.



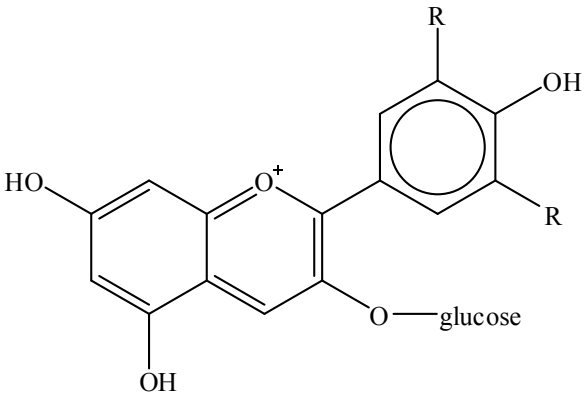
(i) Identify the strongest type of intermolecular force between the two molecules. *[1]*

. . . . . .

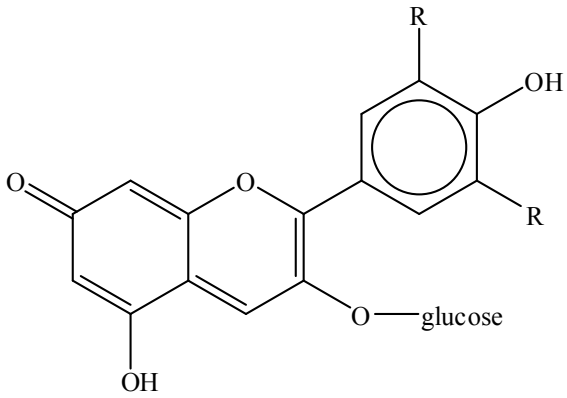
(ii) Annotate the diagram (above) by adding dotted lines to show the strongest intermolecular forces. *[1]*

5.2-S23 Nov 2012 · Paper 3 · QF3 parts (a)

F3. Anthocyanins, the pigments which occur naturally in many flowers and fruits, are water soluble and often change colour as the temperature or pH changes. The diagrams show two structures of the same anthocyanin under different conditions.



A



B

(a) Explain why anthocyanins tend to be soluble in water. *[2]*

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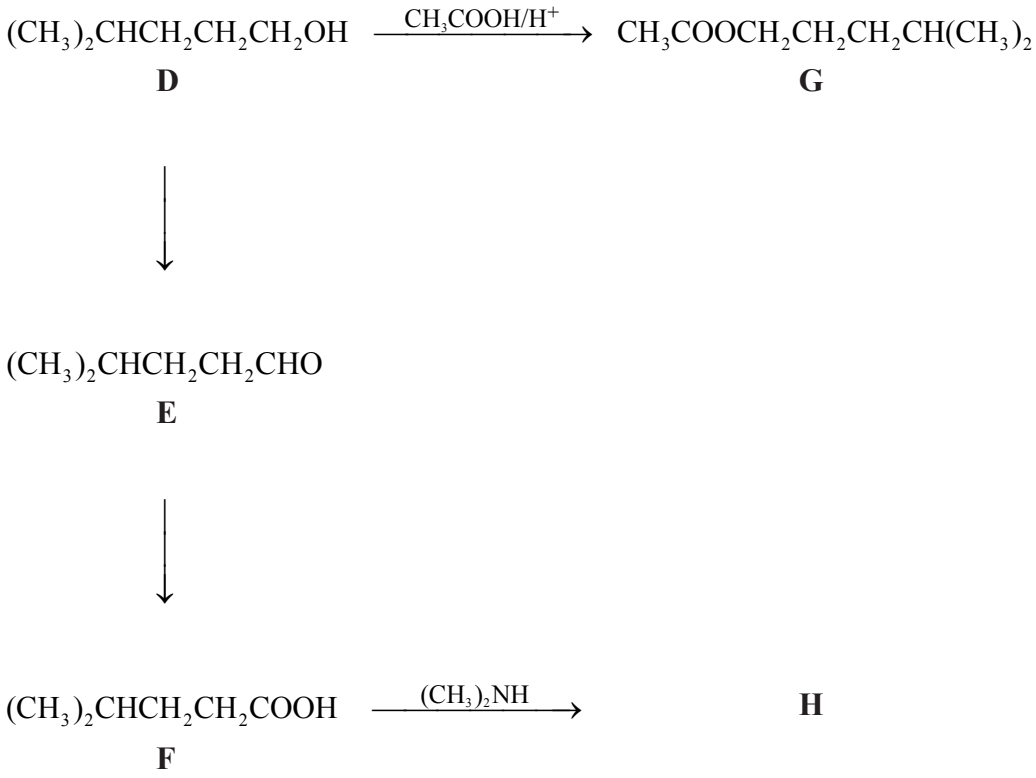
5.2-S24 Nov 2011 · Paper 2 · Q9 parts (b)(iv)

(b) Consider the following reactions.



(iv) Discuss the volatility of **E** compared to **F**. *[2]*

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5.2-S25 May 2011 TZ2 · Paper 2 · Q4

4. Methoxymethane, CH_3OCH_3, and ethanol, C_2H_5OH, have the same relative molecular mass. Explain why methoxymethane has a much lower boiling point than ethanol. *[3]*

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5.2-S26 May 2010 TZ2 · Paper 2 · Q3 parts (b), (c) · also 4.3

3. Consider the bonding and structure of the period 3 elements.

(b) Explain why sulfur, S_8, has a higher melting point than phosphorus, P_4. *[2]*

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(c) Explain why silicon has the highest melting point and argon has the lowest melting point. *[2]*

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LESSON 5.3

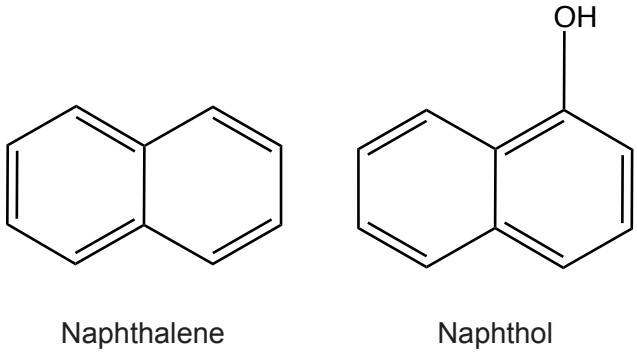
Separation Techniques

2 multiple-choice questions · 20 structured questions

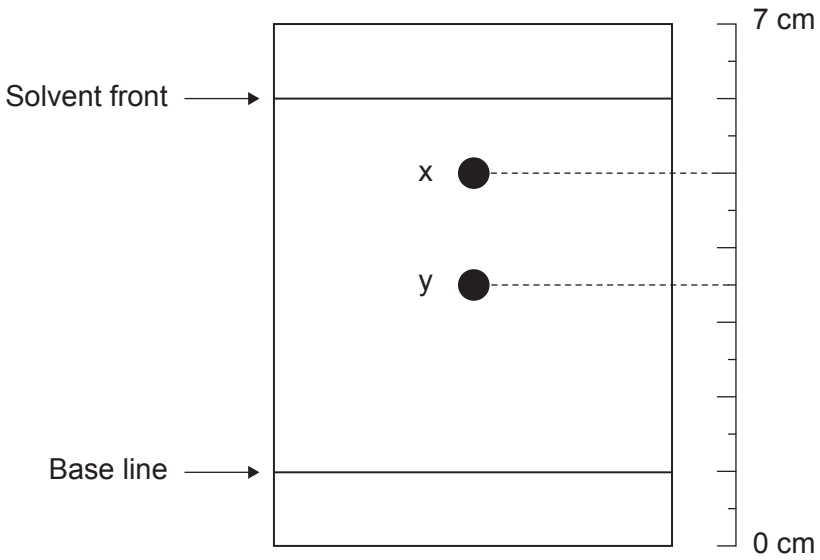
Part A - Multiple choice

5.3-M1 May 2025 TZ1 · Paper 1A · Q12

12. The following thin-layer chromatogram was obtained from a mixture of naphthalene, $C_{10}H_8$, and naphthol, $C_{10}H_8O$. A polar silica was used as the stationary phase, and a non-polar hexane as the mobile phase.



Naphthalene Naphthol



What are the position and retardation factor (R_F) of naphthalene?

A. y and 0.50

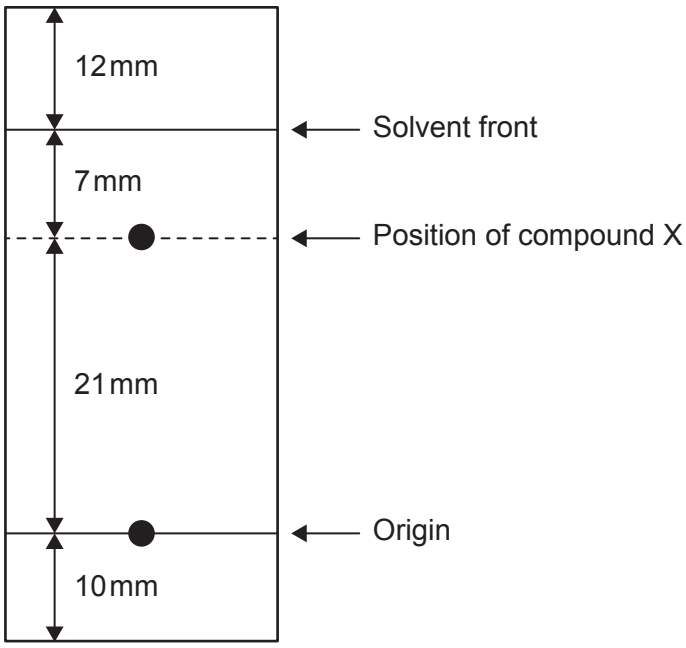
B. y and 0.58

C. x and 0.80

D. x and 0.83

5.3-M2 May 2025 TZ2 · Paper 1A · Q8

8. What is the retardation factor, R_F, of compound X according to this paper chromatogram?



A. 0.55

B. 0.75

C. 0.82

D. 1.33

Part B - Structured questions

5.3-S1 May 2025 TZ1 · Paper 1B · Q2 parts (a), (e)

2. A student was given a mixture to separate and collect the individual components.
The mixture contained sand, $SiO_2(s)$, sodium chloride, NaCl(s), and iron filings, Fe(s).
The student observed the original mixture and made the following hypothesis.

The iron would have the lowest percent by mass because it appeared to be present in the smallest quantity.

(a) Suggest a set of experimental steps required to obtain pure samples of each component of the mixture. [4]

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(e) In a different experiment, the students needed to separate a mixture of methanol and butan-1-ol. State the best method to separate this mixture and the physical property used for the separation. [2]

Separation method: . . . . . .

. . . . . .

Physical property: . . . . . .

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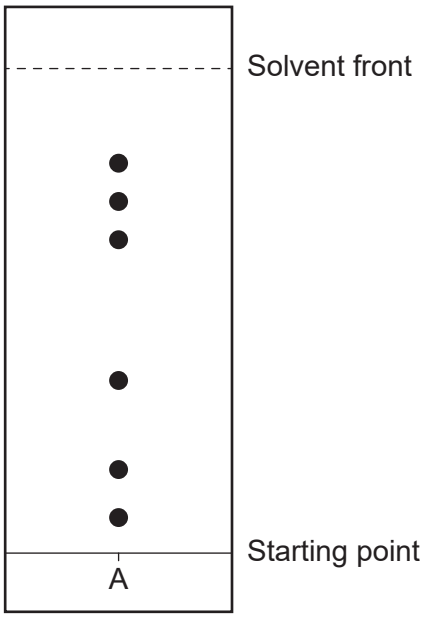
5.3-S2 May 2024 TZ1 · Paper 3 · Q9 parts (a)

9. Green algae are a rich source of pigments such as chlorophylls and carotenoids.

Thin-layer chromatography (TLC) was used to investigate the presence of the carotenoid lutein in an algae extract. A chromatogram with 6 coloured spots was obtained.

R_f	Pigment
0.08	Xantophyll Y
0.17	Xantophyll X
0.36	Lutein
0.64	Chlorophyll b
0.72	Chlorophyll a
0.80	β-carotene



(a) Determine if lutein is present by calculating R_f values of the spots and comparing them with the data in the table. [1]

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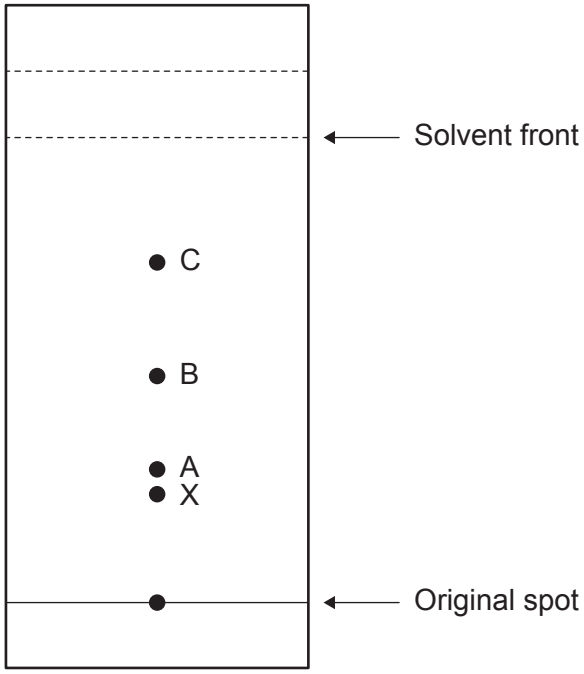
5.3-S3 May 2023 TZ1 · Paper 3 · Q8 parts (a)(i)

8. The amino acids in a protein can be separated using paper chromatography. The R_f values using a solvent of butanol and ethanoic acid are given.

Amino acid	R_f value
Lysine	0.14
Glutamic acid	0.30
Threonine	0.35
Tyrosine	0.45
Asparagine	0.5
Methionine	0.55
Valine	0.61
Tryptophan	0.66
Leucine	0.73

(a) The following diagram shows a chromatogram.



(i) Determine the identity of the amino acid creating spot C by calculating the R_f value from the chromatogram. [1]

Identity of spot C: . . . . . .

5.3-S4 May 2023 TZ2 · Paper 3 · Q7 parts (b)

7. Proteins are large polymers of 2-amino acids.

(b) Explain how paper chromatography can separate and identify mixtures of amino acids. [2]

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5.3-S5 Nov 2020 · Paper 3 · Q1 parts (a), (b), (c)

1. In order to determine the oil content of different types of potato crisps (chips), a student weighed 5.00 g of crushed crisps and mixed them with 20.0 cm^3 of non-polar solvent.

She assumed all the oil in the crisps dissolved in the solvent.

The student then filtered the mixture to remove any solids, and gently heated the solution on a hot plate to evaporate the solvent.

She measured the mass of the oil that remained from each type of crisps.

(a) Suggest why a non-polar solvent was needed. [1]

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(b) State one reason why the mixture was not heated strongly. [1]

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(c) Non-polar solvents can be toxic. Suggest a modification to the experiment which allows the evaporated solvent to be collected. [1]

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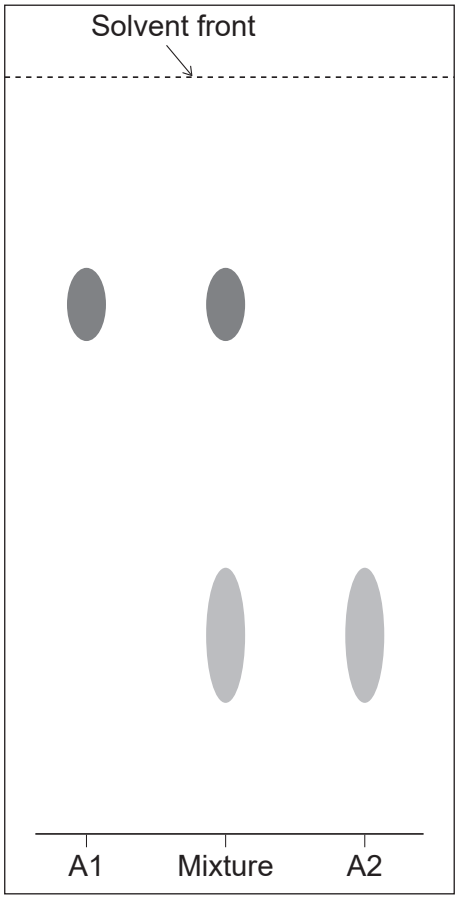
5.3-S6 Nov 2020 · Paper 3 · Q6 parts (a)(i)

6. Proteins are polymers of amino acids.

(a) (i) A paper chromatogram of two amino acids, A1 and A2, is obtained using a non-polar solvent.



Determine the R_f value of A1. [1]

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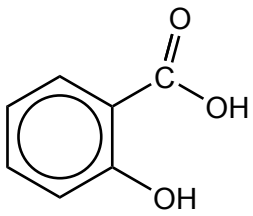
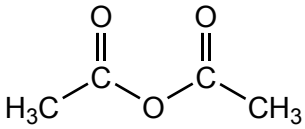
5.3-S7 Nov 2020 · Paper 3 · Q15 parts (b)

15. Aspirin is formed by reacting salicylic acid with ethanoic anhydride. The structure of aspirin is given in section 37 of the data booklet.



Salicylic acid



Ethanoic anhydride

(b) Aspirin crystals are rinsed with water after recrystallization to remove impurities. Suggest why **cold** water is used. [1]

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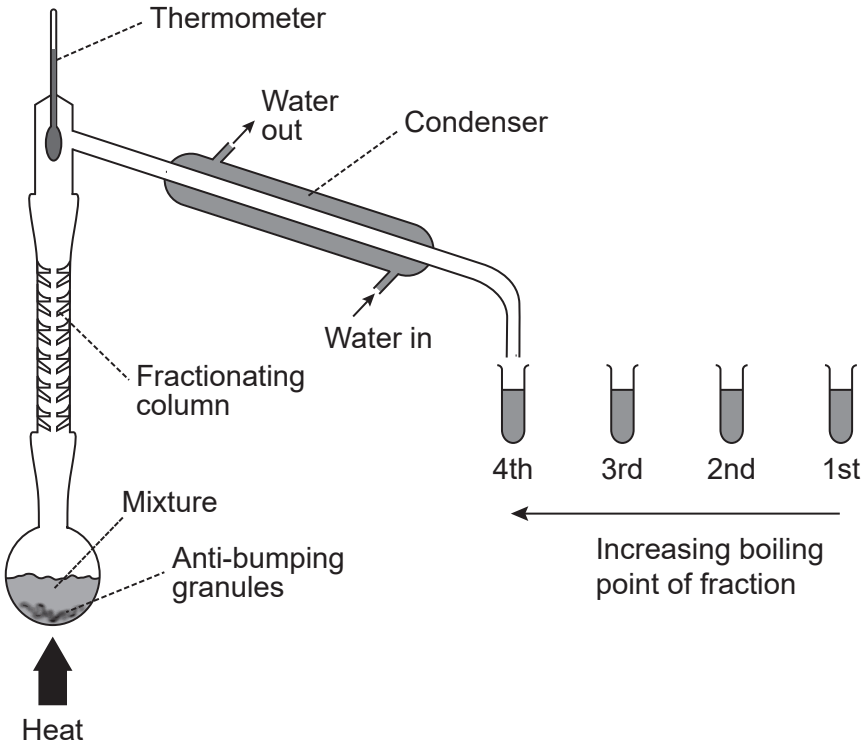
5.3-S8 Nov 2020 · Paper 3 · Q18 parts (b)

18. A mixture of 0.100 mol ethanal, 0.100 mol ethanol and 0.200 mol ethanoic acid is fractionally distilled.



(b) Describe how this mixture is separated by fractional distillation. [2]

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5.3-S9 May 2019 TZ1 · Paper 3 · Q15 parts (a), (c)(i), (c)(ii) · also 7.3

15. Natural gas is an energy source composed mainly of methane.

(a) Calculate the specific energy of methane, in MJ kg^{-1}, using sections 1, 6 and 13 of the data booklet. [1]

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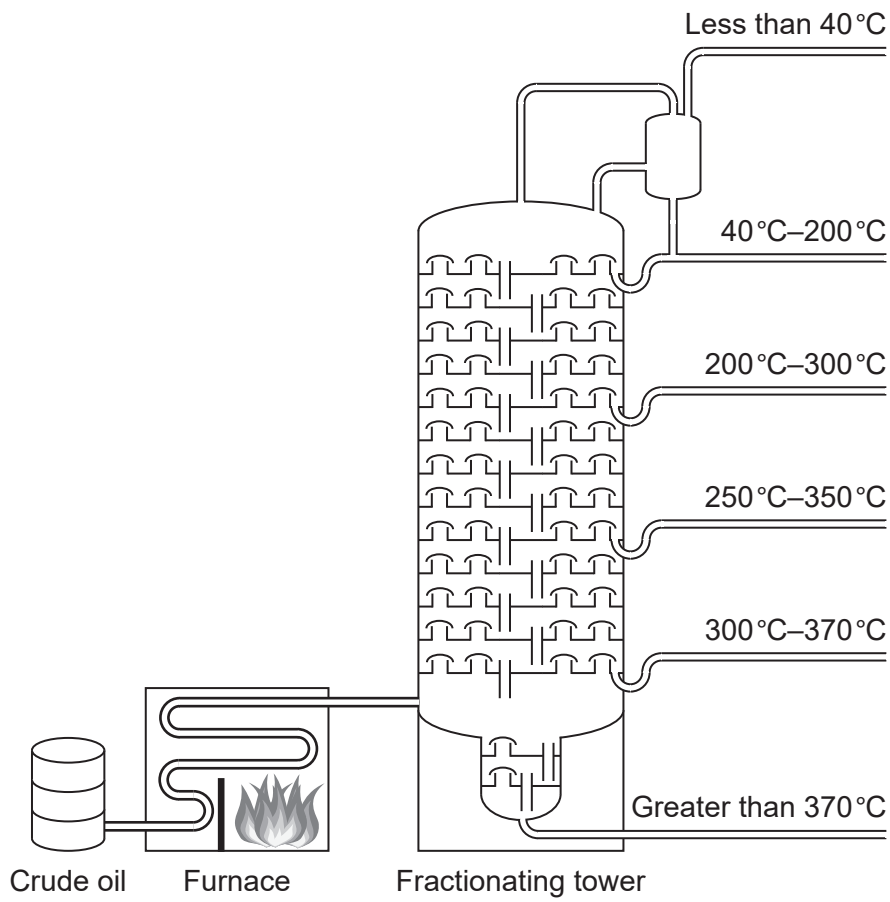
(c) (i) Methane can also be obtained by fractional distillation of crude oil.



[Source: Image used with kind permission of science-resources.co.uk]

Draw a circle on the diagram to show where the methane fraction is withdrawn. [1]

(ii) List the following products, which are also obtained by fractional distillation, according to **decreasing** volatility: asphalt, diesel, gasoline, lubricating motor oil. [1]

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5.3-S10 May 2018 TZ1 · Paper 3 · Q6 parts (c)(ii)

6. Insulin was the first protein to be sequenced. It was determined that the end of one chain had the primary structure Phe–Val–Asn–Gln.

(c) Paper chromatography can be used to identify the amino acids in insulin.

(ii) Outline how the amino acids may be identified from a paper chromatogram. [1]

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5.3-S11 May 2018 TZ1 · Paper 3 · Q20 parts (b)

20. Organic solvents are commonly used in the pharmaceutical industry.

(b) Explain how hexane and propanone may be separated by fractional distillation. [3]

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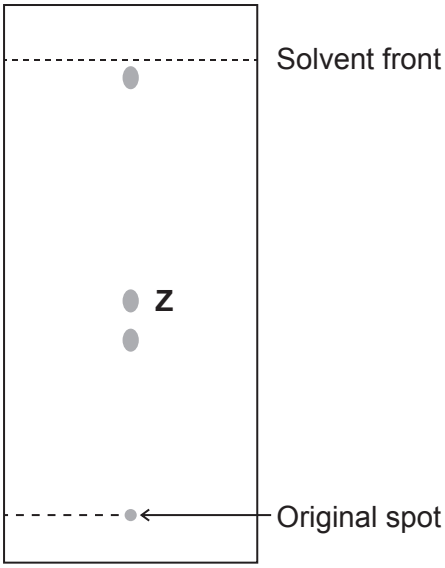
End of Option D

5.3-S12 May 2016 · Paper 3 · Q13 parts (b)

13. Spinach is an excellent source of vitamins A and C.

(b) The pigments from spinach were separated using chromatography.
Identify **Z** by calculating its R_f value and using the data table. [1]



Pigment	R_f value
Xanthophyll	0.35
Chlorophyll a	0.60
Chlorophyll b	0.50
Carotene	0.95

R_f:

. . . . . .

. . . . . .

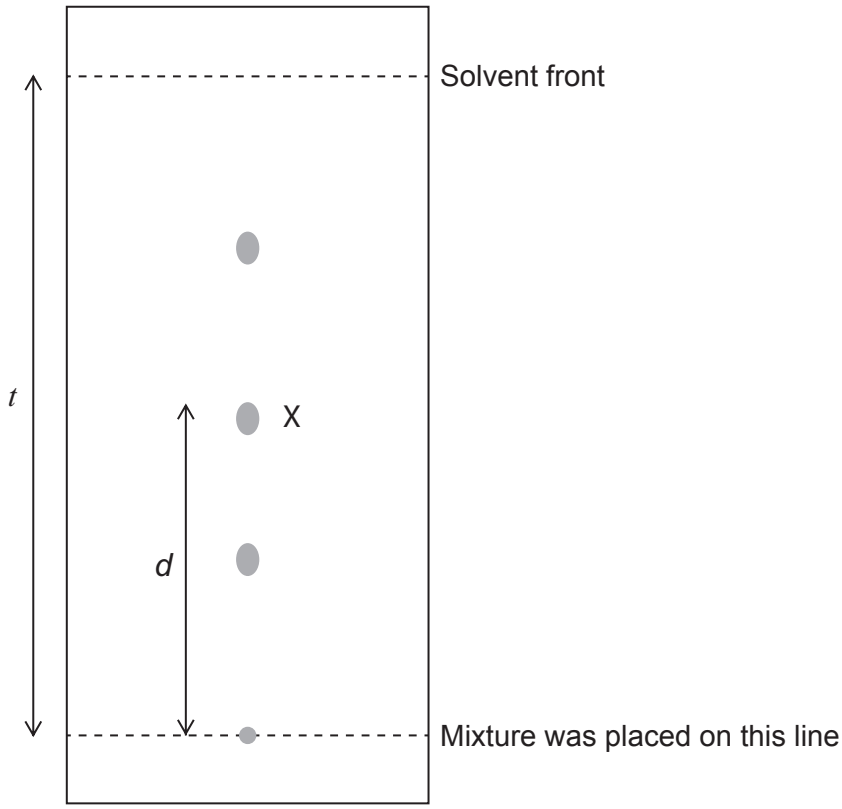
Z:

. . . . . .

5.3-S13 May 2015 TZ2 · Paper 3 · Q6 parts (a)(ii)

6. Proteins are made of long chains of amino acids.

(ii) A mixture of amino acids was spotted onto chromatography paper and eluted with a solvent mixture. The following spots were seen after the paper had been developed with ninhydrin.



Determine the R_f value of the amino acid marked as X. [1]

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5.3-S14 Nov 2014 · Paper 3 · Q3 parts (a), (b)

3. The principles of chromatography can be demonstrated using paper chromatography to analyse the ink of a pen, using propanone as the mobile phase.

(a) Explain how paper chromatography separates the components. *[2]*

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(b) State **one** factor that would alter the R_f value of a particular component. *[1]*

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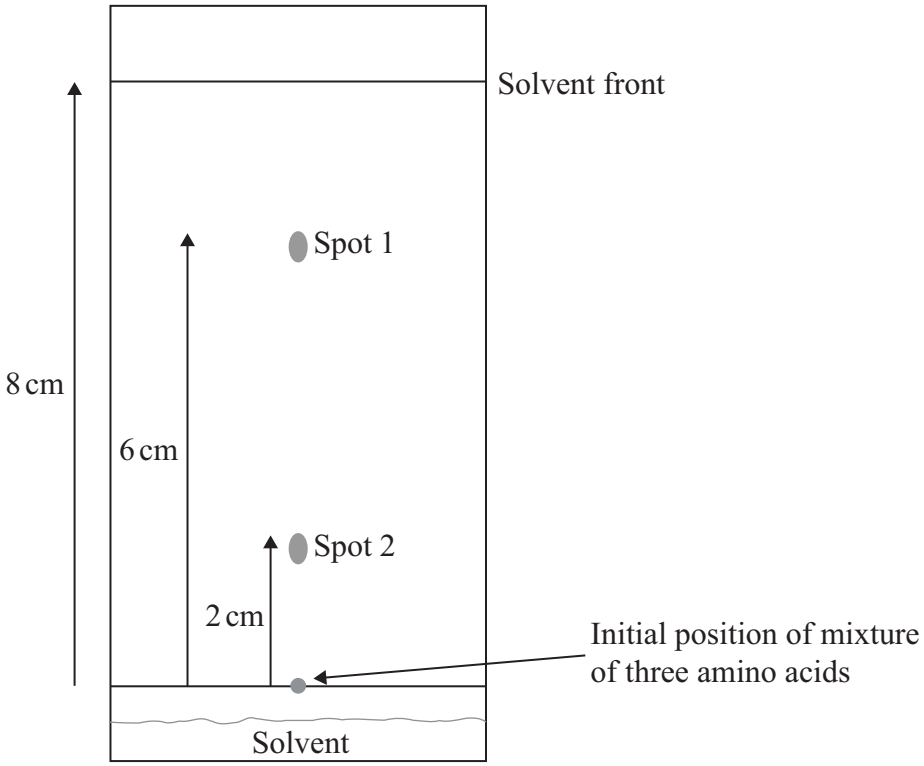
5.3-S15 May 2014 TZ1 · Paper 3 · Q4 parts (a), (b), (c)

4. A sample is known to contain three different amino acids. After carrying out paper chromatography using a solvent made up of propan-1-ol, water and ammonia, the following chromatogram was obtained once the spots had been developed with ninhydrin.



(a) Calculate the R_f values for the two spots. *[1]*

Spot 1:

. . . . . .

Spot 2:

. . . . . .

(b) Suggest a reason why only two spots are present. *[1]*

. . . . . .
. . . . . .

(c) Suggest how the chromatography experiment with the same sample could be altered in order to obtain three spots. *[1]*

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5.3-S16 May 2014 TZ2 · Paper 3 · Q2 parts (a)(i), (a)(ii)

2. Thin-layer chromatography (TLC) is an example of adsorption chromatography. Gas-liquid chromatography (GLC) can be used to separate and identify small samples of gases and volatile liquids.

(a) A mixture of two organic compounds was separated by TLC using a non-polar solvent.

Compound	Distance travelled / mm
A	22
B	65
Solvent	80

(i) Calculate the R_f values of A and B. *[1]*

Compound	R_f value
A	
B	

(ii) Outline why compound B has travelled the greater distance. *[1]*

. . . . . .

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5.3-S17 May 2013 TZ2 · Paper 3 · QA1 parts (a), (b)

A1. Paper chromatography is a simple method used to separate and identify the components in a mixture. To aid identification, the retention factor, R_f, of an unknown component can be compared with the R_f values of pure samples of the possible components.

(a) State the meaning of the term *retention factor*. *[1]*

. . . . . .

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(b) Explain why the value of the retention factor for the same component can be very different if different solvents (eluents) are used for the mobile phase. *[2]*

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5.3-S18 May 2012 TZ1 · Paper 3 · QA3 parts (a), (b)

A3. Chromatography is a technique frequently used to separate mixtures of substances.

(a) Describe how you would show that a sugar solution was a mixture of two substances using the technique of thin-layer chromatography (TLC). *[3]*

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(b) Given solutions of glucose and fructose, explain how you would show that these were the two sugars present in the mixture. *[2]*

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5.3-S19 May 2012 TZ2 · Paper 3 · QA2 parts (a)

A2. Alumina, Al_2O_3, can be used as the stationary phase in both thin-layer chromatography (TLC) and column chromatography (CC). A solvent such as propanone can be used as the mobile phase.

(a) Explain the functions of both the stationary phase and the mobile phase in chromatography and explain their role in allowing a mixture to be separated into its individual components. *[3]*

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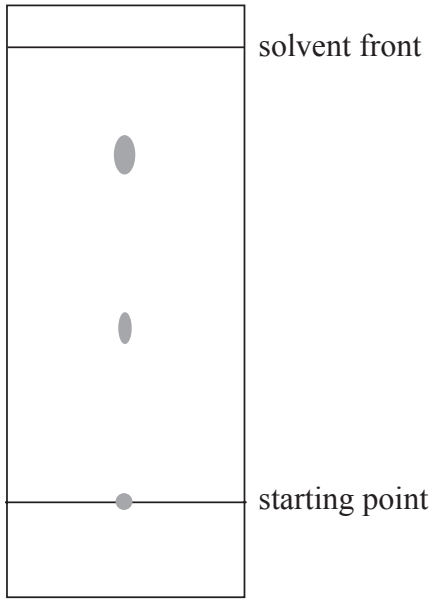
5.3-S20 May 2011 TZ1 · Paper 3 · QA3 parts (a), (b)

A3. A sample of food colouring was analysed using thin-layer chromatography to check whether it contained a banned substance. The R_f value of the banned substance is 0.25 under the same conditions.



(a) Identify a stationary phase commonly used in thin-layer chromatography. *[1]*

. . . . . .

(b) Predict where you would expect the banned dye to appear on the chromatogram and mark this spot with a circle on the diagram above. *[1]*

LESSON 6.1

Redox Reactions & Oxidation Numbers

52 multiple-choice questions · 18 structured questions

Part A - Multiple choice

6.1-M1 May 2025 TZ1 · Paper 1A · Q34

34. Ammonia, NH_3 and nitrous acid, HNO_2, are compounds of nitrogen.
What are the oxidation states of nitrogen in these compounds?

	NH_3	HNO_2
A.	+3	+3
B.	–3	–3
C.	+3	–3
D.	–3	+3

6.1-M2 May 2025 TZ2 · Paper 1A · Q14

14. What is the oxidation state of nitrogen in N_2O_4?

A. –4

B. –2

C. +2

D. +4

6.1-M3 May 2025 TZ2 · Paper 1A · Q36

36. In which compound does sulfur have an oxidation state of +4?

A. H_2S

B. SO_3

C. H_2SO_3

D. H_2SO_4

6.1-M4 May 2025 TZ2 · Paper 1A · Q38

38. Which change represents oxidation of the functional group?

A. $CH_3COOH \rightarrow CH_3CHO$

B. $H_2CO \rightarrow HCOOH$

C. $HCCH \rightarrow H_2CCH_2$

D. $H_2CO \rightarrow CH_3OH$

6.1-M5 May 2025 TZ3 · Paper 1A · Q15

15. Which compounds contain an atom in a +3 oxidation state?

I. NO_2^-

II. $FeCl_3$

III. BrF_3

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.1-M6 Nov 2024 · Paper 1 · Q28

28. In which reaction does H_2 act as an oxidizing agent?

A. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

B. $2Na(s) + H_2(g) \rightarrow 2NaH(s)$

C. $Cl_2(g) + H_2(g) \rightarrow 2HCl(g)$

D. $H_2CCH_2(g) + H_2(g) \rightarrow H_3CCH_3(g)$

6.1-M7 May 2024 TZ1 · Paper 1 · Q8

8. What is the oxidation state of cobalt in the complex ion $[CoCl_4(NH_3)_2]^-$?

A. +2

B. +3

C. +5

D. +6

6.1-M8 May 2024 TZ1 · Paper 1 · Q28

28. In which pair of species does Br have the same oxidation state?

A. BrO_4^- and $HBrO_3$

B. HBr and Br_2O

C. $HBrO_2$ and Br^-

D. BrO^- and Br_2O

6.1-M9 May 2024 TZ2 · Paper 1 · Q10

10. In which species is the oxidation state of chromium different?

A. $[Cr(H_2O)_4Cl_2]^+$

B. $[Cr(H_2O)_3(OH)_3]$

C. $[Cr(NH_3)_6]^{3+}$

D. CrO_3

6.1-M10 May 2024 TZ2 · Paper 1 · Q32

32. Which of the following occurs during the Winkler test used to determine the concentration of dissolved oxygen in water?

Step I: $2Mn^{2+}(ag) + O_2(g) + 4OH^-(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$

Step II: $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$

Step III: $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

A. Manganese(II) species is oxidized.

B. Iodine is reduced by oxygen to iodide ions.

C. Oxygen is oxidized.

D. Thiosulfate ions are reduced.

6.1-M11 May 2023 TZ1 · Paper 1 · Q28

28. Which change involves oxidation of N?

A. NH_3 to N_2

B. NO_2 to NO

C. N_2 to AlN

D. NO_2 to N_2O_4

6.1-M12 May 2023 TZ2 · Paper 1 · Q28

28. Which species is the oxidizing agent?

$$14H^+(aq) + 2Mn^{2+}(aq) + 5BiO_3^-(aq) \rightarrow 2MnO_4^-(aq) + 5Bi^{3+}(aq) + 7H_2O(l)$$

A. $H^+(aq)$

B. $Mn^{2+}(aq)$

C. $BiO_3^-(aq)$

D. $MnO_4^-(aq)$

6.1-M13 May 2022 TZ1 · Paper 1 · Q28

28. In which of the following species would sulfur be reduced if converted to SCl_2?

A. $S_2O_3^{2-}$

B. H_2S

C. S

D. SO_2

6.1-M14 Nov 2021 · Paper 1 · Q8

8. Which complex ion contains a central ion with an oxidation state of +3?

A. $[PtCl_6]^{2-}$

B. $[Cu(H_2O)_4(OH)_2]$

C. $[Ni(NH_3)_4(H_2O)_2]^{2+}$

D. $[Co(NH_3)_4Cl_2]^{+}$

6.1-M15 Nov 2021 · Paper 1 · Q28

28. What is the change in the oxidation state of oxygen?

$$2Fe^{2+}(aq) + H_2O_2(aq) + 2H^{+}(aq) \rightarrow 2H_2O(l) + 2Fe^{3+}(aq)$$

A. +1

B. 0

C. –1

D. –2

6.1-M16 May 2021 TZ1 · Paper 1 · Q9

9. What is the overall charge, x, of the chromium (III) complex?

$$[Cr(H_2O)_4Cl_2]^{x}$$

A. 0

B. 1+

C. 2–

D. 3+

6.1-M17 May 2021 TZ1 · Paper 1 · Q29

29. What is correct for this redox reaction?

$$MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$$

	Reduced	Reducing agent
A.	$MnO_2(s)$	$I^-(aq)$
B.	$I^-(aq)$	$H^+(aq)$
C.	$I^-(aq)$	$MnO_2(s)$
D.	$H^+(aq)$	$I^-(aq)$

6.1-M18 May 2021 TZ2 · Paper 1 · Q28

28. What is the oxidation state of oxygen in H_2O_2?

A. –2

B. –1

C. +1

D. +2

6.1-M19 Nov 2020 · Paper 1 · Q28

28. What are the oxidation states of oxygen?

	O_2	OF_2	H_2O_2
A.	–2	–2	–2
B.	0	–2	–1
C.	0	+2	–1
D.	–2	+2	–2

6.1-M20 Nov 2019 · Paper 1 · Q29

29. In which species does sulfur have the same oxidation state as in SO_3^{2-}?

A. $S_2O_3^{2-}$

B. SO_4^{2-}

C. H_2S

D. $SOCl_2$

6.1-M21 May 2019 TZ2 · Paper 1 · Q8

8. What is the oxidation state of the metal ion and charge of the complex ion in $[Co(NH_3)_4Cl_2]Cl$?

	Oxidation state of metal ion	Charge of complex ion
A.	+1	2+
B.	+2	1+
C.	+3	1+
D.	+3	0

6.1-M22 May 2019 TZ2 · Paper 1 · Q28

28. Which compound contains sulfur with an oxidation state of +6?

A. SO_2

B. H_2S

C. H_2SO_3

D. H_2SO_4

6.1-M23 Nov 2018 · Paper 1 · Q8

8. Which is correct for the complex ion in $[Fe(H_2O)_5Cl]SO_4$?

	Oxidation state of iron	Overall charge of the complex ion
A.	+2	2+
B.	+2	0
C.	+3	1+
D.	+3	2+

6.1-M24 Nov 2018 · Paper 1 · Q28

28. Which is correct for the reaction?

$$P_4(s) + 3H_2O(l) + 3OH^-(aq) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

	Oxidizing agent	Reducing agent
A.	H_2O	P_4
B.	P_4	OH^-
C.	OH^-	P_4
D.	P_4	P_4

6.1-M25 May 2018 TZ1 · Paper 1 · Q28

28. Which equation shows oxygen undergoing reduction?

A. $2F_2 + O_2 \rightarrow 2F_2O$

B. $Na_2O + H_2O \rightarrow 2NaOH$

C. $H_2O_2 + 2HI \rightarrow 2H_2O + I_2$

D. $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$

6.1-M26 May 2018 TZ2 · Paper 1 · Q28

28. Which can describe oxidation?

A. Loss of hydrogen

B. Decrease in oxidation number

C. Gain of electrons

D. Loss of oxygen

6.1-M27 Nov 2017 · Paper 1 · Q29

29. Which of the following is a redox reaction?

A. $3Mg(s) + 2AlCl_3(aq) \rightarrow 2Al(s) + 3MgCl_2(aq)$

B. $SiO_2(s) + 2NaOH(aq) \rightarrow Na_2SiO_3(aq) + H_2O(l)$

C. $KCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + KNO_3(aq)$

D. $2NaHCO_3(aq) \rightarrow Na_2CO_3(aq) + CO_2(g) + H_2O(l)$

6.1-M28 Nov 2017 · Paper 1 · Q30

30. Consider the following half-equations:

$$\underset{\text{(brown)}}{I_2(s)} + 2e^- \rightleftharpoons \underset{\text{(colourless)}}{2I^-(aq)} \qquad E^\ominus = +0.54\,V$$

$$\underset{\text{(purple)}}{MnO_4^-(aq)} + 8H^+(aq) + 5e^- \rightleftharpoons \underset{\text{(colourless)}}{Mn^{2+}(aq)} + 4H_2O(l) \qquad E^\ominus = +1.51\,V$$

Which statement is correct for the reaction between $KMnO_4(aq)$ and $KI(aq)$ in acidic conditions?

A. MnO_4^- reduces I^- to I_2.

B. I^- reduces MnO_4^- to Mn^{2+}.

C. The colour changes from brown to purple.

D. MnO_4^- is oxidized to Mn^{2+}.

6.1-M29 Nov 2017 · Paper 1 · Q32

32. What are the oxidation states of chromium in $(NH_4)_2Cr_2O_7(s)$ and $Cr_2O_3(s)$?

	$(NH_4)_2Cr_2O_7(s)$	$Cr_2O_3(s)$
A.	+7	+3
B.	+6	+3
C.	+6	+6
D.	+7	+6

6.1-M30 May 2017 TZ1 · Paper 1 · Q8

8. What is the charge on the iron(III) complex ion in $[Fe(OH)_2(H_2O)_4]Br$?

A. 0

B. 1+

C. 2+

D. 3+

6.1-M31 May 2017 TZ1 · Paper 1 · Q28

28. Which change represents oxidation?

A. $HClO_4$ to $HClO_3$

B. N_2 to NH_3

C. N_2O to NO

D. SO_4^{2-} to SO_3^{2-}

6.1-M32 May 2017 TZ1 · Paper 1 · Q29

29. A reaction takes place when a rechargeable battery is used:

$$Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$

Which statements are correct?

I. H^+ is reduced
II. The oxidation state of Pb metal changes from 0 to +2
III. PbO_2 is the oxidising agent

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.1-M33 May 2017 TZ2 · Paper 1 · Q28

28. Which element is reduced in the following decomposition?

$$(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + Cr_2O_3(s) + 4H_2O(g)$$

A. N

B. H

C. Cr

D. O

6.1-M34 May 2017 TZ2 · Paper 1 · Q29

29. Which of the following is **not** a redox reaction?

A. $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

B. $C(s) + O_2(g) \rightarrow CO_2(g)$

C. $2CO(g) \rightarrow CO_2(g) + C(s)$

D. $CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(l)$

6.1-M35 Nov 2016 · Paper 1 · Q30

30. Which is a correct statement for the reaction below?

$$2MnO_4^-(aq) + 6H^+(aq) + 5NO_2^-(aq) \rightarrow 2Mn^{2+}(aq) + 5NO_3^-(aq) + 3H_2O(l)$$

A. MnO_4^- is the reducing agent and the oxidation number of Mn increases.

B. MnO_4^- is the oxidizing agent and the oxidation number of Mn decreases.

C. NO_2^- is the reducing agent and the oxidation number of N decreases.

D. NO_2^- is the oxidizing agent and the oxidation number of N increases.

6.1-M36 Nov 2015 · Paper 1 · Q30

30. Which element undergoes reduction in the following reaction?

$$(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(l) + Cr_2O_3(s)$$

A. Cr

B. H

C. N

D. O

6.1-M37 Nov 2015 · Paper 1 · Q31

31. Which best describes reduction?

A. Increase in oxidation number and gain of electrons

B. Increase in oxidation number and loss of electrons

C. Decrease in oxidation number and gain of electrons

D. Decrease in oxidation number and loss of electrons

6.1-M38 May 2015 TZ1 · Paper 1 · Q30

30. Which represents a redox reaction?

A. $NaH(s) + H_2O(l) \rightarrow NaOH(aq) + H_2(g)$

B. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

C. $CuCl_2(aq) + K_2S(aq) \rightarrow CuS(s) + 2KCl(aq)$

D. $HCl(aq) + NH_3(aq) \rightarrow NH_4^+Cl^-(aq)$

6.1-M39 May 2015 TZ2 · Paper 1 · Q30

30. Which is a redox reaction?

A. $[Cu(H_2O)_4]^{2+}(aq) + 4Cl^-(aq) \rightarrow [CuCl_4]^{2-}(aq) + 4H_2O(l)$

B. $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

C. $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

D. $2K_2CrO_4(aq) + 2HCl(aq) \rightarrow K_2Cr_2O_7(aq) + H_2O(l) + 2KCl(aq)$

6.1-M40 Nov 2014 · Paper 1 · Q30

30. Which statement is correct for the following reaction?

$$2ClO_3^-(aq) + SO_2(aq) + H^+(aq) \rightarrow 2ClO_2(g) + HSO_4^-(aq)$$

A. ClO_3^- is the oxidizing agent and it undergoes reduction.

B. ClO_3^- is the reducing agent and it undergoes oxidation.

C. SO_2 is the oxidizing agent and it undergoes oxidation.

D. SO_2 is the reducing agent and it undergoes reduction.

6.1-M41 May 2014 TZ1 · Paper 1 · Q31

31. What are the correct names for $KMnO_4$ and $K_2Cr_2O_7$, using oxidation numbers?

A. Potassium permanganate and potassium dichromate

B. Potassium manganate(IV) and potassium chromate(VII)

C. Potassium permanganate(IV) and potassium dichromate(VII)

D. Potassium manganate(VII) and potassium dichromate(VI)

6.1-M42 May 2014 TZ2 · Paper 1 · Q31

31. Which species are the oxidizing and reducing agents in the following reaction?

$$SO_3^{2-}(aq) + PbO_2(s) + H_2O(l) \rightarrow SO_4^{2-}(aq) + Pb(OH)_2(s)$$

	Oxidizing agent	Reducing agent
A.	PbO_2	H_2O
B.	SO_3^{2-}	PbO_2
C.	H_2O	SO_3^{2-}
D.	PbO_2	SO_3^{2-}

6.1-M43 Nov 2013 · Paper 1 · Q32

32. Consider the following reaction.

$$2Cr(OH)_3(s) + 6ClO^-(aq) \rightarrow 2CrO_4^{2-}(aq) + 3Cl_2(g) + 2OH^-(aq) + 2H_2O(l)$$

Which statement is correct?

A. $Cr(OH)_3$ is the oxidizing agent and the oxidation number of chromium changes from +3 to +6.

B. $Cr(OH)_3$ is the reducing agent and undergoes reduction.

C. ClO^- is the oxidizing agent and the oxidation number of chlorine changes from +1 to 0.

D. ClO^- is the reducing agent and the oxidation number of chlorine changes from –1 to 0.

6.1-M44 May 2013 TZ1 · Paper 1 · Q30

30. Which statement is correct about a reducing agent?

A. It is reduced by gaining electrons.

B. It is oxidized by gaining electrons.

C. It is oxidized by losing electrons.

D. It is reduced by losing electrons.

6.1-M45 May 2013 TZ2 · Paper 1 · Q30

30. Which is the oxidizing agent in the following reaction?

$$5SO_2(g) + 2IO_3^-(aq) + 4H_2O(l) \rightarrow 5SO_4^{2-}(aq) + I_2(aq) + 8H^+(aq)$$

A. SO_2

B. IO_3^-

C. H_2O

D. SO_4^{2-}

6.1-M46 May 2012 TZ2 · Paper 1 · Q28

28. Consider the following reaction:

$$3Sn^{2+}(aq) + Cr_2O_7^{2-}(aq) + 2H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3SnO_2(s) + H_2O(l)$$

Which statement is correct?

A. Sn^{2+} is the oxidizing agent because it undergoes oxidation.

B. Sn^{2+} is the reducing agent because it undergoes oxidation.

C. $Cr_2O_7^{2-}$ is the oxidizing agent because it undergoes oxidation.

D. $Cr_2O_7^{2-}$ is the reducing agent because it undergoes oxidation.

6.1-M47 May 2011 TZ1 · Paper 1 · Q31

31. Which species could be reduced to form NO_2?

A. N_2O

B. NO_3^-

C. HNO_2

D. NO

6.1-M48 May 2011 TZ2 · Paper 1 · Q30

30. What happens to iodine when iodate ions, IO_3^-, are converted to iodine molecules, I_2?

A. It undergoes reduction and its oxidation number changes from –1 to 0

B. It undergoes oxidation and its oxidation number changes from –1 to 0

C. It undergoes reduction and its oxidation number changes from +5 to 0

D. It undergoes oxidation and its oxidation number changes from +5 to 0

6.1-M49 Nov 2010 · Paper 1 · Q8

8. In which complexes does iron have an oxidation number of +3?

I. $[Fe(H_2O)_6]^{3+}$

II. $[Fe(H_2O)_5(CN)]^{2+}$

III. $[Fe(CN)_6]^{3-}$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.1-M50 Nov 2010 · Paper 1 · Q30

30. Consider the following reaction.

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 5Fe^{3+}(aq) + 4H_2O(l)$$

Which statement is correct?

A. MnO_4^- is the oxidizing agent and it loses electrons.

B. MnO_4^- is the reducing agent and it loses electrons.

C. MnO_4^- is the oxidizing agent and it gains electrons.

D. MnO_4^- is the reducing agent and it gains electrons.

6.1-M51 May 2010 TZ1 · Paper 1 · Q30

30. Consider the following reaction.

$$2FeSO_4(aq) + H_2O_2(aq) + H_2SO_4(aq) \rightarrow Fe_2(SO_4)_3(aq) + 2H_2O(l)$$

Which species is the oxidizing agent and which is the reducing agent?

	Oxidizing agent	Reducing agent
A.	$H_2O_2(aq)$	$H_2SO_4(aq)$
B.	$H_2O_2(aq)$	$FeSO_4(aq)$
C.	$FeSO_4(aq)$	$H_2O_2(aq)$
D.	$H_2SO_4(aq)$	$H_2O_2(aq)$

6.1-M52 May 2010 TZ2 · Paper 1 · Q30

30. What is the reducing agent in the reaction below?

$$2MnO_4^-(aq) + Br^-(aq) + H_2O(l) \rightarrow 2MnO_2(s) + BrO_3^-(aq) + 2OH^-(aq)$$

A. Br^-

B. BrO_3^-

C. MnO_4^-

D. MnO_2

Part B - Structured questions

6.1-S1 May 2024 TZ1 · Paper 2 · Q4 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (b)(vi) · also 3.1, 4.1, 4.2, 5.1, 9.3

4. The water-gas shift reaction is another way to manufacture hydrogen.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g) \qquad \Delta H = -41\,kJ\,mol^{-1}$$

(a) (i) State the oxidation state of carbon in carbon monoxide and carbon dioxide. [1]

carbon monoxide: . . . . . .

carbon dioxide: . . . . . .

(ii) Identify the oxidising and reducing agents, and the species oxidised and reduced, in the forward reaction. [2]

	CO (g)	**H_2O (g)**
oxidising or reducing agent?		
species oxidised or reduced?		

(b) (i) Draw the Lewis structure of carbon dioxide. [1]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(ii) Annotate the Lewis structure in (b)(i) to show the polarity of the bonds by adding the symbols $\delta+$ and $\delta-$ as appropriate. [1]

(iii) Explain the molecular geometry and polarity of the carbon dioxide molecule. [2]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(iv) State the hybridization of the carbon atom in carbon monoxide and in carbon dioxide. [1]

carbon monoxide: . . . . . .

carbon dioxide: . . . . . .

(v) State the number of σ and π bonds in a carbon monoxide molecule. [1]

σ bonds: . . . . . .

π bonds: . . . . . .

(vi) Determine the bond enthalpy of carbon monoxide, in $kJ\,mol^{-1}$. Use section 11 of the data booklet and the equation:

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g) \qquad \Delta H = -41\,kJ\,mol^{-1}$ [2]

. . . . . .

. . . . . .

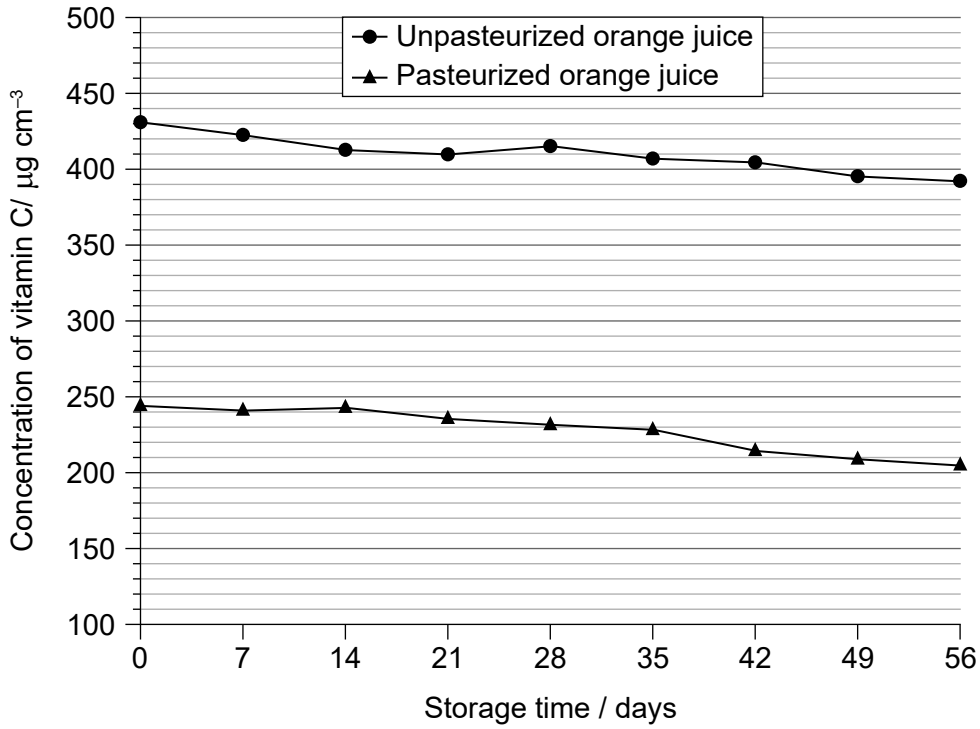
. . . . . .

. . . . . .

. . . . . .

6.1-S2 Nov 2023 TZ1 · Paper 3 · Q1 parts (d)(i)

1. Pasteurization is used to eliminate pathogenic bacteria. The concentration of vitamin C was monitored over a period of time in pasteurized and unpasteurized orange juice.



(d) The concentration of vitamin C (ascorbic acid) can be measured by performing a redox titration using acidified iodate, IO_3^-, and iodide ions. Starch reacts with excess iodine once the vitamin C is consumed to produce a dark-blue complex.

Reaction 1: $IO_3^-(aq) + 5I^-(aq) + 6H^+(aq) \rightarrow 3I_2(aq) + 3H_2O(l)$

Reaction 2: ascorbic acid (aq) + $I_2(aq) \rightarrow 2I^-(aq)$ + dehydroascorbic acid (aq)

(i) Identify the oxidizing agent in reaction 1. [1]

. . . . . .

6.1-S3 May 2023 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b), (c) · also 6.3, 7.1, 7.2

2. Metals A, B, C, D, and E were placed in sulfate solutions.

(a) The results are given where ✓ = reaction occurred and × = no reaction.

Metal	$ASO_4(aq)$	$BSO_4(aq)$	$CSO_4(aq)$	$DSO_4(aq)$	$ESO_4(aq)$
A	—	✓	×	✓	✓
B	×	—	×	✓	✓
C	✓	✓	—	✓	✓
D	×	×	×	—	✓
E	×	×	×	×	—

(i) Identify the oxidation state of metal A in ASO_4. [1]

. . . . . .

. . . . . .

(ii) Deduce the activity series of the metals. [1]

. . . . . .

. . . . . .

(b) Sketch the voltaic cell that produces the greatest electromotive force, EMF, using the information in part (a). Label the direction of electron flow, salt bridge, anode, cathode, the metals (A–E) and solutions used in each half-cell. [4]

(c) Deduce **one** metal that could be electroplated, with high efficiency, on to a metal key using an aqueous salt of the metal. Use section 24 of the data booklet. [1]

. . . . . .

. . . . . .

6.1-S4 May 2023 TZ1 · Paper 3 · Q3 parts (a)(i)

3. New materials have brought many benefits to society but come with associated risks.

(a) High-pressure carbon monoxide disproportionation (HiPco) produces carbon atoms that react with nano catalysts to produce carbon nanotubes.

(i) Write the equation for the disproportionation of carbon monoxide to produce carbon atoms. [1]

. . . . . .

. . . . . .

6.1-S5 May 2021 TZ1 · Paper 2 · Q3 parts (a), (c), (d), (e) · also 7.1, 7.2, 8.2

3. Magnetite, Fe_3O_4, is another ore of iron that contains both Fe^{2+} and Fe^{3+}.

(a) Deduce the ratio of Fe^{2+}:Fe^{3+} in Fe_3O_4. [1]

. . . . . .

. . . . . .

(c) Iron has a relatively small specific heat capacity; the temperature of a 50 g sample rises by 44.4°C when it absorbs 1 kJ of heat energy.

Determine the specific heat capacity of iron, in $J\ g^{-1}\ K^{-1}$. Use section 1 of the data booklet. [1]

. . . . . .

. . . . . .

. . . . . .

(d) A voltaic cell is set up between the $Fe^{2+}(aq) \mid Fe(s)$ and $Fe^{3+}(aq) \mid Fe^{2+}(aq)$ half-cells.

Deduce the equation and the cell potential of the spontaneous reaction. Use section 24 of the data booklet. [2]

Equation:

. . . . . .

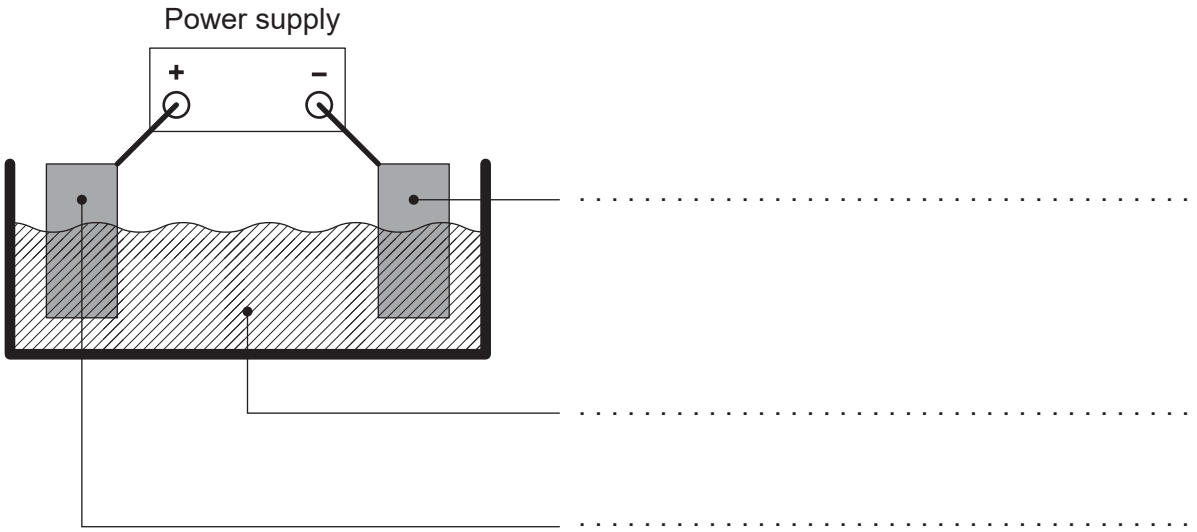
. . . . . .

Cell potential:

. . . . . .

. . . . . .

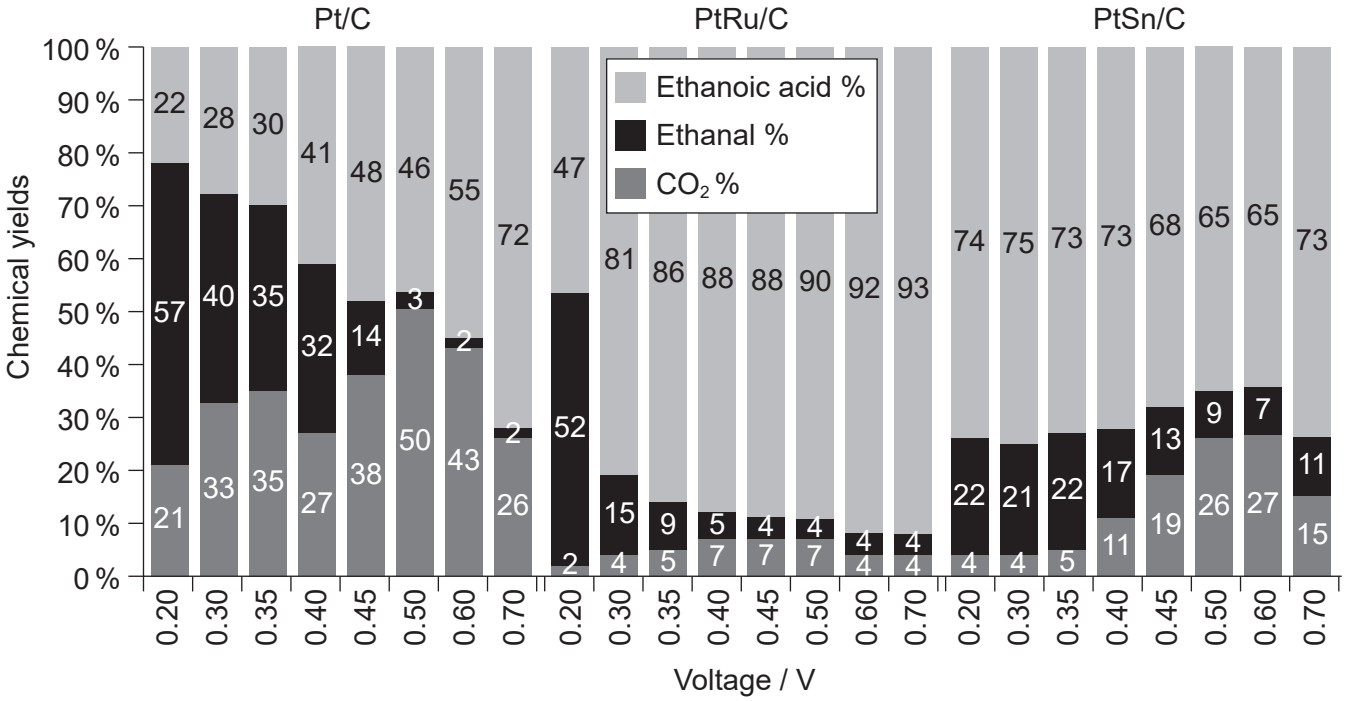
(e) The figure shows an apparatus that could be used to electroplate iron with zinc. Label the figure with the required substances. [2]



6.1-S6 Nov 2019 · Paper 3 · Q2 parts (a)(ii), (a)(iii), (b)

2. Ethanol was electrolysed at different voltages. The products at the anode, ethanoic acid, ethanal and carbon dioxide, were collected and analysed.

The percentages of products obtained using three different catalysts mounted on a carbon anode, platinum (Pt/C), platinum and ruthenium alloy (PtRu/C) and platinum and tin alloy (PtSn/C) are shown.



Chemical yields of ethanoic acid, ethanal and carbon dioxide as a function of voltage for oxidation of 0.100 mol dm^{-3} ethanol at Pt/C, PtRu/C and PtSn/C anodes at 80 °C.

[Source: Product Distributions and Efficiencies for Ethanol Oxidation in a Proton Exchange Membrane Electrolysis Cell, Rakan M. Altarawneh and Peter G. Pickup, *Journal of the Electrochemical Society*, 2017, volume **164**, issue 7, http://jes.ecsdl.org/. Distributed under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/)]

(ii) Determine the change in the average oxidation state of carbon. [2]

From ethanol to ethanal:

. . . . . .

. . . . . .

From ethanol to carbon dioxide:

. . . . . .

. . . . . .

(iii) List the three products at the anode from the least to the most oxidized. [1]

. . . . . .

. . . . . .

(b) Deduce, giving your reason, which catalyst is most effective at fully oxidizing ethanol. [1]

. . . . . .

. . . . . .

. . . . . .

6.1-S7 Nov 2018 · Paper 2 · Q6 parts (a)(iii), (d) · also 5.2

6. Butanoic acid, $CH_3CH_2CH_2COOH$, is a weak acid and ethylamine, $CH_3CH_2NH_2$, is a weak base.

(iii) Deduce the average oxidation state of carbon in butanoic acid. [1]

. . . . . .

. . . . . .

(d) Explain why butanoic acid is a liquid at room temperature while ethylamine is a gas at room temperature. [2]

. . . . . .

. . . . . .

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6.1-S8 Nov 2016 · Paper 3 · Q1 parts (a)(i)

1. In order to provide safe drinking water, a water supply is often treated with disinfectants, which aim to inactivate disease-causing bacteria in the water.

To compare the effectiveness of different disinfectants, a **CT value** is used as a measure of the dosage of disinfectant needed to achieve a certain level of inactivation of specific bacteria.

CT value ($mg\,min\,dm^{-3}$) = C ($mg\,dm^{-3}$) concentration of disinfectant × T (min) contact time with water

(a) The table below compares the CT values of different disinfectants necessary to achieve 99 % inactivation of two types of bacteria, listed as **A** and **B**.

Disinfectant	**CT value / $mg\,min\,dm^{-3}$ for 99 % inactivation of bacteria**	
	Bacterium A	**Bacterium B**
Hypochlorous acid, HOCl	4×10^{-2}	8×10^{-2}
Hypochlorite ion, OCl^-	9.2×10^{-1}	3.3
Chlorine dioxide, ClO_2	1.8×10^{-1}	1.3×10^{-1}
Monochloramine, NH_2Cl	64	94

(i) Deduce the oxidation state of chlorine in the following disinfectants. [1]

HOCl:

. . . . . .

ClO_2:

. . . . . .

6.1-S9 Nov 2015 · Paper 3 · Q9 parts (a)(ii)

9. (a) Aerobic and anaerobic respiration in human beings can be represented by the following overall equations.

Aerobic: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

Anaerobic: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$

Identify the process which: [1]

. . . . . .

(ii) is not shown by a redox equation.

. . . . . .

6.1-S10 May 2015 TZ1 · Paper 3 · Q9 parts (a)(i), (a)(ii) · also 6.2

9. The large-scale production of iron is important for the industrial development of many countries.

(a) (i) Magnetite, Fe_3O_4, is a common ore of iron. Calculate the average oxidation state of iron in the compound and comment on your answer. [2]

. . . . . .
. . . . . .

(ii) State the equation for the reduction of this ore to iron with carbon monoxide. [1]

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. . . . . .

6.1-S11 Nov 2014 · Paper 2 · Q10 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b), (c), (f)(i), (f)(ii), (g) · also 2.1, 5.2, 6.2, 7.2

Iron rusts in the presence of oxygen and water. Rusting is a redox process involving several steps that produces hydrated iron(III) oxide, $Fe_2O_3 \bullet nH_2O$, as the final product.
The half-equations involved for the first step of rusting are given below.

Half-equation 1: $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

Half-equation 2: $O_2(aq) + 4e^- + 2H_2O(l) \rightarrow 4OH^-(aq)$

(a) (i) Identify whether half-equation 1 represents oxidation or reduction, giving a reason for your answer. *[1]*

. . . . . .
. . . . . .

(ii) Identify the oxidation number of each atom in the three species in half-equation 2. *[2]*

$O_2(aq) + 4e^- + 2H_2O(l) \rightarrow 4OH^-(aq)$

☐ ☐☐ ☐☐

(iii) Deduce the overall redox equation for the first step of rusting by combining half-equations 1 and 2. *[1]*

. . . . . .
. . . . . .

(iv) Identify the reducing agent in the redox equation in part (iii). *[1]*

. . . . . .
. . . . . .

(b) The oxygen in half-equation 2 is atmospheric oxygen that is found dissolved in water in very small concentrations. Explain, in terms of intermolecular forces, why oxygen is not very soluble in water. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(c) State the relationship between the electron arrangement of an element and its group and period in the periodic table. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(f) (i) An aqueous solution of sodium chloride is electrolysed using inert electrodes. Explain which product is obtained at the positive electrode (anode) if the concentration of sodium chloride is high. *[3]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) State the half-equations occurring at the electrodes during the electrolysis of the **concentrated** aqueous solution of sodium chloride. *[2]*

Negative electrode (cathode):

. . . . . .

Positive electrode (anode):

. . . . . .

(g) Describe how electrolysis can be used to electroplate a bracelet with a layer of silver metal. Include the choice of electrodes and electrolyte needed in your description. *[3]*

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. . . . . .
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. . . . . .
. . . . . .

6.1-S12 Nov 2013 · Paper 3 · Q9 parts (b)(iv)

9. Enzymes are catalysts that increase the rate of all biochemical reactions, including those involved in respiration.

(b) Cytochrome oxidase is a complex enzyme that catalyses the reduction of oxygen in the final stage of aerobic respiration. This enzyme is inhibited both by nitrogen(II) oxide, NO, and separately by cyanide ions, CN^-. It has been suggested that NO acts competitively while CN^- acts non-competitively in inhibiting the enzyme. Experiments were carried out to test this hypothesis.

(iv) The reducing agent in the cytochrome oxidase reaction is a species that can be denoted as XH_2 in the reduced form. Using this notation, deduce an equation for the reaction of XH_2 and O_2, and outline, using oxidation numbers, why it is a redox reaction. *[2]*

. . . . . .
. . . . . .
. . . . . .

6.1-S13 May 2013 TZ1 · Paper 2 · Q4 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 6.2, 7.1

4. Ethanedioic acid (oxalic acid), $(COOH)_2$, reacts with acidified potassium permanganate solution, $KMnO_4$, according to the following equation.

$$5(COOH)_2(aq) + 2MnO_4^-(aq) + 6H^+(aq) \rightarrow 10CO_2(g) + 2Mn^{2+}(aq) + 8H_2O(l)$$

The reaction is a redox reaction.

(a) Define *oxidation* in terms of electron transfer. *[1]*

. . . . . .
. . . . . .

(b) Calculate the change in oxidation numbers of carbon and manganese. *[2]*

Carbon:

. . . . . .
. . . . . .

Manganese:

. . . . . .
. . . . . .

(c) Identify the oxidizing and reducing agents. *[1]*

Oxidizing agent:

. . . . . .

Reducing agent:

. . . . . .

(d) Deduce the half-equation involving ethanedioic acid. *[1]*

. . . . . .
. . . . . .

(e) (i) The standard electrode potential for the half-equation involving ethanedioic acid is $E^{\ominus} = -0.49\,\text{V}$. Using Table 14 of the Data Booklet, calculate the standard electrode potential for the equation on page 10. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Explain the sign of the calculated standard electrode potential. *[1]*

. . . . . .
. . . . . .

6.1-S14 Nov 2012 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii) · also 6.2, 7.2

4. Arsenic and nitrogen play a significant role in environmental chemistry. Arsenous acid, H_3AsO_3, can be found in oxygen-poor (anaerobic) water, and nitrogen-containing fertilizers can contaminate water.

(a) (i) Define *oxidation* and *reduction* in terms of electron loss or gain. *[1]*

Oxidation:

. . . . . .

Reduction:

. . . . . .

(ii) Deduce the oxidation numbers of arsenic and nitrogen in each of the following species. *[4]*

As_2O_3: . . . . . .

NO_3^-: . . . . . .

H_3AsO_3: . . . . . .

N_2O_3: . . . . . .

(iii) Distinguish between the terms *oxidizing agent* and *reducing agent*. *[1]*

. . . . . .

. . . . . .

. . . . . .

(iv) In the removal of arsenic from contaminated groundwater, H_3AsO_3 is often first oxidized to arsenic acid, H_3AsO_4.

The following **unbalanced** redox reaction shows another method of forming H_3AsO_4.

$$As_2O_3(s) + NO_3^-(aq) \rightarrow H_3AsO_4(aq) + N_2O_3(aq)$$

Deduce the balanced redox equation in **acid**, and then identify both the oxidizing and reducing agents. *[3]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(b) The electrolysis of aqueous copper(II) sulfate is an example of an electrolysis process where the nature of the electrodes can determine which products form. Platinum electrodes were used in **process 1** and copper electrodes in **process 2**.

(i) Draw an annotated diagram of the electrolytic cell in **process 1** and identify the direction of electron flow. *[2]*

(ii) For **process 1** (platinum electrodes), state the half-equations occurring at the positive electrode (anode) and negative electrode (cathode). **Include state symbols for all species**. Describe what is observed at each electrode and comment on any change in the colour and the acidity of the solution. *[7]*

Half-equation at positive electrode (anode):

. . . . . .

. . . . . .

Half-equation at negative electrode (cathode):

. . . . . .

. . . . . .

Observation at positive electrode (anode):

. . . . . .

. . . . . .

Observation at negative electrode (cathode):

. . . . . .

. . . . . .

Change in colour (if any) of the solution:

. . . . . .

. . . . . .

Change in acidity (if any) of the solution:

. . . . . .

. . . . . .

(iii) For **process 2** (copper electrodes), state the half-equations occurring at the positive electrode (anode) and negative electrode (cathode). **Include state symbols for all species**. Describe what is observed at each electrode and comment on any change in the colour and the acidity of the solution. *[7]*

Half-equation at positive electrode (anode):

. . . . . .

. . . . . .

Half-equation at negative electrode (cathode):

. . . . . .

. . . . . .

Observation at positive electrode (anode):

. . . . . .

. . . . . .

Observation at negative electrode (cathode):

. . . . . .

. . . . . .

Change in colour (if any) of the solution:

. . . . . .

. . . . . .

Change in acidity (if any) of the solution:

. . . . . .

. . . . . .

6.1-S15 Nov 2012 · Paper 3 · QD2 parts (b)(i)

D2. Physiological effects of drugs can be significantly reduced, enhanced or altered by other drugs or foods. The problem of drug interactions is particularly important for patients who consume excessive amounts of ethanol.

(b) Detection of ethanol and accurate measurement of its levels in the human body is important for the effective treatment of patients and is required by law in certain cases (such as road accidents).

(i) With reference to the half-equations below, explain in terms of electron transfer, whether dichromate(VI) ions and ethanol are reduced or oxidized in the breathalyser. *[2]*

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$C_2H_5OH + H_2O - 4e^- \rightarrow CH_3COOH + 4H^+$$

Dichromate(VI) ions:

. . . . . .

. . . . . .

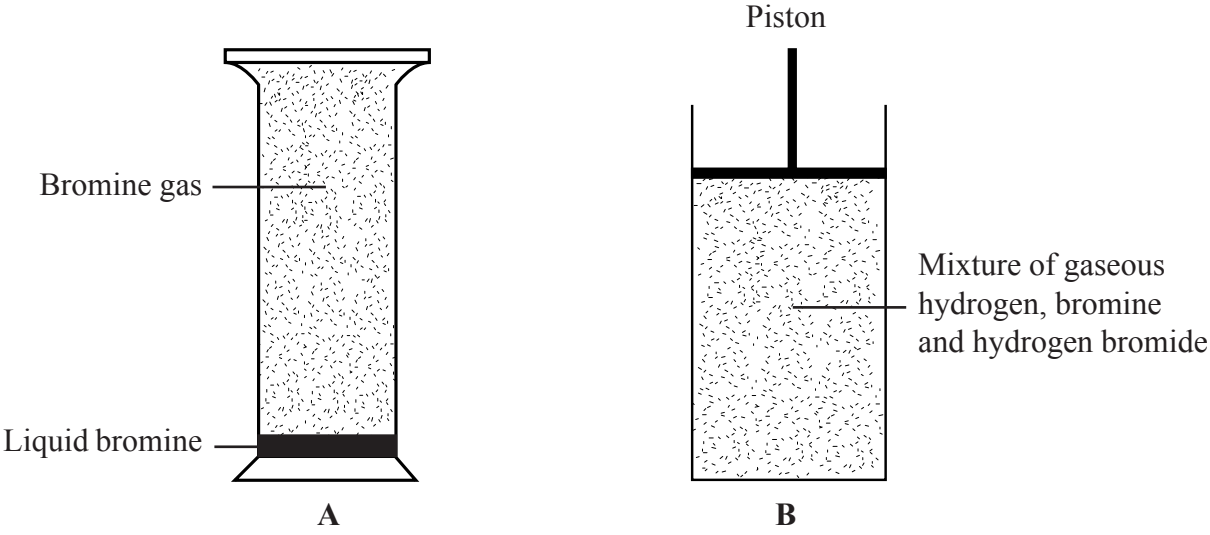
Ethanol:

. . . . . .

. . . . . .

6.1-S16 May 2012 TZ1 · Paper 2 · Q6 parts (e)(i), (e)(ii), (f)(i)

6. Consider the two equilibrium systems involving bromine gas illustrated below.



(e) When bromine dissolves in water, 1 % of the original bromine molecules react according to the following equation.

$$Br_2(aq) + H_2O(l) \rightleftharpoons HOBr(aq) + HBr(aq)$$

(i) Deduce the oxidation numbers of bromine in the reactant **and** products. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Explain the changes in the oxidation numbers of bromine. *[1]*

. . . . . .
. . . . . .

(f) Fluorine reacts with water to produce oxygen.

$$2F_2(g) + 2H_2O(l) \rightarrow 4HF(g) + O_2(g)$$

(i) Identify the oxidizing agent in the reaction. *[1]*

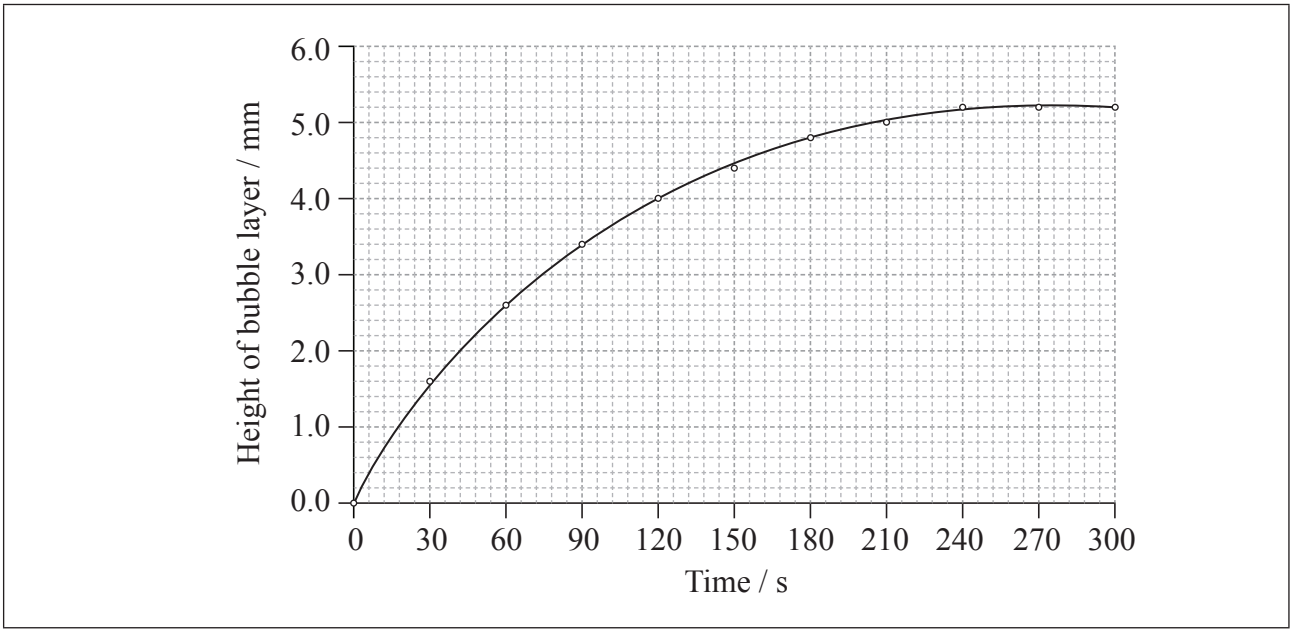
. . . . . .

6.1-S17 May 2012 TZ2 · Paper 2 · Q1 parts (c)(i), (c)(ii) · also 6.2

1. Hydrogen peroxide, $H_2O_2(aq)$, releases oxygen gas, $O_2(g)$, as it decomposes according to the equation below.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

50.0 cm^3 of hydrogen peroxide solution was placed in a boiling tube, and a drop of liquid detergent was added to create a layer of bubbles on the top of the hydrogen peroxide solution as oxygen gas was released. The tube was placed in a water bath at 75 °C and the height of the bubble layer was measured every thirty seconds. A graph was plotted of the height of the bubble layer against time.



(c) The decomposition of hydrogen peroxide to form water and oxygen is a redox reaction.

(i) Deduce the oxidation numbers of oxygen present in each of the species below. *[2]*

Species	Oxidation number of oxygen
H_2O_2	
H_2O	
O_2	

(ii) State two half-equations for the decomposition of hydrogen peroxide. *[2]*

Oxidation:

. . . . . .

Reduction:

. . . . . .

6.1-S18 May 2011 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i) · also 1.2

2. The element antimony, Sb, is usually found in nature as its sulfide ore, stibnite, Sb_2S_3. This ore was used two thousand years ago by ancient Egyptian women as a cosmetic to darken their eyes and eyelashes.

(ii) Deduce the oxidation number of antimony in stibnite. *[1]*

. . . . . .

(b) One method of extracting antimony from its sulfide ore is to roast the stibnite in air. This forms antimony oxide and sulfur dioxide. The antimony oxide is then reduced by carbon to form the free element.

(i) Deduce the chemical equations for these **two** reactions. *[2]*

. . . . . .

. . . . . .

LESSON 6.2

Balancing Redox Reactions

5 multiple-choice questions · 10 structured questions

Part A - Multiple choice

6.2-M1 Nov 2022 · Paper 1 · Q29

29. What is the coefficient for H^+ when the equation below is balanced?

$$__Pb(s) + __NO_3^-(aq) + __H^+(aq) \rightarrow __Pb^{2+}(aq) + __NO(g) + __H_2O(l)$$

A. 2

B. 4

C. 6

D. 8

6.2-M2 May 2022 TZ1 · Paper 1 · Q29

29. How many electrons are needed when the following half-equation is balanced using the lowest possible whole numbers?

$$__ NO_3^-(aq) + __ H^+(aq) + __ e^- \rightarrow __ NO(g) + __ H_2O(l)$$

A. 1

B. 2

C. 3

D. 5

6.2-M3 May 2015 TZ2 · Paper 1 · Q31

31. What is the coefficient for I^- when the following equation is balanced using the smallest possible whole numbers?

$$IO_3^-(aq) + ___ I^-(aq) + ___ H^+(aq) \rightarrow ___ I_2(aq) + ___ H_2O(l)$$

A. 1

B. 2

C. 3

D. 5

6.2-M4 May 2014 TZ1 · Paper 1 · Q30

30. At which side of the equation are electrons, H^+ ions and H_2O needed to complete the half-equation?

$$MnO_4^-(aq) \rightarrow Mn^{2+}(aq)$$

	Electrons	H^+ ions	H_2O
A.	reactant side	reactant side	product side
B.	reactant side	product side	reactant side
C.	product side	reactant side	product side
D.	product side	product side	reactant side

6.2-M5 May 2012 TZ1 · Paper 1 · Q30

30. The equation for the redox reaction between acidified dichromate and iodide ions is shown below.

$$Cr_2O_7^{2-}(aq) + 6I^-(aq) + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3I_2(aq) + 7H_2O(l)$$

Which is the reduction half-equation?

A. $6I^-(aq) + 6e^- \rightarrow 3I_2(aq)$

B. $6I^-(aq) \rightarrow 3I_2(aq) + 6e^-$

C. $Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$

D. $Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l) + 6e^-$

Part B - Structured questions

6.2-S1 May 2018 TZ2 · Paper 2 · Q4 parts (a)

4. (a) In acidic solution, bromate ions, $BrO_3^-(aq)$, oxidize iodide ions, $I^-(aq)$.

$$BrO_3^-(aq) + 6H^+(aq) + 6e^- \rightleftharpoons Br^-(aq) + 3H_2O(l)$$

$$2I^-(aq) \rightleftharpoons I_2(s) + 2e^-$$

Formulate the equation for the redox reaction. [1]

. . . . . .

. . . . . .

6.2-S2 Nov 2017 · Paper 3 · Q14 parts (a), (b) · also 6.1

14. The heme groups in cytochromes contain iron ions that are involved in the reduction of molecular oxygen.

(a) State the half-equation for the reduction of molecular oxygen to water in acidic conditions. [1]

. . . . . .

. . . . . .

(b) Outline the change in oxidation state of the iron ions in heme groups that occurs when molecular oxygen is converted to water. [1]

. . . . . .

. . . . . .

6.2-S3 May 2017 TZ2 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i) · also 1.2, 7.1

2. (a) An acidic sample of a waste solution containing $Sn^{2+}(aq)$ reacted completely with $K_2Cr_2O_7$ solution to form $Sn^{4+}(aq)$.

(i) State the oxidation half-equation. [1]

. . . . . .

(ii) Deduce the overall redox equation for the reaction between acidic $Sn^{2+}(aq)$ and $Cr_2O_7^{2-}(aq)$, using section 24 of the data booklet. [1]

. . . . . .
. . . . . .
. . . . . .

(iii) 13.239 g of $K_2Cr_2O_7(s)$ were dissolved in distilled water to form a 0.100 dm^3 solution. Calculate its molar concentration. [1]

. . . . . .
. . . . . .
. . . . . .

(iv) 10.0 cm^3 of the waste sample required 13.24 cm^3 of the $K_2Cr_2O_7$ solution. Calculate the molar concentration of $Sn^{2+}(aq)$ in the waste sample. [2]

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(b) (i) Corrosion of iron is similar to the processes that occur in a voltaic cell. The initial steps involve the following half-equations:

$$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$$

$$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$$

Calculate $E^{\ominus}$, in V, for the spontaneous reaction using section 24 of the data booklet. [1]

. . . . . .

. . . . . .

6.2-S4 May 2016 · Paper 3 · Q3 parts (a)

3. Iron may be extracted from an ore containing Fe_2O_3 in a blast furnace by reaction with coke, limestone and air. Aluminium is obtained by electrolysis of an ore containing Al_2O_3.

(a) State the overall redox equation when carbon monoxide reduces Fe_2O_3 to Fe. [1]

. . . . . .

. . . . . .

6.2-S5 May 2015 TZ1 · Paper 2 · Q7 parts (a)(ii), (a)(iii), (a)(iv), (f) · also 4.1, 6.1

(ii) Determine the oxidation number of carbon in ethanol and ethanal. [2]

Ethanol:

. . . . . .

Ethanal:

. . . . . .

(iii) Deduce the half-equation for the oxidation of ethanol to ethanal. [1]

. . . . . .

. . . . . .

(iv) Deduce the overall redox equation for the reaction of ethanol to ethanal with acidified potassium dichromate(VI). [2]

. . . . . .
. . . . . .
. . . . . .

(f) Describe the formation of σ and π bonds in an alkene. [2]

. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

6.2-S6 May 2015 TZ2 · Paper 3 · Q19 parts (a)(i), (a)(ii) · also 6.1

19. A variety of techniques can be used to determine the ethanol concentration of the breath, blood or urine.

(a) (i) The breathalyser, one of the earliest tests, uses the reaction between ethanol and acidified potassium dichromate(VI). Ethanol is first oxidized to ethanal. Deduce the half-equation for the reaction of ethanol to ethanal. [1]

. . . . . .

(ii) Outline why the colour changes from orange to green. [1]

. . . . . .
. . . . . .

6.2-S7 Nov 2014 · Paper 3 · Q24 parts (a)

24. (a) In an acidic soil, nitrate ions may undergo reduction to form ammonium ions. Deduce a half-equation for the reaction. *[3]*

. . . . . .
. . . . . .

6.2-S8 May 2012 TZ2 · Paper 3 · QF3 parts (a)

F3. The American chemist Linus Pauling, who won two Nobel prizes, promoted the taking of vitamin C as a way of preventing the common cold. One of the functions of vitamin C in the body is as an antioxidant. During the process ascorbic acid, $C_6H_8O_6$, is converted into dehydroascorbic acid, $C_6H_6O_6$.

(a) Deduce the half-equation to show how vitamin C acts as an antioxidant. *[2]*

. . . . . .

. . . . . .

6.2-S9 May 2011 TZ1 · Paper 3 · QB5 parts (a), (b)

B5. In aerobic respiration, the metabolism of glucose takes place using the processes of oxidation and reduction.

(a) Identify the molecule that undergoes oxidation and state the half-equation for the process. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(b) Identify the molecule that undergoes reduction and state the half-equation for the process. *[2]*

. . . . . .

. . . . . .

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6.2-S10 May 2010 TZ1 · Paper 3 · QC1 parts (b)(i), (b)(ii)

(b) Both carbon monoxide and hydrogen can be used to reduce iron ores. State the equations for the reduction of magnetite, Fe_3O_4, with

(i) carbon monoxide. *[1]*

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(ii) hydrogen. *[1]*

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LESSON 6.3

Reactivity

14 multiple-choice questions · 3 structured questions

Part A - Multiple choice

6.3-M1 May 2025 TZ1 · Paper 1A · Q35

35. Below is a reactivity series of selected elements, arranged from highest activity to lowest.

Zn
Fe
Cd
Co
Ni
Sn
Pb
Cu

Which reaction is spontaneous?

A. $Cd^{2+}(aq) + Pb(s) \rightarrow Pb^{2+}(aq) + Cd(s)$

B. $Fe^{2+}(aq) + Sn(s) \rightarrow Sn^{2+}(aq) + Fe(s)$

C. $Zn^{2+}(aq) + Cu(s) \rightarrow Cu^{2+}(aq) + Zn(s)$

D. $Ni^{2+}(aq) + Co(s) \rightarrow Co^{2+}(aq) + Ni(s)$

6.3-M2 May 2025 TZ3 · Paper 1A · Q36

36. The following occurs when tin is added to separate solutions of lead(II) nitrate, $Pb(NO_3)_2$, and manganese(II) nitrate, $Mn(NO_3)_2$.

$$Sn(s) + Pb(NO_3)_2(aq) \rightarrow Sn(NO_3)_2(aq) + Pb(s)$$

$$Sn(s) + Mn(NO_3)_2(aq) \rightarrow \text{no reaction}$$

What is the order of decreasing reactivity of the metals?

A. Sn > Pb > Mn

B. Sn > Mn > Pb

C. Mn > Sn > Pb

D. Pb > Sn > Mn

6.3-M3 Nov 2023 TZ1 · Paper 1 · Q29

29. The acid H_2S reacts with an active metal, M. Which combination shows the correct role of H_2S, and product formed from the reaction?

	Role of H_2S	Product from H_2S reaction
A.	Oxidizing agent	$H_2(g)$
B.	Oxidizing agent	$S(s)$
C.	Reducing agent	$H_2(g)$
D.	Reducing agent	$S(s)$

6.3-M4 May 2021 TZ1 · Paper 1 · Q28

28. A student performed displacement reactions using metals W and X and solutions of salts of metals W, X, Y and Z. The results are summarized in the table.

		Salt solution			
		W^{2+}	X^{2+}	Y^{2+}	Z^{2+}
Metal	W		No reaction	No reaction	No reaction
	X	Reaction		Reaction	No reaction

Which of the four metals is most reactive?

A. W

B. X

C. Y

D. Z

6.3-M5 Nov 2019 · Paper 1 · Q30

30. The following occurs when metal **X** is added to **Y** sulfate solution and **Z** sulfate solution. (**X**, **Y** and **Z** represent metal elements but not their symbols.)

$$\mathbf{X}(s) + \mathbf{Y}SO_4(aq) \rightarrow \mathbf{X}SO_4(aq) + \mathbf{Y}(s)$$

$$\mathbf{X}(s) + \mathbf{Z}SO_4(aq): \text{ no reaction}$$

What is the order of increasing reactivity?

A. $X < Y < Z$

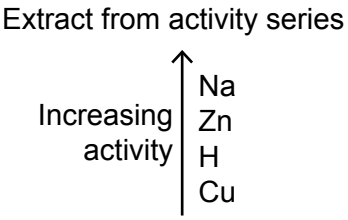
B. $Y < X < Z$

C. $Z < Y < X$

D. $Z < X < Y$

6.3-M6 May 2017 TZ2 · Paper 1 · Q24

24. Which of the following does **not** react with dilute $HCl(aq)$?



A. Na_2CO_3

B. Cu

C. Zn

D. CuO

6.3-M7 May 2013 TZ1 · Paper 1 · Q7

7. Which statements are correct?

I. Fluorine will react with potassium chloride solution to produce chlorine.

II. Iodine will react with sodium chloride solution to produce chlorine.

III. Bromine will react with lithium iodide solution to produce iodine.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.3-M8 Nov 2012 · Paper 1 · Q7

7. Which equation represents a reaction that occurs under normal conditions?

A. $2LiBr(aq) + I_2(aq) \rightarrow 2LiI(aq) + Br_2(aq)$

B. $2KF(aq) + Cl_2(aq) \rightarrow 2KCl(aq) + F_2(aq)$

C. $2LiCl(aq) + I_2(aq) \rightarrow 2LiI(aq) + Cl_2(aq)$

D. $2KBr(aq) + Cl_2(aq) \rightarrow 2KCl(aq) + Br_2(aq)$

6.3-M9 May 2012 TZ1 · Paper 1 · Q25

25. An aqueous solution **X** reacts with a solid **Y**, to produce a flammable gas. Which of the following suggestions could substances **X** and **Y** be?

	X	**Y**
A.	nitric acid, HNO_3	calcium carbonate, $CaCO_3$
B.	sulfuric acid, H_2SO_4	zinc, Zn
C.	hydrochloric acid, HCl	copper, Cu
D.	sodium hydroxide solution, NaOH	aluminum oxide, Al_2O_3

6.3-M10 Nov 2011 · Paper 1 · Q8

8. Which of the following redox reactions take place?

I. $Cl_2(aq) + 2NaI(aq) \rightarrow I_2(aq) + 2NaCl(aq)$

II. $Br_2(aq) + 2NaI(aq) \rightarrow I_2(aq) + 2NaBr(aq)$

III. $I_2(aq) + 2NaBr(aq) \rightarrow Br_2(aq) + 2NaI(aq)$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.3-M11 Nov 2011 · Paper 1 · Q30

30. What is the correct **decreasing** order of reactivity of the metals X, Y and Z based on the following equations?

$$XCl + Y \rightarrow YCl + X$$
$$YCl + Z \rightarrow YCl + Z$$
$$ZCl + X \rightarrow XCl + Z$$

A. X > Y > Z

B. Y > Z > X

C. Z > Y > X

D. Y > X > Z

6.3-M12 May 2011 TZ1 · Paper 1 · Q8

8. Which statement about the elements in group 7 is correct?

A. Br_2 will oxidize Cl^-.

B. F_2 has the least tendency to be reduced.

C. Cl_2 will oxidize I^-.

D. I_2 is a stronger oxidizing agent than F_2.

6.3-M13 May 2011 TZ2 · Paper 1 · Q31

31. Consider the following reactions of three unknown metals X, Y and Z.

$$2XNO_3(aq) + Y(s) \rightarrow 2X(s) + Y(NO_3)_2(aq)$$

$$Y(NO_3)_2(aq) + Z(s) \rightarrow \text{No reaction}$$

$$2XNO_3(aq) + Z(s) \rightarrow 2X(s) + Z(NO_3)_2(aq)$$

What is the order of **increasing** reactivity of the metals (least reactive first)?

A. X < Y < Z

B. X < Z < Y

C. Z < Y < X

D. Y < Z < X

6.3-M14 Nov 2010 · Paper 1 · Q31

31. The following equations indicate reactions that occur spontaneously.

$$Fe(s) + NiCl_2(aq) \rightarrow FeCl_2(aq) + Ni(s)$$

$$Zn(s) + FeCl_2(aq) \rightarrow ZnCl_2(aq) + Fe(s)$$

$$Ni(s) + PbCl_2(aq) \rightarrow NiCl_2(aq) + Pb(s)$$

Which is the **increasing** order of the reactivity of the metals?

A. Fe < Ni < Zn < Pb

B. Pb < Ni < Fe < Zn

C. Ni < Zn < Pb < Fe

D. Zn < Fe < Ni < Pb

Part B - Structured questions

6.3-S1 May 2022 TZ1 · Paper 2 · Q2 parts (a), (b)(i), (b)(iii) · also 7.1, 7.2

2. Magnesium is a reactive metal often found in alloys.

(a) Suggest an experiment that shows that magnesium is more reactive than zinc, giving the observation that would confirm this. [2]

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(b) Magnesium is sometimes used as a sacrificial anode to protect steel from corrosion.

(i) Calculate the standard potential, in V, of a cell formed by magnesium and steel half-cells. Use section 24 of the data booklet and assume steel has the standard electrode potential of iron. [1]

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(iii) This cell causes the electrolytic reduction of water on the steel. State the half-equation for this reduction. [1]

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6.3-S2 May 2017 TZ2 · Paper 3 · Q5 parts (b)(i), (b)(ii)

5. Rhodium and palladium are often used together in catalytic converters. Rhodium is a good reduction catalyst whereas palladium is a good oxidation catalyst.

(b) (i) Nickel is also used as a catalyst. It is processed from an ore until nickel(II) chloride solution is obtained. Identify **one** metal, using sections 24 and 25 of the data booklet, which will not react with water and can be used to extract nickel from the solution. [1]

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(ii) Deduce the redox equation for the reaction of nickel(II) chloride solution with the metal identified in (b)(i). [1]

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6.3-S3 Nov 2015 · Paper 2 · Q7 parts (d)(i), (d)(ii), (e) · also 7.2

(d) Consider the following spontaneous reactions.

$$Fe(s) + Ni^{2+}(aq) \rightarrow Fe^{2+}(aq) + Ni(s)$$

$$Zn(s) + Fe^{2+}(aq) \rightarrow Zn^{2+}(aq) + Fe(s)$$

$$Ni(s) + Pb^{2+}(aq) \rightarrow Ni^{2+}(aq) + Pb(s)$$

(i) Deduce the order of **increasing** reactivity of the metals based on the reactions above. [2]

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(ii) Identify the strongest oxidizing agent in the reactions above. [1]

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(e) Deduce the half-equations for the formation of the major product at the positive electrode (anode) when the following aqueous solutions are electrolysed. [2]

Dilute sodium chloride:

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Concentrated sodium chloride:

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LESSON 7.1

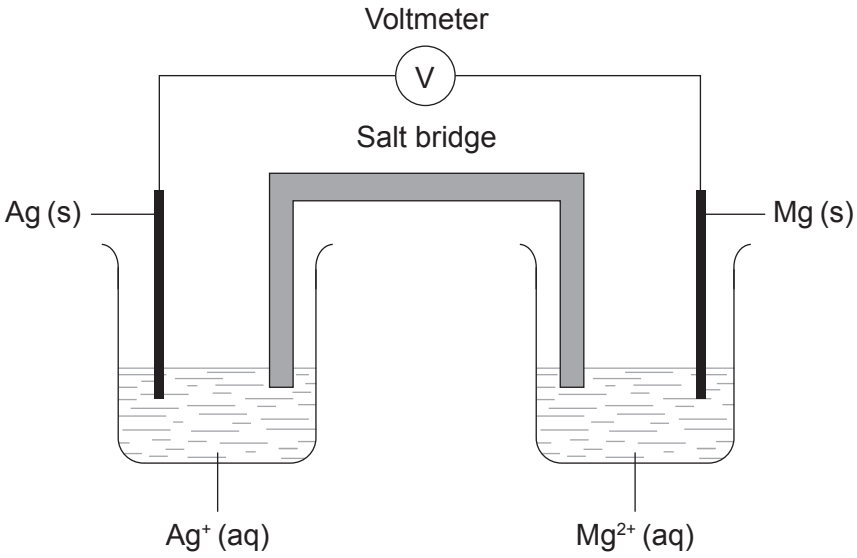
Primary & Secondary Cells

57 multiple-choice questions · 36 structured questions

Part A - Multiple choice

7.1-M1 May 2025 TZ2 · Paper 1A · Q37

37. What is the standard cell potential, $E^{\ominus}_{cell}$, of the following cell?



$Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s)$ $\quad E^{\ominus} = +0.80\,V$

$Mg^{2+}(aq) + 2e^{-} \rightleftharpoons Mg(s)$ $\quad E^{\ominus} = -2.37\,V$

A. $-3.17\,V$

B. $-1.57\,V$

C. $+1.57\,V$

D. $+3.17\,V$

7.1-M2 May 2025 TZ3 · Paper 1A · Q39

39. Consider the following standard electrode potentials:

	$E^{\ominus}$ (V)
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.45
$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$	−0.26

Which species will react with each other spontaneously under standard conditions?

A. Zn and Fe^{2+}

B. Fe and Zn^{2+}

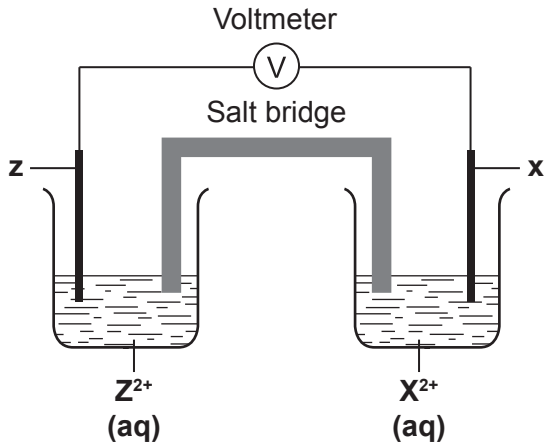
C. Ni and Zn^{2+}

D. Ni and Fe^{2+}

7.1-M3 Nov 2024 · Paper 1 · Q29

29. A voltaic cell is constructed from half-cells using metals **X** and **Z** as electrodes. What will occur when this cell produces electricity?

$\mathbf{X}^{2+}(aq) + 2e^- \rightleftharpoons \mathbf{X}(s)$ $E^{\ominus} = +0.25$ v
$\mathbf{Z}^{2+}(aq) + 2e^- \rightleftharpoons \mathbf{Z}(s)$ $E^{\ominus} = -0.60$ v



A. Positive ions flow through the salt bridge to **Z** half-cell.

B. Electrons flow from **X** half-cell to **Z** half-cell.

C. The concentration of $\mathbf{X}^{2+}$ increases.

D. Mass of electrode **Z** decreases.

7.1-M4 Nov 2024 · Paper 1 · Q31

31. Which statement is correct for the value of $E^{\ominus}$ of the standard hydrogen electrode (SHE)?

A. It was determined experimentally.

B. It depends on the temperature.

C. It was arbitrarily set.

D. It depends on the $E^{\ominus}$ of the second electrode.

7.1-M5 May 2024 TZ1 · Paper 1 · Q29

29. Which statement is correct for a voltaic cell?

A. Reduction occurs at the negative electrode.

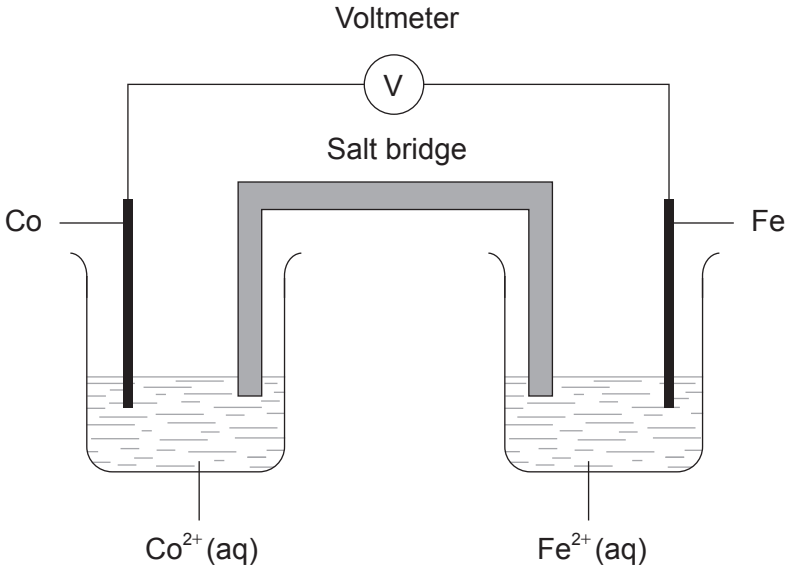
B. Electrical energy is converted to chemical energy.

C. The anode is the positive electrode.

D. Reduction occurs at the cathode.

7.1-M6 May 2024 TZ2 · Paper 1 · Q31

31. Iron is a more reactive metal than cobalt. Which statement is correct about the voltaic cell below?



A. Electrons flow from cobalt to iron in the wire.

B. Negative ions flow through the salt bridge to the iron half-cell.

C. The mass of the cobalt electrode decreases.

D. Reduction occurs at the iron electrode.

7.1-M7 May 2023 TZ1 · Paper 1 · Q30

30. What is the standard potential difference of this cell?

$$Fe(s) \mid Fe^{2+}(aq) \parallel Ag^{+}(aq) \mid Ag(s)$$

Reaction	$E^{\ominus}$
$Fe^{2+}(aq) + 2e^{-} \rightleftharpoons Fe(s)$	$-0.45\,V$
$Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s)$	$+0.80\,V$

A. $0.45 + 2(0.80)$

B. $0.45 + 0.80$

C. $-0.45 - 2(0.80)$

D. $-0.45 - 0.80$

7.1-M8 May 2023 TZ1 · Paper 1 · Q31

31. Which statement is correct about the ions in a cell assembled from these half-cells?

Reaction	$E^{\ominus}$
$Ni^{2+}(aq) + 2e^{-} \rightleftharpoons Ni(s)$	$-0.26\,V$
$Zn^{2+}(aq) + 2e^{-} \rightleftharpoons Zn(s)$	$-0.76\,V$

A. Negative ions flow into the zinc half-cell from the salt bridge.

B. Negative ions flow into the nickel half-cell from the salt bridge.

C. Zn^{2+} ions are reduced to Zn.

D. The concentration of Ni^{2+} ions increases.

7.1-M9 May 2023 TZ2 · Paper 1 · Q29

29. Which chemical process would produce a voltaic cell?

A. spontaneous redox reaction

B. spontaneous non-redox reaction

C. non-spontaneous redox reaction

D. non-spontaneous non-redox reaction

7.1-M10 May 2023 TZ2 · Paper 1 · Q30

30. Which combination of potential (voltage) is correct regarding the reduction and oxidation of the standard hydrogen electrode (SHE)?

	Reduction potential	Oxidation potential
A.	zero	zero
B.	zero	positive
C.	zero	negative
D.	negative	positive

7.1-M11 Nov 2022 · Paper 1 · Q28

28. What occurs during the operation of a voltaic cell based on the given reaction?

$$2Cr(s) + 3Fe^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 3Fe(s)$$

	External circuit	Ion movement in solution
A.	Electrons move from Cr to Fe	$Fe^{2+}(aq)$ move away from $Fe(s)$
B.	Electrons move from Cr to Fe	$Fe^{2+}(aq)$ move toward $Fe(s)$
C.	Electrons move from Fe to Cr	$Cr^{3+}(aq)$ move away from $Cr(s)$
D.	Electrons move from Fe to Cr	$Cr^{3+}(aq)$ move toward $Cr(s)$

7.1-M12 Nov 2022 · Paper 1 · Q30

30. The standard electrode potentials for three half-cells involving chromium are shown.

$$Cr^{3+}(aq) + e^- \rightleftharpoons Cr^{2+}(aq) \qquad E^\ominus = -0.407\,V$$

$$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s) \qquad E^\ominus = -0.744\,V$$

$$Cr^{2+}(aq) + 2e^- \rightleftharpoons Cr(s) \qquad E^\ominus = -0.914\,V$$

Which statement is correct?

A. $Cr^{3+}(aq)$ can oxidize $Cr^{2+}(aq)$ but not $Cr(s)$.

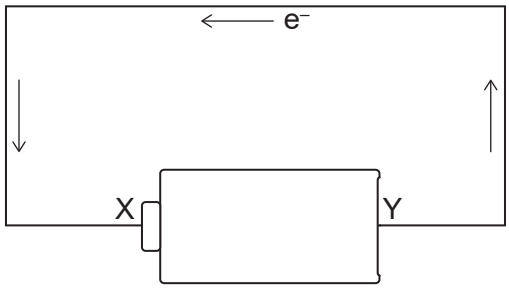
B. $Cr^{3+}(aq)$ can oxidize $Cr(s)$ but not $Cr^{2+}(aq)$.

C. $Cr^{3+}(aq)$ can oxidize both $Cr^{2+}(aq)$ and $Cr(s)$.

D. $Cr^{3+}(aq)$ can oxidize $Cr(s)$ and reduce $Cr^{2+}(aq)$.

7.1-M13 May 2022 TZ2 · Paper 1 · Q29

29. The arrows represent electron flow in the diagram. What does terminal X on the battery represent?



A. Anode and positive terminal

B. Anode and negative terminal

C. Cathode and positive terminal

D. Cathode and negative terminal

7.1-M14 May 2022 TZ2 · Paper 1 · Q30

30. Which $E^{\ominus}$ value, in V, for the reaction $Mn(s) + Zn^{2+}(aq) \rightarrow Mn^{2+}(aq) + Zn(s)$ can be deduced from the following equations?

$$Mn(s) + 2Ag^{+}(aq) \rightarrow Mn^{2+}(aq) + 2Ag(s) \qquad E^{\ominus} = 1.98\,V$$

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s) \qquad E^{\ominus} = 1.10\,V$$

$$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s) \qquad E^{\ominus} = 0.46\,V$$

A. 0.42

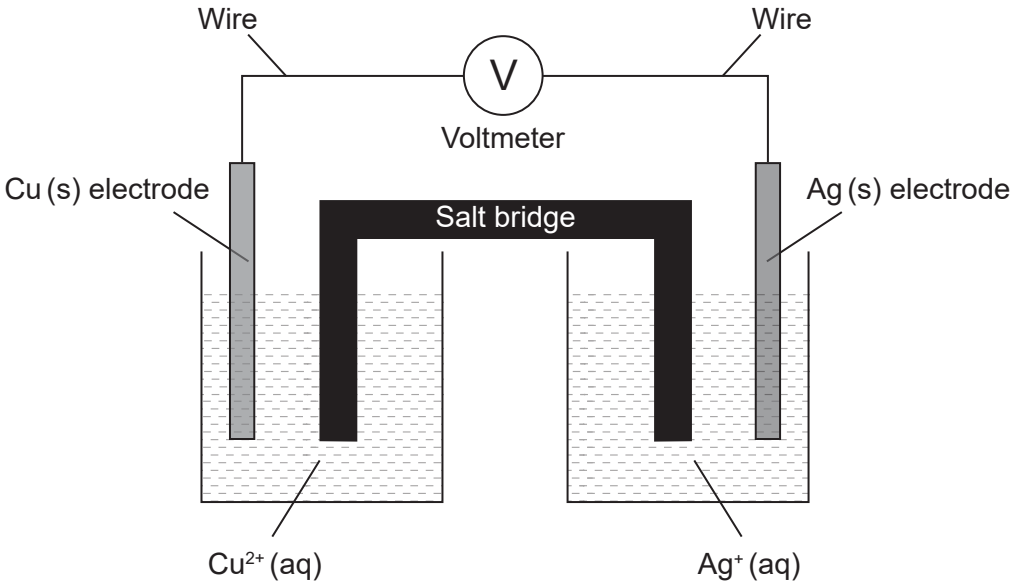
B. 1.34

C. 2.62

D. 3.54

7.1-M15 Nov 2021 · Paper 1 · Q29

29. Consider this voltaic cell, where Cu is a more reactive metal than Ag:



Which combination describes the movement of charge in this cell?

	Flow of electrons in wire	Flow of negative ions in salt bridge
A.	Ag (s) to Cu (s)	Toward Ag^+ (aq)
B.	Cu (s) to Ag (s)	Toward Ag^+ (aq)
C.	Ag (s) to Cu (s)	Toward Cu^{2+} (aq)
D.	Cu (s) to Ag (s)	Toward Cu^{2+} (aq)

7.1-M16 Nov 2021 · Paper 1 · Q30

30. Consider the following standard electrode potentials:

Half-equation	$E^\ominus$ / V
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76
$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s)$	−0.13
$\frac{1}{2}Br_2(l) + e^- \rightleftharpoons Br^-(aq)$	+1.09

Which species will react with each other spontaneously under standard conditions?

A. $Zn^{2+}(aq) + Pb(s)$

B. $Pb^{2+}(aq) + Br_2(l)$

C. $Zn(s) + Br^-(aq)$

D. $Pb(s) + Br_2(l)$

7.1-M17 May 2021 TZ1 · Paper 1 · Q30

30. Which gives the equation and cell potential of the spontaneous reaction?

	$E^{\ominus}$ / V
$Mn^{2+}(aq) + 2e^{-} \rightleftharpoons Mn(s)$	−1.18
$Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s)$	+0.80

		$E^{\ominus}$ / V
A.	$Mn^{2+}(aq) + 2Ag(s) \rightarrow Mn(s) + 2Ag^{+}(aq)$	−1.98
B.	$Mn^{2+}(aq) + 2Ag(s) \rightarrow Mn(s) + 2Ag^{+}(aq)$	+0.38
C.	$Mn(s) + 2Ag^{+}(aq) \rightarrow Mn^{2+}(aq) + 2Ag(s)$	−0.38
D.	$Mn(s) + 2Ag^{+}(aq) \rightarrow Mn^{2+}(aq) + 2Ag(s)$	+1.98

7.1-M18 May 2021 TZ2 · Paper 1 · Q30

30. What would be the electrode potential, $E^{\ominus}$, of the $Mn^{2+}(aq)|Mn(s)$ half-cell if $Fe^{3+}(aq)|Fe^{2+}(aq)$ is used as the reference standard?

$$Mn^{2+}(aq) + 2e^{-} \rightleftharpoons Mn(s) \qquad E^{\ominus} = -1.18\,V$$

$$Fe^{3+}(aq) + e^{-} \rightleftharpoons Fe^{2+}(aq) \qquad E^{\ominus} = +0.77\,V$$

A. −1.95 V

B. −0.41 V

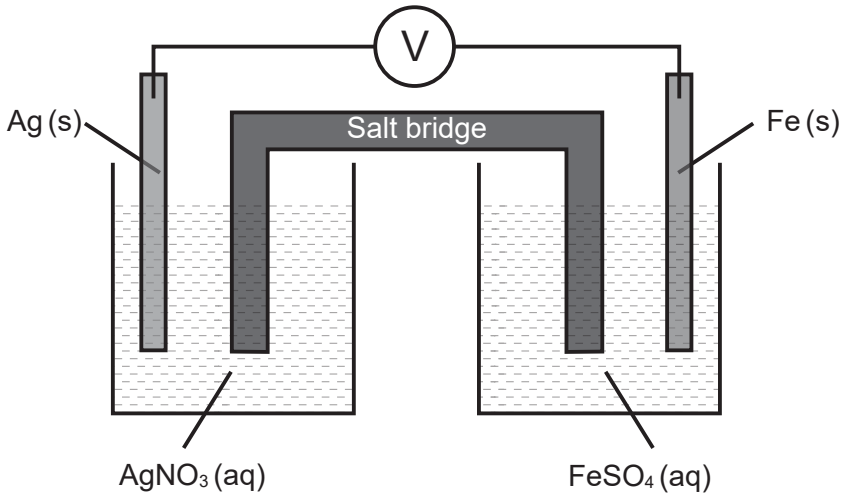
C. +0.41 V

D. +1.95 V

7.1-M19 Nov 2020 · Paper 1 · Q29

29. Iron is a stronger reducing agent than silver.

What is correct when this voltaic cell is in operation?



	Anode (negative electrode)	Cathode (positive electrode)	Direction of electron flow in wire
A.	Ag	Fe	right to left
B.	Ag	Fe	left to right
C.	Fe	Ag	left to right
D.	Fe	Ag	right to left

7.1-M20 May 2019 TZ1 · Paper 1 · Q29

29. Where does oxidation occur in a voltaic cell?

A. positive electrode and anode

B. negative electrode and anode

C. positive electrode and cathode

D. negative electrode and cathode

7.1-M21 May 2019 TZ1 · Paper 1 · Q31

31. Which is **not** a requirement of the standard hydrogen electrode (SHE)?

A. $V = 1\,\text{dm}^3$

B. $p(\text{H}_2) = 100\,\text{kPa}$

C. use of platinum as the electrode material

D. $[\text{H}_3\text{O}^+] = 1\,\text{mol dm}^{-3}$

7.1-M22 May 2019 TZ2 · Paper 1 · Q29

29. The following reaction occurs in a voltaic (galvanic) cell.

$$Mg(s) + 2Ag^{+}(aq) \rightarrow Mg^{2+}(aq) + 2Ag(s)$$

Which reaction takes place at each electrode?

	Anode (negative electrode)	Cathode (positive electrode)
A.	$Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$	$Mg^{2+}(aq) + 2e^{-} \rightarrow Mg(s)$
B.	$Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$	$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-}$
C.	$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-}$	$Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$
D.	$Mg^{2+}(aq) + 2e^{-} \rightarrow Mg(s)$	$Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$

7.1-M23 May 2019 TZ2 · Paper 1 · Q30

30. Consider the following table of standard electrode potentials.

Reaction	$E^{\ominus}$ / V
$Al^{3+}(aq) + 3e^{-} \rightleftharpoons Al(s)$	−1.66
$Pb^{2+}(aq) + 2e^{-} \rightleftharpoons Pb(s)$	−0.13

Which is the strongest oxidizing agent?

A. Pb^{2+}

B. Pb

C. Al^{3+}

D. Al

7.1-M24 Nov 2018 · Paper 1 · Q29

29. Which describes the flow of electrons in a voltaic cell?

A. From the cathode (positive electrode) to the anode (negative electrode) through the external circuit

B. From the anode (negative electrode) to the cathode (positive electrode) through the external circuit

C. From the oxidizing agent to the reducing agent through the salt bridge

D. From the reducing agent to the oxidizing agent through the salt bridge

7.1-M25 Nov 2018 · Paper 1 · Q31

31. Consider the standard electrode potentials:

$$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s) \qquad E^\ominus = -0.74\,V$$
$$Hg^{2+}(aq) + 2e^- \rightleftharpoons Hg(l) \qquad E^\ominus = +0.85\,V$$

What is the cell potential, in V, for the voltaic cell?

$$2Cr(s) + 3Hg^{2+}(aq) \rightarrow 3Hg(l) + 2Cr^{3+}(aq)$$

A. −1.59

B. +0.11

C. +1.07

D. +1.59

7.1-M26 May 2017 TZ1 · Paper 1 · Q30

30. Which statement is correct for the overall reaction in a voltaic cell?

$$2AgNO_3(aq) + Ni(s) \rightarrow 2Ag(s) + Ni(NO_3)_2(aq) \qquad E^\ominus = +1.06\,V$$

A. Electrons flow from Ag electrode to Ni electrode.

B. Ni is oxidized to Ni^{2+} at the cathode (negative electrode).

C. Ag^+ is reduced to Ag at the anode (positive electrode).

D. Ag has a more positive standard electrode potential value than Ni.

7.1-M27 May 2017 TZ2 · Paper 1 · Q30

30. What is the standard half-cell potential of copper if the “zero potential reference electrode” is changed from the standard hydrogen electrode to a standard zinc electrode?

	$E^\ominus$ / V with respect to the standard hydrogen electrode
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34

A. −1.1

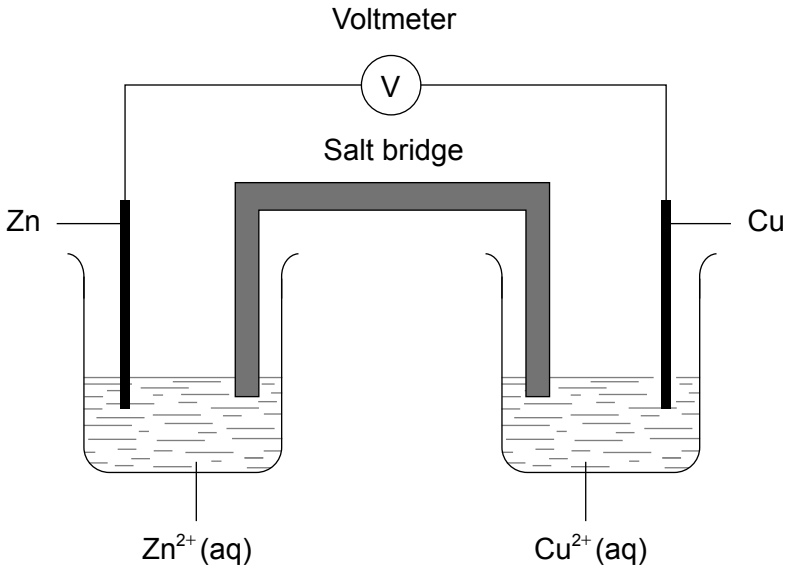
B. −0.34

C. +0.34

D. +1.1

7.1-M28 Nov 2016 · Paper 1 · Q31

31. A voltaic cell is constructed from zinc and copper half-cells. Zinc is more reactive than copper. Which statement is correct when this cell produces electricity?



A. Electrons flow from the copper half-cell to the zinc half-cell.

B. The concentration of Cu^{2+} (aq) increases.

C. Electrons flow through the salt bridge.

D. Negative ions flow through the salt bridge from the copper half-cell to the zinc half-cell.

7.1-M29 May 2016 · Paper 1 · Q31

31. Which statement is correct for a voltaic but **not** for an electrolytic cell?

A. An electrolyte is required.

B. The anode is where oxidation occurs.

C. Ions move in the electrolyte.

D. Electrons flow from the negative electrode to the positive electrode.

7.1-M30 Nov 2015 · Paper 1 · Q32

32. What is $E^{\ominus}$, in V, for the following reaction?

$$VO^{2+}(aq) + 2H^{+}(aq) + V^{2+}(aq) \rightarrow 2V^{3+}(aq) + H_2O(l)$$

	Standard electrode potential, $E^{\ominus}$ / V
$V^{2+}(aq) + 2e^{-} \rightleftharpoons V(s)$	−1.18
$V^{3+}(aq) + e^{-} \rightleftharpoons V^{2+}(aq)$	−0.26
$VO^{2+}(aq) + 2H^{+}(aq) + e^{-} \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34
$VO_2^{+}(aq) + 2H^{+}(aq) + e^{-} \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+1.00

A. −0.60

B. +0.08

C. +0.60

D. +1.26

7.1-M31 May 2015 TZ1 · Paper 1 · Q31

31. Two half-cells are connected via a salt bridge to make a voltaic cell. Which statement about this cell is correct?

A. Oxidation occurs at the positive electrode (cathode).

B. It is also known as an electrolytic cell.

C. Ions flow through the salt bridge.

D. It requires a power supply to operate.

7.1-M32 May 2015 TZ1 · Paper 1 · Q33

33. Consider the standard electrode potentials:

$$Fe^{2+}(aq) + 2e^{-} \rightleftharpoons Fe(s) \qquad E^{\ominus} = -0.45\,V$$

$$\tfrac{1}{2}Cl_2(g) + e^{-} \rightleftharpoons Cl^{-}(aq) \qquad E^{\ominus} = +1.36\,V$$

What is the standard cell potential, in V, for the reaction?

$$Cl_2(g) + Fe(s) \rightarrow 2Cl^{-}(aq) + Fe^{2+}(aq)$$

A. +0.91

B. +1.81

C. +2.27

D. +3.17

7.1-M33 May 2015 TZ2 · Paper 1 · Q32

32. The standard electrode potentials for three reactions involving copper and copper ions are:

$$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq) \qquad E^\ominus = +0.15\,V$$
$$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \qquad E^\ominus = +0.34\,V$$
$$Cu^+(aq) + e^- \rightleftharpoons Cu(s) \qquad E^\ominus = +0.52\,V$$

Which statement is correct?

A. Cu^{2+} ions are a better oxidizing agent than Cu^+ ions.

B. Copper metal is a better reducing agent than Cu^+ ions.

C. Cu^+ ions will spontaneously form copper metal and Cu^{2+} ions in solution.

D. Copper metal can be spontaneously oxidized by Cu^{2+} ions to form Cu^+ ions.

7.1-M34 Nov 2014 · Paper 1 · Q32

32. Consider the following standard electrode potentials.

$$Sn^{2+}(aq) + 2e^- \rightleftharpoons Sn(s) \qquad E^\ominus = -0.14\,V$$
$$H^+(aq) + e^- \rightleftharpoons \tfrac{1}{2}H_2(g) \qquad E^\ominus = 0.00\,V$$
$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\ominus = +0.77\,V$$

Which species will reduce $H^+(aq)$ to $H_2(g)$ under standard conditions?

A. $Fe^{2+}(aq)$

B. $Sn^{2+}(aq)$

C. $Sn(s)$

D. $Fe^{3+}(aq)$

7.1-M35 May 2014 TZ1 · Paper 1 · Q33

33. The overall equation of a voltaic cell is:

$$Ni(s) + 2Ag^+(aq) \rightleftharpoons Ni^{2+}(aq) + 2Ag(s) \qquad E^\ominus = 1.06\,V$$

The standard electrode potential for $Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$, is $-0.26\,V$. What is the standard electrode potential for the silver half-cell, $Ag^+(aq) + e^- \rightleftharpoons Ag(s)$, in V?

A. −1.32

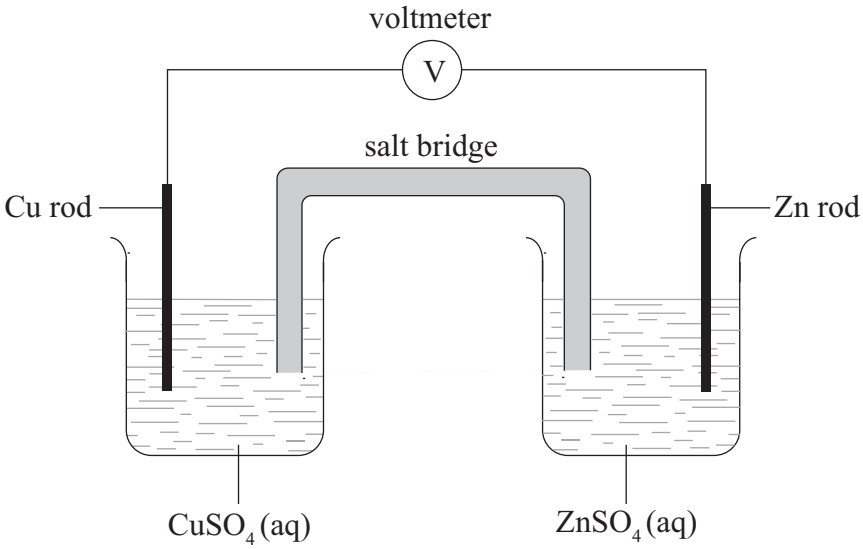
B. −0.80

C. +0.80

D. +1.32

7.1-M36 May 2014 TZ2 · Paper 1 · Q32

32. Zinc is more reactive than copper. In this voltaic cell, which species is reduced and in which direction do negative ions flow in the salt bridge?



	Species reduced	Direction of negative ion flow in salt bridge
A.	Cu^{2+}	from copper half-cell to zinc half-cell
B.	Cu^{2+}	from zinc half-cell to copper half-cell
C.	Zn^{2+}	from copper half-cell to zinc half-cell
D.	Zn^{2+}	from zinc half-cell to copper half-cell

7.1-M37 May 2014 TZ2 · Paper 1 · Q33

33. Which components are used to make the standard hydrogen electrode?

A. $H_2(g)$, $H^+(aq)$, $Pt(s)$

B. $H_2(g)$, $H^+(aq)$, $Ni(s)$

C. $H_2(g)$, $HO^-(aq)$, $Pt(s)$

D. $H_2(g)$, $HO^-(aq)$, $Ni(s)$

7.1-M38 May 2014 TZ2 · Paper 1 · Q34

34. What is the cell potential, in V, of the reaction below?

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

$$\tfrac{1}{2}S_4O_6^{2-}(aq) + e^- \rightleftharpoons S_2O_3^{2-}(aq) \qquad E^\ominus = +0.09\,V$$

$$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq) \qquad E^\ominus = +0.54\,V$$

A. $+0.63$

B. $+0.45$

C. -0.45

D. -0.63

7.1-M39 Nov 2013 · Paper 1 · Q33

33. Consider the following two standard electrode potentials at 298 K.

$$Sn^{2+}(aq) + 2e^- \rightleftharpoons Sn(s) \qquad E^\ominus = -0.14\,V$$

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\ominus = +0.77\,V$$

What is the equation and cell potential for the spontaneous reaction that occurs?

A. $2Fe^{2+}(aq) + Sn^{2+}(aq) \rightarrow 2Fe^{3+}(aq) + Sn(s) \qquad E^\ominus = -0.91\,V$

B. $2Fe^{3+}(aq) + Sn(s) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq) \qquad E^\ominus = +0.91\,V$

C. $2Fe^{2+}(aq) + Sn^{2+}(aq) \rightarrow 2Fe^{3+}(aq) + Sn(s) \qquad E^\ominus = +0.91\,V$

D. $2Fe^{3+}(aq) + Sn(s) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq) \qquad E^\ominus = +1.68\,V$

7.1-M40 May 2013 TZ1 · Paper 1 · Q32

32. Which are correct statements about a voltaic cell?

I. A spontaneous redox reaction occurs which converts chemical energy to electrical energy.

II. Oxidation occurs at the negative electrode (anode).

III. Electricity is conducted by the movement of electrons through the salt bridge.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M41 May 2013 TZ1 · Paper 1 · Q33

33. The standard electrode potentials of some half-reactions are given below.

$$\mathrm{Sn^{4+}(aq) + 2e^- \rightleftharpoons Sn^{2+}(aq)} \qquad E^{\ominus} = +0.15\ \mathrm{V}$$
$$\mathrm{\tfrac{1}{2}I_2(s) + e^- \rightleftharpoons I^-(aq)} \qquad E^{\ominus} = +0.54\ \mathrm{V}$$
$$\mathrm{Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)} \qquad E^{\ominus} = +0.77\ \mathrm{V}$$

Which of the following reactions will occur spontaneously?

A. Iodine reduces $\mathrm{Fe^{3+}}$ to $\mathrm{Fe^{2+}}$

B. Iodine reduces $\mathrm{Sn^{4+}}$ to $\mathrm{Sn^{2+}}$

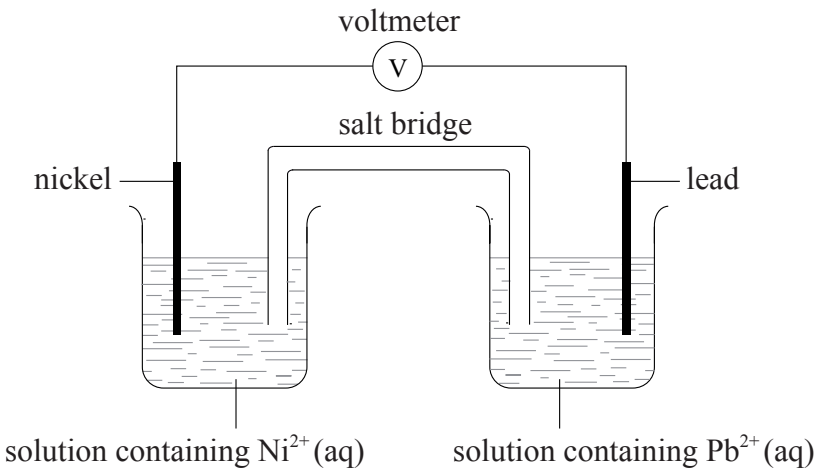
C. Iodine oxidizes $\mathrm{Fe^{2+}}$ to $\mathrm{Fe^{3+}}$

D. Iodine oxidizes $\mathrm{Sn^{2+}}$ to $\mathrm{Sn^{4+}}$

7.1-M42 May 2013 TZ2 · Paper 1 · Q31

31. The overall reaction in the voltaic cell below is:

$$\mathrm{Ni(s) + Pb^{2+}(aq) \rightarrow Ni^{2+}(aq) + Pb(s)}$$



Which statement is correct for the nickel half-cell?

A. Nickel is the positive electrode (cathode) and is reduced.

B. Nickel is the negative electrode (anode) and is reduced.

C. Nickel is the positive electrode (cathode) and is oxidized.

D. Nickel is the negative electrode (anode) and is oxidized.

7.1-M43 Nov 2012 · Paper 1 · Q31

31. A voltaic cell is made by connecting zinc and lead half-cells. The overall equation for the reaction occurring in the cell is shown below.

$$Zn(s) + Pb^{2+}(aq) \rightarrow Pb(s) + Zn^{2+}(aq)$$

Which statements are correct when the cell produces electricity?

I. The zinc is oxidized.

II. Electrons move from zinc to lead in the external circuit.

III. The mass of the lead electrode increases.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M44 Nov 2012 · Paper 1 · Q32

32. Consider the following standard electrode potential values:

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\ominus = +0.77\ V$$

$$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l) \qquad E^\ominus = +1.51\ V$$

What is the cell potential, in V, for this reaction?

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq)$$

A. –2.28

B. –0.74

C. +0.74

D. +2.28

7.1-M45 Nov 2012 · Paper 1 · Q33

33. Consider the following standard electrode potential values:

$$\begin{aligned} &Ca^{2+}(aq) + 2e^- \rightleftharpoons Ca(s) && E^{\ominus} = -2.87\ V \\ &Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s) && E^{\ominus} = -0.45\ V \\ &Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s) && E^{\ominus} = -0.26\ V \\ &Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) && E^{\ominus} = +0.77\ V \end{aligned}$$

Which reaction is spontaneous?

A. $Ca^{2+}(aq) + Ni(s) \rightarrow Ca(s) + Ni^{2+}(aq)$

B. $3Fe^{2+}(aq) \rightarrow Fe(s) + 2Fe^{3+}(aq)$

C. $Fe(s) + 2Fe^{3+}(aq) \rightarrow 3Fe^{2+}(aq)$

D. $Fe^{2+}(aq) + Ni(s) \rightarrow Fe(s) + Ni^{2+}(aq)$

7.1-M46 May 2012 TZ1 · Paper 1 · Q31

31. The equation for the overall reaction in a voltaic cell is:

$$Cu^{2+}(aq) + Zn(s) \rightarrow Cu(s) + Zn^{2+}(aq)$$

Which statements are correct for this cell?

I. Cu is the positive electrode.

II. Negative ions flow from the zinc solution to the copper solution.

III. Chemical energy is converted into electrical energy during this reaction.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M47 May 2012 TZ1 · Paper 1 · Q32

32. Which statement is correct for the following spontaneous reaction?

$$2Fe^{3+}(aq) + 2Br^{-}(aq) \rightarrow 2Fe^{2+}(aq) + Br_2(aq)$$

A. $\Delta E^{\ominus} < 0$ and Br_2 is a better oxidizing agent than Fe^{3+}.

B. $\Delta E^{\ominus} < 0$ and Br^{-} is a better reducing agent than Fe^{2+}.

C. $\Delta E^{\ominus} > 0$ and Fe^{3+} is a better oxidizing agent than Br_2.

D. $\Delta E^{\ominus} > 0$ and Fe^{2+} is a better reducing agent than Br^{-}.

7.1-M48 May 2012 TZ2 · Paper 1 · Q29

29. What occurs during the operation of a voltaic cell based on the following overall reaction?

$$2Ag^{+}(aq) + Cu(s) \rightarrow 2Ag(s) + Cu^{2+}(aq)$$

	External circuit	Ion movement in solution
A.	electrons move from Cu(s) to Ag(s)	$Ag^{+}(aq)$ move towards Cu(s)
B.	electrons move from Ag(s) to Cu(s)	$Ag^{+}(aq)$ move towards Ag(s)
C.	electrons move from Cu(s) to Ag(s)	$Ag^{+}(aq)$ move towards Ag(s)
D.	electrons move from Ag(s) to Cu(s)	$Cu^{2+}(aq)$ move towards Cu(s)

7.1-M49 May 2012 TZ2 · Paper 1 · Q30

30. Consider the following standard electrode potentials:

$$Sn^{4+}(aq) + 2e^{-} \rightleftharpoons Sn^{2+}(aq) \qquad E^{\ominus} = +0.13\ V$$
$$Pb^{2+}(aq) + 2e^{-} \rightleftharpoons Pb(s) \qquad E^{\ominus} = -0.13\ V$$

What is the value of the cell potential, in V, for the spontaneous reaction that occurs when the two half-cells are connected together?

A. –0.26

B. 0.00

C. +0.13

D. +0.26

7.1-M50 Nov 2011 · Paper 1 · Q32

32. Which is **not** a requirement for the standard hydrogen electrode?

A. Pressure of 101 kPa

B. 1 mol dm^{-3} sulfuric acid

C. Temperature of 298 K

D. An inert electrode such as platinum

7.1-M51 May 2011 TZ1 · Paper 1 · Q30

30. Consider the following standard electrode potentials.

$$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Zn}(\text{s}) \qquad E^\ominus = -0.76\ \text{V}$$

$$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq}) \qquad E^\ominus = +1.36\ \text{V}$$

$$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Mg}(\text{s}) \qquad E^\ominus = -2.37\ \text{V}$$

What will happen when zinc powder is added to an aqueous solution of magnesium chloride?

A. No reaction will take place.

B. Chlorine gas will be produced.

C. Magnesium metal will form.

D. Zinc chloride will form.

7.1-M52 May 2011 TZ1 · Paper 1 · Q32

32. What are the features of a standard hydrogen electrode?

I. A temperature of 298 K

II. A carbon electrode

III. Hydrogen gas at 1.01×10^5 Pa (1 atm) pressure

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M53 May 2011 TZ2 · Paper 1 · Q32

32. The standard electrode potentials for two metals are given below.

$$Al^{3+}(aq) + 3e^- \rightleftharpoons Al(s) \qquad E^{\ominus} = -1.66\ V$$
$$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s) \qquad E^{\ominus} = -0.23\ V$$

What is the equation and cell potential for the spontaneous reaction that occurs?

A. $2Al^{3+}(aq) + 3Ni(s) \rightarrow 2Al(s) + 3Ni^{2+}(aq) \qquad E^{\ominus} = 1.89\ V$

B. $2Al(s) + 3Ni^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Ni(s) \qquad E^{\ominus} = 1.89\ V$

C. $2Al^{3+}(aq) + 3Ni(s) \rightarrow 2Al(s) + 3Ni^{2+}(aq) \qquad E^{\ominus} = 1.43\ V$

D. $2Al(s) + 3Ni^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Ni(s) \qquad E^{\ominus} = 1.43\ V$

7.1-M54 Nov 2010 · Paper 1 · Q32

32. A voltaic cell is made by connecting two half-cells represented by the half-equations below.

$$Mn^{2+}(aq) + 2e^- \rightarrow Mn(s) \qquad E^{\ominus} = -1.19\ V$$
$$Pb^{2+}(aq) + 2e^- \rightarrow Pb(s) \qquad E^{\ominus} = -0.13\ V$$

Which statement is correct about this voltaic cell?

A. Mn is oxidized and the voltage of the cell is 1.06 V.

B. Pb is oxidized and the voltage of the cell is 1.06 V.

C. Mn is oxidized and the voltage of the cell is 1.32 V.

D. Pb is oxidized and the voltage of the cell is 1.32 V.

7.1-M55 May 2010 TZ1 · Paper 1 · Q31

31. What conditions are necessary to directly measure a standard electrode potential $(E^{\ominus})$?

I. A half-cell with an electrode in a 1.0 mol dm^{-3} solution of its ions.

II. Connection to a standard hydrogen electrode.

III. A voltmeter between half-cells to measure potential difference.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M56 May 2010 TZ2 · Paper 1 · Q31

31. Which changes could take place at the positive electrode (cathode) in a voltaic cell?

I. $Zn^{2+}(aq)$ to $Zn(s)$

II. $Cl_2(g)$ to $Cl^-(aq)$

III. $Mg(s)$ to $Mg^{2+}(aq)$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.1-M57 May 2010 TZ2 · Paper 1 · Q32

32. Consider the following standard electrode potentials.

$$Cr^{3+}(aq) + 3e^- \rightarrow Cr(s) \qquad E^{\ominus} = -0.74\ V$$

$$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq) \qquad E^{\ominus} = +0.77\ V$$

What will be the cell potential, in V, of a voltaic cell in which the following reaction takes place?

$$Cr(s) + 3Fe^{3+}(aq) \rightarrow 3Fe^{2+}(aq) + Cr^{3+}(aq)$$

A. –1.51

B. –0.03

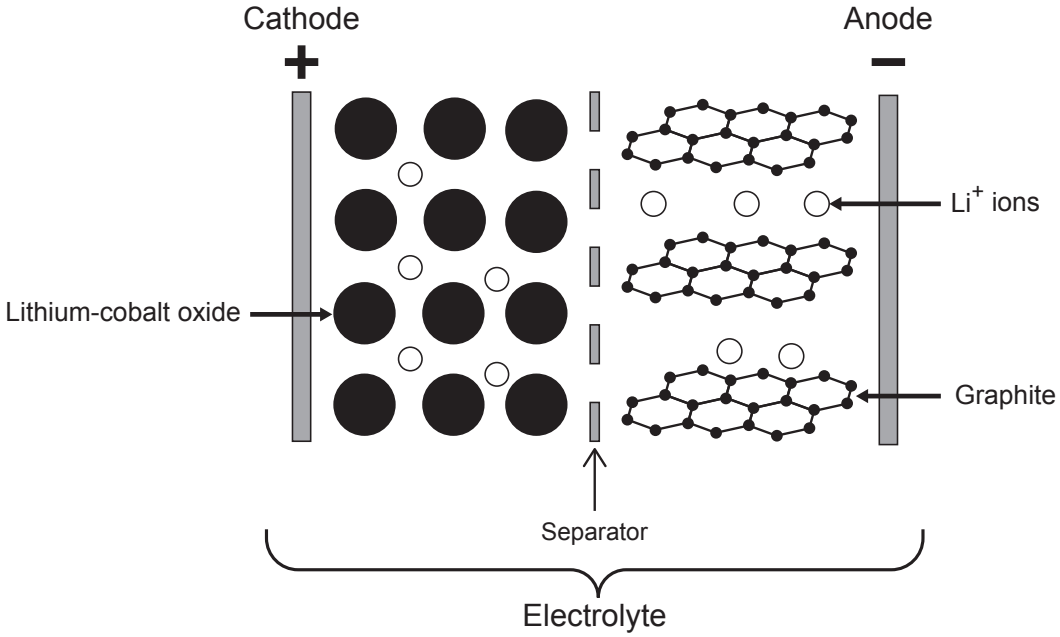
C. +0.03

D. +1.51

Part B - Structured questions

7.1-S1 May 2025 TZ3 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b)

6. A lithium-ion battery consists of lithium ions on lithium-graphite (LiC_6) and lithium-cobalt oxide ($LiCoO_2$) in an electrolyte.



Lithium-ion battery

(a) The lithium ions are produced at the anode from the lithium atoms embedded in graphite.

(i) State a half-equation for the reaction occurring at the anode during discharge. [1]

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(ii) Comment on the sign and value of the standard reduction potential of lithium that make it suitable to use in the battery. Use section 19 of the data booklet. [2]

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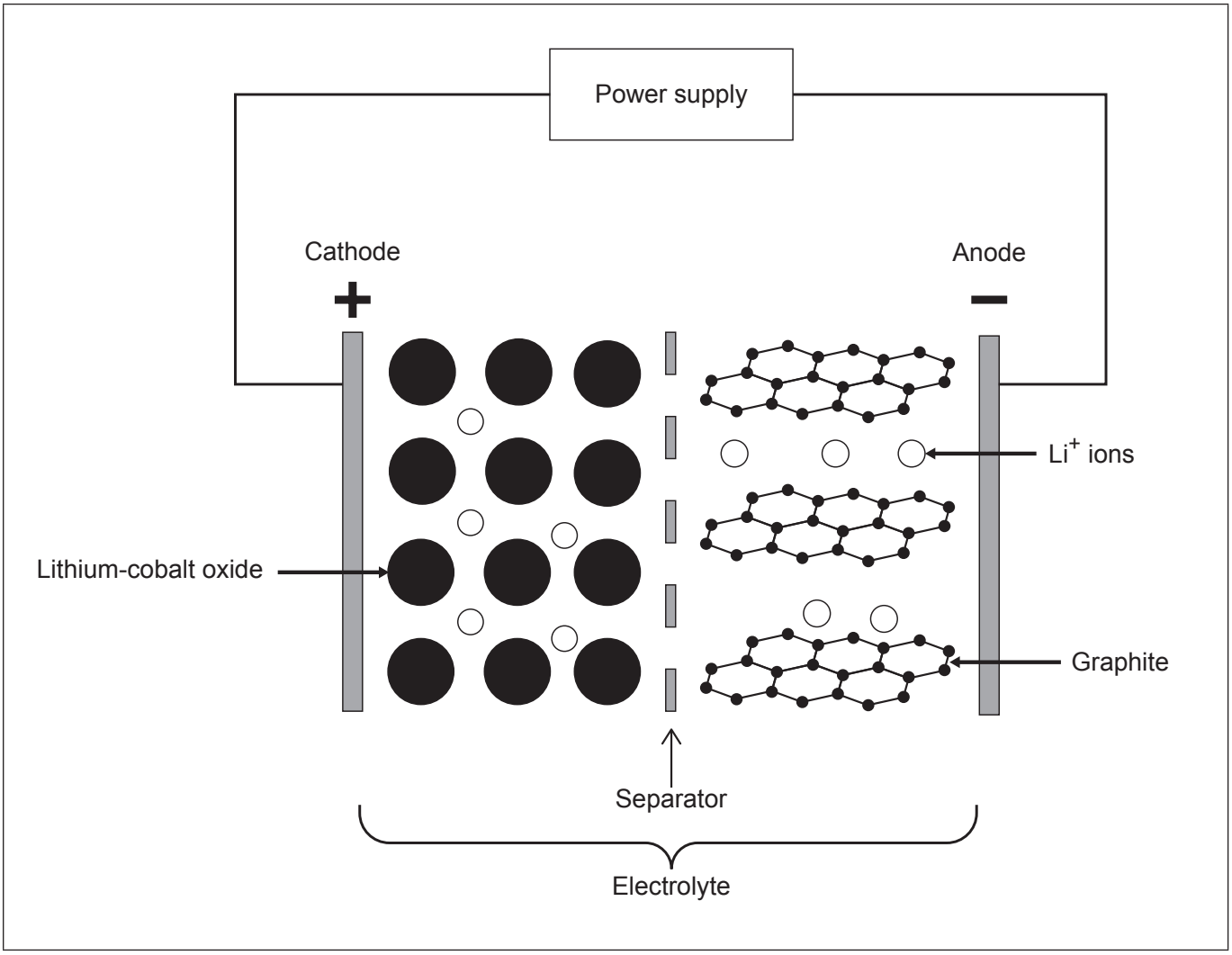
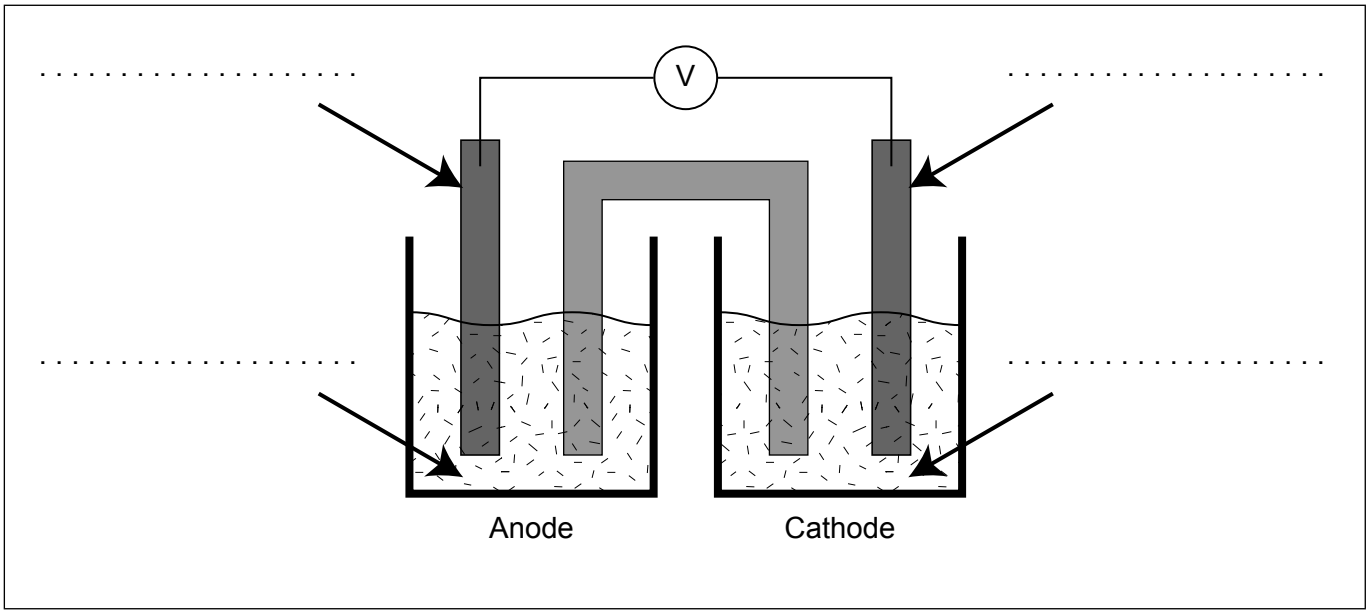
(b) The lithium-ion battery is rechargeable. Annotate the diagram below to show the direction of the flow of electrons in the wire while the battery is recharging. [1]



7.1-S2 Nov 2024 · Paper 2 · Q8 parts (a), (b)

8. A voltaic cell is made using a nickel electrode, Ni (s), a silver electrode, Ag (s), and solutions of nickel nitrate, $Ni(NO_3)_2$ (aq), and silver nitrate, $AgNO_3$ (aq).

(a) Label the diagram with the electrodes and electrolytes, and the direction of electron flow. Use section 24 or 25 of the data booklet. [2]



(b) Outline why a salt bridge must be included to connect the two half cells. [1]

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7.1-S3 Nov 2024 · Paper 3 · Q15 parts (a), (c)

15. Batteries are a portable source of energy.

(a) Outline how a rechargeable battery differs from a primary cell. [1]

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(c) Explain how the chemical nature and quantities of reactants in a voltaic cell affect the electrical output of the cell. [2]

Chemical nature: . . . . . .

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Quantities: . . . . . .

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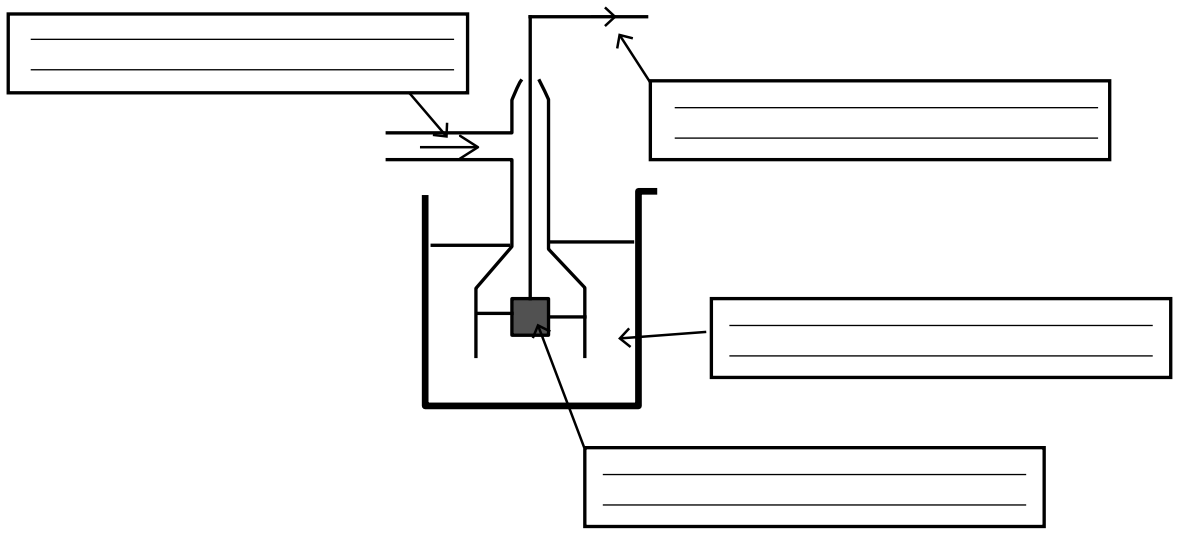
7.1-S4 May 2024 TZ1 · Paper 2 · Q7 parts (a), (b)(i)

7. (a) Label the figure with the essential features of the standard hydrogen electrode, by writing in the boxes. [2]



(b) A standard hydrogen electrode is connected to a copper-copper (II) nitrate half-cell.

(i) Formulate a cell diagram for the spontaneous reaction, representing phase boundaries with "|" and the salt bridge with "||". Use section 24 of the data booklet. [1]

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7.1-S5 May 2024 TZ1 · Paper 3 · Q16 parts (b)(i), (b)(ii)

16. The use of photovoltaic devices is being encouraged to combat global warming.

(b) At night, rechargeable batteries can be used to power installations connected to photovoltaic systems.

(i) Identify the substances which are converted to lead (II) sulfate, $PbSO_4$, during discharge of a lead-acid cell. [2]

Positive electrode (cathode):

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Negative electrode (anode):

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(ii) Outline why lithium ion batteries are now generally preferred to lead-acid batteries. [1]

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7.1-S6 May 2024 TZ2 · Paper 3 · Q13 parts (a), (b)(i), (c)(i), (c)(ii), (c)(iii) · also 7.3

13. Batteries and fuels provide portable sources of energy.

(a) Suggest, with a reason, if specific energy or energy density is a better measure of a fuel's usefulness as an everyday portable energy source. [1]

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(b) (i) Ethylbenzene, $C_6H_5CH_2CH_3$, is an aromatic compound that is used to increase the octane rating in petrol (gasoline). It has a specific energy of $4.135 \times 10^7\,J\,kg^{-1}$.

Calculate the enthalpy of combustion of ethylbenzene, in $kJ\,mol^{-1}$, using section 6 of the data booklet. [2]

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(c) (i) Lithium ion and lead acid are both types of rechargeable batteries.

Outline the mass and voltage advantages of a lithium-ion battery, using sections 6 and 24 of the data booklet. [2]

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(ii) Graphite and a lithium cobalt oxide complex, $LiCoO_2$ (s), are the electrodes used in a lithium-ion cell.

Deduce the half-equations for the reactions occurring at each electrode during the charging of a lithium-ion cell. [2]

Graphite:

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$LiCoO_2$ (s):

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(iii) State **one** disadvantage of the lithium-ion battery. [1]

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7.1-S7 Nov 2023 TZ1 · Paper 3 · Q15 parts (b)

15. Electrochemical cells generate electricity from a spontaneous redox reaction.

(b) Describe, in detail, how both fuel cells and secondary cells can be reused. [2]

Fuel cells:

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Secondary cells:

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7.1-S8 May 2023 TZ1 · Paper 2 · Q4 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1

4. Redox reactions can be used to produce electricity.

(a) State the oxidation state of sulfur in copper (II) sulfate. [1]

. . . . . .

(b) A voltaic cell was constructed using a copper (II) sulfate/copper half-cell and a zinc sulfate/zinc half-cell.

(i) Outline why electrons flow from zinc to copper when these half cells are connected with a wire. Use section 25 of the data booklet. [1]

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(ii) Formulate equations for the reactions taking place at each electrode. [2]

Anode (negative electrode): . . . . . .

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Cathode (positive electrode): . . . . . .

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(c) (i) Calculate the standard cell potential for the voltaic cell in part (b). Use section 24 of the data booklet. [1]

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7.1-S9 Nov 2022 · Paper 2 · Q3 parts (a), (c), (d), (f)(i), (f)(ii), (f)(iii), (f)(iv), (f)(v) · also 2.1

3. Consider the following reaction:

$$Cu^{2+}(aq) + Fe(s) \rightarrow Fe^{2+}(aq) + Cu(s)$$

(a) State the ground-state electron configuration for Fe^{2+}. [1]

. . . . . .

. . . . . .

(c) Predict, with a reason, whether Cu or Cu^{2+} has the greater ionization energy. [1]

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(d) Determine the frequency, in s^{-1}, of a photon that will cause the first ionization of copper. Use sections 1, 2 and 8 of the data booklet. [2]

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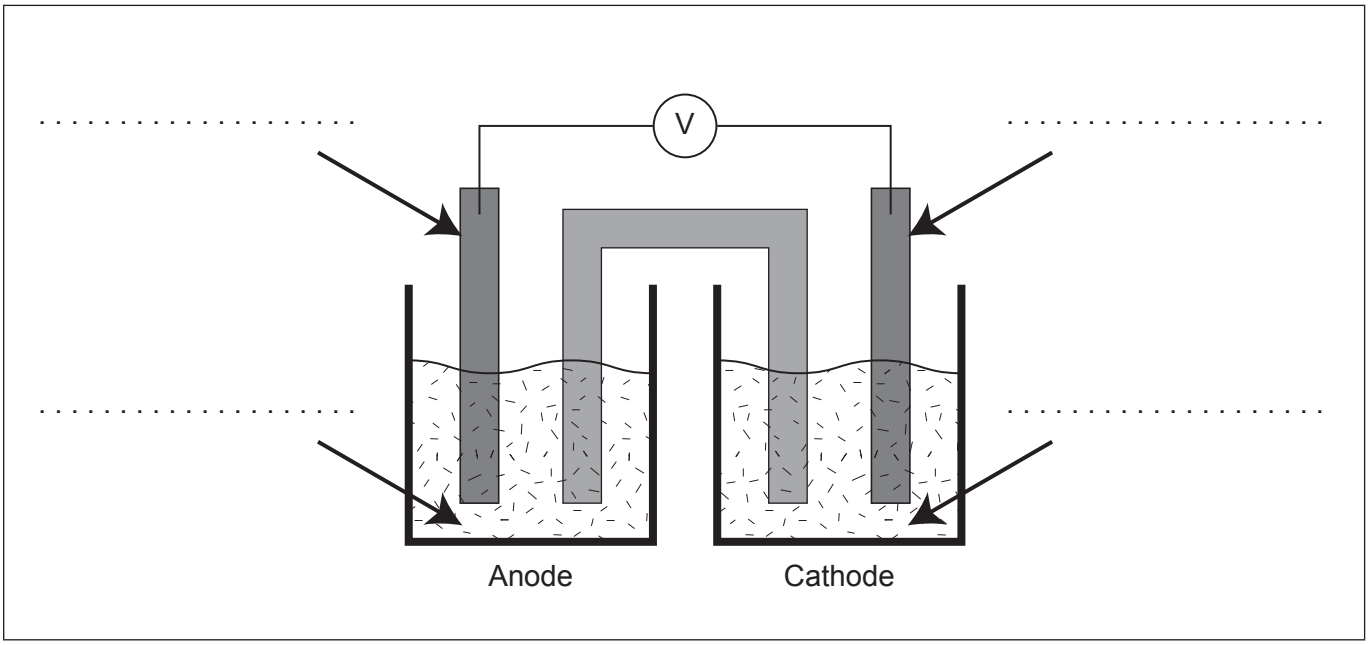
(f) The diagram shows an unlabelled voltaic cell for the reaction:

$$Cu^{2+}(aq) + Fe(s) \rightarrow Fe^{2+}(aq) + Cu(s)$$



(i) Label the diagram with the species from the equation and the direction of electron flow. [2]

(ii) Write the half-equation for the reaction occurring at the anode (negative electrode). [1]

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(iii) The diagram includes a salt bridge that is filled with a saturated solution of KNO_3. Outline the function of the salt bridge. [1]

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(iv) Predict the movement of **all** ionic species through the salt bridge. [2]

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(v) Calculate the standard cell potential, in V, for this cell. Use section 24 of the data booklet. [1]

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7.1-S10 May 2022 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii) · also 7.2

3. Standard electrode potential values, $E^{\ominus}$, can be used to predict spontaneity.

(a) (i) Iron(II) is oxidized by bromine.

$$2Fe^{2+}(aq) + Br_2(l) \rightleftharpoons 2Fe^{3+}(aq) + 2Br^{-}(aq)$$

Calculate the $E^{\ominus}_{cell}$, in V, for the reaction using section 24 of the data booklet. [1]

. . . . . .

. . . . . .

(ii) Determine, giving a reason, if iodine will also oxidize iron(II). [1]

. . . . . .

. . . . . .

. . . . . .

(b) (i) Molten zinc chloride undergoes electrolysis in an electrolytic cell at 450 °C.

Deduce the half-equations for the reaction at each electrode. [2]

Cathode (negative electrode):

. . . . . .

. . . . . .

Anode (positive electrode):

. . . . . .

. . . . . .

(ii) Deduce the overall cell reaction including state symbols. Use section 7 of the data booklet. [2]

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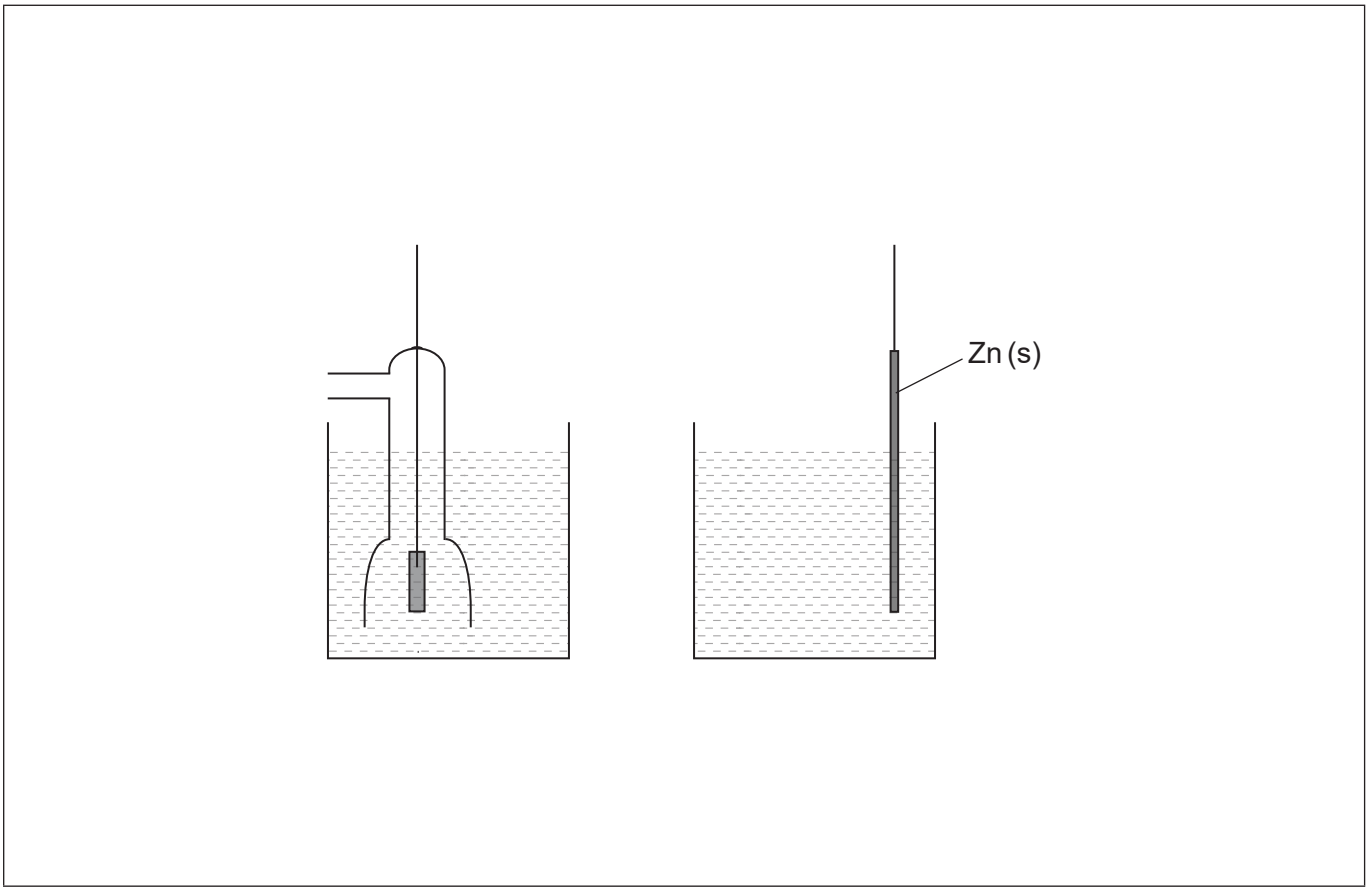
7.1-S11 Nov 2021 · Paper 2 · Q8

8. The standard electrode potential of zinc can be measured using a standard hydrogen electrode (SHE).

Draw and annotate the diagram to show the complete apparatus required to measure the standard electrode potential of zinc. [4]



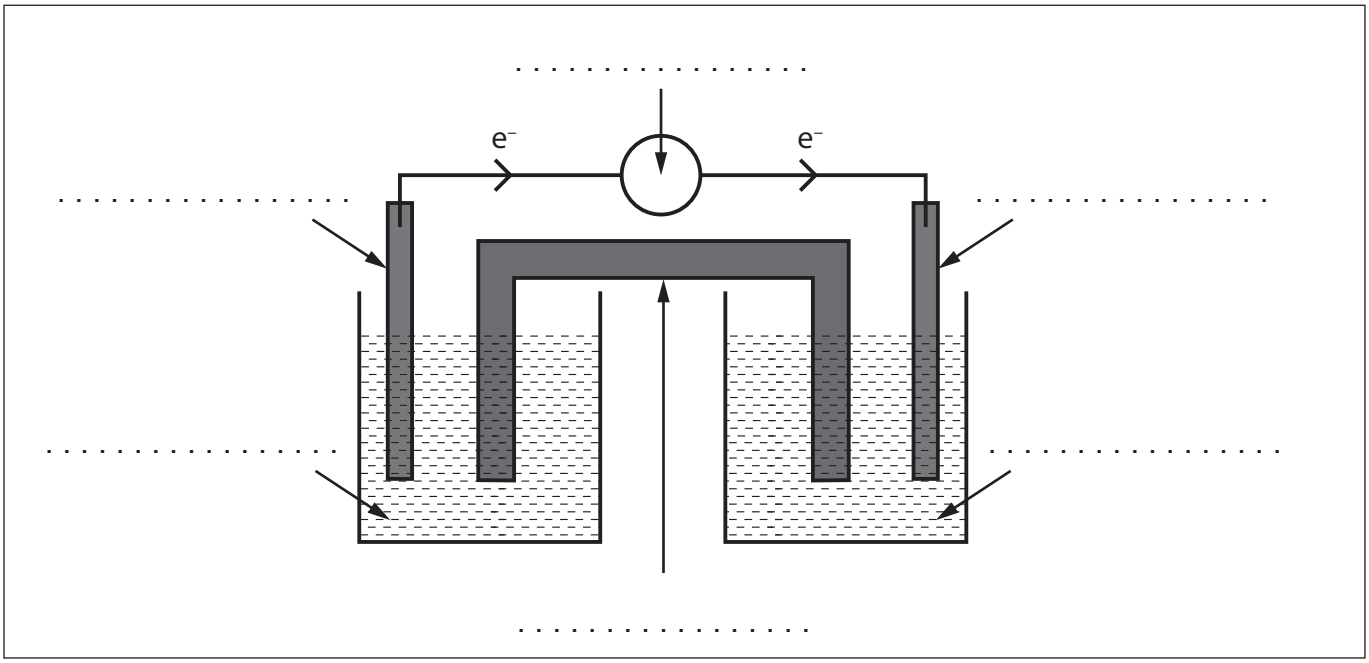
7.1-S12 May 2021 TZ2 · Paper 2 · Q3 parts (a), (b), (c)

3. Oxidation and reduction reactions can have a variety of commercial uses.

(a) A student decides to build a voltaic cell consisting of an aluminium electrode, Al(s), a tin electrode, Sn(s), and solutions of aluminium nitrate, $Al(NO_3)_3(aq)$ and tin(II) nitrate, $Sn(NO_3)_2(aq)$.

Electron flow is represented on the diagram.

Label each line in the diagram using section 25 of the data booklet. [3]

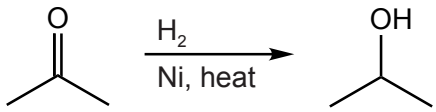


(b) Write the equation for the expected overall chemical reaction in (a). [1]

(c) Calculate the cell potential using section 24 of the data booklet. [1]

7.1-S13 Nov 2020 · Paper 2 · Q4 parts (c), (d)(i), (d)(ii), (d)(iv) · also 5.2

4. Nickel catalyses the conversion of propanone to propan-2-ol.



(c) Discuss, referring to intermolecular forces present, the relative volatility of propanone and propan-2-ol. [3]

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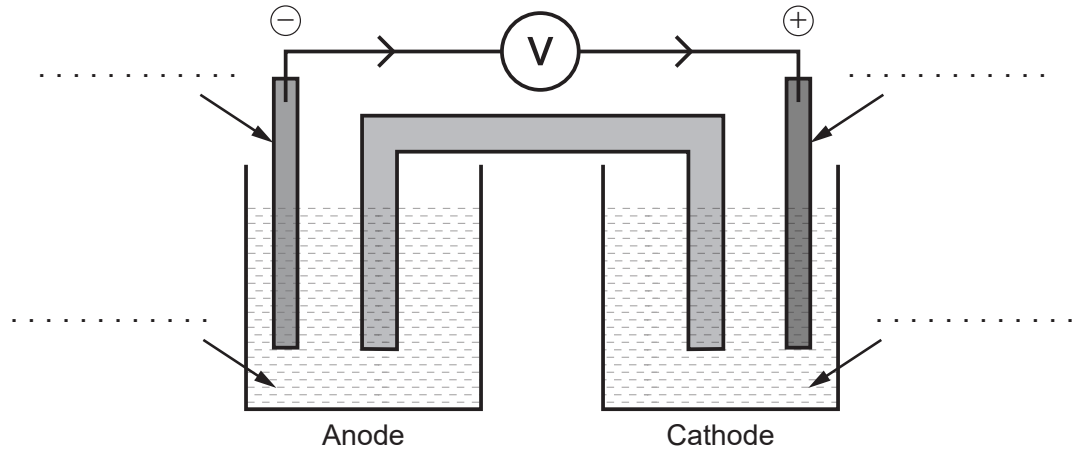
(d) (i) The diagram shows an unlabelled voltaic cell for the reaction.

$$Pb^{2+}(aq) + Ni(s) \rightarrow Ni^{2+}(aq) + Pb(s)$$

Label the diagram with the species in the equation. [1]



(ii) Calculate the standard cell potential, in V, for the cell at 298 K. Use section 24 of the data booklet. [1]

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(iv) Suggest a metal that could replace nickel in a new half-cell and reverse the electron flow. Use section 25 of the data booklet. [1]

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7.1-S14 Nov 2020 · Paper 3 · Q13 parts (a)

13. A voltaic cell is made up of nickel and magnesium half-cells.

$$Mg\ (s)\ |\ Mg^{2+}\ (aq)\ |\ |\ Ni^{2+}\ (aq)\ |\ Ni\ (s)$$

(a) Write the balanced equation for the reaction in this voltaic cell. [1]

. . . . . .

. . . . . .

7.1-S15 May 2019 TZ1 · Paper 2 · Q6 parts (a), (e), (f)(i) · also 2.1

6. This question is about iron.

(a) Deduce the **full** electron configuration of Fe^{2+}. [1]

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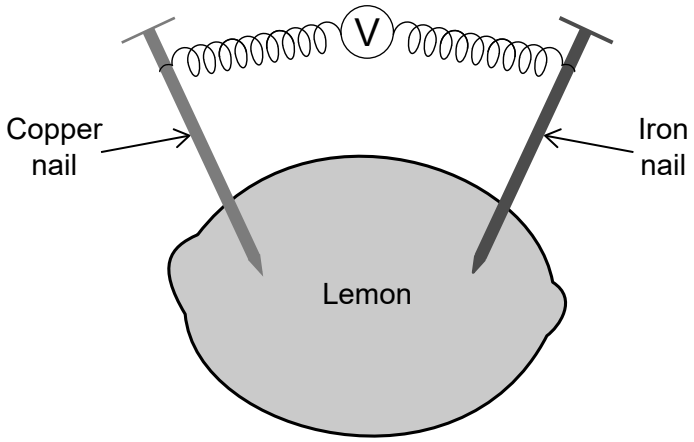
. . . . . .

(e) An iron nail and a copper nail are inserted into a lemon.



Explain why a potential is detected when the nails are connected through a voltmeter. [2]

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(f) (i) Calculate the standard electrode potential, in V, when the $Fe^{2+}(aq) \mid Fe(s)$ and $Cu^{2+}(aq) \mid Cu(s)$ standard half-cells are connected at 298 K. Use section 24 of the data booklet. [1]

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7.1-S16 May 2019 TZ2 · Paper 3 · Q19 parts (c)

19. Electricity can be generated in a variety of ways.

(c) Outline **one** difference between a primary and a secondary cell. [1]

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7.1-S17 Nov 2018 · Paper 3 · Q15 parts (a), (b)

15. Chemical energy from redox reactions can be used as a source of electrical energy.

(a) Outline how a rechargeable battery differs from a primary cell. [1]

. . . . . .

. . . . . .

(b) Formulate half-equations for the reactions at the anode (negative electrode) and cathode (positive electrode) during discharge of a lithium-ion battery. [2]

Anode (negative electrode):

. . . . . .

. . . . . .

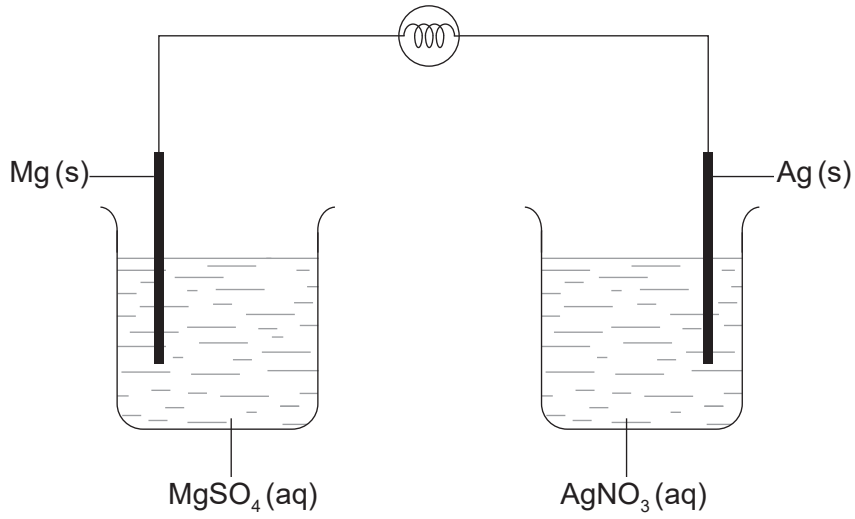
Cathode (positive electrode):

. . . . . .

. . . . . .

7.1-S18 May 2018 TZ1 · Paper 2 · Q6 parts (a), (b), (c), (d), (e) · also 1.2

6. The diagram shows an incomplete voltaic cell with a light bulb in the circuit.



(a) Identify the missing component of the cell and its function. [2]

. . . . . .

. . . . . .

(b) Deduce the half-equations for the reaction at each electrode when current flows. [2]

Positive electrode (cathode):

. . . . . .

Negative electrode (anode):

. . . . . .

(c) Annotate the diagram with the location and direction of electron movement when current flows. [1]

(d) Calculate the cell potential, in V, using section 24 of the data booklet. [1]

. . . . . .

. . . . . .

(e) Determine the loss in mass of one electrode if the mass of the other electrode increases by 0.10 g. [2]

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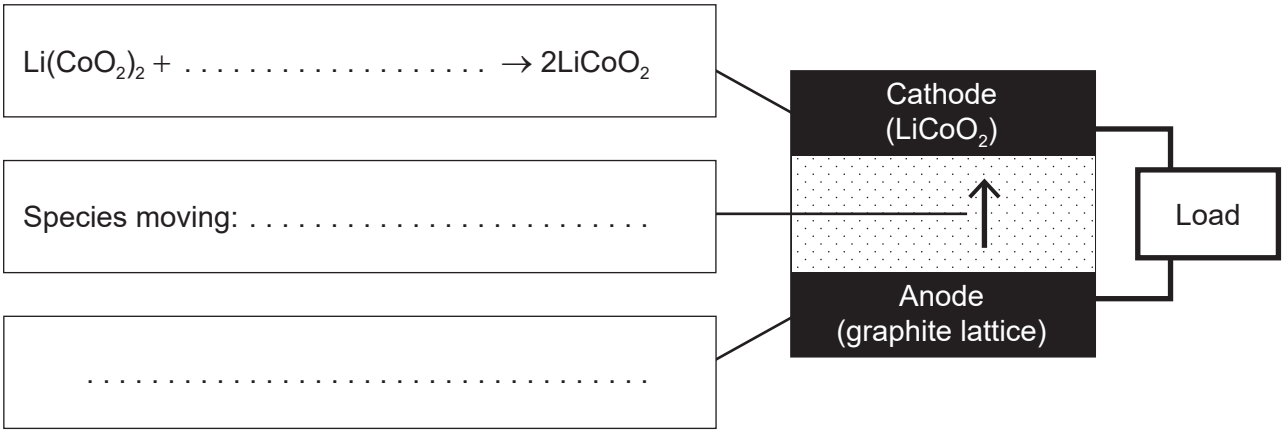
7.1-S19 May 2018 TZ1 · Paper 3 · Q14 parts (a)(i)

14. Modern electric cars store their energy in lithium ion batteries.

(a) The diagram represents a cell in such a battery delivering a current.

(i) Complete the half-equations on the diagram and identify the species moving between the electrodes. [3]



7.1-S20 Nov 2017 · Paper 2 · Q7 parts (a), (b), (c), (d), (e) · also 7.2

7. Consider the following half-cell reactions and their standard electrode potentials.

	$E^{\ominus}$ / V
$Mn^{2+}(aq) + 2e^- \rightleftharpoons Mn(s)$	−1.18
$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$	−0.26
$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$	+0.54

(a) Deduce a balanced equation for the overall reaction when the standard nickel and iodine half-cells are connected. [1]

. . . . . .

. . . . . .

(b) Predict, giving a reason, the direction of movement of electrons when the standard nickel and manganese half-cells are connected. [2]

. . . . . .

. . . . . .

(c) Calculate the cell potential, in V, when the standard iodine and manganese half-cells are connected. [1]

. . . . . .

. . . . . .

(d) Identify the best reducing agent in the table above. [1]

. . . . . .

. . . . . .

(e) State and explain the products of electrolysis of a concentrated aqueous solution of sodium chloride using inert electrodes. Your answer should include half-equations for the reaction at each electrode. [4]

Positive electrode (anode):

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. . . . . .

. . . . . .

Negative electrode (cathode):

. . . . . .

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. . . . . .

7.1-S21 May 2017 TZ1 · Paper 2 · Q3 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1, 6.2

3. Vanadium has a number of different oxidation states.

(a) Determine the oxidation state of vanadium in each of the following species. [2]

V_2O_5:

. . . . . .

VO^{2+}:

. . . . . .

(b) Electrode potentials for the reactions of vanadium and other species are shown below.

$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq)$	−0.26 V
$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34 V
$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+1.0 V

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76 V
$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s)$	−0.13 V
$SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightleftharpoons H_2SO_3(aq) + H_2O(l)$	+0.17 V
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80 V
$\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$	+1.36 V

(i) Identify, from the table, a non-vanadium species that can reduce $VO^{2+}(aq)$ to $V^{3+}(aq)$ but no further. [1]

. . . . . .

(ii) Identify, from the table, a non-vanadium species that could convert $VO_2^+(aq)$ to $V^{2+}(aq)$. [1]

. . . . . .

(c) (i) Formulate an equation for the reaction between $VO^{2+}(aq)$ and $V^{2+}(aq)$ in acidic solution to form $V^{3+}(aq)$. [1]

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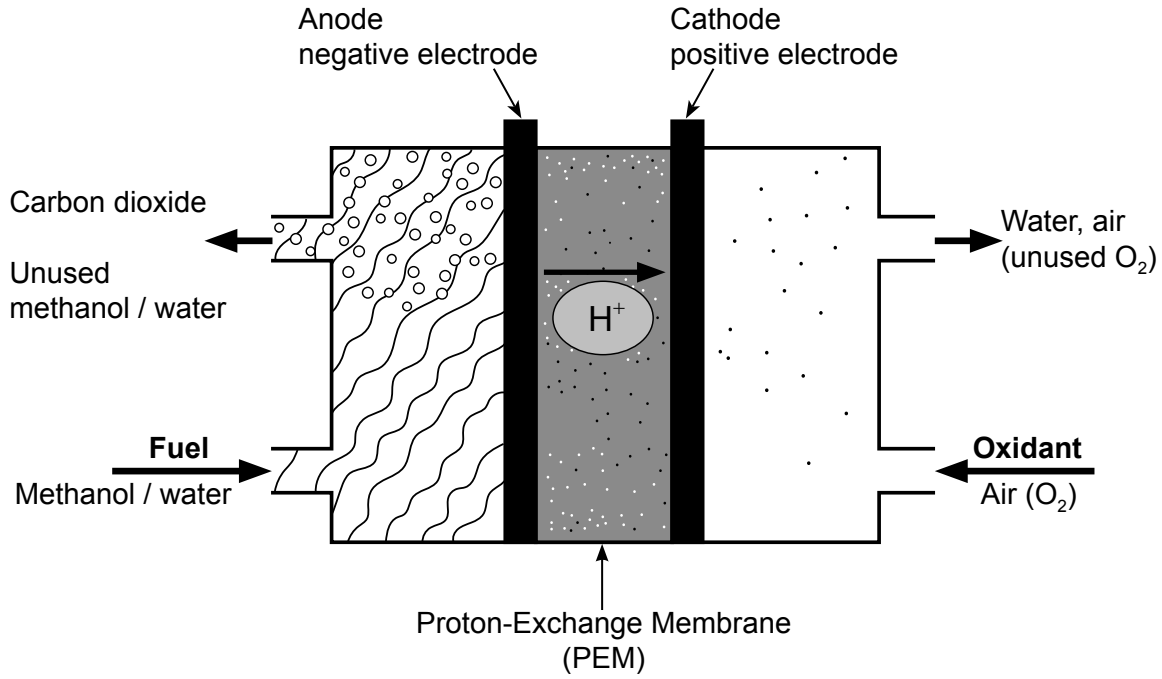
7.1-S22 May 2017 TZ1 · Paper 3 · Q22 parts (a), (b)(i), (b)(ii) · also 7.3

22. As well as being burnt, methanol can also be used to provide electricity through a fuel cell. A schematic diagram of such a fuel cell, that depends on the transfer of hydrogen ions between the electrodes, is shown below.



[Source: adapted from http://greenbigtruck.com]

(a) Deduce half-equations for the reactions at the two electrodes and hence the equation for the overall reaction. [3]

Anode (negative electrode):

. . . . . .

Cathode (positive electrode):

. . . . . .

Overall:

. . . . . .

(b) Even though fuel cells, primary cells and rechargeable cells have similar fundamental characteristics, there are important differences between them.

(i) Suggest a way in which they are similar. [1]

. . . . . .

. . . . . .

(ii) Outline the difference between primary and rechargeable cells. [1]

. . . . . .

. . . . . .

7.1-S23 Nov 2016 · Paper 2 · Q4 parts (d)(ii), (i)(i), (i)(ii), (i)(iii), (j), (k)(i), (k)(ii) · also 2.1, 7.2

4. Magnesium is a group 2 metal which exists as a number of isotopes and forms many compounds.

(ii) State what can be determined from the frequency of the convergence limit. [1]

. . . . . .

. . . . . .

(i) Magnesium chloride can be electrolysed.

(i) Deduce the half-equations for the reactions at each electrode when **molten** magnesium chloride is electrolysed, showing the state symbols of the products. The melting points of magnesium and magnesium chloride are 922 K and 987 K respectively. [2]

Anode (positive electrode):

. . . . . .

Cathode (negative electrode):

. . . . . .

(ii) Identify the type of reaction occurring at the cathode (negative electrode). [1]

. . . . . .

(iii) State the products when a very **dilute** aqueous solution of magnesium chloride is electrolysed. [2]

Anode (positive electrode):

. . . . . .

Cathode (negative electrode):

. . . . . .

(j) Standard electrode potentials are measured relative to the standard hydrogen electrode. Describe a standard hydrogen electrode. [2]

. . . . . .

. . . . . .

. . . . . .

(k) A magnesium half-cell, $Mg(s)/Mg^{2+}(aq)$, can be connected to a copper half-cell, $Cu(s)/Cu^{2+}(aq)$.

(i) Formulate an equation for the spontaneous reaction that occurs when the circuit is completed. [1]

. . . . . .

. . . . . .

(ii) Determine the standard cell potential, in V, for the cell. Refer to section 24 of the data booklet. [1]

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7.1-S24 Nov 2015 · Paper 3 · Q12 parts (a), (b), (c)

12. The nickel–cadmium (NiCad) battery is rechargeable.

During discharge, the following half-reactions take place:

$$Cd(s) + 2OH^-(aq) \rightarrow Cd(OH)_2(s) + 2e^-$$

$$NiO(OH)(s) + H_2O(l) + e^- \rightarrow Ni(OH)_2(s) + OH^-(aq)$$

(a) State the name of the substance used as the negative electrode (anode) during discharge and the name of the substance used as the electrolyte. [2]

Negative electrode (anode):

. . . . . .

Electrolyte:

. . . . . .

(b) When the NiCad battery is recharged, the electrodes are connected to a power supply and electrolysis occurs. State the half-equations for the chemical reactions during recharging. [1]

Negative electrode (cathode):

. . . . . .

Positive electrode (anode):

. . . . . .

(c) Compare rechargeable batteries and fuel cells. [3]

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. . . . . .
. . . . . .

7.1-S25 May 2015 TZ2 · Paper 2 · Q8 parts (a)(i), (b)(i), (c)(i), (d)(i), (d)(ii), (e)(i), (e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi) · also 1.1, 2.1, 6.1, 6.2

8. Chromium is a transition metal with many uses.

(a) (i) Draw an orbital diagram (using the arrow-in-box notation) showing the electrons in the 4s and 3d sub-levels in chromium metal. [1]

(b) (i) State the name of Cr_2O_3. [1]

. . . . . .

(c) Chromium forms the complex ion $[Cr(NH_3)_4Cl_2]^+$.

(i) Deduce the oxidation number of chromium in this complex. [1]

. . . . . .
. . . . . .
. . . . . .

(d) (i) The dichromate ion, $Cr_2O_7^{2-}(aq)$, and the iodide ion, $I^-(aq)$, react together in the presence of an acid to form $Cr^{3+}(aq)$ and $IO_3^-(aq)$ ions. Deduce the half-equation for the reaction of I^- to IO_3^- and the overall equation for this reaction. [2]

Half-equation:

. . . . . .
. . . . . .

Overall equation:

. . . . . .
. . . . . .

(ii) Explain in terms of oxidation numbers whether iodine is oxidized or reduced in part (d) (i). [1]

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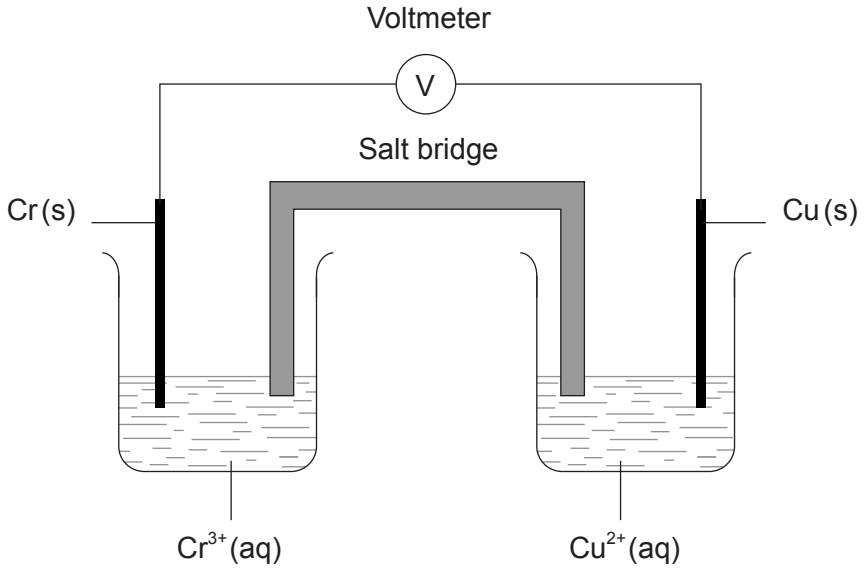
. . . . . .

(e) A voltaic cell is constructed as follows. One half-cell contains a chromium electrode immersed in a solution containing $Cr^{3+}(aq)$ ions. The other half-cell contains a copper electrode immersed in a solution containing $Cu^{2+}(aq)$ ions. The two electrodes are connected to a voltmeter and the two solutions by a salt bridge.



(i) Define the term *standard electrode potential*. [1]

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. . . . . .
. . . . . .

(ii) Calculate the cell potential, in V, under standard conditions, for this voltaic cell, using table 14 of the data booklet and $E^{\ominus}_{Cr^{3+}/Cr} = -0.74\,V$. [1]

. . . . . .
. . . . . .

(iii) Predict the balanced equation for the spontaneous reaction which will produce a current in this voltaic cell. [1]

. . . . . .
. . . . . .

(iv) Identify the negative and the positive electrodes in this cell. [1]

. . . . . .
. . . . . .

(v) Predict the direction of movement of electrons in the external circuit. [1]

. . . . . .
. . . . . .

(vi) State the directions in which the negative ions (anions) and the positive ions (cations) flow in the salt bridge. [1]

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. . . . . .
. . . . . .

7.1-S26 May 2014 TZ2 · Paper 2 · Q2 parts (b), (d)(i), (d)(ii) · also 2.1

2. There are only two isotopes, $^{63}_{29}Cu$ and $^{65}_{29}Cu$, in naturally occurring copper.

(b) State the **full** electronic configuration of a copper atom. *[1]*

. . . . . .

(d) A chemist considered preparing a copper(I) salt by reacting copper metal with the corresponding copper(II) salt according to the equation below.

$$Cu^{2+}(aq) + Cu(s) \rightarrow 2Cu^{+}(aq)$$

(i) Using data from Table 14 of the Data Booklet, calculate the cell potential for this reaction. *[2]*

. . . . . .
. . . . . .
. . . . . .

(ii) Use this result to predict, with a reason, whether this reaction will be spontaneous. *[1]*

. . . . . .
. . . . . .

7.1-S27 May 2014 TZ2 · Paper 2 · Q5 parts (a)(ii), (b)(iv), (c)(i), (c)(ii), (d)(i), (d)(ii) · also 5.2, 6.1, 6.2, 6.3

5. Bleaches in which chlorine is the active ingredient are the most common, although some environmental groups have concerns about their use.

(ii) Outline, with the help of a chemical equation, why this reaction occurs. *[2]*

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. . . . . .
. . . . . .
. . . . . .

(b) In aqueous chlorine the equilibrium below produces chloric(I) acid (hypochlorous acid), HOCl, the active bleach.

$$Cl_2(aq) + H_2O(l) \rightleftharpoons HOCl(aq) + H^{+}(aq) + Cl^{-}(aq)$$

(iv) Suggest why a covalent molecule, such as chloric(I) acid, is readily soluble in water. *[2]*

. . . . . .
. . . . . .
. . . . . .

(c) Aqueous sodium chlorate(I), NaOCl, the most common active ingredient in chlorine based bleaches, oxidizes coloured materials to colourless products while being reduced to the chloride ion. It will also oxidize sulfur dioxide to the sulfate ion.

(i) Deduce a balanced equation for the reaction between the chlorate(I) ion and sulfur dioxide from the appropriate half-equations. *[4]*

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. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) State the initial and final oxidation numbers of both chlorine and sulfur in the final equation. *[2]*

Element	Initial oxidation number	Final oxidation number
Chlorine		
Sulfur		

(d) The standard electrode potential for the reduction of the chlorate(V) ion to the chloride ion is +1.49 V.

(i) Define the term *standard electrode potential*. *[1]*

. . . . . .
. . . . . .

(ii) Referring to Table 14 of the Data Booklet, deduce, giving a reason, whether the oxidation of the chromium(III) ion to the dichromate(VI) ion by the chlorate(V) ion is energetically feasible. *[2]*

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7.1-S28 May 2014 TZ2 · Paper 3 · Q11 parts (a)(i), (b) · also 7.3

11. Modern society is very dependent on electrical power for portable devices.

(a) Two common rechargeable batteries are lead-acid and nickel-cadmium (NiCad) batteries.

(i) State equations for the reactions that occur at each electrode in a **lead-acid battery** when it delivers a current. *[2]*

Positive electrode (cathode):

. . . . . .

Negative electrode (anode):

. . . . . .

(b) Another source of power for portable devices is the fuel cell. Compare fuel cells with **lead-acid** rechargeable batteries, stating **one** similarity and **two** differences. *[3]*

Similarity:

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Differences:

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7.1-S29 Nov 2013 · Paper 2 · Q6 parts (a), (b)(i), (c), (d)(i), (d)(ii), (d)(iii), (e)(iii), (f)(i), (f)(ii) · also 3.1, 4.2, 6.1

6. In acidic solution, ions containing titanium can react according to the half-equation below.

$$TiO^{2+}(aq) + 2H^{+}(aq) + e^{-} \rightleftharpoons Ti^{3+}(aq) + H_2O(l) \qquad E^{\ominus} = -0.06\,V$$

(a) Define the term *standard electrode potential*, $E^{\ominus}$. *[1]*

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(b) (i) State the initial and final oxidation numbers of titanium and hence deduce whether it is oxidized or reduced in this change. *[2]*

Initial oxidation number	Final oxidation number	Oxidized / reduced

(c) In the two experiments below, predict whether a reaction would occur and deduce an equation for any reaction that takes place. Refer to Table 14 of the Data Booklet if necessary. *[3]*

KI(aq) is added to a solution containing $Ti^{3+}(aq)$ ions:

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Zn(s) is added to a solution containing $TiO^{2+}(aq)$ and $H^{+}(aq)$ ions:

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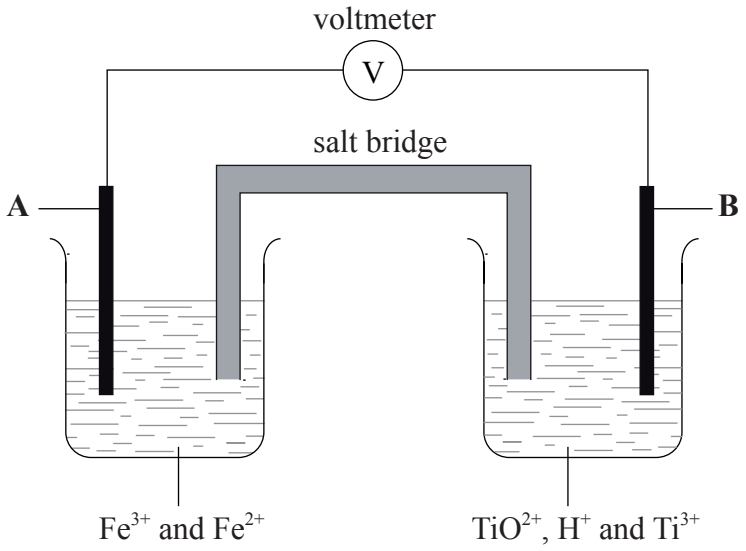
(d) In the diagram below, **A** and **B** are inert electrodes and, in the aqueous solutions, all ions have a concentration of $1\,mol\,dm^{-3}$.



(i) Using Table 14 of the Data Booklet, state the balanced half-equation for the reaction that occurs at electrode **A** and whether it involves oxidation or reduction. *[2]*

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(ii) Calculate the cell potential in V. *[1]*

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(iii) On the diagram above label with an arrow

- the direction of electron flow in the wire
- the direction in which the positive ions flow in the salt bridge. *[1]*

(e) Sodium, silicon and sulfur are elements in period 3 of the periodic table that all form oxides.

(iii) As well as the oxide above, sodium forms a peroxide that contains the peroxide ion, O_2^{2-}. Draw the Lewis (electron dot) structure of the peroxide ion. *[2]*

(f) Although carbon and silicon both belong to group 4 of the periodic table, carbon dioxide and silicon dioxide are different in many ways.

(i) Describe the differences in the hybridization of these group 4 elements and the precise nature of the bonds that they form with the oxygen atoms. *[3]*

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(ii) Xenon, although a noble gas, forms an oxide, XeO_2, that has a structure related to that of SiO_2. Compare the geometry around the silicon atoms in SiO_2 with the geometry around the xenon atoms in XeO_2, using the valence shell electron pair repulsion (VSEPR) theory. *[3]*

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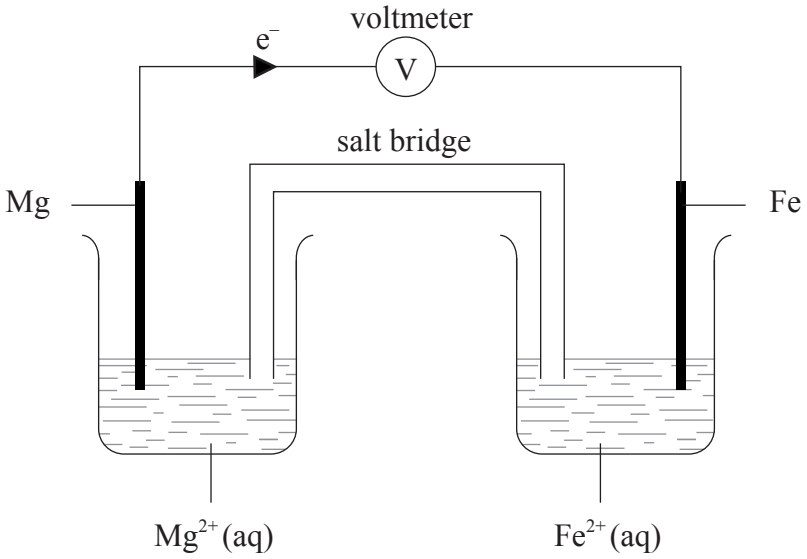
7.1-S30 May 2012 TZ1 · Paper 2 · Q4 parts (a), (b), (c)(i), (c)(ii), (d)(i), (d)(iv) · also 7.2

4. Chemical energy can be converted to electrical energy in the voltaic cell below.



(a) State the half-equation which describes the change at the Mg electrode and deduce which metal is the positive electrode (cathode) of the cell. *[2]*

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(b) Deduce the equation for the overall reaction occurring in the cell. *[1]*

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(c) (i) Define the term *standard electrode potential*. *[1]*

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(ii) Calculate the standard cell potential ($E^{\ominus}$), in V, for the spontaneous reaction in (b), using Table 14 of the Data Booklet. *[1]*

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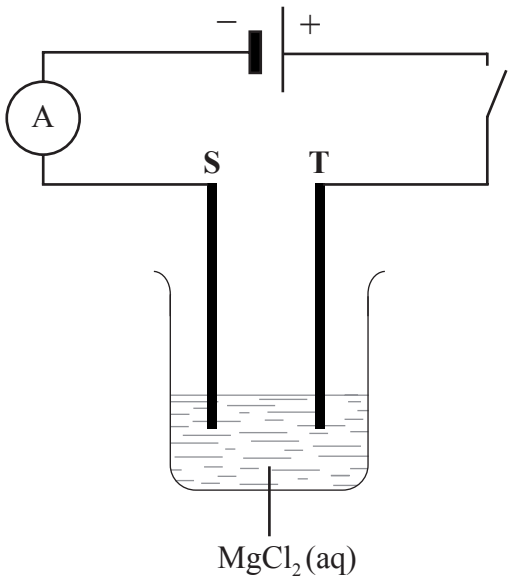
(d) A different chemical change occurs when a saturated aqueous solution of magnesium chloride is electrolysed using inert electrodes, **S** and **T**, in the circuit below.



Different gases are produced at the electrodes **S** and **T**.

(i) State the half-equations for the reactions at each electrode. *[2]*

S: . . . . . .

T: . . . . . .

(iv) Another gas is produced when $MgCl_2$ is at a lower concentration. State the name of this gas and deduce the half-equation for this reaction. *[2]*

Name:

. . . . . .

Equation:

. . . . . .

7.1-S31 May 2012 TZ2 · Paper 3 · QC2 parts (b), (c) · also 7.3

(b) Electricity can also be generated from a lead–acid storage battery. The electrolyte is a solution of sulfuric acid and the electrodes are made of lead and lead(IV) oxide. State the equations for the reactions that occur at the positive and negative electrodes. *[2]*

(+) electrode (cathode):

. . . . . .

(–) electrode (anode):

. . . . . .

(c) Lead–acid batteries are heavy. Much lighter rechargeable cells are nickel–cadmium batteries used in electronic equipment.

Other than their chemical composition, discuss **two** major differences between fuel cells and nickel–cadmium cells. *[2]*

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7.1-S32 Nov 2011 · Paper 2 · Q7 parts (a), (b), (c)(i), (c)(ii), (d)(i), (d)(ii), (d)(iii), (e)(iv), (f) · also 1.1, 2.1, 6.1, 6.2, 7.2

7. Chromium is a typical transition metal with many uses.

(a) Distinguish between the terms *oxidation* and *reduction* in terms of oxidation numbers. *[1]*

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(b) State the names of Cr_2O_3 and CrO_3. *[2]*

Cr_2O_3: . . . . . .

CrO_3: . . . . . .

(c) (i) Define the term *oxidizing agent*. *[1]*

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(ii) $Cr_2O_7^{2-}(aq)$ and $I^-(aq)$ ions react together in the **presence of acid** to form $Cr^{3+}(aq)$ and $IO_3^-(aq)$ ions. Deduce the balanced chemical equation for this redox reaction and identify the species that acts as the oxidizing agent. *[3]*

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(d) A voltaic cell is constructed as follows. One half-cell contains a platinum electrode in a solution containing $K_2Cr_2O_7$ and H_2SO_4. The other half-cell contains an iron electrode in a solution containing Fe^{2+} ions. The two electrodes are connected to a voltmeter and the two solutions by a salt bridge.

(i) Draw a diagram of the voltaic cell, labelling the positive and negative electrodes (cathode and anode) and showing the direction of movement of the electrons and ions. Deduce an equation for the reaction occurring in each of the half-cells, and the equation for the overall cell reaction. *[5]*

(ii) Define the term *standard electrode potential*. *[1]*

. . . . . .

. . . . . .

(iii) Calculate the cell potential, in V, under standard conditions, using information from Table 14 of the Data Booklet. *[1]*

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(iv) Draw an orbital box diagram (arrow-in-box notation) showing the electrons in the 4s and 3d sub-levels in chromium metal. *[1]*

(f) Chromium is often used in electroplating. State what is used as the positive electrode (anode), the negative electrode (cathode) and the electrolyte in the chromium electroplating process. *[3]*

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7.1-S33 May 2011 TZ2 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (d)(ii), (d)(iii), (d)(iv), (d)(v), (e)(ii) · also 6.1, 6.2, 7.2

7. (a) The standard electrode potential for a half-cell made from iron metal in a solution of iron(II) ions, $Fe^{2+}(aq)$, has the value -0.45 V.

(i) Define *standard electrode potential*. *[2]*

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(ii) Explain the significance of the minus sign in -0.45 V. *[1]*

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(b) Consider the following table of standard electrode potentials.

	$E^{\ominus}$ / V
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.45
$Sn^{2+}(aq) + 2e^- \rightleftharpoons Sn(s)$	−0.14
$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$	0.00
$Sn^{4+}(aq) + 2e^- \rightleftharpoons Sn^{2+}(aq)$	+0.15
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80
$\frac{1}{2}Br_2(l) + e^- \rightleftharpoons Br^-(aq)$	+1.07

From the list above:

(i) State the species which is the strongest oxidizing agent. *[1]*

. . . . . .

. . . . . .

(ii) Deduce which species can reduce $Sn^{4+}(aq)$ to $Sn^{2+}(aq)$ but will not reduce $Sn^{2+}(aq)$ to $Sn(s)$ under standard conditions. *[1]*

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. . . . . .

(iii) Deduce which species can reduce $Sn^{2+}(aq)$ to $Sn(s)$ under standard conditions. *[1]*

. . . . . .

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(c) (i) Draw a labelled diagram of a voltaic cell made from an $Fe(s)$ / $Fe^{2+}(aq)$ half-cell connected to an $Ag(s)$ / $Ag^{+}(aq)$ half-cell operating under standard conditions. In your diagram identify the positive electrode (cathode), the negative electrode (anode) and the direction of electron flow in the external circuit. *[5]*

(ii) Deduce the equation for the chemical reaction occurring when the cell in part (c) (i) is operating under standard conditions and calculate the voltage produced by the cell. *[2]*

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(d) An acidified solution of potassium dichromate is often used as an oxidizing agent in organic chemistry. During the oxidation reaction of ethanol to ethanal the dichromate ion is reduced to chromium(III) ions according to the following **unbalanced** half-equation.

$$Cr_2O_7^{2-}(aq) + H^+(aq) + e^- \rightarrow Cr^{3+}(aq) + H_2O(l)$$

(ii) Deduce the oxidation number of chromium in $Cr_2O_7^{2-}$. *[1]*

. . . . . .

. . . . . .

(iii) State the balanced half-equation for the reduction of dichromate ions to chromium(III) ions. *[1]*

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(iv) Deduce the half-equation for the oxidation of ethanol to ethanal and hence the overall redox equation for the oxidation of ethanol to ethanal by acidified dichromate ions. *[3]*

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(v) Explain why it is necessary to carry out the reaction under acidic conditions. *[1]*

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(e) Sodium metal can be obtained by the electrolysis of molten sodium chloride.

(ii) Explain why an aqueous solution of sodium chloride cannot be used to obtain sodium metal by electrolysis. *[2]*

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7.1-S34 Nov 2010 · Paper 3 · QC4 parts (b)

(b) A different type of cell has the half-equation below.

$$Li^+(polymer) + MnO_2(s) + e^- \rightarrow LiMnO_2(s)$$

Identify this type of cell. *[1]*

. . . . . .

7.1-S35 May 2010 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 7.2

6. (a) (i) Draw an annotated diagram of a voltaic cell composed of a magnesium electrode in 1.0 mol dm^{-3} magnesium nitrate solution and a silver electrode in 1.0 mol dm^{-3} silver nitrate solution. State the direction of electron flow on your diagram. *[4]*

(ii) Deduce half-equations for the oxidation and reduction reactions. *[2]*

(iii) Using Table 14 of the Data Booklet, calculate the cell potential for this cell. *[2]*

(b) The standard electrode potentials for three other electrode systems are given below.

	$E^\ominus$ / V
$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$	+1.51
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77
$Cd^{2+}(aq) + 2e^- \rightleftharpoons Cd(s)$	−0.40

(i) Identify which species in the table above is the best reducing agent. *[1]*

(ii) Deduce the equation for the overall reaction with the greatest cell potential. *[2]*

(c) These values were obtained using a standard hydrogen electrode. Describe the materials and conditions used in the standard hydrogen electrode. (A suitably labelled diagram is acceptable). *[4]*

(d) (i) Solid sodium chloride does not conduct electricity but molten sodium chloride does. Explain this difference. *[2]*

(ii) Outline what happens in an electrolytic cell during the electrolysis of molten sodium chloride using inert electrodes. Deduce equations for the reactions occurring at each electrode. *[4]*

7.1-S36 May 2010 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(ii), (c) · also 6.1, 6.3, 7.2

8. The word *redox* comes from a combination of the terms *reduction* and *oxidation*. Redox reactions affect our daily lives.

(a) The overall reaction that takes place in a voltaic cell is shown below.

$$Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$

(i) Determine the oxidation number of lead in Pb, PbO_2 and $PbSO_4$. *[1]*

(ii) Deduce the oxidation and reduction half-equations taking place at the negative lead electrode (anode) and the positive lead(IV) oxide electrode (cathode). Deduce the oxidizing and reducing agents and state the direction of the electron flow between the electrodes. *[4]*

(iii) In order to determine the position of three metals in a reactivity series, the metals were placed in different solutions of metal ions. The table below summarizes whether or not a reaction occurred.

	$Ag^+(aq)$	**$Cu^{2+}(aq)$**	**$Pb^{2+}(aq)$**
Ag (s)		No reaction	No reaction
Cu (s)	Reaction		No reaction
Pb (s)	Reaction	Reaction	

State the equations for the **three** reactions that take place. Use this information to place the metals Ag, Cu and Pb in a reactivity series, with the strongest reducing agent first, and explain your reasoning. *[5]*

(iv) Use information from Table 14 of the Data Booklet to deduce the oxidizing agent that can oxidize chloride ions but not fluoride ions. State the redox equation for the reaction and determine its cell potential. *[4]*

(ii) Predict and explain the products of electrolysis of a **concentrated** solution of NaCl(aq) using inert electrodes. Your answer should include half-equations with state symbols for the reaction at **each** electrode. *[4]*

(c) Electroplating is an important application of electrolytic cells with commercial implications. Copper may be plated using an electrolytic cell with an aqueous acidified copper(II) sulfate electrolyte.

For the copper plating of tin to make jewellery, state the half-equation at **each** electrode. Assume the other electrode is also inert. Suggest **two** observations that you would be able to make as the electroplating progresses. *[4]*

LESSON 7.2

Electrolytic Cells

29 multiple-choice questions · 29 structured questions

Part A - Multiple choice

7.2-M1 May 2025 TZ2 · Paper 1A · Q39

39. What are the products of the electrolysis of dilute aqueous copper(II) sulfate when using inert electrodes?

	Product at anode	Product at cathode
A.	oxygen gas	hydrogen gas
B.	oxygen gas	copper metal
C.	copper metal	oxygen gas
D.	hydrogen gas	copper metal

7.2-M2 May 2025 TZ3 · Paper 1A · Q37

37. What are the products of the electrolysis of molten aluminium oxide, $Al_2O_3(l)$?

	Product at anode	Product at cathode
A.	O_2	Al
B.	O_3	Al
C.	Al	O_2
D.	Al	O_3

7.2-M3 May 2024 TZ1 · Paper 1 · Q30

30. What are the major products of the electrolysis of dilute aqueous magnesium chloride, $MgCl_2(aq)$?

A. magnesium metal and oxygen gas

B. magnesium metal and chlorine gas

C. hydrogen gas and oxygen gas

D. hydrogen gas and chlorine gas

7.2-M4 Nov 2023 TZ1 · Paper 1 · Q30

30. What is formed at the cathode in the electrolysis of $2.0\,mol\,dm^{-3}$ sodium chloride solution?

A. $Cl_2(g)$

B. $H_2(g)$

C. $Na(l)$

D. $O_2(g)$

7.2-M5 May 2023 TZ1 · Paper 1 · Q29

29. Which combination describes an electrolytic cell?

	Energy change	Spontaneity
A.	chemical to electrical	non-spontaneous
B.	electrical to chemical	non-spontaneous
C.	chemical to electrical	spontaneous
D.	electrical to chemical	spontaneous

7.2-M6 May 2023 TZ2 · Paper 1 · Q31

31. Which will eventually yield the greatest mass of deposited copper in the electrolysis of a fixed volume of 1 mol dm^{-3} $CuSO_4(aq)$?

A. copper anode and inert cathode

B. inert anode and copper cathode

C. inert anode and inert cathode

D. zinc anode and zinc cathode

7.2-M7 May 2022 TZ1 · Paper 1 · Q30

30. What are the products when dilute aqueous copper(II) nitrate is electrolysed using platinum electrodes?

$E^{\ominus}(Cu \mid Cu^{2+}) = -0.34\,V$.

	Anode (positive electrode)	Cathode (negative electrode)
A.	$O_2(g)$	$Cu(s)$
B.	$O_2(g)$	$H_2(g)$
C.	$Cu(s)$	$O_2(g)$
D.	$H_2(g)$	$Cu(s)$

7.2-M8 Nov 2021 · Paper 1 · Q31

31. Which aqueous solutions produce oxygen gas during electrolysis?

I. Dilute $CuCl_2(aq)$ with inert electrodes

II. Dilute $FeSO_4(aq)$ with inert electrodes

III. Dilute $CuCl_2(aq)$ with copper electrodes

The standard electrode potentials are provided in the table:

Half-equation	$E^{\ominus}$ / V
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.45
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O(l)$	+1.23
$\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$	+1.36

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.2-M9 May 2021 TZ1 · Paper 1 · Q31

31. What are the products when concentrated aqueous copper(II) chloride is electrolysed using platinum electrodes?

	Anode (positive electrode)	Cathode (negative electrode)
A.	$O_2(g)$	$Cu(s)$
B.	$Cl_2(g)$	$H_2(g)$
C.	$Cl_2(g)$	$Cu(s)$
D.	$O_2(g)$	$H_2(g)$

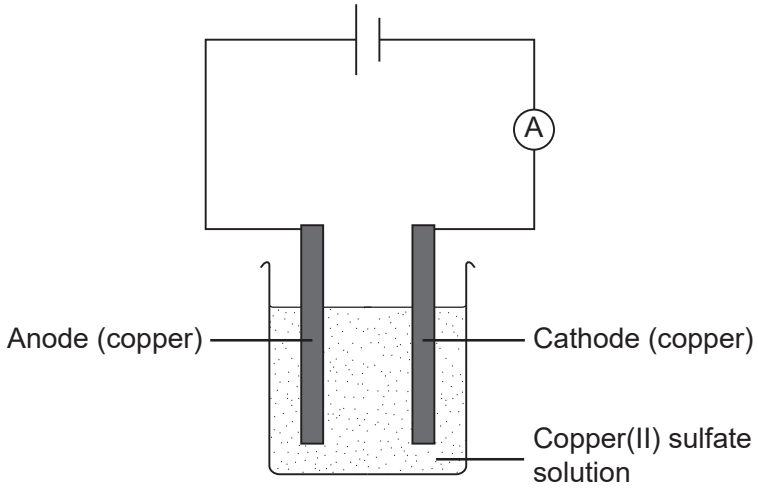
7.2-M10 May 2021 TZ2 · Paper 1 · Q29

29. What are the products of the electrolysis of molten potassium chloride, KCl (l)?

	Anode (positive electrode)	Cathode (negative electrode)
A.	K	Cl
B.	Cl_2	K
C.	Cl	K
D.	K	Cl_2

7.2-M11 May 2021 TZ2 · Paper 1 · Q31

31. What happens to the mass of each copper electrode when aqueous copper(II) sulfate solution is electrolysed?



	Anode (positive electrode)	Cathode (negative electrode)
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

7.2-M12 Nov 2020 · Paper 1 · Q31

31. Which statement is correct when a zinc spoon is electroplated with silver?

A. The cathode (negative electrode) is made of silver.

B. The anode (positive electrode) is the zinc spoon.

C. The anode (positive electrode) is made of silver.

D. The electrolyte is zinc sulfate solution.

7.2-M13 Nov 2019 · Paper 1 · Q31

31. What are the products of electrolysis of concentrated aqueous sodium bromide?

	Positive electrode	Negative electrode
A.	Br_2	Na
B.	O_2	H_2
C.	O_2	Na
D.	Br_2	H_2

7.2-M14 May 2019 TZ1 · Paper 1 · Q28

28. Which product will be obtained at the anode (positive electrode) when molten NaCl is electrolysed

A. $Na(l)$

B. $Cl(g)$

C. $Cl_2(g)$

D. $Na(s)$

7.2-M15 May 2019 TZ2 · Paper 1 · Q31

31. What are the products when concentrated $KBr(aq)$ is electrolysed?

	Anode (positive electrode)	Cathode (negative electrode)
A.	O_2	K
B.	O_2	H_2
C.	Br_2	K
D.	Br_2	H_2

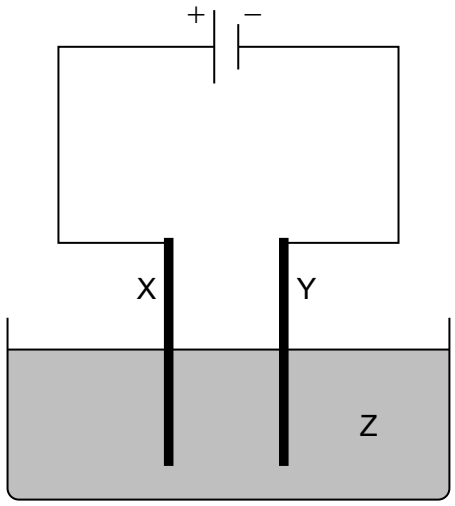
7.2-M16 May 2018 TZ1 · Paper 1 · Q29

29. What are the products of electrolysis when concentrated calcium bromide solution is electrolysed using graphite electrodes?

	Product at cathode (negative electrode)	Product at anode (positive electrode)
A.	hydrogen	bromine
B.	calcium	oxygen
C.	calcium	bromine
D.	hydrogen	oxygen

7.2-M17 May 2018 TZ1 · Paper 1 · Q30

30. Which combination would electroplate an object with copper?



	X	Z	Y
A.	object	$CuSO_4$ (aq)	copper
B.	copper	$CuSO_4$ (aq)	object
C.	object	H_2SO_4 (aq)	copper
D.	copper	H_2O (l)	object

7.2-M18 May 2018 TZ2 · Paper 1 · Q29

29. What are the products of the electrolysis of molten zinc bromide?

	Negative electrode (cathode)	Positive electrode (anode)
A.	zinc	bromine
B.	hydrogen	bromine
C.	bromine	zinc
D.	bromine	hydrogen

7.2-M19 May 2018 TZ2 · Paper 1 · Q31

31. What are the major products of electrolysing concentrated aqueous potassium iodide, KI (aq)?

	Negative electrode (cathode)	Positive electrode (anode)
A.	potassium	iodine
B.	hydrogen	iodine
C.	hydrogen	oxygen
D.	potassium	oxygen

7.2-M20 Nov 2017 · Paper 1 · Q31

31. What are the products when an aqueous solution of copper(II) sulfate is electrolysed using inert graphite electrodes?

	Cathode (negative electrode)	Anode (positive electrode)
A.	$Cu(s)$	$H_2(g)$
B.	$O_2(g)$	$Cu(s)$
C.	$Cu(s)$	$O_2(g)$
D.	$H_2(g)$	$O_2(g)$

7.2-M21 Nov 2016 · Paper 1 · Q33

33. An iron rod is electroplated with silver. Which is a correct condition for this process?

A. The silver electrode is the positive electrode.

B. The iron rod is the positive electrode.

C. The electrolyte is iron(II) sulfate.

D. Oxidation occurs at the negative electrode.

7.2-M22 May 2016 · Paper 1 · Q32

32. Which compound forms both hydrogen and oxygen at the electrodes when a concentrated aqueous solution is electrolyzed?

A. KI

B. NaCl

C. H_2SO_4

D. $AgNO_3$

7.2-M23 Nov 2015 · Paper 1 · Q33

33. What product is formed at the positive electrode (anode) when 0.001 mol dm^{-3} H_2SO_4(aq) is electrolysed?

A. Hydrogen

B. Oxygen

C. Sulfur

D. Sulfur dioxide

7.2-M24 Nov 2014 · Paper 1 · Q31

31. Which species are produced at each electrode during the electrolysis of molten lead(II) bromide, $PbBr_2$(l)?

	Negative electrode (cathode)	Positive electrode (anode)
A.	Br^-(l)	Pb^{2+}(l)
B.	Pb^{2+}(l)	Br^-(l)
C.	Br_2(g)	Pb (l)
D.	Pb (l)	Br_2(g)

7.2-M25 May 2014 TZ1 · Paper 1 · Q32

32. What is an industrial use of electrolysis?

A. The production of graphite

B. The production of iron

C. The production of electric energy

D. Electroplating

7.2-M26 Nov 2013 · Paper 1 · Q34

34. What happens during the electrolysis of concentrated aqueous potassium chloride?

I. Reduction takes place at the negative electrode (cathode).
II. Hydrogen gas is evolved at the negative electrode (cathode).
III. The pH of the electrolyte increases.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.2-M27 May 2013 TZ2 · Paper 1 · Q32

32. Which statement is correct for electroplating an object with gold?

A. The object must be the negative electrode (cathode).

B. The negative electrode (cathode) must be gold.

C. The object must be the positive electrode (anode).

D. The gold electrode must be pure.

7.2-M28 May 2012 TZ1 · Paper 1 · Q33

33. Which combination of electrodes and electrolyte could be used to plate a spoon with silver?

	Negative electrode (cathode)	Positive electrode (anode)	Electrolyte
A.	silver	spoon	sulfuric acid solution
B.	spoon	silver	sulfuric acid solution
C.	silver	spoon	silver nitrate solution
D.	spoon	silver	silver nitrate solution

7.2-M29 May 2010 TZ1 · Paper 1 · Q32

32. What condition is necessary for the electroplating of silver, Ag, onto a steel spoon?

A. The spoon must be the positive electrode.

B. The silver electrode must be the negative electrode.

C. The spoon must be the negative electrode.

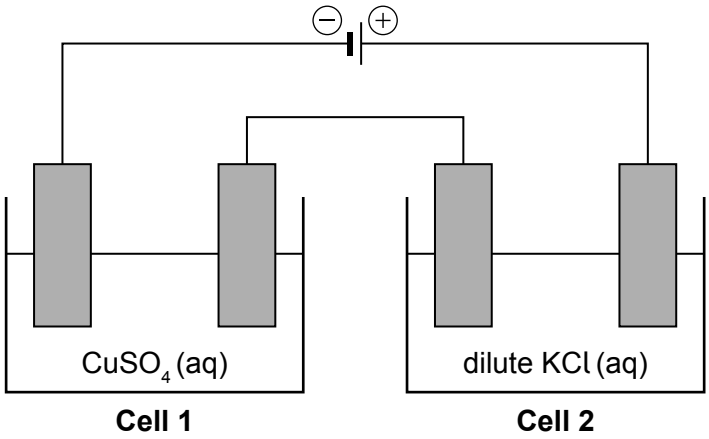
D. The electrolyte must be acidified.

Part B - Structured questions

7.2-S1 Nov 2024 · Paper 2 · Q3 parts (b)(i), (b)(ii), (d), (e)(i) · also 2.1, 9.2

3. Potassium, K, and potassium chloride, KCl, both form lattice structures in the solid state.

(b) Two electrolytic cells are connected in series. All electrodes are inert.



(i) State the half equation for the reaction occurring at each electrode in Cell 2. Use section 24 of the data booklet. [2]

Anode (positive electrode): . . . . . .

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Cathode (negative electrode): . . . . . .

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(ii) Identify the product at the anode in Cell 1. [1]

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(d) Outline the evidence that the electrons in a potassium atom occupy four main energy levels. [2]

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(e) Potassium reacts with water to produce potassium hydroxide.

$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$

(i) Calculate the enthalpy of reaction, in $kJ\,mol^{-1}$, when 1 mol of potassium reacts with water. Use section 12 of the data booklet. ΔH_f of $KOH(aq)$ is $-481.8\,kJ\,mol^{-1}$. [3]

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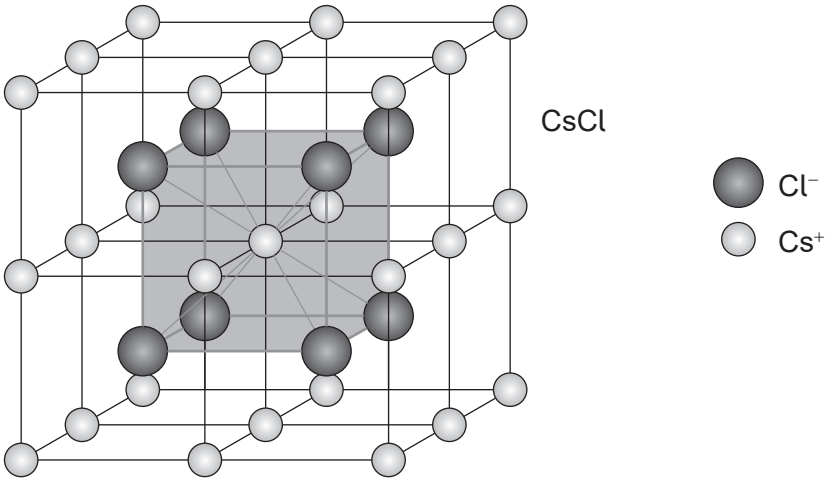
7.2-S2 May 2024 TZ1 · Paper 3 · Q3 parts (c)(ii)

3. Caesium chloride, CsCl, has the ionic lattice structure shown.



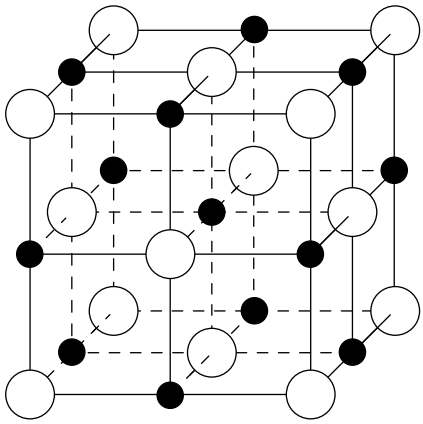
(c) Caesium metal is produced by electrolysis of molten caesium chloride.

(ii) State the half-equation for the formation of caesium by electrolysis. [1]

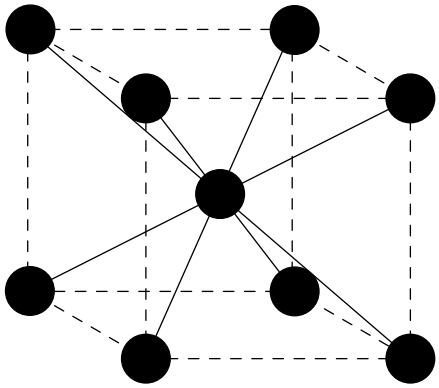
. . . . . .

7.2-S3 May 2024 TZ2 · Paper 2 · Q7 parts (c), (d), (e), (f), (h) · also 6.3, 7.1, 9.4

7. The structures of sodium bromide and sodium metal are shown below.



Sodium bromide



Sodium metal

(c) Write the half-equations for the formation of the products at the positive electrode (anode) and negative electrode (cathode) when molten sodium bromide is electrolysed. [2]

Positive electrode (anode): . . . . . .

. . . . . .

Negative electrode (cathode): . . . . . .

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(d) Determine the products formed at each electrode during the electrolysis of an aqueous solution of sodium bromide. Use section 24 in the data booklet. [2]

Positive electrode (anode): . . . . . .

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Negative electrode (cathode): . . . . . .

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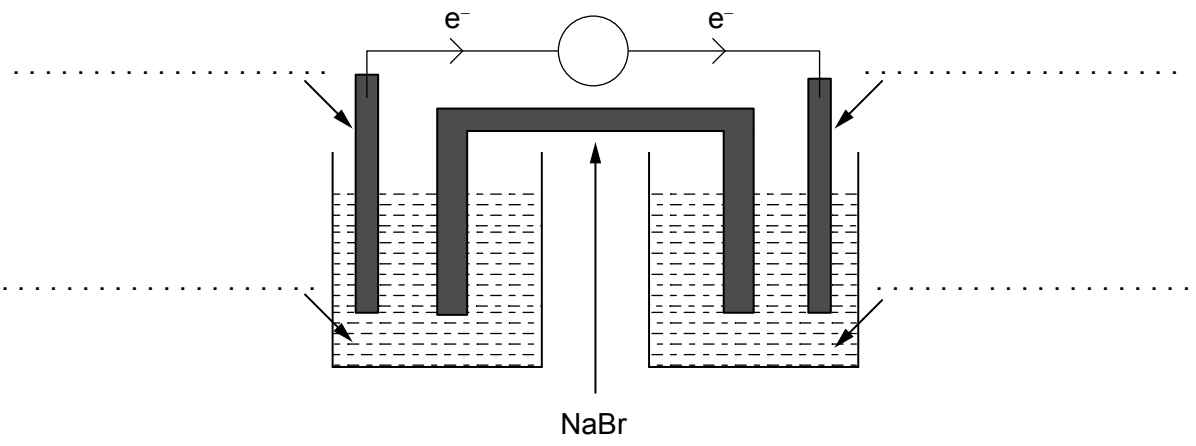
(e) A sodium bromide solution can be used for a salt bridge in a voltaic cell. Annotate the diagram of the magnesium, Mg, and zinc, Zn, voltaic cell shown. Use section 24 of the data booklet. [2]



(f) Determine the lattice enthalpy of sodium bromide using the data given below and sections 8 and 11 of the data booklet. [3]

$Na(s) \rightarrow Na(g)$ $\Delta H = +107 \text{ kJ mol}^{-1}$

$Br_2(l) \rightarrow Br_2(g)$ $\Delta H = +30 \text{ kJ mol}^{-1}$

$Na(s) + \frac{1}{2}Br_2(l) \rightarrow NaBr(s)$ $\Delta H = -361.5 \text{ kJ mol}^{-1}$

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(h) Predict, giving a reason, the result of reacting aqueous sodium bromide separately with iodine and chlorine. [2]

Iodine: . . . . . .

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Chlorine: . . . . . .

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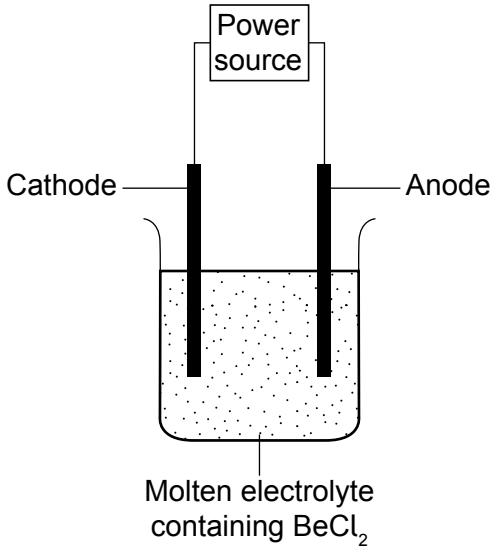
7.2-S4 Nov 2023 TZ1 · Paper 2 · Q5 parts (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (d)(i), (f)(i), (f)(ii), (g), (h) · also 2.1, 3.1, 4.2, 7.1

5. Beryllium is a low-density metal that is used in specialized lightweight alloys.

(b) The production of beryllium is illustrated in the diagram.



(i) Outline why molten $BeCl_2$ is considered an electrolyte. [1]

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(ii) Identify the electrode at which beryllium will be produced **and** the polarity of that electrode. [1]

Electrode: . . . . . .

Polarity: . . . . . .

(iii) Write a balanced equation for the reaction occurring at the other electrode, to the one you identified in 5(b)(ii). [1]

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(c) Beryllium forms a chloride, $BeCl_2$.

(i) Draw the Lewis (electron dot) structure of the $BeCl_2$ molecule. [1]

(ii) Outline how the Lewis (electron dot) structure of the $BeCl_2$ molecule differs from most Lewis (electron dot) structures. [1]

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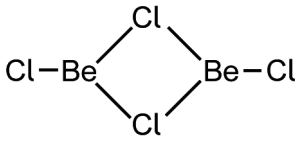
(d) Beryllium chloride, $BeCl_2$, partially dimerizes in the gas phase to produce this molecule:



(i) Identify the hybridization of the beryllium atom in the dimer, Be_2Cl_4. [1]

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(f) The standard electrode potential, E°, of

$$Be^{2+}(aq) + 2e^{-} \rightleftharpoons Be(s)$$

is $-1.85\,V$.

(i) Calculate the cell potential for the reaction

$$Be(s) + 2H_2O(l) \rightarrow Be^{2+}(aq) + 2OH^{-}(aq) + H_2(g)$$

Use section 24 of the data booklet. [1]

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(ii) Deduce, giving a reason, whether this reaction is thermodynamically spontaneous. [1]

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(g) Explain, in terms of nuclear charge, electron subshells and the shielding provided by filled electron shells, why the first ionization energy increases from Li to Be, but decreases from Be to B. [4]

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(h) Outline how the first ionization energy of beryllium could be found from its atomic emission spectrum. [1]

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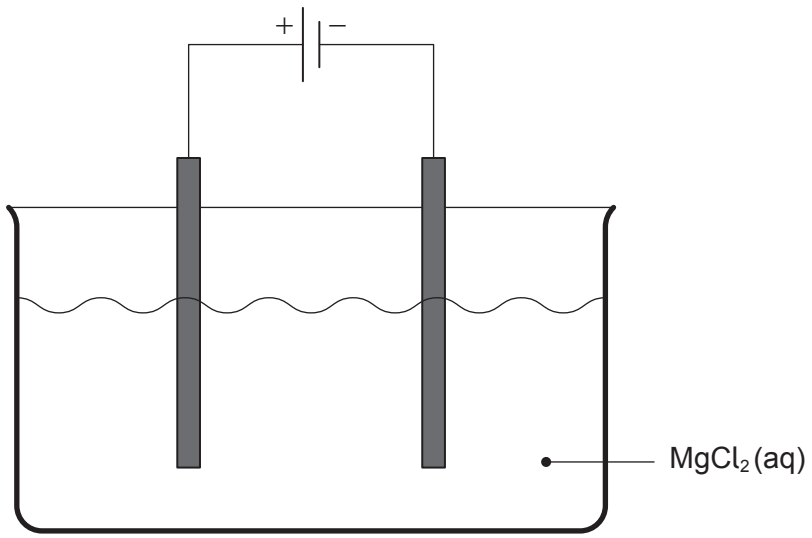
7.2-S5 May 2023 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (a)(iii), (b)(i) · also 1.2, 4.3

3. Electrolysis and Winkler titrations are both applications of redox reactions.

(a) An electrolytic cell was set up using inert electrodes and a dilute aqueous solution of magnesium chloride, $MgCl_2(aq)$.

(i) Annotate the diagram to show the movement of particles that conduct electricity in this cell. [2]



(ii) Deduce the half-equation for the reaction at each electrode. Use section 24 of the data booklet. [2]

Positive electrode: . . . . . .

. . . . . .

Negative electrode: . . . . . .

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(iii) Graphite rods are sometimes used as inert electrodes. Describe the structure of graphite and explain why graphite conducts electricity. [2]

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(b) Winkler titrations can be used to determine the biochemical oxygen demand, BOD, of a water sample. One set of equations for the reactions occurring is:

$$2Mn^{2+}(aq) + O_2(aq) + 4OH^- \rightarrow 2MnO(OH)_2(s)$$

$$MnO(OH)_2(s) + 2I^-(aq) + 4H^+ \rightarrow Mn^{2+}(aq) + I_2(aq) + 3H_2O$$

$$I^-(aq) + I_2(aq) \rightarrow I_3^-(aq)$$

$$2S_2O_3^{2-}(aq) + I_3^-(aq) \rightarrow S_4O_6^{2-}(aq) + 3I^-(aq)$$

$150\,cm^3$ of a water sample was tested using a Winkler titration. $36.0\,cm^3$ of $0.00500\,mol\,dm^{-3}$ sodium thiosulfate solution, $Na_2S_2O_3(aq)$, was required to reach the end point.

(i) Determine the concentration, in mol dm^{-3}, of oxygen dissolved in the water sample. [3]

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7.2-S6 May 2023 TZ2 · Paper 3 · Q3 parts (b)(i), (b)(iii), (d) · also 4.3, 7.3

3. Most metals have to be extracted from an ore. The way in which this is carried out depends on the reactivity of the metal.

(b) Aluminium is produced by electrolytic reduction of a solution of aluminium oxide, Al_2O_3, in molten cryolite, Na_3AlF_6.

(i) Write the half equation for the reaction at the electrode where aluminium is formed. [1]

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(iii) Suggest **one** factor, other than atom economy, that indicates the production of aluminium from its ore has a significant environmental impact. [1]

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(d) An aluminium matrix can be reinforced with carbon nanotubes. Outline why carbon nanotubes are so strong and rigid. [1]

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7.2-S7 Nov 2019 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (c)(iv), (e)(i), (e)(ii) · also 1.2, 2.1, 7.1, 9.2

6. Copper forms two chlorides, copper(I) chloride and copper(II) chloride.

(a) (i) State the electron configuration of the Cu^+ ion. [1]

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(ii) Copper(II) chloride is used as a catalyst in the production of chlorine from hydrogen chloride.

$$4HCl(g) + O_2(g) \rightarrow 2Cl_2(g) + 2H_2O(g)$$

Calculate the standard enthalpy change, $\Delta H^{\ominus}$, in kJ, for this reaction, using section 12 of the data booklet. [2]

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(b) Solid copper(II) chloride absorbs moisture from the atmosphere to form a hydrate of formula $CuCl_2 \bullet xH_2O$.

A student heated a sample of hydrated copper(II) chloride, in order to determine the value of x. The following results were obtained:

Mass of crucible = 16.221 g
Initial mass of crucible and hydrated copper(II) chloride = 18.360 g
Final mass of crucible and anhydrous copper(II) chloride = 17.917 g

Determine the value of x. [3]

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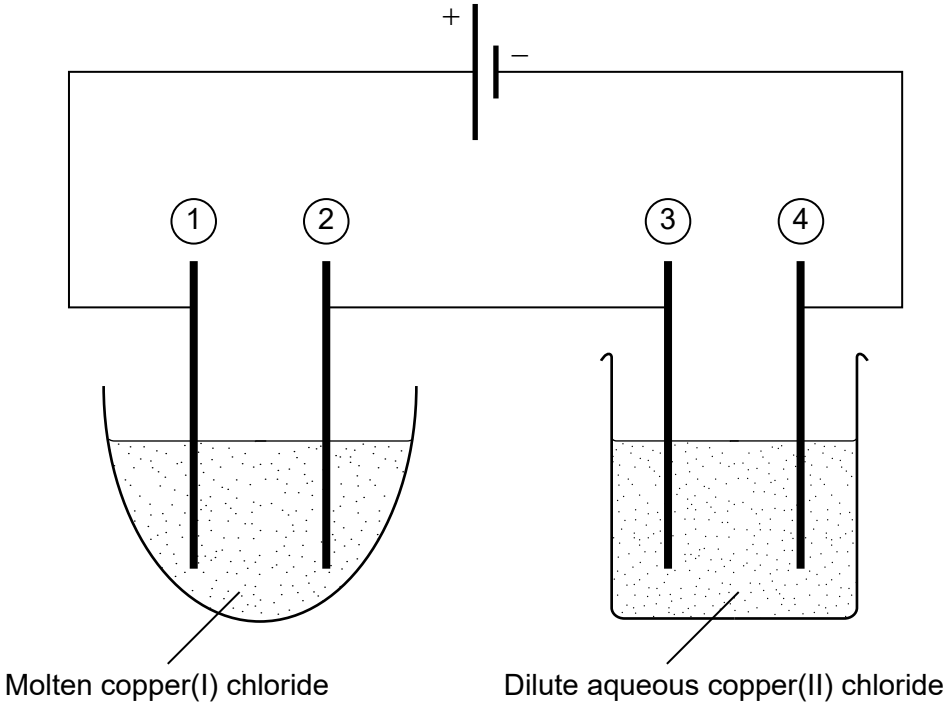
(c) Two electrolysis cells were assembled using graphite electrodes and connected in series as shown.



(i) State how current is conducted through the wires and through the electrolyte. [2]

Wires:

. . . . . .

Electrolyte:

. . . . . .

(ii) Write the half-equation for the formation of gas bubbles at electrode 1. [1]

. . . . . .

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(iii) Bubbles of gas were also observed at another electrode. Identify the electrode and the gas. [1]

Electrode number (on diagram):

. . . . . .

Name of gas:

. . . . . .

(iv) Deduce the half-equation for the formation of the gas identified in (c)(iii). [1]

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(e) Copper(I) chloride undergoes a disproportionation reaction, producing copper(II) chloride and copper.

$$2Cu^{+}(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$$

(i) Calculate the cell potential at 298 K for the disproportionation reaction, in V, using section 24 of the data booklet. [1]

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(ii) Comment on the spontaneity of the disproportionation reaction at 298 K. [1]

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7.2-S8 Nov 2019 · Paper 3 · Q6 parts (b)(ii)

6. Metals are extracted from their ores by various means.

(b) Aluminium is produced by the electrolysis of alumina (aluminium oxide) dissolved in cryolite.

(ii) Write half-equations for the electrolysis of molten alumina using graphite electrodes, deducing the state symbols of the products. [3]

	Melting point / K
Alumina	2345
Cryolite	1285
Aluminium	933
Graphite	3500

Anode (positive electrode):

. . . . . .

. . . . . .

Cathode (negative electrode):

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7.2-S9 May 2019 TZ1 · Paper 2 · Q7

7. An aqueous solution of silver nitrate, $AgNO_3(aq)$, can be electrolysed using platinum electrodes.

Formulate the half-equations for the reaction at each electrode during electrolysis. [2]

Cathode (negative electrode):

. . . . . .

. . . . . .

Anode (positive electrode):

. . . . . .

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7.2-S10 May 2019 TZ2 · Paper 2 · Q4 parts (b)(ii), (c), (d)(i), (e)(ii), (e)(iii) · also 1.1, 6.2, 6.3, 7.1

4. Rhenium, Re, was the last element with a stable isotope to be isolated.

(b) Before its isolation, scientists predicted the existence of rhenium and some of its properties.

(ii) A scientist wants to investigate the catalytic properties of a thin layer of rhenium metal on a graphite surface.

Describe an electrochemical process to produce a layer of rhenium on graphite. [2]

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(c) Describe how the relative reactivity of rhenium, compared to silver, zinc, and copper, can be established using pieces of rhenium and solutions of these metal sulfates. [2]

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(d) One chloride of rhenium has the empirical formula $ReCl_3$.

(i) State the name of this compound, applying IUPAC rules. [1]

. . . . . .

(e) Rhenium forms salts containing the perrhenate(VII) ion, ReO_4^-.

(ii) Deduce the coefficients required to complete the half-equation. [1]

$$____ ReO_4^-(aq) + ____ H^+(aq) + ____ e^- \rightleftharpoons [Re(OH)_2]^{2+}(aq) + ____ H_2O(l) \qquad E^\ominus = +0.36\,V$$

(iii) Predict, giving a reason, whether the reduction of ReO_4^- to $[Re(OH)_2]^{2+}$ would oxidize Fe^{2+} to Fe^{3+} in aqueous solution. Use section 24 of the data booklet. [1]

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7.2-S11 Nov 2018 · Paper 3 · Q4 parts (c)

4. While heating solid cholesteryl benzoate, Reinitzer discovered the liquid crystal phase.

(c) Arc discharge, consisting of two inert metal electrodes in a liquid solvent, is one method of producing carbon nanotubes (CNTs).

Predict, giving a reason, the electrode at which the solvent cyclohexane, C_6H_{12}, will decompose to form CNTs. [2]

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7.2-S12 May 2018 TZ2 · Paper 2 · Q3 parts (a)(iii), (a)(iv), (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v) · also 2.1

3. The emission spectrum of an element can be used to identify it.

(iii) Hydrogen spectral data give the frequency of $3.28 \times 10^{15}\,s^{-1}$ for its convergence limit.

Calculate the ionization energy, in J, for a single atom of hydrogen using sections 1 and 2 of the data booklet. [1]

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(iv) Calculate the wavelength, in m, for the electron transition corresponding to the frequency in (a)(iii) using section 1 of the data booklet. [1]

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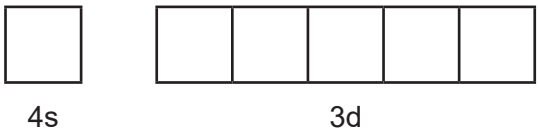
(c) (i) Copper is widely used as an electrical conductor.

Draw arrows in the boxes to represent the electronic configuration of copper in the 4s and 3d orbitals. [1]



(ii) Impure copper can be purified by electrolysis. In the electrolytic cell, impure copper is the anode (positive electrode), pure copper is the cathode (negative electrode) and the electrolyte is copper(II) sulfate solution.

Formulate the half-equation at each electrode. [2]

Anode (positive electrode):

. . . . . .

Cathode (negative electrode):

. . . . . .

(iii) Outline where and in which direction the electrons flow during electrolysis. [1]

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(iv) Deduce any change in the colour of the electrolyte during electrolysis. [1]

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(v) Deduce the gas formed at the anode (positive electrode) when graphite is used in place of copper. [1]

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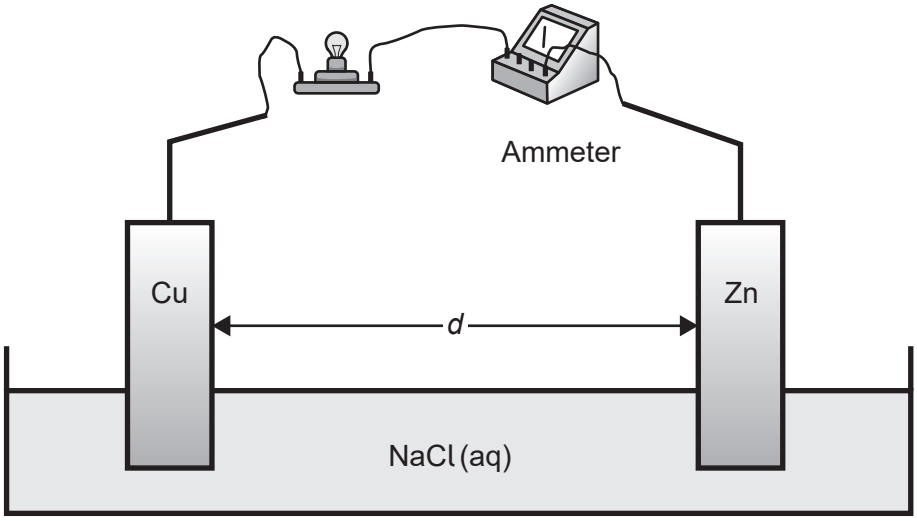
7.2-S13 Nov 2017 · Paper 3 · Q1 parts (b)(iii)

1. A student set up a simple voltaic cell consisting of a copper electrode and a zinc electrode dipped in sodium chloride solution.



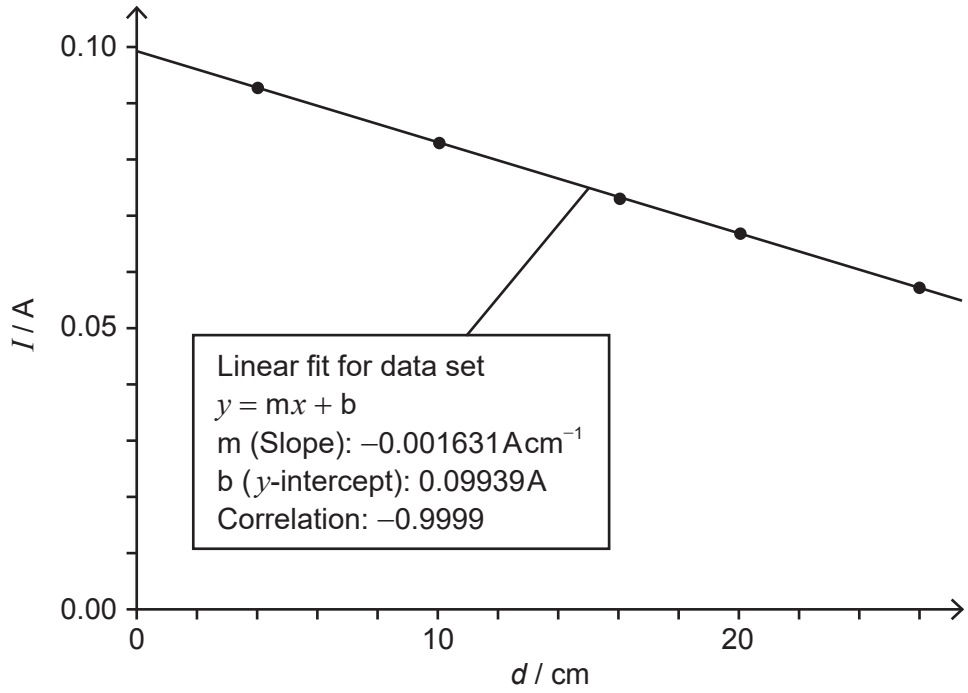
The student gradually increased the distance, *d*, between the electrodes to study the effect on the initial current, I, passing through the light bulb.

The student hypothesized that the initial current would be inversely proportional to the distance between the electrodes.

(b) The following data was collected over five trials.

***d* / ±0.1 cm**	**Average *I* / ±0.04 A**
4.0	0.093
10.0	0.083
16.0	0.073
20.0	0.067
26.0	0.057

The data did not support the student's hypothesis. He investigated other possible relationships by plotting a graph of the average current against the distance between the electrodes. He obtained the following best-fit line with a correlation coefficient (r) of −0.9999.



(iii) Outline how current flows in the sodium chloride solution. [1]

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7.2-S14 May 2015 TZ1 · Paper 2 · Q6 parts (d)(i), (d)(ii), (e), (f)(i), (f)(ii), (g) · also 6.1, 7.1, 9.4

6. Bromomethane was used as a pesticide until it was found to be ozone-depleting.

(d) (i) Bromine can be produced by the electrolysis of **molten** sodium bromide. Deduce the half-equation for the reaction at each electrode. [2]

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

(ii) Predict the products formed at the electrodes during the electrolysis of concentrated **aqueous** sodium bromide. [2]

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

(e) Bromine reacts with aqueous sodium iodide.

$$Br_2(aq) + 2NaI(aq) \rightarrow I_2(aq) + 2NaBr(aq)$$

Identify the oxidizing agent in this reaction. [1]

. . . . . .

(f) (i) Define the term *standard electrode potential,* $E^{\ominus}$. [1]

. . . . . .
. . . . . .

(ii) Draw a labelled diagram for the voltaic cell in which the following reaction occurs.

$$Mg(s) + Cu^{2+}(aq) \rightarrow Mg^{2+}(aq) + Cu(s)$$

Include in your answer the direction of electron flow and the polarity of the electrodes. [4]

(g) Determine the standard enthalpy change of formation, $\Delta H_f^{\ominus}$, of NaCl (s), in $kJ\,mol^{-1}$, using a Born-Haber cycle and tables 7, 10 and 13 of the data booklet. The standard enthalpy change of atomization (standard enthalpy change of sublimation), $\Delta H_{at}^{\ominus}$, of Na (s) is $+108\,kJ\,mol^{-1}$. [4]

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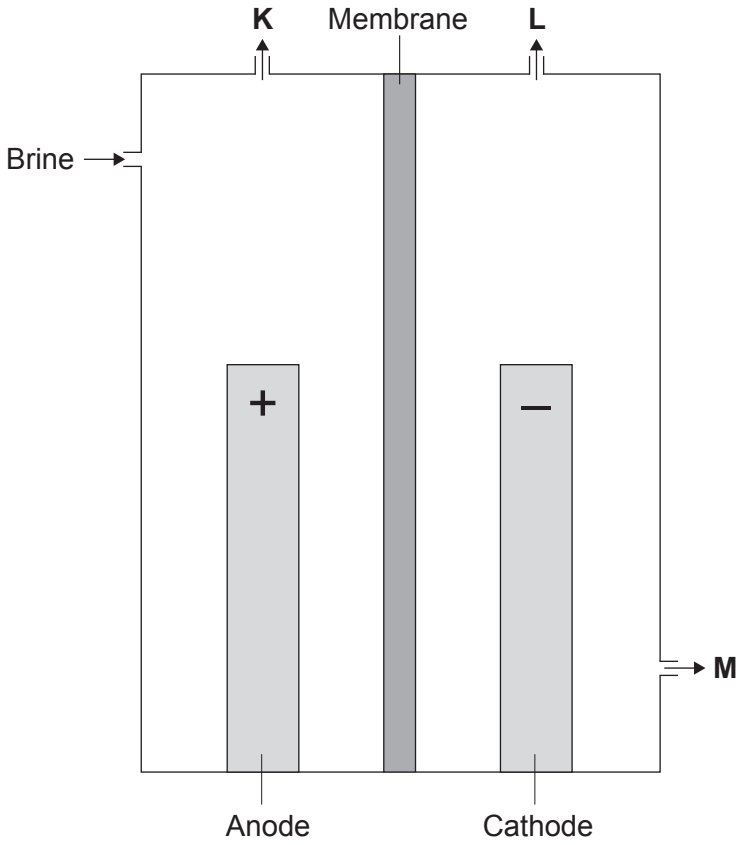
7.2-S15 May 2015 TZ1 · Paper 3 · Q12 parts (b), (c)

12. The membrane chlor-alkali electrolysis cell, shown below, is increasingly used in preference to both mercury and diaphragm cells. It has greater efficiency and less impact on the environment.



(b) Identify the products **K** and **M**. [1]

K:

. . . . . .

M:

. . . . . .

(c) State the half-equations for the reactions at each electrode. [2]

Negative electrode (cathode):

. . . . . .

Positive electrode (anode):

. . . . . .

7.2-S16 May 2015 TZ2 · Paper 2 · Q6 parts (a)

6. Electrolysis is an important industrial process used to obtain very reactive elements from their common ores.

(a) Molten magnesium chloride can be electrolysed using inert graphite electrodes at 800 °C.

Deduce the half-equations, including state symbols, for the reactions occurring at each electrode. (The melting points of $MgCl_2$ and Mg are 714 °C and 649 °C respectively.) [3]

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

7.2-S17 May 2015 TZ2 · Paper 3 · Q13 parts (b)

13. The chlor-alkali industry involves the large-scale electrolysis of aqueous sodium chloride.

(b) Explain, using equations, the reactions that occur at the electrodes in the membrane cell. [2]

Negative electrode (cathode):

. . . . . .

. . . . . .

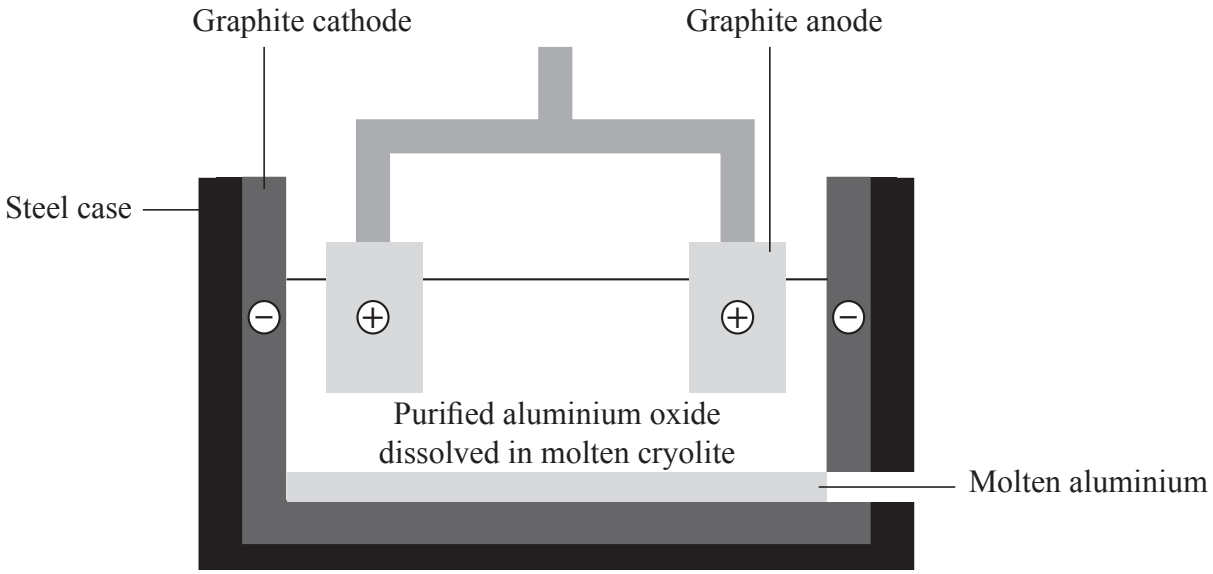
Positive electrode (anode):

. . . . . .

. . . . . .

7.2-S18 Nov 2014 · Paper 3 · Q9 parts (a)(ii), (a)(iii)

9. Aluminium is chemically reactive so it has to be extracted by the electrolysis of aluminium oxide dissolved in molten cryolite.



(ii) Deduce an equation for the discharge of the ions at each electrode. *[2]*

Positive electrode (anode):

. . . . . .

. . . . . .

Negative electrode (cathode):

. . . . . .

. . . . . .

(iii) Suggest why the anodes have to be replaced at regular intervals. *[1]*

. . . . . .

. . . . . .

7.2-S19 May 2014 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(v), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i), (c)(ii), (d)(ii), (d)(iii) · also 6.2, 6.3, 7.1, 7.3

6. Oxidation and reduction can be defined in terms of electron transfer or oxidation numbers.

(a) Alcohols with the molecular formula C_4H_9OH occur as four structural isomers. Three of the isomers can be oxidized with acidified potassium dichromate solution to form compounds with the molecular formula C_4H_8O.

(i) Deduce the half-equation for the oxidation of the alcohol C_4H_9OH. *[1]*

. . . . . .

(ii) Deduce the overall equation for the redox reaction. *[1]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(v) All isomers of the alcohol C_4H_9OH undergo complete combustion. State an equation for the complete combustion of C_4H_9OH. *[2]*

. . . . . .
. . . . . .

(b) A reactivity series can be experimentally determined by adding the metals W, X, Y and Z to solutions of these metal ions. The following reactions were observed:

$$W^{2+}(aq) + X(s) \rightarrow W(s) + X^{2+}(aq)$$
$$Y(s) + W^{2+}(aq) \rightarrow Y^{2+}(aq) + W(s)$$
$$Z^{2+}(aq) + W(s) \rightarrow Z(s) + W^{2+}(aq)$$
$$Y(s) + X^{2+}(aq) \rightarrow Y^{2+}(aq) + X(s)$$

(i) Deduce the order of reactivity of these four metals, from the least to the most reactive. *[1]*

. . . . . .

(ii) A voltaic cell is made by connecting a half-cell of X in $XCl_2(aq)$ to a half-cell of Z in $ZCl_2(aq)$. Deduce the overall equation for the reaction taking place when the cell is operating. *[1]*

. . . . . .

(iii) The standard electrode potential for $Z^{2+}(aq) + 2e^- \rightleftharpoons Z(s)$ is +0.20 V. State which species is oxidized when this half-cell is connected to a standard hydrogen electrode. *[1]*

. . . . . .

(iv) Describe the standard hydrogen electrode including a fully labelled diagram. *[3]*

. . . . . .
. . . . . .
. . . . . .

(c) A student carries out the electrolysis of aqueous potassium iodide, KI, using inert electrodes.

(i) State the half-equation for the reaction that occurs at each electrode. *[2]*

Positive electrode (anode):

. . . . . .

. . . . . .

Negative electrode (cathode):

. . . . . .

. . . . . .

(ii) Suggest, giving a reason, what would happen if the electrodes were changed to aluminium. *[2]*

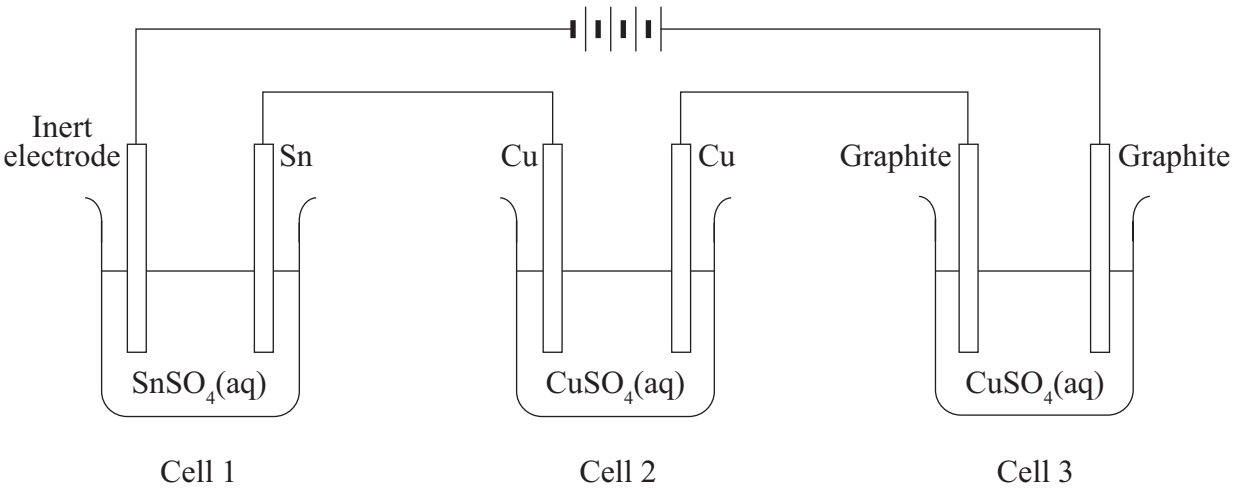
. . . . . .

. . . . . .

. . . . . .

. . . . . .

(d) Three electrolytic cells were set up in series (one cell after the other), as shown below. All of the solutions had a concentration of $1.00\,\text{mol}\,\text{dm}^{-3}$.



(ii) Compare the colour and the pH of the solutions in cells 2 and 3 after the current has been flowing for one hour. *[2]*

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. . . . . .
. . . . . .
. . . . . .

(iii) Explain your answer given for part (d) (ii). *[2]*

Colour:

. . . . . .
. . . . . .
. . . . . .

pH:

. . . . . .
. . . . . .
. . . . . .

7.2-S20 May 2014 TZ2 · Paper 2 · Q8 parts (a), (b)(i), (b)(ii), (c), (d), (g)(i), (g)(ii), (g)(iii), (g)(iv) · also 2.1, 9.4

8. Magnesium, a reactive metal found in many common minerals, is also an essential nutrient for both plants and animals.

(a) Define the term *first ionization energy*. *[2]*

. . . . . .
. . . . . .
. . . . . .

(b) Successive ionization energies of magnesium are given in the table below.

	First	**Second**	**Third**
Energy required / kJ mol^{-1}	738	1450	7730

(i) Explain why the second ionization energy is greater than the first ionization energy. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Explain why the third ionization energy is **much** greater than the second ionization energy. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(c) Although magnesium is usually found as Mg^{2+} in its compounds, it is possible to use the Born–Haber cycle to investigate the possibility of Mg^{+} being able to form stable compounds.

Use the ionization energy data from part (b), along with the other data provided below, to determine the enthalpy change of formation of MgCl(s). Assume that, because Mg^{+} would be similar in size to Na^{+}, MgCl would have a similar lattice enthalpy to NaCl.

Enthalpy of atomization of Mg	+146 kJ mol^{-1}
Bond enthalpy in Cl_2	+243 kJ mol^{-1}
Electron affinity of Cl	–349 kJ mol^{-1}
Lattice enthalpy of NaCl	+790 kJ mol^{-1}

[3]

. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

(d) Consider the lattice enthalpies of MgF_2, $MgCl_2$ and $CaCl_2$. List these from the most endothermic to the least endothermic and explain your order. *[3]*

. . . . . .

Most endothermic ⟶ Least endothermic

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(g) Magnesium is usually produced by the electrolysis of molten magnesium chloride.

(i) Draw a labelled diagram of a suitable apparatus for the electrolysis. *[2]*

(ii) State equations for the reactions that take place at the electrodes. *[2]*

Negative electrode (cathode) reaction:

. . . . . .

Positive electrode (anode) reaction:

. . . . . .

(iii) When **dilute** aqueous magnesium chloride is used as the electrolyte, the reactions at both electrodes are different. State equations for the reactions that occur in aqueous solution. *[2]*

Negative electrode (cathode) reaction:

. . . . . .

Positive electrode (anode) reaction:

. . . . . .

(iv) Outline why magnesium metal is not produced in the electrolysis of aqueous magnesium chloride. *[1]*

. . . . . .

. . . . . .

7.2-S21 May 2014 TZ2 · Paper 3 · Q10 parts (a), (b)

10. Both chlorine and sodium hydroxide are important intermediates in the modern chemical industry.

(a) Outline the process used to produce these substances. *[1]*

. . . . . .
. . . . . .
. . . . . .

(b) State equations for the reactions taking place at the positive electrode (anode) and the negative electrode (cathode), for the process outlined in (a). *[2]*

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

7.2-S22 May 2013 TZ1 · Paper 2 · Q7 parts (a)(ii), (b)(iv), (b)(v), (b)(vi), (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v) · also 2.1, 6.1, 6.3

7. (a) Bromine is a member of group 7, the halogens.

(ii) Deduce, using equations where appropriate, if bromine reacts with sodium chloride solution and with sodium iodide solution. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(b) Iron is a transition metal.

(iv) Deduce the oxidation number of iron in $[Fe(CN)_6]^{3-}$. *[1]*

. . . . . .

. . . . . .

(v) Draw the abbreviated orbital diagram for an **iron atom** using the arrow-in-box notation to represent electrons. *[1]*

(vi) Draw the abbreviated orbital diagram for the **iron ion in $[Fe(CN)_6]^{3-}$** using the arrow-in-box notation to represent electrons. *[1]*

(c) Freshly prepared iron(II) bromide can be electrolysed both in the liquid state and in aqueous solution.

(i) Describe, using a diagram, the essential components of an electrolytic cell. *[3]*

(ii) Describe the **two** ways in which current is conducted in an electrolytic cell. *[2]*

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. . . . . .
. . . . . .
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(iii) Predict and explain the products of electrolysis of a **dilute** iron(II) bromide solution. *[4]*

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. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .
. . . . . .

(iv) Identify another product that is formed if the solution of iron(II) bromide is **concentrated**. *[1]*

. . . . . .

(v) Explain why this other product is formed. *[1]*

. . . . . .
. . . . . .
. . . . . .

7.2-S23 May 2013 TZ2 · Paper 2 · Q7 parts (a), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii), (e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi), (e)(vii) · also 6.1, 6.2, 7.1

7. (a) Define *oxidation* in terms of oxidation number. *[1]*

. . . . . .
. . . . . .

(b) (i) Deduce the balanced chemical equation for the redox reaction of copper, Cu(s), with nitrate ions, $NO_3^-(aq)$, **in acid**, to produce copper(II) ions, $Cu^{2+}(aq)$, and nitrogen(IV) oxide, $NO_2(g)$. *[2]*

. . . . . .
. . . . . .
. . . . . .
. . . . . .

(ii) Deduce the oxidizing and reducing agents in this reaction. *[1]*

Oxidizing agent:

. . . . . .

Reducing agent:

. . . . . .

(c) A voltaic cell was set up, using the standard hydrogen electrode as a reference electrode and a standard $Cu^{2+}(aq)/Cu(s)$ electrode.

(i) Describe the standard hydrogen electrode including a fully labelled diagram. *[3]*

. . . . . .

. . . . . .

. . . . . .

(ii) Define the term *standard electrode potential*, $E^{\ominus}$. *[1]*

. . . . . .

. . . . . .

(iii) Deduce a balanced chemical equation, including state symbols, for the overall reaction which will occur spontaneously when the two half-cells are connected. *[2]*

. . . . . .
. . . . . .
. . . . . .

(d) Another voltaic cell was set up, using a $Sn^{2+}(aq)/Sn(s)$ half-cell and a $Cu^{2+}(aq)/Cu(s)$ half-cell under standard conditions.

(i) Draw a fully labelled diagram of the voltaic cell, showing the positive electrode (cathode), the negative electrode (anode) and the direction of electron movement through the external circuit. *[3]*

(ii) Using Table 14 of the Data Booklet, calculate the cell potential, $E^{\ominus}_{cell}$, in V, when the two half-cells are connected. *[1]*

. . . . . .
. . . . . .
. . . . . .

(e) Water in a beaker at a pressure of $1.01 \times 10^5\,Pa$ and a temperature of $298\,K$ will not spontaneously decompose. However, decomposition of water can be induced by means of electrolysis.

(ii) State why dilute sulfuric acid needs to be added in order for the current to flow in the electrolytic cell. *[1]*

. . . . . .
. . . . . .

(iii) State why copper electrodes cannot be used in the electrolysis of water. Suggest instead suitable **metallic** electrodes for this electrolysis process. *[1]*

. . . . . .
. . . . . .

(iv) Deduce the half-equations for the reactions occurring at the positive electrode (anode) and the negative electrode (cathode). *[2]*

Positive electrode (anode):

. . . . . .
. . . . . .

Negative electrode (cathode):

. . . . . .
. . . . . .

(v) Deduce the overall cell reaction, including state symbols. *[1]*

. . . . . .
. . . . . .

(vi) Draw a fully labelled diagram of the electrolytic cell, showing the positive electrode (anode) and the negative electrode (cathode). *[2]*

(vii) Comment on what is observed at both electrodes. *[1]*

. . . . . .

. . . . . .

7.2-S24 May 2013 TZ2 · Paper 3 · QC1 parts (a)(iii)

C1. (a) The main ore used to produce aluminium by electrolysis is bauxite. Bauxite is mainly aluminium hydroxide, and contains iron(III) oxide and titanium(IV) oxide as impurities.

(iii) State the half-equations for the reactions taking place at the positive and negative electrodes during the production of aluminium by electrolysis. *[2]*

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

7.2-S25 Nov 2011 · Paper 3 · QC5 parts (a)(ii)

C5. The chlor-alkali industry manufactures two important products (sodium hydroxide and chlorine) from the electrolysis of aqueous sodium chloride. The membrane cell is increasingly used in this process.

(ii) State the half-equation for the reaction taking place at each electrode in the membrane cell. *[2]*

Positive electrode (anode):

. . . . . .

Negative electrode (cathode):

. . . . . .

7.2-S26 May 2011 TZ1 · Paper 2 · Q9 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c), (e)(i), (e)(ii) · also 6.1, 6.2, 7.1

9. (a) The conditions used in an electrolytic cell can determine the products formed.

(i) Draw an electrolytic cell illustrating the electrolysis of molten nickel(II) bromide, $NiBr_2$. Include in the diagram the direction of the electron flow, the polarity of electrodes and state the half-equations for the product formed at each electrode. *[5]*

. . . . . .

. . . . . .

(ii) Deduce the equations for the formation of the major product at the positive electrode (anode) when the following aqueous solutions are electrolysed.

- dilute sodium chloride
- concentrated sodium chloride *[2]*

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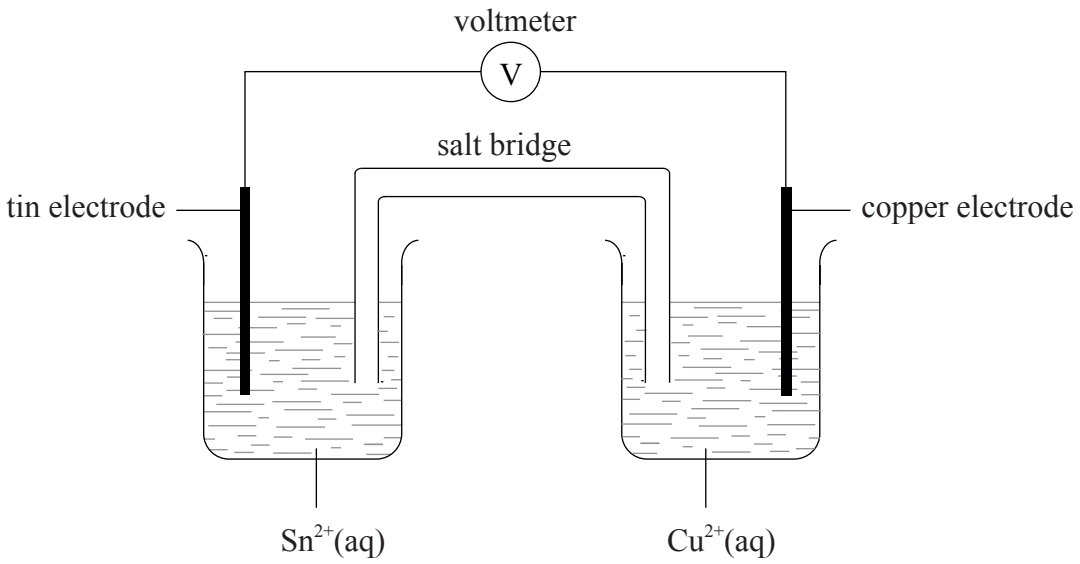
. . . . . .

(b) A voltaic cell is constructed from two half-cells as illustrated below.



(i) Use Table 14 of the Data Booklet to deduce the equation for the spontaneous reaction occurring in this cell. *[1]*

. . . . . .

. . . . . .

(ii) Calculate the standard potential for this cell. *[1]*

. . . . . .

. . . . . .

(iii) State the conditions necessary for the potential of the cell to equal that calculated in part (b) (ii) using the data from Table 14. *[1]*

. . . . . .

. . . . . .

(c) Using the data below and data from Table 14 of the Data Booklet, predict and explain which metal, cadmium or chromium, may be obtained by electrolysis of separate aqueous solutions of $Cd^{2+}(aq)$ ions and $Cr^{2+}(aq)$ ions. *[2]*

	$E^{\ominus}$ / V
$Cd^{2+}(aq) + 2e^{-} \rightarrow Cd(s)$	-0.40
$Cr^{2+}(aq) + 2e^{-} \rightarrow Cr(s)$	-0.91

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(e) Nitrogen monoxide may be removed from industrial emissions via a reaction with ammonia as shown by the equation below.

$$4NH_3(g) + 6NO(g) \rightarrow 5N_2(g) + 6H_2O(l)$$

(i) Deduce the oxidation number of the nitrogen in the reactants and product. *[3]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

(ii) Deduce the oxidation and reduction half-equations and identify the oxidizing agent for the reaction. *[3]*

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7.2-S27 May 2011 TZ1 · Paper 3 · QC4 parts (b)

C4. The chlor-alkali process, converting sodium chloride into chlorine and sodium hydroxide, is an important industrial process which is often carried out in a membrane cell.

For the membrane cell:

(b) state equations for the reactions at each electrode. *[2]*

. . . . . .

. . . . . .

. . . . . .

. . . . . .

7.2-S28 May 2011 TZ2 · Paper 3 · QC1 parts (a)(ii)

C1. Aluminium and its alloys are widely used in industry.

(a) Aluminium metal is obtained by the electrolysis of alumina dissolved in molten cryolite.

(ii) State the half-equation for the reaction that takes place at the positive electrode (anode). *[1]*

. . . . . .

7.2-S29 Nov 2010 · Paper 3 · QC5 parts (a)(iii)

C5. Consider the membrane chlor-alkali electrolysis cell, in which sodium chloride solution is electrolysed.

(a) For this process state the following.

(iii) The half-equation for the reaction that takes place at the positive electrode (anode). *[1]*

. . . . . .

LESSON 7.3

Energy From Fuels

6 multiple-choice questions · 52 structured questions

Part A - Multiple choice

7.3-M1 May 2025 TZ2 · Paper 1A · Q20

20. What is the coefficient of oxygen when the equation for the complete combustion of ethanol is balanced using the smallest whole numbers?

$$__C_2H_5OH + __O_2 \rightarrow __CO_2 + __H_2O$$

A. 1

B. 2

C. 3

D. 7

7.3-M2 May 2019 TZ2 · Paper 1 · Q2

2. What is the sum of the integer coefficients when propene undergoes complete combustion?

$$__C_3H_6(g) + __O_2(g) \rightarrow __CO_2(g) + __H_2O(l)$$

A. 11

B. 17

C. 21

D. 23

7.3-M3 May 2017 TZ2 · Paper 1 · Q1

1. What is the sum of the coefficients when the equation is balanced with whole numbers?

$$__C_8H_{18}(g) + __O_2(g) \rightarrow __CO(g) + __H_2O(l)$$

A. 26.5

B. 30

C. 53

D. 61

7.3-M4 May 2015 TZ2 · Paper 1 · Q2

2. Which expression gives the sum of all the coefficients for the general equation for the complete combustion of hydrocarbons?

$$___\ C_xH_y(g) + ___\ O_2(g) \rightarrow ___\ CO_2(g) + ___\ H_2O(l)$$

A. $1 + x + \frac{y}{4}$

B. $1 + x + \frac{y}{2}$

C. $1 + 2x + \frac{3y}{4}$

D. $1 + 2x + \frac{3y}{2}$

7.3-M5 May 2013 TZ2 · Paper 1 · Q33

33. What are possible products of the incomplete combustion of propan-2-ol?

A. carbon monoxide, hydrogen and carbon

B. carbon dioxide, carbon and hydrogen

C. carbon, carbon monoxide and water

D. carbon dioxide and water only

7.3-M6 May 2010 TZ1 · Paper 1 · Q35

35. Which substances are possible products of the incomplete combustion of octane?

A. Carbon dioxide and hydrogen gas

B. Carbon monoxide and water vapour

C. Carbon monoxide and hydrogen gas

D. Methane and hydrogen gas

Part B - Structured questions

7.3-S1 May 2025 TZ3 · Paper 2 · Q9 parts (b)(i), (b)(ii), (c) · also 8.2, 9.3

9. Ethanol is a biofuel that is made from glucose. Glucose is formed by the biological fixation of carbon through photosynthesis.

(b) (i) Outline **two** advantages of using ethanol as a fuel instead of gasoline (petrol). [2]

(ii) Calculate the enthalpy change for the combustion of ethanol in $kJ\,mol^{-1}$. Use section 12 of the data booklet. [3]

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$$

(c) Calculate the increase in temperature in a sample of $500.0\,cm^3$ of water heated by the combustion of 4.00 g of ethanol. Use sections 2 and 14 of the data booklet. [2]

7.3-S2 Nov 2024 · Paper 3 · Q14 parts (a)(i), (a)(ii)

14. Adding bioethanol to gasoline increases the octane rating of the fuel and lowers the carbon footprint caused by fuel consumption.

(a) (i) Calculate the carbon footprint for octane, C_8H_{18}, in terms of kg CO_2 produced per kg of octane burned. [2]

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. . . . . .

. . . . . .

. . . . . .

. . . . . .

(ii) State a reason why adding bioethanol to gasoline lowers the carbon footprint. [1]

. . . . . .

. . . . . .

. . . . . .

7.3-S3 May 2024 TZ1 · Paper 3 · Q13 parts (a), (b), (c), (f)

13. Methane clathrate looks like ice but burns when ignited because it turns back to water and methane.

(a) Write an equation for the complete combustion of methane. [1]

. . . . . .

. . . . . .

(b) The formula of methane clathrate is $(CH_4)_4(H_2O)_{23}$ and its molar mass is $478.7\,g\,mol^{-1}$. Calculate the specific energy of methane clathrate. Use sections 1, 6 and 13 of the data booklet. [2]

. . . . . .

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. . . . . .

. . . . . .

. . . . . .

(c) Suggest why methane clathrate has a lower specific energy but higher energy density than methane. [2]

Lower specific energy:

. . . . . .

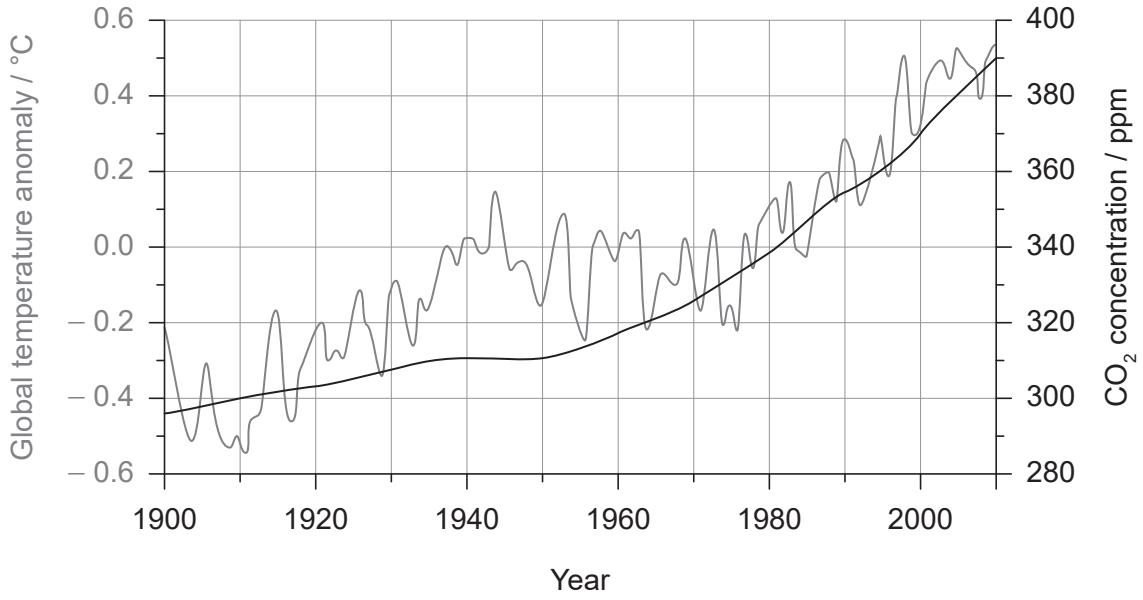
. . . . . .

Higher energy density:

. . . . . .

. . . . . .

(f) The graph shows the increase in temperature and the level of carbon dioxide in the atmosphere.



Discuss how this data could be used to both support and refute the theory that carbon dioxide is a cause of global warming. [2]

Support:

. . . . . .

. . . . . .

. . . . . .

Refute:

. . . . . .

. . . . . .

. . . . . .

7.3-S4 May 2024 TZ1 · Paper 3 · Q14 parts (a)(i), (a)(ii)

14. Methane, gasoline and biodiesel are used as fuels for transport.

(a) (i) Outline why the combustion of methane has a lower environmental impact than gasoline. [2]

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. . . . . .

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(ii) Suggest why gasoline is more widely used than methane as a car fuel, even though it is more expensive in most countries. [1]

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7.3-S5 Nov 2023 TZ1 · Paper 3 · Q12 parts (c)(ii)

12. The spontaneity of nuclear fission and fusion reactions can be explained by changes in nuclear binding energy.

(ii) Explain whether the energy density, in $MJ\,dm^{-3}$, or specific energy, in $MJ\,kg^{-1}$, of hydrogen has a higher value at standard conditions of temperature and pressure. [1]

. . . . . .

. . . . . .

7.3-S6 Nov 2023 TZ1 · Paper 3 · Q13 parts (b)

13. Many molecules interact with light.

(b) Upper atmosphere temperatures recorded by satellites are becoming lower over time.

Suggest how greenhouse gases could be responsible for this trend. [2]

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. . . . . .

7.3-S7 Nov 2023 TZ1 · Paper 3 · Q14 parts (b), (c)

14. Natural gas is a fossil fuel.

(b) State the main component of natural gas. [1]

. . . . . .

(c) Outline **one** advantage and **one** disadvantage, apart from cost, of using natural gas over other fossil fuels. [2]

Advantage:

. . . . . .

. . . . . .

Disadvantage:

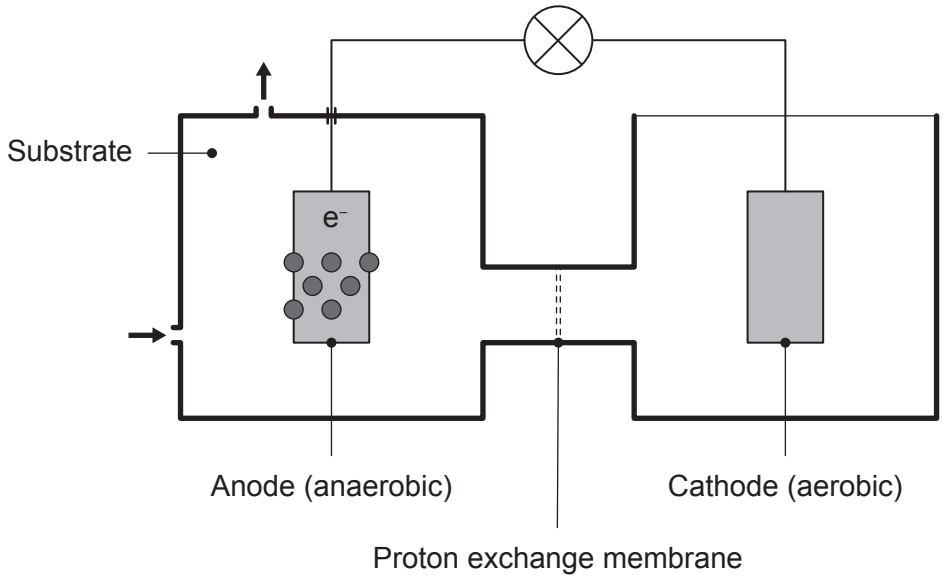
. . . . . .

. . . . . .

7.3-S8 May 2023 TZ1 · Paper 3 · Q13 parts (d)(i)

13. Biofuels can be synthesized or converted to make them more usable.

(d) The Geobacter bacterium anaerobically oxidizes substances present in the substrate of microbial fuel cells. For example, ethanoate ions, $CH_3COO^-(aq)$, are oxidized to carbon dioxide gas.



(i) Outline the function of the proton exchange membrane. [1]

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7.3-S9 May 2023 TZ2 · Paper 2 · Q6 parts (d)(i), (d)(ii) · also 9.3

6. Bromine, $Br_2(l)$, and methanoic acid, $HCOOH(aq)$, react in the presence of sulfuric acid.

$$Br_2(l) + HCOOH(aq) \rightarrow 2HBr(aq) + CO_2(g)$$

(d) (i) Write the equation for the complete combustion of ethanol. [1]

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(ii) Determine the enthalpy change for the combustion of ethanol, in kJ mol^{-1}, using section 11 of the data booklet. [3]

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7.3-S10 May 2023 TZ2 · Paper 3 · Q15 parts (d)(i)

(d) Both ethanol and glucose can be used to generate energy through fuel cells.

(i) Outline **one** way fuel cells differ from primary cells. [1]

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7.3-S11 May 2023 TZ2 · Paper 3 · Q16 parts (a)(ii), (c)(i)

16. Geological transformations produce fossil fuels.

(a) Combustion of coal emits particulates into the atmosphere.

(ii) State the major form of energy produced by the combustion of coal. [1]

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(c) The equation for the combustion of octane is:

$$C_8H_{18}(l) + 12\frac{1}{2}O_2 \rightarrow 8CO_2(g) + 9H_2O(l)$$

(i) Determine the mass of carbon dioxide, in g, produced when 1 kJ of energy is produced. Use section 13 of the data booklet. [3]

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7.3-S12 May 2021 TZ1 · Paper 2 · Q5 parts (c)

5. Ethanol is obtained by the hydration of ethene, C_2H_4.

(c) Suggest **two** possible products of the incomplete combustion of ethene that would not be formed by complete combustion. [1]

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7.3-S13 May 2021 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii) · also 9.2

5. Ethanol and methanoic acid are important industrial products.

(a) Ethanol is used as a fuel.

(i) Write the chemical equation for the complete combustion of ethanol. [1]

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(ii) Deduce the change in enthalpy, ΔH, in kJ, when 56.00 g of ethanol is burned. Use section 13 in the data booklet. [2]

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7.3-S14 Nov 2020 · Paper 3 · Q11 parts (a), (c), (d)

11. Gasoline (petrol), biodiesel and ethanol are fuels.

	Gasoline (petrol)	**Biodiesel**	**Ethanol**
Chemical structure	Mainly hydrocarbons of chain length C_4–C_{12}	Methyl esters of fatty acids of chain lengths C_{12}–C_{22}	CH_3CH_2OH
Energy density / kJ dm^{-3}	31 800	33 400	21 200

(a) Calculate the energy released, in kJ, from the complete combustion of 5.00 dm^3 of ethanol. [1]

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(c) Outline the advantages and disadvantages of using biodiesel instead of gasoline as fuel for a car. Exclude any discussion of cost. [4]

Advantages:

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Disadvantages:

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(d) A mixture of gasoline and ethanol is often used as a fuel. Suggest an advantage of such a mixture over the use of pure gasoline. Exclude any discussion of cost. [1]

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7.3-S15 Nov 2019 · Paper 2 · Q3 parts (c)(i)

3. Propene is an important starting material for many products. The following shows some compounds which can be made from propene, C_3H_6.

$$\text{Propene } (C_3H_6) \rightarrow C_3H_7Cl \rightarrow C_3H_8O \rightarrow C_3H_6O$$

(c) (i) Write an equation for the complete combustion of the compound C_3H_8O formed in (a)(iv). [1]

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7.3-S16 Nov 2019 · Paper 3 · Q16 parts (a)

16. Consider the following data for butane and pentane at STP.

	Specific energy / MJ kg^{-1}	**Energy density / MJ m^{-3}**
Butane	49.5	128
Pentane	48.6	30 400

(a) Discuss the data. [3]

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7.3-S17 Nov 2019 · Paper 3 · Q19 parts (a)

19. Ethanol is a biofuel that can be mixed with gasoline.

(a) Write the equation for the complete combustion of ethanol. [1]

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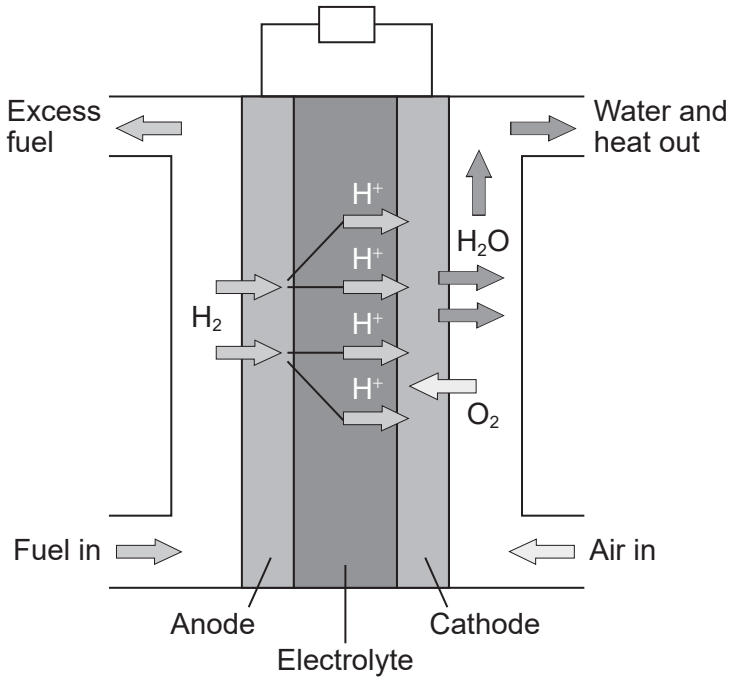
7.3-S18 Nov 2019 · Paper 3 · Q20 parts (a), (b)(i), (c) · also 7.1

20. A proton-exchange membrane (PEM) fuel cell uses pure hydrogen gas as the fuel and a proton exchange membrane as the electrolyte.



[Source: https://en.wikipedia.org/wiki/File:Proton_Exchange_Fuel_Cell_Diagram.svg]

(a) Deduce the half-equations for the reactions occurring at the electrodes. [2]

Anode (negative electrode):

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Cathode (positive electrode):

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(b) (i) Calculate the cell potential, $E^{\ominus}$, in V, using section 24 of the data booklet. [1]

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(c) Suggest an advantage of the PEM fuel cell over the lead-acid battery for use in cars. [1]

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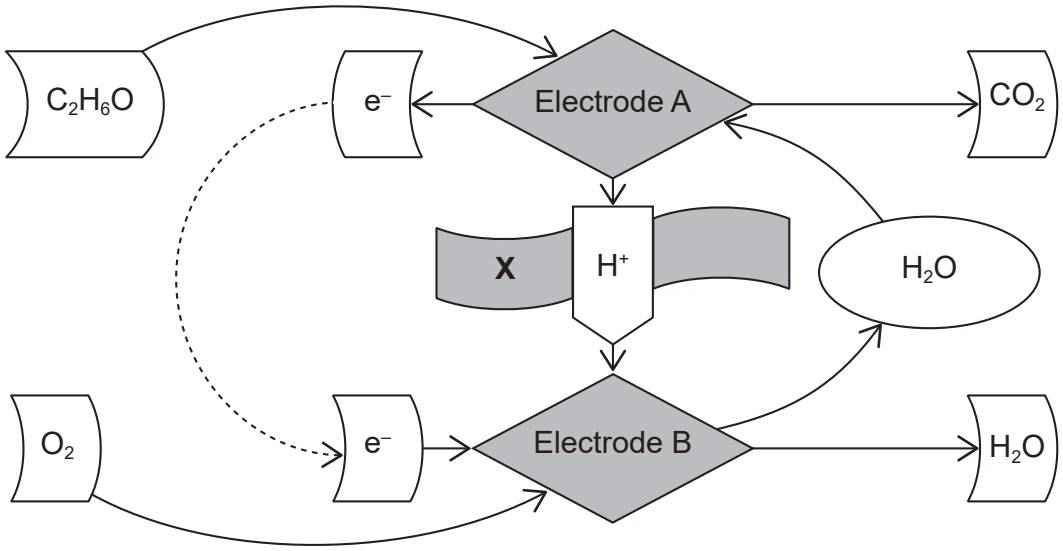
7.3-S19 May 2019 TZ1 · Paper 3 · Q17 parts (b)(i), (b)(ii), (b)(iii), (c)

17. E10 is composed of 10 % ethanol and 90 % normal unleaded fuel.

(b) (i) Ethanol can be used in a direct-ethanol fuel cell (DEFC) as illustrated by the flow chart.



[Source: https://commons.wikimedia.org/wiki/File:Direct_Ethanol_Fuel_Cell.svg, by Fordi]

Deduce the half-equations occurring at electrodes A and B. [2]

Electrode A:

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Electrode B:

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(ii) State the name and function of **X** in the diagram in (b)(i). [2]

Name:

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Function:

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(iii) Outline why aqueous ethanol, rather than pure ethanol, is used in a DEFC. [1]

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(c) Biodiesel containing ethanol can be made from renewable resources.

Suggest **one** environmental disadvantage of producing biodiesel from renewable resources. [1]

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7.3-S20 May 2019 TZ2 · Paper 3 · Q15 parts (b)

15. This question is about fuel for engines.

(b) Determine the specific energy, in $kJ\,g^{-1}$, and energy density, in $kJ\,cm^{-3}$, of hexane, C_6H_{14}. Give both answers to three significant figures.

Hexane: $M_r = 86.2$; $\Delta H_c = -4163\,kJ\,mol^{-1}$; density $= 0.660\,g\,cm^{-3}$ [2]

Specific energy:

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Energy density:

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7.3-S21 May 2019 TZ2 · Paper 3 · Q17

17. This question is about biofuel.

Evaluate the use of biodiesel in place of diesel from crude oil. [2]

Strength:

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Limitation:

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7.3-S22 Nov 2018 · Paper 3 · Q13 parts (b), (c)

13. Coal can be converted to clean-burning synthetic natural gas.

(b) Automobile companies use hydrogen as an alternative to fossil fuels. Some properties of fuels are shown.

Compound	Molar mass / $g\,mol^{-1}$	Density at STP / $g\,dm^{-3}$	ΔH_c / $kJ\,mol^{-1}$	Energy density at STP / $kJ\,dm^{-3}$	Specific energy / $kJ\,g^{-1}$
Hydrogen	2.02	0.0890	–286	12.6	141.6
Methane	16.05	0.707	–891	39.3	55.5

Comment on the specific energies of hydrogen and methane. [1]

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(c) Calculate the mass, in kg, of carbon dioxide produced by the complete combustion of $72.0\,dm^3$ octane, C_8H_{18}.

Density of $C_8H_{18} = 703\,g\,dm^{-3}$

$C_8H_{18}(l) + 12.5O_2(g) \rightarrow 8CO_2(g) + 9H_2O(g)$ [2]

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7.3-S23 May 2018 TZ1 · Paper 3 · Q12 parts (a), (b)(i), (b)(ii) · also 5.3

12. The increased concentration of carbon dioxide in the atmosphere is thought to result from the increased combustion of fossil fuels such as petroleum.

(a) Petroleum contains many hydrocarbons. Explain how these are separated by fractional distillation. [3]

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(b) (i) Determine the specific energy and energy density of petrol (gasoline), using data from sections 1 and 13 of the data booklet. Assume petrol is pure octane, C_8H_{18}. Octane: molar mass = 114.26 g mol^{-1}, density = 0.703 g cm^{-3}. [2]

Specific energy in kJ g^{-1}:

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Energy density in kJ cm^{-3}:

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(ii) Outline why the energy available from an engine will be less than these theoretical values. [1]

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7.3-S24 May 2018 TZ2 · Paper 3 · Q13 parts (a)

13. Crude oil is a useful energy resource.

(a) Outline **two** reasons why oil is one of the world's significant energy sources. [2]

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7.3-S25 Nov 2017 · Paper 3 · Q7 parts (e)

7. Low density polyethene (LDPE) and high density polyethene (HDPE) are both addition polymers.

(e) Many plastics used to be incinerated. Deduce an equation for the complete combustion of two repeating units of PVC, $(–C_2H_3Cl–)_2$. [2]

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7.3-S26 Nov 2017 · Paper 3 · Q16 parts (a), (b), (c)

16. One method of comparing fuels is by considering their specific energies.

(a) Calculate the specific energy of octane, C_8H_{18}, in $kJ\,kg^{-1}$ using sections 1, 6 and 13 of the data booklet. [2]

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(b) A typical wood has a specific energy of $17 \times 10^3\,kJ\,kg^{-1}$. Comment on the usefulness of octane and wood for powering a moving vehicle, using your answer to (a).
If you did not work out an answer for (a), use $45 \times 10^3\,kJ\,kg^{-1}$ but this is not the correct answer. [1]

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(c) State the name of **one** renewable source of energy other than wood. [1]

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7.3-S27 Nov 2017 · Paper 3 · Q17 parts (c)

17. Much of our energy needs are still provided by the refined products of crude oil.

(c) Climate change or global warming is a consequence of increased levels of carbon dioxide in the atmosphere. Explain how the greenhouse effect warms the surface of the earth. [3]

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7.3-S28 Nov 2017 · Paper 3 · Q20 parts (a), (b)

20. A fuel cell converts chemical energy directly to electrical energy.

(a) Deduce the half-equations and the overall equation for the reactions taking place in a direct methanol fuel cell (DMFC) under acidic conditions. [3]

Negative electrode (anode):

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Positive electrode (cathode):

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Overall equation:

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(b) Outline **one** advantage and **one** disadvantage of the methanol cell (DMFC) compared with a hydrogen-oxygen fuel cell. [2]

Advantage:

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Disadvantage:

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7.3-S29 May 2017 TZ1 · Paper 2 · Q4 parts (a), (c)(ii) · also 9.2

4. Millerite, a nickel sulfide mineral, is an important source of nickel. The first step in extracting nickel is to roast the ore in air.

(a) Formulate an equation for the oxidation of nickel(II) sulfide to nickel(II) oxide. [1]

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(c) The reaction for the formation of liquid tetracarbonylnickel is shown below:

$$Ni(s) + 4CO(g) \rightarrow Ni(CO)_4(l)$$

(ii) Calculate a value for $\Delta H^{\ominus}$ in kJ. [1]

	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$CO(g)$	−110.5
$Ni(CO)_4(l)$	−633.0

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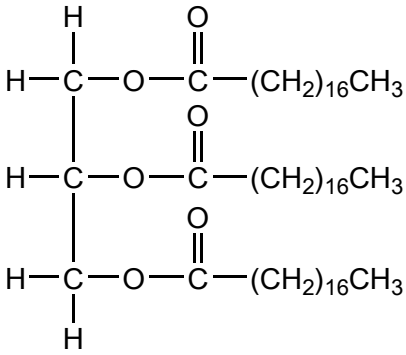
7.3-S30 May 2017 TZ1 · Paper 3 · Q20 parts (c)

20. Vegetable oils, such as that shown, require conversion to biodiesel for use in current internal combustion engines.



(c) Determine the specific energy, in $kJ\,g^{-1}$, and energy density, in $kJ\,cm^{-3}$, of a particular biodiesel using the following data and section 1 of the data booklet.

Density $= 0.850\,g\,cm^{-3}$; Molar mass $= 299\,g\,mol^{-1}$;
Enthalpy of combustion $= 12.0\,MJ\,mol^{-1}$. [2]

Specific energy:

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Energy density:

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7.3-S31 May 2017 TZ2 · Paper 3 · Q17 parts (a), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii)

17. There are many sources of energy available.

(a) State **one** advantage and **one** disadvantage for each energy source in the table. [4]

Energy Source	Advantage	Disadvantage
Biofuels	. . . . .	. . . . .
Fossil fuels	. . . . .	. . . . .

(b) (i) Calculate the specific energy of hydrogen, stating its units. Refer to sections 1, 6 and 13 of the data booklet. [2]

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(ii) Hydrogen has a higher specific energy than petrol (gasoline) but is not used as a primary fuel source in cars. Discuss the disadvantages of using hydrogen. [2]

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(c) Methanol fuel cells provide a portable energy source. The process can be represented by the overall equation $CH_3OH(aq) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$.

(i) Deduce the half-cell equations occurring at each electrode during discharge. [2]

Anode (negative electrode):

. . . . . .

Cathode (positive electrode):

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(ii) Outline the function of the proton-exchange membrane (PEM) in the fuel cell. [1]

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(iii) Explain how the flow of ions allows for the operation of the fuel cell. [2]

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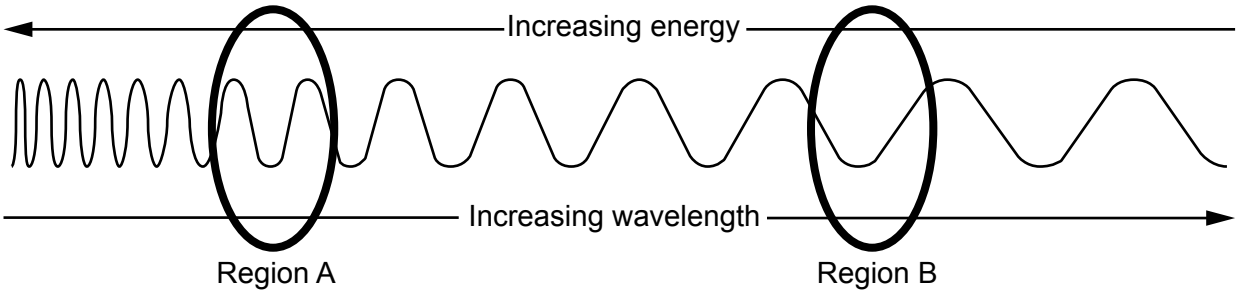
7.3-S32 May 2017 TZ2 · Paper 3 · Q18 parts (a)(i)

18. The combustion of fossil fuels produces large amounts of CO_2, a greenhouse gas.
The diagram below illustrates a range of wavelengths in the electromagnetic spectrum.



(a) (i) Identify which region, **A** or **B**, corresponds to each type of radiation by completing the table. [1]

Type of radiation	Region
Incoming radiation from sun	. . . . .
Re-radiated from Earth's surface	. . . . .
Absorbed by CO_2 in the atmosphere	. . . . .

7.3-S33 Nov 2016 · Paper 3 · Q6 parts (c)(i)

6. Polymers are made up of repeating monomer units which can be manipulated in various ways to give structures with desired properties.

(c) (i) Suggest why incomplete combustion of plastic, such as polyvinyl chloride, is common in industrial and house fires. [1]

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7.3-S34 Nov 2016 · Paper 3 · Q17 parts (a)(i), (a)(ii), (b)(i)

17. Chemical energy from redox reactions can be used as a portable source of electrical energy. A hybrid car uses a lithium ion battery in addition to gasoline as fuel.

(a) (i) Calculate the specific energy of the lithium ion battery, in $MJ\,kg^{-1}$, when 80.0 kg of fuel in the battery releases 1.58×10^7 J. Use section 1 of the data booklet. [1]

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(ii) The specific energy of gasoline is $46.0\,MJ\,kg^{-1}$. Suggest why gasoline may be considered a better energy source than the lithium ion battery based on your answer to part (a) (i). [1]

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(b) (i) The energy density of gasoline is $34.3\,MJ\,dm^{-3}$. Calculate the volume of gasoline, in dm^3, that is equivalent to the energy in 80.0 kg of fuel in the lithium ion battery. Use section 1 of the data booklet. [1]

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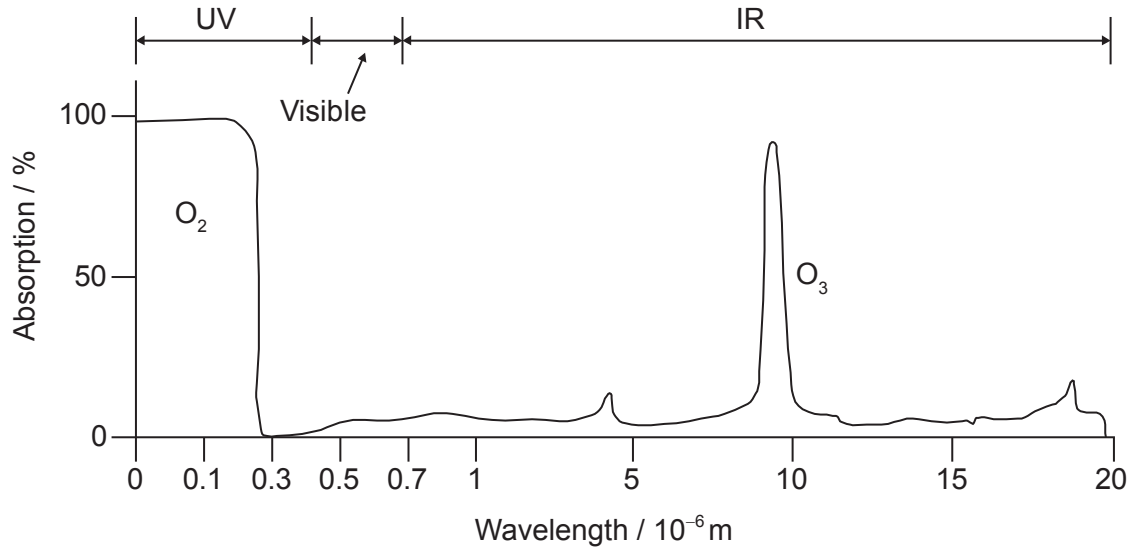
7.3-S35 May 2016 · Paper 3 · Q1 parts (a), (b)(ii) · also 5.1

1. The absorption of infrared (IR) radiation by molecules in the atmosphere affects global temperatures.

Graph of IR absorbances for oxygen and ozone molecules



[Source: adapted from 2007 Thomson Higher Education, www.acs.org]

(a) Using the graph, state, giving your reasons, whether or not oxygen and ozone are greenhouse gases. [2]

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(b) The following data has been compiled for a range of molecules that may be found in the atmosphere.

Molecule	Integrated IR intensity* / km mol^{-1}	Molecular dipole moment / Debyes	GWP** over 100 years
CO_2	25.7	0	1
CCl_4	443.7	0	1400
CCl_3F	705.2	0.45	4750
CCl_2F_2	970.1	0.51	10900
$CClF_3$	1199	0.50	14400
CF_4			

[Sources: “Identifying the Molecular Origin of Global Warming”, Partha P Bera, Joseph S Francisco and Timothy J Lee. Published in J. Phys. Chem. A, Vol. 113, No. 45, 2009 and accessed from www.r744.com]

*Integrated IR intensity is a measure of the extent to which the molecule absorbs infrared radiation passing through the atmosphere.

**GWP: The global warming potential (GWP) is a relative measure of the total contribution of the compound to global warming over the specified time period. It is compared to the same mass of CO_2, which has a GWP of 1.

(ii) Explain the increase in molecular dipole moment as one chlorine atom in CCl_4 is replaced with fluorine to produce CCl_3F. [2]

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7.3-S36 May 2016 · Paper 3 · Q15 parts (b)(i), (b)(ii)

15. Hexane, C_6H_{14}, is not a suitable fuel for internal combustion engines as it has a tendency to auto-ignite, a cause of "knocking".

(b) (i) Octane, C_8H_{18}, can undergo complete combustion under suitable conditions. Calculate the specific energy of octane, in $kJ\,g^{-1}$, using sections 1, 6 and 13 of the data booklet. [1]

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(ii) The specific energy of ethanol is $29.7\,kJ\,g^{-1}$. Evaluate the addition of ethanol to octane (or its isomers) for use as a fuel in motor vehicles, giving **one** advantage and **one** disadvantage. [2]

Advantage:

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Disadvantage:

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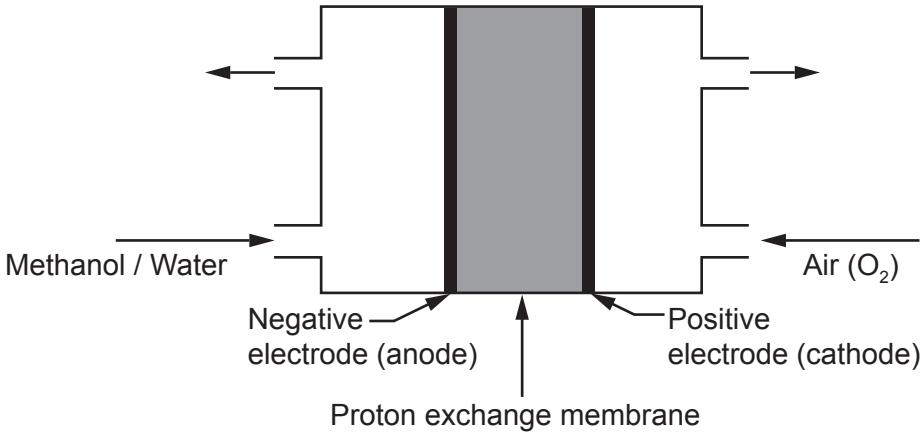
7.3-S37 May 2016 · Paper 3 · Q20 parts (a), (b)

20. Fuel cells and rechargeable batteries are useful sources of energy.

(a) One type of fuel cell contains a proton exchange membrane between electrodes and uses aqueous methanol as the fuel.



State half-equations for the reactions which occur at the negative and positive electrodes. [2]

Negative electrode (anode):

. . . . . .

Positive electrode (cathode):

. . . . . .

(b) Suggest **one** advantage and **one** disadvantage of a fuel cell over a lead–acid battery as an energy source in a motor vehicle. [2]

Advantage:

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Disadvantage:

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7.3-S38 Nov 2015 · Paper 2 · Q3 parts (a), (b) · also 9.3

Propane, $C_3H_8(g)$, undergoes complete combustion to form carbon dioxide, $CO_2(g)$, and water, $H_2O(g)$.

(a) State an equation for the complete combustion of propane, $C_3H_8(g)$. [1]

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(b) Calculate the standard enthalpy change for the reaction in part (a) using bond enthalpy values given in table 10 of the data booklet. [3]

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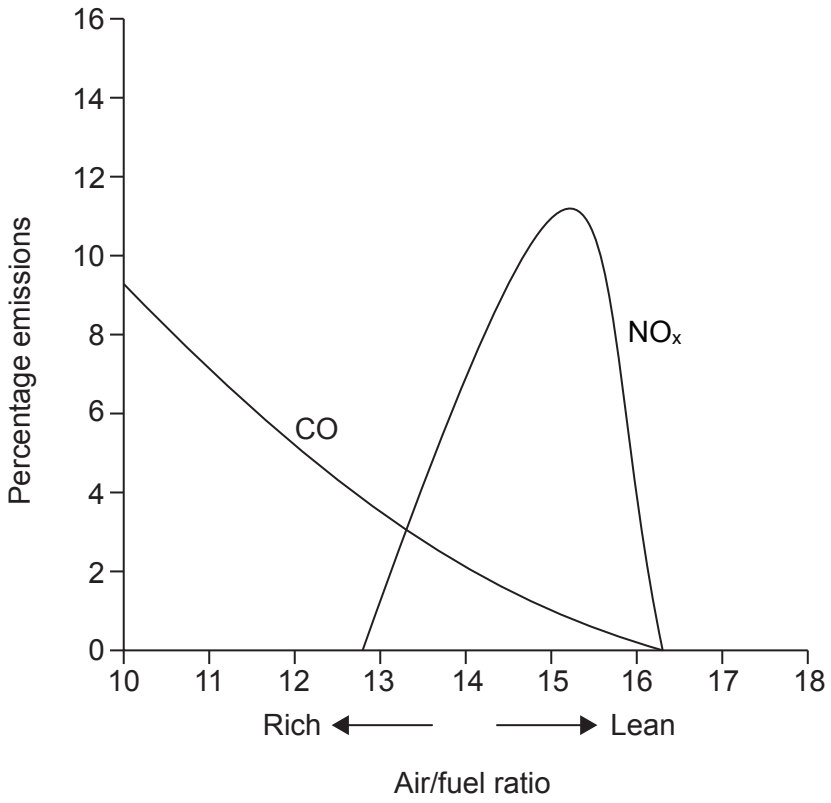
7.3-S39 Nov 2015 · Paper 3 · Q23 parts (b)(i)

23. Lowering air pollutant emissions is a major concern for car manufacturers.

(b) A car manufacturer conducted tests in which the air/fuel ratio in the engine was altered and the amounts of pollutants released measured. The results are represented below.



(i) State and explain the effect of increasing the air/fuel ratio on CO emissions. [2]

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7.3-S40 May 2015 TZ2 · Paper 3 · Q14 parts (a)

14. There has been a shift in the use of crude oil (petroleum) away from its use as an energy source and towards its use as a chemical feedstock.

(a) Suggest **two** reasons for this shift. [2]

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7.3-S41 May 2015 TZ2 · Paper 3 · Q22 parts (b)(i)

22. Climate change is a current global topic of debate.

(b) (i) Describe on a molecular level how the greenhouse effect occurs. [3]

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7.3-S42 Nov 2014 · Paper 3 · Q21 parts (a)

21. (a) Describe how the greenhouse effect causes the atmosphere of the Earth to increase in temperature. *[3]*

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7.3-S43 May 2014 TZ1 · Paper 3 · Q9 parts (a), (b)(iii), (b)(iv), (b)(v)

9. Although fossil fuels are considered significant sources of energy, the energy conversion associated with the production of electricity is a very inefficient process, often in the region of only 40 % of total possible energy conversion.

Fuel cells provide a much more efficient process, often with a 70 % conversion factor.

(a) State the energy change conversion involved in a fuel cell. *[1]*

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(iii) One commercial version of the hydrogen-oxygen fuel cell (with alkaline electrolyte) operates at a temperature of 353 K. The electrodes of the fuel cell are made of graphite but both are covered with a thin layer of platinum. State the function of the platinum. *[1]*

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(iv) Outline the function of the thin polymer membrane used in the corresponding hydrogen-oxygen fuel cell with an **acidic** electrolyte. *[1]*

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(v) Other than cost, state **one** disadvantage of a fuel cell. *[1]*

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7.3-S44 May 2014 TZ1 · Paper 3 · Q15 parts (c)(ii)

15. Although methane is a greenhouse gas produced by anaerobic decomposition of organic matter, methane produced from landfill is now being used in New Jersey, USA, to generate electricity for homes in New York City.

(ii) State the equation, including state symbols, for the oxidation of methane in the atmosphere to produce a primary pollutant which is toxic to humans. *[1]*

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7.3-S45 May 2012 TZ1 · Paper 2 · Q2 parts (b)

2. In 1921 Thomas Midgley discovered that the addition of a lead compound could improve the combustion of hydrocarbons in automobile (car) engines. This was the beginning of the use of leaded gasoline (petrol).

The percentage composition, by mass, of the lead compound used by Midgley is shown below.

	Pb	C	H
Mass composition / %	64.052	29.703	6.245

(b) The combustion of unleaded gasoline still produces pollution with both local and global consequences. Identify **one** exhaust gas which causes local pollution and **one** exhaust gas which causes global pollution. *[2]*

Local pollutant:

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Global pollutant:

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7.3-S46 May 2012 TZ1 · Paper 3 · QC2 parts (a), (c)

C2. Fuel cells and rechargeable batteries are both convenient ways of providing portable electric power.

(a) Compare fuel cells and rechargeable batteries giving **one** similarity and **one** difference. *[2]*

Similarity:

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Difference:

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(c) A common type of fuel cell uses hydrogen and oxygen with an acidic electrolyte. State the half-equations for the reactions at the two electrodes. *[2]*

Positive electrode:

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Negative electrode:

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7.3-S47 Nov 2011 · Paper 3 · QC1 parts (a)(i)

C1. The Industrial Revolution was the result of large-scale extraction of iron from its ore and had significant impact worldwide.

(a) In a blast furnace, a large volume of air is introduced under pressure near the bottom while a mixture of limestone, coke and iron(III) oxide is introduced at the top.

(i) State the equation for the reaction of coke with air in the blast furnace. *[1]*

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7.3-S48 Nov 2011 · Paper 3 · QC2 parts (d)

C2. Since the accidental discovery of polyethene in the 1930s, polymers have played an essential role in daily life because of their wide range of properties and uses.

(d) State **one** environmental impact of the disposal of these polyethenes by using incineration. *[1]*

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7.3-S49 Nov 2011 · Paper 3 · QE2 parts (a)

E2. The temperature of the Earth is increasing. There is considerable scientific evidence to suggest this is due to an increase in the concentration of greenhouse gases as a result of human activity.

(a) Explain how this enhanced greenhouse effect causes the average temperature of the Earth to increase. *[3]*

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7.3-S50 May 2011 TZ1 · Paper 3 · QE3 parts (a)

E3. (a) Outline a condition that leads to the production of carbon monoxide, CO, in an internal combustion engine. *[1]*

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7.3-S51 Nov 2010 · Paper 3 · QC1

C1. "Oil should not be used as a source of energy because it has more important uses." Suggest **two** arguments that support the continued use of oil as an energy source, and **two** against. *[4]*

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7.3-S52 May 2010 TZ2 · Paper 3 · QE1 parts (a), (b)

E1. The greenhouse effect maintains the Earth's average temperature at a habitable level. The components of the Earth's atmosphere responsible for this effect are called greenhouse gases.

(a) Major greenhouse gases are water vapour and carbon dioxide. State **two** other greenhouse gases. *[2]*

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(b) Describe how greenhouse gases cause the greenhouse effect. *[3]*

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LESSON 8.1

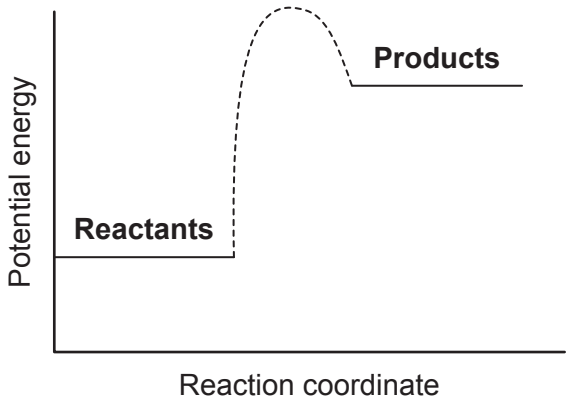
Thermochemical Equations

26 multiple-choice questions · 9 structured questions

Part A - Multiple choice

8.1-M1 May 2025 TZ2 · Paper 1A · Q18

18. What is the correct interpretation of this energy profile?



	Relative stability	Temperature of surroundings
A.	reactants more stable than products	decreases
B.	reactants more stable than products	increases
C.	reactants less stable than products	decreases
D.	reactants less stable than products	increases

8.1-M2 May 2025 TZ3 · Paper 1A · Q20

20. Which statement about an exothermic reaction is correct?

A. Temperature decreases and the products have higher enthalpy than the reactants.

B. Temperature decreases and the products have lower enthalpy than the reactants.

C. Temperature increases and the products have higher enthalpy than the reactants.

D. Temperature increases and the products have lower enthalpy than the reactants.

8.1-M3 May 2024 TZ1 · Paper 1 · Q1

1. What happens as ice melts at 0 °C?

I. Molecules gain kinetic energy and temperature increases.
II. Added energy overcomes hydrogen bonds between molecules.
III. Molecules gain sufficient energy to move from fixed positions.

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

8.1-M4 Nov 2023 TZ1 · Paper 1 · Q14

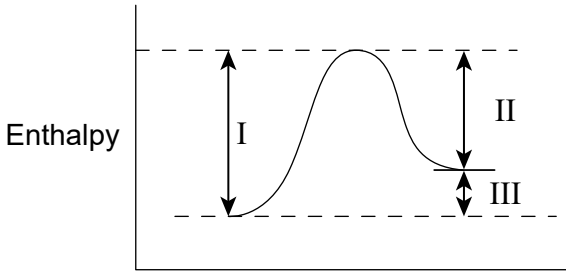
14. Which reactions release heat?

I. $C(s) + O_2(g) \rightarrow CO_2(g)$
II. $Na^+(g) + e^- \rightarrow Na(g)$
III. $NH_3(g) \rightarrow NH_3(l)$

A. I and II only
B. I and III only
C. II and III only
D. I, II and III

8.1-M5 Nov 2023 TZ1 · Paper 1 · Q16

16. Which expression represents the calculation of ΔH?



A. I–II
B. II–I
C. I–III
D. II–III

8.1-M6 Nov 2022 · Paper 1 · Q2

2. What is the change of state for a gas to a solid?

A. Condensation

B. Deposition

C. Freezing

D. Sublimation

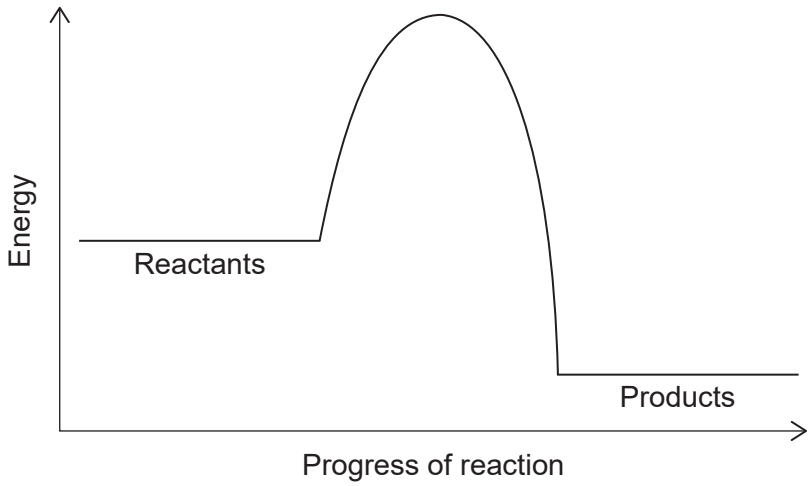
8.1-M7 May 2021 TZ2 · Paper 1 · Q14

14. Which describes an exothermic reaction?

	Heat transfer	Enthalpy
A.	from surroundings to system	reactants > products
B.	from surroundings to system	products > reactants
C.	from system to surroundings	products > reactants
D.	from system to surroundings	reactants > products

8.1-M8 May 2021 TZ2 · Paper 1 · Q15

15. The potential energy profile of a reaction is shown.



What can be determined about stability and energy change from the potential energy profile shown

	More stable	Reaction
A.	reactants	exothermic
B.	reactants	endothermic
C.	products	exothermic
D.	products	endothermic

8.1-M9 Nov 2020 · Paper 1 · Q15

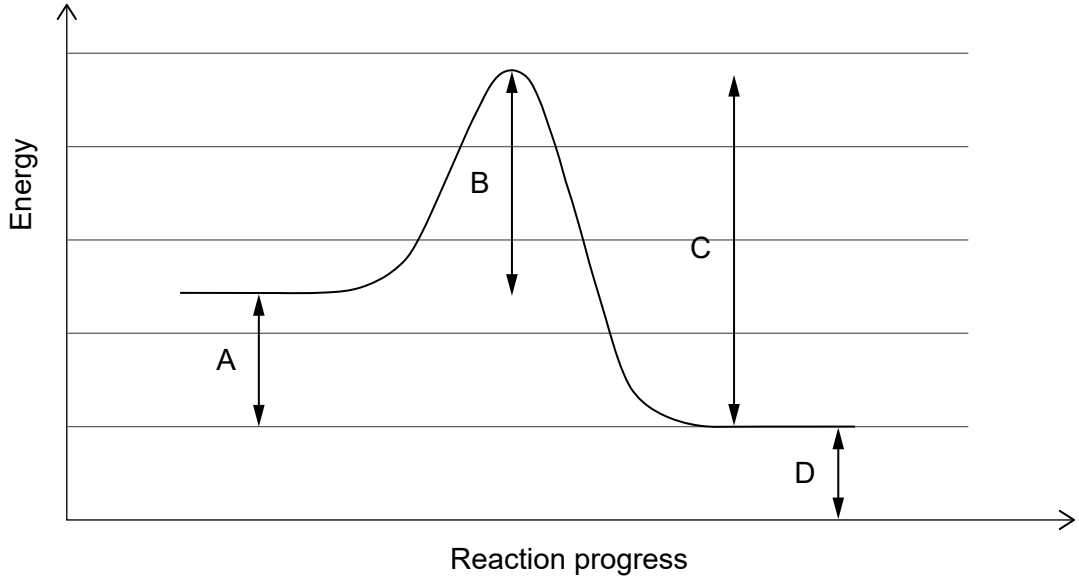
15. Which statements about bond strength and activation energy are correct for this reaction?

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H^{\ominus} = -891\,kJ$$

	Relative bond strength	Relative magnitude of activation energy, E_a
A.	products < reactants	forward > reverse
B.	products > reactants	forward < reverse
C.	products > reactants	forward > reverse
D.	products < reactants	forward < reverse

8.1-M10 May 2019 TZ2 · Paper 1 · Q15

15. What is the activation energy of the reverse reaction?



8.1-M11 Nov 2018 · Paper 1 · Q15

15. Which is correct when $Ba(OH)_2$ reacts with NH_4Cl?

$$Ba(OH)_2(s) + 2NH_4Cl(s) \rightarrow BaCl_2(aq) + 2NH_3(g) + 2H_2O(l) \qquad \Delta H^{\ominus} = +164\,kJ\,mol^{-1}$$

	Temperature	Enthalpy	Stability
A.	increases	products have lower enthalpy than the reactants	products are less stable than the reactants
B.	decreases	products have lower enthalpy than the reactants	products are more stable than the reactants
C.	decreases	products have higher enthalpy than the reactants	products are less stable than the reactants
D.	increases	products have higher enthalpy than the reactants	products are more stable than the reactants

8.1-M12 May 2018 TZ1 · Paper 1 · Q15

15. Which statement is correct?

A. In an exothermic reaction, the products have more energy than the reactants.

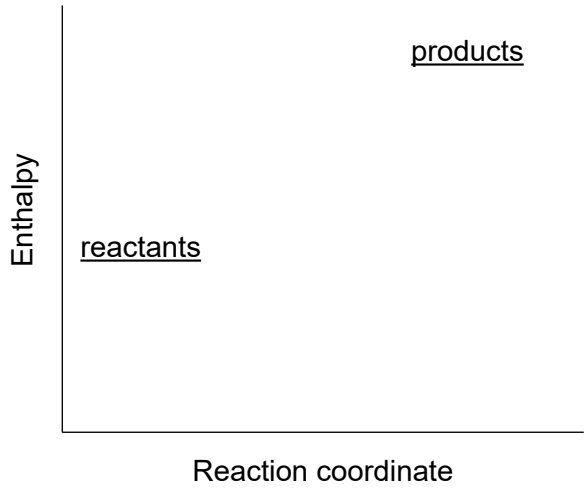
B. In an exothermic reversible reaction, the activation energy of the forward reaction is greater than that of the reverse reaction.

C. In an endothermic reaction, the products are more stable than the reactants.

D. In an endothermic reversible reaction, the activation energy of the forward reaction is greater than that of the reverse reaction.

8.1-M13 May 2018 TZ2 · Paper 1 · Q14

14. Which describes the reaction shown in the potential energy profile?



A. The reaction is endothermic and the products have greater enthalpy than the reactants.

B. The reaction is endothermic and the reactants have greater enthalpy than the products.

C. The reaction is exothermic and the products have greater enthalpy than the reactants.

D. The reaction is exothermic and the reactants have greater enthalpy than the products.

8.1-M14 May 2017 TZ1 · Paper 1 · Q14

14. Which expression gives the mass, in g, of ethanol required to produce 683.5 kJ of heat upon complete combustion?

(M_r for ethanol = 46.0, $\Delta H_c^{\ominus} = -1367\,\text{kJ mol}^{-1}$)

A. $\dfrac{683.5}{1367 \times 46.0}$

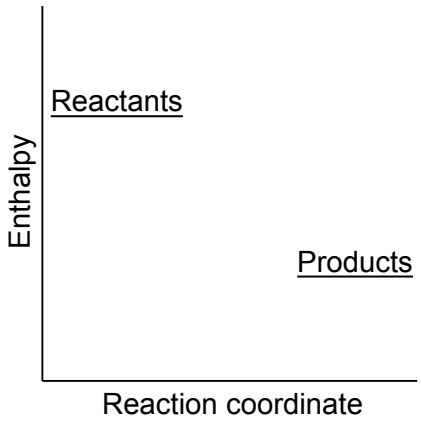
B. $\dfrac{1367}{683.5 \times 46.0}$

C. $\dfrac{683.5 \times 46.0}{1367}$

D. $\dfrac{1367 \times 46.0}{683.5}$

8.1-M15 May 2017 TZ2 · Paper 1 · Q14

14. What can be deduced from this reaction profile?



A. The reactants are less stable than the products and the reaction is exothermic.

B. The reactants are less stable than the products and the reaction is endothermic.

C. The reactants are more stable than the products and the reaction is exothermic.

D. The reactants are more stable than the products and the reaction is endothermic.

8.1-M16 May 2016 · Paper 1 · Q1

1. Which equation represents sublimation?

A. $2Al(s) + 3I_2(g) \rightarrow 2AlI_3(s)$

B. $HgCl_2(s) \rightarrow HgCl_2(g)$

C. $I_2(g) \rightarrow I_2(s)$

D. $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

8.1-M17 Nov 2015 · Paper 1 · Q14

14. Which of the following changes are exothermic?

I. $H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

II. $2C_8H_{18}(g) + 17O_2(g) \rightarrow 16CO(g) + 18H_2O(g)$

III. $C_8H_{18}(g) \rightarrow C_8H_{18}(l)$

A. I and II only

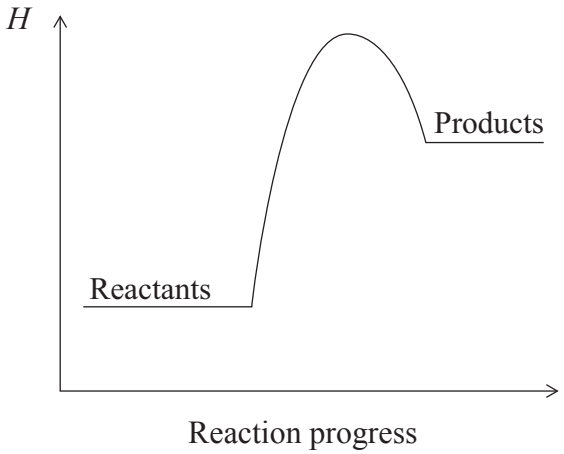
B. I and III only

C. II and III only

D. I, II and III

8.1-M18 May 2014 TZ1 · Paper 1 · Q14

14. Which statement is correct for the reaction with this enthalpy level diagram?



A. Heat energy is released during the reaction and the reactants are more stable than the products.

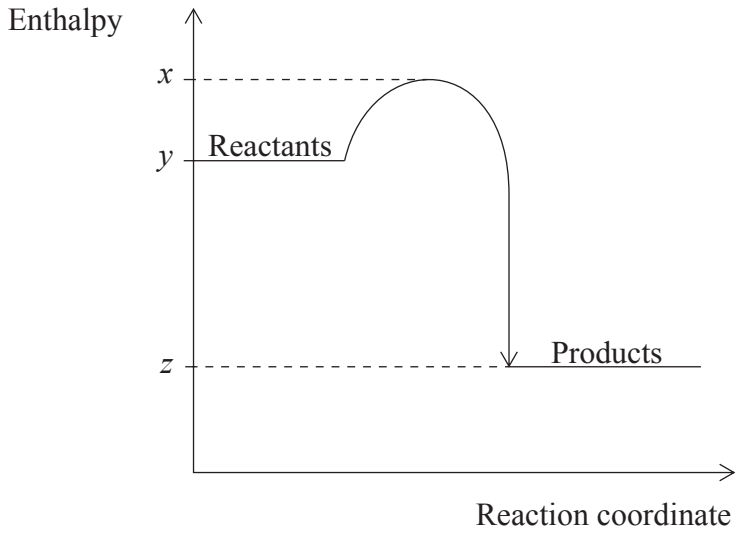
B. Heat energy is absorbed during the reaction and the reactants are more stable than the products.

C. Heat energy is released during the reaction and the products are more stable than the reactants.

D. Heat energy is absorbed during the reaction and the products are more stable than the reactants.

8.1-M19 May 2014 TZ2 · Paper 1 · Q14

14. What is the value of ΔH for the exothermic reaction represented by the diagram below?



A. $y - z$

B. $z - y$

C. $x - z$

D. $z - x$

8.1-M20 Nov 2013 · Paper 1 · Q15

15. Which processes are exothermic?

I. $CH_3CH_2CH_3(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

II. $Cl_2(g) \rightarrow 2Cl(g)$

III. $CH_3CH_2COOH(aq) + NaOH(aq) \rightarrow CH_3CH_2COONa(aq) + H_2O(l)$

A. I and II only

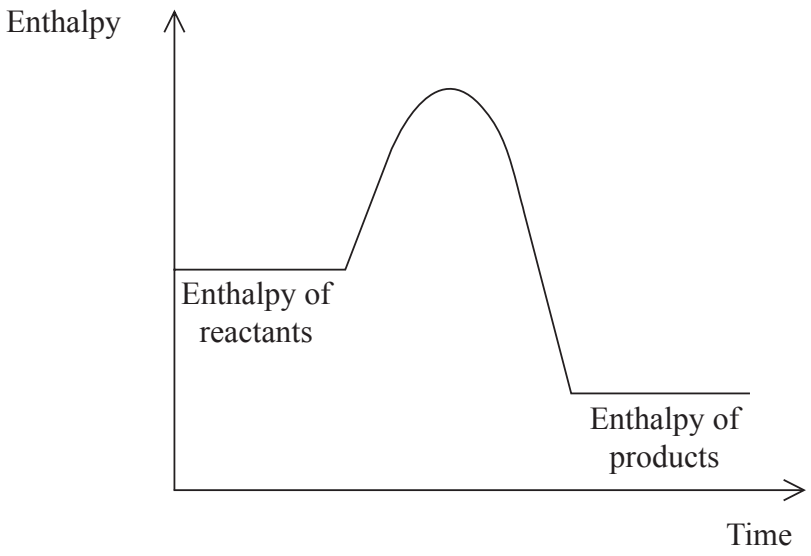
B. I and III only

C. II and III only

D. I, II and III

8.1-M21 May 2013 TZ1 · Paper 1 · Q14

14. Which statement is correct for the enthalpy level diagram shown?



A. The reaction is exothermic and the products are more stable than the reactants.

B. The reaction is exothermic and the sign of the enthalpy change is positive.

C. The reaction is endothermic and the sign of the enthalpy change is negative.

D. The reaction is endothermic and the products are more stable than the reactants.

8.1-M22 May 2013 TZ1 · Paper 1 · Q15

15. Which process is endothermic?

A. $2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(g)$

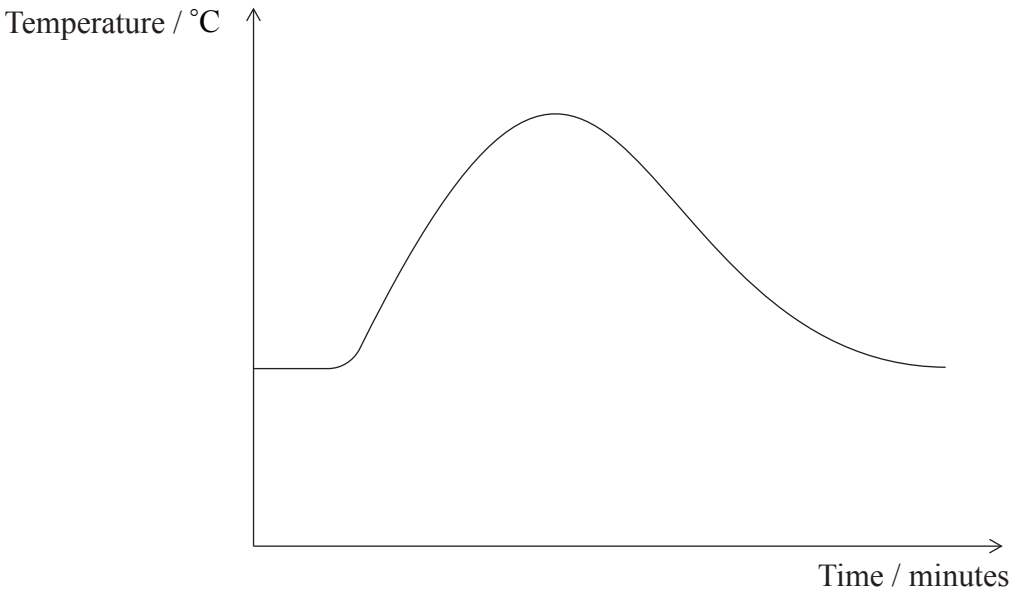
B. $Na(g) \rightarrow Na^+(g) + e^-$

C. $H_2SO_4(aq) + 2KOH(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$

D. $NH_3(g) \rightarrow NH_3(l)$

8.1-M23 Nov 2011 · Paper 1 · Q15

15. A student measured the temperature of a reaction mixture over time using a temperature probe. By considering the graph, which of the following deductions can be made?



I. The reaction is exothermic.

II. The products are more stable than the reactants.

III. The reactant bonds are stronger than the product bonds.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

8.1-M24 May 2011 TZ1 · Paper 1 · Q18

18. Which statement about bonding is correct?

A. Bond breaking is endothermic and requires energy.

B. Bond breaking is endothermic and releases energy.

C. Bond making is exothermic and requires energy.

D. Bond making is endothermic and releases energy.

8.1-M25 May 2010 TZ1 · Paper 1 · Q15

15. Which types of reaction are always exothermic?

I. Neutralization
II. Decomposition
III. Combustion

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

8.1-M26 May 2010 TZ2 · Paper 1 · Q14

14. Which processes are exothermic?

I. Ice melting
II. Neutralization
III. Combustion

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Part B - Structured questions

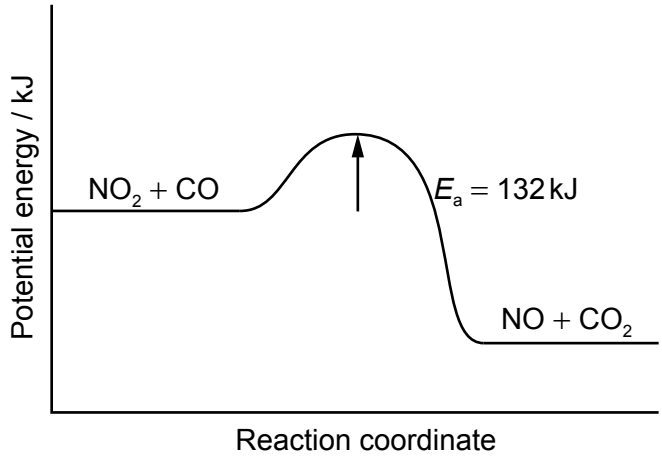
8.1-S1 May 2017 TZ2 · Paper 2 · Q5 parts (b)(iv)

(b) Nitrogen dioxide and carbon monoxide react according to the following equation:

$$NO_2(g) + CO(g) \rightleftharpoons NO(g) + CO_2(g) \qquad \Delta H = -226\,kJ$$

Experimental data shows the reaction is second order with respect to NO_2 and zero order with respect to CO.

(iv) Calculate the activation energy for the reverse reaction. [1]



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8.1-S2 May 2016 · Paper 3 · Q11 parts (b)

11. Glucose, $C_6H_{12}O_6$, is a monosaccharide that our body can use as a source of energy.

(b) Calculate the energy, in kJ, produced from 15.0 g of glucose if its enthalpy of combustion is $-2803\,kJ\,mol^{-1}$. [2]

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8.1-S3 May 2015 TZ2 · Paper 2 · Q7 parts (a)(iii)

7. When nitrogen gas and hydrogen gas are allowed to react in a closed container the following equilibrium is established.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92.6\,kJ$$

(iii) Define the term *activation energy*, E_a. [1]

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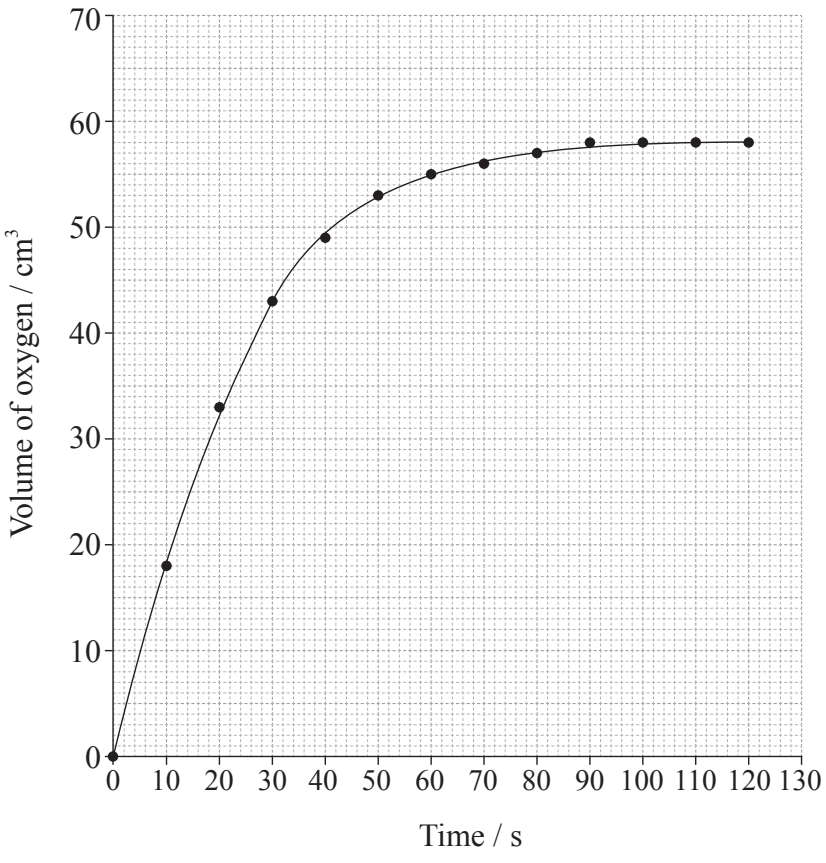
8.1-S4 Nov 2014 · Paper 2 · Q11 parts (e)(i), (e)(ii), (e)(iii), (e)(iv), (f)(ii) · also 2.1, 8.2, 9.1

Hydrogen peroxide decomposes according to the equation below.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

The rate of the decomposition can be monitored by measuring the volume of oxygen gas released. The graph shows the results obtained when a solution of hydrogen peroxide decomposed in the presence of a CuO catalyst.



(e) Hydrochloric acid neutralizes sodium hydroxide, forming sodium chloride and water.

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l) \qquad \Delta H^{\ominus} = -57.9\,kJ\,mol^{-1}$$

(i) Define *standard enthalpy change of reaction*, $\Delta H^{\ominus}$. *[2]*

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(ii) Determine the amount of energy released, in kJ, when $50.0\,cm^3$ of $1.00\,mol\,dm^{-3}$ sodium hydroxide solution reacts with $50.0\,cm^3$ of $1.00\,mol\,dm^{-3}$ hydrochloric acid solution. *[2]*

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(iii) In an experiment, 2.50 g of solid sodium hydroxide was dissolved in $50.0\,cm^3$ of water. The temperature rose by 13.3 °C. Calculate the standard enthalpy change, in $kJ\,mol^{-1}$, for dissolving one mole of solid sodium hydroxide in water.

$$NaOH(s) \rightarrow NaOH(aq)$$ *[3]*

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(iv) Using relevant data from previous question parts, determine $\Delta H^{\ominus}$, in $kJ\,mol^{-1}$, for the reaction of solid sodium hydroxide with hydrochloric acid.

$$NaOH(s) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$ *[2]*

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(ii) Explain why Fe^{3+} is a more stable ion than Fe^{2+} by reference to their electron configurations. *[3]*

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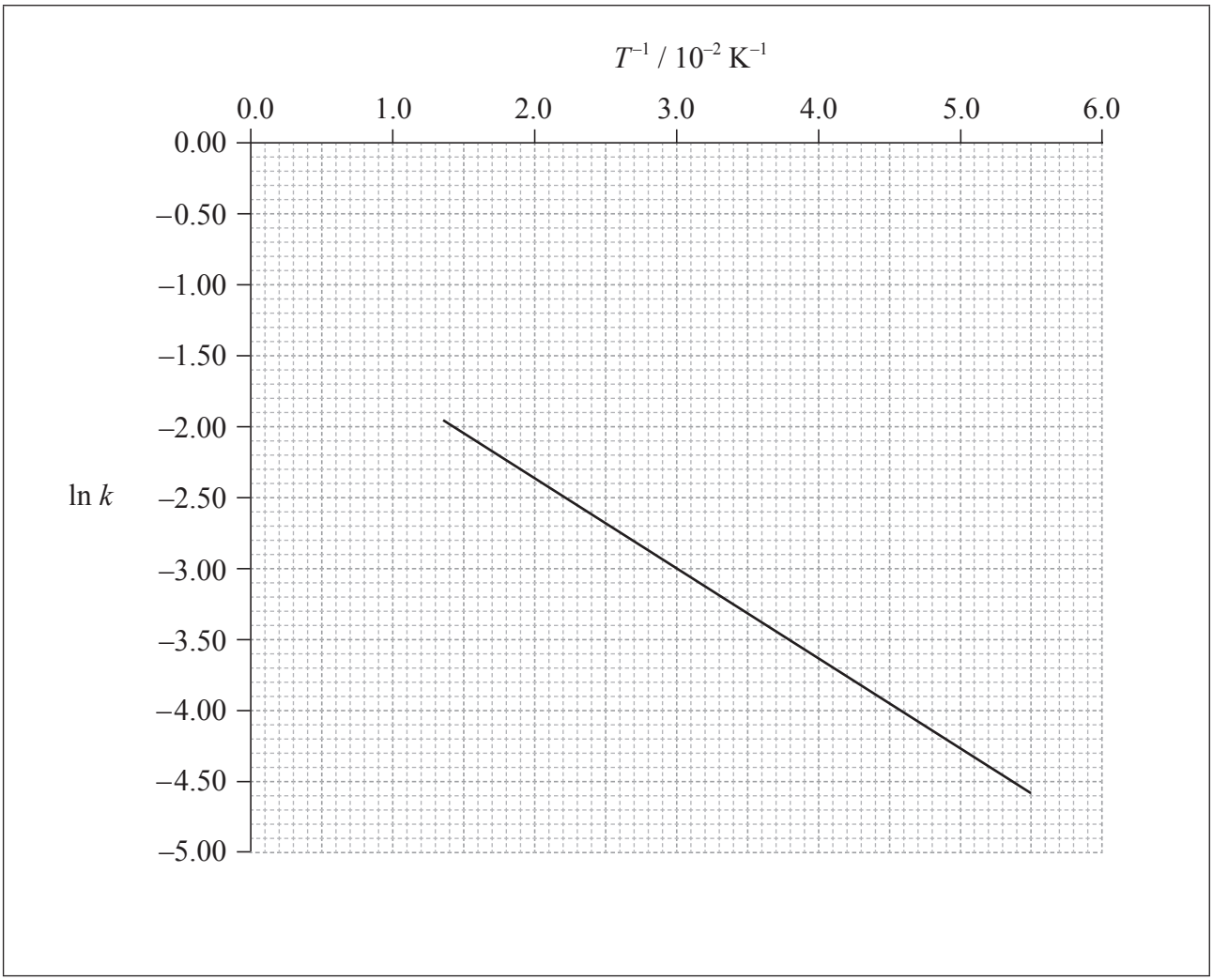
8.1-S5 May 2013 TZ1 · Paper 2 · Q3 parts (a)

3. To determine the activation energy of a reaction, the rate of reaction was measured at different temperatures. The rate constant, k, was determined and ln k was plotted against the inverse of the temperature in Kelvin, T^{-1}. The following graph was obtained.



(a) Define the term *activation energy*, E_a. *[1]*

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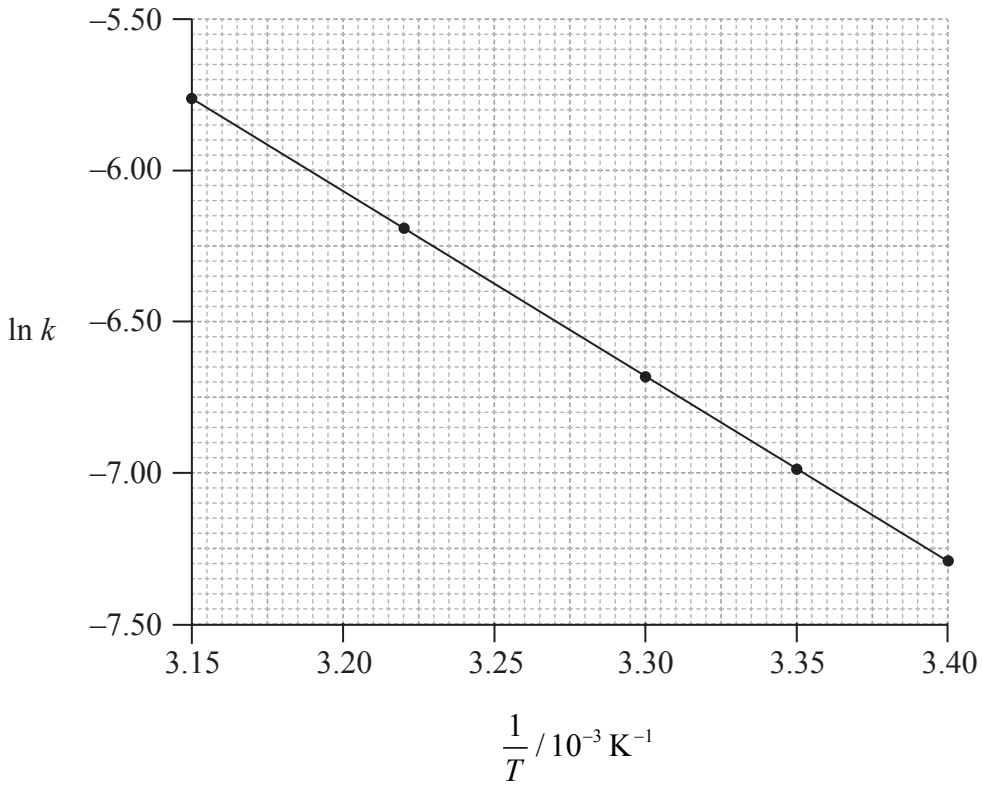
8.1-S6 May 2013 TZ2 · Paper 2 · Q2 parts (a)

2. Consider the following graph of ln k against $\frac{1}{T}$.



(a) A catalyst provides an alternative pathway for a reaction, lowering the activation energy, E_a. Define the term *activation energy*, E_a. *[1]*

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8.1-S7 Nov 2012 · Paper 2 · Q6 parts (b)(i)

6. Chemical kinetics involves an understanding of how the molecular world changes with time.

(b) A catalyst provides an alternative pathway for a reaction, lowering the activation energy, E_a.

(i) Define the term *activation energy*, E_a. *[1]*

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8.1-S8 May 2012 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (c) · also 8.2

8. (a) The equation for the reaction between sodium hydroxide, NaOH, and nitric acid, HNO_3, is shown below.

$$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l) \qquad \Delta H = -57.6\ kJ\ mol^{-1}$$

(i) Sketch and label an enthalpy level diagram for this reaction. *[3]*

(ii) Deduce whether the reactants or the products are more energetically stable, stating your reasoning. *[1]*

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(iii) Calculate the change in heat energy, in kJ, when $50.0\ cm^3$ of $2.50\ mol\ dm^{-3}$ sodium hydroxide solution is added to excess nitric acid. *[2]*

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(c) When 5.35 g ammonium chloride, $NH_4Cl(s)$, is added to 100.0 cm^3 of water, the temperature of the water decreases from 19.30 °C to 15.80 °C. Determine the enthalpy change, in kJ mol^{-1}, for the dissolving of ammonium chloride in water. *[3]*

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8.1-S9 Nov 2011 · Paper 2 · Q8 parts (a), (c)(i), (e)(i) · also 3.1

8. (a) Define the term *activation energy*, E_a. *[1]*

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(c) (i) Define the term *endothermic reaction*. *[1]*

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(e) $BF_3(g)$ reacts with $NH_3(g)$ to form $F_3BNH_3(g)$ according to the equation below.

$$BF_3(g) + NH_3(g) \rightarrow F_3BNH_3(g)$$

(i) Identify the type of bond present between BF_3 and NH_3 in $F_3BNH_3(g)$ and state another example of a compound with this type of bonding. *[2]*

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LESSON 8.2

Calorimetry

26 multiple-choice questions · 20 structured questions

Part A - Multiple choice

8.2-M1 May 2025 TZ1 · Paper 1A · Q23

23. 0.050 mol of sodium hydrogencarbonate was added to 50 cm^3 of 2.0 mol dm^{-3} solution of citric acid.

$$C_6H_8O_7(aq) + 3NaHCO_3(s) \rightarrow C_6H_5O_7Na_3(aq) + 3CO_2(g) + 3H_2O(l)$$

The initial temperature of the solution was 298 K and the final temperature was 297 K.
What is the enthalpy of this reaction in kJ mol^{-1}? ($c_w = 4.18$ J g^{-1} K^{-1}, $Q = mc\Delta T$)

A. +4.18

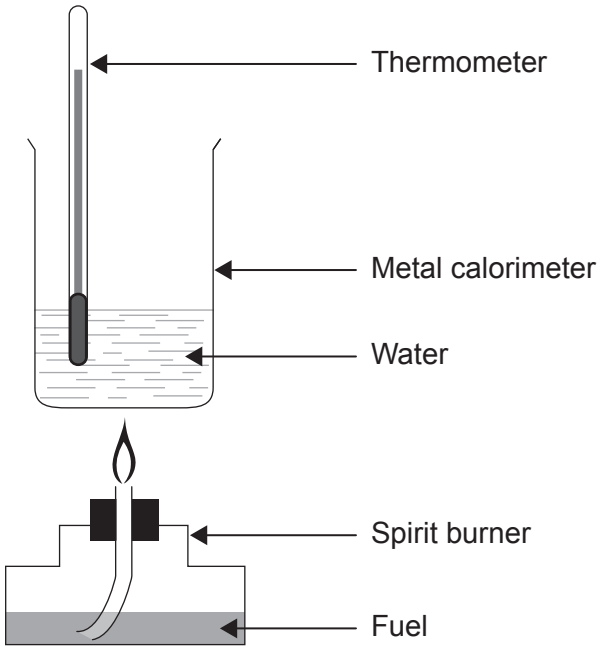
B. –4.18

C. +12.5

D. –12.5

8.2-M2 May 2025 TZ2 · Paper 1A · Q19

19. What is the standard enthalpy change, $\Delta H^{\ominus}_{\text{combustion}}$, according to the data?



Amount of fuel burned = 0.110 mol
Mass of water = 200 g
Initial temperature of water = 21.0 °C
Final temperature of water = 25.0 °C
Specific heat capacity of water, $c_w = 4.18\,\text{J g}^{-1}\,\text{K}^{-1}$
$Q = mc\Delta T$

A. $-2110\,\text{kJ mol}^{-1}$

B. $-30.4\,\text{kJ mol}^{-1}$

C. $+30.4\,\text{kJ mol}^{-1}$

D. $+2110\,\text{kJ mol}^{-1}$

8.2-M3 May 2025 TZ3 · Paper 1A · Q21

21. The energy from burning 0.321 g of methanol causes the temperature of 200 cm^3 of water to rise by 5.00 K. What is the enthalpy of combustion of the methanol in kJ mol^{-1}?

Specific heat capacity of water: $4.18\,\text{J g}^{-1}\,\text{K}^{-1}$, M_r methanol = $32.05\,\text{g mol}^{-1}$

$$Q = mc\Delta T$$

A. −417

B. −579

C. −23 200

D. −417 000

8.2-M4 Nov 2024 · Paper 1 · Q18

18. What is the enthalpy change, in $kJ\,mol^{-1}$, when 107 g of solid ammonium chloride, NH_4Cl, are added to water to form 50.0 cm^3 of solution, producing a maximum decrease of 28 °C?
$M_r\ NH_4Cl = 53.5$
Specific heat capacity of water= $4.18\,J\,g^{-1}\,K^{-1}$

A. $\Delta H = \dfrac{-50.0 \times 4.18 \times (28 + 273)}{2 \times 1000}$

B. $\Delta H = \dfrac{-50.0 \times 4.18 \times 28}{2 \times 1000}$

C. $\Delta H = \dfrac{50.0 \times 4.18 \times (28 + 273)}{2 \times 1000}$

D. $\Delta H = \dfrac{50.0 \times 4.18 \times 28}{2 \times 1000}$

8.2-M5 May 2024 TZ1 · Paper 1 · Q14

14. How much heat energy, in J, does a 2.00 g block of copper metal at 65.0 °C lose when it is dropped into 100.0 cm^3 of water and cools to 15.0 °C?

The specific heat capacity of copper is $0.385\,J\,g^{-1}\,K^{-1}$ and the specific heat capacity of water is $4.18\,J\,g^{-1}\,K^{-1}$.

A. $2.00 \times 0.385 \times (65.0 - 15.0)$

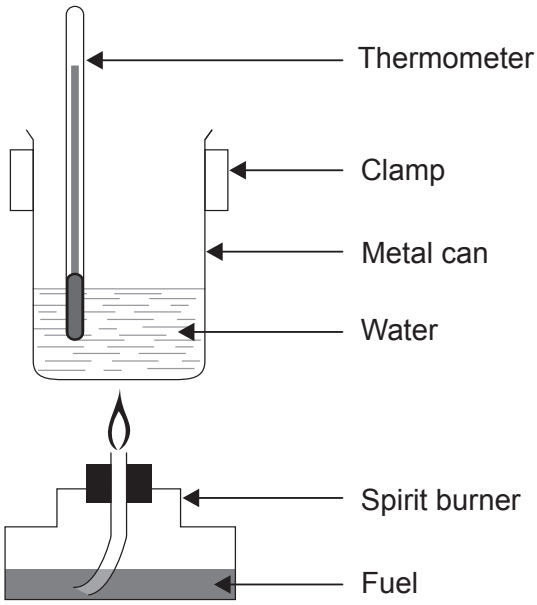
B. $2.00 \times 0.385 \times (65.0 - 15.0 + 273)$

C. $100.0 \times 4.18 \times (65.0 - 15.0)$

D. $100.0 \times 4.18 \times (65.0 - 15.0 + 273)$

8.2-M6 May 2024 TZ2 · Paper 1 · Q16

16. A student determined the enthalpy change of combustion of a fuel by burning it in a spirit burner placed under a metal can containing 100 cm^3 of water.



Which modification can improve the accuracy of the experiment?

A. Placing a lid on the metal can containing water

B. Covering the spirit burner with aluminium foil

C. Increasing the distance between the metal can and the spirit burner

D. Using warm water instead of water at room temperature in the metal can

8.2-M7 May 2023 TZ1 · Paper 1 · Q38

38. What is the percentage error if the enthalpy of combustion of a substance is determined experimentally to be –2100 kJ mol^{-1}, but the literature value is –3500 kJ mol^{-1}?

A. 80 %

B. 60 %

C. 40 %

D. 20 %

8.2-M8 May 2023 TZ2 · Paper 1 · Q15

15. When 100 cm^3 of 1.0 mol dm^{-3} HCl is mixed with 100 cm^3 of 1.0 mol dm^{-3} NaOH, the temperature of the resulting solution increases by 5.0 °C. What will be the temperature change, in °C, when 50 cm^3 of 2.0 mol dm^{-3} HCl is mixed with 50 cm^{-3} of 2.0 mol dm^{-3} NaOH?

A. 2.5

B. 5.0

C. 10

D. 20

8.2-M9 May 2022 TZ1 · Paper 1 · Q14

14. The energy from burning 0.250 g of ethanol causes the temperature of 150 cm^3 of water to rise by 10.5 °C. What is the enthalpy of combustion of ethanol, in kJ mol^{-1}?

Specific heat capacity of water: 4.18 $J\,g^{-1}\,K^{-1}$.

A. $\dfrac{150 \times 4.18 \times 10.5}{\dfrac{0.250}{46.08}}$

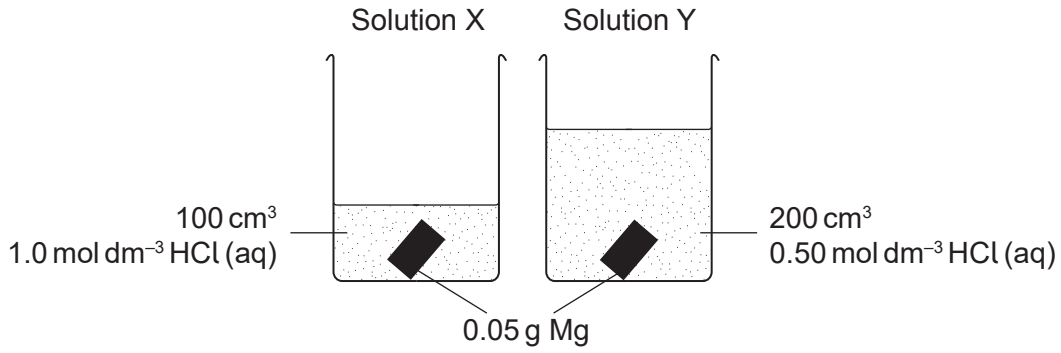
B. $\dfrac{150 \times 4.18 \times 10.5}{\dfrac{0.250}{46.08} \times 1000}$

C. $\dfrac{150 \times 4.18 \times (273 + 10.5)}{\dfrac{0.250}{46.08}}$

D. $\dfrac{150 \times 4.18 \times (273 + 10.5)}{\dfrac{0.250}{46.08} \times 1000}$

8.2-M10 May 2022 TZ2 · Paper 1 · Q13

13. Which statement is correct about identical pieces of magnesium added to two solutions, X and Y, containing hydrochloric acid at the same temperature?



A. Solution X will reach a higher maximum temperature.

B. Solution Y will reach a higher maximum temperature.

C. Solutions X and Y will have the same temperature rise.

D. It is not possible to predict whether X or Y will have the higher maximum temperature because we cannot identify the limiting reactant.

8.2-M11 May 2021 TZ1 · Paper 1 · Q14

14. What is the enthalpy change, **in J**, when 5 g of water is heated from 10°C to 18°C?

Specific heat capacity of water: $4.18\,\text{kJ}\,\text{kg}^{-1}\,\text{K}^{-1}$

A. $5 \times 4.18 \times 8$

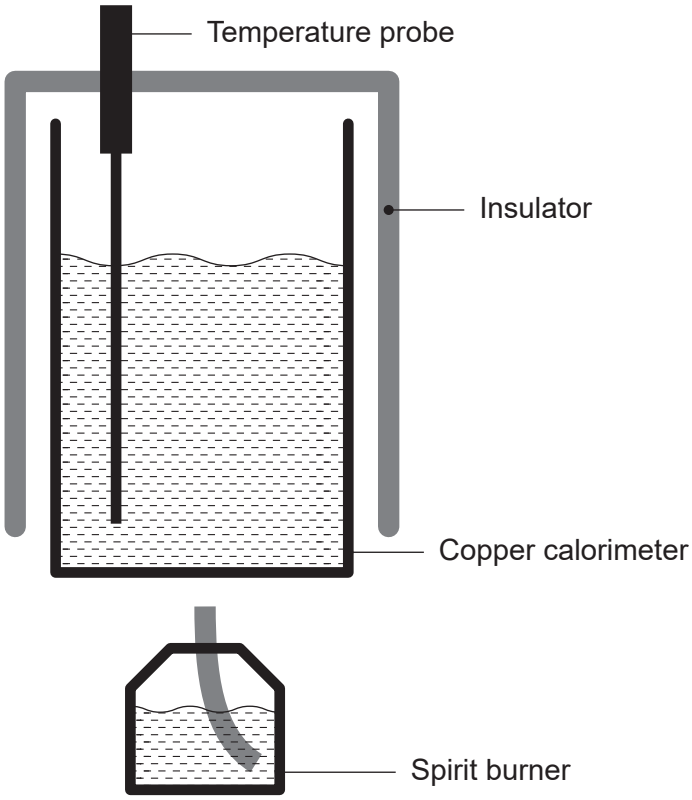
B. $5 \times 10^{-3} \times 4.18 \times 8$

C. $5 \times 4.18 \times (273 + 8)$

D. $5 \times 10^{-3} \times 4.18 \times (273 + 8)$

8.2-M12 May 2021 TZ1 · Paper 1 · Q38

38. The enthalpy of combustion of a fuel was determined using the calorimeter shown. The final result was lower than the literature value.



Which factors could have contributed to this error?

I. Not all heat from the combustion was transferred to the calorimeter.

II. Incomplete combustion occurred.

III. The temperature probe touched the bottom of the calorimeter.

A. I and II only

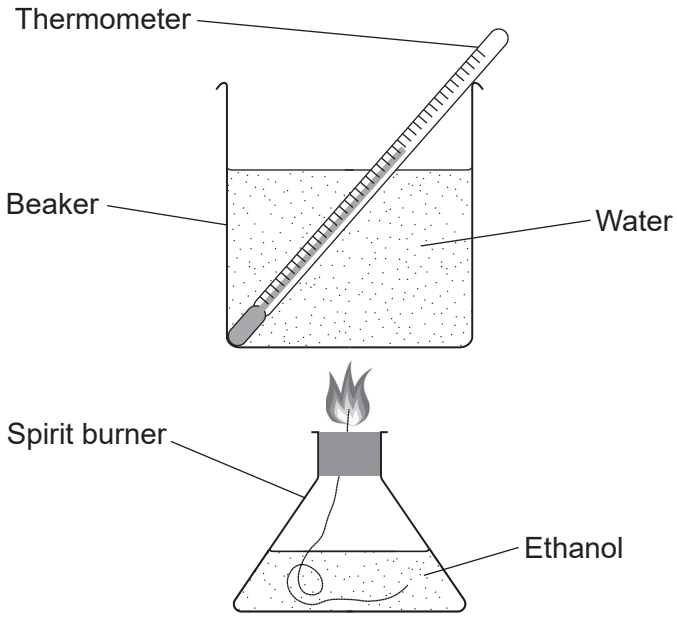
B. I and III only

C. II and III only

D. I, II and III

8.2-M13 Nov 2019 · Paper 1 · Q14

Questions 14 and 15 are about an experiment to measure the enthalpy of combustion, ΔH_c, of ethanol, using the apparatus and setup shown.



14. What is the enthalpy of combustion, ΔH_c, of ethanol in $kJ\,mol^{-1}$?

Maximum temperature of water: 30.0 °C
Initial temperature of water: 20.0 °C
Mass of water in beaker: 100.0 g
Loss in mass of ethanol: 0.230 g
M_r (ethanol): 46.08
Specific heat capacity of water: $4.18\,J\,g^{-1}\,K^{-1}$
$q = mc\Delta T$

A. $-\dfrac{100.0 \times 4.18 \times (10.0 + 273)}{\dfrac{0.230}{46.08} \times 1000}$

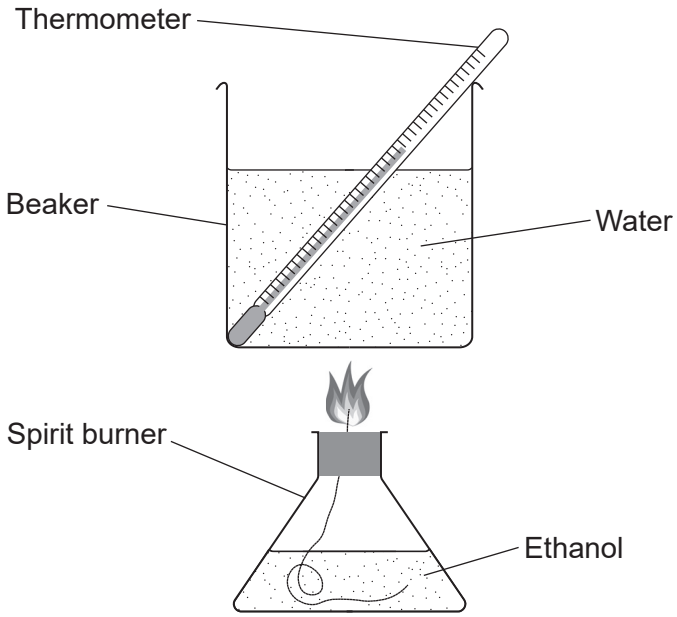
B. $-\dfrac{0.230 \times 4.18 \times 10.0}{\dfrac{100.0}{46.08} \times 1000}$

C. $-\dfrac{100.0 \times 4.18 \times 10.0}{\dfrac{0.230}{46.08} \times 1000}$

D. $-\dfrac{100.0 \times 4.18 \times 10.0}{\dfrac{0.230}{46.08}}$

8.2-M14 Nov 2019 · Paper 1 · Q15

Questions 14 and 15 are about an experiment to measure the enthalpy of combustion, ΔH_c, of ethanol, using the apparatus and setup shown.



15. Which quantity is likely to be the most inaccurate due to the sources of error in this experiment?

A. Mass of ethanol burnt

B. Molecular mass of ethanol

C. Mass of water

D. Temperature change

8.2-M15 May 2019 TZ1 · Paper 1 · Q14

14. When equal masses of X and Y absorb the same amount of energy, their temperatures rise by 5 °C and 10 °C respectively. Which is correct?

A. The specific heat capacity of X is twice that of Y.

B. The specific heat capacity of X is half that of Y.

C. The specific heat capacity of X is one fifth that of Y.

D. The specific heat capacity of X is the same as Y.

8.2-M16 May 2018 TZ2 · Paper 1 · Q15

15. Two 100 cm^3 aqueous solutions, one containing 0.010 mol NaOH and the other 0.010 mol HCl, are at the same temperature.

When the two solutions are mixed the temperature rises by y °C.

Assume the density of the final solution is 1.00 g cm^{-3}.
Specific heat capacity of water = 4.18 J g^{-1} K^{-1}

What is the enthalpy change of neutralization in kJ mol^{-1}?

A. $\frac{200 \times 4.18 \times y}{1000 \times 0.020}$

B. $\frac{200 \times 4.18 \times y}{1000 \times 0.010}$

C. $\frac{100 \times 4.18 \times y}{1000 \times 0.010}$

D. $\frac{200 \times 4.18 \times (y + 273)}{1000 \times 0.010}$

8.2-M17 May 2017 TZ1 · Paper 1 · Q38

38. The molar mass of a gas, determined experimentally, is 32 g mol^{-1}. Its literature molar mass is 40 g mol^{-1}.

What is the percentage error?

A. 80 %

B. 25 %

C. 20 %

D. 8 %

8.2-M18 Nov 2016 · Paper 1 · Q17

17. 5.35 g of solid ammonium chloride, $NH_4Cl(s)$, was added to water to form 25.0 g of solution. The maximum decrease in temperature was 14 K. What is the enthalpy change, in kJ mol^{-1}, for this reaction? (Molar mass of NH_4Cl = 53.5 g mol^{-1}; the specific heat capacity of the solution is 4.18 J g^{-1} K^{-1})

A. $\Delta H = +\frac{25.0 \times 4.18 \times (14 + 273)}{0.1 \times 1000}$

B. $\Delta H = -\frac{25.0 \times 4.18 \times 14}{0.1 \times 1000}$

C. $\Delta H = +\frac{25.0 \times 4.18 \times 14}{0.1 \times 1000}$

D. $\Delta H = +\frac{25.0 \times 4.18 \times 14}{1000}$

8.2-M19 May 2015 TZ1 · Paper 1 · Q14

14. The same amount of heat energy is added to 1.00 g of each substance.

Substance	Specific heat capacity / $J\,g^{-1}\,K^{-1}$
Copper	0.39
Aluminium	0.90
Sodium chloride	0.90
Water	4.18

Which statement is correct if all the substances are at the same temperature before the heat energy is added?

A. Copper will reach the highest temperature.

B. Water will reach the highest temperature.

C. All four substances will reach the same temperature.

D. Aluminium will reach a higher temperature than sodium chloride.

8.2-M20 May 2015 TZ1 · Paper 1 · Q15

15. The heat change in a neutralization reaction can be determined by mixing equal volumes of HCl (aq) and NaOH (aq) of the same concentration in a glass beaker. The maximum temperature change is recorded using an alcohol thermometer.

What is the biggest source of error in this experiment?

A. Heat absorbed by the glass thermometer

B. Random error in the thermometer reading

C. Heat loss to the surroundings

D. Systematic error in measuring the volumes of HCl (aq) and NaOH (aq) using burettes

8.2-M21 May 2014 TZ2 · Paper 1 · Q19

19. What is the temperature rise when 2100 J of energy is supplied to 100 g of water? (Specific heat capacity of water $= 4.2\,J\,g^{-1}\,K^{-1}$.)

A. 5 °C

B. 278 K

C. 0.2 °C

D. 20 °C

8.2-M22 May 2013 TZ2 · Paper 1 · Q14

14. The specific heat capacity of aluminium is $0.900\,J\,g^{-1}\,K^{-1}$. What is the heat energy change, in J, when 10.0 g of aluminium is heated and its temperature increases from 15.0 °C to 35.0 °C?

A. +180

B. +315

C. +1800

D. +2637

8.2-M23 Nov 2012 · Paper 1 · Q14

14. A 5.00 g sample of a substance was heated from 25.0 °C to 35.0 °C using 2.00×10^2 J of energy. What is the specific heat capacity of the substance in J g^{-1} K^{-1}?

A. 4.00×10^{-3}

B. 2.50×10^{-1}

C. 2.00

D. 4.00

8.2-M24 May 2012 TZ1 · Paper 1 · Q14

14. What are the units for specific heat capacity?

A. kJ kg K

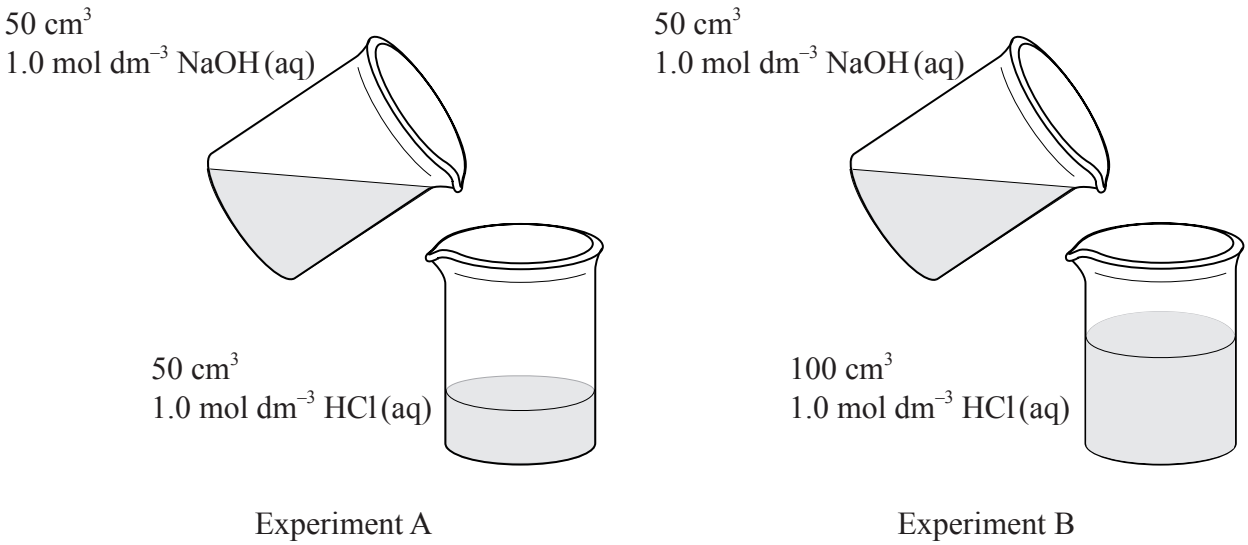
B. kJ kg K^{-1}

C. kJ kg^{-1} K

D. kJ kg^{-1} K^{-1}

8.2-M25 May 2012 TZ1 · Paper 1 · Q15

15. In each of two different experiments, A and B, a solution of sodium hydroxide is added to a solution of hydrochloric acid. The initial temperature of each solution is 25 °C.



Experiment A Experiment B

Which statement is correct?

A. The highest recorded temperature of experiment A is lower than the highest recorded temperature of experiment B.

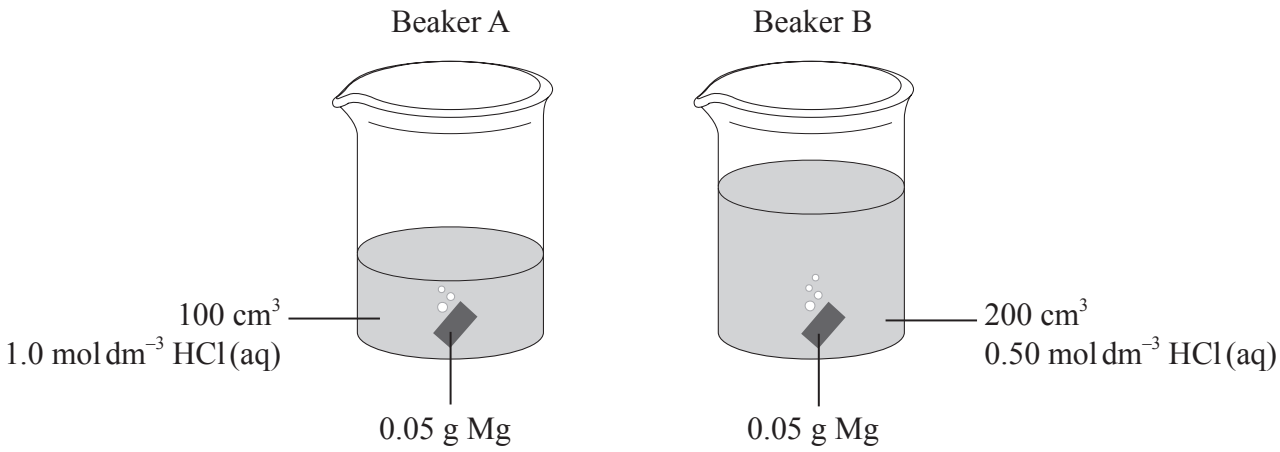
B. The highest recorded temperature of both experiments is equal.

C. The heat produced in experiment A is lower than the heat produced in experiment B.

D. The heat produced in both experiments is equal.

8.2-M26 Nov 2010 · Paper 1 · Q14

14. Identical pieces of magnesium are added to two beakers, A and B, containing hydrochloric acid. Both acids have the same initial temperature but their volumes and concentrations differ.



Which statement is correct?

A. The maximum temperature in A will be higher than in B.

B. The maximum temperature in A and B will be equal.

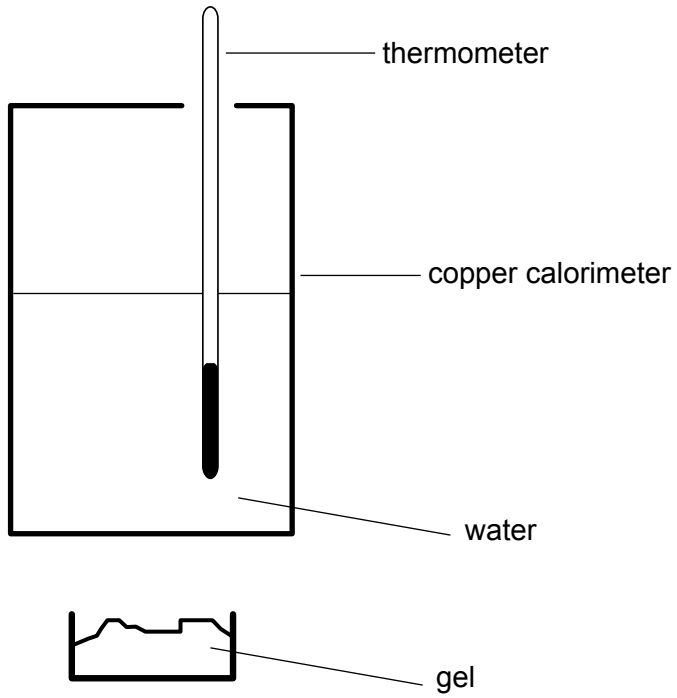
C. It is not possible to predict whether A or B will have the higher maximum temperature.

D. The temperature in A and B will increase at the same rate.

Part B - Structured questions

8.2-S1 May 2024 TZ1 · Paper 2 · Q2 parts (a)(i), (b)(i), (b)(ii), (b)(iii), (c), (d)(ii), (d)(iii), (g)(iii) · also 1.2, 5.2, 7.3

2. A student investigated the use of hand sanitising gel containing propan-1-ol as a camping fuel.



Mass of water / g ± 0.02 g	400.00
Initial temperature of water / °C ± 0.5 °C	19.0
Final temperature of water / °C ± 0.5 °C	40.0
Initial mass of gel / g ± 0.01 g	20.00
Final mass of gel / g ± 0.01 g	18.20

(a) (i) Calculate the heat energy absorbed by the water, in J. Use sections 1 and 2 of the data booklet. [1]

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(b) (i) Calculate the enthalpy of combustion of propan-1-ol, in $kJ\,mol^{-1}$, stating **one** assumption.

If you did not obtain an answer to (a)(i), use 30 000 J, though this is not the correct value. [3]

Calculation: . . . . . .

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Assumption: . . . . . .

(ii) Calculate the percentage error, using section 13 of the data booklet. [1]

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(iii) Suggest the main source of error, and a way to reduce it. [1]

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(c) The student also investigates a second gel containing ethanol, with the same percentage by mass as the propan-1-ol in the first gel.

Determine whether the second gel would release more or less energy per gram than the first gel. Use section 13 of the data booklet. [3]

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(d) Ethanol and propan-1-ol are members of a homologous series.

(ii) State the strongest intermolecular force present in ethanol and propan-1-ol. [1]

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(iii) Predict an intermolecular force which would be stronger in the next member of the homologous series, butan-1-ol. [1]

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(g) Ethyl ethanoate can be produced from ethene via a three-step synthetic route.

step 1: $CH_2CH_2 + H_2O \rightarrow CH_3CH_2OH$

step 2: $CH_3CH_2OH + [O] \rightarrow CH_3COOH$

step 3: $CH_3CH_2OH + CH_3COOH \rightarrow CH_3COOCH_2CH_3 +$ ____

(iii) Identity the second product formed in step 3. [1]

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8.2-S2 May 2023 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(iii), (b)(i), (b)(ii), (b)(v) · also 5.1, 9.2

6. The element sulfur has many industrial uses.

(a) (i) Determine the standard enthalpy of reaction ($\Delta H^{\ominus}_r$), in kJ mol^{-1}, for the oxidation of SO_2 to SO_3. [1]

Substance	Enthalpy of formation, $\Delta H^{\ominus}_f$ (kJ mol^{-1})
SO_2	–296.8
SO_3	–395.8

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(iii) Explain the polarity of the S—O bond. Use section 8 of the data booklet. [2]

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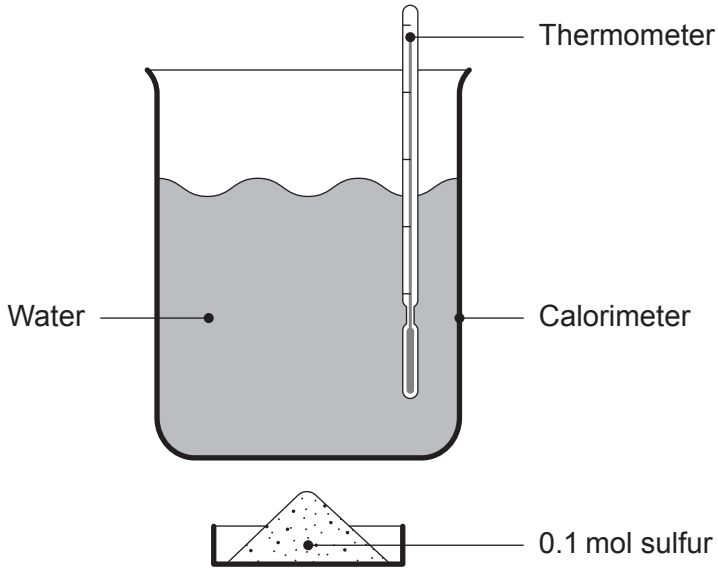
(b) The combustion of 0.1 moles of sulfur (S) was demonstrated in a school laboratory using the following apparatus in a fume cupboard.



(i) Calculate the enthalpy of combustion of sulfur, ΔH_c, in kJ mol^{-1} from this data. Use sections 1 and 2 of the data booklet. [2]

Mass of water (g) ±0.01	50.00
Initial temperature of water (°C) ±0.5	20.0
Final temperature of water (°C) ±0.5	35.0

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(ii) Suggest the major source of systematic error in this experiment and an improvement to reduce this error. [2]

Source of systematic error: . . . . . .

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Improvement: . . . . . .

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(v) Calculate the overall percentage error of this experiment. Use part (b)(i) and section 13 of the data booklet. (If you did not obtain an answer for part (b)(i) use $-50.0\,kJ\,mol^{-1}$, but this is not the correct value.) [1]

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8.2-S3 May 2023 TZ2 · Paper 2 · Q4 parts (c)

4. The periodic table provides information about electron configuration, and physical and chemical properties of elements.

(c) An 11.98 g block of pure aluminium was heated. Calculate the heat energy absorbed, in J, to increase its temperature from 18.0 °C to 40.0 °C. The specific heat capacity of aluminium is $0.902\,J\,g^{-1}\,K^{-1}$. [1]

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8.2-S4 Nov 2022 · Paper 2 · Q1 parts (d)(i), (d)(ii), (d)(iv), (e), (f)(i), (f)(iii), (f)(v) · also 1.2, 3.1, 7.2, 9.2

1. Ammonium nitrate, NH_4NO_3, is used as a high nitrogen fertilizer.

(d) Cold packs contain ammonium nitrate and water separated by a membrane.

(i) The mass of the contents of the cold pack is 25.32 g and its initial temperature is 25.2 °C. Once the contents are mixed, the temperature drops to 0.8 °C.

Calculate the energy, in J, absorbed by the dissolution of ammonium nitrate in water within the cold pack. Assume the specific heat capacity of the solution is $4.18\,J\,g^{-1}\,K^{-1}$. Use section 1 of the data booklet. [1]

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(ii) Determine the mass of ammonium nitrate in the cold pack using your answer obtained in (d)(i) and sections 6 and 19 of the data booklet.

If you did not obtain an answer in (d)(i), use 3.11×10^3 J, although this is not the correct answer. [2]

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(iv) The cold pack contains 9.50 g of ammonium nitrate. Calculate the percentage error in the experimentally determined mass of ammonium nitrate obtained in (d)(ii).

If you did not obtain an answer in (d)(ii), use 6.55 g, although this is not the correct answer. [1]

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(e) Predict, using the given values, the reaction that would take place at the anode and cathode for the electrolysis of an aqueous solution of ammonium nitrate using graphite electrodes. [2]

	$E^{\ominus}$ / V
$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$	+ 1.23
$NO_3^-(aq) + 4H^+(aq) + 3e^- \rightarrow NO(g) + 2H_2O(l)$	+ 0.96
$H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$	0.00

Anode: . . . . . .

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Cathode: . . . . . .

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(f) Solid ammonium nitrate can decompose to gaseous dinitrogen monoxide and liquid water.

(i) Write the chemical equation for this decomposition. [1]

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(iii) Calculate the standard enthalpy change, $\Delta H^{\ominus}$, of the reaction. Use section 12 of the data booklet. [2]

$\Delta H_f^{\ominus}$ ammonium nitrate $= -366\,\text{kJ mol}^{-1}$

$\Delta H_f^{\ominus}$ dinitrogen monoxide $= 82\,\text{kJ mol}^{-1}$

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(v) Deduce the Lewis (electron dot) structure, including formal charges, and shape for dinitrogen monoxide showing nitrogen as the central atom. [3]

Lewis structure:

Shape: . . . . . .

8.2-S5 Nov 2021 · Paper 2 · Q7 parts (a)

7. Alkanes undergo combustion and substitution.

(a) Determine the molar enthalpy of combustion of an alkane if 8.75×10^{-4} moles are burned, raising the temperature of 20.0 g of water by 57.3 °C. [2]

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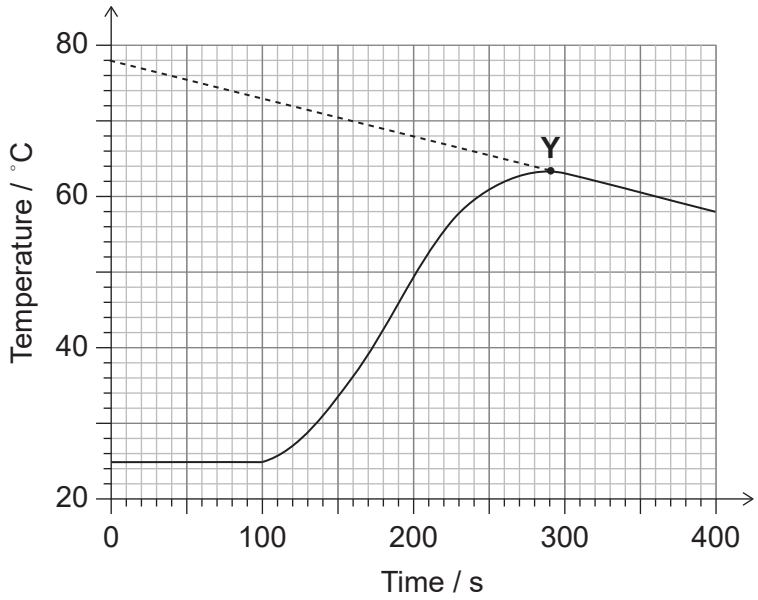
8.2-S6 May 2019 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv)

2. Powdered zinc was reacted with 25.00 cm^3 of 1.000 mol dm^{-3} copper(II) sulfate solution in an insulated beaker. Temperature was plotted against time.



[Source: © International Baccalaureate Organization 2019]

(a) (i) Estimate the time at which the powdered zinc was placed in the beaker. [1]

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(ii) State what point **Y** on the graph represents. [1]

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(b) (i) The maximum temperature used to calculate the enthalpy of reaction was chosen at a point on the extrapolated (dotted) line.

State the maximum temperature which should be used and outline **one** assumption made in choosing this temperature on the extrapolated line. [2]

Maximum temperature:

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Assumption:

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(ii) To determine the enthalpy of reaction the experiment was carried out five times. The same volume and concentration of copper(II) sulfate was used but the mass of zinc was different each time. Suggest, with a reason, if zinc or copper(II) sulfate should be in excess for each trial. [1]

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(iii) The formula $q = mc\Delta T$ was used to calculate the energy released. The values used in the calculation were $m = 25.00\,\text{g}$, $c = 4.18\,\text{J g}^{-1}\text{K}^{-1}$.

State an assumption made when using these values for m and c. [2]

Value	Assumption
$m = 25.00\,\text{g}$	. . . . .
$c = 4.18\,\text{J g}^{-1}\text{K}^{-1}$	. . . . .

(iv) Predict, giving a reason, how the final enthalpy of reaction calculated from this experiment would compare with the theoretical value. [1]

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8.2-S7 May 2019 TZ1 · Paper 3 · Q8 parts (c)

8. Starch is a natural polymer of glucose.

(c) Calculate the energy released, in kJ g^{-1}, when 3.49 g of starch are completely combusted in a calorimeter, increasing the temperature of 975 g of water from 21.0 °C to 36.0 °C. Use section 1 of the data booklet. [2]

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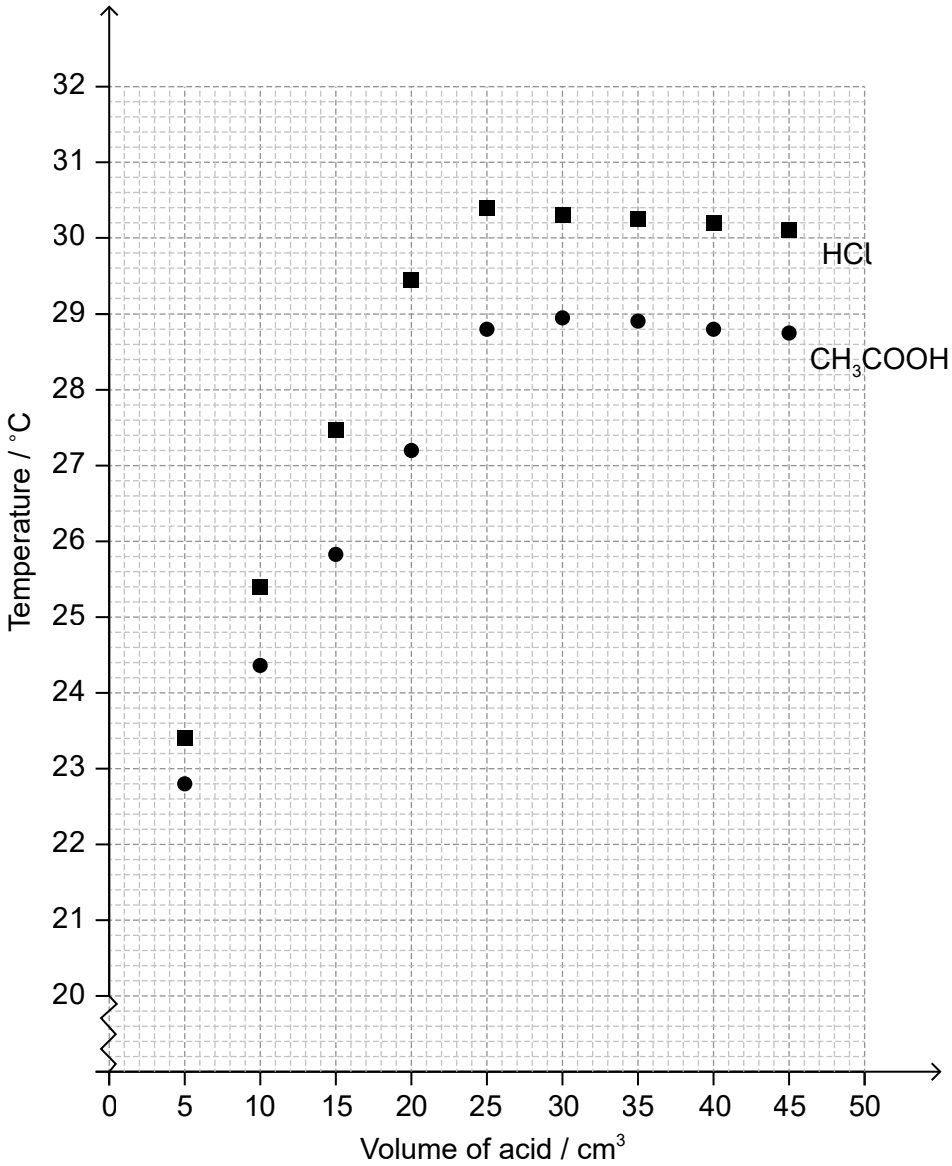
8.2-S8 Nov 2017 · Paper 2 · Q1 parts (a), (b), (c), (d)(i), (d)(ii) · also 1.2

1. A student titrated two acids, hydrochloric acid, HCl (aq) and ethanoic acid, CH_3COOH (aq), against 50.0 cm^3 of 0.995 mol dm^{-3} sodium hydroxide, NaOH (aq), to determine their concentration. The temperature of the reaction mixture was measured after each acid addition and plotted against the volume of each acid.



(a) Using the graph, estimate the initial temperature of the solutions. [1]

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(b) Determine the maximum temperature reached in each experiment by analysing the graph. [2]

HCl:

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CH_3COOH:

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(c) Calculate the concentration of ethanoic acid, CH_3COOH, in $mol\,dm^{-3}$. [2]

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(d) (i) Determine the heat change, q, in kJ, for the neutralization reaction between ethanoic acid and sodium hydroxide.
Assume the specific heat capacities of the solutions and their densities are those of water. [2]

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(ii) Calculate the enthalpy change, ΔH, in $kJ\,mol^{-1}$, for the reaction between ethanoic acid and sodium hydroxide. [2]

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8.2-S9 May 2017 TZ1 · Paper 3 · Q2 parts (a), (b)

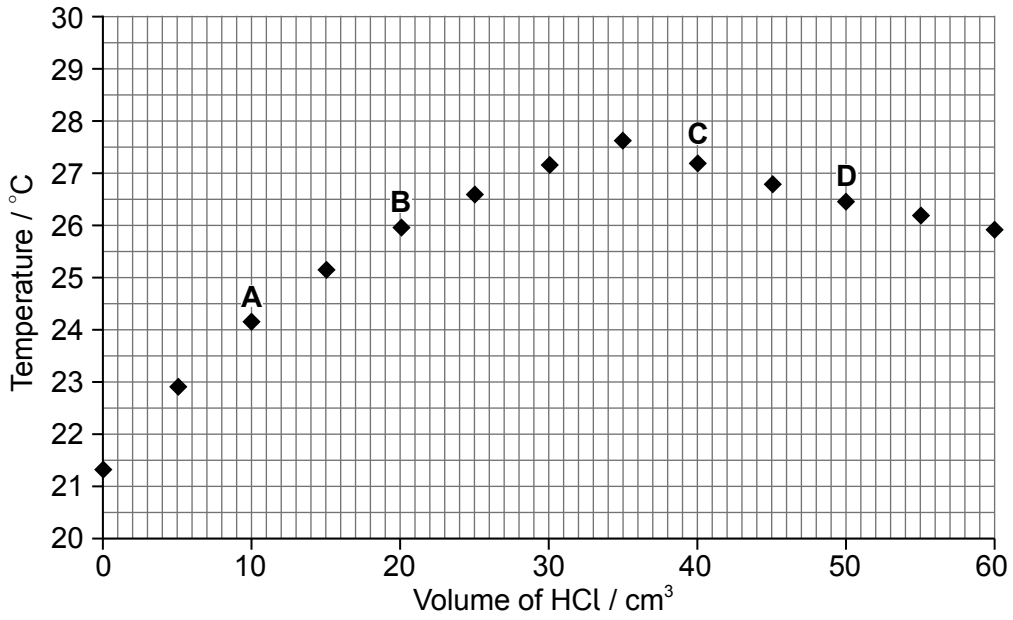
Answer **all** questions. Write your answers in the boxes provided.

A class was determining the concentration of aqueous sodium hydroxide by titrating it with hydrochloric acid, whilst monitoring the pH of the solution. The sodium hydroxide solution was added into a glass beaker from a measuring cylinder and the hydrochloric acid added using a burette. One group of students accidentally used a temperature probe rather than a pH probe. Their results are given below.

Volume of aqueous NaOH = $25.0 \pm 0.5\,\text{cm}^3$

Concentration of HCl = $1.00 \pm 0.01\,\text{mol}\,\text{dm}^{-3}$

Volume HCl $\pm$ 0.1 / cm^3	0.0	5.0	10.0	15.0	20.0	25.0	30.0	35.0	40.0	45.0	50.0	55.0	60.0
Temperature $\pm$ 0.1 / °C	21.3	22.9	24.2	25.1	25.9	26.6	27.2	27.6	27.2	26.8	26.5	26.2	25.9



2. (a) Deduce why more heat was produced in mixture **B** than in mixture **A**. [1]

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(b) Deduce why the temperature is higher in mixture **C** than in mixture **D**. [1]

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8.2-S10 May 2017 TZ1 · Paper 3 · Q3

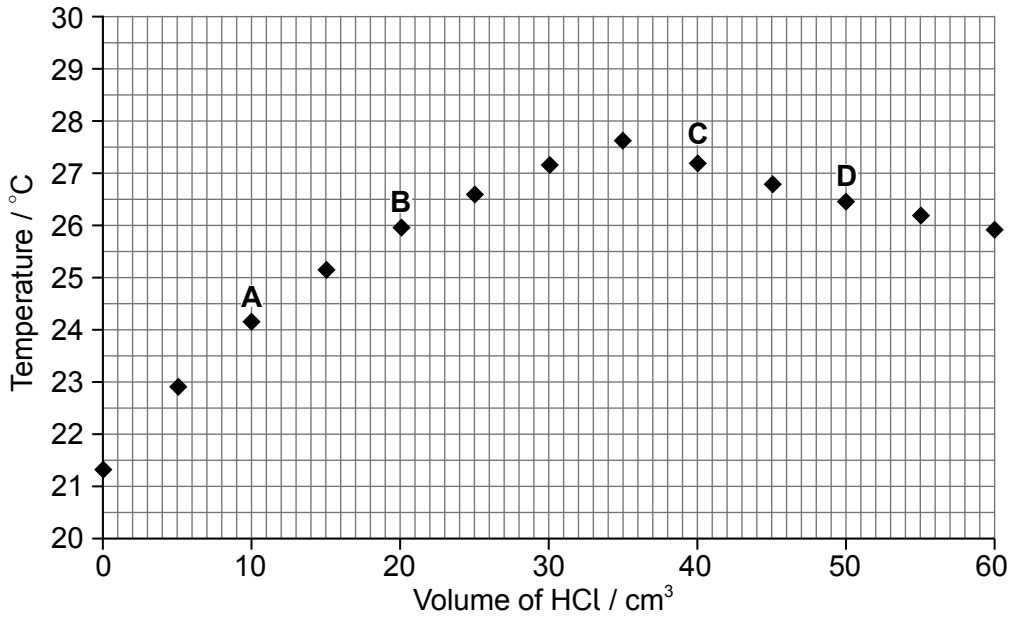
Answer **all** questions. Write your answers in the boxes provided.

A class was determining the concentration of aqueous sodium hydroxide by titrating it with hydrochloric acid, whilst monitoring the pH of the solution. The sodium hydroxide solution was added into a glass beaker from a measuring cylinder and the hydrochloric acid added using a burette. One group of students accidentally used a temperature probe rather than a pH probe. Their results are given below.

Volume of aqueous NaOH = $25.0 \pm 0.5\,\text{cm}^3$

Concentration of HCl = $1.00 \pm 0.01\,\text{mol dm}^{-3}$

Volume HCl $\pm$ 0.1 / cm^3	0.0	5.0	10.0	15.0	20.0	25.0	30.0	35.0	40.0	45.0	50.0	55.0	60.0
Temperature $\pm$ 0.1 / °C	21.3	22.9	24.2	25.1	25.9	26.6	27.2	27.6	27.2	26.8	26.5	26.2	25.9



3. Suggest how the end point of the titration might be estimated from the graph. [1]

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8.2-S11 May 2017 TZ1 · Paper 3 · Q5 parts (a), (b), (c), (d), (f)

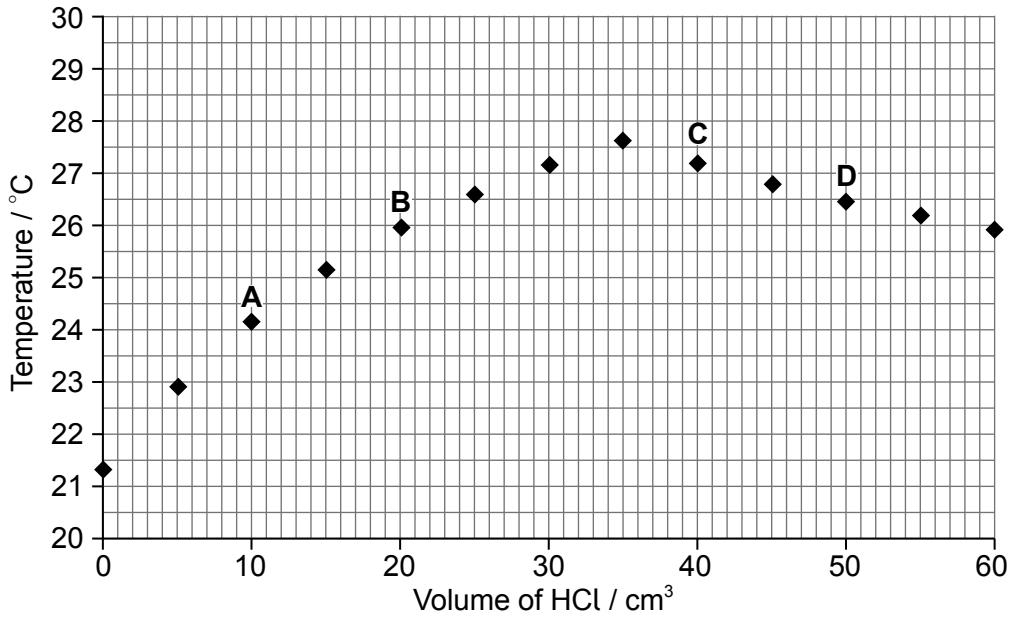
Answer **all** questions. Write your answers in the boxes provided.

A class was determining the concentration of aqueous sodium hydroxide by titrating it with hydrochloric acid, whilst monitoring the pH of the solution. The sodium hydroxide solution was added into a glass beaker from a measuring cylinder and the hydrochloric acid added using a burette. One group of students accidentally used a temperature probe rather than a pH probe. Their results are given below.

Volume of aqueous NaOH = $25.0 \pm 0.5\,cm^3$

Concentration of HCl = $1.00 \pm 0.01\,mol\,dm^{-3}$

Volume HCl $\pm$ 0.1 / cm^3	0.0	5.0	10.0	15.0	20.0	25.0	30.0	35.0	40.0	45.0	50.0	55.0	60.0
Temperature $\pm$ 0.1 / °C	21.3	22.9	24.2	25.1	25.9	26.6	27.2	27.6	27.2	26.8	26.5	26.2	25.9



5. The graph of temperature against titre can be used to calculate the concentration of alkali without knowing the concentration of the hydrochloric acid, using the enthalpy of neutralization.

(a) Explain how the concentration may be calculated in this way. [2]

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(b) Heat losses would make this method less accurate than the pH probe method. Outline why the thermometric method would always give a lower, not a higher, concentration. [2]

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(c) Suggest how heat loss could be reduced. [1]

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(d) State **one** other assumption that is usually made in the calculation of the heat produced. [1]

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(f) Outline why the thermochemical method would not be appropriate for $0.001\,\text{mol}\,\text{dm}^{-3}$ hydrochloric acid and aqueous sodium hydroxide of a similar concentration. [1]

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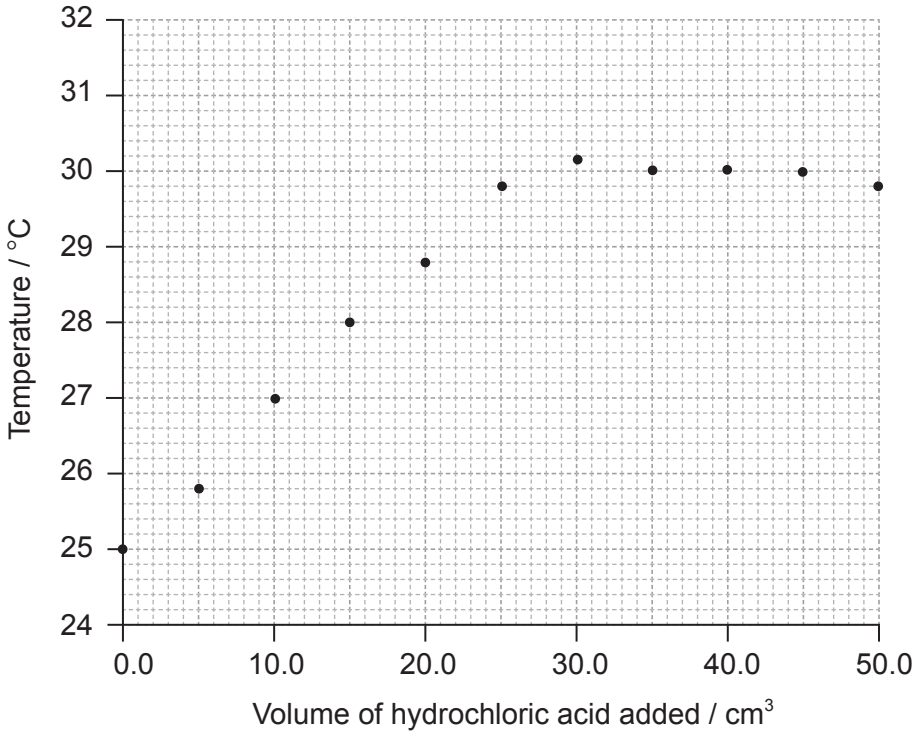
8.2-S12 May 2015 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv) · also 1.2

1. A student carried out an experiment to determine the concentration of a hydrochloric acid solution and the enthalpy change of the reaction between aqueous sodium hydroxide and this acid by thermometric titration.

She added 5.0 cm^3 portions of hydrochloric acid to 25.0 cm^3 of 1.00 $mol\,dm^{-3}$ sodium hydroxide solution in a glass beaker until the total volume of acid added was 50.0 cm^3, measuring the temperature of the mixture each time. Her results are plotted in the graph below.



The initial temperature of both solutions was the same.

(a) (i) By drawing appropriate lines, determine the volume of hydrochloric acid required to completely neutralize the 25.0 cm^3 of sodium hydroxide solution. [2]

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(ii) Determine the concentration of the hydrochloric acid, including units. [2]

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(b) (i) Determine the change in temperature, ΔT. [1]

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(ii) Calculate the enthalpy change, in $kJ\,mol^{-1}$, for the reaction of hydrochloric acid and sodium hydroxide solution. [3]

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(iii) The accepted theoretical value from the literature of this enthalpy change is $-58\,kJ\,mol^{-1}$. Calculate the percentage error correct to **two** significant figures. [1]

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(iv) Suggest the major source of error in the experimental procedure **and** an improvement that could be made to reduce it. [2]

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8.2-S13 May 2014 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(ii), (d)(iii), (d)(iv) · also 1.2, 3.1, 4.1, 4.2, 9.1

1. (a) Two chemistry students wished to determine the enthalpy of hydration of anhydrous magnesium sulfate. They measured the initial and the highest temperature reached when anhydrous magnesium sulfate, $MgSO_4(s)$, was dissolved in water. They presented their results in the table below.

mass of anhydrous magnesium sulfate / g	3.01
volume of water / cm^3	50.0
initial temperature / °C	17.0
highest temperature / °C	26.7

(i) Calculate the amount, in mol, of anhydrous magnesium sulfate. *[1]*

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(ii) Calculate the enthalpy change, ΔH_1, for anhydrous magnesium sulfate dissolving in water, in $kJ\,mol^{-1}$. State your answer to the correct number of significant figures. *[2]*

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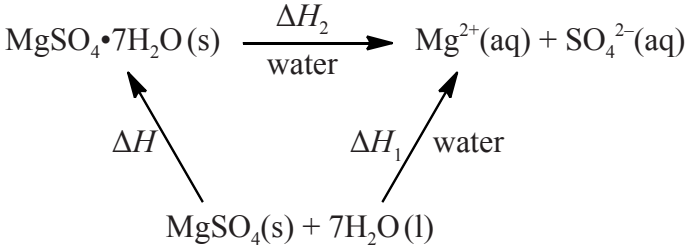
(b) The students repeated the experiment using 6.16 g of solid hydrated magnesium sulfate, $MgSO_4 \cdot 7H_2O(s)$, and $50.0\,cm^3$ of water. They found the enthalpy change, ΔH_2, to be $+18\,kJ\,mol^{-1}$.

The enthalpy of hydration of solid anhydrous magnesium sulfate is difficult to determine experimentally, but can be determined using the diagram below.



(i) Determine the enthalpy change, ΔH, in $kJ\,mol^{-1}$, for the hydration of solid anhydrous magnesium sulfate, $MgSO_4$. *[1]*

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(ii) The literature value for the enthalpy of hydration of anhydrous magnesium sulfate is $-103\,kJ\,mol^{-1}$. Calculate the percentage difference between the literature value and the value determined from experimental results, giving your answer to **one** decimal place. (If you did not obtain an answer for the experimental value in (b)(i) then use the value of $-100\,kJ\,mol^{-1}$, but this is **not** the correct value.) *[1]*

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(c) Another group of students experimentally determined an enthalpy of hydration of $-95\,kJ\,mol^{-1}$. Outline two reasons which may explain the variation between the experimental and literature values. *[2]*

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(d) Magnesium sulfate is one of the products formed when acid rain reacts with dolomitic limestone. This limestone is a mixture of magnesium carbonate and calcium carbonate.

(ii) Deduce the Lewis (electron dot) structure of the carbonate ion, giving the shape and the oxygen-carbon-oxygen bond angle. *[3]*

Lewis (electron dot) structure:

Shape:

. . . . . .

Bond angle:

. . . . . .

(iii) There are three possible Lewis structures that can be drawn for the carbonate ion, which lead to a resonance structure. Explain, with reference to the electrons, why all carbon-oxygen bonds have the same length. *[1]*

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(iv) Deduce the hybridization of the carbon atom in the carbonate ion. *[1]*

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8.2-S14 May 2013 TZ1 · Paper 2 · Q8 parts (a)(i), (a)(ii), (b)(i) · also 9.2

8. To determine the enthalpy change of combustion of methanol, CH_3OH, 0.230 g of methanol was combusted in a spirit burner. The heat released increased the temperature of 50.0 cm^3 of water from 24.5 °C to 45.8 °C.

(a) (i) Calculate the enthalpy change of combustion of methanol. *[4]*

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(ii) Using the theoretical value in Table 12 of the Data Booklet, discuss the experimental result, including **one** improvement that could be made. *[3]*

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(b) Methanol can be produced according to the following equation.

$$CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$$

(i) Calculate the standard enthalpy change of this reaction, using the values of enthalpy of combustion in Table 12 of the Data Booklet. *[3]*

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8.2-S15 Nov 2011 · Paper 3 · QB1

B1. A student carried out an experiment to determine the energy value of 100.00 g of a food product by burning some of it. A 5.00 g sample was burned and the heat produced was used to heat water in a glass beaker.

She recorded the following data:

Mass of water heated = 100.00 g

Initial temperature of water = 19.2 °C

Highest temperature of water = 28.6 °C

Other data:

Heat capacity of the glass beaker = 90.2 J K^{-1}

Specific heat capacity of water = 4.18 J g^{-1} K^{-1}

Calculate the energy value for 100.00 g of the food product, in kJ, showing your working. *[3]*

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8.2-S16 May 2011 TZ1 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 9.2

1. Methanol is made in large quantities as it is used in the production of polymers and in fuels. The enthalpy of combustion of methanol can be determined theoretically or experimentally.

$$CH_3OH(l) + 1\tfrac{1}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

	$CH_3OH(l)$	$O_2(g)$	$CO_2(g)$	$H_2O(g)$
Standard enthalpy of formation, $\Delta H_f^{\ominus}$ / kJ mol^{-1}	–239	0	–394	–242
Entropy, $S^{\ominus}$ /J K^{-1} mol^{-1}	240	205	214	189

(a) Using the information from the table above, determine the theoretical enthalpy of combustion of methanol. *[2]*

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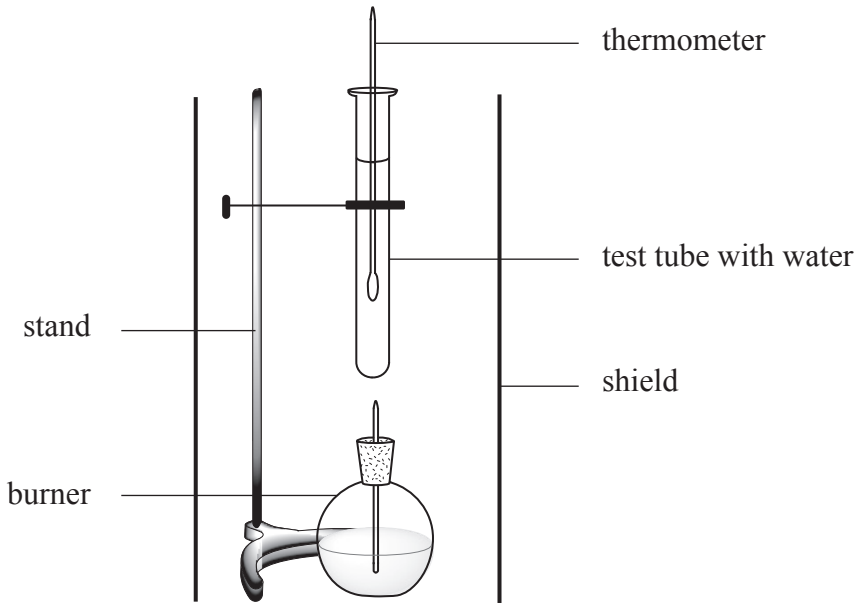
(b) The enthalpy of combustion of methanol can also be determined experimentally in a school laboratory. A burner containing methanol was weighed and used to heat water in a test tube as illustrated below.



The following data were collected.

Initial mass of burner and methanol / g	80.557
Final mass of burner and methanol / g	80.034
Mass of water in test tube / g	20.000
Initial temperature of water / °C	21.5
Final temperature of water / °C	26.4

(i) Calculate the amount, in mol, of methanol burned. *[2]*

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(ii) Calculate the heat absorbed, in kJ, by the water. *[3]*

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(iii) Determine the enthalpy change, in kJ mol^{-1}, for the combustion of methanol. *[2]*

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(c) The Data Booklet value for the enthalpy of combustion of methanol is –726 kJ mol^{-1}. Suggest why this value differs from the values calculated in parts (a) and (b).

(i) Part (a) *[1]*

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. . . . . .

(ii) Part (b) *[1]*

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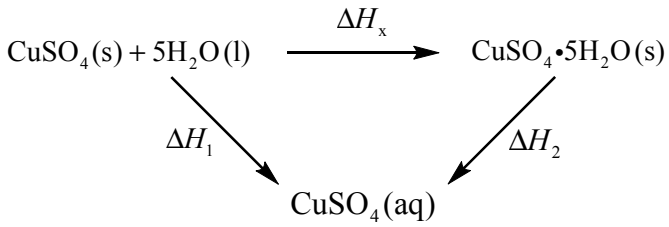
8.2-S17 May 2011 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 1.2, 9.1

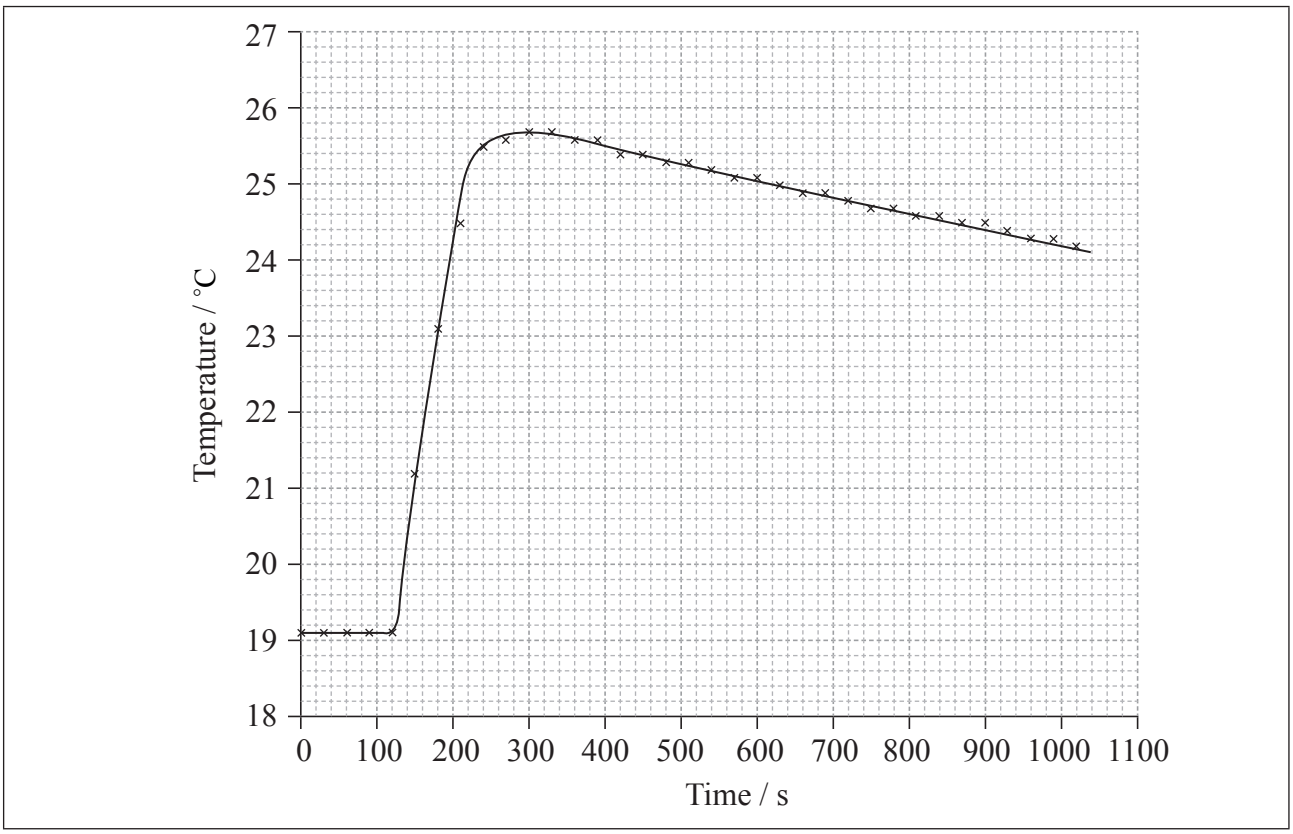
1. If white anhydrous copper(II) sulfate powder is left in the atmosphere it slowly absorbs water vapour giving the blue pentahydrated solid.

$$\underset{\text{(anhydrous)}}{CuSO_4(s)} + 5H_2O(l) \rightarrow \underset{\text{(pentahydrated)}}{CuSO_4 \bullet 5H_2O(s)}$$

It is difficult to measure the enthalpy change for this reaction directly. However, it is possible to measure the heat changes directly when both anhydrous and pentahydrated copper(II) sulfate are separately dissolved in water, and then use an energy cycle to determine the required enthalpy change value, ΔH_x, indirectly.



(a) To determine ΔH_1 a student placed 50.0 g of water in a cup made of expanded polystyrene and used a data logger to measure the temperature. After two minutes she dissolved 3.99 g of anhydrous copper(II) sulfate in the water and continued to record the temperature while continuously stirring. She obtained the following results.



(i) Calculate the amount, in mol, of anhydrous copper(II) sulfate dissolved in the 50.0 g of water. *[1]*

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(ii) Determine what the temperature rise would have been, in °C, if no heat had been lost to the surroundings. *[2]*

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(iii) Calculate the heat change, in kJ, when 3.99 g of anhydrous copper(II) sulfate is dissolved in the water. *[2]*

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(iv) Determine the value of ΔH_1 in kJ mol^{-1}. *[1]*

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(b) To determine ΔH_2, 6.24 g of pentahydrated copper(II) sulfate was dissolved in 47.75 g of water. It was observed that the temperature of the solution decreased by 1.10 °C.

(i) Calculate the amount, in mol, of water in 6.24 g of pentahydrated copper(II) sulfate. *[2]*

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(ii) Determine the value of ΔH_2 in kJ mol^{-1}. *[2]*

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(iii) Using the values obtained for ΔH_1 in (a) (iv) and ΔH_2 in (b) (ii), determine the value for ΔH_x in kJ mol^{-1}. *[1]*

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(c) The magnitude (the value without the + or – sign) found in a data book for ΔH_x is 78.0 kJ mol^{-1}.

(i) Calculate the percentage error obtained in this experiment. (If you did not obtain an answer for the experimental value of ΔH_x then use the value 70.0 kJ mol^{-1}, but this is **not** the true value.) *[1]*

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. . . . . .

(ii) The student recorded in her qualitative data that the anhydrous copper(II) sulfate she used was pale blue rather than completely white. Suggest a reason why it might have had this pale blue colour and deduce how this would have affected the value she obtained for ΔH_x. *[2]*

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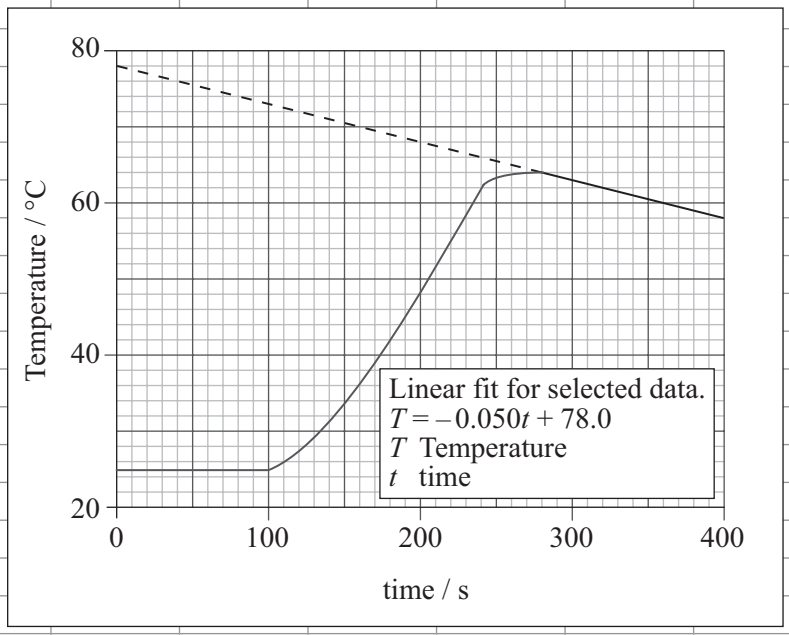
8.2-S18 Nov 2010 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (c), (d) · also 1.2

1. The data below are from an experiment to measure the enthalpy change for the reaction of aqueous copper(II) sulfate, $CuSO_4(aq)$ and zinc, $Zn(s)$.

$$Cu^{2+}(aq) + Zn(s) \rightarrow Cu(s) + Zn^{2+}(aq)$$

50.0 cm^3 of 1.00 $mol\,dm^{-3}$ copper(II) sulfate solution was placed in a polystyrene cup and zinc powder was added after 100 seconds. The temperature-time data was taken from a data-logging software program. The table shows the initial 23 readings.

	A	B	C	D	E	F	G	H
1	**time / s**	**Temperature / °C**						
2	0.0	24.8						
3	1.0	24.8						
4	2.0	24.8						
5	3.0	24.8						
6	4.0	24.8						
7	5.0	24.8						
8	6.0	24.8						
9	7.0	24.8						
10	8.0	24.8						
11	9.0	24.8						
12	10.0	24.8						
13	11.0	24.8						
14	12.0	24.8						
15	13.0	24.8						
16	14.0	24.8						
17	15.0	24.8						
18	16.0	24.8						
19	17.0	24.8						
20	18.0	24.8						
21	19.0	24.8						
22	20.0	24.8						
23	21.0	24.8						
24	22.0	24.8						



A straight line has been drawn through some of the data points. The equation for this line is given by the data logging software as

$$T = -0.050t + 78.0$$

where T is the Temperature at time t.

(a) The heat produced by the reaction can be calculated from the temperature change, ΔT, using the expression below.

$$\text{Heat change} = \text{Volume of } CuSO_4(aq) \times \text{Specific heat capacity of } H_2O \times \Delta T$$

Describe **two** assumptions made in using this expression to calculate heat changes. *[2]*

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(b) (i) Use the data presented by the data logging software to deduce the temperature change, ΔT, which would have occurred if the reaction had taken place instantaneously with no heat loss. *[2]*

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(ii) Calculate the heat, in kJ, produced during the reaction using the expression given in part (a). *[1]*

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(c) The colour of the solution changed from blue to colourless. Deduce the amount, in moles, of zinc which reacted in the polystyrene cup. *[1]*

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(d) Calculate the enthalpy change, in $kJ\,mol^{-1}$, for this reaction. *[1]*

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8.2-S19 Nov 2010 · Paper 3 · QB1 parts (a)

B1. Foods such as rice, bread and potatoes are rich in carbohydrates. There are three main types of carbohydrate – monosaccharides, disaccharides and polysaccharides.

(a) Glucose, $C_6H_{12}O_6$, is a monosaccharide. When 0.85 g of glucose was completely combusted in a calorimeter, the temperature of 200.10 g of water increased from 20.20 °C to 27.55 °C. Calculate the energy value of glucose in J g^{-1}. *[3]*

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8.2-S20 May 2010 TZ2 · Paper 3 · QB1

B1. The energy value of food may be determined using a food calorimeter.

The combustion of 2.00 g of dried bread in a food calorimeter raised the temperature of 600 cm^3 water from 20.5 °C to 29.0 °C. Calculate the energy content of bread in kJ per 100 g. Specific heat capacity of water = 4.18 J g^{-1} K^{-1}. *[4]*

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LESSON 9.1

Hess's Law

16 multiple-choice questions · 2 structured questions

Part A - Multiple choice

9.1-M1 Nov 2024 · Paper 1 · Q16

16. Consider the following equations:

$$H_2(g) + F_2(g) \rightarrow 2HF(g) \qquad \Delta H^{\ominus} = -x\,kJ$$

$$C(s) + 2F_2(g) \rightarrow CF_4(g) \qquad \Delta H^{\ominus} = -y\,kJ$$

$$2C(s) + 2H_2(g) \rightarrow C_2H_4(g) \qquad \Delta H^{\ominus} = +z\,kJ$$

What is the $\Delta H^{\ominus}$ of the reaction?

$$C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$$

A. $-x - y - z$

B. $-x - y + z$

C. $-2x + 2y - z$

D. $-2x - 2y - z$

9.1-M2 Nov 2023 TZ1 · Paper 1 · Q15

15. Which expression represents the calculation used to obtain the $\Delta H^{\ominus}$ value for the conversion of oxygen to one mole of ozone (O_3)?

		$\Delta H^{\ominus}$, kJ
Eqn (i)	$2CO_2 \rightarrow 2CO + O_2$	+566
Eqn (ii)	$3CO + O_3 \rightarrow 3CO_2$	–992

A. $-566 - 992$

B. $-566 + 992$

C. $1.5 \times (-566) + 992$

D. $1.5 \times (-566) - 992$

9.1-M3 May 2023 TZ1 · Paper 1 · Q14

14. What is the enthalpy change for the following reaction?

$$CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$$

Reaction	$\Delta H^{\ominus}$
$2C_{(graphite)} + O_2(g) \rightarrow 2CO(g)$	$-222\,kJ\ mol^{-1}$
$C_{(graphite)} + 2H_2(g) \rightarrow CH_4(g)$	$-74\,kJ\ mol^{-1}$
$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$	$-484\,kJ\ mol^{-1}$

A. $-74 - 242 + 111$

B. $+74 + 242 - 111$

C. $-74 - 484 - 222$

D. $+74 + 484 - 222$

9.1-M4 May 2019 TZ1 · Paper 1 · Q15

15. What is the enthalpy change of reaction for the following equation?

$$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$$

$$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l) \qquad \Delta H = x$$

$$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H = y$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = z$$

A. $x + y + z$

B. $-x - y + z$

C. $x - y - z$

D. $x - y + z$

9.1-M5 Nov 2018 · Paper 1 · Q14

14. Consider the following reactions:

$$Fe_2O_3(s) + CO(g) \rightarrow 2FeO(s) + CO_2(g) \qquad \Delta H^{\ominus} = -3\,kJ$$
$$Fe(s) + CO_2(g) \rightarrow FeO(s) + CO(g) \qquad \Delta H^{\ominus} = +11\,kJ$$

What is the $\Delta H^{\ominus}$ value, in kJ, for the following reaction?

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$$

A. -25

B. -14

C. $+8$

D. $+19$

9.1-M6 May 2018 TZ1 · Paper 1 · Q14

14. What is the enthalpy of combustion of butane in **kJ mol^{-1}**?

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = x\ kJ$$
$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = y\ kJ$$
$$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g) \qquad \Delta H = z\ kJ$$

A. $4x + 5y - z$

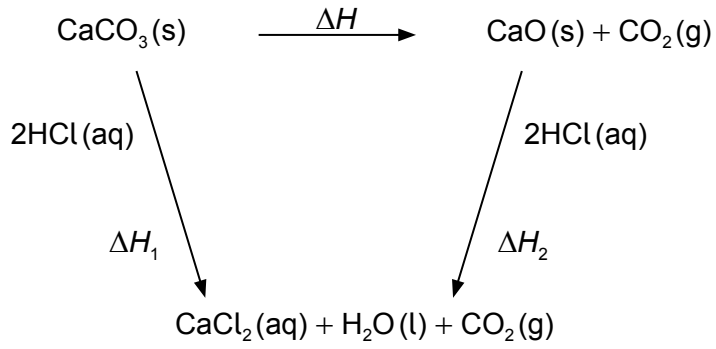
B. $4x + 5y + z$

C. $8x + 10y - 2z$

D. $8x + 5y + 2z$

9.1-M7 May 2017 TZ1 · Paper 1 · Q15

15. Which expression gives the enthalpy change, ΔH, for the thermal decomposition of calcium carbonate?



A. $\Delta H = \Delta H_1 - \Delta H_2$

B. $\Delta H = 2\Delta H_1 - \Delta H_2$

C. $\Delta H = \Delta H_1 - 2\Delta H_2$

D. $\Delta H = \Delta H_1 + \Delta H_2$

9.1-M8 May 2015 TZ2 · Paper 1 · Q14

14. When four moles of aluminium and four moles of iron combine with oxygen to form their oxides, the enthalpy changes are –3338 kJ and –1644 kJ respectively.

$$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s) \qquad \Delta H = -3338\,kJ$$
$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s) \qquad \Delta H = -1644\,kJ$$

What is the enthalpy change, in kJ, for the reduction of one mole of iron(III) oxide by aluminium?

$$Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(s) + Al_2O_3(s)$$

A. +1694

B. +847

C. –847

D. –1694

9.1-M9 Nov 2014 · Paper 1 · Q14

14. Consider the following equations.

$$2\text{Fe}(\text{s}) + 1\tfrac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{Fe}_2\text{O}_3(\text{s}) \qquad \Delta H^{\ominus} = x$$
$$\text{CO}(\text{g}) + \tfrac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \qquad \Delta H^{\ominus} = y$$

What is the enthalpy change of the reaction below?

$$\text{Fe}_2\text{O}_3(\text{s}) + 3\text{CO}(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 2\text{Fe}(\text{s})$$

A. $3y - x$

B. $3y + x$

C. $-3y - x$

D. $-3y + x$

9.1-M10 May 2014 TZ1 · Paper 1 · Q15

15. The enthalpy changes of three reactions are given below.

$$2\text{HCOOH}(\text{l}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \qquad \Delta H = a$$
$$\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l}) \qquad \Delta H = b$$
$$2\text{HCOOC}_2\text{H}_5(\text{l}) + 7\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l}) \qquad \Delta H = c$$

What is the enthalpy change for the following reaction?

$$\text{HCOOH}(\text{l}) + \text{C}_2\text{H}_5\text{OH}(\text{l}) \rightarrow \text{HCOOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$$

A. $a + b + c$

B. $a + 2b - c$

C. $\tfrac{1}{2}a + b + \tfrac{1}{2}c$

D. $\tfrac{1}{2}a + b - \tfrac{1}{2}c$

9.1-M11 Nov 2013 · Paper 1 · Q16

16. Consider the following two equations.

$$2Ca(s) + O_2(g) \rightarrow 2CaO(s) \qquad \Delta H^{\ominus} = +x \text{ kJ}$$
$$Ca(s) + 0.5O_2(g) + CO_2(g) \rightarrow CaCO_3(s) \qquad \Delta H^{\ominus} = +y \text{ kJ}$$

What is $\Delta H^{\ominus}$, in kJ, for the following reaction?

$$CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$$

A. $y - 0.5x$

B. $y - x$

C. $0.5 - y$

D. $x - y$

9.1-M12 May 2013 TZ2 · Paper 1 · Q15

15. Enthalpy changes of reaction are provided for the following reactions.

$$2C(s) + 2H_2(g) \rightarrow C_2H_4(g) \qquad \Delta H^{\ominus} = +52 \text{ kJ mol}^{-1}$$
$$2C(s) + 3H_2(g) \rightarrow C_2H_6(g) \qquad \Delta H^{\ominus} = -85 \text{ kJ mol}^{-1}$$

What is the enthalpy change, in kJ mol^{-1}, for the reaction between ethene and hydrogen?

$$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$$

A. –137

B. –33

C. +33

D. +137

9.1-M13 Nov 2012 · Paper 1 · Q15

15. Using the equations below:

$$\mathrm{C(s) + O_2(g) \rightarrow CO_2(g)} \qquad \Delta H^{\ominus} = -390\ \mathrm{kJ}$$
$$\mathrm{H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)} \qquad \Delta H^{\ominus} = -286\ \mathrm{kJ}$$
$$\mathrm{CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)} \qquad \Delta H^{\ominus} = -890\ \mathrm{kJ}$$

what is $\Delta H^{\ominus}$, in kJ, for the following reaction?

$$\mathrm{C(s) + 2H_2(g) \rightarrow CH_4(g)}$$

A. –214

B. –72

C. +72

D. +214

9.1-M14 May 2011 TZ1 · Paper 1 · Q17

17. Consider the two reactions involving iron and oxygen.

$$\mathrm{2Fe(s) + O_2(g) \rightarrow 2FeO(s)} \qquad \Delta H^{\ominus} = -544\ \mathrm{kJ}$$
$$\mathrm{4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)} \qquad \Delta H^{\ominus} = -1648\ \mathrm{kJ}$$

What is the enthalpy change, in kJ, for the reaction below?

$$\mathrm{4FeO(s) + O_2(g) \rightarrow 2Fe_2O_3(s)}$$

A. –1648 – 2(–544)

B. –544 – (–1648)

C. –1648 – 544

D. –1648 – 2(544)

9.1-M15 Nov 2010 · Paper 1 · Q15

15. Consider the equations below.

$$\mathrm{CH_4(g) + O_2(g) \rightarrow HCHO(l) + H_2O(l)} \qquad \Delta H^{\ominus} = x$$

$$\mathrm{HCHO(l) + \tfrac{1}{2}O_2(g) \rightarrow HCOOH(l)} \qquad \Delta H^{\ominus} = y$$

$$\mathrm{2HCOOH(l) + \tfrac{1}{2}O_2(g) \rightarrow (COOH)_2(s) + H_2O(l)} \qquad \Delta H^{\ominus} = z$$

What is the enthalpy change of the reaction below?

$$\mathrm{2CH_4(g) + 3\tfrac{1}{2}O_2(g) \rightarrow (COOH)_2(s) + 3H_2O(l)}$$

A. $x + y + z$

B. $2x + y + z$

C. $2x + 2y + z$

D. $2x + 2y + 2z$

9.1-M16 May 2010 TZ2 · Paper 1 · Q15

15. The standard enthalpy changes for the combustion of carbon and carbon monoxide are shown below.

$$\mathrm{C(s) + O_2(g) \rightarrow CO_2(g)} \qquad \Delta H_c^{\ominus} = -394\ \mathrm{kJ\,mol^{-1}}$$

$$\mathrm{CO(g) + \tfrac{1}{2}O_2(g) \rightarrow CO_2(g)} \qquad \Delta H_c^{\ominus} = -283\ \mathrm{kJ\,mol^{-1}}$$

What is the standard enthalpy change, in kJ, for the following reaction?

$$\mathrm{C(s) + \tfrac{1}{2}O_2(g) \rightarrow CO(g)}$$

A. –677

B. –111

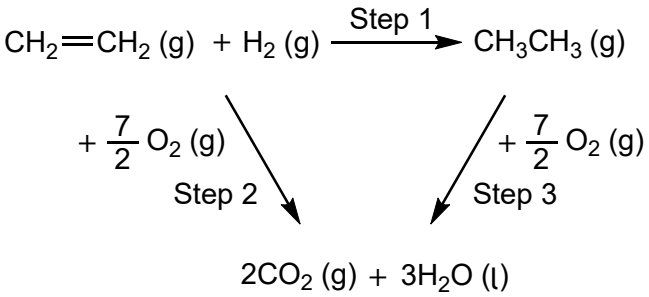
C. +111

D. +677

Part B - Structured questions

9.1-S1 Nov 2018 · Paper 2 · Q7 parts (b), (c), (d) · also 9.2

7. Consider the following Hess's law cycle:



(b) Calculate the standard enthalpy change, $\Delta H^{\ominus}$, of step 2 using section 13 of the data booklet. [1]

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(c) Determine the standard enthalpy change, $\Delta H^{\ominus}$, of step 1. [1]

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(d) Suggest one reason why the calculated value of $\Delta H^{\ominus}$ using Hess's Law in part (c) can be considered accurate and one reason why it can be considered approximate. [2]

Accurate:

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Approximate:

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9.1-S2 May 2017 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii) · also 8.2

9. The Bombardier beetle sprays a mixture of hydroquinone and hydrogen peroxide to fight off predators. The reaction equation to produce the spray can be written as:

$$\underset{\text{hydroquinone}}{C_6H_4(OH)_2(aq)} + H_2O_2(aq) \rightarrow \underset{\text{quinone}}{C_6H_4O_2(aq)} + 2H_2O(l)$$

(a) (i) Calculate the enthalpy change, in kJ, for the spray reaction, using the data below. [2]

$C_6H_4(OH)_2(aq) \rightarrow C_6H_4O_2(aq) + H_2(g)$ $\Delta H^{\ominus} = +177.0\,kJ$

$2H_2O(l) + O_2(g) \rightarrow 2H_2O_2(aq)$ $\Delta H^{\ominus} = +189.2\,kJ$

$H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$ $\Delta H^{\ominus} = +285.5\,kJ$

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(ii) The energy released by the reaction of one mole of hydrogen peroxide with hydroquinone is used to heat 850 cm^3 of water initially at 21.8°C. Determine the highest temperature reached by the water.

Specific heat capacity of water $= 4.18\,kJ\,kg^{-1}K^{-1}$.

(If you did not obtain an answer to part (i), use a value of 200.0 kJ for the energy released, although this is not the correct answer.) [2]

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LESSON 9.2

Enthalpies of Formation & Combustion

21 multiple-choice questions · 13 structured questions

Part A - Multiple choice

9.2-M1 May 2025 TZ1 · Paper 1A · Q22

22. Which equation correctly shows **both** enthalpy of formation, $\Delta H^{\ominus}_{f}$, of a compound and enthalpy of combustion, $\Delta H^{\ominus}_{c}$, of its elements?

A. $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

B. $\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$

C. $2S(s) + 3O_2(g) \rightarrow 2SO_3(g)$

D. $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$

9.2-M2 May 2025 TZ1 · Paper 1A · Q24

24. The table shows enthalpy of combustion data for several substances.

Substance	$\Delta H^{\ominus}_{c}$ (kJ mol^{-1})
C_2H_6	−1561
C_2H_4	−1411
H_2	−286

Which expression would give the enthalpy for the following reaction?

$$C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$$

A. $-1561 + 286 - 1411$

B. $-1411 - 286 + 1561$

C. $-1411 + 286 + 1561$

D. $-1561 - 286 - 1411$

9.2-M3 May 2024 TZ2 · Paper 1 · Q17

17. Standard enthalpy changes of reaction are provided for the following reactions.

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H^{\ominus} = -394\,kJ$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H^{\ominus} = -286\,kJ$$

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H^{\ominus} = -1367\,kJ$$

What is the standard enthalpy change, in kJ, of the following reaction?

$$2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$$

A. $(-394 \times 2) - (286 \times 3) - 1367$

B. $(394 \times 2) + (286 \times 3) + 1367$

C. $(394 \times 2) + (286 \times 3) - 1367$

D. $(-394 \times 2) - (286 \times 3) + 1367$

9.2-M4 May 2023 TZ2 · Paper 1 · Q14

14. The enthalpy of formation of ammonia gas is $-46\,kJ\ mol^{-1}$.

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

What is the energy released, in kJ, in the reaction?

A. 23

B. 46

C. 69

D. 92

9.2-M5 Nov 2022 · Paper 1 · Q14

14. What is the value for enthalpy of formation of methane from the given enthalpies of combustion?

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = -394\,kJ\ mol^{-1}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = -286\,kJ\ mol^{-1}$$

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = -891\,kJ\ mol^{-1}$$

A. $(-394 - 286 - 891)\,kJ\ mol^{-1}$

B. $(-394 - (2 \times 286) - 891)\,kJ\ mol^{-1}$

C. $(-394 - 286 + 891)\,kJ\ mol^{-1}$

D. $(-394 - (2 \times 286) + 891)\,kJ\ mol^{-1}$

9.2-M6 May 2022 TZ1 · Paper 1 · Q15

15. What is the enthalpy change of the following reaction?

$$CH_2CHCH_2CH_3 + HBr \rightarrow CH_3CHBrCH_2CH_3$$

Substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$CH_2CHCH_2CH_3$	0.1
HBr	−36.3
$CH_3CHBrCH_2CH_3$	−156.0

A. −119.6 kJ

B. +119.6 kJ

C. −119.8 kJ

D. +119.8 kJ

9.2-M7 Nov 2021 · Paper 1 · Q15

15. Which equation represents the standard enthalpy of formation of lithium oxide?

A. $4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$

B. $2Li(s) + \frac{1}{2}O_2(g) \rightarrow Li_2O(s)$

C. $Li(s) + \frac{1}{4}O_2(g) \rightarrow \frac{1}{2}Li_2O(s)$

D. $Li(g) + \frac{1}{4}O_2(g) \rightarrow \frac{1}{2}Li_2O(g)$

9.2-M8 May 2021 TZ1 · Paper 1 · Q15

15. What is the enthalpy change of the reaction, in kJ?

$$2C(\text{graphite}) + O_2(g) \rightarrow 2CO(g)$$

Substance	$\Delta H^{\ominus}_{\text{combustion}}$ / kJ mol^{-1}
C (graphite)	−394
CO (g)	−283

A. $-394 - 283$

B. $2(-394) + 2(-283)$

C. $-394 + 283$

D. $2(-394) + 2(283)$

9.2-M9 Nov 2020 · Paper 1 · Q14

14. Which equation shows the enthalpy of formation, ΔH_f, of ethanol?

A. $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(g)$

B. $4C(s) + 6H_2(g) + O_2(g) \rightarrow 2C_2H_5OH(g)$

C. $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$

D. $4C(s) + 6H_2(g) + O_2(g) \rightarrow 2C_2H_5OH(l)$

9.2-M10 Nov 2019 · Paper 1 · Q16

16. What is the enthalpy change of the reaction?

$$C_6H_{14}(l) \rightarrow C_2H_4(g) + C_4H_{10}(g)$$

	Enthalpy of combustion / kJ mol^{-1}
$C_6H_{14}(l)$	–4163
$C_2H_4(g)$	–1411
$C_4H_{10}(g)$	–2878

A. $+ 1411 + 2878 + 4163$

B. $+ 1411 - 2878 - 4163$

C. $+ 1411 + 2878 - 4163$

D. $- 1411 - 2878 + 4163$

9.2-M11 Nov 2017 · Paper 1 · Q16

16. What is the standard enthalpy of formation, in kJ mol^{-1}, of IF (g)?

$$IF_7(g) + I_2(s) \rightarrow IF_5(g) + 2IF(g) \qquad \Delta H^{\ominus} = -89\,kJ$$

$\Delta H_f^{\ominus}(IF_7) = -941\,kJ\,mol^{-1}$
$\Delta H_f^{\ominus}(IF_5) = -840\,kJ\,mol^{-1}$

A. –190

B. –95

C. +6

D. +95

9.2-M12 Nov 2016 · Paper 1 · Q16

16. Hydrazine reacts with oxygen.

$$N_2H_4(l) + O_2(g) \rightarrow N_2(g) + 2H_2O(l) \qquad \Delta H^{\ominus} = -623\,kJ$$

What is the standard enthalpy of formation of $N_2H_4(l)$ in kJ? The standard enthalpy of formation of $H_2O(l)$ is $-286\,kJ$.

A. $-623 - 286$

B. $-623 + 572$

C. $-572 + 623$

D. $-286 + 623$

9.2-M13 May 2016 · Paper 1 · Q16

16. The equation for the formation of ethyne is:

$$2C(s) + H_2(g) \rightarrow C_2H_2(g)$$

What is the enthalpy change, in kJ, for this reaction using the enthalpy of combustion data below?

Reaction	$\Delta H^{\ominus}$ / kJ
$C(s) + O_2(g) \rightarrow CO_2(g)$	−394
$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	−572
$2C_2H_2(g) + 5O_2(g) \rightarrow 4CO_2(g) + 2H_2O(l)$	−2602

A. $2 \times (-394) + \frac{1}{2}(-572) - \frac{1}{2}(-2602)$

B. $2 \times (-394) + (-572) - (-2602)$

C. $2 \times (-394) + \frac{1}{2}(-572) + \frac{1}{2}(-2602)$

D. $2 \times (-394) + (-572) + (-2602)$

9.2-M14 Nov 2015 · Paper 1 · Q15

15. Which change represents the standard enthalpy change of formation?

A. The formation of 1 mol of a compound in its standard state from its gaseous atoms

B. The formation of 1 mol of a compound in its standard state from its elements

C. The formation of 1 mol of a compound in its standard state from its gaseous atoms in their standard states

D. The formation of 1 mol of a compound in its standard state from its elements in their standard states

9.2-M15 May 2015 TZ1 · Paper 1 · Q16

16. Which equation represents the standard enthalpy of formation of liquid methanol?

A. $C(g) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3OH(l)$

B. $C(g) + 4H(g) + O(g) \rightarrow CH_3OH(l)$

C. $C(s) + 4H(g) + O(g) \rightarrow CH_3OH(l)$

D. $C(s) + 2H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3OH(l)$

9.2-M16 May 2015 TZ2 · Paper 1 · Q16

16. Which equation represents the standard enthalpy change of formation, $\Delta H_f^{\ominus}$, of tetrachloromethane?

A. $C(g) + 4Cl(g) \rightarrow CCl_4(g)$

B. $C(s) + 4Cl(g) \rightarrow CCl_4(l)$

C. $C(g) + 2Cl_2(g) \rightarrow CCl_4(g)$

D. $C(s) + 2Cl_2(g) \rightarrow CCl_4(l)$

9.2-M17 May 2013 TZ2 · Paper 1 · Q16

16. Which reaction has an enthalpy change equal to the standard enthalpy change of combustion?

A. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

B. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

C. $2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$

D. $C_5H_{12}(g) + 8O_2(g) \rightarrow 5CO_2(g) + 6H_2O(g)$

9.2-M18 Nov 2012 · Paper 1 · Q17

17. Which is the best definition of the standard state?

A. The standard state of a solid is the most pure form of the solid.

B. The standard state of a solid is the most pure form of the solid at $298\,^{\circ}C$.

C. The standard state of a gas is the most pure form of the gas at $298\,^{\circ}C$.

D. The standard state of a gas is the most pure form of the gas at a pressure of 100 kPa.

9.2-M19 Nov 2011 · Paper 1 · Q16

16. Consider the following enthalpy of combustion data.

$$\mathrm{C(s) + O_2(g) \rightarrow CO_2(g)} \qquad \Delta H^{\ominus} = -x \text{ kJ mol}^{-1}$$

$$\mathrm{H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)} \qquad \Delta H^{\ominus} = -y \text{ kJ mol}^{-1}$$

$$\mathrm{C_2H_6(g) + 3\tfrac{1}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)} \qquad \Delta H^{\ominus} = -z \text{ kJ mol}^{-1}$$

What is the enthalpy of formation of ethane in kJ mol^{-1}?

$$\mathrm{2C(s) + 3H_2(g) \rightarrow C_2H_6(g)}$$

A. $[(-x) + (-y)] - (-z)$

B. $(-z) - [(-x) + (-y)]$

C. $[(-2x) + (-3y)] - (-z)$

D. $(-z) - [(-2x) + (-3y)]$

9.2-M20 Nov 2010 · Paper 1 · Q16

16. Given the enthalpy change for the reaction below:

$$\mathrm{2H_2(g) + O_2(g) \rightarrow 2H_2O(l)} \qquad \Delta H^{\ominus} = -572 \text{ kJ}$$

which statement is correct?

A. The standard enthalpy change of combustion of $\mathrm{H_2(g)}$ is –286 kJ mol^{-1}.

B. The standard enthalpy change of combustion of $\mathrm{H_2(g)}$ is +286 kJ mol^{-1}.

C. The standard enthalpy change of formation of $\mathrm{H_2O(l)}$ is –572 kJ mol^{-1}.

D. The standard enthalpy change of formation of $\mathrm{H_2O(l)}$ is +572 kJ mol^{-1}.

9.2-M21 May 2010 TZ1 · Paper 1 · Q16

16. Which reaction has an enthalpy change equal to a standard enthalpy change of formation, $\Delta H_f^{\ominus}$? All reactions occur at 298 K and 1.01×10^5 Pa.

A. $\mathrm{C_4H_8(g) + H_2O(g) \rightarrow C_4H_9OH(l)}$

B. $\mathrm{4CO_2(g) + 5H_2O(g) \rightarrow C_4H_9OH(l) + \tfrac{13}{2}O_2(g)}$

C. $\mathrm{4C(s) + 5H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow C_4H_9OH(l)}$

D. $\mathrm{8C(s) + 10H_2(g) + O_2(g) \rightarrow 2C_4H_9OH(l)}$

Part B - Structured questions

9.2-S1 May 2025 TZ3 · Paper 2 · Q5 parts (a)(i), (c)(i), (c)(ii) · also 3.1, 4.2

5. Ethyne, C_2H_2, is a useful raw material in the synthesis of organic compounds.

(a) (i) Calculate the standard enthalpy change, in $kJ\,mol^{-1}$, of the complete hydrogenation of ethyne to produce ethane. Use section 14 of the data booklet. [2]

$$C_2H_2(g) + 2\,H_2(g) \rightarrow C_2H_6(g)$$

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(c) (i) Draw the Lewis formula of C_2H_2. [1]

(ii) State the hybridization of the carbon atom in C_2H_2. [1]

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9.2-S2 Nov 2023 TZ1 · Paper 2 · Q4 parts (a)(i), (b) · also 3.1

4. Carbon disulfide, CS_2, undergoes gas phase hydrolysis according to the overall equation

$$CS_2(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 2H_2S(g)$$

(a) (i) Calculate the enthalpy change in this reaction from section 12 of the data booklet and the given values: [2]

	$CS_2(g)$	$H_2S(g)$
$\Delta H_f^\ominus$	$+88.7\,kJ\,mol^{-1}$	$-20.6\,kJ\,mol^{-1}$

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(b) Deduce the molecular geometries of CS_2 and H_2S, and the reason why they are different. [2]

Molecular geometry CS_2: . . . . . .

Molecular geometry H_2S: . . . . . .

Reason for difference: . . . . . .

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9.2-S3 May 2023 TZ1 · Paper 3 · Q7 parts (a)(ii), (d)(i) · also 7.3

7. Lipids and carbohydrates play an essential role in the body.

(ii) Fructose is a carbohydrate. Determine the energy, in kJ, released by the respiration of 10.5 g of fructose, $C_6H_{12}O_6$. [2]

Enthalpy of ($\Delta H_c^\ominus$) of fructose = $-2810\,kJ\,mol^{-1}$

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(d) Lipids and carbohydrates both release energy in the body.

(i) Write balanced equations for the complete oxidation of fructose, $C_6H_{12}O_6$, and the triglyceride tristearin, $C_{57}H_{110}O_6$. [2]

$C_6H_{12}O_6$: . . . . . .

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$C_{57}H_{110}O_6$: . . . . . .

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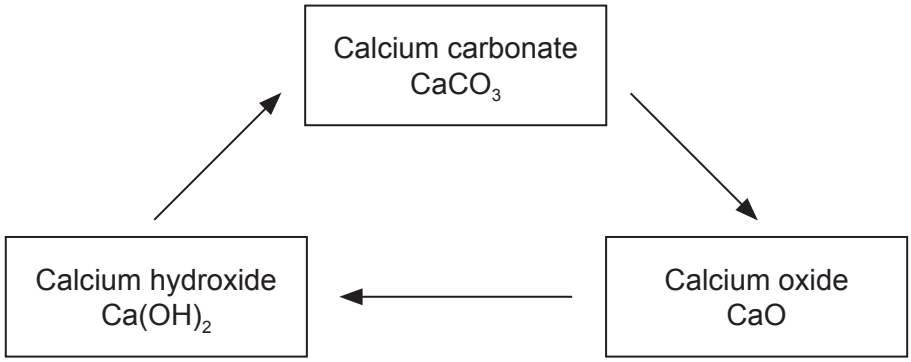
9.2-S4 May 2021 TZ2 · Paper 2 · Q1 parts (b)(i), (b)(iv), (d)(ii) · also 1.2, 8.1

1. Limestone can be converted into a variety of useful commercial products through the lime cycle. Limestone contains high percentages of calcium carbonate, $CaCO_3$.



(b) Thermodynamic data for the decomposition of calcium carbonate is given.

Substance	$\Delta H_f^\ominus$ / kJ mol^{-1}	$S^\ominus$ / J K^{-1} mol^{-1}
$CaCO_3$(s)	−1207	93
CaO(s)	−635	40
CO_2(g)	−393.5	214

(i) Calculate the enthalpy change of reaction, ΔH, in kJ, for the decomposition of calcium carbonate. [2]

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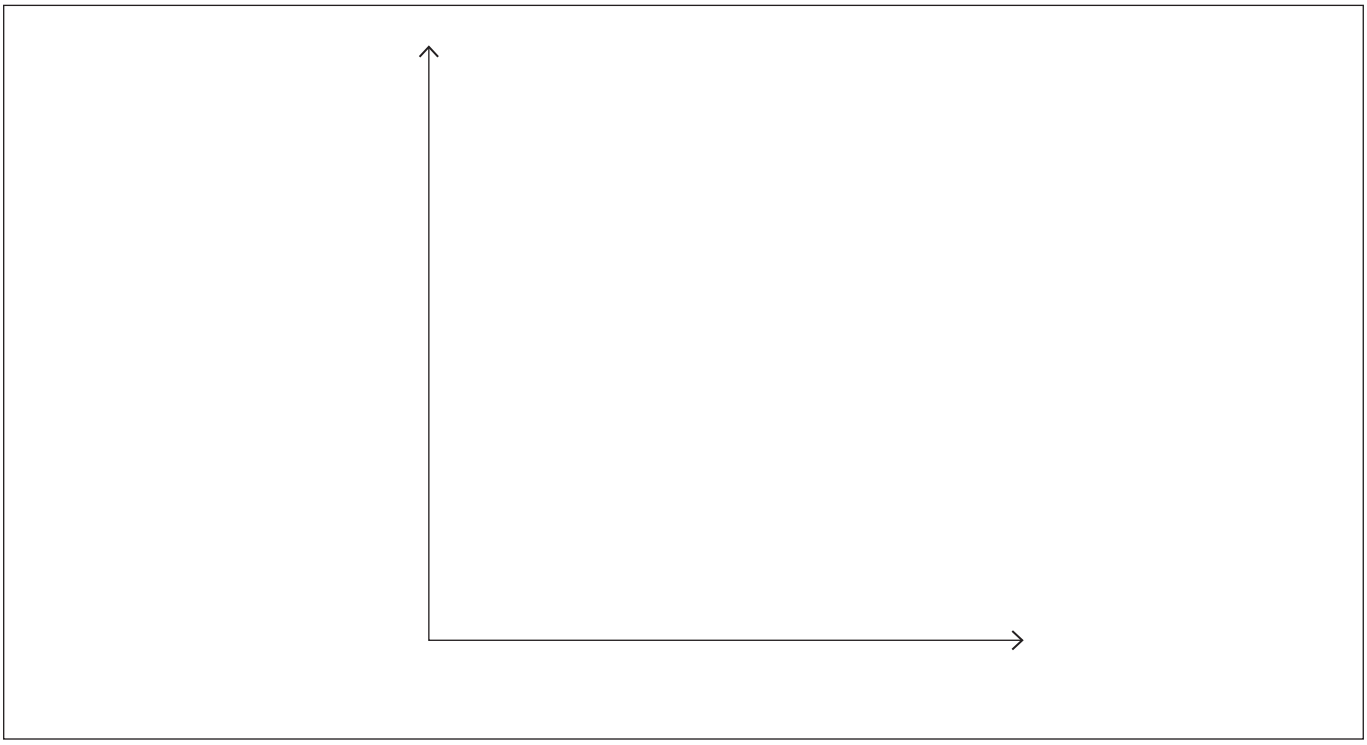
(iv) Sketch an energy profile for the decomposition of calcium carbonate based on your answer to b(i), labelling the axes and activation energy, E_a. [3]



(d) Calcium hydroxide reacts with carbon dioxide to reform calcium carbonate.

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

(ii) 2.85 g of $CaCO_3$ was collected in the experiment in d(i). Calculate the percentage yield of $CaCO_3$.

(If you did not obtain an answer to d(i), use 4.00 g, but this is not the correct value.) [1]

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9.2-S5 May 2018 TZ2 · Paper 3 · Q7 parts (e)

7. Lipids provide energy and are an important part of a balanced diet.

(e) Determine, to the correct number of significant figures, the energy produced by the respiration of 29.9 g of $C_5H_{10}O_5$.

$\Delta H_c\,(C_5H_{10}O_5) = 205.9\,\text{kJ mol}^{-1}$ [2]

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9.2-S6 Nov 2017 · Paper 2 · Q5 parts (a)

5. Ethane-1,2-diol, $HOCH_2CH_2OH$, reacts with thionyl chloride, $SOCl_2$, according to the reaction below.

$$HOCH_2CH_2OH\,(l) + 2SOCl_2\,(l) \rightarrow ClCH_2CH_2Cl\,(l) + 2SO_2\,(g) + 2HCl\,(g)$$

(a) Calculate the standard enthalpy change for this reaction using the following data. [2]

	$HOCH_2CH_2OH\,(l)$	$SOCl_2\,(l)$	$ClCH_2CH_2Cl\,(l)$	$SO_2\,(g)$	$HCl\,(g)$
$\Delta H_f^{\ominus}$ / kJ mol^{-1}	−454.7	−245.7	−165.2	−296.9	−92.3

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9.2-S7 May 2017 TZ2 · Paper 2 · Q6 parts (b)

6. The photochemical chlorination of methane can occur at low temperature.

(b) The overall equation for monochlorination of methane is:

$$CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$$

Calculate the standard enthalpy change for the reaction, $\Delta H^{\ominus}$, using section 12 of the data booklet. [2]

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9.2-S8 May 2016 · Paper 2 · Q2 parts (a)(iii)

2. Phosgene, $COCl_2$, is usually produced by the reaction between carbon monoxide and chlorine according to the equation:

$$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

(iii) The standard enthalpy change of formation of phosgene, $\Delta H_f^{\ominus}$, is $-220.1\,kJ\,mol^{-1}$. Determine the standard enthalpy change, $\Delta H^{\ominus}$, for the forward reaction of the equilibrium, in kJ, using section 12 of the data booklet. [2]

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9.2-S9 May 2015 TZ2 · Paper 2 · Q3 parts (a)

3. Carbon monoxide reacts with hydrogen to produce methanol.

$$CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$$

Substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}	$\Delta G_f^{\ominus}$ / kJ mol^{-1}	$S^{\ominus}$ / J mol^{-1} K^{-1}
CO (g)	– 110.5	– 137.2	+ 197.6
CH_3OH (l)	– 239.0	– 166.0	+ 126.8

(a) Calculate the standard enthalpy change, $\Delta H^{\ominus}$, in kJ mol^{-1}, for the reaction. [1]

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9.2-S10 May 2014 TZ1 · Paper 2 · Q4 parts (b)

4. Buta-1,3-diene can be hydrogenated to produce butane, according to the reaction below.

$$C_4H_6(g) + 2H_2(g) \rightarrow C_4H_{10}(g)$$

(b) Determine the standard enthalpy change of reaction, $\Delta H^{\ominus}$, in kJ mol^{-1}, at 298 K for the hydrogenation reaction, using Table 11 of the Data Booklet. *[1]*

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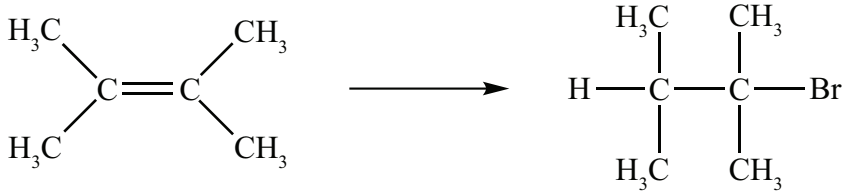
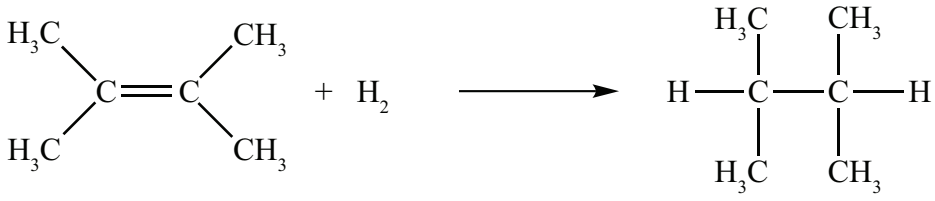
9.2-S11 May 2014 TZ2 · Paper 2 · Q7 parts (e)(iv), (e)(vi), (e)(vii) · also 5.2, 8.2

7. Alkenes, such as **A** (shown below), are important intermediates in the petrochemical industry because they undergo addition reactions to produce a wide variety of products, such as the conversion shown below.



A **B**

(e) In the gas phase, **A** reacts with hydrogen to form **D**:



A **D**

(iv) Given that the enthalpy change of formation of compounds **A** and **D**, in the gas phase, is $-68\,kJ\,mol^{-1}$ and $-178\,kJ\,mol^{-1}$, respectively, calculate the enthalpy change for the reaction of **A** with hydrogen. *[2]*

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(vi) The standard enthalpy change of combustion of **A** is $-4000\,kJ\,mol^{-1}$. Calculate the amount of **A**, in mol, that would have to be burned to raise the temperature of 1 dm^3 of water from 20 °C to 100 °C. *[2]*

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(vii) Compound **D** is isomeric with hexane. Predict, giving reasons, how the boiling points of these compounds would compare. *[2]*

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9.2-S12 Nov 2011 · Paper 2 · Q5 parts (b)(i), (b)(ii)

(b) Consider the reaction:

$$CuS(s) + H_2(g) \rightarrow Cu(s) + H_2S(g)$$

Given:

	$H_2S(g)$	$CuS(s)$
$\Delta G_f^{\ominus}$ / kJ mol^{-1}	–33.6	–53.6
$\Delta H_f^{\ominus}$ / kJ mol^{-1}	–20.6	–53.1

(i) Suggest why the $\Delta H_f^{\ominus}$ values for $H_2(g)$ and $Cu(s)$ are not given in the table. *[1]*

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(ii) Determine the standard enthalpy change at 298 K for the reaction. *[1]*

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9.2-S13 May 2011 TZ2 · Paper 2 · Q6 parts (b)(i), (b)(ii)

(b) Propene can be hydrogenated in the presence of a nickel catalyst to form propane. Use the data below to answer the questions that follow.

Compound	Formula	$\Delta H^{\ominus}_f$ / kJ mol^{-1}	$S^{\ominus}$ / J K^{-1} mol^{-1}
hydrogen	$H_2(g)$	0	+131
propane	$C_3H_8(g)$	−104	+270
propene	$C_3H_6(g)$	+20.4	+267

(i) Outline why the value for the standard enthalpy change of formation of hydrogen is zero. *[1]*

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(ii) Calculate the standard enthalpy change for the hydrogenation of propene. *[2]*

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LESSON 9.3

Bond Enthalpies

8 multiple-choice questions · 15 structured questions

Part A - Multiple choice

9.3-M1 May 2024 TZ1 · Paper 1 · Q15

15. What is the correct formula for calculating enthalpy change of reaction, $\Delta H^{\ominus}$ reaction?

A. $\Delta H^{\ominus}$ reaction $= \Sigma \Delta H_f^{\ominus}$ reactants $- \Sigma \Delta H_f^{\ominus}$ products

B. $\Delta H^{\ominus}$ reaction $= \Sigma \Delta H_c^{\ominus}$ products $- \Sigma \Delta H_c^{\ominus}$ reactants

C. $\Delta H^{\ominus}$ reaction $= \Sigma \Delta H_{bonds}^{\ominus}$ reactants $- \Sigma \Delta H_{bonds}^{\ominus}$ products

D. $\Delta H^{\ominus}$ reaction $= \Sigma \Delta H_{bonds}^{\ominus}$ products $- \Sigma \Delta H_{bonds}^{\ominus}$ reactants

9.3-M2 May 2022 TZ2 · Paper 1 · Q16

16. Which equation represents the bond enthalpy for H–Br in hydrogen bromide?

A. $HBr(g) \rightarrow H^{+}(g) + Br^{-}(g)$

B. $HBr(g) \rightarrow H(g) + Br(g)$

C. $HBr(g) \rightarrow \frac{1}{2}H_2(g) + \frac{1}{2}Br_2(l)$

D. $HBr(g) \rightarrow \frac{1}{2}H_2(g) + \frac{1}{2}Br_2(g)$

9.3-M3 Nov 2021 · Paper 1 · Q14

14. Which combustion reaction releases the **least** energy per mole of C_3H_8?

	Approximate bond enthalpy / kJ mol^{-1}
O=O	500
C=O	800
C≡O	1000

A. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

B. $C_3H_8(g) + \frac{9}{2}O_2(g) \rightarrow 2CO_2(g) + CO(g) + 4H_2O(g)$

C. $C_3H_8(g) + 4O_2(g) \rightarrow CO_2(g) + 2CO(g) + 4H_2O(g)$

D. $C_3H_8(g) + \frac{7}{2}O_2(g) \rightarrow 3CO(g) + 4H_2O(g)$

9.3-M4 May 2019 TZ2 · Paper 1 · Q14

14. Methane undergoes incomplete combustion.

$$2CH_4(g) + 3O_2(g) \rightarrow 2CO(g) + 4H_2O(g)$$

What is the enthalpy change, in kJ, using the bond enthalpy data given below?

Bond	Average bond enthalpy / kJ mol^{-1}
C–H	414
O–H	463
O=O	498
C≡O	1077

A. $[2(1077) + 4(463)] - [2(414) + 3(498)]$

B. $[2(414) + 3(498)] - [2(1077) + 4(463)]$

C. $[8(414) + 3(498)] - [2(1077) + 8(463)]$

D. $[2(1077) + 8(463)] - [8(414) + 3(498)]$

9.3-M5 May 2016 · Paper 1 · Q17

17. Which equation represents the average bond enthalpy of the Si–H bond in SiH_4?

A. $SiH_4(g) \rightarrow SiH_3(g) + H(g)$

B. $\frac{1}{4}SiH_4(g) \rightarrow \frac{1}{4}Si(g) + H(g)$

C. $SiH_4(g) \rightarrow SiH_3(g) + \frac{1}{2}H_2(g)$

D. $SiH_4(g) \rightarrow Si(g) + 4H(g)$

9.3-M6 May 2015 TZ2 · Paper 1 · Q15

15. Which enthalpy changes can be calculated using **only** bond enthalpy data?

I. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

II. $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

III. $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

9.3-M7 Nov 2014 · Paper 1 · Q15

15. Consider the following bond enthalpy data.

Bond	Bond enthalpy / kJ mol^{-1}
H–H	436
Cl–Cl	243
H–Cl	432

What is the enthalpy change, in kJ mol^{-1}, of this reaction?

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

A. +247

B. –247

C. –185

D. +185

9.3-M8 May 2011 TZ2 · Paper 1 · Q17

17. Which equation represents the bond enthalpy for the H–Br bond in hydrogen bromide?

A. $HBr(g) \rightarrow H(g) + Br(g)$

B. $HBr(g) \rightarrow H(g) + Br(l)$

C. $HBr(g) \rightarrow H(g) + \frac{1}{2}Br_2(l)$

D. $HBr(g) \rightarrow H(g) + \frac{1}{2}Br_2(g)$

Part B - Structured questions

9.3-S1 May 2025 TZ1 · Paper 2 · Q8 parts (c)(v), (f)(i), (f)(ii), (f)(iii) · also 5.2, 9.2

8. Organic compounds have many industrial applications.

(c) An alkene such as ethene can be used as starting material for a range of compounds.

(v) Explain, in terms of the intermolecular forces present, the trend in the boiling points of the first four alkenes. [2]

Alkene	Boiling point / K
ethene	169
propene	225
but-1-ene	267
pent-1-ene	303

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(f) Ethene reacts with steam to produce ethanol.

$$C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(g)$$

(i) Calculate the enthalpy, in kJ, of the reaction using section 12 of the data booklet. [3]

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(ii) Calculate the enthalpy of the reaction, in kJ. Use section 13 of the data booklet and $\Delta H_f^{\ominus}$ of $CH_3CH_2OH(g) = -235\,kJ\,mol^{-1}$. [2]

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(iii) Outline why the enthalpies calculated in (i) and (ii) are different. [1]

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9.3-S2 May 2025 TZ2 · Paper 2 · Q3 parts (a)(iii), (a)(iv) · also 8.1

3. Phosgene (carbonyl dichloride, Cl_2CO) is an important industrial intermediate.

(a) Phosgene may be formed by the free-radical reaction between carbon monoxide, CO, and chlorine, Cl_2, which is initiated by UV light.

(iii) Determine the enthalpy change for the reaction between carbon monoxide and chlorine from bond enthalpies. Use section 12 of the data booklet. [3]

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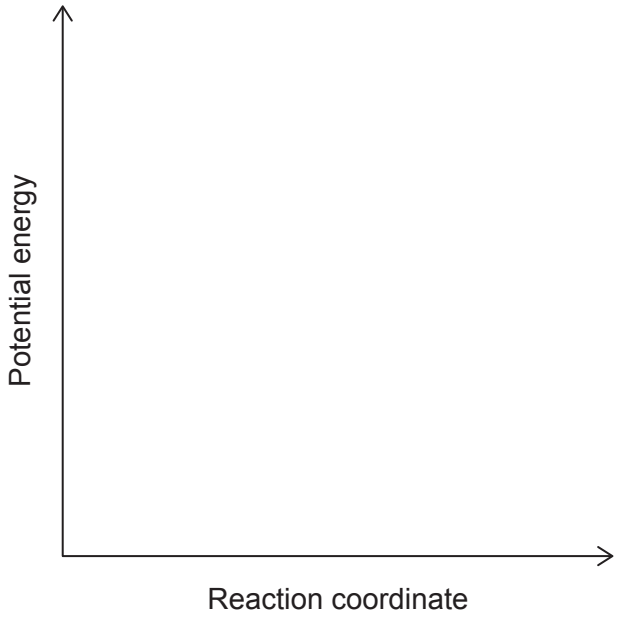
(iv) Sketch an energy profile for this reaction and label the “Reactants”, “Products” and “ΔH”. [2]



9.3-S3 Nov 2023 TZ1 · Paper 2 · Q2 parts (c)(i), (c)(iii), (g) · also 6.1

2. Methanoic acid can be produced by the hydrogenation of carbon dioxide according to the equilibrium

$$CO_2(g) + H_2(g) \rightleftharpoons HCOOH(g)$$

(c) Bond enthalpies are a useful way of finding approximate enthalpy changes for reactions.

(i) Determine the enthalpy change, $\Delta H^{\ominus}$, of this reaction, using section 11 of the data booklet. [3]

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(iii) Bond enthalpies are usually only approximate values. Identify which of the bond enthalpies you have just used is actually an exact value, and give a reason for your choice. [1]

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(g) Determine the oxidation state of carbon in methanoic acid. [1]

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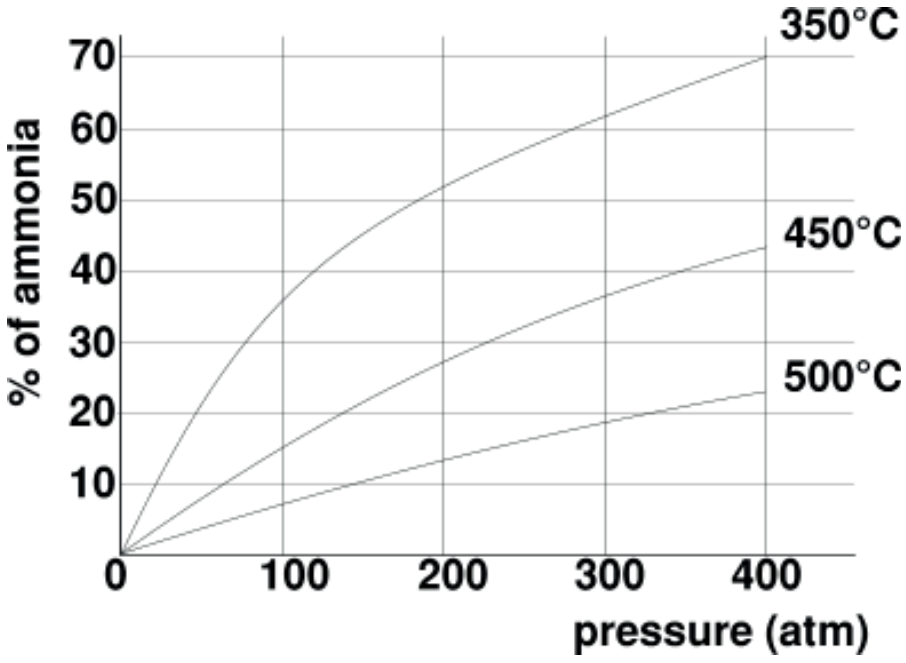
9.3-S4 May 2022 TZ1 · Paper 2 · Q3 parts (b)(i), (b)(ii)

3. Ammonia is produced by the Haber–Bosch process which involves the equilibrium:

$$N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$$

The percentage of ammonia at equilibrium under various conditions is shown:



(b) One factor affecting the position of equilibrium is the enthalpy change of the reaction.

(i) Determine the enthalpy change, ΔH, for the Haber–Bosch process, in kJ. Use Section 11 of the data booklet. [3]

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(ii) Outline why the value obtained in (b)(i) might differ from a value calculated using ΔH_f data. [1]

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9.3-S5 May 2021 TZ1 · Paper 2 · Q4 parts (c)(i), (d)(i), (d)(ii) · also 1.2

4. Hydrogen peroxide can react with methane and oxygen to form methanol. This reaction can occur below 50°C if a gold nanoparticle catalyst is used.

(c) Methanol is usually manufactured from methane in a two-stage process.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$
$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(l)$$

(i) Determine the overall equation for the production of methanol. [1]

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(d) Consider the first stage of the reaction.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$$

(i) Determine the enthalpy change, ΔH, in kJ. Use section 11 of the data booklet.

Bond enthalpy of CO = $1077\,kJ\,mol^{-1}$. [3]

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(ii) State **one** reason why you would expect the value of ΔH calculated from the $\Delta H_f^{\ominus}$ values, given in section 12 of data booklet, to differ from your answer to (d)(i). [1]

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9.3-S6 May 2021 TZ2 · Paper 2 · Q4 parts (b), (c), (d) · also 4.1, 4.2

4. Organic chemistry can be used to synthesize a variety of products.

(b) Determine the change in enthalpy, ΔH, for the combustion of but-2-ene, using section 11 of the data booklet. [3]

$$CH_3CH{=}CHCH_3(g) + 6O_2(g) \rightarrow 4CO_2(g) + 4H_2O(g)$$

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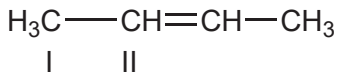
(c) State the hybridization of the carbon I and II atoms in but-2-ene. [1]



Carbon	**I**	**II**
Hybridization	. . . . .	. . . . .

(d) Draw diagrams to show how sigma (σ) and pi (π) bonds are formed between atoms. [2]

Sigma (σ):

Pi (π):

9.3-S7 Nov 2020 · Paper 2 · Q3 parts (a), (b) · also 9.2

3. An equation for the combustion of propane is given below.

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$$

(a) Determine the standard enthalpy change, $\Delta H^{\ominus}$, for this reaction, using section 11 of the data booklet. [3]

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(b) Calculate the standard enthalpy change, $\Delta H^{\ominus}$, for this reaction using section 12 of the data booklet. [2]

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9.3-S8 May 2018 TZ1 · Paper 2 · Q3 parts (c)(i), (c)(ii), (c)(iii), (d) · also 4.1, 9.2

3. This question is about ethene, C_2H_4, and ethyne, C_2H_2.

(c) (i) Under certain conditions, ethyne can be converted to benzene.

Determine the standard enthalpy change, $\Delta H^{\ominus}$, for the reaction stated, using section 11 of the data booklet. [2]

$$3C_2H_2(g) \rightarrow C_6H_6(g)$$

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(ii) Determine the standard enthalpy change, $\Delta H^{\ominus}$, for the following similar reaction, using ΔH_f values in section 12 of the data booklet. [2]

$$3C_2H_2(g) \rightarrow C_6H_6(l)$$

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(iii) Explain, giving two reasons, the difference in the values for (c)(i) and (ii). If you did not obtain answers, use −475 kJ for (i) and −600 kJ for (ii). [2]

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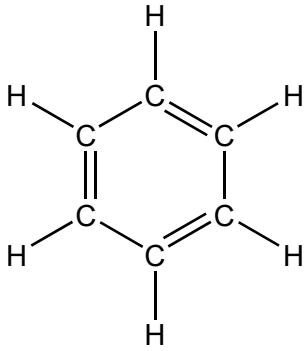
(d) One possible Lewis structure for benzene is shown.



State one piece of physical evidence that this structure is **incorrect**. [1]

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9.3-S9 May 2018 TZ2 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii) · also 9.2

5. Enthalpy changes depend on the number and type of bonds broken and formed.

(a) Hydrogen gas can be formed industrially by the reaction of natural gas with steam.

$$CH_4(g) + H_2O(g) \rightarrow 3H_2(g) + CO(g)$$

Determine the enthalpy change, ΔH, for the reaction, in kJ, using section 11 of the data booklet.

Bond enthalpy for C≡O: 1077 kJ mol^{-1} [3]

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(b) The table lists the standard enthalpies of formation, $\Delta H_f^{\ominus}$, for some of the species in the reaction above.

	$CH_4(g)$	$H_2O(g)$	CO (g)	$H_2(g)$
$\Delta H_f^{\ominus}$ / kJ mol^{-1}	–74.0	–242	–111	

(i) Outline why no value is listed for $H_2(g)$. [1]

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(ii) Determine the value of $\Delta H^{\ominus}$, in kJ, for the reaction using the values in the table. [1]

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(iii) Outline why the value of enthalpy of reaction calculated from bond enthalpies is less accurate. [1]

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9.3-S10 Nov 2016 · Paper 2 · Q1 parts (a)(iii), (b)(i), (b)(ii), (c), (d) · also 5.2, 6.1, 9.2

1. Ethane-1,2-diol, $HOCH_2CH_2OH$, has a wide variety of uses including the removal of ice from aircraft and heat transfer in a solar cell.

(a) Ethane-1,2-diol can be formed according to the following reaction.

$$2CO(g) + 3H_2(g) \rightleftharpoons HOCH_2CH_2OH(g)$$

(iii) Calculate the enthalpy change, $\Delta H^{\ominus}$, in kJ, for this reaction using section 11 of the data booklet. The bond enthalpy of the carbon–oxygen bond in $CO(g)$ is $1077\,kJ\,mol^{-1}$. [3]

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(b) (i) Calculate $\Delta H^{\ominus}$, in kJ, for this similar reaction below using $\Delta H_f^{\ominus}$ data from section 12 of the data booklet. $\Delta H_f^{\ominus}$ of $HOCH_2CH_2OH(l)$ is $-454.8\,kJ\,mol^{-1}$. [1]

$$2CO(g) + 3H_2(g) \rightleftharpoons HOCH_2CH_2OH(l)$$

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(ii) Deduce why the answers to (a)(iii) and (b)(i) differ. [1]

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(c) Determine the average oxidation state of carbon in ethene and in ethane-1,2-diol. [2]

Ethene:

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Ethane-1,2-diol:

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(d) Explain why the boiling point of ethane-1,2-diol is significantly greater than that of ethene. [2]

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9.3-S11 May 2015 TZ1 · Paper 2 · Q5 parts (b)(i), (b)(ii), (c), (g) · also 4.3, 8.2

5. Ethanol has many industrial uses.

(b) (i) Define the term *average bond enthalpy*. [2]

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(ii) Ethanol can be used as a fuel. Determine the enthalpy of combustion of ethanol at 298 K, in $kJ\,mol^{-1}$, using the values in table 10 of the data booklet, assuming all reactants and products are gaseous. [4]

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(c) Students can also measure the enthalpy of combustion of ethanol in the laboratory using calorimetry. Suggest the major source of systematic error in these procedures. [1]

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(g) Identify **three** allotropes of carbon and describe their structures. [4]

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9.3-S12 Nov 2014 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (c) · also 9.2

The reaction between ethene and steam is used in the industrial production of ethanol.

$$C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(g)$$

The enthalpy change of the reaction can be calculated either by using average bond enthalpies or by using standard enthalpies of formation.

(a) Determine the enthalpy change of the reaction, in $kJ\,mol^{-1}$, using the average bond enthalpies in Table 10 of the Data Booklet. *[3]*

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(b) (i) Define the term *standard enthalpy change of formation*. *[2]*

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(ii) Determine the enthalpy change of the reaction, in $kJ\,mol^{-1}$, between ethene and steam using the enthalpy change of formation values given below.

Compound	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$C_2H_5OH(g)$	–235
$C_2H_4(g)$	+52
$H_2O(g)$	–242

[2]

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(c) Comment on which of the values obtained in (a) and (b)(ii) is more accurate, giving a reason. *[1]*

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9.3-S13 Nov 2012 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (c)(ii), (c)(iii) · also 3.1, 4.1, 4.2, 9.2

5. The strength of a covalent bond is measured in terms of its bond enthalpy.

(a) Define the term *average bond enthalpy*. *[2]*

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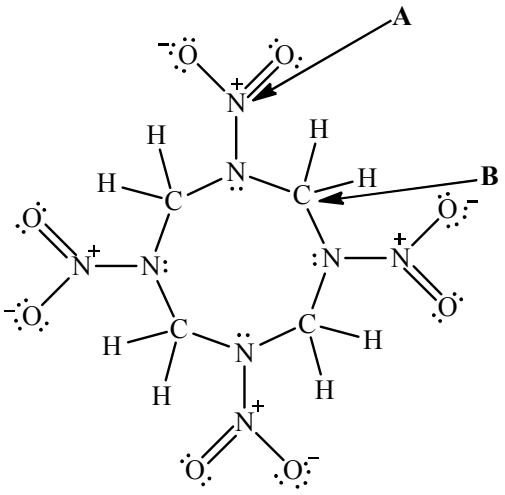
(b) 1,3,5,7-tetranitro-1,3,5,7-tetrazocane, shown below, can be used as an explosive.



The following equation represents the thermal decomposition of the compound.

$$C_4H_8N_8O_8\,(s) \rightarrow 4N_2(g) + 4CO(g) + 4H_2O(g)$$

(i) Calculate the enthalpy change when 10.0 g of the compound decomposes, using average bond enthalpy data from Table 10 of the Data Booklet and the following additional average bond enthalpy data at 298 K. *[4]*

Bond	**ΔH / kJ mol^{-1}**
C≡O	1072
N–O	201
N=O	607

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(ii) The CO molecule has dative covalent bonding. Identify a nitrogen-containing positive ion which also has this type of bonding. *[1]*

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(iii) Describe in words and with the aid of a suitable diagram the difference between sigma (σ) and pi (π) bonds. *[3]*

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(iv) Determine the number of σ and π bonds in 1,3,5,7-tetranitro-1,3,5,7-tetrazocane, using the Lewis structure shown on page 16. *[2]*

σ bonds: . . . . . .

π bonds: . . . . . .

(v) Explain the term *hybridization* and deduce the hybridization (sp, sp^2 or sp^3) of the atoms labelled **A** and **B** in the diagram on page 16. *[3]*

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(c) Methanol reacts with carbon monoxide to form ethanoic acid, $CH_3COOH\,(l)$.

$$CH_3OH\,(l) + CO\,(g) \rightarrow CH_3COOH\,(l)$$

(ii) Define the term *standard enthalpy change of formation*, $\Delta H_f^{\ominus}$. *[2]*

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(iii) The standard enthalpy change of formation of $CO\,(g)$ is –111 kJ mol^{-1}. Using Table 11 of the Data Booklet, determine the enthalpy change of the reaction, in kJ mol^{-1}. *[1]*

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9.3-S14 Nov 2011 · Paper 2 · Q3 parts (a), (b), (c), (d) · also 5.2, 7.3

3. (a) Define the term *average bond enthalpy*. *[2]*

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(b) Deduce the balanced chemical equation for the complete combustion of butan-1-ol. *[1]*

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(c) Determine the standard enthalpy change, in kJ mol^{-1}, for the complete combustion of butan-1-ol, using the information from Table 10 of the Data Booklet. *[3]*

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(d) Based on the types of intermolecular force present, explain why butan-1-ol has a higher boiling point than butanal. *[2]*

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9.3-S15 May 2010 TZ1 · Paper 2 · Q5 parts (a)(i)

5. (a) The production of ammonia is an important industrial process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

(i) Using the average bond enthalpy values in Table 10 of the Data Booklet, determine the standard enthalpy change for this reaction. *[3]*

LESSON 9.4

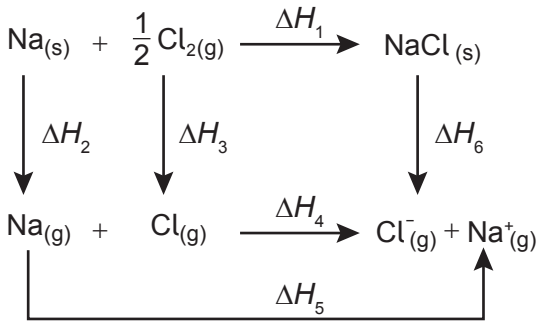
Born-Haber Cycles

35 multiple-choice questions · 4 structured questions

Part A - Multiple choice

9.4-M1 May 2025 TZ3 · Paper 1A · Q22

22. Which energy change on the diagram below represents the electron affinity of chlorine?



A. ΔH_2

B. ΔH_3

C. ΔH_4

D. ΔH_5

9.4-M2 May 2024 TZ1 · Paper 1 · Q16

16. Which change represents electron affinity?

A. $O_2(g) \rightarrow 2O(g)$

B. $O(g) \rightarrow O^+(g) + e^-$

C. $2O(g) + e^- \rightarrow O_2(g)$

D. $O^-(g) + e^- \rightarrow O^{2-}(g)$

9.4-M3 May 2024 TZ2 · Paper 1 · Q18

18. Which compound has the smallest lattice enthalpy?

A. RbBr

B. SrO

C. $MgBr_2$

D. Na_2O

9.4-M4 May 2022 TZ1 · Paper 1 · Q16

16. Which compound has the largest value of lattice enthalpy?

A. Na_2O

B. K_2O

C. Na_2S

D. K_2S

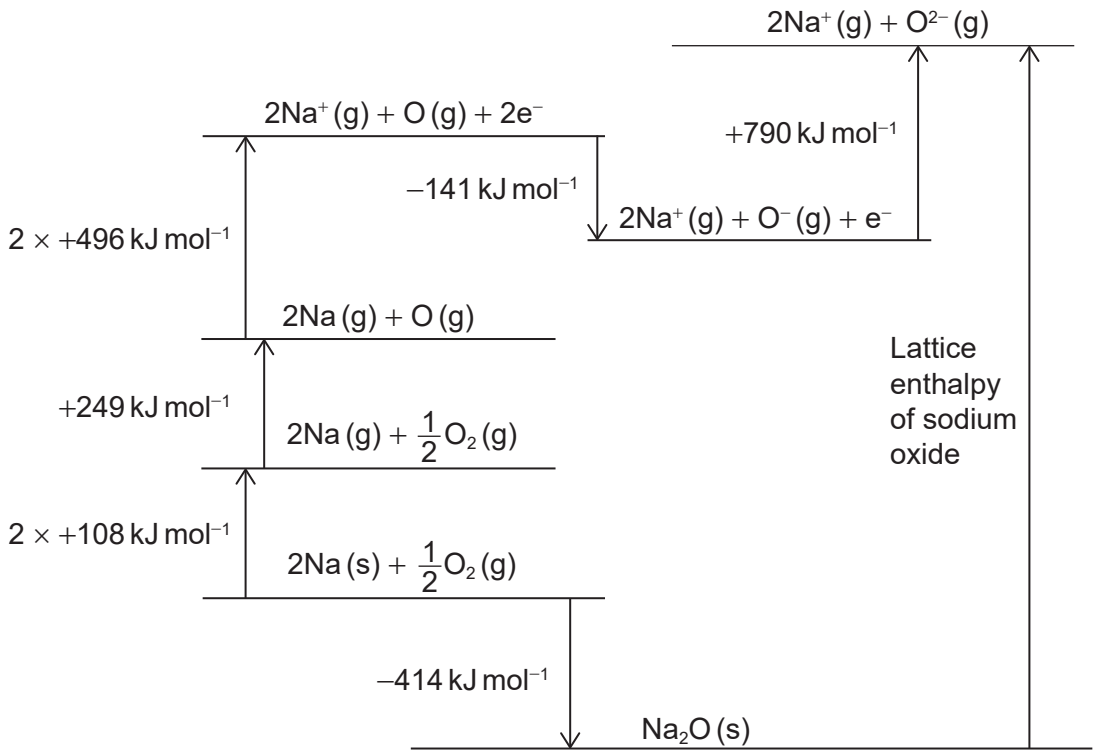
9.4-M5 Nov 2021 · Paper 1 · Q9

9. Which combination would create the strongest ionic bond?

	Ionic radius	Charges on ions
A.	large	high
B.	large	low
C.	small	high
D.	small	low

9.4-M6 Nov 2021 · Paper 1 · Q16

16. Consider the Born–Haber cycle for the formation of sodium oxide:



What is the lattice enthalpy, in $kJ\,mol^{-1}$, of sodium oxide?

A. $414 + 2(108) + 249 + 2(496) - 141 + 790$

B. $414 + 2(108) + 249 + 2(496) + 141 + 790$

C. $-414 + 2(108) + 249 + 2(496) - 141 + 790$

D. $-414 - 2(108) - 249 - 2(496) + 141 - 790$

9.4-M7 May 2021 TZ1 · Paper 1 · Q17

17. Which substance has the highest lattice enthalpy?

A. KCl

B. $CaCl_2$

C. KF

D. CaF_2

9.4-M8 May 2021 TZ2 · Paper 1 · Q16

16. Which represents electron affinity?

A. $Al^{2+}(g) \rightarrow Al^{3+}(g) + e^{-}$

B. $C(g) + e^{-} \rightarrow C^{-}(g)$

C. $Cl_2(g) \rightarrow 2Cl(g)$

D. $S(s) \rightarrow S^{+}(g) + e^{-}$

9.4-M9 May 2019 TZ1 · Paper 1 · Q17

17. Which equation represents the standard enthalpy of atomization of bromine, Br_2?

A. $\frac{1}{2}Br_2(l) \rightarrow Br(g)$

B. $Br_2(l) \rightarrow 2Br(g)$

C. $Br_2(l) \rightarrow 2Br(l)$

D. $\frac{1}{2}Br_2(l) \rightarrow Br(l)$

9.4-M10 May 2019 TZ2 · Paper 1 · Q16

16. Which equation represents lattice enthalpy?

A. $NaCl(g) \rightarrow Na^{+}(g) + Cl^{-}(g)$

B. $NaCl(s) \rightarrow Na^{+}(g) + Cl^{-}(g)$

C. $NaCl(s) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$

D. $NaCl(s) \rightarrow Na^{+}(s) + Cl^{-}(s)$

9.4-M11 Nov 2018 · Paper 1 · Q17

17. Which change is exothermic?

A. $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$

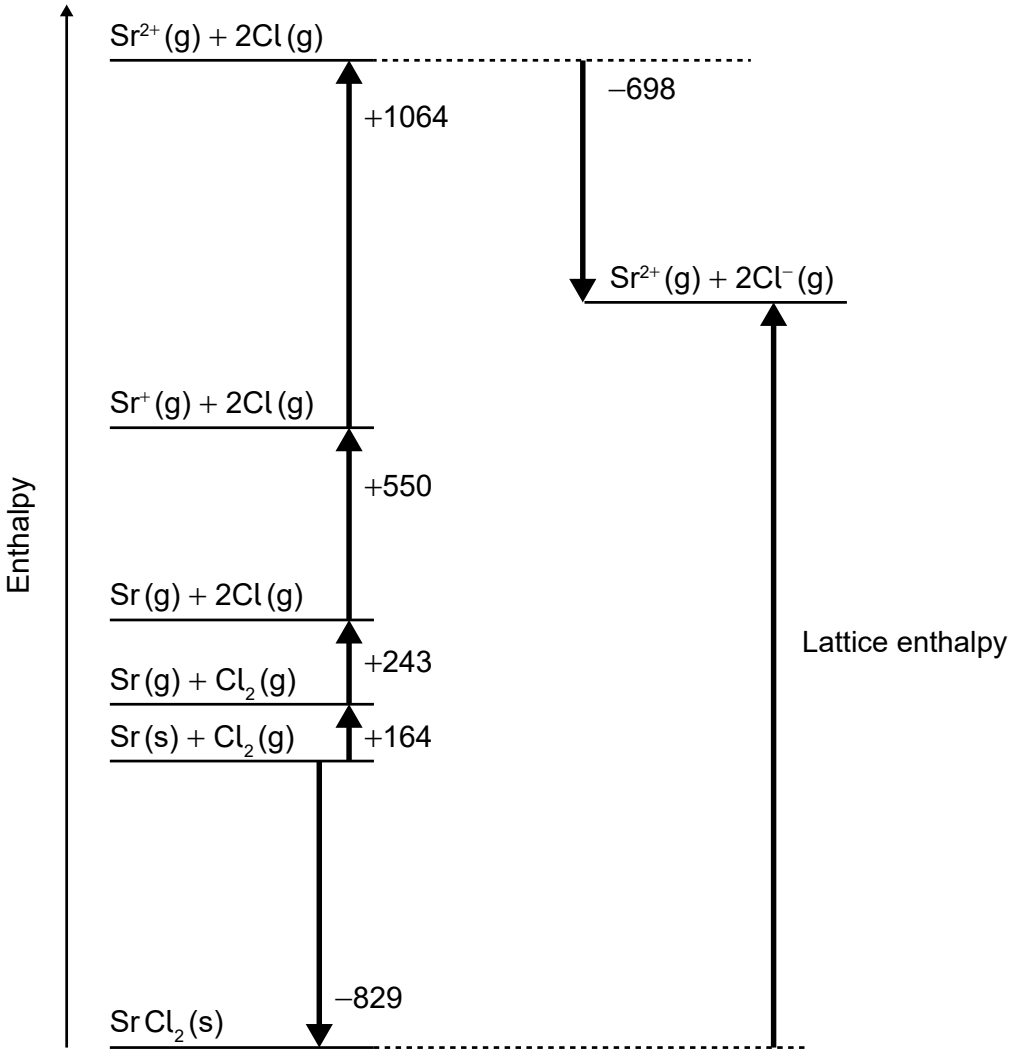
B. $K(g) \rightarrow K^{+}(g) + e^{-}$

C. $KCl(s) \rightarrow K^{+}(g) + Cl^{-}(g)$

D. $Cl(g) + e^{-} \rightarrow Cl^{-}(g)$

9.4-M12 May 2018 TZ2 · Paper 1 · Q16

16. Which value represents the lattice enthalpy, in kJ mol^{-1}, of strontium chloride, $SrCl_2$?



A. $-(-829) + 164 + 243 + 550 + 1064 - (-698)$

B. $-829 + 164 + 243 + 550 + 1064 - 698$

C. $-(-829) + 164 + 243 + 550 + 1064 - 698$

D. $-829 + 164 + 243 + 550 + 1064 - (-698)$

9.4-M13 Nov 2017 · Paper 1 · Q18

18. Which equation represents the lattice enthalpy of magnesium sulfide?

A. $MgS(s) \rightarrow Mg(g) + S(g)$

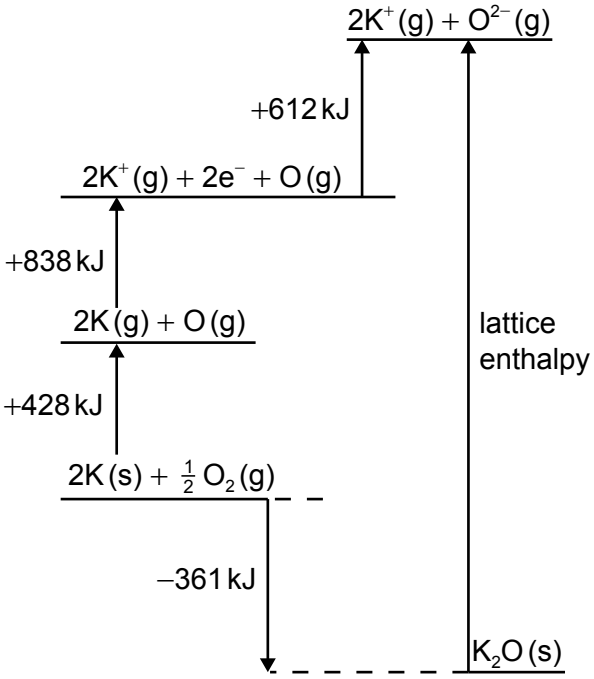
B. $MgS(s) \rightarrow Mg^{+}(g) + S^{-}(g)$

C. $MgS(s) \rightarrow Mg^{2+}(g) + S^{2-}(g)$

D. $MgS(s) \rightarrow Mg(s) + S(s)$

9.4-M14 May 2017 TZ2 · Paper 1 · Q16

16. The Born-Haber cycle for potassium oxide is shown below:



Which expression represents the lattice enthalpy in $kJ\,mol^{-1}$?

A. $-361 + 428 + 838 + 612$

B. $-(-361) + 428 + 838 + 612$

C. $-361 + 428 + 838 - 612$

D. $-(-361) + 428 + 838 - 612$

9.4-M15 Nov 2016 · Paper 1 · Q19

19. Which ionic compound has the largest value of lattice enthalpy?

A. MgS

B. MgO

C. $CaBr_2$

D. NaF

9.4-M16 Nov 2015 · Paper 1 · Q16

16. Which equation represents electron affinity?

A. $C(g) + e^- \rightarrow C^-(g)$

B. $Na^+(aq) + e^- \rightarrow Na(s)$

C. $\frac{1}{2}Cl_2(g) + e^- \rightarrow Cl^-(g)$

D. $B(g) + e^- \rightarrow B^+(g) + 2e^-$

9.4-M17 Nov 2015 · Paper 1 · Q17

17. Which combination results in an ionic compound having the **greatest** magnitude of lattice enthalpy?

	Sum of ionic radii	Ionic charges
A.	small	large
B.	large	large
C.	large	small
D.	small	small

9.4-M18 May 2015 TZ2 · Paper 1 · Q17

17. What is the correct order for **increasing** lattice enthalpy?

A. $MgO < MgCl_2 < NaCl < CsCl$

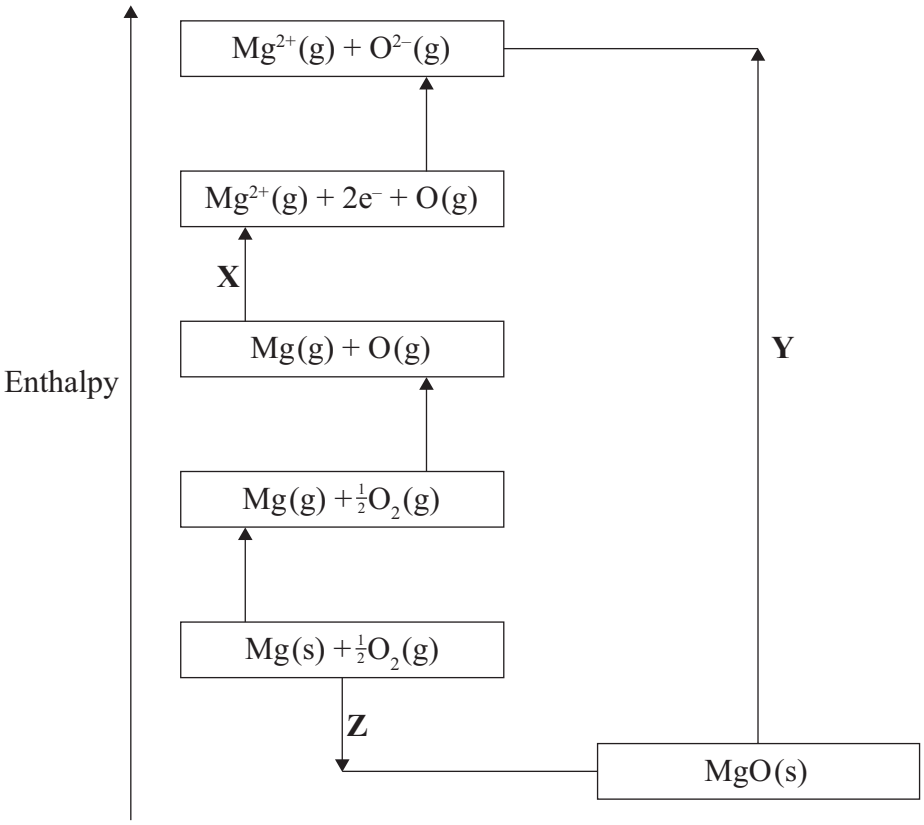
B. $CsCl < NaCl < MgCl_2 < MgO$

C. $NaCl < CsCl < MgO < MgCl_2$

D. $NaCl < CsCl < MgCl_2 < MgO$

9.4-M19 Nov 2014 · Paper 1 · Q17

17. The Born–Haber cycle for the formation of magnesium oxide is shown below.



What is a correct description of the steps **X**, **Y** and **Z** in this cycle?

	Step X	Step Y	Step Z
A.	2nd ionization energy of Mg	enthalpy of formation of MgO	lattice enthalpy of MgO
B.	2nd ionization energy of Mg	lattice enthalpy of MgO	enthalpy of formation of MgO
C.	sum of the 1st and 2nd ionization energies of Mg	lattice enthalpy of MgO	enthalpy of formation of MgO
D.	sum of 1st and 2nd ionization energies of Mg	enthalpy of formation of MgO	lattice enthalpy of MgO

9.4-M20 May 2014 TZ1 · Paper 1 · Q16

16. What is the correct definition of lattice enthalpy?

A. Enthalpy change when one mole of a solid ionic compound is separated into gaseous ions.

B. Enthalpy change when one mole of a solid ionic compound is separated into its ions in their standard state.

C. Enthalpy change when one mole of a solid ionic compound is formed from gaseous elements.

D. Enthalpy change when one mole of a compound is formed from the elements in their standard states.

9.4-M21 May 2014 TZ2 · Paper 1 · Q16

16. Which equation represents the lattice enthalpy of calcium chloride?

A. $CaCl(s) \rightarrow Ca^{+}(g) + Cl^{-}(g)$

B. $CaCl_2(s) \rightarrow Ca^{2+}(g) + 2Cl^{-}(g)$

C. $CaCl_2(g) \rightarrow Ca^{2+}(g) + 2Cl^{-}(g)$

D. $CaCl_2(s) \rightarrow Ca^{2+}(aq) + 2Cl^{-}(aq)$

9.4-M22 May 2014 TZ2 · Paper 1 · Q18

18. Which equation represents the second electron affinity of oxygen?

A. $\frac{1}{2}O_2(g) + 2e^{-} \rightarrow O^{2-}(g)$

B. $O(g) + 2e^{-} \rightarrow O^{2-}(g)$

C. $O_2(g) + 4e^{-} \rightarrow 2O^{2-}(g)$

D. $O^{-}(g) + e^{-} \rightarrow O^{2-}(g)$

9.4-M23 Nov 2013 · Paper 1 · Q17

17. Which ionic compound has the most endothermic lattice enthalpy?

A. Sodium chloride

B. Sodium oxide

C. Magnesium chloride

D. Magnesium oxide

9.4-M24 May 2013 TZ1 · Paper 1 · Q16

16. Which combination of ions will give the greatest absolute lattice enthalpy?

A. A small positive ion with a high charge and a small negative ion with a high charge

B. A small positive ion with a low charge and a small negative ion with a low charge

C. A large positive ion with a high charge and a large negative ion with a high charge

D. A large positive ion with a low charge and a small negative ion with a low charge

9.4-M25 May 2013 TZ2 · Paper 1 · Q18

18. Which compound has the most positive lattice enthalpy of dissociation?

A. NaCl

B. NaBr

C. $MgCl_2$

D. $MgBr_2$

9.4-M26 Nov 2012 · Paper 1 · Q16

16. Which is the best definition of electron affinity?

A. The ability of an atom to attract the electrons in a covalent bond.

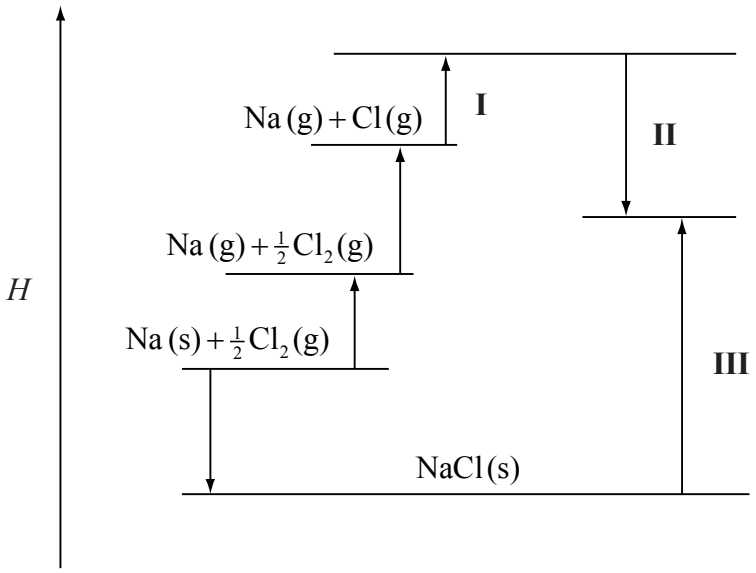
B. The attraction of an atom for an electron.

C. The enthalpy change when an atom gains an electron.

D. The enthalpy change when a gaseous atom gains an electron.

9.4-M27 May 2012 TZ1 · Paper 1 · Q16

16. The diagram represents the Born–Haber cycle for the lattice enthalpy of sodium chloride.



What is the name of the enthalpy changes **I**, **II** and **III**?

	I	II	III
A.	ionization energy of Na	electron affinity of Cl	lattice enthalpy of NaCl
B.	lattice enthalpy of NaCl	ionization energy of Na	electron affinity of Cl
C.	electron affinity of Cl	ionization energy of Na	lattice enthalpy of NaCl
D.	ionization energy of Na	lattice enthalpy of NaCl	electron affinity of Cl

9.4-M28 May 2012 TZ2 · Paper 1 · Q16

16. Which equation represents the electron affinity of chlorine?

A. $Cl(g) + e^- \rightarrow Cl^-(g)$

B. $Cl(g) + e^- \rightarrow Cl\bullet(g)$

C. $Cl_2(g) + 2e^- \rightarrow 2Cl^-(g)$

D. $Cl(g) \rightarrow Cl^+(g) + e^-$

9.4-M29 Nov 2011 · Paper 1 · Q17

17. Which row of the table correctly represents the equations for the lattice enthalpy of substance XY and the electron affinity of atom Y?

	Lattice enthalpy	Electron affinity
A.	$X^+(g) + Y^-(g) \rightarrow XY(g)$	$Y^-(g) + e^- \rightarrow Y^{2-}(g)$
B.	$X^+(g) + Y^-(g) \rightarrow XY(s)$	$Y(g) + e^- \rightarrow Y^-(g)$
C.	$X^+(g) + Y^-(g) \rightarrow XY(s)$	$Y(s) + e^- \rightarrow Y^-(s)$
D.	$X^+(g) + Y^-(g) \rightarrow XY(g)$	$Y(g) + e^- \rightarrow Y^-(g)$

9.4-M30 May 2011 TZ1 · Paper 1 · Q19

19. Which equation corresponds to the lattice enthalpy for silver iodide, AgI?

A. $AgI(s) \rightarrow Ag(s) + I(g)$

B. $AgI(s) \rightarrow Ag(s) + \frac{1}{2}I_2(g)$

C. $AgI(s) \rightarrow Ag^+(aq) + I^-(aq)$

D. $AgI(s) \rightarrow Ag^+(g) + I^-(g)$

9.4-M31 May 2011 TZ2 · Paper 1 · Q16

16. Which ionic compound has the greatest lattice enthalpy?

A. MgO

B. CaO

C. NaF

D. KF

9.4-M32 Nov 2010 · Paper 1 · Q17

17. Which is a correct definition of lattice enthalpy?

A. It is the enthalpy change that occurs when an electron is removed from 1 mol of gaseous atoms.

B. It is the enthalpy change that occurs when 1 mol of a compound is formed from its elements.

C. It is the enthalpy change that occurs when 1 mol of solid crystal changes into a liquid.

D. It is the enthalpy change that occurs when 1 mol of solid crystal is formed from its gaseous ions.

9.4-M33 May 2010 TZ1 · Paper 1 · Q17

17. Which process has an enthalpy change that represents the lattice enthalpy of sodium chloride?

A. $NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$

B. $NaCl(s) \rightarrow Na(s) + \frac{1}{2}Cl_2(g)$

C. $NaCl(g) \rightarrow Na^+(g) + Cl^-(g)$

D. $NaCl(s) \rightarrow Na(s) + Cl(g)$

9.4-M34 May 2010 TZ1 · Paper 1 · Q18

18. Which is the correct order of increasing magnitude of lattice enthalpy (lowest first)?

A. $NaCl < KCl < MgS < MgO$

B. $MgO < MgS < KCl < NaCl$

C. $KCl < NaCl < MgS < MgO$

D. $MgO < NaCl < KCl < MgS$

9.4-M35 May 2010 TZ2 · Paper 1 · Q20

20. Which combination of ionic radius and ionic charge would result in the highest lattice enthalpy for an ionic compound?

	Ionic radius	Ionic charge
A.	small	high
B.	large	high
C.	small	low
D.	large	low

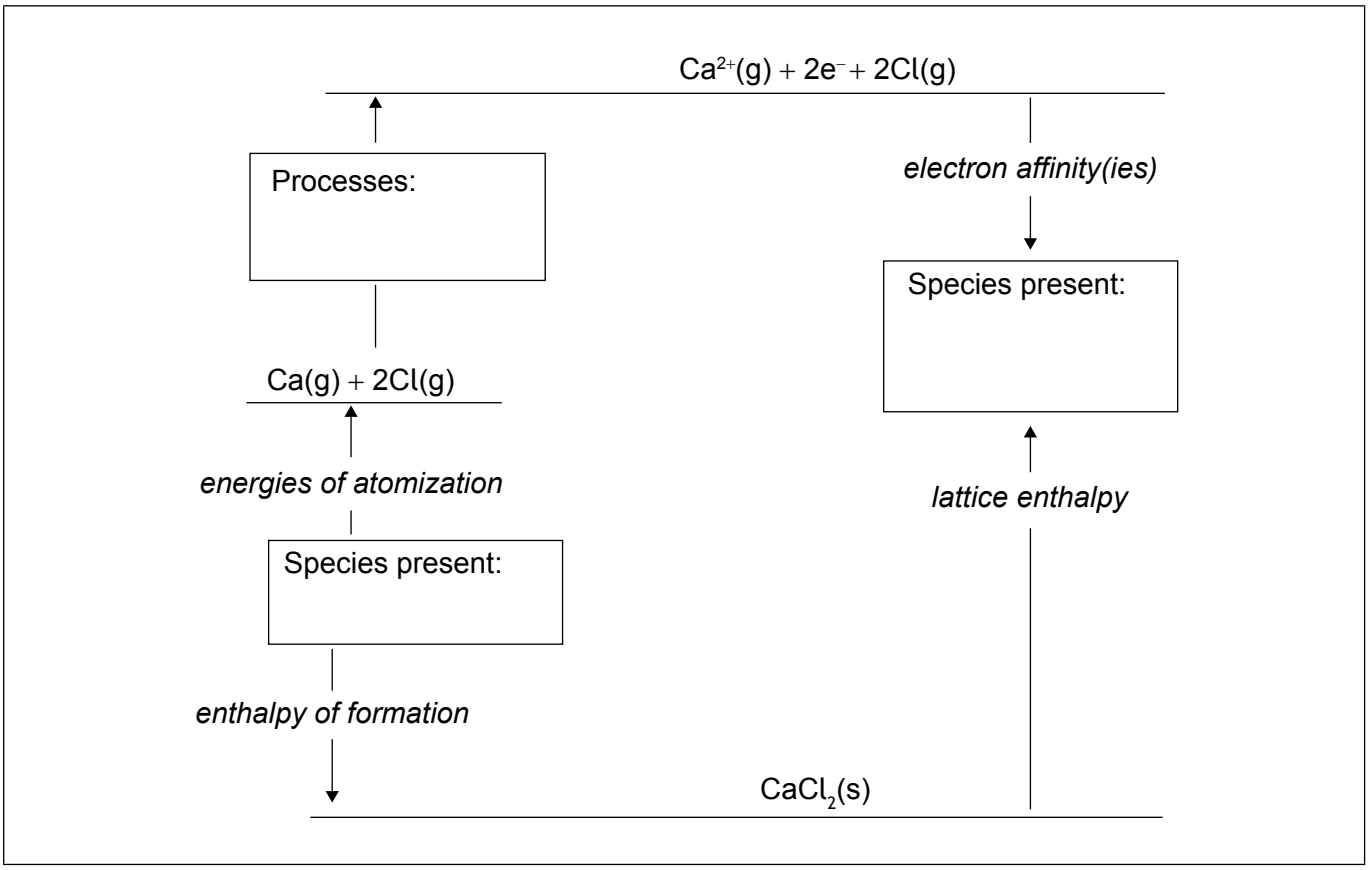
Part B - Structured questions

9.4-S1 May 2025 TZ1 · Paper 2 · Q4 parts (c)(i), (c)(ii)

4. 3.162 g of calcium carbonate, $CaCO_3(s)$, is reacted with 20.0 cm^3 of 4.00 mol dm^{-3} hydrochloric acid, HCl(aq).

(c) Calcium carbonate and calcium chloride are ionic compounds.

(i) Annotate the Born–Haber cycle for calcium chloride, $CaCl_2$, by filling the names of processes and formulas of species, including state symbols, in the boxes. [3]



(ii) Outline why the lattice enthalpy of calcium chloride is lower than the lattice enthalpy of magnesium chloride. [1]

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9.4-S2 May 2023 TZ2 · Paper 2 · Q8 parts (b), (c)(i) · also 2.1

8. Solid ionic compounds form crystal lattices.

(b) Explain why the lattice enthalpy of barium chloride, $BaCl_2$, is lower than that of magnesium chloride. [2]

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(c) Cobalt also forms chlorides with the formula $CoCl_2$.

(i) State the full electron configuration of the cobalt(II) ion, Co^{2+}. [1]

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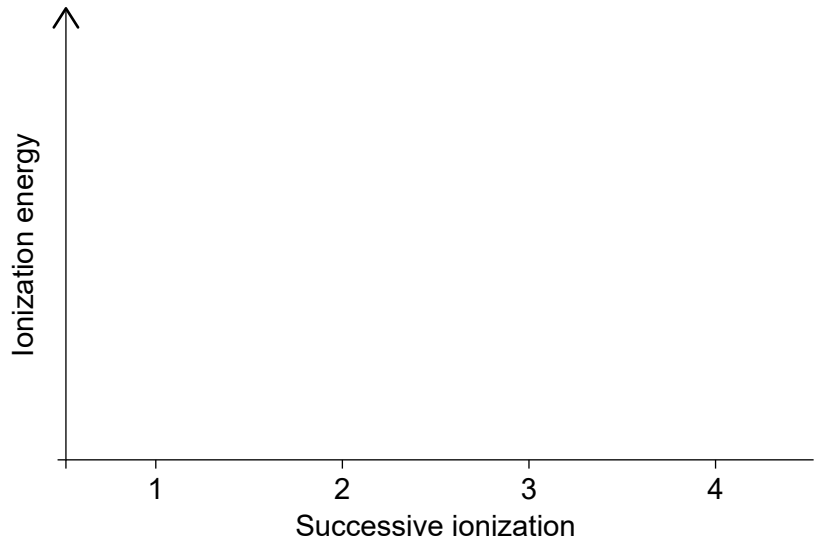
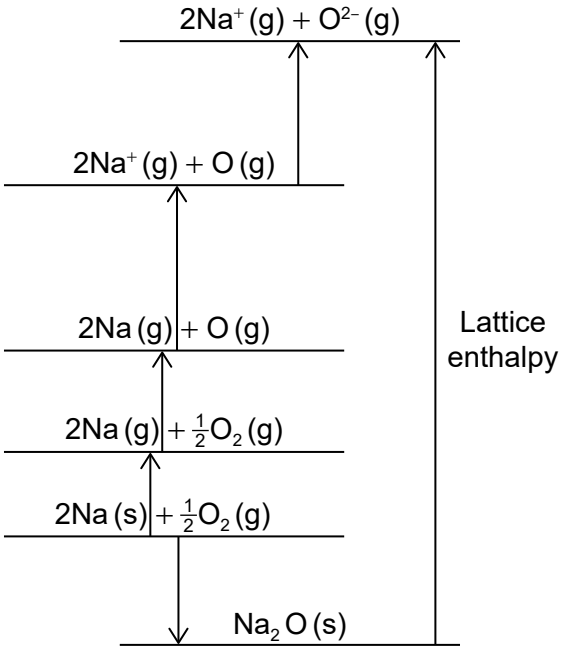
9.4-S3 May 2019 TZ1 · Paper 2 · Q3 parts (a), (d)(i), (d)(ii), (d)(iii), (f), (g)(i), (g)(ii), (h), (k) · also 1.2, 2.1, 4.1, 6.1, 9.2, 9.3

3. This question is about sodium and its compounds.

(a) Plot the relative values of the first four ionization energies of sodium. [1]



(d) The Born-Haber cycle for sodium oxide is shown (not to scale).



(i) Calculate values for the following changes using section 8 of the data booklet. [2]

$$\Delta H_{\text{atomisation}}\,(\text{Na}) = 107\,\text{kJ mol}^{-1}$$

$$\Delta H_{\text{atomisation}}\,(\text{O}) = 249\,\text{kJ mol}^{-1}$$

$\frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{O}^{2-}(\text{g})$:

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$\text{Na}(\text{s}) \rightarrow \text{Na}^{+}(\text{g})$:

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(ii) The standard enthalpy of formation of sodium oxide is $-414\,\text{kJ mol}^{-1}$. Determine the lattice enthalpy of sodium oxide, in kJ mol^{-1}, using section 8 of the data booklet and your answers to (d)(i). [2]

(If you did not get answers to (d)(i), use $+850\,\text{kJ mol}^{-1}$ and $+600\,\text{kJ mol}^{-1}$ respectively, but these are not the correct answers.)

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(iii) Justify why K_2O has a lower lattice enthalpy (absolute value) than Na_2O. [1]

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(f) Sodium peroxide, Na_2O_2, is formed by the reaction of sodium oxide with oxygen.

$$2Na_2O(s) + O_2(g) \rightarrow 2Na_2O_2(s)$$

Calculate the percentage yield of sodium peroxide if 5.00 g of sodium oxide produces 5.50 g of sodium peroxide. [2]

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(g) Sodium peroxide is used in diving apparatus to produce oxygen from carbon dioxide.

$$2Na_2O_2(s) + 2CO_2(g) \rightarrow 2Na_2CO_3(s) + O_2(g)$$

(i) Determine the enthalpy change, ΔH, in kJ, for this reaction using data from the table and section 12 of the data booklet. [3]

	ΔH_f / kJ mol^{-1}
$Na_2O_2(s)$	–510.9
$Na_2CO_3(s)$	–1130.7

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(ii) Outline why bond enthalpy values are not valid in calculations such as that in (g)(i). [1]

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(h) An allotrope of molecular oxygen is ozone. Compare, giving a reason, the bond enthalpies of the O to O bonds in O_2 and O_3. [1]

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(k) State the oxidation number of carbon in sodium carbonate, Na_2CO_3. [1]

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9.4-S4 May 2012 TZ2 · Paper 2 · Q3 parts (a), (b), (c)(i), (c)(ii)

3. Lattice enthalpies can be determined experimentally using a Born–Haber cycle and theoretically using calculations based on electrostatic principles.

(a) The experimental lattice enthalpies of the chlorides of lithium, LiCl, sodium, NaCl, potassium, KCl, and rubidium, RbCl, are given in Table 13 of the Data Booklet. Explain the trend in the values. *[2]*

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(b) Explain why magnesium chloride, $MgCl_2$, has a much greater lattice enthalpy than sodium chloride, NaCl. *[2]*

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(c) (i) Identify the process labelled **a** on the Born–Haber cycle for the determination of the standard enthalpy of formation of lithium fluoride, LiF. *[1]*

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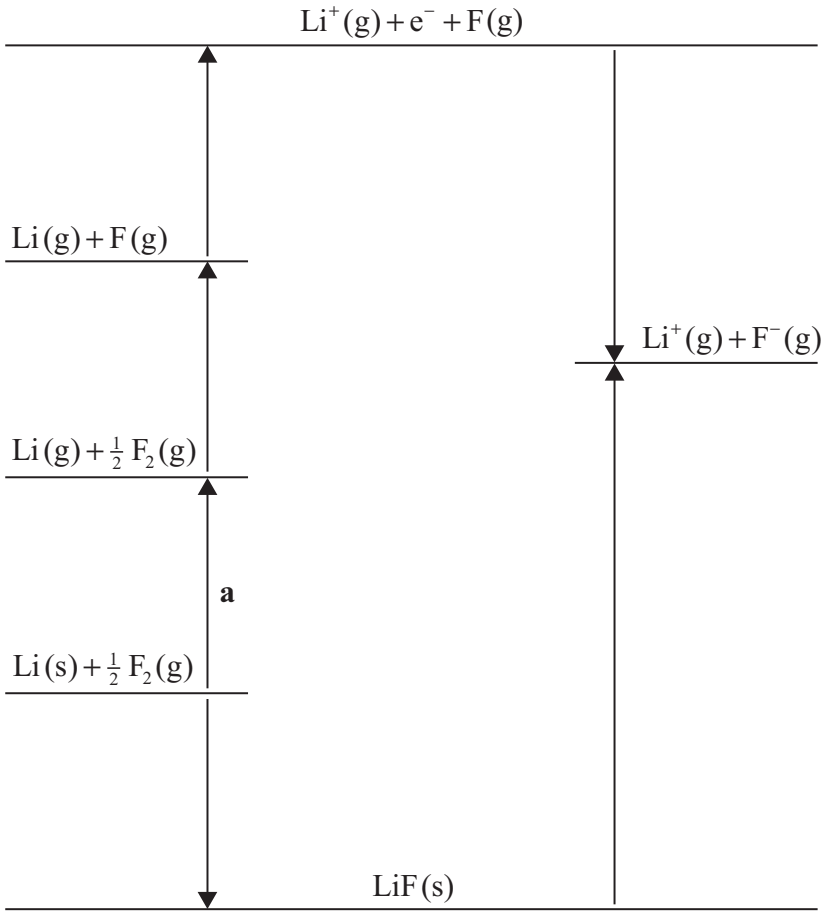

(ii) The enthalpy change for process **a** is $+159\ \text{kJ mol}^{-1}$. Calculate the standard enthalpy of formation of lithium fluoride, LiF, using this and other values from the Data Booklet. *[2]*

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INDEX

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2.1-S10	Nov 2021 · Paper 2 · Q2	(a)	127
2.1-S11	Nov 2021 · Paper 2 · Q9	(b)(ii)	128
2.1-S12	May 2021 TZ1 · Paper 2 · Q1	(c), (d)(iii)	129
2.1-S13	May 2021 TZ2 · Paper 2 · Q2	(a)(ii), (b)(i)	130
1.2-S20	Nov 2020 · Paper 2 · Q1	(a)(i)	49
2.1-S14	Nov 2020 · Paper 2 · Q6	(a)	131
7.2-S7	Nov 2019 · Paper 2 · Q6	(a)(i)	485
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(a)	693
7.1-S15	May 2019 TZ1 · Paper 2 · Q6	(a)	432
2.1-S15	May 2019 TZ1 · Paper 3 · Q1	(b)(iv)	132
2.1-S16	May 2019 TZ2 · Paper 2 · Q3	(c)	133

3.1-S14	Nov 2018 · Paper 2 · Q3	(a)(i), (a)(ii)	186
2.1-S17	Nov 2018 · Paper 2 · Q4	(c)	134
4.1-S6	May 2018 TZ1 · Paper 2 · Q2	(e)	231
7.2-S12	May 2018 TZ2 · Paper 2 · Q3	(a)(iii), (a)(iv), (c)(i)	491
2.1-S18	Nov 2017 · Paper 2 · Q3	(a)	134
2.1-S19	May 2017 TZ1 · Paper 2 · Q2	(d)(i), (d)(iii)	135
4.1-S9	May 2017 TZ2 · Paper 2 · Q4	(c)	234
7.1-S23	Nov 2016 · Paper 2 · Q4	(d)(ii)	440
2.1-S20	May 2016 · Paper 2 · Q4	(e)(i), (e)(ii), (e)(iii)	136
2.1-S21	Nov 2015 · Paper 2 · Q2	(a)	137
2.1-S22	May 2015 TZ1 · Paper 2 · Q4	(a)	137
2.1-S23	May 2015 TZ1 · Paper 2 · Q8	(e)	138
2.1-S24	May 2015 TZ1 · Paper 3 · Q1	(a)	138
2.1-S25	May 2015 TZ1 · Paper 3 · Q18	(a)(i)	139
7.1-S25	May 2015 TZ2 · Paper 2 · Q8	(a)(i)	443
2.1-S26	Nov 2014 · Paper 2 · Q8	(b)(i), (b)(ii)	140
6.1-S11	Nov 2014 · Paper 2 · Q10	(c)	364
8.1-S4	Nov 2014 · Paper 2 · Q11	(f)(ii)	573
2.1-S27	May 2014 TZ1 · Paper 2 · Q5	(d)(i), (e)	142
7.1-S26	May 2014 TZ2 · Paper 2 · Q2	(b)	445
7.2-S20	May 2014 TZ2 · Paper 2 · Q8	(a), (b)(i), (b)(ii)	503
2.1-S28	Nov 2013 · Paper 2 · Q4	(a)	144
3.1-S23	May 2013 TZ1 · Paper 2 · Q6	(c)(i)	197
7.2-S22	May 2013 TZ1 · Paper 2 · Q7	(b)(v), (b)(vi)	507
2.1-S29	May 2013 TZ2 · Paper 2 · Q3	(e)	144
2.1-S30	Nov 2012 · Paper 2 · Q2	(b)(ii)	145
2.1-S31	May 2012 TZ1 · Paper 2 · Q8	(b)(i), (b)(ii)	145
2.1-S32	May 2012 TZ2 · Paper 2 · Q9	(a)(i), (a)(ii)	146
2.1-S33	Nov 2011 · Paper 2 · Q2	(c)	147
7.1-S32	Nov 2011 · Paper 2 · Q7	(e)(iv)	456
2.1-S34	May 2011 TZ1 · Paper 2 · Q3	(c)	147
2.1-S35	May 2011 TZ2 · Paper 2 · Q3	(a)(i), (a)(ii), (a)(iii)	148
2.1-S36	Nov 2010 · Paper 2 · Q3	(d)	148
2.1-S37	Nov 2010 · Paper 2 · Q4	(a)(i), (a)(ii), (a)(iii)	149
1.2-S53	May 2010 TZ1 · Paper 2 · Q1	(c)(i)	87
2.1-S38	May 2010 TZ2 · Paper 2 · Q6	(d), (e)(i), (e)(ii)	150

3.1 VSEPR Theory

3.1-S1	May 2025 TZ1 · Paper 2 · Q6	(b)(ii), (b)(iii)	169
3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(a)(i), (a)(iii)	171
9.2-S1	May 2025 TZ3 · Paper 2 · Q5	(c)(i)	646
3.1-S3	May 2025 TZ3 · Paper 2 · Q10	(a)(ii), (a)(iii)	173
3.1-S4	Nov 2024 · Paper 2 · Q4	(a)(i), (a)(ii)	174
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(b)(i)	354
2.1-S4	May 2024 TZ1 · Paper 2 · Q5	(b)(i)	119
3.1-S5	May 2024 TZ2 · Paper 2 · Q2	(b)(i), (b)(ii), (b)(iii)	176
9.2-S2	Nov 2023 TZ1 · Paper 2 · Q4	(b)	647
7.2-S4	Nov 2023 TZ1 · Paper 2 · Q5	(c)(i), (c)(ii)	480
2.1-S5	May 2023 TZ1 · Paper 2 · Q2	(c)(i), (c)(ii)	120
3.1-S6	May 2023 TZ1 · Paper 2 · Q5	(a)(i)	177
3.1-S7	May 2023 TZ1 · Paper 2 · Q9	(b)(iii), (b)(iv)	178
3.1-S8	May 2023 TZ2 · Paper 2 · Q2	(b), (c)(ii)	179
8.2-S4	Nov 2022 · Paper 2 · Q1	(f)(v)	599
1.2-S16	Nov 2022 · Paper 2 · Q5	(c)	44
2.1-S7	May 2022 TZ1 · Paper 2 · Q6	(a)(ii)	125
3.1-S9	May 2022 TZ2 · Paper 2 · Q6	(c)(i), (c)(ii)	181
3.1-S10	Nov 2021 · Paper 2 · Q3	(a)(i), (b)(i)	181
3.1-S11	May 2021 TZ1 · Paper 2 · Q2	(a)(i), (a)(ii)	183
4.1-S3	May 2021 TZ1 · Paper 2 · Q7	(a)(i)	228
2.1-S13	May 2021 TZ2 · Paper 2 · Q2	(c)	130
3.1-S12	Nov 2020 · Paper 2 · Q2	(a)	184
4.1-S4	Nov 2019 · Paper 2 · Q1	(a), (c)	229
3.1-S13	May 2019 TZ1 · Paper 2 · Q5	(c)(i), (c)(iii), (c)(iv)	185

5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(b)(i)	302
2.1-S16	May 2019 TZ2 · Paper 2 · Q3	(d)(ii)	133
3.1-S14	Nov 2018 · Paper 2 · Q3	(b)(i), (b)(ii), (c)	186
5.2-S11	May 2018 TZ1 · Paper 2 · Q1	(b)	305
3.1-S15	May 2018 TZ2 · Paper 2 · Q7	(b), (c)(i), (c)(ii)	188
3.1-S16	May 2018 TZ2 · Paper 3 · Q2	(a)	190
3.1-S17	Nov 2017 · Paper 2 · Q4	(a)	191
3.1-S18	May 2017 TZ1 · Paper 2 · Q5	(a)	192
3.1-S19	May 2017 TZ2 · Paper 2 · Q3	(b)	194
4.1-S9	May 2017 TZ2 · Paper 2 · Q4	(b)(ii)	234
1.2-S35	Nov 2016 · Paper 2 · Q2	(d)	60
3.1-S20	Nov 2016 · Paper 3 · Q22	(c)(i)	195
1.2-S38	May 2016 · Paper 2 · Q1	(a)(i)	65
5.2-S15	Nov 2015 · Paper 2 · Q6	(c)(iii)	310
1.2-S41	May 2015 TZ1 · Paper 2 · Q1	(d)	70
2.1-S23	May 2015 TZ1 · Paper 2 · Q8	(f)	138
4.1-S13	May 2015 TZ2 · Paper 2 · Q9	(c)(i), (c)(ii)	238
2.1-S26	Nov 2014 · Paper 2 · Q8	(c)(ii)	140
8.2-S13	May 2014 TZ1 · Paper 2 · Q1	(d)(ii)	613
3.1-S21	May 2014 TZ2 · Paper 2 · Q4	(a), (b)	195
3.1-S22	Nov 2013 · Paper 2 · Q3	(a)(ii)	197
1.2-S44	Nov 2013 · Paper 2 · Q5	(h)	74
7.1-S29	Nov 2013 · Paper 2 · Q6	(e)(iii), (f)(ii)	449
3.1-S23	May 2013 TZ1 · Paper 2 · Q6	(c)(ii), (c)(iii)	197
3.1-S24	May 2013 TZ2 · Paper 2 · Q5	(b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii)	199
1.2-S47	Nov 2012 · Paper 2 · Q1	(e)	79
9.3-S13	Nov 2012 · Paper 2 · Q5	(b)(ii)	673
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(a)(i), (a)(ii)	202
3.1-S26	May 2012 TZ2 · Paper 2 · Q4	whole question	205
3.1-S27	Nov 2011 · Paper 2 · Q6	(c)(i), (c)(ii)	206
8.1-S9	Nov 2011 · Paper 2 · Q8	(e)(i)	579
3.1-S28	May 2011 TZ2 · Paper 2 · Q5	(a)(i), (a)(ii)	207
3.1-S29	Nov 2010 · Paper 2 · Q7	(a)(i), (a)(ii), (d)(ii)	208
1.2-S55	May 2010 TZ2 · Paper 2 · Q4	(b)	89

4.1 Sigma & Pi Bonds and Delocalization

3.1-S1	May 2025 TZ1 · Paper 2 · Q6	(b)(iv)	169
4.2-S1	Nov 2024 · Paper 2 · Q2	(a)(ii)	261
3.1-S4	Nov 2024 · Paper 2 · Q4	(a)(iii)	174
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(b)(v)	354
3.1-S5	May 2024 TZ2 · Paper 2 · Q2	(c), (d)	176
5.2-S4	Nov 2023 TZ1 · Paper 2 · Q1	(d)(iv)	299
3.1-S8	May 2023 TZ2 · Paper 2 · Q2	(c)(i)	179
4.2-S2	Nov 2022 · Paper 2 · Q2	(d)	261
5.2-S6	May 2022 TZ1 · Paper 2 · Q5	(a)(iii)	300
2.1-S7	May 2022 TZ1 · Paper 2 · Q6	(a)(iii)	125
4.1-S1	May 2022 TZ2 · Paper 2 · Q7	(b)(i), (b)(ii)	226
4.1-S2	Nov 2021 · Paper 2 · Q10	(a)(i)	227
4.1-S3	May 2021 TZ1 · Paper 2 · Q7	(a)(ii), (b)	228
9.3-S6	May 2021 TZ2 · Paper 2 · Q4	(d)	664
3.1-S12	Nov 2020 · Paper 2 · Q2	(c)	184
4.1-S4	Nov 2019 · Paper 2 · Q1	(b), (d), (e)(i), (e)(ii)	229
4.1-S5	May 2019 TZ1 · Paper 2 · Q2	(c)	230
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(h)	693
5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(b)(ii)	302
2.1-S16	May 2019 TZ2 · Paper 2 · Q3	(d)(i)	133
5.2-S11	May 2018 TZ1 · Paper 2 · Q1	(i)	305
4.1-S6	May 2018 TZ1 · Paper 2 · Q2	(g)(i), (g)(ii)	231
9.3-S8	May 2018 TZ1 · Paper 2 · Q3	(d)	666
3.1-S15	May 2018 TZ2 · Paper 2 · Q7	(d)	188
4.1-S7	May 2018 TZ2 · Paper 2 · Q8	(a)	233
4.1-S8	May 2017 TZ1 · Paper 2 · Q7	(a), (b)	233
4.1-S9	May 2017 TZ2 · Paper 2 · Q4	(a)(i), (b)(i)	234

1.2-S35	Nov 2016 · Paper 2 · Q2	(e)	60
4.1-S10	Nov 2016 · Paper 2 · Q5	(b)(i)	236
4.1-S11	May 2016 · Paper 2 · Q3	(c)(i), (c)(ii)	236
5.2-S15	Nov 2015 · Paper 2 · Q6	(c)(ii)	310
1.2-S41	May 2015 TZ1 · Paper 2 · Q1	(e)	70
6.2-S5	May 2015 TZ1 · Paper 2 · Q7	(f)	381
4.1-S12	May 2015 TZ1 · Paper 3 · Q20	(a)	237
4.1-S13	May 2015 TZ2 · Paper 2 · Q9	(d)(i), (d)(iii), (d)(iv)	238
4.1-S14	May 2015 TZ2 · Paper 3 · Q23	(a)	240
4.1-S15	May 2015 TZ2 · Paper 3 · Q30	(a), (b)	241
4.1-S16	Nov 2014 · Paper 3 · Q22	(a)	241
8.2-S13	May 2014 TZ1 · Paper 2 · Q1	(d)(iii)	613
4.1-S17	May 2014 TZ1 · Paper 3 · Q16	(d)	242
3.1-S21	May 2014 TZ2 · Paper 2 · Q4	(d), (e)	195
3.1-S22	Nov 2013 · Paper 2 · Q3	(b)	197
1.2-S44	Nov 2013 · Paper 2 · Q5	(g)	74
4.1-S18	Nov 2013 · Paper 3 · Q27	(a)(i), (a)(ii)	242
3.1-S23	May 2013 TZ1 · Paper 2 · Q6	(e)	197
4.1-S19	May 2013 TZ1 · Paper 3 · QE1	(c)	243
4.1-S20	May 2013 TZ1 · Paper 3 · QG1	(a)	244
4.1-S21	May 2013 TZ2 · Paper 3 · QG2	(a)	244
9.3-S13	Nov 2012 · Paper 2 · Q5	(b)(iii), (b)(iv)	673
4.1-S22	May 2012 TZ1 · Paper 3 · QE1	(a)	245
4.1-S23	May 2012 TZ2 · Paper 2 · Q7	(a)(vi), (a)(vii)	245
4.1-S24	May 2012 TZ2 · Paper 3 · QE3	(a)	247
3.1-S27	Nov 2011 · Paper 2 · Q6	(d)(i)	206
4.3-S10	May 2011 TZ1 · Paper 2 · Q6	(c), (d)	277
4.1-S25	May 2011 TZ1 · Paper 3 · QE4	(a)	247
4.1-S26	May 2011 TZ2 · Paper 3 · QG1	(b)	248
4.1-S27	Nov 2010 · Paper 2 · Q2	(a)(i)	248
4.1-S28	Nov 2010 · Paper 3 · QE4	(a)(i)	249
4.1-S29	Nov 2010 · Paper 3 · QG1	(a), (b)	250
4.1-S30	May 2010 TZ2 · Paper 3 · QE3	(d)	250

4.2 Hybridization

3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(a)(ii)	171
9.2-S1	May 2025 TZ3 · Paper 2 · Q5	(c)(ii)	646
4.2-S1	Nov 2024 · Paper 2 · Q2	(a)(i)	261
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(b)(iv)	354
3.1-S5	May 2024 TZ2 · Paper 2 · Q2	(b)(iv)	176
7.2-S4	Nov 2023 TZ1 · Paper 2 · Q5	(d)(i)	480
4.2-S2	Nov 2022 · Paper 2 · Q2	(b)	261
5.2-S6	May 2022 TZ1 · Paper 2 · Q5	(a)(iv)	300
4.1-S1	May 2022 TZ2 · Paper 2 · Q7	(b)(iii)	226
4.1-S2	Nov 2021 · Paper 2 · Q10	(a)(ii)	227
9.3-S6	May 2021 TZ2 · Paper 2 · Q4	(c)	664
3.1-S12	Nov 2020 · Paper 2 · Q2	(b)	184
3.1-S13	May 2019 TZ1 · Paper 2 · Q5	(c)(ii)	185
2.1-S16	May 2019 TZ2 · Paper 2 · Q3	(d)(iii)	133
3.1-S17	Nov 2017 · Paper 2 · Q4	(c)	191
3.1-S18	May 2017 TZ1 · Paper 2 · Q5	(b)	192
1.2-S38	May 2016 · Paper 2 · Q1	(a)(ii)	65
5.2-S15	Nov 2015 · Paper 2 · Q6	(c)(i)	310
4.1-S13	May 2015 TZ2 · Paper 2 · Q9	(d)(ii)	238
2.1-S26	Nov 2014 · Paper 2 · Q8	(d)(i), (d)(ii)	140
8.2-S13	May 2014 TZ1 · Paper 2 · Q1	(d)(iv)	613
3.1-S21	May 2014 TZ2 · Paper 2 · Q4	(c)	195
7.1-S29	Nov 2013 · Paper 2 · Q6	(f)(i)	449
3.1-S23	May 2013 TZ1 · Paper 2 · Q6	(c)(iv)	197
4.2-S3	May 2013 TZ2 · Paper 2 · Q8	(b)	263
9.3-S13	Nov 2012 · Paper 2 · Q5	(b)(v)	673
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(a)(iii)	202
4.1-S23	May 2012 TZ2 · Paper 2 · Q7	(b)(i), (b)(ii)	245

4.3-S10	May 2011 TZ1 · Paper 2 · Q6	(e)	277
4.1-S27	Nov 2010 · Paper 2 · Q2	(a)(ii)	248
3.1-S29	Nov 2010 · Paper 2 · Q7	(d)(iii)	208

4.3 Covalent Network Structures

5.2-S1	May 2025 TZ3 · Paper 2 · Q7	(c)	295
4.3-S1	Nov 2023 TZ1 · Paper 3 · Q3	(b)	269
7.2-S5	May 2023 TZ2 · Paper 2 · Q3	(a)(iii)	483
4.3-S2	May 2023 TZ2 · Paper 3 · Q1	(c)	270
7.2-S6	May 2023 TZ2 · Paper 3 · Q3	(d)	484
4.3-S3	May 2022 TZ2 · Paper 2 · Q8	(a)(i), (a)(ii)	270
4.3-S4	Nov 2020 · Paper 3 · Q4	(a)	273
4.3-S5	May 2019 TZ1 · Paper 3 · Q6	(a), (b)	273
3.1-S15	May 2018 TZ2 · Paper 2 · Q7	(e)	188
4.3-S6	May 2018 TZ2 · Paper 3 · Q1	(a)(i), (a)(ii), (b), (c)	274
4.3-S7	Nov 2015 · Paper 3 · Q13	(a)	275
9.3-S11	May 2015 TZ1 · Paper 2 · Q5	(g)	670
4.3-S8	May 2015 TZ2 · Paper 3 · Q16	(c)	275
4.3-S9	Nov 2014 · Paper 2 · Q5	(a), (b)	276
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(b)(iii)	202
4.1-S23	May 2012 TZ2 · Paper 2 · Q7	(b)(iii)	245
1.2-S50	Nov 2011 · Paper 2 · Q1	(c)(ii)	85
4.3-S10	May 2011 TZ1 · Paper 2 · Q6	(a), (b)(i), (b)(ii)	277
5.2-S26	May 2010 TZ2 · Paper 2 · Q3	(c)	319

5.1 Molecular Polarity

3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(a)(iv)	171
3.1-S3	May 2025 TZ3 · Paper 2 · Q10	(a)(iv)	173
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(b)(ii), (b)(iii)	354
8.2-S2	May 2023 TZ1 · Paper 2 · Q6	(a)(iii)	596
1.2-S16	Nov 2022 · Paper 2 · Q5	(f)	44
3.1-S10	Nov 2021 · Paper 2 · Q3	(b)(ii)	181
3.1-S15	May 2018 TZ2 · Paper 2 · Q7	(a)(i)	188
3.1-S17	Nov 2017 · Paper 2 · Q4	(b)	191
1.2-S38	May 2016 · Paper 2 · Q1	(a)(iv)	65
7.3-S35	May 2016 · Paper 3 · Q1	(b)(ii)	549
5.1-S1	Nov 2015 · Paper 2 · Q5	(a)(i)	283
5.1-S2	May 2015 TZ1 · Paper 3 · Q10	(b)(i)	283
4.1-S13	May 2015 TZ2 · Paper 2 · Q9	(c)(iii)	238
2.1-S26	Nov 2014 · Paper 2 · Q8	(d)(iii)	140
1.2-S44	Nov 2013 · Paper 2 · Q5	(i)	74
3.1-S24	May 2013 TZ2 · Paper 2 · Q5	(c)(iv)	199
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(a)(iv), (a)(v)	202
3.1-S27	Nov 2011 · Paper 2 · Q6	(c)(iii)	206
2.1-S38	May 2010 TZ2 · Paper 2 · Q6	(b)(i)	150

5.2 Intermolecular Forces

9.3-S1	May 2025 TZ1 · Paper 2 · Q8	(c)(v)	659
3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(a)(v)	171
5.2-S1	May 2025 TZ3 · Paper 2 · Q7	(b)(i), (b)(ii)	295
5.2-S2	Nov 2024 · Paper 2 · Q1	(a)(i)	296
8.2-S1	May 2024 TZ1 · Paper 2 · Q2	(d)(ii), (d)(iii)	594
5.2-S3	May 2024 TZ2 · Paper 2 · Q5	(b)	297
5.2-S4	Nov 2023 TZ1 · Paper 2 · Q1	(c)(i), (c)(ii)	299
2.1-S5	May 2023 TZ1 · Paper 2 · Q2	(c)(iii)	120
5.2-S5	May 2023 TZ1 · Paper 3 · Q9	(b)	300
5.2-S6	May 2022 TZ1 · Paper 2 · Q5	(a)(ii)	300
4.1-S1	May 2022 TZ2 · Paper 2 · Q7	(c)	226
2.1-S13	May 2021 TZ2 · Paper 2 · Q2	(d)	130
7.1-S13	Nov 2020 · Paper 2 · Q4	(c)	430
5.2-S7	Nov 2019 · Paper 3 · Q15	(c)	301
4.1-S5	May 2019 TZ1 · Paper 2 · Q2	(h)	230
5.2-S8	May 2019 TZ1 · Paper 3 · Q10	(a)	301

5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(b)(iii), (d)(iii)	302
5.2-S10	May 2019 TZ2 · Paper 3 · Q11	(c)	304
6.1-S7	Nov 2018 · Paper 2 · Q6	(d)	362
5.2-S11	May 2018 TZ1 · Paper 2 · Q1	(e)(i), (e)(ii)	305
5.2-S12	May 2018 TZ1 · Paper 3 · Q1	(a)(i), (a)(ii)	307
3.1-S15	May 2018 TZ2 · Paper 2 · Q7	(a)(ii)	188
5.2-S13	May 2017 TZ1 · Paper 3 · Q12	(a)(i)	309
4.1-S9	May 2017 TZ2 · Paper 2 · Q4	(a)(ii)	234
9.3-S10	Nov 2016 · Paper 2 · Q1	(d)	669
5.2-S14	Nov 2016 · Paper 3 · Q16	(b)	309
1.2-S38	May 2016 · Paper 2 · Q1	(a)(v)	65
5.2-S15	Nov 2015 · Paper 2 · Q6	(d)(i), (d)(ii)	310
1.2-S41	May 2015 TZ1 · Paper 2 · Q1	(c)	70
5.2-S16	May 2015 TZ1 · Paper 3 · Q19	(b)(i)	312
5.2-S17	May 2015 TZ2 · Paper 2 · Q10	(f)	312
6.1-S11	Nov 2014 · Paper 2 · Q10	(b)	364
2.1-S27	May 2014 TZ1 · Paper 2 · Q5	(c)(i), (c)(ii)	142
5.2-S18	May 2014 TZ1 · Paper 3 · Q6	(a)	314
1.2-S43	May 2014 TZ2 · Paper 2 · Q1	(g)	73
7.1-S27	May 2014 TZ2 · Paper 2 · Q5	(b)(iv)	446
9.2-S11	May 2014 TZ2 · Paper 2 · Q7	(e)(vii)	652
5.2-S19	Nov 2013 · Paper 2 · Q8	(d)	315
3.1-S23	May 2013 TZ1 · Paper 2 · Q6	(c)(v)	197
5.2-S20	May 2013 TZ2 · Paper 2 · Q4	(b), (c)(i), (c)(ii)	315
1.2-S47	Nov 2012 · Paper 2 · Q1	(b)	79
5.2-S21	Nov 2012 · Paper 3 · QC3	(a)(i)	316
5.2-S22	Nov 2012 · Paper 3 · QC4	(b)(i), (b)(ii)	317
5.2-S23	Nov 2012 · Paper 3 · QF3	(a)	317
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(a)(vi)	202
9.3-S14	Nov 2011 · Paper 2 · Q3	(d)	677
5.2-S24	Nov 2011 · Paper 2 · Q9	(b)(iv)	318
5.2-S25	May 2011 TZ2 · Paper 2 · Q4	whole question	319
4.1-S27	Nov 2010 · Paper 2 · Q2	(d)(iv)	248
3.1-S29	Nov 2010 · Paper 2 · Q7	(b)	208
5.2-S26	May 2010 TZ2 · Paper 2 · Q3	(b)	319

5.3 Separation Techniques

5.3-S1	May 2025 TZ1 · Paper 1B · Q2	(a), (e)	322
1.2-S9	Nov 2024 · Paper 3 · Q16	(a)(iii)	39
5.3-S2	May 2024 TZ1 · Paper 3 · Q9	(a)	323
5.3-S3	May 2023 TZ1 · Paper 3 · Q8	(a)(i)	323
5.3-S4	May 2023 TZ2 · Paper 3 · Q7	(b)	324
5.3-S5	Nov 2020 · Paper 3 · Q1	(a), (b), (c)	325
5.3-S6	Nov 2020 · Paper 3 · Q6	(a)(i)	326
5.3-S7	Nov 2020 · Paper 3 · Q15	(b)	327
5.3-S8	Nov 2020 · Paper 3 · Q18	(b)	327
5.3-S9	May 2019 TZ1 · Paper 3 · Q15	(c)(i), (c)(ii)	329
5.3-S10	May 2018 TZ1 · Paper 3 · Q6	(c)(ii)	330
7.3-S23	May 2018 TZ1 · Paper 3 · Q12	(a)	540
5.3-S11	May 2018 TZ1 · Paper 3 · Q20	(b)	330
1.2-S37	Nov 2016 · Paper 3 · Q25	(b)	64
5.3-S12	May 2016 · Paper 3 · Q13	(b)	331
5.3-S13	May 2015 TZ2 · Paper 3 · Q6	(a)(ii)	332
5.3-S14	Nov 2014 · Paper 3 · Q3	(a), (b)	333
5.3-S15	May 2014 TZ1 · Paper 3 · Q4	(a), (b), (c)	333
5.3-S16	May 2014 TZ2 · Paper 3 · Q2	(a)(i), (a)(ii)	335
5.3-S17	May 2013 TZ2 · Paper 3 · QA1	(a), (b)	335
5.3-S18	May 2012 TZ1 · Paper 3 · QA3	(a), (b)	336
5.3-S19	May 2012 TZ2 · Paper 3 · QA2	(a)	337
5.3-S20	May 2011 TZ1 · Paper 3 · QA3	(a), (b)	337

6.1 Redox Reactions & Oxidation Numbers

3.1-S1	May 2025 TZ1 · Paper 2 · Q6	(a)(i)	169
3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(c)(ii)	171
3.1-S4	Nov 2024 · Paper 2 · Q4	(d)(iv)	174
1.2-S7	Nov 2024 · Paper 2 · Q7	(a)	37
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(a)(i), (a)(ii)	354
9.3-S3	Nov 2023 TZ1 · Paper 2 · Q2	(g)	661
6.1-S2	Nov 2023 TZ1 · Paper 3 · Q1	(d)(i)	356
7.1-S8	May 2023 TZ1 · Paper 2 · Q4	(a)	423
6.1-S3	May 2023 TZ1 · Paper 3 · Q2	(a)(i)	357
6.1-S4	May 2023 TZ1 · Paper 3 · Q3	(a)(i)	358
1.2-S17	May 2022 TZ1 · Paper 2 · Q1	(d)(iii)	45
1.2-S18	May 2022 TZ2 · Paper 2 · Q1	(c)	47
2.1-S9	May 2022 TZ2 · Paper 2 · Q5	(b)	127
2.1-S12	May 2021 TZ1 · Paper 2 · Q1	(e)(ii)	129
6.1-S5	May 2021 TZ1 · Paper 2 · Q3	(a)	359
1.2-S20	Nov 2020 · Paper 2 · Q1	(b)(v), (b)(vi)	49
6.1-S6	Nov 2019 · Paper 3 · Q2	(a)(ii), (a)(iii), (b)	361
4.1-S5	May 2019 TZ1 · Paper 2 · Q2	(g)	230
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(k)	693
5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(d)(ii)	302
6.1-S7	Nov 2018 · Paper 2 · Q6	(a)(iii)	362
2.1-S18	Nov 2017 · Paper 2 · Q3	(d)(ii)	134
6.2-S2	Nov 2017 · Paper 3 · Q14	(b)	379
7.1-S21	May 2017 TZ1 · Paper 2 · Q3	(a)	437
4.1-S9	May 2017 TZ2 · Paper 2 · Q4	(a)(iii), (a)(iv)	234
9.3-S10	Nov 2016 · Paper 2 · Q1	(c)	669
6.1-S8	Nov 2016 · Paper 3 · Q1	(a)(i)	363
1.2-S38	May 2016 · Paper 2 · Q1	(b)(iii), (b)(iv)	65
1.2-S39	Nov 2015 · Paper 2 · Q1	(a)(i), (a)(ii)	68
6.1-S9	Nov 2015 · Paper 3 · Q9	(a)(ii)	363
7.2-S14	May 2015 TZ1 · Paper 2 · Q6	(e)	495
6.2-S5	May 2015 TZ1 · Paper 2 · Q7	(a)(ii)	381
6.1-S10	May 2015 TZ1 · Paper 3 · Q9	(a)(i)	364
7.1-S25	May 2015 TZ2 · Paper 2 · Q8	(c)(i), (d)(ii)	443
6.2-S6	May 2015 TZ2 · Paper 3 · Q19	(a)(ii)	382
6.1-S11	Nov 2014 · Paper 2 · Q10	(a)(i), (a)(ii), (a)(iv)	364
7.1-S27	May 2014 TZ2 · Paper 2 · Q5	(c)(ii)	446
7.1-S29	Nov 2013 · Paper 2 · Q6	(b)(i)	449
6.1-S12	Nov 2013 · Paper 3 · Q9	(b)(iv)	366
6.1-S13	May 2013 TZ1 · Paper 2 · Q4	(a), (b), (c)	367
7.2-S22	May 2013 TZ1 · Paper 2 · Q7	(b)(iv)	507
1.1-S4	May 2013 TZ1 · Paper 3 · QD1	(b)(iii)	6
1.2-S46	May 2013 TZ2 · Paper 2 · Q1	(e)(i), (e)(ii)	77
7.2-S23	May 2013 TZ2 · Paper 2 · Q7	(a), (b)(ii)	510
6.1-S14	Nov 2012 · Paper 2 · Q4	(a)(i), (a)(ii), (a)(iii)	368
6.1-S15	Nov 2012 · Paper 3 · QD2	(b)(i)	373
6.1-S16	May 2012 TZ1 · Paper 2 · Q6	(e)(i), (e)(ii), (f)(i)	373
6.1-S17	May 2012 TZ2 · Paper 2 · Q1	(c)(i)	375
7.1-S32	Nov 2011 · Paper 2 · Q7	(a), (c)(i)	456
7.2-S26	May 2011 TZ1 · Paper 2 · Q9	(e)(i)	516
6.1-S18	May 2011 TZ2 · Paper 2 · Q2	(a)(ii)	376
7.1-S33	May 2011 TZ2 · Paper 2 · Q7	(d)(ii)	458
1.2-S53	May 2010 TZ1 · Paper 2 · Q1	(a)(i), (a)(ii)	87
7.1-S36	May 2010 TZ2 · Paper 2 · Q8	(a)(i)	463

6.2 Balancing Redox Reactions

3.1-S1	May 2025 TZ1 · Paper 2 · Q6	(a)(iv)	169
1.2-S19	Nov 2021 · Paper 2 · Q6	(c)(iv)	48
7.2-S10	May 2019 TZ2 · Paper 2 · Q4	(e)(ii)	489
3.1-S14	Nov 2018 · Paper 2 · Q3	(d)(i), (d)(ii)	186
6.2-S1	May 2018 TZ2 · Paper 2 · Q4	(a)	379

6.2-S2	Nov 2017 · Paper 3 · Q14	(a)	379
7.1-S21	May 2017 TZ1 · Paper 2 · Q3	(c)(i)	437
6.2-S3	May 2017 TZ2 · Paper 2 · Q2	(a)(i), (a)(ii)	380
6.2-S4	May 2016 · Paper 3 · Q3	(a)	381
6.2-S5	May 2015 TZ1 · Paper 2 · Q7	(a)(iii), (a)(iv)	381
6.1-S10	May 2015 TZ1 · Paper 3 · Q9	(a)(ii)	364
7.1-S25	May 2015 TZ2 · Paper 2 · Q8	(d)(i)	443
6.2-S6	May 2015 TZ2 · Paper 3 · Q19	(a)(i)	382
6.1-S11	Nov 2014 · Paper 2 · Q10	(a)(iii)	364
6.2-S7	Nov 2014 · Paper 3 · Q24	(a)	382
7.2-S19	May 2014 TZ1 · Paper 2 · Q6	(a)(i), (a)(ii)	500
7.1-S27	May 2014 TZ2 · Paper 2 · Q5	(c)(i)	446
6.1-S13	May 2013 TZ1 · Paper 2 · Q4	(d)	367
7.2-S23	May 2013 TZ2 · Paper 2 · Q7	(b)(i)	510
6.1-S14	Nov 2012 · Paper 2 · Q4	(a)(iv)	368
6.1-S17	May 2012 TZ2 · Paper 2 · Q1	(c)(ii)	375
6.2-S8	May 2012 TZ2 · Paper 3 · QF3	(a)	383
7.1-S32	Nov 2011 · Paper 2 · Q7	(c)(ii)	456
7.2-S26	May 2011 TZ1 · Paper 2 · Q9	(e)(ii)	516
6.2-S9	May 2011 TZ1 · Paper 3 · QB5	(a), (b)	383
7.1-S33	May 2011 TZ2 · Paper 2 · Q7	(d)(iii), (d)(iv), (d)(v)	458
6.2-S10	May 2010 TZ1 · Paper 3 · QC1	(b)(i), (b)(ii)	384

6.3 Reactivity

2.1-S3	May 2025 TZ2 · Paper 2 · Q2	(b)(vii)	116
7.2-S3	May 2024 TZ2 · Paper 2 · Q7	(h)	477
6.1-S3	May 2023 TZ1 · Paper 3 · Q2	(a)(ii)	357
6.3-S1	May 2022 TZ1 · Paper 2 · Q2	(a)	390
7.2-S10	May 2019 TZ2 · Paper 2 · Q4	(c)	489
6.3-S2	May 2017 TZ2 · Paper 3 · Q5	(b)(i), (b)(ii)	390
6.3-S3	Nov 2015 · Paper 2 · Q7	(d)(i), (d)(ii)	391
2.1-S27	May 2014 TZ1 · Paper 2 · Q5	(b)(ii)	142
7.2-S19	May 2014 TZ1 · Paper 2 · Q6	(b)(i)	500
7.1-S27	May 2014 TZ2 · Paper 2 · Q5	(a)(ii)	446
7.2-S22	May 2013 TZ1 · Paper 2 · Q7	(a)(ii)	507
1.2-S49	May 2012 TZ2 · Paper 2 · Q2	(a)	84
7.1-S36	May 2010 TZ2 · Paper 2 · Q8	(a)(iii)	463

7.1 Primary & Secondary Cells

3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(c)(v)	171
7.1-S1	May 2025 TZ3 · Paper 2 · Q6	(a)(i), (a)(ii), (b)	415
7.1-S2	Nov 2024 · Paper 2 · Q8	(a), (b)	417
7.1-S3	Nov 2024 · Paper 3 · Q15	(a), (c)	418
7.1-S4	May 2024 TZ1 · Paper 2 · Q7	(a), (b)(i)	418
7.1-S5	May 2024 TZ1 · Paper 3 · Q16	(b)(i), (b)(ii)	419
7.2-S3	May 2024 TZ2 · Paper 2 · Q7	(e)	477
7.1-S6	May 2024 TZ2 · Paper 3 · Q13	(c)(i), (c)(ii), (c)(iii)	420
7.2-S4	Nov 2023 TZ1 · Paper 2 · Q5	(f)(i), (f)(ii)	480
7.1-S7	Nov 2023 TZ1 · Paper 3 · Q15	(b)	422
7.1-S8	May 2023 TZ1 · Paper 2 · Q4	(b)(i), (b)(ii), (c)(i)	423
6.1-S3	May 2023 TZ1 · Paper 3 · Q2	(b)	357
7.1-S9	Nov 2022 · Paper 2 · Q3	(f)(i), (f)(ii), (f)(iii), (f)(iv), (f)(v)	424
6.3-S1	May 2022 TZ1 · Paper 2 · Q2	(b)(i)	390
7.1-S10	May 2022 TZ2 · Paper 2 · Q3	(a)(i), (a)(ii)	427
7.1-S11	Nov 2021 · Paper 2 · Q8	whole question	428
6.1-S5	May 2021 TZ1 · Paper 2 · Q3	(d)	359
7.1-S12	May 2021 TZ2 · Paper 2 · Q3	(a), (b), (c)	429
7.1-S13	Nov 2020 · Paper 2 · Q4	(d)(i), (d)(ii), (d)(iv)	430
7.1-S14	Nov 2020 · Paper 3 · Q13	(a)	431
7.2-S7	Nov 2019 · Paper 2 · Q6	(e)(i), (e)(ii)	485
7.3-S18	Nov 2019 · Paper 3 · Q20	(b)(i)	535
7.1-S15	May 2019 TZ1 · Paper 2 · Q6	(e), (f)(i)	432
7.2-S10	May 2019 TZ2 · Paper 2 · Q4	(e)(iii)	489

7.1-S16	May 2019 TZ2 · Paper 3 · Q19	(c)	433
7.1-S17	Nov 2018 · Paper 3 · Q15	(a), (b)	433
7.1-S18	May 2018 TZ1 · Paper 2 · Q6	(a), (b), (c), (d)	434
7.1-S19	May 2018 TZ1 · Paper 3 · Q14	(a)(i)	435
7.1-S20	Nov 2017 · Paper 2 · Q7	(a), (b), (c), (d)	436
7.1-S21	May 2017 TZ1 · Paper 2 · Q3	(b)(i), (b)(ii)	437
7.1-S22	May 2017 TZ1 · Paper 3 · Q22	(b)(i), (b)(ii)	439
6.2-S3	May 2017 TZ2 · Paper 2 · Q2	(b)(i)	380
7.1-S23	Nov 2016 · Paper 2 · Q4	(j), (k)(i), (k)(ii)	440
2.1-S20	May 2016 · Paper 2 · Q4	(b)(ii)	136
7.1-S24	Nov 2015 · Paper 3 · Q12	(a), (b), (c)	442
7.2-S14	May 2015 TZ1 · Paper 2 · Q6	(f)(i), (f)(ii)	495
7.1-S25	May 2015 TZ2 · Paper 2 · Q8	(e)(i), (e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi)	443
7.2-S19	May 2014 TZ1 · Paper 2 · Q6	(b)(ii), (b)(iii), (b)(iv)	500
7.1-S26	May 2014 TZ2 · Paper 2 · Q2	(d)(i), (d)(ii)	445
7.1-S27	May 2014 TZ2 · Paper 2 · Q5	(d)(i), (d)(ii)	446
7.1-S28	May 2014 TZ2 · Paper 3 · Q11	(a)(i)	448
7.1-S29	Nov 2013 · Paper 2 · Q6	(a), (c), (d)(i), (d)(ii), (d)(iii)	449
6.1-S13	May 2013 TZ1 · Paper 2 · Q4	(e)(i), (e)(ii)	367
7.2-S23	May 2013 TZ2 · Paper 2 · Q7	(c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii)	510
7.1-S30	May 2012 TZ1 · Paper 2 · Q4	(a), (b), (c)(i), (c)(ii)	452
7.1-S31	May 2012 TZ2 · Paper 3 · QC2	(b)	455
7.1-S32	Nov 2011 · Paper 2 · Q7	(d)(i), (d)(ii), (d)(iii)	456
7.2-S26	May 2011 TZ1 · Paper 2 · Q9	(b)(i), (b)(ii), (b)(iii)	516
7.1-S33	May 2011 TZ2 · Paper 2 · Q7	(a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii)	458
7.1-S34	Nov 2010 · Paper 3 · QC4	(b)	462
7.1-S35	May 2010 TZ1 · Paper 2 · Q6	(a)(i), (a)(ii), (a)(iii), (b)(i), (b)(ii), (c)	462
7.1-S36	May 2010 TZ2 · Paper 2 · Q8	(a)(ii), (a)(iv)	463

7.2 Electrolytic Cells

2.1-S3	May 2025 TZ2 · Paper 2 · Q2	(b)(viii)	116
7.2-S1	Nov 2024 · Paper 2 · Q3	(b)(i), (b)(ii)	475
7.2-S2	May 2024 TZ1 · Paper 3 · Q3	(c)(ii)	476
7.2-S3	May 2024 TZ2 · Paper 2 · Q7	(c), (d)	477
7.2-S4	Nov 2023 TZ1 · Paper 2 · Q5	(b)(i), (b)(ii), (b)(iii)	480
6.1-S3	May 2023 TZ1 · Paper 3 · Q2	(c)	357
7.2-S5	May 2023 TZ2 · Paper 2 · Q3	(a)(i), (a)(ii)	483
7.2-S6	May 2023 TZ2 · Paper 3 · Q3	(b)(i)	484
8.2-S4	Nov 2022 · Paper 2 · Q1	(e)	599
6.3-S1	May 2022 TZ1 · Paper 2 · Q2	(b)(iii)	390
7.1-S10	May 2022 TZ2 · Paper 2 · Q3	(b)(i), (b)(ii)	427
6.1-S5	May 2021 TZ1 · Paper 2 · Q3	(e)	359
2.1-S14	Nov 2020 · Paper 2 · Q6	(c)	131
7.2-S7	Nov 2019 · Paper 2 · Q6	(c)(i), (c)(ii), (c)(iii), (c)(iv)	485
7.2-S8	Nov 2019 · Paper 3 · Q6	(b)(ii)	488
7.2-S9	May 2019 TZ1 · Paper 2 · Q7	whole question	489
7.2-S10	May 2019 TZ2 · Paper 2 · Q4	(b)(ii)	489
1.2-S26	Nov 2018 · Paper 2 · Q1	(d)	55
7.2-S11	Nov 2018 · Paper 3 · Q4	(c)	491
7.2-S12	May 2018 TZ2 · Paper 2 · Q3	(c)(ii), (c)(iii), (c)(iv), (c)(v)	491
7.1-S20	Nov 2017 · Paper 2 · Q7	(e)	436
7.2-S13	Nov 2017 · Paper 3 · Q1	(b)(iii)	493
7.1-S23	Nov 2016 · Paper 2 · Q4	(i)(i), (i)(ii), (i)(iii)	440
6.3-S3	Nov 2015 · Paper 2 · Q7	(e)	391
7.2-S14	May 2015 TZ1 · Paper 2 · Q6	(d)(i), (d)(ii)	495
7.2-S15	May 2015 TZ1 · Paper 3 · Q12	(b), (c)	497
7.2-S16	May 2015 TZ2 · Paper 2 · Q6	(a)	498
7.2-S17	May 2015 TZ2 · Paper 3 · Q13	(b)	498
6.1-S11	Nov 2014 · Paper 2 · Q10	(f)(i), (f)(ii), (g)	364
7.2-S18	Nov 2014 · Paper 3 · Q9	(a)(ii), (a)(iii)	499
7.2-S19	May 2014 TZ1 · Paper 2 · Q6	(c)(i), (c)(ii), (d)(ii), (d)(iii)	500
7.2-S20	May 2014 TZ2 · Paper 2 · Q8	(g)(i), (g)(ii), (g)(iii), (g)(iv)	503
7.2-S21	May 2014 TZ2 · Paper 3 · Q10	(a), (b)	507

7.2-S22	May 2013 TZ1 · Paper 2 · Q7	(c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v)	507
7.2-S23	May 2013 TZ2 · Paper 2 · Q7	(e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi), (e)(vii)	510
7.2-S24	May 2013 TZ2 · Paper 3 · QC1	(a)(iii)	514
6.1-S14	Nov 2012 · Paper 2 · Q4	(b)(i), (b)(ii), (b)(iii)	368
7.1-S30	May 2012 TZ1 · Paper 2 · Q4	(d)(i), (d)(iv)	452
7.1-S32	Nov 2011 · Paper 2 · Q7	(f)	456
7.2-S25	Nov 2011 · Paper 3 · QC5	(a)(ii)	515
7.2-S26	May 2011 TZ1 · Paper 2 · Q9	(a)(i), (a)(ii), (c)	516
7.2-S27	May 2011 TZ1 · Paper 3 · QC4	(b)	519
7.1-S33	May 2011 TZ2 · Paper 2 · Q7	(e)(ii)	458
7.2-S28	May 2011 TZ2 · Paper 3 · QC1	(a)(ii)	520
2.1-S37	Nov 2010 · Paper 2 · Q4	(b)(i), (b)(ii), (b)(iii)	149
7.2-S29	Nov 2010 · Paper 3 · QC5	(a)(iii)	520
7.1-S35	May 2010 TZ1 · Paper 2 · Q6	(d)(i), (d)(ii)	462
7.1-S36	May 2010 TZ2 · Paper 2 · Q8	(b)(ii), (c)	463

7.3 Energy From Fuels

7.3-S1	May 2025 TZ3 · Paper 2 · Q9	(b)(i)	523
7.3-S2	Nov 2024 · Paper 3 · Q14	(a)(i), (a)(ii)	524
8.2-S1	May 2024 TZ1 · Paper 2 · Q2	(c)	594
7.3-S3	May 2024 TZ1 · Paper 3 · Q13	(a), (b), (c), (f)	525
7.3-S4	May 2024 TZ1 · Paper 3 · Q14	(a)(i), (a)(ii)	527
7.1-S6	May 2024 TZ2 · Paper 3 · Q13	(a), (b)(i)	420
7.3-S5	Nov 2023 TZ1 · Paper 3 · Q12	(c)(ii)	527
7.3-S6	Nov 2023 TZ1 · Paper 3 · Q13	(b)	528
7.3-S7	Nov 2023 TZ1 · Paper 3 · Q14	(b), (c)	528
9.2-S3	May 2023 TZ1 · Paper 3 · Q7	(d)(i)	647
7.3-S8	May 2023 TZ1 · Paper 3 · Q13	(d)(i)	529
7.3-S9	May 2023 TZ2 · Paper 2 · Q6	(d)(i)	530
7.2-S6	May 2023 TZ2 · Paper 3 · Q3	(b)(iii)	484
7.3-S10	May 2023 TZ2 · Paper 3 · Q15	(d)(i)	530
7.3-S11	May 2023 TZ2 · Paper 3 · Q16	(a)(ii), (c)(i)	530
1.2-S17	May 2022 TZ1 · Paper 2 · Q1	(a)(i)	45
7.3-S12	May 2021 TZ1 · Paper 2 · Q5	(c)	531
7.3-S13	May 2021 TZ2 · Paper 2 · Q5	(a)(i)	532
7.3-S14	Nov 2020 · Paper 3 · Q11	(a), (c), (d)	532
7.3-S15	Nov 2019 · Paper 2 · Q3	(c)(i)	533
7.3-S16	Nov 2019 · Paper 3 · Q16	(a)	534
7.3-S17	Nov 2019 · Paper 3 · Q19	(a)	534
7.3-S18	Nov 2019 · Paper 3 · Q20	(a), (c)	535
4.1-S5	May 2019 TZ1 · Paper 2 · Q2	(f)(ii)	230
5.3-S9	May 2019 TZ1 · Paper 3 · Q15	(a)	329
7.3-S19	May 2019 TZ1 · Paper 3 · Q17	(b)(i), (b)(ii), (b)(iii), (c)	536
5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(a)	302
7.3-S20	May 2019 TZ2 · Paper 3 · Q15	(b)	537
7.3-S21	May 2019 TZ2 · Paper 3 · Q17	whole question	538
7.3-S22	Nov 2018 · Paper 3 · Q13	(b), (c)	539
5.2-S11	May 2018 TZ1 · Paper 2 · Q1	(f)	305
7.3-S23	May 2018 TZ1 · Paper 3 · Q12	(b)(i), (b)(ii)	540
7.3-S24	May 2018 TZ2 · Paper 3 · Q13	(a)	541
7.3-S25	Nov 2017 · Paper 3 · Q7	(e)	541
7.3-S26	Nov 2017 · Paper 3 · Q16	(a), (b), (c)	542
7.3-S27	Nov 2017 · Paper 3 · Q17	(c)	543
7.3-S28	Nov 2017 · Paper 3 · Q20	(a), (b)	543
7.3-S29	May 2017 TZ1 · Paper 2 · Q4	(a)	544
7.3-S30	May 2017 TZ1 · Paper 3 · Q20	(c)	545
7.1-S22	May 2017 TZ1 · Paper 3 · Q22	(a)	439
7.3-S31	May 2017 TZ2 · Paper 3 · Q17	(a), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii)	545
7.3-S32	May 2017 TZ2 · Paper 3 · Q18	(a)(i)	547
7.3-S33	Nov 2016 · Paper 3 · Q6	(c)(i)	547
7.3-S34	Nov 2016 · Paper 3 · Q17	(a)(i), (a)(ii), (b)(i)	548
7.3-S35	May 2016 · Paper 3 · Q1	(a)	549
7.3-S36	May 2016 · Paper 3 · Q15	(b)(i), (b)(ii)	551

7.3-S37	May 2016 · Paper 3 · Q20	(a), (b)	552
7.3-S38	Nov 2015 · Paper 2 · Q3	(a)	553
7.3-S39	Nov 2015 · Paper 3 · Q23	(b)(i)	554
5.2-S17	May 2015 TZ2 · Paper 2 · Q10	(g)	312
7.3-S40	May 2015 TZ2 · Paper 3 · Q14	(a)	554
7.3-S41	May 2015 TZ2 · Paper 3 · Q22	(b)(i)	555
7.3-S42	Nov 2014 · Paper 3 · Q21	(a)	555
7.2-S19	May 2014 TZ1 · Paper 2 · Q6	(a)(v)	500
7.3-S43	May 2014 TZ1 · Paper 3 · Q9	(a), (b)(iii), (b)(iv), (b)(v)	555
7.3-S44	May 2014 TZ1 · Paper 3 · Q15	(c)(ii)	556
7.1-S28	May 2014 TZ2 · Paper 3 · Q11	(b)	448
7.3-S45	May 2012 TZ1 · Paper 2 · Q2	(b)	557
7.3-S46	May 2012 TZ1 · Paper 3 · QC2	(a), (c)	557
7.1-S31	May 2012 TZ2 · Paper 3 · QC2	(c)	455
9.3-S14	Nov 2011 · Paper 2 · Q3	(b)	677
7.3-S47	Nov 2011 · Paper 3 · QC1	(a)(i)	558
7.3-S48	Nov 2011 · Paper 3 · QC2	(d)	558
7.3-S49	Nov 2011 · Paper 3 · QE2	(a)	559
7.3-S50	May 2011 TZ1 · Paper 3 · QE3	(a)	559
7.3-S51	Nov 2010 · Paper 3 · QC1	whole question	559
7.3-S52	May 2010 TZ2 · Paper 3 · QE1	(a), (b)	560

8.1 Thermochemical Equations

9.3-S2	May 2025 TZ2 · Paper 2 · Q3	(a)(iv)	660
5.2-S2	Nov 2024 · Paper 2 · Q1	(a)(ii)	296
3.1-S4	Nov 2024 · Paper 2 · Q4	(d)(i)	174
5.2-S3	May 2024 TZ2 · Paper 2 · Q5	(c)(ii)	297
4.3-S3	May 2022 TZ2 · Paper 2 · Q8	(f)(ii)	270
9.2-S4	May 2021 TZ2 · Paper 2 · Q1	(b)(iv)	648
5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(c)(iii)	302
8.1-S1	May 2017 TZ2 · Paper 2 · Q5	(b)(iv)	572
4.1-S11	May 2016 · Paper 2 · Q3	(b)(i)	236
8.1-S2	May 2016 · Paper 3 · Q11	(b)	572
8.1-S3	May 2015 TZ2 · Paper 2 · Q7	(a)(iii)	573
8.1-S4	Nov 2014 · Paper 2 · Q11	(e)(i), (e)(ii)	573
8.1-S5	May 2013 TZ1 · Paper 2 · Q3	(a)	576
8.1-S6	May 2013 TZ2 · Paper 2 · Q2	(a)	577
1.2-S47	Nov 2012 · Paper 2 · Q1	(a)(iv)	79
8.1-S7	Nov 2012 · Paper 2 · Q6	(b)(i)	577
8.1-S8	May 2012 TZ2 · Paper 2 · Q8	(a)(i), (a)(ii), (a)(iii)	578
8.1-S9	Nov 2011 · Paper 2 · Q8	(a), (c)(i)	579

8.2 Calorimetry

3.1-S2	May 2025 TZ2 · Paper 2 · Q1	(b)(vi)	171
7.3-S1	May 2025 TZ3 · Paper 2 · Q9	(c)	523
8.2-S1	May 2024 TZ1 · Paper 2 · Q2	(a)(i), (b)(i), (b)(ii), (b)(iii)	594
8.2-S2	May 2023 TZ1 · Paper 2 · Q6	(b)(i), (b)(ii), (b)(v)	596
8.2-S3	May 2023 TZ2 · Paper 2 · Q4	(c)	598
8.2-S4	Nov 2022 · Paper 2 · Q1	(d)(i), (d)(ii), (d)(iv)	599
8.2-S5	Nov 2021 · Paper 2 · Q7	(a)	601
6.1-S5	May 2021 TZ1 · Paper 2 · Q3	(c)	359
8.2-S6	May 2019 TZ1 · Paper 3 · Q2	(a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv)	602
8.2-S7	May 2019 TZ1 · Paper 3 · Q8	(c)	604
1.2-S26	Nov 2018 · Paper 2 · Q1	(b)(i), (b)(ii)	55
8.2-S8	Nov 2017 · Paper 2 · Q1	(a), (b), (d)(i), (d)(ii)	605
8.2-S9	May 2017 TZ1 · Paper 3 · Q2	(a), (b)	607
8.2-S10	May 2017 TZ1 · Paper 3 · Q3	whole question	608
8.2-S11	May 2017 TZ1 · Paper 3 · Q5	(a), (b), (c), (d), (f)	609
9.1-S2	May 2017 TZ2 · Paper 2 · Q9	(a)(ii)	638
1.2-S38	May 2016 · Paper 2 · Q1	(d)(i)	65
9.3-S11	May 2015 TZ1 · Paper 2 · Q5	(c)	670
8.2-S12	May 2015 TZ2 · Paper 2 · Q1	(a)(i), (b)(i), (b)(ii), (b)(iii), (b)(iv)	611
8.1-S4	Nov 2014 · Paper 2 · Q11	(e)(iii)	573

8.2-S13	May 2014 TZ1 · Paper 2 · Q1	(a)(ii), (b)(ii), (c)	613
9.2-S11	May 2014 TZ2 · Paper 2 · Q7	(e)(vi)	652
8.2-S14	May 2013 TZ1 · Paper 2 · Q8	(a)(i), (a)(ii)	616
8.1-S8	May 2012 TZ2 · Paper 2 · Q8	(c)	578
8.2-S15	Nov 2011 · Paper 3 · QB1	whole question	618
8.2-S16	May 2011 TZ1 · Paper 2 · Q1	(b)(i), (b)(ii), (b)(iii), (c)(ii)	619
8.2-S17	May 2011 TZ2 · Paper 2 · Q1	(a)(ii), (a)(iii), (a)(iv), (b)(ii), (c)(i), (c)(ii)	622
8.2-S18	Nov 2010 · Paper 2 · Q1	(a), (b)(i), (b)(ii), (d)	626
8.2-S19	Nov 2010 · Paper 3 · QB1	(a)	628
8.2-S20	May 2010 TZ2 · Paper 3 · QB1	whole question	628

9.1 Hess's Law

9.1-S1	Nov 2018 · Paper 2 · Q7	(c), (d)	637
3.1-S18	May 2017 TZ1 · Paper 2 · Q5	(g)	192
9.1-S2	May 2017 TZ2 · Paper 2 · Q9	(a)(i)	638
8.1-S4	Nov 2014 · Paper 2 · Q11	(e)(iv)	573
8.2-S13	May 2014 TZ1 · Paper 2 · Q1	(b)(i)	613
8.2-S17	May 2011 TZ2 · Paper 2 · Q1	(b)(iii)	622

9.2 Enthalpies of Formation & Combustion

9.3-S1	May 2025 TZ1 · Paper 2 · Q8	(f)(ii)	659
9.2-S1	May 2025 TZ3 · Paper 2 · Q5	(a)(i)	646
7.2-S1	Nov 2024 · Paper 2 · Q3	(e)(i)	475
5.2-S3	May 2024 TZ2 · Paper 2 · Q5	(c)(i)	297
9.2-S2	Nov 2023 TZ1 · Paper 2 · Q4	(a)(i)	647
8.2-S2	May 2023 TZ1 · Paper 2 · Q6	(a)(i)	596
9.2-S3	May 2023 TZ1 · Paper 3 · Q7	(a)(ii)	647
3.1-S8	May 2023 TZ2 · Paper 2 · Q2	(a)(iv)	179
8.2-S4	Nov 2022 · Paper 2 · Q1	(f)(iii)	599
3.1-S10	Nov 2021 · Paper 2 · Q3	(c)(i)	181
9.2-S4	May 2021 TZ2 · Paper 2 · Q1	(b)(i)	648
7.3-S13	May 2021 TZ2 · Paper 2 · Q5	(a)(ii)	532
9.3-S7	Nov 2020 · Paper 2 · Q3	(b)	666
7.2-S7	Nov 2019 · Paper 2 · Q6	(a)(ii)	485
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(g)(i)	693
9.1-S1	Nov 2018 · Paper 2 · Q7	(b)	637
9.3-S8	May 2018 TZ1 · Paper 2 · Q3	(c)(ii)	666
9.3-S9	May 2018 TZ2 · Paper 2 · Q5	(b)(i), (b)(ii)	668
9.2-S5	May 2018 TZ2 · Paper 3 · Q7	(e)	650
9.2-S6	Nov 2017 · Paper 2 · Q5	(a)	650
7.3-S29	May 2017 TZ1 · Paper 2 · Q4	(c)(ii)	544
9.2-S7	May 2017 TZ2 · Paper 2 · Q6	(b)	651
9.3-S10	Nov 2016 · Paper 2 · Q1	(b)(i)	669
9.2-S8	May 2016 · Paper 2 · Q2	(a)(iii)	651
9.2-S9	May 2015 TZ2 · Paper 2 · Q3	(a)	652
9.3-S12	Nov 2014 · Paper 2 · Q2	(b)(i), (b)(ii)	672
9.2-S10	May 2014 TZ1 · Paper 2 · Q4	(b)	652
9.2-S11	May 2014 TZ2 · Paper 2 · Q7	(e)(iv)	652
8.2-S14	May 2013 TZ1 · Paper 2 · Q8	(b)(i)	616
3.1-S24	May 2013 TZ2 · Paper 2 · Q5	(a)(iii), (a)(iv)	199
9.3-S13	Nov 2012 · Paper 2 · Q5	(c)(ii), (c)(iii)	673
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(b)(i), (b)(ii)	202
9.2-S12	Nov 2011 · Paper 2 · Q5	(b)(i), (b)(ii)	654
8.2-S16	May 2011 TZ1 · Paper 2 · Q1	(a), (c)(i)	619
9.2-S13	May 2011 TZ2 · Paper 2 · Q6	(b)(i), (b)(ii)	655
3.1-S29	Nov 2010 · Paper 2 · Q7	(c)(i)	208

9.3 Bond Enthalpies

9.3-S1	May 2025 TZ1 · Paper 2 · Q8	(f)(i), (f)(iii)	659
9.3-S2	May 2025 TZ2 · Paper 2 · Q3	(a)(iii)	660
7.3-S1	May 2025 TZ3 · Paper 2 · Q9	(b)(ii)	523
6.1-S1	May 2024 TZ1 · Paper 2 · Q4	(b)(vi)	354
9.3-S3	Nov 2023 TZ1 · Paper 2 · Q2	(c)(i), (c)(iii)	661

7.3-S9	May 2023 TZ2 · Paper 2 · Q6	(d)(ii)	530
9.3-S4	May 2022 TZ1 · Paper 2 · Q3	(b)(i), (b)(ii)	662
4.3-S3	May 2022 TZ2 · Paper 2 · Q8	(f)(i)	270
9.3-S5	May 2021 TZ1 · Paper 2 · Q4	(d)(i), (d)(ii)	663
9.3-S6	May 2021 TZ2 · Paper 2 · Q4	(b)	664
9.3-S7	Nov 2020 · Paper 2 · Q3	(a)	666
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(g)(ii)	693
5.2-S9	May 2019 TZ2 · Paper 2 · Q1	(c)(ii)	302
9.3-S8	May 2018 TZ1 · Paper 2 · Q3	(c)(i), (c)(iii)	666
9.3-S9	May 2018 TZ2 · Paper 2 · Q5	(a), (b)(iii)	668
3.1-S18	May 2017 TZ1 · Paper 2 · Q5	(f)	192
9.3-S10	Nov 2016 · Paper 2 · Q1	(a)(iii), (b)(ii)	669
7.3-S38	Nov 2015 · Paper 2 · Q3	(b)	553
9.3-S11	May 2015 TZ1 · Paper 2 · Q5	(b)(i), (b)(ii)	670
9.3-S12	Nov 2014 · Paper 2 · Q2	(a), (c)	672
1.2-S44	Nov 2013 · Paper 2 · Q5	(c)	74
9.3-S13	Nov 2012 · Paper 2 · Q5	(a), (b)(i)	673
3.1-S25	May 2012 TZ1 · Paper 2 · Q5	(c)	202
9.3-S14	Nov 2011 · Paper 2 · Q3	(a), (c)	677
3.1-S29	Nov 2010 · Paper 2 · Q7	(c)(ii), (c)(iii)	208
9.3-S15	May 2010 TZ1 · Paper 2 · Q5	(a)(i)	678
1.2-S55	May 2010 TZ2 · Paper 2 · Q4	(c)	89

9.4 Born-Haber Cycles

9.4-S1	May 2025 TZ1 · Paper 2 · Q4	(c)(i), (c)(ii)	692
2.1-S3	May 2025 TZ2 · Paper 2 · Q2	(b)(iii)	116
7.2-S3	May 2024 TZ2 · Paper 2 · Q7	(f)	477
2.1-S6	May 2023 TZ1 · Paper 3 · Q1	(a)	123
9.4-S2	May 2023 TZ2 · Paper 2 · Q8	(b)	693
9.4-S3	May 2019 TZ1 · Paper 2 · Q3	(d)(i), (d)(ii), (d)(iii)	693
2.1-S20	May 2016 · Paper 2 · Q4	(a)(i)	136
7.2-S14	May 2015 TZ1 · Paper 2 · Q6	(g)	495
7.2-S20	May 2014 TZ2 · Paper 2 · Q8	(c), (d)	503
9.4-S4	May 2012 TZ2 · Paper 2 · Q3	(a), (b), (c)(i), (c)(ii)	697
2.1-S37	Nov 2010 · Paper 2 · Q4	(c)(i), (c)(ii), (c)(iv)	149